BOUNDEDNESS OF THREEFOLDS OF GENERAL TYPE IN LARGE CHARACTERISTIC

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Abstract. We prove the analogues of theorems of Hacon, McKernan, and Xu in characteristic zero for threefold pairs with coefficients in a fixed DCC set, over algebraically closed fields of large characteristic: effective birationality, the descending chain condition for volumes, and boundedness of stable pairs of fixed volume. The bound on the characteristic depends neither on the coefficient set nor on the volume, and the multiple of the log canonical divisor that defines a birational map depends only on the coefficient set.

Contents

1. Introduction 1
2. Preliminaries 8
3. A uniform bound on anti-canonical volumes 10
4. Shrinking reduced boundaries 21
5. Regular surfaces over arbitrary fields 24
6. Birational linear systems from one non-zero section 41
7. Birational linear systems from pencils of surfaces 52
8. The first non-zero section on models with bounded denominators 57
9. DCC of volumes and log birational boundedness 98
10. Effective birationality for reduced boundaries 119
11. Effective birationality for DCC coefficients 128
12. The descending chain condition for volumes 134
13. Boundedness at a fixed volume 135
References 160

1. Introduction

We work over an algebraically closed field k of characteristic $p > 0$.

A *DCC set* is a set of real numbers that contains no infinite strictly decreasing sequence. In 1994, Kollár proposed the following conjecture for surfaces.

Conjecture 1.1 ([Kol94, Conjecture 1.3]). *Let $I \subset [0,1]$ be a DCC set, and let $\mathcal{C}$ be the set of pairs (X, B) such that X is a proper surface with quotient singularities, B is an $\mathbb{R}$-divisor on X, (X, B) is lc, $B \in I$, and $K_X + B$ is ample. Then:*

(1) The set $\{(K_X + B)^2 \mid (X, B) \in \mathcal{C}\}$ satisfies the DCC.
(2) For any real number $V > 0$, the pairs $(X, B) \in \mathcal{C}$ with $(K_X + B)^2 = V$ form finitely many deformation types.

He also wrote that the conjectures of [Kol94] can be posed for varieties of any dimension.

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In characteristic zero, Hacon, McKernan and Xu proved the analogue of Conjecture 1.1(1) in any dimension, together with a uniform birationality statement. Let d be a positive integer, $I \subset [0,1]$ a DCC set, and $\mathcal{D}$ the set of projective lc pairs (X,B) of dimension d over an algebraically closed field of characteristic zero such that $B \in I$. By [HMX14, Theorem 1.3], the following hold:

(1) The set $\{\mathrm{vol}(K_X+B) \mid (X,B) \in \mathcal{D}\}$ satisfies the DCC.
(2) There exists a real number $\delta > 0$ depending only on d and I such that $\mathrm{vol}(K_X+B) \geq \delta$ for any $(X,B) \in \mathcal{D}$ with K_X+B big.
(3) There exists a positive integer m depending only on d and I such that $|\lfloor m(K_X+B) \rfloor|$ defines a birational map for any $(X,B) \in \mathcal{D}$ with K_X+B big.

Hacon, McKernan and Xu write that [HMX14, Theorem 1.3] was conjectured by Alexeev [Ale94, 11.3] and Kollár [Kol94]. For surfaces, Alexeev proved (1) for the pairs with K_X+B ample [Ale94, Theorem 8.2]. In dimension 3, [BL26, Theorem 1.6] proves an explicit lower bound for $\mathrm{vol}(K_X+B)$ when $B \neq 0$ is reduced and K_X+B is ample.

Hacon, McKernan and Xu then proved the boundedness counterpart of Conjecture 1.1(2) in any dimension, again in characteristic zero. Let d be a positive integer, V a positive rational number, and $I \subset [0,1]$ a DCC set. By [HMX18, Theorem 1.1], the set of projective slc pairs (X,B) of dimension d such that $B \in I$, K_X+B is an ample $\mathbb{Q}$-divisor, and

$$(K_X+B)^d = V$$

is bounded. In particular, there exists a finite set I_0 such that $B \in I_0$ for any such pair (X,B).

For surfaces, the following statements hold with constants independent of the characteristic and of the algebraically closed ground field. Let $I \subset [0,1] \cap \mathbb{Q}$ be a DCC set and $V > 0$ a real number, and consider 2-dimensional slc models (X,B), that is, projective slc surface pairs with K_X+B ample, over an algebraically closed field of any characteristic such that $B \in I$. Hacon and Kovács [HK19, Theorems 1 and 2] proved (1) and (2) below, attributing them to Alexeev [Ale94, AM04], who proved them over a fixed algebraically closed field:

(1) There exists a positive integer r depending only on I and V such that $r(K_X+B)$ is very ample for any such (X,B) with $(K_X+B)^2 = V$.
(2) The set of the numbers $(K_X+B)^2$ of all such (X,B) satisfies the DCC and has a positive lower bound depending only on I.

The moduli stack of stable surfaces of fixed volume is proper over $\mathbb{Z}[1/30]$, assuming the existence of locally-stable reduction [ABP24, Theorem 1.1]. Over $\mathbb{C}$, $K_X^2 \geq 1/48983$ for any klt stable surface X [LL26, Theorem 1.5], $(K_X+B)^2 \geq 1/462$ for any projective lc surface pair (X,B) with $B \neq 0$ reduced and K_X+B ample [LS23, Theorem 1.4], and $|m(K_X+B)|$ defines a birational map for any integer $m \geq 1882$ and any connected projective slc surface pair (X,B) with B reduced and K_X+B ample [LL26, Corollary 11.9].

For varieties of general type, the characteristic-zero statement is the following: for any positive integer d there exists a positive integer r_d such that $|rK_X|$ defines a birational map for any integer $r \geq r_d$ and any smooth projective variety X of dimension d of general type [HM06, Theorem 1.1], [Tak06, Theorem 1.1]. In dimension 3, one may take $r_3 = 57$ [Che18, Corollary 1.2]. Over $\mathbb{C}$, if X is a normal projective variety of dimension d such that K_X is Cartier, nef, and big, then $|rK_X|$ defines a birational map for any integer $r \geq d+3$ [Liu26, Theorem 1.6]. Das pointed out in 2020 that the analogue of the existence of r_d in positive characteristic was unknown even in dimension 3 [Das22, p. 66]. In dimension 3, Zhang [Zha22, Theorem 1.3] proved the following partial result. Let X be a minimal terminal threefold of general type over an algebraically closed field of characteristic p. For a positive integer n, let

$$S^0_-(X, K_X+nK_X) \subset H^0(X, \mathcal{O}_X((n+1)K_X))$$

be the subspace of Frobenius-stable sections [Zha22, Section 2.2]. Then $S^0_-(X, K_X+nK_X)$ defines a birational map in each of the following cases:

(1) $\dim \mathrm{Pic}^0(X) > 0$ and $n \geq 21$, or $\dim \mathrm{Pic}^0(X) > 0$, $p > 2$, and $n \geq 17$.
(2) $\dim \mathrm{Pic}^0(X) = 0$, X is Gorenstein, and $n \geq 3n_0(p) + 3$, where $n_0(2) = 13$, $n_0(3) = 10$, $n_0(5) = 9$, and $n_0(p) = 8$ for $p \geq 7$.

For lc pairs over algebraically closed fields of any characteristic, Das proposed the following conjecture.

Conjecture 1.2 ([Das22, Conjecture 1.1]). *Let d be a positive integer and $I \subset [0,1] \cap \mathbb{Q}$ a DCC set. Then there exists a positive integer N depending only on d and I satisfying the following.*

Let (X, B) be a projective lc pair of dimension d over an algebraically closed field. Assume that $B \in I$ and $K_X + B$ is big. Then $||\lfloor m(K_X + B)\rfloor||$ defines a birational map for any integer $m \geq N$.

When $d = 2$, Conjecture 1.2 holds over any algebraically closed field [Das22, Theorem 1.2].

The proofs in this paper depend on the minimal model program for threefolds in characteristic $p > 5$ [HX15, Bir16, BW17, HNT20], and on abundance for projective lc threefold pairs (X, B) over algebraically closed fields of characteristic $p > 5$ such that B is a $\mathbb{Q}$-divisor, which follows from [Xu24, Theorems 1.4 and 1.6], [Xu26, Theorem 1.1], and the claimed result [OAI26, Theorem 1.1] (see Lemma 2.7).

The goal of this paper is to prove, in dimension 3 and over algebraically closed fields of characteristic $p > P$, the analogues of [HMX14, Theorem 1.3] and [HMX18, Theorem 1.1]: a degree m such that $||\lfloor m(K_X + B)\rfloor||$ defines a birational map, the DCC and a positive lower bound for volumes, and the boundedness of stable pairs of fixed volume V. Here P is a positive integer fixed first: P depends neither on the DCC set I nor on V, and m depends only on I, not on p.

Main results. We prove the following three theorems.

Theorem 1.3. *There exists a positive integer P with the following property. Let $I \subset [0,1]$ be a DCC set. Then there exists a positive integer m depending only on I with the following property. Let (X, B) be a projective lc pair of dimension 3 over an algebraically closed field of characteristic $p > P$. Assume that $B \in I$ and $K_X + B$ is big. Then $||\lfloor m(K_X + B)\rfloor||$ defines a birational map and*

$$\mathrm{vol}(K_X + B) \geq m^{-3}.$$

Theorem 1.3 for $I = \{0\}$ gives an absolute constant m_0 such that $|am_0K_X|$ defines a birational map for any positive integer a and any projective lc threefold X over an algebraically closed field of characteristic $p > P$ such that K_X is big (Lemma 10.2). In particular, $|am_0K_X|$ defines a birational map for any positive integer a and any smooth projective threefold X of general type over an algebraically closed field of characteristic $p > P$.

Theorem 1.3 does not prove Conjecture 1.2 for $d = 3$. For $p > P$, Theorem 1.3 and Lemma 10.2(1) show that $||\lfloor am(K_X + B)\rfloor||$ defines a birational map for any positive integer a, while Conjecture 1.2 asks for the birationality of $||\lfloor m'(K_X + B)\rfloor||$ for any integer $m' \geq N$ and in any characteristic. On the other hand, Theorem 1.3 allows $I \not\subset \mathbb{Q}$.

Theorem 1.4. *There exists a positive integer P with the following property. Let $I \subset [0,1]$ be a DCC set. Then there exists a real number $c > 0$ depending only on I with the following properties.*

(1) Let $\mathcal{V}_{\mathrm{lc}}$ be the set of volumes $\mathrm{vol}(K_X + B)$ of all projective lc pairs (X, B) of dimension 3 over all algebraically closed fields of characteristic $> P$ such that $B \in I$ and $K_X + B$ is big. Then $\mathcal{V}_{\mathrm{lc}}$ is a DCC set and

$$\mathcal{V}_{\mathrm{lc}} \subset [c, +\infty).$$

(2) Let $\mathcal{V}_{\mathrm{slc}}$ be the set of volumes $\mathrm{vol}(K_X + B)$ of all projective slc pairs (X, B) of dimension 3 (see Definition 12.1) over all algebraically closed fields of characteristic $> P$ such that $B \in I$ and $\mathrm{vol}(K_X + B) > 0$. Then $\mathcal{V}_{\mathrm{slc}}$ is a DCC set and

$$\mathcal{V}_{\mathrm{slc}} \subset [c, +\infty).$$

(3) *Let $V > 0$ be a real number. Then there exists a finite set $\mathcal{W}$ of tuples of real numbers, depending only on I and V, with the following property. Let (X, B) be a pair as in (2) such that $K_X + B$ is ample and $(K_X + B)^3 = V$. Let $\nu : Y \to X$ be the normalization of X, D_Y the conductor divisor on Y (see Definition 12.1), $B_Y := \nu_*^{-1}B$, and $Y_1, \dots, Y_n$ the connected components of Y. For any $i \in \{1, \dots, n\}$, let*

$$D_i := D_Y|_{Y_i} \quad \text{and} \quad B_i := B_Y|_{Y_i}.$$

Then $n \le V/c$,

$$\sum_{i=1}^{n} \mathrm{vol}(K_{Y_i} + D_i + B_i) = V, \quad \text{and} \quad \big(\mathrm{vol}(K_{Y_i} + D_i + B_i)\big)_{1 \le i \le n} \in \mathcal{W}.$$

Theorem 1.5. *There exists a positive integer P with the following property. Let $I \subset [0, 1]$ be a DCC set and $V > 0$ a real number. Let $\mathcal{S}$ be the set of projective slc pairs (X, B) of dimension 3 (see Definition 12.1) over algebraically closed fields of characteristic $> P$ such that $B \in I$, $K_X + B$ is ample, and*

$$(K_X + B)^3 = V.$$

Then:

(1) *There exists a finite set $I_0 \subset I$ such that $\mathcal{S}$ is bounded over $\mathbb{Z}[1/30]$ with coefficients in I_0 (see Definition 13.1).*
(2) *If $I \subset \mathbb{Q}$, then there exists a positive integer N such that $N(K_X + B)$ is a very ample Cartier divisor for any $(X, B) \in \mathcal{S}$.*

The proofs show that one may take $P = P_{\mathrm{red}}$ in all three theorems, where $P_{\mathrm{red}} \ge 11$ is the bound on the characteristic above which $|n_{\mathrm{red}}(K_X + B)|$ defines a birational map for a fixed integer n_{red} and any projective lc pair (X, B) of dimension 3 such that $B \in \{0, 1\}$ and $K_X + B$ is big (Definition-Theorem 10.4). The proofs do not give an explicit value of P_{red}.

For $B = 0$, Theorem 1.5 implies that the projective slc threefolds X with K_X ample and $K_X^3 = V$ over algebraically closed fields of characteristic $p > P$ form a bounded family over $\mathbb{Z}[1/30]$.

We are not aware of earlier proofs of Theorems 1.3, 1.4, and 1.5.

Sketch of the proof. Theorem 1.3 is proved in Section 11 with $P := P_{\mathrm{red}}$, from two statements which play the roles of [HMX14, Lemmas 7.3 and 7.4]. Let $I \subset [0, 1]$ be a DCC set, let (X, B) be a projective lc pair of dimension 3 over an algebraically closed field k of characteristic $p > P_{\mathrm{red}}$ such that $B \in I$ and $K_X + B$ is big, let $f : X' \to X$ be a log resolution of (X, B), and let

$$B' := f_*^{-1}B + \mathrm{Exc}(f).$$

First, $K_{X'} + \beta(I)B'$ is pseudo-effective for a rational number $\beta(I) \in [0, 1)$ depending only on I (see Theorem 11.3). Second, for $\alpha \in (0, 1]$ and $\beta \in [0, 1)$ there exists a positive integer $m(\alpha, \beta)$ such that if $\mathrm{mult}_S B \ge \alpha$ for any prime divisor $S \subset \mathrm{Supp}\, B$ and $K_{X'} + \beta B'$ is pseudo-effective, then $|\lfloor m(\alpha, \beta)(K_X + B)\rfloor|$ defines a birational map and $\mathrm{vol}(K_X + B) \ge m(\alpha, \beta)^{-3}$ (see Proposition 11.4). Thus Theorem 1.3 holds with

$$m := m(\alpha_I, \beta(I)), \quad \text{where} \quad \alpha_I := \min\big((I \cup \{1\}) \backslash \{0\}\big).$$

The proofs of Theorem 11.3 and Proposition 11.4 use $p > P_{\mathrm{red}}$ only through the integer n_{red} for reduced boundaries, and otherwise only $p \ge 11$. Hence P does not depend on I.

Theorem 11.3 is proved by contradiction, as in [HMX14, Proof of Lemma 7.3]. If Theorem 11.3 fails for I, then there exist projective lc pairs (X_i, B_i) of dimension 3 over algebraically closed fields of characteristic $p_i > P_{\mathrm{red}}$ with $B_i \in I \cup \{1\}$ and $K_{X_i} + B_i$ big, and log resolutions $g_i : X_i' \to X_i$ such that $K_{X_i'} + (1 - \frac{1}{i})B_i'$ is not pseudo-effective for $B_i' := (g_i)_*^{-1}B_i + \mathrm{Exc}(g_i)$. The log smooth pairs (X_i', B_i') lead to $\mathbb{Q}$-factorial klt Fano threefolds $\overline{X}_i$ of Picard number 1 and lc pairs $(\overline{X}_i, \overline{B}_i)$, with coefficients in a DCC set $J \subset [0, 1] \cap \mathbb{Q}$, such that $K_{\overline{X}_i} + \lambda_i \overline{B}_i \equiv 0$ for

strictly increasing rational numbers $\lambda_i \in [\frac{1}{2}, 1)$ with $\lim_{i\to+\infty} \lambda_i = 1$ (see Lemma 11.2). Since $(-K_{\overline{X}_i})^3 \le c_0$ for a real number c_0 independent of i (see Theorem 3.8, a version of [Das21, Theorem 1.3] whose constant does not depend on the ground field),

$$0 < v_i := \mathrm{vol}(K_{\overline{X}_i} + \overline{B}_i) = \Big(\frac{1-\lambda_i}{\lambda_i}\Big)^3 (-K_{\overline{X}_i})^3 \to 0.$$

On the other hand, for a log resolution $f_i : Z_i \to \overline{X}_i$ of $(\overline{X}_i, \overline{B}_i)$ and $\Sigma_i := (f_i)_*^{-1} \operatorname{Supp} \overline{B}_i + \operatorname{Exc}(f_i)$, we have

$$\mathrm{vol}(K_{Z_i} + \Sigma_i) \le \Big(\frac{2}{\alpha_J} - 1\Big)^3 c_0, \quad \text{where} \quad \alpha_J := \min(J \backslash \{0\}),$$

since $\operatorname{Supp} \overline{B}_i \le \frac{1}{\alpha_J} \overline{B}_i$, $\lambda_i \ge \frac{1}{2}$, and

$$K_{\overline{X}_i} + \frac{1}{\alpha_J} \overline{B}_i \equiv \Big(\frac{1}{\alpha_J \lambda_i} - 1\Big)(-K_{\overline{X}_i}).$$

Moreover, $|n_{\mathrm{red}}(K_{Z_i} + \Sigma_i)|$ defines a birational map. Thus the pairs $(\overline{X}_i, \overline{B}_i)$, whose ground fields may have different characteristics, are log birationally bounded over $\mathbb{Z}[1/30]$, that is, by one family over a scheme of finite type over $\mathbb{Z}[1/30]$ (see Definition 9.1 and Proposition 9.11), and $\{v_i \mid i \in \mathbb{N}^+\}$ is a DCC set (see Theorem 9.9), a contradiction. In [HMX14, Proof of Lemma 7.3], the log birational boundedness is deduced from [HMX14, Lemma 7.2] for the coefficients $\frac{r-1}{r}$, and here from n_{red}.

For Proposition 11.4, let q be the least integer with $q > \frac{4}{\alpha(1-\beta)}$ and let $\theta := \frac{2\beta}{1+\beta}$. Rounding the coefficients of $\frac{3+\beta}{4} B'$ down to $\frac{1}{q}\mathbb{Z}$ similarly to [HMX14, Proof of Lemma 7.4] and running an MMP, we obtain a $\frac{1}{q}$-lc log minimal model (Y, B_Y) with coefficients in $I_q := \{\frac{j}{q} \mid j = 0, 1, \dots, q\}$ such that $D := K_Y + B_Y$ is nef and big, $K_Y + \theta B_Y$ is pseudo-effective, and

$$H^0(Y, \mathcal{O}_Y(\lfloor mD \rfloor)) \subset H^0(X, \mathcal{O}_X(\lfloor m(K_X + B) \rfloor)) \quad \text{for any integer } m \ge 0$$

(see Lemma 10.3). Thus $D^2 \cdot B_Y \le \frac{D^3}{1-\theta}$, and on a log resolution $g : Z \to Y$ of (Y, B_Y) with reduced boundary $\Sigma_Z := \operatorname{Supp}(g_*^{-1} B_Y + \operatorname{Exc}(g))$,

$$(g^*D)^2 \cdot n_{\mathrm{red}}(K_Z + \Sigma_Z) \le c_1 D^3, \quad \text{where} \quad c_1 := n_{\mathrm{red}}\Big(1 + \frac{q-1}{1-\theta}\Big).$$

A pencil in $|n_{\mathrm{red}}(K_Z + \Sigma_Z)|$ yields a contraction $h : Y' \to C$ from a resolution Y' of Y onto a smooth projective curve and a base point free divisor H of positive degree on C with $(D_{Y'})^2 \cdot h^*H \le c_1 D^3$, where $D_{Y'}$ is the pullback of D to Y'. If $p \ge 11$, then $|\lfloor mD \rfloor|$, and hence $|\lfloor m(K_X + B) \rfloor|$, defines a birational map for any integer m divisible by q with $m > 12(a_q + b_q c_1) + 1$, where a_q and b_q depend only on q (see Theorem 7.2).

By [Zha22, Theorem 3.1(ii)], it suffices that Frobenius-stable sections [Zha22, Section 2.2] define a birational map on the generic fiber of h, which is a regular surface over the field $k(C)$, not algebraically closed. Over any field of characteristic at least 11, for ϵ-lc surface pairs with big adjoint divisor and boundary coefficients bounded below by a positive number, Section 5 bounds the number of boundary components needed to keep the adjoint divisor big and the Cartier index of the adjoint divisor of the minimal model (see Proposition 5.7 and Lemma 5.2). The bound 11 is that of [BT22, Proposition 5.2]: a regular del Pezzo surface over a field F of characteristic at least 11 whose ring of global functions is F is smooth over F (see Lemma 5.5). Thus the Frobenius-stable sections of a multiple of the adjoint divisor bounded in terms of q define a birational map on the generic fiber (see **Step 5** of the proof of Theorem 7.2). In [HMX14, Proof of Lemma 7.4], the birationality is deduced instead from [HMX14, Lemma 7.2] for the rounded boundary.

Definition-Theorem 10.4 is proved in Section 10. Let (X, B) be a projective lc pair of dimension 3 over k such that B is reduced and $K_X + B$ is big. If a resolution of X admits a non-constant morphism to an abelian variety and $p \ge 11$, then $|m_A(K_X + B)|$ defines a birational map for an integer m_A independent of (X, B) and p (see Theorem 10.1, whose proof

uses [Zha22, Theorem 3.4]). Otherwise, if $p > 5$, then $K_{X'} + (1-\eta)B'$ is pseudo-effective for $B' := f_*^{-1}B + \mathrm{Exc}(f)$ on a log resolution $f : X' \to X$ and a rational number $\eta \in (0, \frac{1}{2})$ independent of (X, B) and p (see Proposition 4.1 for $J = \{0\}$). This is Theorem 11.3 for reduced boundaries, proved for any $p > 5$ without n_{red}: counterexamples lead to Fano threefolds $\overline{X}_i$ of Picard number 1 with $K_{\overline{X}_i} + \lambda_i S_i \equiv 0$, where S_i is reduced and $\lim_{i\to+\infty} \lambda_i = 1$, and [Das22, Theorem 1.2] on a component Γ_i of S_i implies $(1-\lambda_i)^2 S_i^2 \cdot \Gamma_i \geq N^{-2}$ for an integer N independent of i, so

$$(-K_{\overline{X}_i})^3 \geq \lambda_i^3 S_i^2 \cdot \Gamma_i \geq \frac{\lambda_i^3}{N^2(1-\lambda_i)^2} \to +\infty,$$

which contradicts Theorem 3.8. For $q := \lfloor \frac{4}{\eta} \rfloor + 1$, there exists a log minimal model (Y, B_Y) as in Set-up 8.1, that is, a projective $\mathbb{Q}$-factorial $\frac{1}{q}$-lc pair with $B_Y \in I_q \backslash \{1\}$, nef and big $D := K_Y + B_Y$, and a resolution with trivial Albanese variety, such that the sections of $\lfloor mD \rfloor$ are sections of $m(K_X + B)$ for any $m \geq 0$ (see Lemma 10.3 for $\alpha = 1$ and $\beta = 1 - \eta$).

Let

$$b_0 := \min\{j \in \mathbb{N}^+ \mid H^0(Y, \mathcal{O}_Y(\lfloor jD \rfloor)) \neq 0\}.$$

On the pairs (Y, B_Y),

$$b_0 \leq Q_q r \quad \text{for any integer } r \geq \max\{2, J_q\} \text{ with } r^3 D^3 > 216, \text{ if } p > P_q^\dagger$$

(see Theorem 8.18), and $|\lfloor mD \rfloor|$ defines a birational map for any integer m divisible by q with $m \geq 4qn_0 + 2$, where $n_0 := c_q(2c_q(qb_0 + 1) + 1)$, if $p > 5$ (see Theorem 6.6). Here $J_q := \mathrm{lcm}(1, 2, \ldots, \lfloor 4q/3 \rfloor)$, and Q_q, $P_q^\dagger$, and c_q depend only on q. Thus if $D^3 < 1$ and $p > P_q^\dagger$, there exists an integer m such that $|\lfloor mD \rfloor|$ defines a birational map and $m^3 D^3$ is bounded by a constant depending only on q, so the pairs (Y, B_Y) with $D^3 < 1$ are log birationally bounded (see Proposition 9.11), and $D^3 \geq v_q$ for a real number $v_q \in (0, 1]$ depending only on q (see Theorem 9.9). Hence there exists an integer m_q depending only on q such that $|m_A m_q (K_X + B)|$ defines a birational map in either case if $p > \max\{11, P_q^\dagger\}$. The argument is that of [HMX14, Proof of Lemma 7.2], with Theorems 8.18 and 6.6 in place of [HMX14, Theorem 3.5.5], and with Proposition 9.11 and Theorem 9.9 in place of [HMX14, Theorem 3.5.2]. Theorem 8.18 is used only for this q, as its bound $P_q^\dagger$ depends on q.

For a non-zero $u \in H^0(Y, \mathcal{O}_Y(\lfloor b_0 D \rfloor))$, the proof of Theorem 6.6 raises to 1 the coefficients of the components of $\mathrm{div}(u^q) + qb_0 D$ and of the exceptional divisors on a log resolution, which yields a component S of coefficient 1 in the boundary of the ample model, and lifts the birational linear systems on the normalization of S, given by [Das22, Theorem 1.2] (see Definition-Lemma 6.3), as Frobenius-stable sections through the trace of Frobenius, instead of a vanishing theorem (see Proposition 6.4 and Lemma 6.5).

Theorem 8.18 is proved by contradiction. Let (X, B) be a pair as in Set-up 8.1 with $b_0 > Q_q r$ and $p > P_q^\dagger$, let $D := K_X + B$, and let $L := RD$, where $R \geq 4qr$ is an integer depending on b_0 and p (see Set-up 8.12). Let $X_{k(X)} := X \times_k \mathrm{Spec}\, k(X)$ and let $\xi \in X_{k(X)}$ be the $k(X)$-rational point defined by the generic point of X. We use ξ since X has no very general closed point if k is countable. The sections of the multiples of L that vanish to high order at ξ have a stable base locus which is a curve near ξ (see Construction-Proposition 8.13 and Lemma 8.9): for the divisor of zeros E of such a section, $\mathrm{mult}_V E \leq \frac{1}{2}\mathrm{mult}_\xi E$ for any prime divisor $V \subset X_{k(X)}$ with $\xi \in V$ (see Corollary 7.4, which needs the trivial Albanese variety), and Hasse–Schmidt derivatives of order $\mathrm{mult}_V E$, which replace partial derivatives as the p-th iterate of a partial derivative is zero, remove V from the base locus and keep the order at ξ at least $\frac{1}{2}\mathrm{mult}_\xi E$ (see Lemma 8.11). Thus X is covered by curves C_t, parametrized by a smooth surface T (see Construction-Proposition 8.15), with

$$\lambda := D \cdot C_t < \frac{1}{8R}.$$

Let $g : W \to X$ be the morphism from the normalized total space of the family, with fibers F_t over T. If g is birational, then qJ_qD is Cartier near C_t (see Theorem 8.5), so $\lambda \geq \frac{1}{qJ_q} > \frac{1}{8R}$ (see Corollary 8.8), a contradiction.

If g is not birational, then $k(W)/k(X)$ is purely inseparable of degree p^e with $e \geq 1$, and $g = \pi \circ \varphi$, where $\varphi : W \to Y_e$ is a composite of the quotients by regular foliations $\mathcal{F}_0, \dots, \mathcal{F}_{e-1}$ of rank 1 [Pos26, Sections 2.4 and 2.5] and $\pi : Y_e \to X$ is birational, and $\deg(\mathcal{F}_i|_{F_t}) \leq -p^i$ (see Proposition 8.16). Thus the π-exceptional negative part $B_{Y_e}^{<0} \geq 0$ (see Proposition 8.16(2)) of the $\mathbb{Q}$-divisor B_{Y_e} with $K_{Y_e} + B_{Y_e} = \pi^*D$ and $\pi_*K_{Y_e} = K_X$ satisfies

$$(\varphi^* B_{Y_e}^{<0}) \cdot F_t \geq p - 3 - \lambda > 0.$$

On the other hand, the curves C_t are pairwise disjoint (see **Step 6** of the proof of Theorem 8.18), so g is quasi-finite and $B_{Y_e}^{<0} = 0$, a contradiction.

Theorem 1.4 is proved in Section 12 as in [HMX14, Proof of Theorem 1.3], with Proposition 9.11 and Theorem 9.9 in place of [HMX14, Theorem 3.5.2]. For the integer m of Theorem 1.3 for $I \cup \{1\}$ and a real number $w > 0$, the lc pairs (X, B) of Theorem 1.4(1) with $\operatorname{vol}(K_X + B) \leq w$ satisfy that $||\lfloor m(K_X + B) \rfloor||$ defines a birational map and $m^3 \operatorname{vol}(K_X + B) \leq m^3 w$, so they form a log birationally bounded set, whose volumes form a DCC set (see Proposition 9.11 and Theorem 9.9). For an slc pair, $\operatorname{vol}(K_X + B)$ is the sum of the volumes of the lc pairs $(Y_i, D_i + B_i)$ on the components of the normalization (see Definition 12.1 and Lemma 12.2), so (2) and (3) follow from (1) with $c := m^{-3}$ (see Lemma 11.1(3)).

In characteristic 0, [HMX13, Proof of Theorem 1.9] moves all pairs of a log birationally bounded set to one fiber of a family by the deformation invariance of volumes [HMX13, Theorem 1.8]. The families in Theorem 9.9 are over $\mathbb{Z}[1/30]$, with geometric fibers of characteristic 0 or $p > 5$, and the proof uses instead that for a family $\mathcal{Y} \to T$ with $\Sigma = \sum_{j=1}^n \Sigma_j$ as in Set-up 9.2 and real numbers $b_1, \dots, b_n \in [0, 1]$, the volume $\operatorname{vol}(K_{\mathcal{Y}_t} + \sum_j b_j \Sigma_{j,t})$ does not depend on t in a non-empty open subset of T (see Lemma 9.3). If the adjoint divisor is pseudo-effective on the geometric generic fiber, an MMP there writes the boundary as a convex combination of rational boundaries whose adjoint divisors are semi-ample on the output (see Lemma 2.7), so the volume is an intersection number on a flat family. Otherwise the volume is 0. With Lemma 9.3, the argument of [HMX13, Proposition 5.1] shows that the volumes of lc pairs (X, B) with coefficients in a DCC set and with birational maps $f : \mathcal{Y}_t \dashrightarrow X$ such that $\operatorname{Supp} \Sigma_t$ contains $\operatorname{Supp} f_*^{-1} B$ and the prime divisors contracted by f form a DCC set (see Proposition 9.8).

Theorem 1.5 is proved in Section 13 along the route of [HMX18, Proposition 7.3 and Proof of Theorem 1.1]. For a projective slc pair (X, B) of dimension 3 over k with $p > P_{\mathrm{red}}$, $B \in I$, $K_X + B$ ample, and $(K_X + B)^3 = V$, the tuple of the volumes of the lc pairs $(Y_i, D_i + B_i)$ on the components of the normalization belongs to a finite set (see Theorem 1.4(3)). For $w > 0$, the lc pairs $(Y_i, D_i + B_i)$ with $(K_{Y_i} + D_i + B_i)^3 = w$ are log birationally bounded (see Theorem 1.3 and Proposition 9.11), hence bounded over $\mathbb{Z}[1/30]$ with coefficients in a finite set (see Theorem 13.4), and they are fibers of finitely many families on whose fibers the adjoint divisor is $\mathbb{R}$-linearly equivalent to $\sum_k c_k H_k$ for real numbers $c_k > 0$ and Cartier divisors H_k whose invertible sheaves are restrictions of invertible sheaves generated by global sections over the base (see Proposition 13.6). Gluing along the conductors, by [Pos24, Theorem 5.3.6] for $p > 5$ and [Kol16, Theorem 23] in characteristic 0, proves Theorem 1.5 (see Theorem 13.8).

The normal case of [HMX18, Proposition 7.3] uses the deformation invariance of volumes and [HMX18, Corollary 1.3] on good minimal models in families. Theorem 13.4 uses instead Lemma 9.3 and [FKL16, Theorem A], by which $\operatorname{vol}(D') = \operatorname{vol}(D)$ for big $\mathbb{R}$-divisors $D' \leq D$ implies $H^0(\lfloor mD' \rfloor) = H^0(\lfloor mD \rfloor)$ for any $m \in \mathbb{N}^+$. Theorem 13.8 writes a real boundary as a convex combination of rational boundaries whose differents are invariant under the involution τ of the normalized conductor, since being lc, being $\mathbb{R}$-Cartier, and the τ-invariance of the different are rational polyhedral conditions on the coefficients (see Lemma 2.6).

Structure of the paper. Section 2 recalls basic facts on sections and volumes, and abundance for threefolds in characteristic $p > 5$. Section 3 proves, in characteristic $p > 5$, for coefficients in a fixed DCC set and with constants independent of the field, the global ACC and a numerical rigidity for lc surface pairs, the existence of a real number $b > 0$ such that (X, B) is lc whenever $(X, (1-b)B)$ is lc, for $\mathbb{Q}$-factorial threefolds X and $\mathbb{Q}$-divisors $B \geq 0$ with coefficients in a fixed DCC set of rational numbers, and an upper bound for $(-K_X)^3$ for projective klt pairs (X, B) of dimension 3 such that $-K_X$ is ample, $K_X + B \equiv 0$, and B has coefficients in a fixed DCC set of rational numbers. Section 4 shrinks reduced boundaries uniformly. Section 5 studies regular surfaces over arbitrary fields. Section 6 obtains birational linear systems from one non-zero section. Section 7 obtains non-zero sections and birational linear systems from pencils of surfaces over curves. Section 8 bounds the degree of the first non-zero section on models whose boundary coefficients have bounded denominators. Section 9 proves the DCC for volumes of log birationally bounded lc pairs with coefficients in a fixed DCC set and a criterion for log birational boundedness. Section 10 proves effective birationality for reduced boundaries. Section 11 proves effective birationality and the lower bound for volumes for DCC coefficients. Section 12 studies volumes of slc pairs and proves the DCC for volumes. Section 13 proves boundedness at a fixed volume.

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2. Preliminaries

We adopt the standard notation and definitions for the MMP from [Sho92, KM98, BCHM10] and use them freely.

2.1. Pairs.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \operatorname{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B).

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

For a set $I \subset \mathbb{R}$ and an $\mathbb{R}$-divisor B, we write $B \in I$ if $\operatorname{mult}_P B \in I$ for every prime divisor $P \subset \operatorname{Supp} B$.

Definition 2.2. Let $\epsilon \geq 0$ be a real number. A pair (X, B) is *ϵ-lc* if $a(E, X, B) \geq \epsilon$ for any prime divisor E over X.

2.2. Sections and volumes.

Definition 2.3. Let X be a normal variety over k, D an $\mathbb{R}$-divisor on X, and m an integer. Then $\mathcal{O}_X(\lfloor mD \rfloor)$ is the divisorial sheaf of the $\mathbb{Z}$-divisor $\lfloor mD \rfloor$. Since $\operatorname{div}_X(s)$ is a $\mathbb{Z}$-divisor for any $s \in k(X)\backslash\{0\}$, we have

$$H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) = \{0\} \cup \{s \in k(X)\backslash\{0\} \mid \operatorname{div}_X(s) + mD \geq 0\}.$$

Assume that $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) \neq 0$, and let $s_0, \ldots, s_N$ be a basis of $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor))$. The complete linear system $|\lfloor mD \rfloor|$ defines the rational map $[s_0 : \cdots : s_N] : X \dashrightarrow \mathbb{P}^N_k$, and the image

of X has function field $k(s_1/s_0, \dots, s_N/s_0) \subset k(X)$. We say that $||\lfloor mD \rfloor||$ *defines a birational map* if

$$k(s_1/s_0, \dots, s_N/s_0) = k(X),$$

and that $||\lfloor mD \rfloor||$ *defines a generically finite map* if

$$\operatorname{trdeg}_k k(s_1/s_0, \dots, s_N/s_0) = \dim X.$$

More generally, for a k-subspace $\Lambda \subset k(X)$ with $0 < \dim \Lambda < +\infty$, let $k(\Lambda) \subset k(X)$ be the subfield generated over k by the quotients s/s' with $s, s' \in \Lambda$ and $s' \neq 0$. We say that Λ *defines a birational map* if $k(\Lambda) = k(X)$, and *defines a generically finite map* if $\operatorname{trdeg}_k k(\Lambda) = \dim X$.

For a normal projective variety X of dimension d over k and an $\mathbb{R}$-divisor D on X, the *volume* of D is

$$\operatorname{vol}(D) := \limsup_{m \to +\infty} \frac{h^0(X, \mathcal{O}_X(\lfloor mD \rfloor))}{m^d/d!}.$$

By [FKL16, Theorem 3.5(i)(iv)], the lim sup above is a limit, and $\operatorname{vol}(D) = \operatorname{vol}(D')$ for any $\mathbb{R}$-divisor D' on X such that $D - D'$ is a numerically trivial $\mathbb{R}$-Cartier $\mathbb{R}$-divisor. If D is $\mathbb{R}$-Cartier, then $\operatorname{vol}(D)$ depends only on the numerical class of D, and the induced function on $N^1(X)_{\mathbb{R}}$ is continuous and satisfies $\operatorname{vol}(\lambda D) = \lambda^d \operatorname{vol}(D)$ for any real number $\lambda > 0$ [FKL16, Theorem 3.5(ii)], [Cut15, Theorem 2.5 and (3)].

Lemma 2.4. *Let $f : X' \to X$ be a birational morphism between normal projective varieties, D an $\mathbb{R}$-divisor on X, D' an $\mathbb{R}$-divisor on X', and $m \geq 0$ an integer. We regard $H^0(X', \mathcal{O}_{X'}(\lfloor mD' \rfloor))$ and $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor))$ as subspaces of $k(X) = k(X')$. Then the following hold.*

(1) If $f_*D' \leq D$, then

$$H^0(X', \mathcal{O}_{X'}(\lfloor mD' \rfloor)) \subset H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)).$$

(2) If D is $\mathbb{R}$-Cartier and $D' \geq f^*D$, then

$$H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) \subset H^0(X', \mathcal{O}_{X'}(\lfloor mD' \rfloor)).$$

In particular, if D is $\mathbb{R}$-Cartier, $f_*D' \leq D$, and $D' \geq f^*D$, then

$$H^0(X', \mathcal{O}_{X'}(\lfloor mD' \rfloor)) = H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) \quad \text{and} \quad \operatorname{vol}(D') = \operatorname{vol}(D).$$

Proof. (1) Let $s \in H^0(X', \mathcal{O}_{X'}(\lfloor mD' \rfloor))$ be non-zero, so $\operatorname{div}_{X'}(s) + mD' \geq 0$. Since $m \geq 0$ and

$$f_* \operatorname{div}_{X'}(s) = \operatorname{div}_X(s),$$

we have

$$\operatorname{div}_X(s) + mD \geq \operatorname{div}_X(s) + mf_*D' = f_*\big(\operatorname{div}_{X'}(s) + mD'\big) \geq 0.$$

(2) Let $s \in H^0(X, \mathcal{O}_X(\lfloor mD \rfloor))$ be non-zero, and let $G := \operatorname{div}_X(s) + mD$. Then $G \geq 0$ and G is $\mathbb{R}$-Cartier. Since f^*G is f-numerically trivial and $f_*f^*G = G \geq 0$, the negativity lemma implies $f^*G \geq 0$. Thus

$$\operatorname{div}_{X'}(s) + mD' \geq \operatorname{div}_{X'}(s) + mf^*D = f^*G \geq 0.$$

The hypotheses of the last claim do not depend on m, so the equality holds for any integer $m \geq 0$, and $\operatorname{vol}(D') = \operatorname{vol}(D)$. The lemma follows. □

Definition 2.5. Let X be a reduced quasi-projective scheme over a field such that X is S_2 and regular or nodal at any codimension 1 point. Let $\operatorname{Weil}^*(X)$ be the free abelian group generated by the prime divisors on X that are not contained in the non-normal locus of X [Kol13, 5.6]. An $\mathbb{R}$-*divisor* (resp. $\mathbb{Q}$-*divisor*) on X is an element of $\operatorname{Weil}^*(X) \otimes_{\mathbb{Z}} \mathbb{R}$ (resp. $\operatorname{Weil}^*(X) \otimes_{\mathbb{Z}} \mathbb{Q}$). An $\mathbb{R}$-divisor (resp. a $\mathbb{Q}$-divisor) on X is $\mathbb{R}$-*Cartier* (resp. $\mathbb{Q}$-*Cartier*) if it is a linear combination with real (resp. rational) coefficients of elements $G \in \operatorname{Weil}^*(X)$ that are Cartier, that is, elements G with invertible divisorial sheaf $\mathcal{O}_X(G)$ [Kol13, (5.6.3)]. If X is normal, then $\operatorname{Weil}^*(X)$ is the group of $\mathbb{Z}$-divisors on X, and Definition 2.5 recovers the usual $\mathbb{R}$-divisors, $\mathbb{Q}$-divisors, $\mathbb{R}$-Cartier $\mathbb{R}$-divisors, and $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors.

Lemma 2.6. *Let X be a reduced quasi-projective scheme over a field such that X is S_2 and regular or nodal at any codimension 1 point (for example a normal variety). Let D be a $\mathbb{Q}$-divisor on X and $S_1, \dots, S_n$ prime divisors on X that are not contained in the non-normal locus of X. Let*

$$V := \left\{ x = (x_1, \dots, x_n) \in \mathbb{R}^n \;\middle|\; D + \sum_{i=1}^{n} x_i S_i \text{ is } \mathbb{R}\text{-Cartier} \right\}.$$

Then the following hold.

1. *Either $V = \emptyset$, or $V = x_0 + W$ for some $x_0 \in \mathbb{Q}^n$ and some linear subspace $W \subset \mathbb{R}^n$ spanned by vectors in $\mathbb{Q}^n$.*
2. *For any $x \in \mathbb{Q}^n$, $D + \sum_{i=1}^{n} x_i S_i$ is $\mathbb{R}$-Cartier if and only if it is $\mathbb{Q}$-Cartier.*

In particular, any $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor on X is $\mathbb{Q}$-Cartier.

Proof. By Definition 2.5, the $\mathbb{R}$-Cartier $\mathbb{R}$-divisors form the $\mathbb{R}$-span of the $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors, so V is the set of real solutions of a finite system Σ of linear equations with rational coefficients, and the set of $x \in \mathbb{Q}^n$ such that $D + \sum_i x_i S_i$ is $\mathbb{Q}$-Cartier is the set of rational solutions of Σ. The lemma follows. □

2.3. **Abundance for threefolds.**

Lemma 2.7. *Let (X, B) be a projective lc pair of dimension 3 over an algebraically closed field k of characteristic $p > 5$. Assume that B is a $\mathbb{Q}$-divisor and $K_X + B$ is nef. Then $K_X + B$ is semi-ample.*

Proof. Let $L := K_X + B$ and let H be an ample divisor on X. The numerical dimension of the nef $\mathbb{Q}$-divisor L is

$$\nu(L) := \max\{i \in \{0, 1, 2, 3\} \mid L^i \cdot H^{3-i} \neq 0\}.$$

Suppose that $\nu(L) = 0$. Since L is nef, L is pseudo-effective, so $\kappa(X, L) \geq 0$ by [Xu24, Theorem 1.4]. Thus there exist a positive integer m and a divisor $D \geq 0$ on X such that $D \sim mL$. Since $\nu(L) = 0$,

$$D \cdot H^2 = mL \cdot H^2 = 0.$$

Since $P \cdot H^2 > 0$ for any prime divisor P on X, $D = 0$. Thus $mL \sim 0$, and L is semi-ample.

Suppose that $\nu(L) = 1$. Then L is semi-ample by [OAI26, Theorem 1.1].

Suppose that $\nu(L) = 2$. Then L is semi-ample by [Xu26, Theorem 1.1].

Suppose that $\nu(L) = 3$. Then L is nef and big, so L is semi-ample by [Xu24, Theorem 1.6]. The lemma follows. □

3. A uniform bound on anti-canonical volumes

The main result of this section is Theorem 3.8: for a DCC set $I \subset [0, 1] \cap \mathbb{Q}$, it bounds the anti-canonical volumes of the klt Fano threefolds X that carry a klt pair (X, B) with $B \in I$ and $K_X + B \equiv 0$, by a constant depending only on I, uniformly over all algebraically closed fields of characteristic $p > 5$.

For a subset $\Lambda \subset [0, 1]$, let $\mathcal{D}(\Lambda)$ be the set consisting of 1 and the real numbers $(r - 1 + f)/r \leq 1$, where $r \in \mathbb{N}^+$ and f is a finite sum of elements of Λ (the empty sum is 0).

Lemma 3.1 (cf. [Bir16, Proposition 11.7])**.** *Let $I \subset [0, 1]$ be a DCC set. Then there exists a finite set $I' \subset I$ depending only on I satisfying the following. Let (X, B) be an lc pair of dimension 2 over an algebraically closed field of characteristic $p > 5$, and let $X \to Z$ be a projective morphism onto a variety Z with $\dim Z \leq 1$. Assume that $B \in I$ and*

$$K_X + B \equiv 0/Z.$$

Then the coefficient of any prime component of B dominating Z belongs to I'.

Proof. Let $\mathcal{H} \subset I$ be the set of $\operatorname{mult}_D B$, where $(X, B) \to Z$ is as in the lemma and D is a prime component of B dominating Z. If $\mathcal{H}$ is finite, then $I' := \mathcal{H}$ satisfies the lemma. Suppose that $\mathcal{H}$ is infinite. Since I is a DCC set, $\mathcal{H}$ contains a strictly increasing sequence $(b_i)_{i\geq 1}$ with $b_1 > 0$. Since $b_i < b_{i+1} \leq 1$, $b_i < 1$ for any i. Thus for any $i \geq 1$ there exist an algebraically closed field k_i of characteristic $p_i > 5$, a pair (X_i, B_i) and a morphism $X_i \to Z_i$ over k_i as in the lemma, and a prime component D_i of B_i dominating Z_i with

$$\operatorname{mult}_{D_i} B_i = b_i.$$

In the proof, "passing to a subsequence" refers to the index i.

$(*)$ Let $J \subset [0, 1]$ be a DCC set and $u > 0$ a real number. For any $i \in \mathbb{N}^+$ let $n_i \in \mathbb{N}^+$, $c_{j,i} \in J\backslash\{0\}$, and $w_{j,i} \in \mathbb{N}^+$ for $1 \leq j \leq n_i$, with $u_i := \sum_{j=1}^{n_i} c_{j,i}w_{j,i} \leq u$. Assume that $\{u_i\}_i$ is finite. Then $(c_{1,i})_i$ is not strictly increasing.

Proof of $(*)$. Let $\beta := \min(J\backslash\{0\})$, which exists as J is a DCC set. Suppose that $(c_{1,i})_i$ is strictly increasing. Since $c_{j,i} \geq \beta$ and $w_{j,i} \geq 1$ for any i, j, $n_i \leq u/\beta$ and $w_{j,i} \leq u/\beta$. Passing to a subsequence, we may assume that $n_i = n$, $w_{j,i} = w_j$, and $u_i = u_0$ do not depend on i, and, since J is a DCC set, that $(c_{j,i})_i$ is increasing for any $2 \leq j \leq n$. Then $\sum_{j=1}^{n} c_{j,i}w_j$ is strictly increasing in i, which contradicts $\sum_{j=1}^{n} c_{j,i}w_j = u_0$.

For a $\mathbb{Q}$-factorial klt surface Y, an $\mathbb{R}$-divisor $\Delta \geq 0$ with (Y, Δ) klt, and a prime divisor S exceptional over Y with $a(S, Y, \Delta) < 1$, let $\mu : Y' \to Y$ be a log resolution on which S is a divisor and F the sum of the other μ-exceptional prime divisors. By the negativity lemma, a $(K_{Y'} + \mu_*^{-1}\Delta + (1 - a(S, Y, \Delta))S + F)$-MMP over Y [Tan18, Theorem 1.1] contracts exactly the components of F. Its result $g : W \to Y$, with W $\mathbb{Q}$-factorial and $\operatorname{Exc}(g) = S$, is called the *extraction* of S with respect to Δ.

Claim 3.2. *Let W be a $\mathbb{Q}$-factorial normal surface over an algebraically closed field, S a prime divisor on W with normalization $\nu : S^\nu \to S$, $C_1, \dots, C_n$ prime divisors on W different from S, and $c_1, \dots, c_n > 0$ real numbers such that $(W, S + \sum_j c_jC_j)$ is lc near S. Let B_{S^ν} be the different of $(W, S + \sum_j c_jC_j)$ on S^ν. Then for any $z \in S^\nu$ with $\nu(z) \in \bigcup_j C_j$ there exist a positive integer r_z and non-negative integers $\lambda_{1,z}, \dots, \lambda_{n,z}$ such that $\lambda_{j,z} \geq 1$ if and only if $\nu(z) \in C_j$, and*

$$\operatorname{mult}_z B_{S^\nu} = \frac{r_z - 1 + \sum_{j=1}^{n} c_j\lambda_{j,z}}{r_z} \leq 1.$$

Proof. Let $w := \nu(z)$ and let Δ_{S^ν} be the different of (W, S) on S^ν. Since W is $\mathbb{Q}$-factorial, $B_{S^\nu} = \Delta_{S^\nu} + \sum_j c_j\nu^*C_j$ [Kol13, (4.2.10)], and the coefficients of B_{S^ν} are at most 1 [Kol13, Proposition 4.5(3)]. If $w \in C_j$, then for a positive integer N_j such that N_jC_j is Cartier near w, $\nu^*(N_jC_j) \geq 0$ is a Cartier divisor near z whose support contains z, so $\operatorname{mult}_z \nu^*C_j > 0$. If $w \notin C_j$, then $\operatorname{mult}_z \nu^*C_j = 0$. As $c_j > 0$ and $w \in \bigcup_j C_j$, $\operatorname{mult}_z \Delta_{S^\nu} < 1$. Since (W, S) is lc at w and $\operatorname{mult}_z \Delta_{S^\nu} < 1$, by [Kol13, 3.35, (3.36.1), and Proposition 4.5(6)], S is regular at w, (W, S) is plt at w, and

$$\operatorname{mult}_z \Delta_{S^\nu} = 1 - \frac{1}{r_z},$$

where, for the minimal resolution $f : W' \to W$ of a neighbourhood of w and the exceptional curves $E_1, \dots, E_m$ of f over w, r_z is the absolute value of the determinant of the intersection matrix $(E_k \cdot E_{k'})_{k,k'}$ ($m = 0$ and $r_z = 1$ if W is regular at w). Write $f^*C_j = f_*^{-1}C_j + \sum_k d_{j,k}E_k$ with $d_{j,k} \geq 0$. Then for any k',

$$\sum_k d_{j,k}(E_k \cdot E_{k'}) = -f_*^{-1}C_j \cdot E_{k'} \in \mathbb{Z},$$

so $d_{j,k} \in \frac{1}{r_z}\mathbb{Z}$ by Cramer's rule. As S is regular at w, ν is an isomorphism near z and $f_*^{-1}S \to S$ is an isomorphism over a neighbourhood of w. Let z' be the point of $f_*^{-1}S$ over w, and let $(\cdot)_{z'}$

denote the local intersection number at z'. Then

$$\operatorname{mult}_z \nu^* C_j = (f^* C_j \cdot f_*^{-1} S)_{z'} = (f_*^{-1} C_j \cdot f_*^{-1} S)_{z'} + \sum_k d_{j,k} (E_k \cdot f_*^{-1} S)_{z'} \in \frac{1}{r_z} \mathbb{N}.$$

Let $\lambda_{j,z} := r_z \operatorname{mult}_z \nu^* C_j \in \mathbb{N}$. Then $\lambda_{j,z} \geq 1$ if and only if $w \in C_j$, and

$$\operatorname{mult}_z B_{S^\nu} = 1 - \frac{1}{r_z} + \sum_j c_j \frac{\lambda_{j,z}}{r_z} \leq 1$$

is the formula of the claim. □

Step 1. In this step we reduce to an extremal contraction. Possibly replacing (X_i, B_i) with a $\mathbb{Q}$-factorial dlt modification [Tan18, Theorem 4.7 and Remark 4.8] and D_i with the strict transform of D_i, we may assume that X_i is $\mathbb{Q}$-factorial and $B_i \in I \cup \{1\}$. Since $K_{X_i} + B_i \equiv 0/Z_i$,

$$K_{X_i} + B_i - b_i D_i \equiv -b_i D_i / Z_i,$$

and $(X_i, B_i - b_i D_i)$ is dlt. Since D_i dominates Z_i, $-D_i$ is not pseudo-effective over Z_i. Run a $(K_{X_i} + B_i - b_i D_i)$-MMP over Z_i [Tan18, Theorem 1.1]. Any step of MMP contracts curves C with $D_i \cdot C > 0$ and $(K_{X_i} + B_i) \cdot C = 0$. Thus D_i is not contracted, and the push-forwards of B_i form lc pairs which are numerically trivial over Z_i. The MMP ends with a Mori fiber space $X_i \to T_i$ over Z_i with $D_i \cdot R_i > 0$ for the contracted ray R_i. Possibly replacing X_i with the outcome of the MMP and Z_i with T_i, we may assume that $\rho(X_i/Z_i) = 1$ and D_i is ample over Z_i, and X_i is klt since $(X_i, B_i - b_i D_i)$ is dlt and the MMP preserves the dlt property.

Step 2. In this step we treat the case $\dim Z_i = 1$. We first show the following.

Claim 3.3. *Let $J \subset [0,1]$ be a DCC set. For any i let (Y_i, B_{Y_i}) be an lc pair of dimension 2 over an algebraically closed field k_i of characteristic $p_i > 5$ and $Y_i \to T_i$ a projective contraction onto a curve such that $K_{Y_i} + B_{Y_i} \equiv 0/T_i$ and $B_{Y_i} \in J$. Then there do not exist prime components P_i of B_{Y_i} dominating T_i such that $(\operatorname{mult}_{P_i} B_{Y_i})_i$ is strictly increasing.*

Proof. Suppose that such P_i exist, and let F_i be a general fiber of $Y_i \to T_i$. Since $p_i > 2$ and

$$-K_{Y_i} \cdot F_i = B_{Y_i} \cdot F_i \geq (\operatorname{mult}_{P_i} B_{Y_i}) P_i \cdot F_i > 0,$$

the generic fiber of $Y_i \to T_i$ is a regular curve geometrically isomorphic to $\mathbb{P}^1$ [Das21, Lemma 5.1], so F_i is a smooth rational curve contained in the non-singular locus of Y_i. Let $C_{1,i} = P_i, C_{2,i}, \dots, C_{n_i,i}$ be the prime components of B_{Y_i} dominating T_i, and let $c_{j,i} := \operatorname{mult}_{C_{j,i}} B_{Y_i}$ and $w_{j,i} := C_{j,i} \cdot F_i$. Then the $w_{j,i}$ are positive integers and

$$\sum_{j=1}^{n_i} c_{j,i} w_{j,i} = B_{Y_i} \cdot F_i = -K_{Y_i} \cdot F_i = 2.$$

Since $(c_{1,i})_i$ is strictly increasing, this contradicts $(*)$ with $u_i = u = 2$. □

Passing to a subsequence, we may assume $\dim Z_i = 1$ for any i. This contradicts Claim 3.3 for $X_i \to Z_i$, $J = I \cup \{1\}$, and $P_i = D_i$.

Step 3. In this step we treat the case where (X_i, B_i) is not klt. By **Step 2**, passing to a subsequence, we may assume that Z_i is a point for any i. Then $\rho(X_i) = 1$ and $K_{X_i} + B_i \equiv 0$. Since D_i is ample and $B_i \geq b_i D_i$, $-K_{X_i} \equiv B_i$ is ample. Passing to a subsequence, assume that (X_i, B_i) is not klt for any i.

Assume that D_i contains an lc center x_i of (X_i, B_i), which is a point as $b_i < 1$. If x_i were an lc center of $(X_i, B_i - b_i D_i)$, then for an lc place E of $(X_i, B_i - b_i D_i)$ with center x_i we would have $a(E, X_i, B_i) = -b_i \operatorname{mult}_E D_i < 0$. Thus for any $t > b_i$, $(X_i, B_i - b_i D_i + t D_i)$ is not lc near x_i, so

$$b_i = \operatorname{lct}(X_i, B_i - b_i D_i; D_i),$$

where for an lc pair (Y, Δ) and a non-zero $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $M \geq 0$ on Y we let

$$\operatorname{lct}(Y, \Delta; M) := \sup\{t \in \mathbb{R}_{\geq 0} \mid (Y, \Delta + tM) \text{ is lc}\}.$$

Since $B_i - b_iD_i \in I \cup \{1\}$, the sequence $(b_i)_i$ is a strictly increasing sequence in a set satisfying the ACC [Das21, Theorem 6.1], a contradiction. Passing to a subsequence, we may assume that D_i contains no lc center of (X_i, B_i).

Assume that B_i has a component S_i with coefficient 1. Then $S_i \neq D_i$. Since $\rho(X_i) = 1$, $D_i \cap S_i$ contains a point x_i. Let $B_{S_i^\nu}$ be the different of (X_i, B_i) on the normalization S_i^ν of S_i, and let $z_i \in S_i^\nu$ be a point over x_i. By [Das21, Proposition 2.9], $B_{S_i^\nu} \in \mathcal{D}(I \cup \{1\})$, which is a DCC set [Das21, Lemma 2.8(6)]. By Claim 3.2 for (X_i, B_i) and S_i, where X_i is $\mathbb{Q}$-factorial by **Step 1**,

$$\operatorname{mult}_{z_i} B_{S_i^\nu} = \frac{r_i - 1 + f_i + \lambda_i b_i}{r_i},$$

where f_i is a finite sum of elements of $I \cup \{1\}$, and r_i, λ_i are positive integers since D_i, with coefficient $b_i > 0$, passes through x_i. Since $K_{S_i^\nu} + B_{S_i^\nu} \equiv 0$ and $B_{S_i^\nu} \neq 0$, $S_i^\nu \cong \mathbb{P}^1$ and $\deg B_{S_i^\nu} = 2$. By $(*)$ with $J = \mathcal{D}(I \cup \{1\})$ and all $w_{j,i} = 1$, the coefficients of $B_{S_i^\nu}$ belong to a finite subset of $\mathcal{D}(I \cup \{1\})$ (otherwise a strictly increasing sequence of them contradicts $(*)$). Passing to a subsequence, $\operatorname{mult}_{z_i} B_{S_i^\nu} = c$ does not depend on i. Since $\lambda_i b_i \leq r_i c$ and $b_i \geq b_1$, $\lambda_i \leq r_i/b_1$. If $c < 1$, then $r_i \leq \frac{1}{1-c}$. Passing to a subsequence, $r_i = r$ and $\lambda_i = \lambda$ do not depend on i, and

$$f_i + \lambda b_i = rc - r + 1.$$

If $c = 1$, then $f_i + \lambda_i b_i = 1$, $\lambda_i \leq 1/b_1$, and passing to a subsequence $\lambda_i = \lambda$. Since $c \leq 1$, $f_i \leq 1$, so f_i is a sum of a bounded number of elements of $I \cup \{1\}$, and passing to a subsequence $(f_i)_i$ is increasing, and $f_i + \lambda b_i$ is strictly increasing, a contradiction.

Thus, passing to a subsequence, the lc centers of (X_i, B_i) are points not on D_i. Let x_i be one of them and E_i' a prime divisor over X_i with center x_i and $a(E_i', X_i, B_i) = 0$. Let $g_i : Y_i \to X_i$ be the extraction of E_i' with respect to $(1-\eta)B_i$ for a real number $0 < \eta \ll 1$ (as X_i is klt by **Step 1**, $(X_i, (1-\eta)B_i)$ is klt, and $a(E_i', X_i, (1-\eta)B_i) = \eta a(E_i', X_i, 0) < 1$), and $B_{Y_i} := g_{i*}^{-1}B_i + E_i'$. Then Y_i is $\mathbb{Q}$-factorial, $\rho(Y_i) = 2$, $K_{Y_i} + B_{Y_i} = g_i^*(K_{X_i} + B_i) \equiv 0$, and $D_i' \cdot E_i' = 0$ for $D_i' := g_{i*}^{-1}D_i$. Since E_i' is a prime divisor, $-E_i'$ is not nef, so the extremal ray R_i of $\overline{\mathrm{NE}}(Y_i)$ other than $\mathbb{R}_{\geq 0}[E_i']$ satisfies $E_i' \cdot R_i > 0$. Since R_i is $(K_{Y_i} + B_{Y_i} - \delta E_i')$-negative for $0 < \delta \ll 1$, let h_i be the contraction of R_i [Tan18, Theorem 4.4]. Any curve C contracted by h_i satisfies $E_i' \cdot C > 0$, so $C \neq D_i'$. If h_i maps onto a curve, any fiber of h_i is irreducible [Tan18, Theorem 4.4(5)] and intersects E_i', so D_i' dominates the base, which contradicts Claim 3.3 for $J = I \cup \{1\}$ and $P_i = D_i'$. If h_i is birational, then $(h_{i*}B_{Y_i}, h_{i*}D_i')$ on $h_i(Y_i)$ satisfies the assumptions of this step and $h_{i*}E_i'$ has coefficient 1, so the previous two paragraphs imply a contradiction. Thus, passing to a subsequence, we may assume that (X_i, B_i) is klt for any i.

Step 4. In this step we show, using the results of [BM24], that the singularities of the surfaces are not bounded from below.

$(**)$ Let $J \subset [0,1]$ be a DCC set. For any i let (X_i, B_i) be a klt pair of dimension 2 over an algebraically closed field k_i of characteristic $p_i > 5$ such that X_i is $\mathbb{Q}$-factorial, $\rho(X_i) = 1$, $K_{X_i} + B_i \equiv 0$, and $B_i \in J$. Assume that B_i has a prime component D_i such that $(\operatorname{mult}_{D_i} B_i)_i$ is strictly increasing. Then there does not exist $\epsilon_0 > 0$ such that X_i is ϵ_0-lc for infinitely many i.

Proof of $(**)$. Suppose that X_i is ϵ_0-lc for any i after passing to a subsequence, and fix a rational number $\epsilon' \in (0, \epsilon_0)$. Since $-K_{X_i} \equiv B_i \neq 0$ and $\rho(X_i) = 1$, X_i is an ϵ'-klt del Pezzo surface over k_i, that is, $-K_{X_i}$ is ample and $a(E, X_i, 0) > \epsilon'$ for any prime divisor E over X_i. By [BM24, Proposition 5.11], the Picard number of the minimal resolution of X_i is bounded above by an integer depending only on ϵ', so by [BM24, Lemma 5.9] there exists a positive integer N depending only on ϵ' such that NK_{X_i} is Cartier for any i. By [BM24, Theorem 5.6],

$$K_{X_i}^2 \leq v := \max\left\{9,\ 8 + \frac{20(1-\epsilon')^2}{\epsilon'}\right\}.$$

Let $C_{1,i} = D_i, C_{2,i}, \dots, C_{n_i,i}$ be the prime components of B_i, and let $c_{j,i} := \operatorname{mult}_{C_{j,i}} B_i$ and $w_{j,i} := -NK_{X_i} \cdot C_{j,i}$. Since $-K_{X_i}$ is ample, the $w_{j,i}$ are positive integers, and

$$\sum_{j=1}^{n_i} c_{j,i} w_{j,i} = -NK_{X_i} \cdot B_i = NK_{X_i}^2 = \frac{(NK_{X_i})^2}{N} \in \frac{1}{N}\mathbb{Z} \cap (0, Nv].$$

The right-hand side takes finitely many values, so $(*)$ with $u = Nv$ implies a contradiction.

We also need two statements on surfaces over arbitrary algebraically closed fields. The first is a version of [Bir16, Lemma 11.3] whose constant does not depend on the field.

Claim 3.4. *Let $\Lambda \subset [0,1]$ be a DCC set. Then there exists $\epsilon_\Lambda \in (0, \frac{1}{2}]$ depending only on Λ satisfying the following. Let (Y, B_Y) be an lc pair of dimension 2 over an algebraically closed field such that Y is $\mathbb{Q}$-factorial and $B_Y \in \Lambda \cup [1-\epsilon_\Lambda, 1]$. Let Γ_Y be the $\mathbb{R}$-divisor on Y with $\operatorname{mult}_C \Gamma_Y = 1$ if $\operatorname{mult}_C B_Y \geq 1 - \epsilon_\Lambda$ and $\operatorname{mult}_C \Gamma_Y = \operatorname{mult}_C B_Y$ otherwise, for any prime divisor C on Y. Then (Y, Γ_Y) is lc.*

Proof. Let $\mathcal{T}$ be the set of the lc thresholds $\operatorname{lct}(Y, \Delta; M)$, where (Y, Δ) is an lc pair of dimension 2 over an algebraically closed field such that Y is $\mathbb{Q}$-factorial and $\Delta \in \Lambda$, and M is a non-zero reduced divisor on Y. By [Das21, Theorem 6.1], $\mathcal{T}$ satisfies the ACC. Thus there exists $\epsilon_\Lambda \in (0, \frac{1}{2}]$ with

$$\mathcal{T} \cap [1 - \epsilon_\Lambda, 1) = \emptyset.$$

Let $\epsilon := \epsilon_\Lambda$. Write $B_Y = \Delta + \sum_{j=1}^n c_j S_j$, where $\Delta \in \Lambda \cap [0, 1-\epsilon)$, $S_1, \dots, S_n$ are distinct prime divisors, and $c_j \geq 1 - \epsilon$ for any j. We may assume $n \geq 1$. Let $M := \sum_{j=1}^n S_j$. Since

$$\Delta + (1-\epsilon)M \leq B_Y,$$

$(Y, \Delta + (1-\epsilon)M)$ is lc, so $\operatorname{lct}(Y, \Delta; M) \geq 1 - \epsilon$. Thus $\operatorname{lct}(Y, \Delta; M) \geq 1$, that is, $(Y, \Delta + M) = (Y, \Gamma_Y)$ is lc. □

The second is a version of [Bir16, Lemma 11.6].

Claim 3.5. *Let $b \in (0, 1]$ be a real number. Let (Y, Δ) be an lc pair of dimension 2 over an algebraically closed field such that Y is $\mathbb{Q}$-factorial and klt, and let $y \in Y$ be a closed point. Then*

$$\#\{C \mid C \text{ is a prime component of } \Delta,\ y \in C,\ \operatorname{mult}_C \Delta \geq b\} \leq \frac{2}{b \min\{b, \frac{1}{2}\}}.$$

Proof. Let $C_1, \dots, C_n$ be the prime components of Δ passing through y with coefficient at least b. We may assume $n \geq 3$. Let $\Delta_0 := b \sum_{j=1}^n C_j$ and let

$$t := \max\{s \in \mathbb{R} \mid (Y, s\Delta_0) \text{ is lc near } y\}.$$

Since $\Delta_0 \leq \Delta$, $t \geq 1$.

Assume that $tb = 1$. Then $(Y, C_1 + C_2 + C_3)$ is lc near y, which contradicts [Kol13, Corollary 3.45] applied to C_1 and the curves C_2 and C_3 with coefficient 1. Thus $tb < 1$.

By the maximality of t, $(Y, t\Delta_0)$ is not klt at y. Since $tb < 1$, possibly shrinking Y around y, y is the only lc center of $(Y, t\Delta_0)$. Let S be a prime divisor over Y with center y and $a(S, Y, t\Delta_0) = 0$, and let $g : W \to Y$ be the extraction of S with respect to $(1-\eta)t\Delta_0$ for a real number $0 < \eta \ll 1$ (as Y is klt, $(Y, (1-\eta)t\Delta_0)$ is klt, and $a(S, Y, (1-\eta)t\Delta_0) = \eta a(S, Y, 0) < 1$), so that W is $\mathbb{Q}$-factorial and $\operatorname{Exc}(g) = S$. Let $C_j' := g_*^{-1} C_j$. Then

$$K_W + S + tb \sum_{j=1}^n C_j' = g^*(K_Y + t\Delta_0),$$

$(W, S + tb \sum_j C_j')$ is lc near S, and the restriction of $K_W + S + tb \sum_j C_j'$ to the normalization S^ν of S is numerically trivial. Let B_{S^ν} be the different of $(W, S + tb \sum_j C_j')$ on S^ν. Then

$\deg B_{S^\nu} = -\deg K_{S^\nu}$. Since $y \in C_j$, $C'_j \cap S \neq \emptyset$ for any j, so $B_{S^\nu} \neq 0$. Thus $S^\nu \cong \mathbb{P}^1$ and $\deg B_{S^\nu} = 2$. By Claim 3.2 for $(W, S + tb\sum_j C'_j)$, for any point $z \in S^\nu$ with image in $\bigcup_j C'_j$,

$$\operatorname{mult}_z B_{S^\nu} = \frac{r_z - 1 + tb\sum_{j=1}^n \lambda_{j,z}}{r_z} \leq 1,$$

where $r_z \in \mathbb{N}^+$, $\lambda_{j,z} \in \mathbb{N}$, and $\lambda_{j,z} \geq 1$ if the image of z belongs to C'_j. Thus

$$\operatorname{mult}_z B_{S^\nu} \geq \min\left\{tb, \frac{1}{2}\right\} \geq \min\left\{b, \frac{1}{2}\right\},$$

and

$$\#\{j \mid \text{the image of } z \text{ belongs to } C'_j\} \leq \sum_{j=1}^n \lambda_{j,z} \leq \frac{1}{tb} \leq \frac{1}{b}.$$

Since $\deg B_{S^\nu} = 2$, at most $2/\min\{b, \frac{1}{2}\}$ points z of S^ν have image on $\bigcup_j C'_j$. Since any C'_j intersects S, the last two bounds give $n \leq \frac{2}{\min\{b,\frac{1}{2}\}} \cdot \frac{1}{b}$. □

Step 5. In this step we extract divisors with small log discrepancies repeatedly, following [Bir16, Construction 11.5], and conclude by Claim 3.5. Let $\epsilon := \epsilon_I$ be the constant of Claim 3.4 for $\Lambda = I$, and let $m := 17$. We show by induction on $j \in \{0, 1, \ldots, m\}$ that, after passing to a subsequence, there exist pairs $(X_i^{(j)}, B_i^{(j)})$ over k_i and a DCC set $J_j \subset I \cup [1-\epsilon, 1)$ satisfying the following.

(1) $(X_i^{(j)}, B_i^{(j)})$ is a klt pair of dimension 2, $X_i^{(j)}$ is $\mathbb{Q}$-factorial, $\rho(X_i^{(j)}) = 1$, and
$$K_{X_i^{(j)}} + B_i^{(j)} \equiv 0.$$
(2) $B_i^{(j)} \in J_j$, and $B_i^{(j)}$ has a prime component $D_i^{(j)}$ such that $(\operatorname{mult}_{D_i^{(j)}} B_i^{(j)})_i$ is strictly increasing.
(3) $B_i^{(j)}$ has at least j prime components with coefficients in $[1-\epsilon, 1)$.

Since (X_i, B_i) is klt, $B_i \in I$. Thus by **Step 3**, $(X_i^{(0)}, B_i^{(0)}) := (X_i, B_i)$, $D_i^{(0)} := D_i$, and $J_0 := I$ satisfy (1)–(3) for $j = 0$.

Assume (1)–(3) for some $0 \leq j \leq m$. By $(**)$, passing to a subsequence, there exists a prime divisor E_i over $X_i^{(j)}$ with

$$a(E_i, X_i^{(j)}, 0) < \min\left\{\epsilon, \frac{1}{i}\right\}.$$

Let $e_i := 1 - a(E_i, X_i^{(j)}, B_i^{(j)})$. Since $B_i^{(j)} \geq 0$ and $(X_i^{(j)}, B_i^{(j)})$ is klt,

$$1 - \min\left\{\epsilon, \frac{1}{i}\right\} < e_i < 1,$$

so passing to a subsequence $(e_i)_i$ is strictly increasing. Let $g_i : Y_i \to X_i^{(j)}$ be the extraction of E_i with respect to $B_i^{(j)}$, where $a(E_i, X_i^{(j)}, B_i^{(j)}) = 1 - e_i < 1$, and let $B_{Y_i} := g_{i*}^{-1} B_i^{(j)} + e_i E_i$. Then Y_i is $\mathbb{Q}$-factorial, $\rho(Y_i) = 2$, (Y_i, B_{Y_i}) is klt, and

$$K_{Y_i} + B_{Y_i} = g_i^*(K_{X_i^{(j)}} + B_i^{(j)}) \equiv 0.$$

As in **Step 3**, the extremal ray R_i of $\overline{\mathrm{NE}}(Y_i)$ other than $\mathbb{R}_{\geq 0}[E_i]$ satisfies $E_i \cdot R_i > 0$. Let h_i be the contraction of R_i. If h_i maps onto a curve, then E_i dominates the base, which contradicts Claim 3.3 for (Y_i, B_{Y_i}), the DCC set $J_j \cup \{e_i\}_i$, and $P_i = E_i$. Thus, passing to a subsequence, h_i is birational and contracts exactly one curve E''_i. Since $E''_i \cdot E_i > 0$, $E''_i \neq E_i$. Suppose that, for some i, $E''_i = g_{i*}^{-1} C$ for a prime component C of $B_i^{(j)}$ with $\operatorname{mult}_C B_i^{(j)} \in [1-\epsilon, 1)$. Let Δ_{Y_i} be the $\mathbb{R}$-divisor on Y_i with $\operatorname{mult}_G \Delta_{Y_i} = 1$ if $\operatorname{mult}_G B_{Y_i} \in [1-\epsilon, 1)$ and $\operatorname{mult}_G \Delta_{Y_i} = \operatorname{mult}_G B_{Y_i}$ otherwise, for any prime divisor G on Y_i. Since $B_i^{(j)} \in J_j$ and $h_{i*} B_{Y_i} \in J_j \cup \{e_i\}_i$, and both

$X_i^{(j)}$ and $h_i(Y_i)$ are $\mathbb{Q}$-factorial klt surfaces, Claim 3.4 for $\Lambda = I$ shows that $(X_i^{(j)}, g_{i*}\Delta_{Y_i})$ and $(h_i(Y_i), h_{i*}\Delta_{Y_i})$ are lc. As E_i and E_i'' have coefficient 1 in Δ_{Y_i},

$$K_{Y_i}+\Delta_{Y_i} = g_i^*(K_{X_i^{(j)}}+g_{i*}\Delta_{Y_i})+a(E_i, X_i^{(j)}, g_{i*}\Delta_{Y_i})E_i = h_i^*(K_{h_i(Y_i)}+h_{i*}\Delta_{Y_i})+a(E_i'', h_i(Y_i), h_{i*}\Delta_{Y_i})E_i''$$

and both log discrepancies are non-negative as $(X_i^{(j)}, g_{i*}\Delta_{Y_i})$ and $(h_i(Y_i), h_{i*}\Delta_{Y_i})$ are lc, so $(K_{Y_i} + \Delta_{Y_i}) \cdot E_i \le 0$ and $(K_{Y_i} + \Delta_{Y_i}) \cdot E_i'' \le 0$. Since $\rho(Y_i) = 2$, $\overline{\mathrm{NE}}(Y_i)$ is spanned by $[E_i]$ and $[E_i'']$, so $-(K_{Y_i} + \Delta_{Y_i})$ is nef. On the other hand $K_{Y_i} + \Delta_{Y_i} \equiv \Delta_{Y_i} - B_{Y_i}$ is effective and non-zero, as its coefficient of E_i is $1 - e_i > 0$, so it has positive intersection with an ample divisor, a contradiction. Thus for any i, $E_i'' \ne g_{i*}^{-1}C$ for any prime component C of $B_i^{(j)}$ with $\mathrm{mult}_C B_i^{(j)} \in [1-\epsilon, 1)$. Let

$$X_i^{(j+1)} := h_i(Y_i), \quad B_i^{(j+1)} := h_{i*}B_{Y_i}, \quad D_i^{(j+1)} := h_{i*}E_i, \quad \text{and} \quad J_{j+1} := J_j \cup \{e_i\}_i.$$

Since $(e_i)_i$ is strictly increasing, J_{j+1} is a DCC set. Since $(K_{Y_i} + B_{Y_i}) \cdot E_i'' = 0$,

$$K_{Y_i} + B_{Y_i} = h_i^*\big(K_{X_i^{(j+1)}} + B_i^{(j+1)}\big),$$

so (1) holds for $j+1$. Condition (2) holds for $j+1$ as $\mathrm{mult}_{D_i^{(j+1)}} B_i^{(j+1)} = e_i$. The images of the j components in (3) and $D_i^{(j+1)}$ are distinct components of $B_i^{(j+1)}$ with coefficients in $[1-\epsilon, 1)$, so (3) holds for $j+1$.

Since (1)–(3) hold for $j = m$, passing to a subsequence, let E_i, g_i, B_{Y_i}, h_i, and E_i'' be as in the paragraph after (1)–(3) for $j = m$, and let $x_i := g_i(E_i)$ and $y_i := h_i(E_i'')$. Let $S_{1,i}, \dots, S_{m,i}$ be distinct components of $B_i^{(m)}$ with coefficients in $[1-\epsilon, 1)$, and let $S'_{j,i} := g_{i*}^{-1}S_{j,i}$. Since $E_i^2 < 0$, $E_i''^2 < 0$, and $E_i \cdot E_i'' > 0$, the classes of E_i and E_i'' form a basis of $N^1(Y_i)_{\mathbb{R}}$. Let $1 \le j \le m$. If $S'_{j,i} = E_i''$, then $S'_{j,i} \cdot E_i > 0$. Otherwise $S'_{j,i} \cdot E_i \ge 0$ and $S'_{j,i} \cdot E_i'' \ge 0$, and since $S'_{j,i} \not\equiv 0$, $S'_{j,i} \cdot (E_i + E_i'') > 0$. Thus for any j,

$$x_i \in S_{j,i} \quad \text{or} \quad y_i \in h_i(S'_{j,i}) \text{ with } S'_{j,i} \ne E_i''.$$

Since $m = 17$, at least 9 of the curves $S_{j,i}$ pass through x_i, or at least 9 of the curves $h_i(S'_{j,i})$ pass through y_i. The surfaces $X_i^{(m)}$ and $h_i(Y_i)$ are $\mathbb{Q}$-factorial and klt, and

$$\mathrm{mult}_{S_{j,i}} B_i^{(m)} \ge 1-\epsilon \ge \frac{1}{2}, \quad \mathrm{mult}_{h_i(S'_{j,i})} h_{i*}B_{Y_i} \ge 1-\epsilon \ge \frac{1}{2}.$$

By Claim 3.5 with $b = 1-\epsilon$, at most $\frac{4}{1-\epsilon} \le 8$ components of $B_i^{(m)}$ (resp. $h_{i*}B_{Y_i}$) with coefficients at least $1-\epsilon$ pass through x_i (resp. y_i), a contradiction. □

Lemma 3.6 (cf. [Das21, Theorem 6.4]). *Let $I \subset [0,1]$ be a DCC set. Then there exists a real number $\delta \in (0,1)$ depending only on I satisfying the following. Let X be a normal projective surface over an algebraically closed field of characteristic $p > 5$, and let B and B' be $\mathbb{R}$-divisors on X. Assume that*

(1) $B \in I$,
(2) $(1-\delta)B \le B' \le B$,
(3) (X, B') is lc, and
(4) $K_X + B' \equiv 0$.

Then $B' = B$.

Proof. Suppose that the lemma does not hold. Then for any $i \ge 2$ there exist an algebraically closed field k_i of characteristic $p_i > 5$ and X_i, B_i, B_i' over k_i satisfying the assumptions with $\delta = \frac{1}{i}$ and $B_i' \ne B_i$.

We show that the set J of the coefficients of all the B_i' is a DCC set. Otherwise there exist prime divisors P_j on X_{i_j} for $j \ge 1$ such that $c_j' := \mathrm{mult}_{P_j} B'_{i_j}$ is strictly decreasing. Since any B_i' has finitely many components, passing to a subsequence we may assume that $(i_j)_j$ is

strictly increasing. Let $c_j := \mathrm{mult}_{P_j} B_{i_j} \in I$. Passing to a subsequence, $(c_j)_j$ is increasing. Let $c := \lim_j c_j$ and $c' := \lim_j c'_j$. Since $0 \leq c_j - c'_j \leq \frac{c_j}{i_j} \leq \frac{1}{i_j}$, $c = c'$. Thus

$$c_j \leq c = c' < c'_j$$

for any j, which contradicts $c'_j \leq c_j$.

By Lemma 3.1 for the DCC set J and the morphisms $X_i \to \operatorname{Spec} k_i$, there exists a finite set $J' \subset J$ containing the coefficients of all the B'_i. Since $B'_i \leq B_i$ and $B'_i \neq B_i$, there exists a prime divisor P_i on X_i with

$$c'_i := \mathrm{mult}_{P_i} B'_i < c_i := \mathrm{mult}_{P_i} B_i.$$

Passing to a subsequence, $c'_i = c'$ does not depend on i and $(c_i)_i$ is increasing. Since $c' < c_i \leq \frac{c'}{1-1/i}$ for any i, $\lim_i c_i = c'$. Since $(c_i)_i$ is increasing, $c_i \leq c'$ for any i, a contradiction. □

Lemma 3.7. *Let $I \subset [0,1] \cap \mathbb{Q}$ be a DCC set. Then there exists a real number $\delta > 0$ depending only on I satisfying the following. Let X be a $\mathbb{Q}$-factorial normal quasi-projective threefold over an algebraically closed field of characteristic $p > 5$, and let $B \geq 0$ be a $\mathbb{Q}$-divisor on X with $B \in I$. Assume that $(X, (1-\delta)B)$ is lc. Then (X, B) is lc.*

Proof. Let $\mathcal{T}$ be the set of the lc thresholds $\mathrm{lct}(X, 0; B)$ for all $\mathbb{Q}$-factorial normal quasi-projective threefolds X over algebraically closed fields of characteristic > 5 such that $(X, 0)$ is lc and all non-zero $\mathbb{Q}$-divisors $B \geq 0$ on X with $B \in I$. By [Wal23, Theorem 1.5], $\mathcal{T}$ satisfies the ACC. Thus either $\mathcal{T} \cap [0,1) = \emptyset$, and we let $\delta := \frac{1}{2}$, or $\mathcal{T} \cap [0,1)$ has a maximum $t_0 < 1$, and we let $\delta := \frac{1-t_0}{2}$. Then δ depends only on I, and

$$\mathcal{T} \cap [1-\delta, 1) = \emptyset.$$

Let X and B be as in the lemma, and assume that $(X, (1-\delta)B)$ is lc. We may assume that $B \neq 0$. Since X is $\mathbb{Q}$-factorial and $0 \leq (1-\delta)B$, the pair $(X, 0)$ is lc, so $\mathrm{lct}(X, 0; B) \in \mathcal{T}$. Since $\mathrm{lct}(X, 0; B) \geq 1 - \delta$, we have $\mathrm{lct}(X, 0; B) \geq 1$, that is, (X, B) is lc. □

Theorem 3.8 (cf. [Das21, Theorem 1.3])**.** *Let $I \subset [0,1] \cap \mathbb{Q}$ be a DCC set. Then there exists a real number $c > 0$ depending only on I satisfying the following. Let (X, B) be a projective klt pair of dimension 3 over an algebraically closed field of characteristic $p > 5$. Assume that $B \in I$, $K_X + B \equiv 0$, and $-K_X$ is $\mathbb{Q}$-Cartier and ample. Then*

$$(-K_X)^3 \leq c.$$

Proof. **Step 1.** In this step we fix the constants. Let $\Lambda_1 := I \cup \{1\}$ and $\Lambda_2 := \mathcal{D}(\Lambda_1)$. Taking $r = 1$, we see $\Lambda_1 \subset \Lambda_2$. Moreover $\Lambda_2 \subset [0,1] \cap \mathbb{Q}$, and Λ_2 is a DCC set [Das21, Lemma 2.8(6)]. We let $\delta_1, \delta_2, \delta_3, \delta_4$ be positive real numbers depending only on I satisfying the following.

(1) For any regular projective curve C over a field and any $\mathbb{Q}$-divisors Δ, Δ' on C such that $\Delta \in \Lambda_2$, $(1-\delta_1)\Delta \leq \Delta' \leq \Delta$, (C, Δ') is lc, and $K_C + \Delta' \equiv 0$, we have $\Delta' = \Delta$ [Das21, Theorem 5.7].
(2) For any normal projective surface Y over a field such that K_Y is $\mathbb{Q}$-Cartier and any $\mathbb{Q}$-divisor $\Delta \geq 0$ on Y with $\Delta \in \Lambda_1$, if $(Y, (1-\delta_2)\Delta)$ is lc, then (Y, Δ) is lc [Das21, Theorem 6.1 and the first part of Theorem 6.4].
(3) δ_3 is the real number given by Lemma 3.7 for the DCC set Λ_1.
(4) $\delta_4 \in (0,1)$ is the real number given by Lemma 3.6 for the DCC set Λ_2.

Let

$$\delta_0 := \min\{\delta_1, \delta_2, \delta_3, \delta_4\} \in (0,1) \quad \text{and} \quad c := \frac{27}{\delta_0^3}.$$

Then c depends only on I.

Step 2. In this step we prove the following claim.

Claim 3.9. *Let δ_0 be as in **Step 1**, let (X, B) be a projective klt pair of dimension 3 over an algebraically closed field k of characteristic $p > 5$, let $\delta \in (0, \delta_0)$ be a rational number, and let $\Phi \geq 0$ be a $\mathbb{Q}$-divisor on X. Assume that*

(i) $B \in I$ and $K_X + B \equiv 0$,
(ii) $K_X + \Phi$ is $\mathbb{Q}$-Cartier and $K_X + \Phi \equiv 0$, and
(iii) $\Phi \geq (1 - \delta)B$.

Then (X, Φ) is klt.

Proof. Suppose that (X, Φ) is not klt.

Since log resolutions of $(X, \mathrm{Supp}(B + \Phi))$ exist [FW24, Proposition 2.4] and (X, B) is klt, there exists a largest number $t_0 \in (0, 1]$ such that $(X, (1 - t_0)B + t_0\Phi)$ is lc. Then $t_0 \in \mathbb{Q}$, and $(X, (1 - t_0)B + t_0\Phi)$ is lc and not klt. Moreover $K_X + (1 - t_0)B + t_0\Phi$ is $\mathbb{Q}$-Cartier, $K_X + (1 - t_0)B + t_0\Phi \equiv 0$, and $(1 - t_0)B + t_0\Phi \geq (1 - \delta)B$. Possibly replacing Φ with $(1 - t_0)B + t_0\Phi$, we may assume that (X, Φ) is lc and not klt.

By [Bir16, Theorem 1.6], there exists a crepant $\mathbb{Q}$-factorial dlt modification $u : U \to X$ of (X, Φ): U is $\mathbb{Q}$-factorial, (U, Ψ) is dlt, where Ψ is the $\mathbb{Q}$-divisor with $K_U + \Psi = u^*(K_X + \Phi)$, and any u-exceptional prime divisor has coefficient 1 in Ψ. Let $S_1, \dots, S_n$ be the distinct components of $\lfloor \Psi \rfloor$. Then $n \geq 1$: if $\lfloor \Psi \rfloor = 0$, then u has no exceptional prime divisor and (U, Ψ) is klt, so (X, Φ) is klt. Since any u-exceptional prime divisor is S_j for some j, we may write

$$u^*(K_X + B) = K_U + B_U + \sum_{j=1}^{n} a_j S_j, \tag{3.1}$$

where $a_1, \dots, a_n \in \mathbb{Q}$, and B_U is the sum of the terms of $u_*^{-1}B$ along the prime divisors different from $S_1, \dots, S_n$. Then $B_U \in I$. Since (X, B) is klt, $a_j < 1$ for any j. Since $\Phi \geq (1 - \delta)B$,

$$\Psi \geq (1 - \delta)B_U + \sum_{j=1}^{n} S_j. \tag{3.2}$$

Since (U, Ψ) is $\mathbb{Q}$-factorial dlt, so is $(U, \Psi - S_1)$. Since $K_U + \Psi \equiv 0$, we have $K_U + \Psi - S_1 \equiv -S_1$, which is not pseudo-effective, as S_1 is a prime divisor on the projective variety U. By [BW17, Theorem 1.7], there exists a $(K_U + \Psi - S_1)$-MMP

$$U = U_0 \dashrightarrow U_1 \dashrightarrow \cdots \dashrightarrow U_N =: U'$$

ending with a Mori fiber space $U' \to T$. For any $i \in \{0, \dots, N\}$, let Ψ_i, B_{U_i}, and $S_{j,i}$ be the pushforwards of Ψ, B_U, and S_j to U_i, and let R_i be the extremal ray contracted by the step $U_i \dashrightarrow U_{i+1}$ of the MMP if $i < N$, and by $U' \to T$ if $i = N$. Let $f_i : U_i \to Z_i$ be the contraction of R_i [BW17, Theorem 1.3]. Thus $Z_N = T$. The pair $(U_i, \Psi_i - S_{1,i})$ is $\mathbb{Q}$-factorial dlt, and $(K_{U_i} + \Psi_i - S_{1,i}) \cdot R_i < 0$.

Claim 3.10. *Let $i \in \{0, \dots, N\}$, and let L be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on U_i such that $L \cdot R_i = 0$. Then there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor L_{Z_i} on Z_i such that $L \sim_{\mathbb{Q}} f_i^* L_{Z_i}$. If moreover f_i is birational, then $f_{i*}L$ is $\mathbb{Q}$-Cartier and $L = f_i^* f_{i*} L$.*

Proof. Since $(U_i, \Psi_i - S_{1,i} - \epsilon\lfloor \Psi_i - S_{1,i} \rfloor)$ is klt and $K_{U_i} + \Psi_i - S_{1,i} - \epsilon\lfloor \Psi_i - S_{1,i} \rfloor$ is negative on R_i for $0 < \epsilon \ll 1$, this follows from [BW17, Theorem 1.2] and the negativity lemma. □

We show by induction on $i \in \{0, \dots, N\}$ that

$$K_{U_i} + \Psi_i \equiv 0, \qquad K_{U_i} + B_{U_i} + \sum_{j=1}^{n} a_j S_{j,i} \equiv 0, \qquad \text{and } (U_i, \Psi_i) \text{ is lc.} \tag{3.3}$$

For $i = 0$, (3.3) follows from $K_U + \Psi = u^*(K_X + \Phi)$, (3.1), and the dlt property of (U, Ψ). Assume (3.3) for some $i < N$. Let $g_i : U_{i+1} \to Z_i$ be the identity if f_i is divisorial, and the flipped contraction if f_i is a flipping contraction. Let L be $K_{U_i} + \Psi_i$ or $K_{U_i} + B_{U_i} + \sum_j a_j S_{j,i}$, and let L_{i+1} be the pushforward of L to U_{i+1}. Since $L \equiv 0$, Claim 3.10 implies that $f_{i*}L$ is $\mathbb{Q}$-Cartier

and $L = f_i^* f_{i*} L$, and $f_{i*} L \equiv 0$ since f_i is surjective. Since g_i is small, $L_{i+1} = g_i^* f_{i*} L \equiv 0$. For $L = K_{U_i} + \Psi_i$, we obtain

$$K_{U_i} + \Psi_i = f_i^*(K_{Z_i} + f_{i*}\Psi_i) \quad \text{and} \quad K_{U_{i+1}} + \Psi_{i+1} = g_i^*(K_{Z_i} + f_{i*}\Psi_i),$$

so $a(E, U_i, \Psi_i) = a(E, U_{i+1}, \Psi_{i+1})$ for any prime divisor E over U_i, and (U_{i+1}, Ψ_{i+1}) is lc. This proves (3.3) for $i+1$.

By (3.3), $(K_{U_i} + \Psi_i) \cdot R_i = 0$, so $S_{1,i} \cdot R_i > 0$ for any i. Thus $S_{1,i}$ is not the exceptional divisor of f_i when f_i is divisorial, so $S_{1,N} \neq 0$, and $S_{1,N}$ is ample over T since $\rho(U'/T) = 1$. Pushing (3.2) forward to U', we have $\Psi_N \geq (1-\delta)B_{U_N} + \sum_j S_{j,N}$.

Let F be the generic fiber of $U' \to T$, a projective variety over the function field $k(T)$ of T. If $\dim T = 0$, then $F = U'$ and $k(T) = k$. For a $\mathbb{Q}$-divisor D on U', let D_F be the restriction of D to F, and let Ψ_F, B_F, and $S_{j,F}$ be the restrictions of Ψ_N, B_{U_N}, and $S_{j,N}$ to F. By [Das21, Lemmas 3.4 and 3.5], F is normal and $\mathbb{Q}$-factorial, $K_F = (K_{U'})_F$, and (F, Ψ_F) is lc. Since $S_{1,N}$ is ample over T and $\dim F = 3 - \dim T \geq 1$, $S_{1,F}$ is ample and non-zero. Thus $S_{1,N}$ dominates T, and $S_{1,F}$ is a prime divisor on F [Das21, Lemma 3.4].

Claim 3.11. *For any $\mathbb{Q}$-divisor D_F on F, there exists a unique $\lambda_{D_F} \in \mathbb{Q}$ with $D_F \sim_{\mathbb{Q}} \lambda_{D_F} S_{1,F}$. If $D_F \geq 0$, then $\lambda_{D_F} \geq 0$, and $\lambda_{D_F} = 0$ if and only if $D_F = 0$.*

Proof. Let $\overline{D_F}$ be the $\mathbb{Q}$-divisor on U' such that $\operatorname{mult}_{\overline{Q}} \overline{D_F} = \operatorname{mult}_Q D_F$ for any prime divisor Q on F, where $\overline{Q}$ is the closure of Q in U', and such that any component of $\overline{D_F}$ is of the form $\overline{Q}$. Since U' is $\mathbb{Q}$-factorial, $\overline{D_F}$ is $\mathbb{Q}$-Cartier. Since $S_{1,N} \cdot R_N > 0$, there exists $\lambda_{D_F} \in \mathbb{Q}$ with $(\overline{D_F} - \lambda_{D_F} S_{1,N}) \cdot R_N = 0$. By Claim 3.10, $\overline{D_F} - \lambda_{D_F} S_{1,N} \sim_{\mathbb{Q}} f_N^* M$ for a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor M on T. Since a multiple of M is Cartier and hence principal near the generic point of T, $(f_N^* M)_F \sim_{\mathbb{Q}} 0$ and $D_F \sim_{\mathbb{Q}} \lambda_{D_F} S_{1,F}$. The rest follows from $D_F \cdot S_{1,F}^{\dim F - 1} = \lambda_{D_F} S_{1,F}^{\dim F}$ and $S_{1,F}^{\dim F} > 0$. □

By (3.3) with $i = N$, we have $(K_{U'} + \Psi_N) \cdot R_N = 0$ and $(K_{U'} + B_{U_N} + \sum_j a_j S_{j,N}) \cdot R_N = 0$. By Claim 3.10, $K_{U'} + \Psi_N$ and $K_{U'} + B_{U_N} + \sum_j a_j S_{j,N}$ are $\mathbb{Q}$-linearly equivalent to pullbacks of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on T, whose restrictions to F are $\mathbb{Q}$-linearly trivial. Thus

$$K_F + \Psi_F \sim_{\mathbb{Q}} 0 \qquad \text{and} \qquad K_F + B_F + \sum_{j=1}^{n} a_j S_{j,F} \sim_{\mathbb{Q}} 0. \tag{3.4}$$

Restricting $\Psi_N \geq (1-\delta)B_{U_N} + \sum_j S_{j,N}$ to F, we have $\Psi_F - (1-\delta)B_F - \sum_j S_{j,F} \geq 0$. By Claim 3.11, let $\lambda_1, \dots, \lambda_n, \mu, \lambda'$ be the non-negative rational numbers with

$$S_{j,F} \sim_{\mathbb{Q}} \lambda_j S_{1,F}, \qquad B_F \sim_{\mathbb{Q}} \mu S_{1,F}, \qquad \text{and} \qquad \Psi_F - (1-\delta)B_F - \sum_{j=1}^{n} S_{j,F} \sim_{\mathbb{Q}} \lambda' S_{1,F}.$$

Then $\lambda_1 = 1$. Let

$$\lambda := \sum_{j=1}^{n} (1 - a_j)\lambda_j \geq 1 - a_1 > 0,$$

and for any $t \in [0,1] \cap \mathbb{Q}$, let $\Gamma(t) := (1-t)B_F + \sum_{j=1}^{n} S_{j,F}$. By (3.4),

$$K_F + \Gamma(t) = \Big(K_F + B_F + \sum_{j=1}^{n} a_j S_{j,F}\Big) + \sum_{j=1}^{n} (1 - a_j) S_{j,F} - t B_F \sim_{\mathbb{Q}} (\lambda - t\mu) S_{1,F}. \tag{3.5}$$

By (3.4),

$$K_F + \Gamma(\delta) = (K_F + \Psi_F) - (\Psi_F - \Gamma(\delta)) \sim_{\mathbb{Q}} -\lambda' S_{1,F},$$

so $\lambda - \delta\mu = -\lambda' \leq 0$ by (3.5) and the uniqueness in Claim 3.11. Thus $\mu \geq \lambda/\delta > 0$, and $\delta' := \lambda/\mu \in (0, \delta]$ satisfies

$$K_F + \Gamma(\delta') \sim_{\mathbb{Q}} 0 \qquad \text{and} \qquad K_F + \Gamma(0) \sim_{\mathbb{Q}} \lambda S_{1,F}. \tag{3.6}$$

Since $0 < \delta' \leq \delta$,

$$(1-\delta)\Gamma(0) \leq \Gamma(\delta') \leq \Gamma(0). \tag{3.7}$$

We have $\Gamma(0) \in \Lambda_1 \subset \Lambda_2$. Since $(1-\delta)\Gamma(0) \leq \Gamma(\delta) \leq \Psi_F$, F is $\mathbb{Q}$-factorial, and (F, Ψ_F) is lc, the pair $(F, (1-\delta)\Gamma(0))$ is lc.

Assume that $\dim F = 1$. Then F is a regular projective curve over $k(T)$, and $(F, \Gamma(\delta'))$ is lc since the coefficients of $\Gamma(\delta')$ are ≤ 1. Since $\delta < \delta_1$, (3.6), (3.7), and the choice of δ_1 imply $\Gamma(\delta') = \Gamma(0)$, which contradicts (3.6), as $\lambda S_{1,F} \not\sim_{\mathbb{Q}} 0$.

Assume that $\dim F \geq 2$. We first show that $(F, \Gamma(0))$ is lc. If $\dim F = 2$, then K_F is $\mathbb{Q}$-Cartier, $(F, (1-\delta_2)\Gamma(0))$ is lc since $\delta_2 > \delta$, and $(F, \Gamma(0))$ is lc by the choice of δ_2. If $\dim F = 3$, then T is a point, $F = U'$ is a $\mathbb{Q}$-factorial normal projective threefold over k, $(F, (1-\delta_3)\Gamma(0))$ is lc since $\delta_3 > \delta$, and $(F, \Gamma(0))$ is lc by the choice of δ_3.

Let S^ν be the normalization of $S_{1,F}$. By [Das21, Proposition 2.9], there exist $\mathbb{Q}$-divisors $\Delta_{S^\nu} \geq 0$ and $\Delta'_{S^\nu} \geq 0$ on S^ν, the differents of $\Gamma(0) - S_{1,F}$ and $\Gamma(\delta') - S_{1,F}$, such that

$$K_{S^\nu} + \Delta_{S^\nu} \sim_{\mathbb{Q}} (K_F + \Gamma(0))|_{S^\nu} \qquad \text{and} \qquad K_{S^\nu} + \Delta'_{S^\nu} \sim_{\mathbb{Q}} (K_F + \Gamma(\delta'))|_{S^\nu}, \tag{3.8}$$

$\Delta_{S^\nu} \in \mathcal{D}(\Lambda_1) = \Lambda_2$, and for any prime divisor P on S^ν, either $\operatorname{mult}_P \Delta_{S^\nu} = \operatorname{mult}_P \Delta'_{S^\nu} = 1$, or there exist $r \in \mathbb{N}^+$ and integers $m_Q \geq 0$, indexed by the prime divisors $Q \neq S_{1,F}$ on F and depending only on F, $S_{1,F}$, P, and Q, with

$$\operatorname{mult}_P \Delta_{S^\nu} = \frac{r - 1 + \sum_Q m_Q \operatorname{mult}_Q \Gamma(0)}{r} \quad \text{and} \quad \operatorname{mult}_P \Delta'_{S^\nu} = \frac{r - 1 + \sum_Q m_Q \operatorname{mult}_Q \Gamma(\delta')}{r}.$$

By (3.7), for any Q,

$$(1-\delta)\operatorname{mult}_Q \Gamma(0) \leq \operatorname{mult}_Q \Gamma(\delta') \leq \operatorname{mult}_Q \Gamma(0).$$

If $\operatorname{mult}_P \Delta_{S^\nu} \neq 1$ or $\operatorname{mult}_P \Delta'_{S^\nu} \neq 1$, then since $r - 1 \geq (1-\delta)(r-1)$, the formulas for $\operatorname{mult}_P \Delta_{S^\nu}$ and $\operatorname{mult}_P \Delta'_{S^\nu}$ imply

$$(1-\delta)\operatorname{mult}_P \Delta_{S^\nu} \leq \operatorname{mult}_P \Delta'_{S^\nu} \leq \operatorname{mult}_P \Delta_{S^\nu}.$$

Thus

$$(1-\delta)\Delta_{S^\nu} \leq \Delta'_{S^\nu} \leq \Delta_{S^\nu}. \tag{3.9}$$

By (3.6) and (3.8),

$$K_{S^\nu} + \Delta'_{S^\nu} \sim_{\mathbb{Q}} 0 \qquad \text{and} \qquad K_{S^\nu} + \Delta_{S^\nu} \sim_{\mathbb{Q}} \lambda S_{1,F}|_{S^\nu}.$$

Since $S^\nu \to F$ is finite, $K_{S^\nu} + \Delta_{S^\nu}$ is ample.

If $\dim F = 2$, then S^ν is a regular projective curve over $k(T)$, and (S^ν, Δ'_{S^ν}) is lc since the coefficients of Δ'_{S^ν} are ≤ 1 by (3.9). Since $\delta < \delta_1$, the choice of δ_1 implies $\Delta'_{S^\nu} = \Delta_{S^\nu}$, which contradicts the ampleness of $K_{S^\nu} + \Delta_{S^\nu}$.

If $\dim F = 3$, then S^ν is a normal projective surface over k. Since $(U', \Gamma(0))$ is lc, (S^ν, Δ_{S^ν}) is lc [Das21, Lemma 2.10]. Thus K_{S^ν} is $\mathbb{Q}$-Cartier [Das21, Lemma 3.7], and (S^ν, Δ'_{S^ν}) is lc by (3.9). Since $\delta < \delta_4$ and $K_{S^\nu} + \Delta'_{S^\nu} \equiv 0$, Lemma 3.6 and the choice of δ_4 imply $\Delta'_{S^\nu} = \Delta_{S^\nu}$, which contradicts the ampleness of $K_{S^\nu} + \Delta_{S^\nu}$.

Thus (X, Φ) is klt. □

Step 3. In this step we bound $(-K_X)^3$. Let (X, B) be a pair satisfying the assumptions of Theorem 3.8, let k be the ground field, and let $V := (-K_X)^3$. Suppose that $V > c$. Then $\delta_0^3 V > 27$, so there exist rational numbers $\delta \in (0, \delta_0)$ and $\tau > 3$ with

$$\delta^3 V > \tau^3.$$

Let $x \in X$ be a smooth point. Since $(-\delta K_X)^3 = \delta^3 V > \tau^3$ and $-K_X$ is ample, comparing $h^0(X, \mathcal{O}_X(-m\delta K_X))$ with $\dim_k \mathcal{O}_{X,x}/\mathfrak{m}_x^{m\tau+1}$ for divisible m gives a $\mathbb{Q}$-divisor $D \geq 0$ with $D \sim_{\mathbb{Q}} -\delta K_X$ and $\operatorname{mult}_x D > \tau > 3$.

Let $\Phi := (1-\delta)B + D$. Then $\Phi \geq (1-\delta)B$, $K_X + \Phi$ is $\mathbb{Q}$-Cartier, and

$$K_X + \Phi \equiv K_X + (1-\delta)B - \delta K_X = (1-\delta)(K_X + B) \equiv 0.$$

By Claim 3.9, (X, Φ) is klt. Let E_1 be the exceptional prime divisor of the blow-up of x. Since x is a smooth point of the threefold X and $B \geq 0$,

$$a(E_1, X, \Phi) = 3 - \operatorname{mult}_x \Phi \leq 3 - \operatorname{mult}_x D < 0,$$

a contradiction. Thus $(-K_X)^3 = V \leq c$. □

4. Shrinking reduced boundaries

In this section, we prove the following proposition.

Proposition 4.1. *Let $J \subset [0,1) \cap \mathbb{Q}$ be a DCC set. There exists a rational number $\eta \in (0, \frac{1}{2})$ depending only on J with the following property. Let $(X, B+S)$ be a projective $\mathbb{Q}$-factorial dlt pair of dimension 3 over an algebraically closed field of characteristic $p > 5$. Assume that*

(i) S is a reduced divisor, $B \in J$, and B and S have no common component, and
(ii) $K_X + B + S$ is big.

Then $K_X + B + (1-\eta)S$ is pseudo-effective.

Proof. We divide the proof into several steps.

Step 1. Suppose that the proposition does not hold. For any integer $i \geq 1$, let

$$t_i := 1 - \frac{1}{i+2}.$$

Since $1 - t_i \in (0, \frac{1}{2}) \cap \mathbb{Q}$ does not have the property of the proposition, there exists a projective $\mathbb{Q}$-factorial dlt pair $(X_i, B_i + S_i)$ of dimension 3 over an algebraically closed field k_i of characteristic $p_i > 5$ satisfying the following.

(1) S_i is a reduced divisor, $B_i \in J$, and B_i and S_i have no common component.
(2) $K_{X_i} + B_i + S_i$ is big.
(3) $K_{X_i} + B_i + t_i S_i$ is not pseudo-effective.

Since $(X_i, B_i + S_i)$ is $\mathbb{Q}$-factorial dlt, $\lfloor B_i + S_i \rfloor = S_i$, and $t_i < 1$,

(4) $(X_i, B_i + t_i S_i)$ is klt.

Step 2. By (4) and [Xu24, Theorem 2.12], we may run a $(K_{X_i} + B_i + t_i S_i)$-MMP. By (3), the MMP ends with a Mori fiber space $\pi_i : X_i' \to Z_i$. Since the MMP does not extract divisors, the pushforward of $K_{X_i} + B_i + S_i$ to X_i' is big. Possibly replacing X_i with X_i' and B_i, S_i with their pushforwards, we may assume that (1), (2), and (4) hold, X_i is $\mathbb{Q}$-factorial, and there exists a contraction $\pi_i : X_i \to Z_i$ satisfying the following.

(5) $\rho(X_i/Z_i) = 1$, $\dim Z_i \leq 2$, and $-(K_{X_i} + B_i + t_i S_i)$ is ample over Z_i.

We let

$$L_i := K_{X_i} + B_i + S_i.$$

Since $\rho(X_i/Z_i) = 1$ and $\dim Z_i < \dim X_i$,

(6) any $\mathbb{Q}$-divisor $D \geq 0$ on X_i is nef over Z_i.

Since L_i is big, $L_i \sim_{\mathbb{Q}} A_i + D_i$ for some ample $\mathbb{Q}$-divisor A_i and some $\mathbb{Q}$-divisor $D_i \geq 0$. By (6), L_i is ample over Z_i. By (5),

$$(1-t_i)S_i = L_i - (K_{X_i} + B_i + t_i S_i)$$

is ample over Z_i. In particular, $S_i \neq 0$.

Step 3. Let $\epsilon > 0$ be the real number of [Jia25, Proposition 4.3] for the DCC set J. In this step, we show that Z_i is a point for any i such that $t_i > 1 - \epsilon$.

Suppose that $t_i > 1-\epsilon$ and $\dim Z_i > 0$. Let A_{Z_i} be an ample Cartier divisor on Z_i such that $-(K_{X_i}+B_i+t_iS_i)+\pi_i^*A_{Z_i}$ is ample. By [Tan17, Theorem 1], there exists a $\mathbb{Q}$-divisor $G_i \geq 0$ with

$$G_i \sim_{\mathbb{Q}} -(K_{X_i}+B_i+t_iS_i)+\pi_i^*A_{Z_i}$$

such that $(X_i, B_i+t_iS_i+G_i)$ is klt.

Write $B_i+t_iS_i=\sum_j d_{ij}P_{ij}$, where the P_{ij} are distinct prime divisors, and let

$$B_i^\sharp := \sum_{d_{ij}\leq 1-\epsilon} d_{ij}P_{ij} + \sum_{d_{ij}>1-\epsilon} P_{ij}.$$

By [Jia25, Proposition 4.3] for $(X_i, B_i+t_iS_i)\to Z_i$ with zero nef part and the $\mathbb{Q}$-divisor G_i, $(X_i, B_i^\sharp)$ is lc, and any $-(K_{X_i}+B_i^\sharp)$-MMP over Z_i ends with a projective threefold X_i'' over Z_i such that $-(K_{X_i''}+B_i''^\sharp)$ is nef over Z_i, where $B_i''^\sharp$ is the pushforward of $B_i^\sharp$.

Since $t_i > 1-\epsilon$, any component of S_i has coefficient 1 in $B_i^\sharp$, so $B_i^\sharp \geq B_i+S_i$. By **Step 2** and (6),

$$K_{X_i}+B_i^\sharp = L_i + (B_i^\sharp - B_i - S_i)$$

is ample over Z_i. Since $\rho(X_i/Z_i)=1$, the cone $\mathrm{NE}(X_i/Z_i)$ is an extremal ray, $-(K_{X_i}+B_i^\sharp)$ is negative on $\mathrm{NE}(X_i/Z_i)\backslash\{0\}$, and the contraction of $\mathrm{NE}(X_i/Z_i)$ is π_i, which is of fiber type. Thus $-(K_{X_i}+B_i^\sharp)$ is not nef over Z_i, and the first step of any $-(K_{X_i}+B_i^\sharp)$-MMP over Z_i is the contraction π_i of fiber type. Hence no $-(K_{X_i}+B_i^\sharp)$-MMP over Z_i ends with a threefold on which the pushforward of $-(K_{X_i}+B_i^\sharp)$ is nef over Z_i, a contradiction. Thus Z_i is a point.

Step 4. Let $\delta>0$ be the real number of Lemma 3.7 for the DCC set $J\cup\{1\}$. Possibly truncating the sequence, we may assume that $t_i > \max\{1-\epsilon, 1-\delta\}$ for any i. By **Step 3**, Z_i is a point and $\rho(X_i)=1$ for any i. By **Step 2**, L_i and S_i are ample, and by (5), $-(K_{X_i}+B_i+t_iS_i)$ is ample.

Since

$$(1-\delta)(B_i+S_i) \leq t_i(B_i+S_i) \leq B_i+t_iS_i$$

and X_i is $\mathbb{Q}$-factorial, $(X_i, (1-\delta)(B_i+S_i))$ is klt by (4). Since $B_i+S_i \in J\cup\{1\}$, (X_i, B_i+S_i) is lc by Lemma 3.7.

Let $C_i\subset X_i$ be a curve, and let

$$\lambda_i := -\frac{(K_{X_i}+B_i)\cdot C_i}{S_i\cdot C_i} \in \mathbb{Q}.$$

Since $\rho(X_i)=1$,

$$K_{X_i}+B_i+\lambda_iS_i \equiv 0,$$

and $(K_{X_i}+B_i+tS_i)\cdot C_i = (t-\lambda_i)S_i\cdot C_i$ for any $t\in\mathbb{R}$. Since $S_i\cdot C_i>0$, $(K_{X_i}+B_i+t_iS_i)\cdot C_i<0$, and $L_i\cdot C_i>0$, we have $t_i<\lambda_i<1$. In particular, $\lim_{i\to+\infty}\lambda_i=1$, and

$$L_i \equiv (1-\lambda_i)S_i.$$

Let $\mu_i := \frac{1-\lambda_i}{1-t_i}\in(0,1)$. Then

$$B_i+\lambda_iS_i = \mu_i(B_i+t_iS_i)+(1-\mu_i)(B_i+S_i).$$

Since log discrepancies are affine in the boundary, $(X_i, B_i+t_iS_i)$ is klt, and (X_i, B_i+S_i) is lc, the pair $(X_i, B_i+\lambda_iS_i)$ is klt. Since $-K_{X_i}\equiv B_i+\lambda_iS_i$, $S_i\neq 0$, and $\rho(X_i)=1$, the $\mathbb{Q}$-Cartier divisor $-K_{X_i}$ is ample.

Possibly passing to a subsequence, we may assume that $(\lambda_i)_{i\geq 1}$ is strictly increasing. Then

$$I := J\cup\{\lambda_i \mid i\geq 1\} \subset [0,1)\cap\mathbb{Q}$$

is a DCC set, and the coefficients of $B_i+\lambda_iS_i$ belong to I for any i. Let $c>0$ be the real number of Theorem 3.8 for I. By Theorem 3.8,

$$(-K_{X_i})^3 \leq c \quad \text{for any } i. \tag{4.1}$$

Step 5. Let R_i be a prime component of S_i, let $\nu_i : R_i^\nu \to X_i$ be the composition of the normalization of R_i and the inclusion $R_i \subset X_i$, and let $B_{R_i^\nu}$ be the different defined by

$$K_{R_i^\nu} + B_{R_i^\nu} := (K_{X_i} + B_i + S_i)|_{R_i^\nu} = \nu_i^* L_i.$$

Since $(X_i, B_i + S_i)$ is lc, the pair $(R_i^\nu, B_{R_i^\nu})$ is lc [Das21, Lemma 2.10]. Since $B_i + S_i \in J \cup \{1\}$, we have $B_{R_i^\nu} \in \mathcal{D}(J \cup \{1\})$ [Bir16, Propositions 4.1–4.2], [Das21, Proposition 2.9], and $\mathcal{D}(J \cup \{1\})$ is a DCC subset of $[0,1] \cap \mathbb{Q}$ [Das21, Lemma 2.8(6)]. Since L_i is ample and ν_i is finite, $K_{R_i^\nu} + B_{R_i^\nu}$ is ample.

Let N be the positive integer of [Das22, Theorem 1.2] for the DCC set $\mathcal{D}(J \cup \{1\})$. It depends only on J. By [Das22, Theorem 1.2], $|\lfloor N(K_{R_i^\nu} + B_{R_i^\nu})\rfloor|$ defines a birational map, so

$$(K_{R_i^\nu} + B_{R_i^\nu})^2 = \mathrm{vol}(K_{R_i^\nu} + B_{R_i^\nu}) \geq \frac{1}{N^2}.$$

Since ν_i is birational onto R_i and $L_i \equiv (1-\lambda_i)S_i$,

$$(1-\lambda_i)^2 S_i^2 \cdot R_i = L_i^2 \cdot R_i = (K_{R_i^\nu} + B_{R_i^\nu})^2 \geq \frac{1}{N^2}.$$

Since S_i is ample, $S_i^2 \cdot T \geq 0$ for any component T of S_i, so $S_i^3 \geq S_i^2 \cdot R_i$. Since $\rho(X_i) = 1$, $B_i \geq 0$, and $S_i \geq 0$, the $\mathbb{Q}$-divisors B_i and S_i are nef. Thus

$$(-K_{X_i})^3 = (B_i + \lambda_i S_i)^3 \geq \lambda_i^3 S_i^3 \geq \frac{\lambda_i^3}{N^2(1-\lambda_i)^2}.$$

Since $\lim_{i\to+\infty} \lambda_i = 1$, the right-hand side tends to $+\infty$, which contradicts (4.1). The proposition follows. □

Corollary 4.2. *Let $I \subset [0,1] \cap \mathbb{Q}$ be a DCC set such that $I \cap [0,a]$ is finite for any $a < 1$. There exist a rational number $\theta \in (0,1)$ and an integer $q \geq 2$ depending only on I with the following property. Let X be a smooth projective threefold over an algebraically closed field of characteristic $p > 5$, and let $B = \sum_i b_i B_i \geq 0$ be a $\mathbb{Q}$-divisor, where the B_i are distinct prime divisors. Assume that B has simple normal crossing support, $B \in I \cup \{1\}$, and $K_X + B$ is big. Let $B^- := \sum_i \min\{b_i, \theta\} B_i$. Then qB^- is a Weil divisor, (X, B^-) is $\frac{1}{q}$-lc, and*

$$\frac{1}{8}\,\mathrm{vol}(K_X + B) \leq \mathrm{vol}(K_X + B^-) \leq \mathrm{vol}(K_X + B).$$

Proof. Let $\eta \in (0, \frac{1}{2})$ be the rational number of Proposition 4.1 for the DCC set $I \backslash \{1\}$, and let

$$\theta := 1 - \frac{\eta}{2} \in \left(\frac{3}{4}, 1\right) \cap \mathbb{Q}.$$

The set $I' := (I \cap [0,\theta]) \cup \{\theta\}$ is a finite subset of $\mathbb{Q}$. Let $q \geq 2$ be an integer such that $qI' \subset \mathbb{Z}$. Then θ and q depend only on I.

Let

$$B' := \sum_{b_i \leq \theta} b_i B_i \quad \text{and} \quad S := \sum_{b_i > \theta} B_i.$$

Then $B^- = B' + \theta S$ and $B^- \in I'$, so qB^- is a Weil divisor. Since X is smooth and qB^- is a divisor, any log discrepancy of (X, B^-) belongs to $\frac{1}{q}\mathbb{Z}$. Since B^- has simple normal crossing support and coefficients at most $\theta < 1$, (X, B^-) is klt. Thus (X, B^-) is $\frac{1}{q}$-lc.

Since $B^- \leq B$, we have $\mathrm{vol}(K_X + B^-) \leq \mathrm{vol}(K_X + B)$. We let

$$L := K_X + B' + S \quad \text{and} \quad L' := K_X + B' + (1-\eta)S.$$

Since $b_i \leq 1$ for any i, $B' + S \geq B$, so L is big. By Proposition 4.1 for the log smooth pair $(X, B' + S)$, L' is pseudo-effective. Since $\theta = 1 - \frac{\eta}{2}$,

$$K_X + B^- = \frac{1}{2}L + \frac{1}{2}L'.$$

Since L' is pseudo-effective, by [Cut15, (3) and Theorem 2.5],

$$\mathrm{vol}(K_X + B^-) = \mathrm{vol}\left(\frac{1}{2}L + \frac{1}{2}L'\right) \geq \mathrm{vol}\left(\frac{1}{2}L\right) = \frac{1}{8}\mathrm{vol}(L) \geq \frac{1}{8}\mathrm{vol}(K_X + B).$$

The corollary follows. □

5. Regular surfaces over arbitrary fields

In this section we prove Lemma 5.2 over an arbitrary field, and Lemma 5.5 and Proposition 5.7 over an arbitrary field of characteristic ≥ 11. Theorems 7.2 and 10.1 use them on the generic fibers of families of surfaces, which are regular surfaces over function fields.

Lemma 5.1. *Let C be an integral curve which is proper over a field F, and let*

$$d_C := \dim_F H^0(C, \mathcal{O}_C).$$

Then $\chi_F(\mathcal{O}_C) \leq d_C$, and $\chi_F(\mathcal{G}) \in d_C\mathbb{Z}$ for any coherent sheaf $\mathcal{G}$ on C. In particular, the degree $\chi_F(\mathcal{L}) - \chi_F(\mathcal{O}_C)$ over F of any invertible sheaf $\mathcal{L}$ on C belongs to $d_C\mathbb{Z}$.

Proof. Since C is integral and proper over F, $H^0(C, \mathcal{O}_C)$ is a field of degree d_C over F, and the cohomology groups of coherent sheaves on C are vector spaces over it. Thus $\chi_F(\mathcal{G}) \in d_C\mathbb{Z}$, and $\chi_F(\mathcal{O}_C) = d_C - \dim_F H^1(C, \mathcal{O}_C) \leq d_C$. □

Lemma 5.2. *Let $\epsilon \in (0, 1]$ be a real number and N a positive integer. Then there exists a positive integer M depending only on ϵ and N satisfying the following for any field F.*

(1) Let (X, B) be an ϵ-lc pair of dimension 2 over F (with X a normal quasi-projective variety over F, see Definitions 2.1 and 2.2 and [Das21, Definition 2.2]). Assume that X is $\mathbb{Q}$-factorial, B is a $\mathbb{Q}$-divisor, and

$$\#\{E \mid E \text{ is a prime divisor exceptional over } X \text{ and } a(E, X, B) < 1\} \leq N.$$

Then MK_X and MG are Cartier for any prime component G of B.

(2) Let (X, B) be a projective terminal ϵ-lc pair of dimension 2 over F and q a positive integer. Assume that X is regular, B has at most N prime components, $K_X + B$ is big, and qB is a Weil divisor. Let $f : X \to X'$ be the composition of the steps of a $(K_X + B)$-MMP ending with a minimal model, and let $B' := f_*B$. Then $qM(K_{X'} + B')$ is nef, big, and Cartier, and

$$H^0(X, \mathcal{O}_X(\lfloor m(K_X + B)\rfloor)) = H^0(X', \mathcal{O}_{X'}(\lfloor m(K_{X'} + B')\rfloor))$$

as subspaces of the function field $F(X) = F(X')$ for any integer $m \geq 0$.

Proof. Let

$$N_1 := \lceil 2/\epsilon \rceil, \quad N_2 := N!N_1^{2N}, \quad \text{and} \quad M := \mathrm{lcm}(1, 2, \ldots, N_2).$$

We show that M satisfies (1) and (2).

Proof of (1). Since X is $\mathbb{Q}$-factorial, any Weil divisor on X is $\mathbb{Q}$-Cartier. It suffices to show that MK_X and MG are Cartier near any closed point x of X for any prime component G of B. If X is regular at x, then any Weil divisor on X is Cartier near x since $\mathcal{O}_{X,x}$ is a UFD. From now on we fix a closed point x at which X is not regular.

Step 1. Since X is normal and excellent, the non-regular locus of X is a finite set of closed points. Let U be an affine open neighborhood of x such that $U\backslash\{x\}$ is regular, and let $f : Y \to U$ be the minimal resolution of U [Kol13, Theorem 2.25] with no boundary (see also [Sat25, Definition 2.11]): Y is regular, f is a projective birational morphism that is an isomorphism over $U\backslash\{x\}$, and K_Y is f-nef. Let $E_1, \ldots, E_n$ be the prime divisors on Y contained in $f^{-1}(x)$. Since U is normal and x is not a regular point of U,

$$f^{-1}(x) = E_1 \cup \cdots \cup E_n \text{ is connected and } n \geq 1.$$

Let $\kappa := \kappa(x)$, and for any i let $d_i := \dim_\kappa H^0(E_i, \mathcal{O}_{E_i})$. For a $\mathbb{Q}$-divisor D_Y on Y we let $D_Y \cdot E_i$ be the degree over κ of $D_Y|_{E_i}$. Since Y is regular and E_i is an integral curve proper over κ, Lemma 5.1 over κ gives $\chi_\kappa(\mathcal{O}_{E_i}) \le d_i$ and

$$D_Y \cdot E_i \in d_i\mathbb{Z} \text{ for any Weil divisor } D_Y \text{ on } Y \text{ and any } i.$$

We write $K_U := K_X|_U$ and B for $B|_U$, choose K_Y with $f_*K_Y = K_U$, and write

$$f^*K_U = K_Y + \sum_{i=1}^n \gamma_i E_i \quad \text{and} \quad K_Y + B_Y = f^*(K_U + B),$$

where $B_Y := f_*^{-1}B + \sum_{i=1}^n \beta_i E_i$. Since K_Y is f-nef, $-\sum_i \gamma_i E_i = K_Y - f^*K_U$ is f-nef, so $\gamma_i \ge 0$ for any i by the negativity lemma [Tan18, Lemma 2.11]. Since $B \ge 0$ is $\mathbb{Q}$-Cartier, $\beta_i - \gamma_i = \mathrm{mult}_{E_i} f^*B \ge 0$. Since $a(E_i, U, B) = 1 - \beta_i \ge \epsilon$, we get

$$0 \le \gamma_i \le \beta_i \le 1 - \epsilon \quad \text{for any } i.$$

Since $(U \ni x, B)$ is ϵ-lc with $\epsilon > 0$, no prime divisor over U has log discrepancy 0 with respect to (U, B), so $(U \ni x)$ is a rational singularity [Kol13, Proposition 2.28], that is, $R^1 f_*\mathcal{O}_Y = 0$. Let $R := \mathcal{O}_{U,x}$ and $f_R : Y_R := Y \times_U \operatorname{Spec} R \to \operatorname{Spec} R$. Then Y_R is regular, f_R is proper and birational, $\dim f_R^{-1}(y) \le 1$ for any $y \in \operatorname{Spec} R$, and $H^1(Y_R, \mathcal{O}_{Y_R}) = (R^1 f_*\mathcal{O}_Y)_x = 0$. The matrix $(E_i \cdot E_j)_{1 \le i,j \le n}$ is negative definite [Lip69, Lemma (14.1)].

Step 2. We show the following.

Claim 5.3. *Let D be a Weil divisor on U such that f^*D is a Weil divisor. Then D is Cartier near x.*

Proof. Since Y is regular, $L := f^*D$ is Cartier, and $L \cdot E_i = 0$ for any i. By [Lip69, Theorem (12.1)(i)], $\mathcal{O}_{Y_R}(L|_{Y_R}) \cong \mathcal{O}_{Y_R}$, so $L|_{Y_R} = \mathrm{div}_{Y_R}(u)$ for a non-zero rational function u on Y. Thus

$$D|_{\operatorname{Spec} R} = (f_R)_*(L|_{Y_R}) = \mathrm{div}_{\operatorname{Spec} R}(u),$$

so no component of $D - \mathrm{div}_U(u)$ contains x, and D is Cartier near x. □

Step 3. Assume that $\beta_j = 0$ for some j. Since $K_Y + B_Y = f^*(K_U + B)$,

$$0 = (K_Y + B_Y) \cdot E_j = K_Y \cdot E_j + f_*^{-1}B \cdot E_j + \sum_{i \ne j} \beta_i E_i \cdot E_j.$$

Each term on the right is non-negative since K_Y is f-nef and $\beta_i \ge 0$, so $\beta_i = 0$ for any i with $E_i \cap E_j \ne \emptyset$. Since $E_1 \cup \cdots \cup E_n$ is connected, $\beta_i = 0$ for any i, so $\gamma_i = 0$ for any i. Thus $f^*K_U = K_Y$ is a Weil divisor. Write $B = \sum_k b_k G_k$, where the G_k are the prime components of B and $b_k > 0$, and write

$$f^*G_k = f_*^{-1}G_k + \sum_{i=1}^n t_{k,i} E_i.$$

Then $t_{k,i} \ge 0$ since $G_k \ge 0$ is $\mathbb{Q}$-Cartier, and $\sum_k b_k t_{k,i} = \beta_i - \gamma_i = 0$ for any i. Thus $t_{k,i} = 0$, so $f^*G_k = f_*^{-1}G_k$ is a Weil divisor for any k. By the Claim, K_U and G_k are Cartier near x for any k, hence so are MK_U and MG_k.

Step 4. Assume that $\beta_i > 0$ for any i. Then $a(E_i, X, B) = 1 - \beta_i < 1$ for any i, and $E_1, \dots, E_n$ are distinct prime divisors exceptional over X, so $n \le N$. For any i, j let

$$m_{ij} := -\frac{E_i \cdot E_j}{d_i} \in \mathbb{Z}.$$

Since $\chi_\kappa(\mathcal{O}_{E_i}) \le d_i$ by **Step 1**, the adjunction formula [Tan18, Corollary 2.9] gives

$$K_Y \cdot E_i \ge -2d_i - E_i^2 = (m_{ii} - 2)d_i.$$

Since $f_*^{-1}B \cdot E_i \geq 0$ and $\beta_j E_j \cdot E_i \geq 0$ for any $j \neq i$,

$$0 = (K_Y + B_Y) \cdot E_i \geq K_Y \cdot E_i + \beta_i E_i^2 \geq \big((1-\beta_i)m_{ii} - 2\big)d_i \geq (\epsilon m_{ii} - 2)d_i.$$

Since $E_i^2 < 0$, we have $m_{ii} \geq 1$, so

$$1 \leq m_{ii} \leq N_1 \quad \text{for any } i.$$

Let $i \neq j$. The matrix

$$\begin{pmatrix} E_i^2 & E_i \cdot E_j \\ E_i \cdot E_j & E_j^2 \end{pmatrix}$$

is negative definite, so $(E_i \cdot E_j)^2 < E_i^2 E_j^2$ and

$$m_{ij}m_{ji} = \frac{(E_i \cdot E_j)^2}{d_i d_j} < m_{ii}m_{jj} \leq N_1^2.$$

Since $m_{ij}, m_{ji} \leq 0$ are integers and $m_{ij} = 0$ if and only if $m_{ji} = 0$, we get $|m_{ij}| \leq N_1^2$. Thus $|m_{ij}| \leq N_1^2$ for any i, j. Since $d_i m_{ij} = -E_i \cdot E_j$ and $(-E_i \cdot E_j)_{i,j}$ is positive definite,

$$\Big(\prod_{i=1}^n d_i\Big) \det(m_{ij}) = \det(-E_i \cdot E_j) > 0.$$

Since $\det(m_{ij})$ is an integer and $n \leq N$, the Leibniz formula implies

$$1 \leq \det(m_{ij}) \leq n! N_1^{2n} \leq N_2,$$

so $\det(m_{ij})$ divides M.

Let D be a Weil divisor on U and write $f^*D = f_*^{-1}D + \sum_j t_j E_j$ with $t_j \in \mathbb{Q}$. Since $f^*D \cdot E_i = 0$ and $f_*^{-1}D$ is a Weil divisor on Y,

$$\sum_{j=1}^n m_{ij} t_j = \frac{f_*^{-1}D \cdot E_i}{d_i} \in \mathbb{Z} \quad \text{for any } i.$$

By Cramer's rule, $\det(m_{ij})t_j \in \mathbb{Z}$, so $Mt_j \in \mathbb{Z}$ for any j. Thus $f^*(MD)$ is a Weil divisor, and MD is Cartier near x by the Claim. In particular, MK_U and MG are Cartier near x for any prime component G of B.

By **Steps 3 and 4**, MK_X and MG are Cartier near x for any prime component G of B, and (1) follows.

Proof of (2). Write the MMP as $X = X_0 \to X_1 \to \cdots \to X_{n_1} = X'$, where $g_j : X_j \to X_{j+1}$ are the steps of the MMP, and let B_j and K_{X_j} be the images of B and K_X on X_j, so that $K_{X'} = f_* K_X$. Since X is regular, X is $\mathbb{Q}$-factorial, so X_j is $\mathbb{Q}$-factorial for any j [Tan18, Theorem 4.4]. For any $j < n_1$, $-(K_{X_j} + B_j)$ is g_j-ample, so $K_{X_j} + B_j - g_j^*(K_{X_{j+1}} + B_{j+1}) \geq 0$ by the negativity lemma [Tan18, Lemma 2.11]. Pulling back to X, we get

$$E := K_X + B - f^*(K_{X'} + B') \geq 0 \quad \text{and} \quad f_* E = 0.$$

Thus $a(P, X', B') = a(P, X, B - E) \geq a(P, X, B)$ for any prime divisor P over X', so (X', B') is ϵ-lc.

Let P be a prime divisor exceptional over X' such that $a(P, X', B') < 1$. If P is exceptional over X, then $a(P, X', B') \geq a(P, X, B) > 1$ since (X, B) is terminal, which is not possible. Thus P is a prime divisor on X with $f_* P = 0$, and

$$1 > a(P, X', B') = a(P, X, B - E) = 1 - \operatorname{mult}_P B + \operatorname{mult}_P E,$$

so $\operatorname{mult}_P B > \operatorname{mult}_P E \geq 0$ and P is a prime component of B. Since B has at most N prime components,

$$\#\{P \mid P \text{ is a prime divisor exceptional over } X' \text{ and } a(P, X', B') < 1\} \leq N.$$

Since qB' is a Weil divisor, B' is a $\mathbb{Q}$-divisor. By (1), $MK_{X'}$ and MG' are Cartier for any prime component G' of B'. Thus

$$qM(K_{X'}+B') = q\cdot MK_{X'} + \sum_{G'}(q\,\mathrm{mult}_{G'}\,B')\cdot MG'$$

is Cartier.

Let u be a non-zero rational function on X and $m\geq 0$ an integer. Since $\mathrm{div}_X(u)$ and $\mathrm{div}_{X'}(u)$ are Weil divisors,

$$\mathrm{div}_X(u)+\lfloor m(K_X+B)\rfloor \geq 0 \iff \mathrm{div}_X(u)+m(K_X+B)\geq 0 \quad \text{and}$$
$$\mathrm{div}_{X'}(u)+\lfloor m(K_{X'}+B')\rfloor \geq 0 \iff \mathrm{div}_{X'}(u)+m(K_{X'}+B')\geq 0.$$

If $\mathrm{div}_X(u)+m(K_X+B)\geq 0$, then pushing forward by f we get $\mathrm{div}_{X'}(u)+m(K_{X'}+B')\geq 0$. Conversely, if $\mathrm{div}_{X'}(u)+m(K_{X'}+B')\geq 0$, then $f^*\big(\mathrm{div}_{X'}(u)+m(K_{X'}+B')\big)\geq 0$ since $\mathrm{div}_{X'}(u)+m(K_{X'}+B')$ is $\mathbb{Q}$-Cartier, so

$$\mathrm{div}_X(u)+m(K_X+B) = f^*\big(\mathrm{div}_{X'}(u)+m(K_{X'}+B')\big)+mE\geq 0.$$

Thus the equality of H^0 in (2) holds. Since K_X+B is big, the equality of H^0 implies that $K_{X'}+B'$ is big. Since (X',B') is a minimal model, $K_{X'}+B'$ is nef. Thus $qM(K_{X'}+B')$ is nef, big, and Cartier. □

Definition-Lemma 5.4. For a regular projective surface Y over a field F, a closed point $x\in Y$, and a $\mathbb{Q}$-divisor $D=\sum_k d_kD_k$ on Y, where the D_k are distinct prime divisors, let $\mu_x(D):=\sum_k d_k\mu_x(D_k)$, where for a local equation f_k of D_k at x,

$$\mu_x(D_k):=\max\{\mu\in\mathbb{N}\mid f_k\in\mathfrak{m}_x^\mu\}.$$

Let $h: Y'\to Y$ be the blow-up of such a surface Y at a closed point x, and let E be the exceptional curve. Then the following hold.

(1) $h^*D=h_*^{-1}D+\mu_x(D)E$ for any $\mathbb{Q}$-divisor D on Y.
(2) $\mu_y(h_*^{-1}D)\leq\mu_x(D)$ for any $\mathbb{Q}$-divisor $D\geq 0$ on Y and any closed point $y\in E$.

Proof. (1) For a prime divisor D this is [Liu02, Proposition 9.2.23], and both sides are linear in D.

(2) Both sides are linear in D, so we may assume that D is a prime divisor. Let $\mu:=\mu_y(h_*^{-1}D)$. A local equation of $h_*^{-1}D$ at y belongs to $\mathfrak{m}_y^\mu$, and $E\not\subset h_*^{-1}D$. Since $E\cong\mathbb{P}^1_{\kappa(x)}$ is regular, $\mathcal{O}_{E,y}$ is a discrete valuation ring, and the image of the local equation in $\mathcal{O}_{E,y}$ has valuation at least μ. Thus, computing intersection numbers with F-dimensions, by (1) and [Liu02, Proposition 9.2.5],

$$\mu[\kappa(y):F]\leq h_*^{-1}D\cdot E=(h^*D-\mu_x(D)E)\cdot E=\mu_x(D)[\kappa(x):F].$$

Since $\kappa(x)\subset\kappa(y)$, $\mu\leq\mu_x(D)$. □

Lemma 5.5. *Let $\epsilon\in(0,1)$ be a real number. There exists a positive integer N depending only on ϵ with the following property. Let X be a regular projective surface over a field F of characteristic ≥ 11 with $H^0(X,\mathcal{O}_X)=F$, and let $B=\sum_i b_iB_i$ be a $\mathbb{Q}$-divisor on X, where the B_i are distinct prime divisors. Assume that $B\neq 0$, (X,B) is ϵ-lc, and $K_X+B\equiv 0$. Then $\rho(X)\leq N$ and $\sum_i b_i\leq N$.*

Proof. Let

$$N_1:=\max\{9,\lceil 2/\epsilon\rceil+4\},\quad N_2:=\frac{4}{\epsilon^2}\Big(9+\frac{2N_1}{\epsilon}\Big),\quad\text{and}\quad N_3:=\lceil 2/\epsilon\rceil.$$

We show that the positive integer $N:=\lceil N_1+N_2(1+N_3^2)\rceil$ satisfies the lemma. Let $p\geq 11$ be the characteristic of F. Removing the terms with $b_i=0$, we may assume that $b_i>0$ for any i. Intersection numbers, degrees, and Euler characteristics χ of coherent sheaves on schemes proper over F are computed with F-dimensions.

Let Y be a regular projective surface over F and C a divisor on Y with $C \geq 0$ and $C \neq 0$. By the adjunction formula [Liu02, Theorem 9.1.37],

$$(K_Y + C) \cdot C = -2\chi(\mathcal{O}_C). \tag{5.1}$$

If C is an integral curve on Y, we let $d_C := \dim_F H^0(C, \mathcal{O}_C)$. Since Y is regular, $\mathcal{O}_Y(D)|_C$ is invertible and $C \cdot D = \chi(\mathcal{O}_Y(D)|_C) - \chi(\mathcal{O}_C)$ for any divisor D on Y, so Lemma 5.1 gives

$$\chi(\mathcal{O}_C) \leq d_C \quad \text{and} \quad \chi(\mathcal{O}_C),\ C \cdot D \in d_C\mathbb{Z} \ \text{ for any divisor } D \text{ on } Y. \tag{5.2}$$

We divide the proof into seven steps.

Step 1. In this step we reduce to the case when F is separably closed. Let F^s be a separable closure of F, $X^s := X \times_F F^s$, and B^s the pullback of B to X^s. For any closed point $x \in X$, the ring $\kappa(x) \otimes_F F^s$ is reduced and 0-dimensional since F^s/F is separable algebraic, so X^s is regular by [Mat89, Theorem 23.7(ii)]. By flat base change, $H^0(X^s, \mathcal{O}_{X^s}) = F^s$, so X^s is connected, hence integral. Any prime divisor over X^s is defined over $X \times_F F'$ for some finite separable extension F'/F, and $X \times_F F' \to X$ is étale and surjective. Thus (X^s, B^s) is ϵ-lc by [Kol13, 2.14(1)–(2)]. The divisor $K_{X^s} + B^s$ is the pullback of $K_X + B$, and pullback to X^s preserves intersection numbers by flat base change, so

$$K_{X^s} + B^s \equiv 0 \quad \text{and} \quad \rho(X) \leq \rho(X^s).$$

Since F^s/F is separable, for any i we have $B_i \times_F F^s = \sum_{k=1}^{n_i} B_{i,k}$ for some integer $n_i \geq 1$ and distinct prime divisors $B_{i,k}$, so

$$B^s = \sum_i \sum_{k=1}^{n_i} b_i B_{i,k} \quad \text{and} \quad \sum_i n_i b_i \geq \sum_i b_i.$$

Possibly replacing F with F^s and (X, B) with (X^s, B^s), we may assume that F is separably closed.

Step 2. In this step we run a K_X-MMP and show that it is crepant for $K_X + B$. Let H_X be an ample divisor on X. Since $B \geq 0$ and $B \neq 0$,

$$K_X \cdot H_X = -B \cdot H_X < 0.$$

Since X is regular, X is $\mathbb{Q}$-factorial. By [Tan18, Theorems 1.1 and 4.4], there exists a K_X-MMP

$$X = X_m \xrightarrow{f_m} X_{m-1} \xrightarrow{f_{m-1}} \cdots \xrightarrow{f_1} X_0 =: X',$$

where for any $j \in \{1, \ldots, m\}$, f_j is a birational morphism whose exceptional locus is a curve E_j with $K_{X_j} \cdot E_j < 0$, and $\rho(X_j) = \rho(X_{j-1}) + 1$. The exceptional locus of f_j is one curve since the classes of two f_j-exceptional curves $C \neq C'$ would be proportional and $C'^2 < 0$ by [Tan18, Lemma 2.11], so $C \cdot C' < 0$ would follow, which is impossible. Moreover, either $K_{X'}$ is nef or there exists a $K_{X'}$-Mori fiber space $X' \to Z$ [Tan18, Theorem 1.1(4)]. Let $g := f_1 \circ \cdots \circ f_m : X \to X'$. Then $\rho(X) = \rho(X') + m$.

Fix a canonical divisor K_X, and for any j let K_{X_j} be its pushforward to X_j. In particular, $K_{X'} = g_* K_X$. We show that X_j is regular by descending induction on j. Assume that X_j is regular for some $j \geq 1$. By the negativity lemma [Tan18, Lemma 2.11], $E_j^2 < 0$, so

$$K_{X_j} - f_j^* K_{X_{j-1}} = \frac{K_{X_j} \cdot E_j}{E_j^2} E_j, \quad \text{and} \quad \frac{K_{X_j} \cdot E_j}{E_j^2} > 0.$$

Since X_j is regular, $(X_j, 0)$ is terminal by [Kol13, Theorem 2.29(1)]. For any exceptional prime divisor P over X_{j-1}, either $P = E_j$ and $a(P, X_{j-1}, 0) = 1 + \frac{K_{X_j} \cdot E_j}{E_j^2} > 1$, or P is exceptional over X_j and $a(P, X_{j-1}, 0) \geq a(P, X_j, 0) > 1$. Thus $(X_{j-1}, 0)$ is terminal, and X_{j-1} is regular by [Kol13, Theorem 2.29(1)]. By [Liu02, Theorem 9.2.2], f_j is a composite of blow-ups of closed points. Since E_j is the only exceptional curve of f_j, f_j is the blow-up of X_{j-1} at the closed point $x_j := f_j(E_j)$.

Since $(X',0)$ is terminal, $K_X - g^*K_{X'} \geq 0$. If $K_{X'}$ were nef, then

$$K_X \cdot H_X = g^*K_{X'} \cdot H_X + (K_X - g^*K_{X'}) \cdot H_X \geq 0,$$

a contradiction. Thus there exists a $K_{X'}$-Mori fiber space $\pi : X' \to Z$. By definition, $\pi_*\mathcal{O}_{X'} = \mathcal{O}_Z$, $\dim Z \leq 1$, $\rho(X'/Z) = 1$, and $-K_{X'}$ is ample over Z.

For any $j \in \{0, \dots, m\}$, let B_{X_j} be the pushforward of B to X_j, and let $B_{X'} := B_{X_0} = g_*B$. Then $g^*(K_{X'} + B_{X'}) - (K_X + B)$ is g-exceptional, so for any curve Γ on X',

$$(K_{X'} + B_{X'}) \cdot \Gamma = g^*(K_{X'} + B_{X'}) \cdot g^*\Gamma = (K_X + B) \cdot g^*\Gamma = 0.$$

Thus $g^*(K_{X'} + B_{X'}) - (K_X + B)$ is numerically trivial, so it is zero by the negativity lemma [Tan18, Lemma 2.11]. Pushing forward to X_j, we obtain

$$K_{X_j} + B_{X_j} = (f_1 \circ \cdots \circ f_j)^*(K_{X'} + B_{X'}) \quad \text{for any } j \in \{0, \dots, m\}. \tag{5.3}$$

Thus for any j, $B_{X_j} \geq 0$, $a(P, X_j, B_{X_j}) = a(P, X, B) \geq \epsilon$ for any prime divisor P over X_j, and $\operatorname{mult}_P B_{X_j} \leq 1 - \epsilon$ for any prime divisor P on X_j. Since $-K_{X'}$ is ample over Z and $K_{X'} + B_{X'} \equiv 0$, $B_{X'} \neq 0$.

For any $k \geq j$, we also write E_j for $(f_{j+1} \circ \cdots \circ f_k)_*^{-1}E_j$ on X_k, and let $c_j := \operatorname{mult}_{E_j} B$. Then $0 \leq c_j \leq 1 - \epsilon$. By [Liu02, Propositions 9.2.5 and 9.2.24], $E_j \cong \mathbb{P}^1_{\kappa(x_j)}$ on X_j, $E_j^2 = -[\kappa(x_j) : F]$ on X_j, and $K_{X_j} = f_j^*K_{X_{j-1}} + E_j$. By Definition-Lemma 5.4(1), $f_j^*D = (f_j)_*^{-1}D + \mu_{x_j}(D)E_j$ for any $\mathbb{Q}$-divisor D on X_{j-1}. Comparing the coefficients of E_j in (5.3) for j and $j-1$, we obtain

$$c_j = \mu_{x_j}(B_{X_{j-1}}) - 1 \quad \text{for any } j \in \{1, \dots, m\}. \tag{5.4}$$

Let $I' := \{i \mid B_i \text{ is not } g\text{-exceptional}\}$, and let $B'_i := g_*B_i$ for any $i \in I'$. Then

$$B_{X'} = \sum_{i \in I'} b_i B'_i,$$

and any g-exceptional B_i is equal to E_j for some j, with $b_i = c_j$. Thus

$$\sum_i b_i \leq \sum_{i \in I'} b_i + \sum_{j=1}^m c_j. \tag{5.5}$$

Step 3. In this step we show that one of the following holds.

(i) $\dim Z = 0$, X' is smooth over F, $\rho(X') = 1$, and $K_{X'}^2 \leq 9$.
(ii) $\dim Z = 1$, $Z \cong \mathbb{P}^1_F$, and π is smooth.
(iii) $\dim Z = 1$, π is smooth, $-K_{X'}$ is nef, $K_{X'}^2 = 0$, and Z is a smooth projective curve over F with $H^0(Z, \mathcal{O}_Z) = F$ and $\dim_F H^1(Z, \mathcal{O}_Z) = 1$.

Assume that $\dim Z = 0$. Then $\rho(X') = 1$ and $-K_{X'}$ is ample. Since $p \geq 11$ and

$$H^0(X', \mathcal{O}_{X'}) = H^0(X, \mathcal{O}_X) = F,$$

X' is smooth over F by [BT22, Proposition 5.2]. Let $\overline{F}$ be an algebraic closure of F. Then $X' \times_F \overline{F}$ is a smooth del Pezzo surface, so $K_{X'}^2 \leq 9$ by [BT22, Lemma 5.1]. Thus (i) holds.

Assume that $\dim Z = 1$. Since $\pi_*\mathcal{O}_{X'} = \mathcal{O}_Z$, Z is a normal, hence regular, projective curve with $H^0(Z, \mathcal{O}_Z) = F$, and F is algebraically closed in $F(Z)$. Let $p_a(Z) := \dim_F H^1(Z, \mathcal{O}_Z)$. Since F is separably closed and $p \neq 2$, π is smooth by [BT22, Proposition 2.18(2)(4)(6)], so for any closed point $z \in Z$, the fiber X'_z is a smooth conic over $\kappa(z)$. By (5.1) and since $X_z'^2 = 0$,

$$-K_{X'} \cdot X'_z = 2[\kappa(z) : F] > 0. \tag{5.6}$$

Since X'_z is integral, any integral curve C on X' with $K_{X'} \cdot C \geq 0$ dominates Z. We first show that

$$K_{X'}^2 = 8(1 - p_a(Z)). \tag{5.7}$$

Let $\mathcal{L} := \omega_{X'/Z}^{-1}$ and $\mathcal{V} := \pi_*\mathcal{L}$. For any closed point $z \in Z$, $\dim_{\kappa(z)} H^0(X'_z, \mathcal{L}|_{X'_z}) = 3$, $H^1(X'_z, \mathcal{L}|_{X'_z}) = 0$, and $\mathcal{L}|_{X'_z}$ embeds X'_z as a conic in $\mathbb{P}^2_{\kappa(z)}$. By cohomology and base change, $\mathcal{V}$

is locally free of rank 3, and $\mathcal{L}$ defines a closed immersion over Z of X' into $\mathbb{P} := \operatorname{Proj}\operatorname{Sym}\mathcal{V}$ with $\mathcal{O}_{\mathbb{P}}(1)|_{X'} \cong \mathcal{L}$. Let $\pi_{\mathbb{P}} : \mathbb{P} \to Z$ be the projection. Then X' is the zero locus of a section of $\mathcal{O}_{\mathbb{P}}(2) \otimes \pi_{\mathbb{P}}^*\mathcal{N}$ for some invertible sheaf $\mathcal{N}$ on Z, that is, of a section of $\operatorname{Sym}^2\mathcal{V} \otimes \mathcal{N}$. Since $p \neq 2$, the section corresponds to a symmetric homomorphism $\mathcal{V}^\vee \to \mathcal{V} \otimes \mathcal{N}$, which is an isomorphism because X'_z is a smooth conic for any closed point $z \in Z$. Taking determinants,

$$\det \mathcal{V}^{-1} \cong \det \mathcal{V} \otimes \mathcal{N}^3.$$

Since

$$\omega_{\mathbb{P}/Z} \cong \mathcal{O}_{\mathbb{P}}(-3) \otimes \pi_{\mathbb{P}}^* \det \mathcal{V},$$

the adjunction formula implies

$$\omega_{X'/Z} \cong \mathcal{L}^{-1} \otimes \pi^*(\det \mathcal{V} \otimes \mathcal{N}).$$

Since $\omega_{X'/Z} = \mathcal{L}^{-1}$ and $\pi^* : \operatorname{Pic}(Z) \to \operatorname{Pic}(X')$ is injective, $\mathcal{N} \cong \det \mathcal{V}^{-1}$. Substituting into $\det \mathcal{V}^{-1} \cong \det \mathcal{V} \otimes \mathcal{N}^3$, we get $\det \mathcal{V}^{-1} \cong \det \mathcal{V}^{-2}$, so $\det \mathcal{V} \cong \mathcal{N} \cong \mathcal{O}_Z$. Thus $X' \sim 2\xi$ on $\mathbb{P}$, where $\xi := c_1(\mathcal{O}_{\mathbb{P}}(1))$, and

$$K_{X'/Z}^2 = \xi^2 \cdot X' = 2\xi^3 = 2 \deg \det \mathcal{V} = 0.$$

Since π is smooth and Z is regular, $\omega_{X'} \cong \omega_{X'/Z} \otimes \pi^*\omega_Z$ [Liu02, Example 6.3.18 and Theorem 6.4.9(a)]. Since $\pi^*K_Z \cdot X'_z = 0$, (5.6) gives $K_{X'/Z} \cdot X'_z = K_{X'} \cdot X'_z = -2[\kappa(z) : F]$ for any closed point $z \in Z$. Since $(\pi^*K_Z)^2 = 0$ and $\deg K_Z = 2p_a(Z) - 2$ by the Riemann–Roch theorem,

$$K_{X'}^2 = K_{X'/Z}^2 + 2K_{X'/Z} \cdot \pi^*K_Z = -4(2p_a(Z) - 2) = 8(1 - p_a(Z)).$$

Claim 5.6. *Let C be an integral curve on X', let $\nu : C^\nu \to C$ be the normalization, and let*

$$L_C := H^0(C^\nu, \mathcal{O}_{C^\nu}).$$

Then $\chi(\mathcal{O}_{C^\nu}) \geq \chi(\mathcal{O}_C)$, and equality holds only if ν is an isomorphism. Moreover, if $\chi(\mathcal{O}_{C^\nu}) > 0$, then $C^\nu \cong \mathbb{P}^1_{L_C}$.

Proof. Since ν is finite and birational, $\nu_*\mathcal{O}_{C^\nu}/\mathcal{O}_C$ has finite length, and it is zero only if ν is an isomorphism. The first sentence follows. Assume that $\chi(\mathcal{O}_{C^\nu}) > 0$. Since L_C is a field and

$$\chi(\mathcal{O}_{C^\nu}) = [L_C : F]\big(1 - \dim_{L_C} H^1(C^\nu, \mathcal{O}_{C^\nu})\big) > 0,$$

$H^1(C^\nu, \mathcal{O}_{C^\nu}) = 0$, so ω_{C^ν} has degree -2 over L_C. Since C^ν is regular and $p \neq 2$, C^ν is a smooth conic over L_C by [BT22, Lemmas 2.16 and 2.17(3)]. Since L_C/F is finite and F is separably closed, L_C is separably closed, so C^ν has an L_C-rational point and $C^\nu \cong \mathbb{P}^1_{L_C}$. □

We show that $p_a(Z) \leq 1$, and that $p_a(Z) = 0$ if $-K_{X'}$ is not nef. If $-K_{X'}$ is nef, then $K_{X'}^2 \geq 0$, so $p_a(Z) \leq 1$ by (5.7). Assume that $-K_{X'}$ is not nef. Then there exists an integral curve C on X' with

$$B_{X'} \cdot C = -K_{X'} \cdot C < 0.$$

Thus C is a component of $B_{X'}$ and C dominates Z. Write $B_{X'} = bC + G$, where $b := \operatorname{mult}_C B_{X'}$. Since $G \geq 0$ and $C \not\subset \operatorname{Supp} G$, $G \cdot C \geq 0$, so $bC^2 \leq B_{X'} \cdot C < 0$ and

$$(K_{X'} + bC) \cdot C = -G \cdot C \leq 0.$$

Since $C^2 < 0$ and $b \leq 1 - \epsilon$, the last inequality and (5.1) imply $(1 - b)(-C^2) \leq 2\chi(\mathcal{O}_C)$, so $\chi(\mathcal{O}_C) > 0$. By (5.2), $\chi(\mathcal{O}_C) = d_C$ and $-C^2 \in d_C\mathbb{Z}$. Since $K_{X'} \cdot C = -C^2 - 2d_C > 0$, we have $-C^2 \geq 3d_C$, so

$$b \geq \frac{K_{X'} \cdot C}{-C^2} = 1 - \frac{2d_C}{-C^2} \geq \frac{1}{3}.$$

Let $m_C := [F(C) : F(Z)]$. For any closed point $z \in Z$, $C \cdot X'_z = m_C[\kappa(z) : F]$ and $G \cdot X'_z \geq 0$, so by (5.6),

$$bm_C[\kappa(z) : F] \leq B_{X'} \cdot X'_z = 2[\kappa(z) : F].$$

Thus $m_C \le 2/b \le 6 < p$, so $F(C)/F(Z)$ is separable. Let $\nu : C^\nu \to C$ be the normalization and $L_C := H^0(C^\nu, \mathcal{O}_{C^\nu}) \subset F(C)$. Since L_C/F is finite and F is separably closed, L_C is purely inseparable over F, hence over $F(Z)$. Since $F(C)/F(Z)$ is separable, $L_C \subset F(Z)$. Since F is algebraically closed in $F(Z)$, $L_C = F$. Since $\chi(\mathcal{O}_{C^\nu}) \ge \chi(\mathcal{O}_C) > 0$, $C^\nu \cong \mathbb{P}^1_F$ by Claim 5.6, and $\pi \circ \nu : C^\nu \to Z$ is a finite separable morphism of degree m_C between regular curves. By the Hurwitz formula for regular curves over an arbitrary field [Liu02, Theorem 7.4.16],

$$-2 = \deg K_{C^\nu} \ge m_C \deg K_Z = m_C(2p_a(Z) - 2),$$

so $p_a(Z) = 0$.

If $p_a(Z) = 0$, then $\deg K_Z = -2$, so Z is smooth over F by [BT22, Lemmas 2.16 and 2.17(3)]. Since F is separably closed, Z has an F-rational point and $Z \cong \mathbb{P}^1_F$, so (ii) holds. If $p_a(Z) = 1$, then $-K_{X'}$ is nef, $K^2_{X'} = 0$ by (5.7), and Z is smooth over F by [CCJS23, Theorem 2.8] since $p \ge 5$, so (iii) holds.

Step 4. In this step we prove the lemma in case (iii). We first show that no integral curve C on X' with $\chi(\mathcal{O}_{C^\nu}) > 0$ dominates Z, where C^ν is the normalization of C. Otherwise, by Claim 5.6, $C^\nu \cong \mathbb{P}^1_{L_C}$, and the dominant morphism $C^\nu \to Z$ induces a non-constant morphism $C^\nu \to Z \times_F L_C$ over L_C. After base change to an algebraic closure $\overline{L}$ of L_C, we obtain a non-constant morphism from $\mathbb{P}^1_{\overline{L}}$ to the smooth projective curve $Z \times_F \overline{L}$ of genus 1, which is impossible by Lüroth's theorem.

Since $-K_{X'}$ is nef and $B_{X'} \equiv -K_{X'}$, $B_{X'}$ is nef and $B^2_{X'} = K^2_{X'} = 0$. Since $B_{X'}$ is nef, $B_{X'} \cdot B'_i \ge 0$ for any $i \in I'$. Since $b_i > 0$ and

$$\sum_{i \in I'} b_i B_{X'} \cdot B'_i = B^2_{X'} = 0,$$

we have $B_{X'} \cdot B'_i = 0$ for any $i \in I'$. Since $B_{X'} \cdot X'_z > 0$ for any closed point $z \in Z$ by (5.6), B'_i dominates Z. Since $K_{X'} \cdot B'_i = -B_{X'} \cdot B'_i = 0$, (5.1) implies $\chi(\mathcal{O}_{B'_i}) = -B'^2_i/2$. Let B'^ν_i be the normalization of B'_i. By the previous paragraph and Claim 5.6,

$$-\frac{B'^2_i}{2} = \chi(\mathcal{O}_{B'_i}) \le \chi(\mathcal{O}_{B'^\nu_i}) \le 0,$$

so $B'^2_i \ge 0$. Since $B'_i \cdot B'_k \ge 0$ for any $i \ne k$ and

$$0 = B^2_{X'} = \sum_{i \in I'} b_i^2 B'^2_i + \sum_{i,k \in I', i \ne k} b_i b_k B'_i \cdot B'_k,$$

we have $B'^2_i = 0$ and $B'_i \cdot B'_k = 0$ for any $i \ne k$. Thus $\chi(\mathcal{O}_{B'_i}) = \chi(\mathcal{O}_{B'^\nu_i}) = 0$, so B'_i is regular by Claim 5.6. Hence the curves B'_i with $i \in I'$ are regular and pairwise disjoint, so $\mu_x(B_{X'}) \in \{0\} \cup \{b_i \mid i \in I'\}$ for any closed point $x \in X'$. In particular, $\mu_x(B_{X'}) < 1$. If $m \ge 1$, then (5.4) for $j = 1$ implies $c_1 = \mu_{x_1}(B_{X'}) - 1 < 0$, a contradiction. Thus $m = 0$, $X = X'$, I' contains any i, and $\rho(X) = 2 \le N$. For any closed point $z \in Z$ and any $i \in I'$, $B'_i \cdot X'_z = m_i[\kappa(z) : F]$, where $m_i \ge 1$ is the degree of $B'_i \to Z$. By (5.6),

$$\sum_i b_i \le \sum_i b_i m_i = \frac{B_{X'} \cdot X'_z}{[\kappa(z) : F]} = 2 \le N.$$

Step 5. From now on we assume that (i) or (ii) holds. In this step we show that there exists an ample Cartier divisor H on X' with

$$B_{X'} \cdot H \le N_1 \quad \text{and} \quad B^2_{X'} \le 9, \tag{5.8}$$

and we bound $(g_*^{-1} B_{X'})^2$.

In case (i), let $H := -K_{X'}$. Then H is an ample Cartier divisor, and

$$B_{X'} \cdot H = B^2_{X'} = K^2_{X'} \le 9 \le N_1.$$

In case (ii), since π is smooth and the fibers of π are smooth conics, X' is a Brauer–Severi scheme of relative dimension 1 over $Z \cong \mathbb{P}^1_F$ [Gro68, I, Corollary 8.3]. Since F is separably closed, $\mathrm{Br}(\mathbb{P}^1_F) = \mathrm{Br}(F) = 0$ [CS21, Theorem 6.1.3], so $X' \cong \mathbb{P}_Z(\mathcal{W})$ for a locally free sheaf $\mathcal{W}$ of rank 2 on Z [Gro68, I, 8.4]. By [HM82, Theorem 4.1], $X' \cong \mathbb{P}_Z(\mathcal{O}_Z \oplus \mathcal{O}_Z(-e))$ for some integer $e \geq 0$. Let $S \subset X'$ be a section of π with $S^2 = -e$, and $z \in Z$ an F-rational point. Then $K_{X'} \cdot S = e - 2$ by (5.1), and

$$-K_{X'} \sim 2S + (e+2)X'_z.$$

Let $b_S := \mathrm{mult}_S B_{X'} \leq 1 - \epsilon$. Since $B_{X'} - b_S S \geq 0$ and $S \not\subset \mathrm{Supp}(B_{X'} - b_S S)$,

$$0 = (K_{X'} + B_{X'}) \cdot S \geq (K_{X'} + b_S S) \cdot S = e - 2 - b_S e \geq \epsilon e - 2,$$

so $e \leq 2/\epsilon$. Let $H := S + (e+1)X'_z$. Then H is an ample Cartier divisor,

$$B_{X'} \cdot H = -K_{X'} \cdot H = e + 4 \leq \lceil 2/\epsilon \rceil + 4 \leq N_1,$$

and $B^2_{X'} = K^2_{X'} = 8$. Thus (5.8) holds in cases (i) and (ii).

For any $i \in I'$, let $d_i := d_{B_i} = \dim_F H^0(B_i, \mathcal{O}_{B_i})$ and let L_i be the field of global functions on the normalization of B'_i. Since B_i and B'_i are birational, L_i is also the field of global functions on the normalization of B_i, so $d_i \leq [L_i : F]$. The pullback of H to the normalization of B'_i has degree $B'_i \cdot H$, which is a positive integral multiple of $[L_i : F]$. Thus $d_i \leq [L_i : F] \leq B'_i \cdot H$ and $1 \leq B'_i \cdot H$, so by (5.8),

$$\sum_{i \in I'} b_i d_i \leq B_{X'} \cdot H \leq N_1 \quad \text{and} \quad \sum_{i \in I'} b_i \leq B_{X'} \cdot H \leq N_1. \tag{5.9}$$

For any $i \in I'$, since $(K_X + B) \cdot B_i = 0$ and $(B - b_i B_i) \cdot B_i \geq 0$, (5.1) and (5.2) imply

$$-2d_i - (1 - b_i)B_i^2 \leq (K_X + B_i) \cdot B_i - (1 - b_i)B_i^2 = (K_X + b_i B_i) \cdot B_i \leq 0.$$

Since $1 - b_i \geq \epsilon$, $B_i^2 \geq -2d_i/\epsilon$. For any $j \in \{0, \ldots, m\}$, let $D_j := (f_1 \circ \cdots \circ f_j)_*^{-1} B_{X'}$. Thus $D_0 = B_{X'}$ and $D_m = \sum_{i \in I'} b_i B_i$. Since $B_i \cdot B_k \geq 0$ for any $i \neq k$ and $b_i \leq 1$, (5.9) implies

$$D_m^2 \geq \sum_{i \in I'} b_i^2 B_i^2 \geq -\frac{2}{\epsilon} \sum_{i \in I'} b_i d_i \geq -\frac{2N_1}{\epsilon}. \tag{5.10}$$

Step 6. In this step we bound the number of blow-ups at points x_j with large $\mu_{x_j}(D_{j-1})$, and we perform the blow-ups at such points first. The division of the blow-ups by multiplicity follows [BM24, Proof of Proposition 5.11]. For any $j \in \{1, \ldots, m\}$, let $\mu_j := \mu_{x_j}(D_{j-1})$. By Definition-Lemma 5.4(1), $D_j = f_j^* D_{j-1} - \mu_j E_j$, so $D_j^2 = D_{j-1}^2 - \mu_j^2 [\kappa(x_j) : F]$ by [Liu02, Proposition 9.2.5]. Thus by (5.8) and (5.10),

$$\sum_{j=1}^{m} \mu_j^2 [\kappa(x_j) : F] = D_0^2 - D_m^2 \leq 9 + \frac{2N_1}{\epsilon}. \tag{5.11}$$

By (5.11),

$$\#\{j \mid \mu_j \geq \epsilon/2\} \leq \sum_{\mu_j \geq \epsilon/2} [\kappa(x_j) : F] \leq \frac{4}{\epsilon^2}\left(9 + \frac{2N_1}{\epsilon}\right) = N_2.$$

Let $1 \leq j < k \leq m$, and assume that the image of x_k in X_{j-1} is x_j. For any $i \in \{j-1, \ldots, k-1\}$ let y_i be the image of x_k in X_i. For any $i \in \{j, \ldots, k-1\}$, either $y_{i-1} \neq x_i$, in which case f_i is an isomorphism near y_i and $\mu_{y_i}(D_i) = \mu_{y_{i-1}}(D_{i-1})$, or $y_{i-1} = x_i$, in which case $y_i \in E_i$ and $\mu_{y_i}(D_i) \leq \mu_{y_{i-1}}(D_{i-1})$ by Definition-Lemma 5.4(2). Since $y_{k-1} = x_k$ and $y_{j-1} = x_j$,

$$\mu_k = \mu_{y_{k-1}}(D_{k-1}) \leq \mu_{y_{j-1}}(D_{j-1}) = \mu_j.$$

In particular, if $\mu_k \geq \epsilon/2$ and the image of x_k in X_{j-1} is x_j, then $\mu_j \geq \epsilon/2$.

Assume that $\mu_j < \epsilon/2 \leq \mu_{j+1}$ for some j. Then by the previous paragraph, the image x' of x_{j+1} in X_{j-1} is not x_j, so $x_{j+1} \notin E_j$ and f_j is an isomorphism near x_{j+1}. Both $f_j \circ f_{j+1}$ and the composite of the blow-up of X_{j-1} at x' with the blow-up at the preimage of x_j are the

blow-up of X_{j-1} at the reduced closed subscheme $\{x_j, x'\}$. We replace $f_j \circ f_{j+1}$ by the second composite. Since the multiplicity of D_{j-1} at x_j (resp. x') does not change under the blow-up at x' (resp. x_j), in the new factorization of g the center x' has multiplicity μ_{j+1}, the preimage of x_j has multiplicity μ_j, and the other centers and their multiplicities do not change. Thus the numbers of i with $\mu_i \geq \epsilon/2$ and with $\mu_i < \epsilon/2$ do not change, and the number of pairs $i < k$ with $\mu_i < \epsilon/2 \leq \mu_k$ decreases. Since the arguments of **Step 2** for (5.3), (5.4), and the properties of B_{X_j} only use that each f_j is the blow-up of a regular surface at a closed point, they hold for the new factorization, and so do (5.11) and the previous paragraph. Possibly repeating the exchange finitely many times, we may assume that there exists an integer $m_1 \leq N_2$ such that $\mu_j \geq \epsilon/2$ for any $j \leq m_1$ and $\mu_j < \epsilon/2$ for any $j > m_1$. We let $W := X_{m_1}$.

Step 7. In this step we bound $m - m_1$ and conclude the proof. We first show by induction on j that for any $j \in \{0, \dots, m\}$, the curves $E_1, \dots, E_j$ on X_j satisfy the following.

(S1) E_k is regular for any $k \leq j$.
(S2) For any $k < k' \leq j$, $E_k \cap E_{k'}$ is empty or one closed point y, and the intersection multiplicity of E_k and $E_{k'}$ at y is 1.
(S3) No point of X_j belongs to three of the curves $E_1, \dots, E_j$.

Let Γ be a regular curve on X_{j-1} through x_j. Then $(f_j)_*^{-1}\Gamma \cong \Gamma$ is regular and, by [Liu02, Propositions 9.2.5 and 9.2.23], $(f_j)_*^{-1}\Gamma \cdot E_j = [\kappa(x_j) : F]$. Since $\kappa(x_j) \subset \kappa(y)$ for any closed point $y \in E_j$, $(f_j)_*^{-1}\Gamma \cap E_j$ is one closed point, with intersection multiplicity 1. If Γ_1 and Γ_2 are regular curves on X_{j-1} whose intersection multiplicity at x_j is 1, then

$$(f_j)_*^{-1}\Gamma_1 \cdot (f_j)_*^{-1}\Gamma_2 = (f_j^*\Gamma_1 - E_j) \cdot (f_j^*\Gamma_2 - E_j) = \Gamma_1 \cdot \Gamma_2 - [\kappa(x_j) : F],$$

which is the contribution of the points of $\Gamma_1 \cap \Gamma_2$ other than x_j. Thus

$$(f_j)_*^{-1}\Gamma_1 \cap (f_j)_*^{-1}\Gamma_2 \cap E_j = \emptyset.$$

By (S1) and (S2) for $j-1$, the curves E_k with $k < j$ and $x_j \in E_k$ are regular, and two of them have intersection multiplicity 1 at x_j. Thus $E_j \cong \mathbb{P}^1_{\kappa(x_j)}$ is regular, $E_k \cap E_j$ is one closed point with intersection multiplicity 1 for any $k < j$ with $x_j \in E_k$, and two of the curves $E_1, \dots, E_{j-1}$ through x_j have no common point on E_j. Since at most two of the curves $E_1, \dots, E_{j-1}$ on X_{j-1} contain x_j, (S1)–(S3) for $j-1$ imply (S1)–(S3) for j.

Let Σ_j be the set of closed points of X_j belonging to two of the curves $E_1, \dots, E_j$. By the previous paragraph, E_j contains exactly one point of Σ_j if x_j belongs to exactly one of $E_1, \dots, E_{j-1}$, and exactly two if $x_j \in \Sigma_{j-1}$. Moreover,

$$\Sigma_j \backslash E_j = f_j^{-1}(\Sigma_{j-1} \backslash \{x_j\}).$$

Hence $\#\Sigma_j \leq \#\Sigma_{j-1} + 1$ for any $j \geq 2$, and $\Sigma_1 = \emptyset$. Thus $\#\Sigma_{m_1} \leq m_1 - 1$ if $m_1 \geq 1$.

Let $j \in \{1, \dots, m\}$ be an integer such that $\mu_j < \epsilon/2$, and let $n := \#\{k < j \mid x_j \in E_k\}$. Then $n \leq 2$ by (S3), and

$$B_{X_{j-1}} = D_{j-1} + \sum_{k=1}^{j-1} c_k E_k.$$

By (S1), $\mu_{x_j}(E_k) = 1$ if $x_j \in E_k$, so by (5.4),

$$0 \leq c_j = \mu_j + \sum_{k<j, x_j \in E_k} c_k - 1 < \frac{\epsilon}{2} + n(1-\epsilon) - 1.$$

Thus $n = 2$, that is, $x_j \in \Sigma_{j-1}$. Let $x_j \in E_k \cap E_{k'}$ with $k, k' < j$, and for any $k'' \in \{1, \dots, m\}$ let $a_{k''} := 1 - c_{k''} = a(E_{k''}, X, B) \in [\epsilon, 1]$. Since $\mu_j < \epsilon/2$,

$$a_j - \frac{\epsilon}{2} = \left(a_k - \frac{\epsilon}{2}\right) + \left(a_{k'} - \frac{\epsilon}{2}\right) + \left(\frac{\epsilon}{2} - \mu_j\right) > \left(a_k - \frac{\epsilon}{2}\right) + \left(a_{k'} - \frac{\epsilon}{2}\right). \tag{5.12}$$

For $j = 1$ we have $n = 0$. Thus if $m \geq 1$, then $\mu_1 \geq \epsilon/2$ and $m_1 \geq 1$.

Since $x_j \in \Sigma_{j-1}$ for any $j > m_1$, we define maps $r_j : \Sigma_j \to \Sigma_{m_1}$ for $j \geq m_1$ by $r_{m_1} := \mathrm{id}_{\Sigma_{m_1}}$, $r_j(y) := r_{j-1}(x_j)$ for $y \in \Sigma_j \cap E_j$, and $r_j(y) := r_{j-1}(f_j(y))$ for $y \in \Sigma_j \backslash E_j$. Fix $w \in \Sigma_{m_1}$, let E_{k_1} and E_{k_2} be the curves among $E_1, \ldots, E_{m_1}$ on W through w, and let

$$J_w := \{j > m_1 \mid r_{j-1}(x_j) = w\}.$$

We let $\lambda_{k_1} := (1, 0)$, $\lambda_{k_2} := (0, 1)$, and $\lambda_j := \lambda_k + \lambda_{k'}$ for any $j \in J_w$ with $x_j \in E_k \cap E_{k'}$. For $y \in r_j^{-1}(w) \cap E_k \cap E_{k'}$, let $\sigma_y \subset \mathbb{R}^2$ be the open cone spanned by λ_k and $\lambda_{k'}$. By induction on $j \geq m_1$, the following hold.

(C1) Any $y \in r_j^{-1}(w)$ belongs to $E_k \cap E_{k'}$ for some $k, k' \in \{k_1, k_2\} \cup J_w$ with $\lambda_k, \lambda_{k'}$ linearly independent.

(C2) The cones σ_y for $y \in r_j^{-1}(w)$ are pairwise disjoint and contain no λ_k with

$$k \in \{k_1, k_2\} \cup (J_w \cap [1, j]).$$

For the inductive step, the blow-up at x_j with $j \in J_w$ and $x_j \in E_k \cap E_{k'}$ replaces σ_{x_j} with the open cones spanned by λ_k, λ_j and by $\lambda_j, \lambda_{k'}$, which are disjoint subsets of σ_{x_j} not containing $\lambda_j \in \sigma_{x_j}$, and for $j \notin J_w$, f_j maps $r_j^{-1}(w)$ bijectively onto $r_{j-1}^{-1}(w)$. Thus $\lambda_j \in \sigma_{x_j}$ is different from λ_k for any $k \in \{k_1, k_2\} \cup (J_w \cap [1, j-1])$, so the vectors λ_j with $j \in J_w$ are pairwise distinct. Write $\lambda_j = (u_j, v_j)$. By (5.12) and induction on j, for any $j \in \{k_1, k_2\} \cup J_w$,

$$1 - \frac{\epsilon}{2} \geq a_j - \frac{\epsilon}{2} \geq (u_j + v_j)\frac{\epsilon}{2},$$

so $u_j + v_j < 2/\epsilon \leq N_3$. For $j \in J_w$, λ_j belongs to the open first quadrant, so $1 \leq u_j, v_j < N_3$. Thus $\#J_w \leq N_3^2$.

If $m \geq 1$, then $1 \leq m_1 \leq N_2$, $\#\Sigma_{m_1} \leq m_1 - 1$, and any $j > m_1$ belongs to J_w for some $w \in \Sigma_{m_1}$, so

$$m \leq m_1 + (m_1 - 1)N_3^2 \leq N_2(1 + N_3^2).$$

The inequality also holds if $m = 0$. Since $\rho(X') \leq 2$,

$$\rho(X) = \rho(X') + m \leq 2 + N_2(1 + N_3^2) \leq N.$$

Since $c_j < 1$ for any j, (5.5) and (5.9) imply

$$\sum_i b_i \leq \sum_{i \in I'} b_i + \sum_{j=1}^{m} c_j \leq N_1 + m \leq N_1 + N_2(1 + N_3^2) \leq N.$$

The lemma follows. □

Proposition 5.7. *Let $\epsilon \in (0, 1)$ and $\beta > 0$ be real numbers. There exists a positive integer N depending only on ϵ and β with the following property. Let X be a regular projective surface over a field F of characteristic ≥ 11 with $H^0(X, \mathcal{O}_X) = F$, and let $B = \sum_{i \in \Lambda} b_i B_i$ be a $\mathbb{Q}$-divisor on X, where the B_i are distinct prime divisors. Assume that the following hold.*

(1) $b_i \geq \beta$ for any $i \in \Lambda$.

(2) (X, B) is ϵ-lc.

(3) $K_X + B$ is big.

Then there exists a subset $\Lambda' \subset \Lambda$ such that $\#\Lambda' \leq N$ and $K_X + \sum_{i \in \Lambda'} b_i B_i$ is big.

Proof. Let N_1 be a positive integer with the property of Lemma 5.5 for ϵ, and let

$$N := \max\left\{10,\ \lceil N_1 + N_1/\beta \rceil + 2,\ \lfloor 2/\beta \rfloor + 1\right\}.$$

Since (X, B) is ϵ-lc, $\beta \leq b_i \leq 1 - \epsilon$ for any $i \in \Lambda$. Let H be an ample Cartier divisor on X. We divide the proof into six steps.

Step 1. We show two positivity facts on a normal projective surface V over F.

(a) Let L be a big $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on V, and let M be a nef $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on V such that $M \cdot A' > 0$ for any ample $\mathbb{R}$-divisor A' on V (for example, a non-zero nef $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $M \geq 0$). Then $L \cdot M > 0$, since $L \equiv A' + L'$ for an ample $\mathbb{R}$-divisor A' and an $\mathbb{R}$-divisor

$L' \geq 0$, so that $L \cdot M \geq A' \cdot M > 0$. In particular, let $V \to Z$ be a contraction onto a curve Z and W an $\mathbb{R}$-divisor on V whose components are contained in fibers of $V \to Z$. Then no $\mathbb{R}$-Cartier $\mathbb{R}$-divisor $L \equiv W$ is big: a fiber $M \geq 0$ of $V \to Z$ is a non-zero nef Cartier divisor, and $L \cdot M = W \cdot M = 0$.

(b) Assume that V is regular. Then any $\mathbb{Q}$-divisor G on V with $G^2 > 0$ and $G \cdot H' > 0$ for an ample divisor H' on V is big, by the Riemann–Roch theorem and Serre duality.

Step 2. In this step we assume that K_X is pseudo-effective. By [CWZ26, Theorems 2.17 and 2.18] (see also [Lip69, Theorem (27.1)] and [Tan18, Theorem 1.1]), there exists a morphism $g : X \to X_{\min}$ onto a regular projective surface $X_{\min}$ which is a composite of contractions of (-1)-curves, such that either $K_{X_{\min}}$ is nef or $X_{\min}$ has a Mori fiber space structure. Since $K_{X_{\min}} = g_*K_X$ is pseudo-effective, $K_{X_{\min}}$ is nef. Since $X_{\min}$ is regular, $(X_{\min}, 0)$ is lc, so $K_{X_{\min}}$ is semi-ample by [Tan20, Theorem 4.10]. By [Liu02, Theorem 9.2.2 and Proposition 9.2.24],

$$K_X = g^*K_{X_{\min}} + E_g$$

for a divisor E_g such that $\operatorname{mult}_C E_g > 0$ for any curve C on X exceptional over $X_{\min}$ and $\operatorname{mult}_C E_g = 0$ for any other prime divisor C on X. Let

$$\Lambda_{\min} := \{i \in \Lambda \mid B_i \text{ is not exceptional over } X_{\min}\}.$$

Then $g_*B = \sum_{i\in\Lambda_{\min}} b_i g_*B_i$, and the g_*B_i with $i \in \Lambda_{\min}$ are distinct prime divisors on $X_{\min}$. Let $H_{X_{\min}}$ be an ample Cartier divisor on $X_{\min}$.

(a) Let $\Lambda' \subset \Lambda_{\min}$ and $B' := \sum_{i\in\Lambda'} b_iB_i$. We show that if $K_{X_{\min}} + \lambda g_*B'$ is big for every rational number $\lambda \in (0, 1]$, then $K_X + B'$ is big. Since $g^*g_*B' - B' \geq 0$ is exceptional over $X_{\min}$, there exists a rational number $\lambda_0 \in (0, 1]$ such that $E_g \geq \lambda_0(g^*g_*B' - B')$. Then

$$K_X + B' = g^*(K_{X_{\min}} + \lambda_0 g_*B') + \big(E_g - \lambda_0(g^*g_*B' - B')\big) + (1 - \lambda_0)B'$$

is big.

(b) If $K_{X_{\min}}$ is big, then $K_X = g^*K_{X_{\min}} + E_g$ is big, and $\Lambda' := \emptyset$ satisfies the proposition.

(c) Suppose that $K_{X_{\min}}$ is neither big nor numerically trivial. Since $K_{X_{\min}}$ is semi-ample and $K^2_{X_{\min}} = 0$, there exist a contraction $\pi_{\min} : X_{\min} \to Z$ onto a regular projective curve Z and an ample $\mathbb{Q}$-divisor A on Z such that $K_{X_{\min}} \sim_{\mathbb{Q}} \pi^*_{\min}A$. Since $K_{X_{\min}} + g_*B = g_*(K_X + B)$ is big, by **Step 1**(a) there exists $i_0 \in \Lambda_{\min}$ such that $\pi_{\min}(g_*B_{i_0}) = Z$, so $K_{X_{\min}} \cdot g_*B_{i_0} > 0$. For any sufficiently small rational number $\delta > 0$,

$$(K_{X_{\min}} + \delta g_*B_{i_0})^2 = 2\delta K_{X_{\min}} \cdot g_*B_{i_0} + \delta^2(g_*B_{i_0})^2 > 0 \quad \text{and} \quad (K_{X_{\min}} + \delta g_*B_{i_0}) \cdot H_{X_{\min}} > 0,$$

so $K_{X_{\min}} + \delta g_*B_{i_0}$ is big by **Step 1**(b). Thus $K_{X_{\min}} + \lambda b_{i_0} g_*B_{i_0}$ is big for any rational number $\lambda > 0$, and $K_X + b_{i_0}B_{i_0}$ is big by **Step 2**(a). Hence $\Lambda' := \{i_0\}$ satisfies the proposition.

(d) Suppose that $K_{X_{\min}} \equiv 0$. Then $g_*B \equiv K_{X_{\min}} + g_*B$ is big. By **Step 2**(a), it suffices to find a subset $\Lambda' \subset \Lambda_{\min}$ such that $\#\Lambda' \leq 10$ and $\sum_{i\in\Lambda'} b_i g_*B_i$ is big.

If $(g_*B_i)^2 > 0$ for some $i \in \Lambda_{\min}$, then g_*B_i is big by **Step 1**(b), and $\Lambda' := \{i\}$ works. If $(g_*B_i)^2 = 0$ for some $i \in \Lambda_{\min}$, then g_*B_i is nef, so $g_*B \cdot g_*B_i > 0$ by **Step 1**(a), and there exists $j \in \Lambda_{\min}\backslash\{i\}$ such that $g_*B_i \cdot g_*B_j > 0$. For a sufficiently small rational number $\delta > 0$,

$$(g_*B_i + \delta g_*B_j)^2 = 2\delta g_*B_i \cdot g_*B_j + \delta^2(g_*B_j)^2 > 0,$$

so $g_*B_i + \delta g_*B_j$ is big by **Step 1**(b), and $\Lambda' := \{i, j\}$ works. Thus we may assume that $(g_*B_i)^2 < 0$ for any $i \in \Lambda_{\min}$.

For $i \in \Lambda_{\min}$ let $d_i := \dim_F H^0(g_*B_i, \mathcal{O}_{g_*B_i})$. Since $X_{\min}$ is regular, Lemma 5.1 gives $\chi_F(\mathcal{O}_{g_*B_i}) \leq d_i$ and

$$\chi_F(\mathcal{O}_{g_*B_i}),\ L \cdot g_*B_i \in d_i\mathbb{Z} \quad \text{for any divisor } L \text{ on } X_{\min}.$$

By the adjunction formula [Liu02, Theorem 9.1.37] and since $K_{X_{\min}} \equiv 0$, $(g_*B_i)^2 = -2\chi_F(\mathcal{O}_{g_*B_i})$. Since $(g_*B_i)^2 < 0$, we get $\chi_F(\mathcal{O}_{g_*B_i}) = d_i$ and

$$(g_*B_i)^2 = -2d_i.$$

Let $\mathcal{I} := (g_*B_i \cdot g_*B_j)_{i,j\in\Lambda_{\min}}$, and let $\mathcal{C} = (c_{ij})_{i,j\in\Lambda_{\min}}$ be the matrix with $c_{ii} = 2$ and

$$c_{ij} = -(g_*B_i \cdot g_*B_j)/d_i \quad \text{for } i \neq j.$$

Then c_{ij} is a non-positive integer for $i \neq j$, $c_{ij} = 0$ if and only if $c_{ji} = 0$, and

$$\operatorname{diag}(d_i)_{i\in\Lambda_{\min}}\mathcal{C} = -\mathcal{I},$$

so $\mathcal{C}$ is a symmetrizable generalized Cartan matrix [CCC+10, Section 2]. For $\Lambda_1 \subset \Lambda_{\min}$, let $\mathcal{I}_{\Lambda_1}$ and $\mathcal{C}_{\Lambda_1}$ be the principal submatrices indexed by Λ_1. We say that Λ_1 is connected if the graph on Λ_1 with an edge between i and j if and only if $g_*B_i \cdot g_*B_j > 0$ is connected. For a connected Λ_1 and an integer $\gamma \gg 0$, $\mathcal{I}_{\Lambda_1} + \gamma \operatorname{Id}$ is an irreducible non-negative symmetric matrix. By the Perron–Frobenius theorem [HJ13, Theorem 8.4.4], the largest eigenvalue of $\mathcal{I}_{\Lambda_1}$ is simple and has an eigenvector with positive entries.

We show that $\mathcal{I}$ has a positive eigenvalue. Suppose not. If $\det \mathcal{I}_{\Lambda_1} = 0$ for a connected component Λ_1 of $\Lambda_{\min}$, then 0 is the largest eigenvalue of $\mathcal{I}_{\Lambda_1}$, so $\ker \mathcal{I}_{\Lambda_1}$ is spanned by a vector $(u_i)_{i\in\Lambda_1}$ with positive entries, which we may take rational since $\mathcal{I}$ is rational. Then

$$L := \sum_{i\in\Lambda_1} u_i g_*B_i$$

satisfies $L \cdot g_*B_j = 0$ for any $j \in \Lambda_{\min}$, so $L \geq 0$ is a non-zero nef $\mathbb{Q}$-divisor with $g_*B \cdot L = 0$, which contradicts **Step 1**(a). Otherwise $\mathcal{I}$ is negative definite. Let

$$v = (v_i)_{i\in\Lambda_{\min}} := -\mathcal{I}^{-1}(H_{X_{\min}} \cdot g_*B_i)_{i\in\Lambda_{\min}} \quad \text{and} \quad R := \sum_{i\in\Lambda_{\min}} v_i g_*B_i,$$

so that $(H_{X_{\min}} + R) \cdot g_*B_i = 0$ for any $i \in \Lambda_{\min}$. Write $v = v^+ - v^-$, where v^+ and v^- have non-negative entries and disjoint supports. Since $(\mathcal{I}v)_i = -H_{X_{\min}} \cdot g_*B_i < 0$ for any $i \in \Lambda_{\min}$ and $g_*B_i \cdot g_*B_j \geq 0$ for any $i \neq j$,

$$0 \geq (v^-)^{\mathrm{T}}\mathcal{I}v = (v^-)^{\mathrm{T}}\mathcal{I}v^+ - (v^-)^{\mathrm{T}}\mathcal{I}v^- \geq -(v^-)^{\mathrm{T}}\mathcal{I}v^- \geq 0,$$

so $v^- = 0$. Since $R \geq 0$ and $(H_{X_{\min}} + R) \cdot g_*B_i = 0$ for any $i \in \Lambda_{\min}$, $H_{X_{\min}} + R$ is nef and $(H_{X_{\min}} + R) \cdot A' \geq H_{X_{\min}} \cdot A' > 0$ for any ample $\mathbb{R}$-divisor A' on $X_{\min}$. Since

$$g_*B \cdot (H_{X_{\min}} + R) = \sum_{i\in\Lambda_{\min}} b_i(H_{X_{\min}} + R) \cdot g_*B_i = 0,$$

this contradicts **Step 1**(a).

Let $\Lambda' \subset \Lambda_{\min}$ be minimal with respect to inclusion such that $\mathcal{I}_{\Lambda'}$ has a positive eigenvalue. Then $\mathcal{I}_{\Lambda_1}$ is negative semidefinite for any $\Lambda_1 \subsetneq \Lambda'$, and Λ' is connected, since otherwise $\mathcal{I}_{\Lambda'}$ is block diagonal with a block having a positive eigenvalue. For any connected $\Lambda_1 \subsetneq \Lambda'$, $\dim\ker \mathcal{I}_{\Lambda_1} \leq 1$, so the matrix

$$-\mathcal{I}_{\Lambda_1} = \operatorname{diag}(d_i)_{i\in\Lambda_1}\mathcal{C}_{\Lambda_1}$$

is positive definite or positive semidefinite of corank 1, while $-\mathcal{I}_{\Lambda'}$ has a negative eigenvalue. By [Kac90, Lemma 4.5], and since the conditions of [Kac90, Theorem 4.3] defining the types do not change under left multiplication by a diagonal matrix with positive entries, $\mathcal{C}_{\Lambda_1}$ is of finite or affine type for any connected $\Lambda_1 \subsetneq \Lambda'$, and $\mathcal{C}_{\Lambda'}$ is neither of finite nor of affine type. Thus the Dynkin diagram of $\mathcal{C}_{\Lambda'}$ is hyperbolic in the sense of [CCC+10, Section 1], and $\#\Lambda' \leq 10$ by [CCC+10, Proposition 3.6]. The largest eigenvalue of $\mathcal{I}_{\Lambda'}$ is positive and has an eigenvector with positive entries. Approximating such an eigenvector, we obtain positive rational numbers u_i ($i \in \Lambda'$) such that $L := \sum_{i\in\Lambda'} u_i g_*B_i$ satisfies $L^2 > 0$. Since $L \cdot H_{X_{\min}} > 0$, L is big by **Step 1**(b), so

$$\sum_{i\in\Lambda'} b_i g_*B_i \geq \Big(\min_{i\in\Lambda'} \frac{b_i}{u_i}\Big) L$$

is big.

Step 3. From now on we assume that K_X is not pseudo-effective. In this step, let $f : X \to X'$ be a birational morphism onto a normal projective surface X', and let $\pi : X' \to Z$ be a contraction onto a regular projective curve Z. An $\mathbb{R}$-divisor on X is called *vertical* if $\pi \circ f$ maps any component of the divisor to a closed point. For a closed point $z \in Z$ with residue field $\kappa(z)$, we write

$$(\pi \circ f)^* z = \sum_j m_{z,j} \Gamma_{z,j},$$

where the $\Gamma_{z,j}$ are distinct prime divisors and $m_{z,j} \in \mathbb{N}^+$. Fix a closed point $z_1 \in Z$ and let $\Phi := (\pi \circ f)^* z_1 / [\kappa(z_1) : F]$. Since $z/[\kappa(z) : F] - z_1/[\kappa(z_1) : F]$ is a $\mathbb{Q}$-divisor of degree 0 on Z, $(\pi \circ f)^* z \equiv [\kappa(z) : F]\Phi$ for any closed point $z \in Z$. Thus Φ is nef, $\Phi^2 = 0$, $\Phi \cdot \Gamma = 0$ for any vertical divisor Γ, and $\Phi \cdot C > 0$ for any curve C on X which is not vertical.

(a) By Zariski's lemma [Liu02, Theorem 9.1.23], for any closed point $z \in Z$ and any real numbers s_j,

$$\Big(\sum_j s_j \Gamma_{z,j} \Big)^2 \le 0,$$

with equality if and only if $s_j/m_{z,j}$ does not depend on j. Since components of different fibers are disjoint, $W'^2 < 0$ for any non-zero vertical $\mathbb{R}$-divisor W' such that for any closed point $z \in Z$, some $\Gamma_{z,j}$ is not a component of W'.

(b) For a vertical $\mathbb{Q}$-divisor $W = \sum_{z,j} w_{z,j} \Gamma_{z,j}$, let

$$\tau_z(W) := \min_j \frac{w_{z,j}}{m_{z,j}}, \quad \tau(W) := \sum_z \tau_z(W)[\kappa(z) : F], \quad \text{and} \quad M_W := (\pi \circ f)^* \Big(\sum_z \tau_z(W) z \Big),$$

where z runs over the closed points of Z (only finitely many $\tau_z(W)$ are non-zero). Then $M_W \equiv \tau(W)\Phi$, $W - M_W \ge 0$, and for any z, some $\Gamma_{z,j}$ is not a component of $W - M_W$. Since $M_W \cdot \Gamma = 0$ for any vertical Γ and $M_W^2 = 0$, (a) implies

$$W^2 = (W - M_W)^2 \le 0,$$

with equality if and only if $W = M_W$. Moreover, $\tau(W) \le \tau(W')$ for any vertical $\mathbb{Q}$-divisor $W' \ge W$, and $\tau(W - M_W) = 0$. For a component $\Gamma_{z,j}$ and a real number c,

$$\tau(W + c\Gamma_{z,j}) = \tau(W) - [\kappa(z) : F]\tau_z(W) + [\kappa(z) : F] \min \Big\{ \frac{w_{z,j} + c}{m_{z,j}}, \min_{j' \ne j} \frac{w_{z,j'}}{m_{z,j'}} \Big\},$$

where the inner minimum is $+\infty$ if $\Gamma_{z,j}$ is the only component of $(\pi \circ f)^* z$. Hence $\tau(W + c\Gamma_{z,j})$ is an increasing piecewise linear function of c with rational coefficients, and if $c_- < c_+$ are rational numbers such that $\tau(W + c_-\Gamma_{z,j}) \le 0 < \tau(W + c_+\Gamma_{z,j})$, then $\tau(W + c_0\Gamma_{z,j}) = 0$ for some rational number $c_0 \in [c_-, c_+)$.

(c) Let B^0 be a $\mathbb{Q}$-divisor on X such that $0 \le B^0 \le B$ and $K_X + B^0 \equiv W$ for a vertical $\mathbb{Q}$-divisor W with $\tau(W) = 0$. We show that there exists a birational morphism $f_0 : X \to X''$ onto a normal projective surface X'' such that

$$D_0 := f_{0*}(K_X + B^0) \equiv 0 \quad \text{and} \quad E_{f_0} := K_X + B^0 - f_0^* D_0 \ge 0.$$

Since $M_W \equiv 0$, $K_X + B^0 \equiv W - M_W \ge 0$ is pseudo-effective. Since (X, B^0) is ϵ-lc, by [Tan18, Theorem 1.1] there exists a $(K_X + B^0)$-MMP $f_0 : X \to X''$ such that D_0 is nef, and $E_{f_0} \ge 0$ since f_0 is a $(K_X + B^0)$-MMP. Since $f_0^* D_0$ is nef,

$$0 \le f_0^* D_0 \cdot \Phi = (W - M_W - E_{f_0}) \cdot \Phi = -E_{f_0} \cdot \Phi \le 0,$$

so $E_{f_0} \cdot \Phi = 0$ and E_{f_0} is vertical. Thus $W' := W - M_W - E_{f_0}$ is a vertical $\mathbb{Q}$-divisor with $W' \equiv f_0^* D_0$, and $\tau(W') \le \tau(W - M_W) = 0$ by (b). By (b),

$$0 \le (f_0^* D_0)^2 = W'^2 = (W' - M_{W'})^2 \le 0,$$

so $W' = M_{W'} \equiv \tau(W')\Phi$ by (b). Since $\Phi \cdot H > 0$ and $0 \le f_0^* D_0 \cdot H = \tau(W')\Phi \cdot H \le 0$, we get $\tau(W') = 0$ and $f_0^* D_0 \equiv 0$. By the projection formula, $D_0 \equiv 0$.

Step 4. Possibly replacing Λ with a subset, we may assume that $K_X + B - b_iB_i$ is not big for any $i \in \Lambda$. Since K_X is not big, $\Lambda \neq \emptyset$. We fix $i_0 \in \Lambda$. Since $K_X + B - (b_{i_0} - c)B_{i_0}$ differs from $K_X + B - (b_{i_0} - c')B_{i_0}$ by $(c - c')B_{i_0} \geq 0$ for $c \geq c'$ and the big cone is open, there exists a real number $t \in [0, b_{i_0})$ such that for any $c \in [0, b_{i_0}]$, $K_X + B - (b_{i_0} - c)B_{i_0}$ is big if and only if $c > t$. Let

$$B_t := B - (b_{i_0} - t)B_{i_0}.$$

Then $K_X + B_t$ is pseudo-effective and not big. Since (X, B_t) is ϵ-lc, by [Tan18, Theorem 1.1] there exists a $(K_X + B_t)$-MMP $f : X \to X'$ onto a normal projective surface X' such that $D := f_*(K_X + B_t)$ is nef, and

$$E := K_X + B_t - f^*D \geq 0$$

since f is a $(K_X + B_t)$-MMP. Since $K_X + B_t = f^*D + E$ is not big, D is not big, so $D^2 = 0$. Since (X', f_*B_t) is lc, by [Tan20, Theorem 1.1], D is semi-ample, that is, $D = \sum_j r_jD_j$ with $r_j > 0$ and D_j base point free Cartier divisors (see [Tan18, Section 2.1]). Since $D^2 = 0$ and $D_j \cdot D_{j'} \geq 0$ for any j, j', we have $D \cdot D_j = 0$ for any j.

We show that $t \in \mathbb{Q}$. If $D \equiv 0$, let D' be an ample Cartier divisor on X'. Otherwise, let $D' := D_j$ for some j with $D_j \not\equiv 0$, which is linearly equivalent to a non-zero effective divisor. In both cases D' is a nef Cartier divisor with $D \cdot D' = 0$ and $D' \cdot A' > 0$ for any ample $\mathbb{R}$-divisor A' on X'. Since f is birational, $f_*(K_X + B) = D + (b_{i_0} - t)f_*B_{i_0}$ is big, so $f_*(K_X + B) \cdot D' > 0$ by **Step 1**(a). By the projection formula,

$$(b_{i_0} - t)B_{i_0} \cdot f^*D' = (K_X + B) \cdot f^*D' > 0.$$

Since f^*D' is Cartier and $K_X + B$ is a $\mathbb{Q}$-divisor, both intersection numbers $B_{i_0} \cdot f^*D'$ and $(K_X + B) \cdot f^*D'$ are rational, so

$$t = b_{i_0} - \frac{(K_X + B) \cdot f^*D'}{B_{i_0} \cdot f^*D'} \in \mathbb{Q}.$$

Thus B_t, D, and E are $\mathbb{Q}$-divisors. Since D is a $\mathbb{Q}$-divisor, qD is base point free for some positive integer q [Tan20, Theorem 4.10]. Since $D^2 = 0$, either $D \equiv 0$, or there exist a contraction $\pi : X' \to Z$ onto a regular projective curve Z and an ample $\mathbb{Q}$-divisor A on Z such that $D \sim_{\mathbb{Q}} \pi^*A$.

Step 5. In this step, let $\Lambda^+ \subset \Lambda$, $B_{\Lambda^+} := \sum_{i \in \Lambda^+} b_iB_i$, $i' \in \Lambda$, and let B^0 be a $\mathbb{Q}$-divisor on X. Assume that the following hold.

(1) $K_X + B_{\Lambda^+}$ is big.
(2) $0 \leq B^0 \leq B_{\Lambda^+}$, and $\operatorname{mult}_{B_i} B^0 = b_i$ for any $i \in \Lambda^+ \backslash \{i_0, i'\}$.
(3) There exists a birational morphism $f_0 : X \to X''$ onto a normal projective surface X'' such that $D_0 := f_{0*}(K_X + B^0) \equiv 0$ and $E_{f_0} := K_X + B^0 - f_0^*D_0 \geq 0$.

We show that there exists a subset $\Lambda' \subset \Lambda^+$ such that

$$\#\Lambda' \leq N_1 + N_1/\beta + 2 \quad \text{and} \quad K_X + \sum_{i \in \Lambda'} b_iB_i \text{ is big.}$$

Since $f_{0*}E_{f_0} = 0$, E_{f_0} is exceptional over X''. Let $\Delta := B^0 - E_{f_0}$. Then Δ is a $\mathbb{Q}$-divisor, $\Delta \leq B^0 \leq B_{\Lambda^+} \leq B$, and $K_X + \Delta = f_0^*D_0 \equiv 0$. Let

$$\Lambda' := \{i \in \Lambda^+ \mid \operatorname{mult}_{B_i} \Delta \geq 0\} \quad \text{and} \quad B' := \sum_{i \in \Lambda'} b_iB_i.$$

For $i \in \Lambda'$, $\operatorname{mult}_{B_i} B' = \operatorname{mult}_{B_i} B_{\Lambda^+} \geq \operatorname{mult}_{B_i} \Delta$. For $i \in \Lambda^+ \backslash \Lambda'$, $\operatorname{mult}_{B_i} \Delta < 0 = \operatorname{mult}_{B_i} B'$. For any prime divisor C on X which is not a component of B_{Λ^+},

$$\operatorname{mult}_C B' = \operatorname{mult}_C B_{\Lambda^+} = 0 \geq \operatorname{mult}_C \Delta.$$

Thus $B' \geq \lambda B_{\Lambda^+} + (1 - \lambda)\Delta$ for a sufficiently small rational number $\lambda > 0$, and

$$K_X + B' = \lambda(K_X + B_{\Lambda^+}) + (1 - \lambda)(K_X + \Delta) + \big(B' - \lambda B_{\Lambda^+} - (1 - \lambda)\Delta\big)$$

is big. It remains to show that $\#\Lambda' \le N_1 + N_1/\beta + 2$.

We construct a regular projective surface Y with morphisms $g_Y : X \to Y$ and $f_Y : Y \to X''$ such that $f_0 = f_Y \circ g_Y$ and K_Y is nef over X''. Let Y' be a regular projective surface with morphisms $X \to Y' \to X''$ whose composite is f_0, and let C be a curve on Y' which is exceptional over X'' such that $K_{Y'} \cdot C < 0$. If $C^2 \ge 0$, then C is nef over X'', so $-C \ge 0$ by the negativity lemma [Tan18, Lemma 2.11(1)], which is impossible. Thus $C^2 < 0$. Let $d := \dim_F H^0(C, \mathcal{O}_C)$. Since Y' is regular, Lemma 5.1 gives $K_{Y'} \cdot C, C^2 \in d\mathbb{Z}$ and $\chi_F(\mathcal{O}_C) \le d$. By the adjunction formula [Liu02, Theorem 9.1.37],

$$K_{Y'} \cdot C + C^2 = -2\chi_F(\mathcal{O}_C) \ge -2d,$$

so $K_{Y'} \cdot C = C^2 = -d$, that is, C is a (-1)-curve. By [CWZ26, Theorem 2.17] (see also [Lip69, Theorem (27.1)]), there exists a contraction $Y' \to Y''$ of C onto a regular projective surface Y'', and $Y' \to X''$ factors through Y'' since C is exceptional over X''. Since $\rho(Y'') = \rho(Y') - 1$, starting from $Y' = X$ and replacing Y' with Y'' as long as such a curve C exists, we obtain Y after finitely many contractions. Let $\Delta_Y := g_{Y*}\Delta$. Then

$$K_Y + \Delta_Y = f_Y^* D_0 \quad \text{and} \quad K_X + \Delta = g_Y^*(K_Y + \Delta_Y).$$

Since $-\Delta_Y \equiv K_Y$ is nef over X'' and $f_{Y*}\Delta_Y = f_{0*}B^0 \ge 0$, $\Delta_Y \ge 0$ by [Tan18, Lemma 2.11(1)].

By [Liu02, Theorem 9.2.2], g_Y is a composite

$$X = X_n \to X_{n-1} \to \cdots \to X_0 = Y,$$

where $h_{k,k-1} : X_k \to X_{k-1}$ is the blow-up of X_{k-1} at a closed point y_k with exceptional curve E_k, and for $0 \le k \le k' \le n$ let $h_{k',k} : X_{k'} \to X_k$ be the induced morphism. Let Δ_{X_k} be the pushforward of Δ to X_k, so that $K_{X_k} + \Delta_{X_k}$ is the pullback of $K_Y + \Delta_Y$. By Definition-Lemma 5.4(1) and [Liu02, Proposition 9.2.24],

$$\operatorname{mult}_{E_k} \Delta_{X_k} = \mu_{y_k}(\Delta_{X_{k-1}}) - 1.$$

We say that y_k is infinitely near to y_j if $j < k$ and $X_{k-1} \to X_{j-1}$ maps y_k to y_j.

We show that if $\operatorname{mult}_{E_k} \Delta_{X_k} \ge 0$, then $\operatorname{mult}_{E_j} \Delta_{X_j} \ge 0$ for any j such that y_k is infinitely near to y_j. Suppose not, and let j be the smallest index such that y_k is infinitely near to y_j and $\operatorname{mult}_{E_j} \Delta_{X_j} < 0$. Any component of $\Delta_{X_{j-1}}$ through y_j is $(h_{j-1,0})_*^{-1}C$ for a component C of Δ_Y, or $(h_{j-1,j'})_*^{-1}E_{j'}$ for some j' such that y_j is infinitely near to $y_{j'}$. For such j', y_k is infinitely near to $y_{j'}$ and $j' < j$, so $\operatorname{mult}_{E_{j'}} \Delta_{X_{j'}} \ge 0$. Thus $\Delta' := \sum_C (\operatorname{mult}_C \Delta_{X_{j-1}})C$, where C runs over the prime divisors on X_{j-1} through y_j, satisfies $\Delta' \ge 0$ and $\mu_{y_j}(\Delta') = \operatorname{mult}_{E_j} \Delta_{X_j} + 1 < 1$. For any k' such that $y_{k'}$ is infinitely near to y_j, any component of $\Delta_{X_{k'-1}}$ through $y_{k'}$ is $(h_{k'-1,j-1})_*^{-1}C$ for a component C of Δ', or $(h_{k'-1,j'})_*^{-1}E_{j'}$ for some j' with $j' = j$ or $y_{j'}$ infinitely near to y_j. By induction on k', $\operatorname{mult}_{E_{j'}} \Delta_{X_{j'}} < 0$ for any such j', so by Definition-Lemma 5.4(2),

$$\mu_{y_{k'}}(\Delta_{X_{k'-1}}) \le \mu_{y_{k'}}\big((h_{k'-1,j-1})_*^{-1}\Delta'\big) \le \mu_{y_j}(\Delta') < 1.$$

Thus $\operatorname{mult}_{E_{k'}} \Delta_{X_{k'}} < 0$, and $k' = k$ yields a contradiction.

Since blow-ups at closed points none of which is infinitely near to another can be performed in any order, possibly reordering indices we may assume that there exists an integer n_0 such that $\operatorname{mult}_{E_k} \Delta_{X_k} \ge 0$ if and only if $k \le n_0$. The prime divisors on X_{n_0} are $(h_{n_0,0})_*^{-1}C$ for the prime divisors C on Y and $(h_{n_0,k})_*^{-1}E_k$ with $k \le n_0$, so $\Delta_{X_{n_0}} \ge 0$. Moreover,

$$K_X + \Delta = h_{n,n_0}^*(K_{X_{n_0}} + \Delta_{X_{n_0}}),$$

and $\operatorname{mult}_{B_i} \Delta \ge 0$ implies that B_i is not exceptional over X_{n_0}.

Since $K_X + \Delta = h_{n,n_0}^*(K_{X_{n_0}} + \Delta_{X_{n_0}})$ and $\Delta \le B$, $(X_{n_0}, \Delta_{X_{n_0}})$ is ϵ-lc and $K_{X_{n_0}} + \Delta_{X_{n_0}} \equiv 0$. If $\Delta_{X_{n_0}} = 0$, then $K_{X_{n_0}} \equiv 0$, and $K_X - h_{n,n_0}^* K_{X_{n_0}} \ge 0$ by [Liu02, Proposition 9.2.24], so K_X is pseudo-effective, a contradiction. Thus $\Delta_{X_{n_0}} \ne 0$. Since $h_{n,n_0*}\mathcal{O}_X = \mathcal{O}_{X_{n_0}}$, $H^0(X_{n_0}, \mathcal{O}_{X_{n_0}}) = F$.

By Lemma 5.5,

$$\rho(X_{n_0}) \le N_1 \quad \text{and} \quad \sum_C \operatorname{mult}_C \Delta_{X_{n_0}} \le N_1,$$

where C runs over the prime divisors on X_{n_0}. We count the elements $i \in \Lambda'$ in three classes.

(1) If B_i is exceptional over X'', then $h_{n,n_0*}B_i$ is a curve on X_{n_0} exceptional over X''. The curves on X_{n_0} exceptional over X'' are linearly independent in $N^1(X_{n_0})$: if C' is an $\mathbb{R}$-linear combination of such curves with $C' \equiv 0$, then $C' \ge 0$ and $-C' \ge 0$ by [Tan18, Lemma 2.11(1)]. Thus the number of such i is at most $\rho(X_{n_0}) \le N_1$.
(2) $\#(\Lambda' \cap \{i_0, i'\}) \le 2$.
(3) If B_i is not exceptional over X'' and $i \notin \{i_0, i'\}$, then

$$\operatorname{mult}_{h_{n,n_0*}B_i} \Delta_{X_{n_0}} = \operatorname{mult}_{B_i} \Delta = \operatorname{mult}_{B_i} B^0 = b_i \ge \beta,$$

where the second equality holds since E_{f_0} is exceptional over X'' and B_i is not. Since $\Delta_{X_{n_0}} \ge 0$, the number of such i is at most N_1/β.

Thus $\#\Lambda' \le N_1 + N_1/\beta + 2$.

Now assume that $D \equiv 0$ in **Step 4**. By this step with $\Lambda^+ := \Lambda$, $B^0 := B_t$, $i' := i_0$, and $f_0 := f$, we obtain $\Lambda' \subset \Lambda$ such that $\#\Lambda' \le N_1 + N_1/\beta + 2 \le N$ and $K_X + \sum_{i\in\Lambda'} b_i B_i$ is big.

Step 6. In this step we assume that $D \sim_{\mathbb{Q}} \pi^* A$ for a contraction $\pi : X' \to Z$ onto a regular projective curve Z and an ample $\mathbb{Q}$-divisor A on Z as in **Step 4**. We use the notation of **Step 3** for f and π. Since

$$f_*(K_X + B) = D + (b_{i_0} - t) f_* B_{i_0} \sim_{\mathbb{Q}} \pi^* A + (b_{i_0} - t) f_* B_{i_0}$$

is big, B_{i_0} is not vertical by **Step 1**(a). Let

$$\Lambda^{\mathrm{h}} := \{i \in \Lambda \mid B_i \text{ is not vertical}\} \quad \text{and} \quad \Lambda^{\mathrm{v}} := \Lambda \backslash \Lambda^{\mathrm{h}},$$

and write $B_t = B_t^{\mathrm{h}} + B_t^{\mathrm{v}}$, where

$$B_t^{\mathrm{h}} := \sum_{i\in\Lambda^{\mathrm{h}}} (\operatorname{mult}_{B_i} B_t) B_i \quad \text{and} \quad B_t^{\mathrm{v}} := \sum_{i\in\Lambda^{\mathrm{v}}} b_i B_i.$$

Let

$$B^{\mathrm{h}} := \sum_{i\in\Lambda^{\mathrm{h}}} b_i B_i = B_t^{\mathrm{h}} + (b_{i_0} - t) B_{i_0}.$$

Since $\pi \circ f$ contracts any curve exceptional over X', E is vertical, and so is

$$W_0 := (\pi \circ f)^* A + E - B_t^{\mathrm{v}}.$$

Moreover, $K_X + B_t^{\mathrm{h}} = f^*D + E - B_t^{\mathrm{v}} \sim_{\mathbb{Q}} W_0$. For any closed point $z \in Z$ and any component C of $\pi^* z$, $f_*^{-1} C$ is a component of $(\pi \circ f)^* z$ which is not a component of E. Since $E \ge 0$, we get

$$\tau(W_0 + B_t^{\mathrm{v}}) = \tau\big((\pi \circ f)^* A + E\big) = \deg A > 0.$$

(a) We bound $\#\Lambda^{\mathrm{h}}$. Let X_η be the generic fiber of $\pi \circ f$, a regular projective curve over the function field $F(Z)$ with $H^0(X_\eta, \mathcal{O}_{X_\eta}) = F(Z)$. Since $K_X + B_t^{\mathrm{h}} \sim_{\mathbb{Q}} W_0$ and W_0 is vertical, $\deg(K_{X_\eta} + B_t^{\mathrm{h}}|_{X_\eta}) = 0$. Since $\deg K_{X_\eta} = -2\chi_{F(Z)}(\mathcal{O}_{X_\eta}) \ge -2$ and $B_i|_{X_\eta}$ is a closed point of degree $[F(B_i) : F(Z)] \ge 1$ for any $i \in \Lambda^{\mathrm{h}}$,

$$\sum_{i\in\Lambda^{\mathrm{h}}} (\operatorname{mult}_{B_i} B_t)[F(B_i) : F(Z)] = \deg(B_t^{\mathrm{h}}|_{X_\eta}) \le 2.$$

Since $\operatorname{mult}_{B_i} B_t = b_i \ge \beta$ for any $i \in \Lambda^{\mathrm{h}} \backslash \{i_0\}$, we get $\#\Lambda^{\mathrm{h}} \le \lfloor 2/\beta \rfloor + 1$.

(b) Let $B^\flat$ be a $\mathbb{Q}$-divisor with $0 \le B^\flat \le B_t^{\mathrm{v}}$ such that $\tau(W_0 + B^\flat) > 0$. We show that $K_X + B^{\mathrm{h}} + B^\flat$ is big. Let $W := W_0 + B^\flat$. Then

$$K_X + B^{\mathrm{h}} + B^\flat \sim_{\mathbb{Q}} M_W + (W - M_W) + (b_{i_0} - t) B_{i_0},$$

where $M_W \equiv \tau(W)\Phi$ and $W - M_W \geq 0$ by **Step 3**(b). Since B_{i_0} is not vertical, $\Phi \cdot B_{i_0} > 0$. For a sufficiently small rational number $\delta \in (0, b_{i_0} - t]$,

$$(M_W + \delta B_{i_0})^2 = 2\delta\tau(W)\Phi \cdot B_{i_0} + \delta^2 B_{i_0}^2 > 0 \quad \text{and} \quad (M_W + \delta B_{i_0}) \cdot H > 0,$$

so $M_W + \delta B_{i_0}$ is big by **Step 1**(b). Adding $(W - M_W) + (b_{i_0} - t - \delta)B_{i_0} \geq 0$, we see that $K_X + B^{\mathrm{h}} + B^{\flat}$ is big.

(c) Write $\Lambda^{\mathrm{v}} = \{i_1, \dots, i_m\}$, and for $0 \leq k \leq m$ let $B_k^{\mathrm{v}} := \sum_{k<j\leq m} b_{i_j} B_{i_j}$. Then $B_0^{\mathrm{v}} = B_t^{\mathrm{v}}$, $B_m^{\mathrm{v}} = 0$, and $\tau(W_0 + B_k^{\mathrm{v}})$ is decreasing in k.

If $\tau(W_0 + B_m^{\mathrm{v}}) > 0$, then $K_X + B^{\mathrm{h}}$ is big by **Step 6**(b), and $\Lambda' := \Lambda^{\mathrm{h}}$ satisfies

$$\#\Lambda' \leq \lfloor 2/\beta \rfloor + 1 \leq N$$

by **Step 6**(a).

Otherwise, let $k := \min\{k' \geq 1 \mid \tau(W_0 + B_{k'}^{\mathrm{v}}) \leq 0\}$. Since

$$\tau(W_0 + B_k^{\mathrm{v}} + b_{i_k} B_{i_k}) = \tau(W_0 + B_{k-1}^{\mathrm{v}}) > 0,$$

by **Step 3**(b) there exists a rational number $c_0 \in [0, b_{i_k})$ such that $\tau(W_0 + B_k^{\mathrm{v}} + c_0 B_{i_k}) = 0$. Let

$$\Lambda^+ := \Lambda^{\mathrm{h}} \cup \{i_k, \dots, i_m\}, \quad B_{\Lambda^+} := \sum_{i\in\Lambda^+} b_i B_i = B^{\mathrm{h}} + B_{k-1}^{\mathrm{v}}, \quad \text{and} \quad B^0 := B_t^{\mathrm{h}} + B_k^{\mathrm{v}} + c_0 B_{i_k}.$$

Then $K_X + B_{\Lambda^+} = K_X + B^{\mathrm{h}} + B_{k-1}^{\mathrm{v}}$ is big by **Step 6**(b), $0 \leq B^0 \leq B_{\Lambda^+}$, and $\mathrm{mult}_{B_i} B^0 = b_i$ for any $i \in \Lambda^+ \backslash \{i_0, i_k\}$. Since

$$K_X + B^0 \sim_{\mathbb{Q}} W_0 + B_k^{\mathrm{v}} + c_0 B_{i_k} \quad \text{and} \quad \tau(W_0 + B_k^{\mathrm{v}} + c_0 B_{i_k}) = 0,$$

condition (3) of **Step 5** holds by **Step 3**(c) applied to B^0. By **Step 5** with $i' := i_k$, there exists $\Lambda' \subset \Lambda^+$ such that $\#\Lambda' \leq N_1 + N_1/\beta + 2 \leq N$ and $K_X + \sum_{i\in\Lambda'} b_i B_i$ is big.

In all cases we obtain $\Lambda' \subset \Lambda$ such that $\#\Lambda' \leq N$ and $K_X + \sum_{i\in\Lambda'} b_i B_i$ is big. The proposition follows. □

6. Birational linear systems from one non-zero section

In this section, we prove Theorem 6.6.

Definition 6.1. Let X be a normal projective scheme of finite type over an F-finite field of characteristic p and D a $\mathbb{Q}$-divisor on X. We denote by $S^0_-(X, K_X + D)$ the subspace of *Frobenius-stable sections* of $H^0(X, \mathcal{O}_X(K_X + \lceil D \rceil))$ defined in [Zha22, Section 2.2].

For any $e \in \mathbb{N}^+$, let $F^e : X \to X$ be the e-th iterate of the absolute Frobenius morphism, which is finite since the ground field is F-finite. Tensoring the trace map $F^e_* \mathcal{O}_X(K_X) \to \mathcal{O}_X(K_X)$ of duality for F^e with $\mathcal{O}_X(\lceil D \rceil)$ and taking reflexive hulls, we obtain a morphism

$$F^e_* \mathcal{O}_X(K_X + p^e \lceil D \rceil) \to \mathcal{O}_X(K_X + \lceil D \rceil).$$

Since $\lceil p^e D \rceil \leq p^e \lceil D \rceil$, composing it with the inclusion

$$F^e_* \mathcal{O}_X(K_X + \lceil p^e D \rceil) \hookrightarrow F^e_* \mathcal{O}_X(K_X + p^e \lceil D \rceil),$$

we obtain

$$\mathrm{Tr}^e : F^e_* \mathcal{O}_X(K_X + \lceil p^e D \rceil) \to \mathcal{O}_X(K_X + \lceil D \rceil).$$

Let

$$S^e(X, K_X + D) := \mathrm{Im}\Big(H^0(X, F^e_* \mathcal{O}_X(K_X + \lceil p^e D \rceil)) \xrightarrow{H^0(\mathrm{Tr}^e)} H^0(X, \mathcal{O}_X(K_X + \lceil D \rceil))\Big).$$

Then $S^{e+1}(X, K_X + D) \subset S^e(X, K_X + D)$ for any $e \in \mathbb{N}^+$, and we let

$$S^0(X, K_X + D) := \bigcap_{e\in\mathbb{N}^+} S^e(X, K_X + D).$$

For a $\mathbb{Q}$-divisor $\Delta \geq 0$ on X, let $E := \lceil D \rceil - \lceil D - \Delta \rceil$, which is a divisor with $E \geq 0$, and let $s_E \in H^0(X, \mathcal{O}_X(E))$ be the section with divisor E. For any $e \in \mathbb{N}^+$, let

$$S^e_\Delta(X, K_X + D) := s_E \cdot S^e(X, K_X + D - \Delta) \subset S^e(X, K_X + D),$$

and let

$$S^0_\Delta(X, K_X + D) := \bigcap_{e \in \mathbb{N}^+} S^e_\Delta(X, K_X + D).$$

If D is a nef and big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, let Θ^{amp}_D be the set of $\mathbb{Q}$-divisors $\Delta \geq 0$ on X such that $D - \Delta$ is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, and let

$$S^0_-(X, K_X + D) := \bigcap_{\Delta \in \Theta^{\mathrm{amp}}_D} \bigcup_{t \in \mathbb{Q}_{>0}} S^0_{t\Delta}(X, K_X + D).$$

If D is not a nef and big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, let $\mathcal{B}^{\mathrm{nef}}_D$ be the set of divisors $R \geq 0$ on X such that $D - R$ is a nef and big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, and let

$$S^0_-(X, K_X + D) := \sum_{R \in \mathcal{B}^{\mathrm{nef}}_D} s_R \cdot S^0_-(X, K_X + D - R) \subset S^0(X, K_X + D),$$

where $s_R \in H^0(X, \mathcal{O}_X(R))$ is the section with divisor R.

Lemma 6.2 (cf. [Zha22, Theorem 3.1]). *Let X be a normal projective variety of dimension d over an algebraically closed field of characteristic $p > 0$, D a nef and big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, and H a $\mathbb{Q}$-Cartier Weil divisor on X. Assume that $|H|$ defines a generically finite map. Then:*

(1) $S^0_-(X, K_X + D + jH) \neq 0$ for any integer $j \geq d$.
(2) $S^0_-(X, K_X + D + jH)$ defines a birational map for any integer $j \geq d + 1$.

Proof. In [Zha22, Theorem 3.1], we take $Y = X$, $f = \mathrm{id}_X$, which is a projective morphism with $f_*\mathcal{O}_X = \mathcal{O}_X$, and $\widetilde{H}$ a very ample Cartier divisor on X. The integer d in [Zha22, Theorem 3.1] is then $\dim X$. The conclusions of [Zha22, Theorem 3.1] used below, namely (i) and the first conclusion of (ii), do not involve $\widetilde{H}$.

The generic fiber of id_X is $\operatorname{Spec} k(X)$, and $K_X + D$ restricts to the zero divisor 0 on $\operatorname{Spec} k(X)$. We show that

$$S^0_-(\operatorname{Spec} k(X), 0) = k(X).$$

Since the ground field is perfect and $k(X)$ is finitely generated over it, $[k(X) : k(X)^p] < +\infty$, that is, $k(X)$ is F-finite. Any divisor on $\operatorname{Spec} k(X)$ is 0, the canonical module of $\operatorname{Spec} k(X)$ is $k(X)$, and the zero divisor is ample. For any $e \in \mathbb{N}^+$, under the isomorphism

$$F^e_* k(X) \cong \mathrm{Hom}_{k(X)}(F^e_* k(X), k(X))$$

of duality for the finite morphism F^e, the trace map $F^e_* k(X) \to k(X)$ is the evaluation map at 1. Since 1 belongs to a basis of the $k(X)$-vector space $F^e_* k(X)$, there exists a $k(X)$-linear map $F^e_* k(X) \to k(X)$ sending 1 to 1, so the trace map is surjective. Thus $S^e(\operatorname{Spec} k(X), 0) = k(X)$ for any $e \in \mathbb{N}^+$, and $S^0(\operatorname{Spec} k(X), 0) = k(X)$. Since any $\mathbb{Q}$-divisor on $\operatorname{Spec} k(X)$ is 0, $\Theta^{\mathrm{amp}}_0 = \{0\}$ and

$$S^0_-(\operatorname{Spec} k(X), 0) = S^0_0(\operatorname{Spec} k(X), 0) = S^0(\operatorname{Spec} k(X), 0) = k(X).$$

In particular, $S^0_-(\operatorname{Spec} k(X), 0) \neq 0$, and $S^0_-(\operatorname{Spec} k(X), 0)$ defines the isomorphism

$$\operatorname{Spec} k(X) \to \mathbb{P}^0_{k(X)},$$

which is birational.

Since $f^*H = H$, (1) and (2) are [Zha22, Theorem 3.1(i) and the first conclusion of Theorem 3.1(ii)], respectively, with $s = j$. □

Definition-Lemma 6.3. Let q be a positive integer and

$$\mathcal{D}_q := \left\{ 1 - \frac{a}{qr} \;\middle|\; a \in \{0, 1, \dots, q\},\ r \in \mathbb{N}^+ \right\}.$$

Then there exists a positive integer N such that for any projective lc pair (X, B) of dimension 2 over an algebraically closed field such that $B \in \mathcal{D}_q$ and $K_X + B$ is big, $|\lfloor N(K_X + B) \rfloor|$ defines a birational map. We denote by N_q the smallest such integer and set

$$c_q := q \left\lceil \frac{3N_q + 2}{q} \right\rceil.$$

Proof. Since $\mathcal{D}_q \subset [0, 1]$ and $\mathcal{D}_q \cap [0, x]$ is finite for any $x < 1$, $\mathcal{D}_q$ is a DCC set. By [Das22, Theorem 1.2], there exists a positive integer N such that for any projective lc pair (X, B) of dimension 2 over an algebraically closed field such that $B \in \mathcal{D}_q$ and $K_X + B$ is big, $|\lfloor m(K_X + B) \rfloor|$ defines a birational map for any integer $m \geq N$. Taking $m = N$, the lemma follows. □

Proposition 6.4. *Let q be a positive integer and $I_q := \{\frac{j}{q} \mid j = 0, 1, \dots, q\}$. Let (X, B) be a projective lc pair of dimension 3 over an algebraically closed field of characteristic $p > 0$ such that $B \in I_q$, $\lfloor B \rfloor \neq 0$, and $A := K_X + B$ is ample. Let $m \geq 3N_q + 2$ be an integer divisible by q, let ϕ be the rational map defined by $|mA|$, and let Z be the closure of the image of ϕ. Then the following hold.*

(1) The map ϕ is defined at the generic point of any prime component S of $\lfloor B \rfloor$, and the composite of $\phi|_S$ with the normalization of S is birational onto the closure of $\phi(S)$.
(2) If $\lfloor B \rfloor$ has at least two prime components, then $\dim Z = 3$.

Proof. Let k be the ground field. Let $S_1, \dots, S_n$ be the prime components of $\lfloor B \rfloor$, and for any $i \in \{1, \dots, n\}$ let $\nu_i : T_i \to S_i$ be the normalization. We also regard $\lfloor B \rfloor$ as a reduced closed subscheme of X.

Step 1: adjunction to T_i. Choose a non-zero rational 3-form ω on X with $\operatorname{mult}_{S_i} \operatorname{div}(\omega) = -1$ for any $i \in \{1, \dots, n\}$, and let $K_X := \operatorname{div}(\omega)$. Since $\operatorname{mult}_{S_i} B = 1$,

$$\operatorname{mult}_{S_i} A = \operatorname{mult}_{S_i} K_X + \operatorname{mult}_{S_i} B = 0 \quad \text{for any } i \in \{1, \dots, n\}. \tag{6.1}$$

In particular, for any integer j and any non-zero $f \in H^0(X, \mathcal{O}_X(\lfloor jA \rfloor))$, we have

$$\operatorname{mult}_{S_i} \operatorname{div}(f) \geq -\operatorname{mult}_{S_i} \lfloor jA \rfloor = 0,$$

so any $f \in H^0(X, \mathcal{O}_X(\lfloor jA \rfloor))$ is regular at the generic point of S_i, and we let $f|_{T_i} \in k(S_i) = k(T_i)$ be the restriction of f.

Since A is an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor, A is $\mathbb{Q}$-Cartier by Lemma 2.6. Let c be a positive integer such that cA is Cartier. By (6.1), local equations of cA are units at the generic point of S_i, so their restrictions to S_i define a Cartier divisor $cA|_{S_i}$ on S_i. We let

$$H_i := \frac{1}{c} \nu_i^* (cA|_{S_i}).$$

Since A is ample and ν_i is finite, H_i is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on T_i. Since X and S_i are smooth at the generic point of S_i and $\operatorname{mult}_{S_i} K_X = -1$, the Poincaré residue ω_i of ω along S_i is a non-zero rational 2-form on T_i. We let $K_{T_i} := \operatorname{div}(\omega_i)$ and $B_{T_i} := H_i - K_{T_i}$, so that

$$K_{T_i} + B_{T_i} = H_i, \tag{6.2}$$

and B_{T_i} is the different of (X, B) on T_i [Kol13, Definition 4.2]. By [Das21, Proposition 2.9 and Lemma 2.10] (see also [Bir16, Propositions 4.1–4.2]), (T_i, B_{T_i}) is lc, and any coefficient of B_{T_i} is either 1 or equal to $\frac{r-1+\beta}{r}$, where r is a positive integer and β is a sum of coefficients of $B - S_i$, repetitions allowed and $\beta = 0$ for the empty sum. Since $B \in I_q$, $q\beta \in \mathbb{N}$. Since (T_i, B_{T_i}) is lc, $\frac{r-1+\beta}{r} \leq 1$, so $\beta \leq 1$. Thus $\ell := q - q\beta \in \{0, 1, \dots, q\}$ and

$$\frac{r - 1 + \beta}{r} = 1 - \frac{\ell}{qr} \in \mathcal{D}_q.$$

Since $1 = 1 - \frac{0}{q} \in \mathcal{D}_q$, $B_{T_i} \in \mathcal{D}_q$. Since H_i is ample, by Definition-Lemma 6.3 and (6.2), $||\lfloor N_q H_i \rfloor||$ defines a birational map for any i.

Step 2: monotonicity of Frobenius-stable sections. Let Y be a normal projective variety of dimension d over k, let ω_Y be a non-zero rational d-form on Y, and let $K_Y := \mathrm{div}(\omega_Y)$. For a positive integer e, let C_Y^e be the e-th iterate of the Cartier operator on rational d-forms on Y [MS11, Section 3.3]. For a $\mathbb{Q}$-divisor D on Y, the trace map of Definition 6.1 at level e is, up to a constant in k^*, the map

$$H^0(Y, \mathcal{O}_Y(K_Y + \lceil p^e D \rceil)) \to H^0(Y, \mathcal{O}_Y(K_Y + \lceil D \rceil))$$

sending u to $C_Y^e(u\omega_Y)/\omega_Y$, since under duality the trace of F^e is C_Y^e up to a constant in k^* [BS13, Theorem 3.2.1 and Proposition 3.3.1]. Since a constant in k^* does not change images,

$$S^0(Y, K_Y + D) = \bigcap_{e \geq 1} \left\{ C_Y^e(u\omega_Y)/\omega_Y \;\middle|\; u \in H^0(Y, \mathcal{O}_Y(K_Y + \lceil p^e D \rceil)) \right\}.$$

Since sections are regarded as elements of $k(Y)$,

$$S^0_{t\Delta}(Y, K_Y + D) = S^0(Y, K_Y + D - t\Delta)$$

for any $\mathbb{Q}$-divisor $\Delta \geq 0$ on Y and any $t \in \mathbb{Q}_{>0}$. If $D_1 \leq D_2$ are $\mathbb{Q}$-divisors on Y, then $\lceil p^e D_1 \rceil \leq \lceil p^e D_2 \rceil$ for any e, so

$$S^0(Y, K_Y + D_1) \subset S^0(Y, K_Y + D_2). \tag{6.3}$$

In particular, $S^0(Y, K_Y + D - t\Delta) \subset S^0(Y, K_Y + D - t'\Delta)$ for any $\mathbb{Q}$-divisor $\Delta \geq 0$ on Y and any rational numbers $t > t' > 0$.

We show that for any nef and big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors D and D' on Y such that $D' - D \geq 0$,

$$S^0_-(Y, K_Y + D) \subset S^0_-(Y, K_Y + D'). \tag{6.4}$$

Let $s \in S^0_-(Y, K_Y + D)$, and let $\Delta \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on Y such that $D' - \Delta$ is ample. Since D is big, there exists a rational number $\epsilon > 0$ such that $D - \epsilon\Delta$ is big. By Kodaira's lemma, there exist an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor H and a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G \geq 0$ on Y such that $D - \epsilon\Delta \sim_{\mathbb{Q}} H + G$. Let $\Delta' := \epsilon\Delta + G$. Then $\Delta' \geq 0$ is $\mathbb{Q}$-Cartier, and $D - \Delta' \sim_{\mathbb{Q}} H$ is ample, so $s \in S^0(Y, K_Y + D - t\Delta')$ for some $t \in \mathbb{Q}_{>0}$. Since $D - t\Delta' \leq D' - t\epsilon\Delta$, by (6.3),

$$s \in S^0(Y, K_Y + D' - t\epsilon\Delta).$$

Since Δ is arbitrary, $s \in S^0_-(Y, K_Y + D')$, which proves (6.4).

Step 3: birational stable systems on T_i. For any i, we let

$$W_i := S^0_-(T_i, K_{T_i} + (m-1)H_i) \subset k(T_i),$$

where $K_{T_i} = \mathrm{div}(\omega_i)$ as in **Step 1**. We show that W_i defines a birational map.

For any

$$u \in H^0(T_i, \mathcal{O}_{T_i}(\lfloor N_q H_i \rfloor)) \backslash \{0\},$$

we have $\mathrm{div}(u) + N_q H_i \geq 0$. Let $g_i : T_i' \to T_i$ be a resolution [Bir16, Section 2.4] such that the mobile part of $g_i^* ||\lfloor N_q H_i \rfloor||$ is base point free. Let F_i be the largest $\mathbb{Q}$-divisor on T_i' such that

$$F_i \leq g_i^*(\mathrm{div}(u) + N_q H_i) \quad \text{for any } u \in H^0(T_i, \mathcal{O}_{T_i}(\lfloor N_q H_i \rfloor)) \backslash \{0\},$$

and let $M_i := N_q g_i^* H_i - F_i$, so that

$$N_q g_i^* H_i = M_i + F_i, \quad F_i \geq 0. \tag{6.5}$$

For any two such u, the $\mathbb{Q}$-divisors $g_i^*(\mathrm{div}(u) + N_q H_i)$ differ by a principal divisor, so the coefficients of F_i and $N_q g_i^* H_i$ at any prime divisor differ by an integer. Thus M_i is a divisor, and the divisors $\mathrm{div}_{T_i'}(u) + M_i$ form the mobile part $\Sigma_i \subset |M_i|$ of $g_i^* ||\lfloor N_q H_i \rfloor||$, which is base point free. Σ_i defines the composite of g_i and the birational map defined by $||\lfloor N_q H_i \rfloor||$ (**Step 1**), so $|M_i|$ defines a birational map. In particular, $|M_i|$ defines a generically finite map, and M_i is a nef and big Cartier divisor.

We let $K_{T_i'} := \mathrm{div}(g_i^*\omega_i)$. Since $m-1-3N_q \geq 1$ and H_i is ample, $(m-1-3N_q)g_i^*H_i$ is nef and big. By Lemma 6.2 for T_i', $d=2$, $D=(m-1-3N_q)g_i^*H_i$, $H=M_i$, and $j=3$,

$$S^0_-(T_i', K_{T_i'} + (m-1-3N_q)g_i^*H_i + 3M_i)$$

defines a birational map. The $\mathbb{Q}$-divisors $(m-1-3N_q)g_i^*H_i + 3M_i$ and $(m-1)g_i^*H_i$ are nef, big, and $\mathbb{Q}$-Cartier, and by (6.5),

$$(m-1)g_i^*H_i - ((m-1-3N_q)g_i^*H_i + 3M_i) = 3F_i \geq 0.$$

By (6.4) and [Zha22, Proposition 2.3(iv-1)],

$$S^0_-(T_i', K_{T_i'} + (m-1-3N_q)g_i^*H_i + 3M_i) \subset S^0_-(T_i', K_{T_i'} + (m-1)g_i^*H_i) \subset W_i$$

as k-subspaces of $k(T_i') = k(T_i)$. Thus W_i defines a birational map. In particular, $W_i \neq 0$.

Step 4: lifting in high degrees. We show that there exist a Cartier divisor $R \geq 0$ on $\lfloor B \rfloor$ containing none of $S_1, \dots, S_n$ and an integer j' satisfying the following: with $R_i := \nu_i^*R$, for any integer $j \geq j'$ and any

$$(u_1, \dots, u_n) \in \bigoplus_{i=1}^n H^0(T_i, \mathcal{O}_{T_i}(\lfloor jH_i \rfloor - R_i)),$$

there exists $f \in H^0(X, \mathcal{O}_X(\lfloor jA \rfloor))$ such that

$$f|_{T_i} = u_i \quad \text{for any } i \in \{1, \dots, n\}. \tag{6.6}$$

For any integer r, restriction to the generic points of $S_1, \dots, S_n$ (**Step 1**) defines a homomorphism

$$\rho_r : \mathcal{O}_X(\lfloor rA \rfloor) \to \bigoplus_{i=1}^n \nu_{i*}\underline{k(T_i)},$$

sending f to $(f|_{T_1}, \dots, f|_{T_n})$, where $\underline{k(T_i)}$ is the constant sheaf on T_i. A non-zero local section f belongs to $\mathrm{Ker}\,\rho_r$ if and only if $\mathrm{mult}_{S_i}\mathrm{div}(f) \geq 1$ for any i, so, since X is normal,

$$\mathrm{Ker}\,\rho_r = \mathcal{O}_X(\lfloor rA \rfloor - \lfloor B \rfloor).$$

Let $\mathcal{F}_r := \rho_r(\mathcal{O}_X(\lfloor rA \rfloor))$. The ideal sheaf of $\lfloor B \rfloor$ annihilates $\mathcal{F}_r$, so $\mathcal{F}_r$ is a coherent $\mathcal{O}_{\lfloor B \rfloor}$-module, and we have an exact sequence

$$0 \to \mathcal{O}_X(\lfloor rA \rfloor - \lfloor B \rfloor) \to \mathcal{O}_X(\lfloor rA \rfloor) \xrightarrow{\rho_r} \mathcal{F}_r \to 0. \tag{6.7}$$

Let

$$\mathcal{P}_r := \bigoplus_{i=1}^n \nu_{i*}\mathcal{O}_{T_i}(\lfloor rH_i \rfloor),$$

a coherent $\mathcal{O}_{\lfloor B \rfloor}$-submodule of $\bigoplus_{i=1}^n \nu_{i*}\underline{k(T_i)}$. Since $\mathcal{O}_{X,S_i}$ is a discrete valuation ring and $\mathrm{mult}_{S_i}\lfloor rA \rfloor = 0$ by (6.1), the stalks of $\mathcal{F}_r$ and $\mathcal{P}_r$ at the generic point of S_i are both equal to $k(T_i)$.

Let $\mathcal{J} \subset \mathcal{O}_{\lfloor B \rfloor}$ be the ideal sheaf of local sections g such that $g\mathcal{P}_r \subset \mathcal{F}_r$ for any $r \in \{0, 1, \dots, c-1\}$. Then $\mathcal{J}$ is the intersection of the annihilators of the coherent modules $(\mathcal{F}_r + \mathcal{P}_r)/\mathcal{F}_r$, so $\mathcal{J}$ is coherent, and $\mathcal{J}$ is equal to $\mathcal{O}_{\lfloor B \rfloor}$ at the generic point of any S_i. Let $\mathcal{O}_{\lfloor B \rfloor}(1)$ be the restriction of an ample line bundle on X. Possibly replacing $\mathcal{O}_{\lfloor B \rfloor}(1)$ with a sufficiently high power, we may assume that $\mathcal{J} \otimes \mathcal{O}_{\lfloor B \rfloor}(1)$ is globally generated. Since k is infinite, a general section

$$\sigma \in H^0(\lfloor B \rfloor, \mathcal{J} \otimes \mathcal{O}_{\lfloor B \rfloor}(1)) \subset H^0(\lfloor B \rfloor, \mathcal{O}_{\lfloor B \rfloor}(1))$$

does not vanish at the generic points of $S_1, \dots, S_n$. Since $\lfloor B \rfloor$ is reduced, σ is locally a nonzerodivisor, so the zero divisor $R \geq 0$ of σ is a Cartier divisor on $\lfloor B \rfloor$ containing none of $S_1, \dots, S_n$, and $\mathcal{O}_{\lfloor B \rfloor}(-R) \subset \mathcal{J}$. By the projection formula,

$$\bigoplus_{i=1}^{n} \nu_{i*}\mathcal{O}_{T_i}(\lfloor rH_i \rfloor - R_i) = \mathcal{O}_{\lfloor B \rfloor}(-R)\mathcal{P}_r \subset \mathcal{F}_r \quad \text{for any } r \in \{0, 1, \dots, c-1\}. \tag{6.8}$$

Let j be an integer, and write $j = ac + r$ with $a \in \mathbb{Z}$ and $r \in \{0, 1, \dots, c-1\}$. Since cA and cH_i are Cartier divisors,

$$\lfloor jA \rfloor = acA + \lfloor rA \rfloor \quad \text{and} \quad \lfloor jH_i \rfloor = acH_i + \lfloor rH_i \rfloor.$$

A local generator of $\mathcal{O}_X(acA)$ restricts to a local generator of $\mathcal{O}_{T_i}(acH_i)$, so (6.7) for j is (6.7) for r tensored with $\mathcal{O}_X(acA)$. By Serre vanishing, there exists an integer a_0 such that

$$H^1(X, \mathcal{O}_X(\lfloor rA \rfloor - \lfloor B \rfloor) \otimes \mathcal{O}_X(acA)) = 0$$

for any $a \geq a_0$ and any $r \in \{0, 1, \dots, c-1\}$. Let $j' := a_0 c$. For $j \geq j'$, the map

$$H^0(X, \mathcal{O}_X(\lfloor jA \rfloor)) \to H^0(\lfloor B \rfloor, \mathcal{F}_r \otimes \mathcal{O}_X(acA))$$

induced by ρ_j is surjective, and by (6.8), $H^0(\lfloor B \rfloor, \mathcal{F}_r \otimes \mathcal{O}_X(acA))$ contains

$$\bigoplus_{i=1}^{n} H^0(T_i, \mathcal{O}_{T_i}(\lfloor rH_i \rfloor + acH_i - R_i)) = \bigoplus_{i=1}^{n} H^0(T_i, \mathcal{O}_{T_i}(\lfloor jH_i \rfloor - R_i)).$$

This proves (6.6).

Step 5: Cartier trace and residue. We use the following properties of the Cartier operator [MS11, Section 3.3, (6)]. Let Y be a variety of dimension d over k and e a positive integer. Then C_Y^e is additive, $C_Y^e(g^{p^e}\eta) = gC_Y^e(\eta)$ for any $g \in k(Y)$ and any rational d-form η, and C_Y^e sends a rational d-form regular at a smooth point of Y to a rational d-form regular at that point. Let ξ be a smooth closed point of Y and $x_1, \dots, x_d$ a regular system of parameters of $\mathcal{O}_{Y,\xi}$. Then there exists an affine open neighbourhood U of ξ with $x_1, \dots, x_d \in \mathcal{O}(U)$ such that $\mathcal{O}(U)$ is a free $\mathcal{O}(U)^p$-module with basis $\{x_1^{a_1} \cdots x_d^{a_d}\}_{0 \leq a_1, \dots, a_d \leq p-1}$, and for any integers $0 \leq a_1, \dots, a_d \leq p-1$,

$$C_Y^1(x_1^{a_1} \cdots x_d^{a_d} dx_1 \wedge \cdots \wedge dx_d) = \begin{cases} dx_1 \wedge \cdots \wedge dx_d & \text{if } a_1 = \cdots = a_d = p-1, \\ 0 & \text{otherwise} \end{cases}$$

[MS11, Section 3.3, (6)]. Since $g \to g^p$ is an isomorphism $\mathcal{O}(U) \to \mathcal{O}(U)^p$, induction on e shows that $\mathcal{O}(U)$ is a free $\mathcal{O}(U)^{p^e}$-module with basis

$$\{x_1^{a_1} \cdots x_d^{a_d}\}_{0 \leq a_1, \dots, a_d \leq p^e - 1},$$

and for any integers $0 \leq a_1, \dots, a_d \leq p^e - 1$,

$$C_Y^e(x_1^{a_1} \cdots x_d^{a_d} dx_1 \wedge \cdots \wedge dx_d) = \begin{cases} dx_1 \wedge \cdots \wedge dx_d & \text{if } a_1 = \cdots = a_d = p^e - 1, \\ 0 & \text{otherwise.} \end{cases} \tag{6.9}$$

For any positive integer e, let $j_e := p^e(m-1) + 1$, and for $f \in k(X)$ let

$$\tau_e(f) := C_X^e(f\omega)/\omega \in k(X).$$

We show that for any $f \in H^0(X, \mathcal{O}_X(\lfloor j_e A \rfloor))$,

$$\tau_e(f) \in H^0(X, \mathcal{O}_X(mA)) \quad \text{and} \quad \tau_e(f)|_{T_i} = C_{T_i}^e(f|_{T_i}\omega_i)/\omega_i \text{ for any } i \in \{1, \dots, n\}. \tag{6.10}$$

Since $q \mid m$ and qB is a divisor, $mA = mK_X + mB$ is a divisor. Since K_X is a divisor and $j_e A = p^e mA - (p^e - 1)K_X - (p^e - 1)B$,

$$\lfloor j_e A \rfloor = p^e mA - (p^e - 1)K_X - \lceil (p^e - 1)B \rceil. \tag{6.11}$$

Let $f \in H^0(X, \mathcal{O}_X(\lfloor j_e A \rfloor))$ be non-zero, and let V be a prime divisor on X. Since X is normal, X is smooth at the generic point of V. Let z be a uniformizer of $\mathcal{O}_{X,V}$, let η be a 3-form generating

the module of 3-forms regular at the generic point of V, and let $\theta \in k(X)$ such that $\omega = \theta\eta$. Let $\kappa := \mathrm{mult}_V \operatorname{div}(\theta) = \mathrm{mult}_V K_X$ and $\lambda := \mathrm{mult}_V A = \kappa + \mathrm{mult}_V B$. Then $m\lambda = \mathrm{mult}_V(mA) \in \mathbb{Z}$, and by (6.11),

$$\mathrm{mult}_V \operatorname{div}(f) \geq -p^e m\lambda + (p^e - 1)\kappa + \lceil (p^e - 1)\,\mathrm{mult}_V B \rceil.$$

Let $w := f z^{p^e m\lambda} \theta^{1-p^e}$. Then $w \in \mathcal{O}_{X,V}$ by the last inequality, and $f\omega = (z^{-m\lambda}\theta)^{p^e} w\eta$. Thus

$$C_X^e(f\omega) = z^{-m\lambda}\theta C_X^e(w\eta),$$

and $C_X^e(w\eta) = w'\eta$ for some $w' \in \mathcal{O}_{X,V}$, so $\tau_e(f) = z^{-m\lambda}w'$ with $m\lambda = \mathrm{mult}_V(mA)$ and $w' \in \mathcal{O}_{X,V}$. Since V is arbitrary and X is normal, $\tau_e(f) \in H^0(X, \mathcal{O}_X(mA))$.

Fix $i \in \{1, \dots, n\}$. Since X and S_i are smooth at the generic point of S_i, $\mathrm{mult}_{S_i} K_X = -1$, and f is regular at the generic point of S_i, there exists a closed point $\xi \in S_i$ such that X and S_i are smooth at ξ, f is regular at ξ, and $\xi \notin \mathrm{Supp}(K_X + S_i)$, that is, no component of K_X other than S_i passes through ξ. Let z, x, y be a regular system of parameters of $\mathcal{O}_{X,\xi}$ such that $S_i = (z = 0)$ near ξ, and let U be the neighbourhood of ξ given by the first paragraph of this step for $Y = X$ and z, x, y. Possibly shrinking U to a principal open subset containing ξ, which keeps the basis given there, there exists $v \in \mathcal{O}(U)^*$ such that $f \in \mathcal{O}(U)$, $S_i \cap U = (z = 0)$ is smooth, and

$$\omega|_U = v\frac{dz}{z} \wedge dx \wedge dy.$$

Since $S_i \cap U$ is smooth, ν_i is an isomorphism over $S_i \cap U$, and we identify $S_i \cap U$ with the open subset $\nu_i^{-1}(S_i \cap U)$ of T_i. Then $\overline{x} := x|_{S_i \cap U}$ and $\overline{y} := y|_{S_i \cap U}$ form a regular system of parameters of $\mathcal{O}_{T_i,\xi}$. Possibly shrinking U again to a principal open subset containing ξ, the parameters $\overline{x}$ and $\overline{y}$ satisfy the conclusion of the first paragraph of this step on $S_i \cap U$ for $Y = T_i$, and $\omega_i = \overline{v}d\overline{x} \wedge d\overline{y}$ on $S_i \cap U$, where $\overline{v} := v|_{S_i \cap U}$. Write

$$fv = \sum_{0 \leq a_1, a_2, a_3 \leq p^e - 1} g_{a_1,a_2,a_3}^{p^e} z^{a_1} x^{a_2} y^{a_3}, \quad g_{a_1,a_2,a_3} \in \mathcal{O}(U).$$

For $a_1 \geq 1$, the term

$$g_{a_1,a_2,a_3}^{p^e} z^{a_1 - 1} x^{a_2} y^{a_3} dz \wedge dx \wedge dy$$

of $f\omega$ has z-exponent $a_1 - 1 \leq p^e - 2$, so C_X^e sends it to 0. For $a_1 = 0$,

$$g_{0,a_2,a_3}^{p^e} z^{-1} x^{a_2} y^{a_3} = (g_{0,a_2,a_3}/z)^{p^e} z^{p^e - 1} x^{a_2} y^{a_3}.$$

Thus only the terms of fv with $a_1 = 0$ contribute to $C_X^e(f\omega)$, and

$$C_X^e(f\omega) = \frac{g_{0,p^e-1,p^e-1}}{z} dz \wedge dx \wedge dy, \quad \text{so} \quad \tau_e(f) = \frac{g_{0,p^e-1,p^e-1}}{v}.$$

On the other hand, restricting fv to $S_i \cap U$ kills the terms with $a_1 \geq 1$, so

$$f|_{T_i}\omega_i = \sum_{0 \leq a_2, a_3 \leq p^e - 1} \overline{g}_{0,a_2,a_3}^{p^e} \overline{x}^{a_2} \overline{y}^{a_3} d\overline{x} \wedge d\overline{y}, \quad \overline{g}_{0,a_2,a_3} := g_{0,a_2,a_3}|_{S_i \cap U},$$

and

$$C_{T_i}^e(f|_{T_i}\omega_i) = \overline{g}_{0,p^e-1,p^e-1} d\overline{x} \wedge d\overline{y}.$$

Therefore

$$C_{T_i}^e(f|_{T_i}\omega_i)/\omega_i = \frac{\overline{g}_{0,p^e-1,p^e-1}}{\overline{v}} = \tau_e(f)|_{T_i}$$

on the dense open subset $S_i \cap U$ of T_i, hence in $k(T_i)$. This proves (6.10).

Step 6: lifting stable sections. We show that

$$\bigoplus_{i=1}^n W_i \subset \left\{ (f|_{T_1}, \dots, f|_{T_n}) \;\middle|\; f \in H^0(X, \mathcal{O}_X(mA)) \right\}. \tag{6.12}$$

Let $(s_1,\dots,s_n)\in\bigoplus_{i=1}^n W_i$. For any i, let $D_i:=(m-1)H_i$, an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on T_i. Since T_i is projective, there exists a Cartier divisor $L_i\geq 0$ on T_i such that

$$\operatorname{Supp} L_i\supset \operatorname{Supp} R_i\cup\operatorname{Supp} H_i\cup\operatorname{Supp} K_{T_i}.$$

Let $\delta>0$ be a rational number such that $D_i-\delta L_i$ is ample for any i. Since $s_i\in W_i$ and $\delta L_i\geq 0$ is a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, $s_i\in S^0(T_i,K_{T_i}+D_i-t\delta L_i)$ for some $t\in\mathbb{Q}_{>0}$ (**Step 2**). By (6.3), we may take one t for all i. Let $\epsilon:=t\delta$, so that

$$s_i\in S^0(T_i,K_{T_i}+D_i-\epsilon L_i)\quad\text{for any } i. \tag{6.13}$$

We claim that for any $e\gg 0$,

$$K_{T_i}+\lceil p^e(D_i-\epsilon L_i)\rceil\leq\lfloor j_eH_i\rfloor-R_i\quad\text{for any } i. \tag{6.14}$$

Let Γ be a prime divisor on T_i. If $\Gamma\not\subset\operatorname{Supp} L_i$, then both sides of (6.14) have coefficient 0 at Γ. If $\Gamma\subset\operatorname{Supp} L_i$, then $\operatorname{mult}_\Gamma L_i\geq 1$. Since $j_eH_i=p^eD_i+H_i$, the coefficient at Γ of the right-hand side minus the left-hand side of (6.14) is at least

$$\begin{aligned}(p^e\operatorname{mult}_\Gamma D_i+\operatorname{mult}_\Gamma H_i-1-\operatorname{mult}_\Gamma R_i)-(\operatorname{mult}_\Gamma K_{T_i}+p^e\operatorname{mult}_\Gamma D_i-p^e\epsilon+1)&\\ =p^e\epsilon+\operatorname{mult}_\Gamma(H_i-K_{T_i}-R_i)-2,&\end{aligned}$$

which is non-negative for $e\gg 0$. Since $\operatorname{Supp} L_i$ has finitely many components, the claim follows.

Fix e such that (6.14) holds and $j_e\geq j'$. By (6.13), for any i there exists a section

$$u_i\in H^0(T_i,\mathcal{O}_{T_i}(K_{T_i}+\lceil p^e(D_i-\epsilon L_i)\rceil))$$

such that $C^e_{T_i}(u_i\omega_i)/\omega_i=s_i$. By (6.14), $u_i\in H^0(T_i,\mathcal{O}_{T_i}(\lfloor j_eH_i\rfloor-R_i))$. By (6.6), there exists $f\in H^0(X,\mathcal{O}_X(\lfloor j_eA\rfloor))$ such that $f|_{T_i}=u_i$ for any i. By (6.10), $\tau_e(f)\in H^0(X,\mathcal{O}_X(mA))$ and $\tau_e(f)|_{T_i}=C^e_{T_i}(u_i\omega_i)/\omega_i=s_i$ for any i. This proves (6.12).

Step 7: proof of (1) and (2). Let $f_0,\dots,f_N$ be a basis of $H^0(X,\mathcal{O}_X(mA))$, so that $\phi=[f_0:\cdots:f_N]:X\dashrightarrow\mathbb{P}^N$.

(1) Fix i. By **Step 3**, $W_i\neq 0$, so by (6.12) there exists $f\in H^0(X,\mathcal{O}_X(mA))$ with $f|_{T_i}\neq 0$. By **Step 1**, each of $f_0,\dots,f_N$ is regular at the generic point of S_i, and one of them is a unit there, so ϕ is defined at the generic point of S_i. The composite of ν_i and the restriction of ϕ to S_i is the rational map $[f_0|_{T_i}:\cdots:f_N|_{T_i}]$, defined by the k-subspace

$$\Lambda_i:=\{f|_{T_i}\mid f\in H^0(X,\mathcal{O}_X(mA))\}\subset k(T_i).$$

By (6.12) with $s_{i'}=0$ for any $i'\neq i$, $W_i\subset\Lambda_i$. Since W_i defines a birational map by **Step 3**, so does Λ_i. Thus the composite is birational onto the closure of $\phi(S_i)$, which proves (1).

(2) Assume that $n\geq 2$. By (1), the closure of $\phi(S_1)$ is a surface contained in Z, so $\dim Z\geq 2$. Suppose that $\dim Z=2$. For $i\in\{1,2\}$, let $\phi_i:=\phi|_{S_i}\circ\nu_i:T_i\dashrightarrow Z$. By (1), $\overline{\phi_i(T_i)}\subset Z$ is a surface, hence $\overline{\phi_i(T_i)}=Z$, so ϕ_i is dominant and induces an injection $\phi_i^*:k(Z)\to k(T_i)$. By **Step 3**, there exist $s_1\in W_1\backslash\{0\}$ and $s_2\in W_2\backslash\{0\}$. By (6.12), there exist $f',f''\in H^0(X,\mathcal{O}_X(mA))$ such that

$$(f'|_{T_1},\dots,f'|_{T_n})=(s_1,0,\dots,0)\quad\text{and}\quad(f''|_{T_1},\dots,f''|_{T_n})=(0,s_2,0,\dots,0).$$

Since $(f'+f'')|_{T_1}=s_1\neq 0$, $f'+f''\neq 0$, so $f'/(f'+f'')=\phi^*\psi$ for some $\psi\in k(Z)$, namely the restriction to Z of the ratio of the linear forms on $\mathbb{P}^N$ corresponding to f' and $f'+f''$. Since $(f'+f'')|_{T_1}=s_1\neq 0$ and $(f'+f'')|_{T_2}=s_2\neq 0$, $f'+f''$ is a unit at the generic points of S_1 and S_2, so

$$\phi_1^*\psi=\frac{f'|_{T_1}}{(f'+f'')|_{T_1}}=\frac{s_1}{s_1}=1\quad\text{and}\quad\phi_2^*\psi=\frac{f'|_{T_2}}{(f'+f'')|_{T_2}}=\frac{0}{s_2}=0.$$

Since ϕ_1^* and ϕ_2^* are injective, $\psi=1$ and $\psi=0$, a contradiction. Thus $\dim Z=3$, which proves (2). □

Lemma 6.5. *Let X be a normal projective threefold over an algebraically closed field of characteristic $p>0$, $B\geq 0$ a $\mathbb{Q}$-divisor on X, and q and n positive integers. Assume that:*

(1) qB is a Weil divisor.
(2) $D := K_X + B$ is $\mathbb{Q}$-Cartier, nef, and big.
(3) $||\lfloor nD \rfloor||$ defines a generically finite map.

Then $||\lfloor mD \rfloor||$ defines a birational map for any positive integer m divisible by q with $m \geq 4qn + 2$.

Proof. Let $H := qnD$ and fix

$$0 \neq s_0 \in H^0(X, \mathcal{O}_X(\lfloor nD \rfloor)).$$

Since K_X and qB are Weil divisors, $H = qnK_X + n \cdot qB$ is a $\mathbb{Q}$-Cartier Weil divisor. Since $q\lfloor nD \rfloor \leq H$, we have $s \cdot s_0^{q-1} \in H^0(X, \mathcal{O}_X(H))$ for any $s \in H^0(X, \mathcal{O}_X(\lfloor nD \rfloor))$, and $(s \cdot s_0^{q-1})/(s' \cdot s_0^{q-1}) = s/s'$ for $s' \neq 0$. Thus $|H|$ defines a generically finite map.

Let m be a positive integer divisible by q with $m \geq 4qn + 2$. Since $m - 1 - 4qn \geq 1$, $(m - 1 - 4qn)D$ is a nef and big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. Since

$$(m - 1 - 4qn)D + 4H = (m - 1)D,$$

by Lemma 6.2(2) for $(m - 1 - 4qn)D$, H, $d = 3$, and $j = 4$, the subspace

$$S^0_-(X, K_X + (m - 1)D) \subset H^0(X, \mathcal{O}_X(K_X + \lceil (m - 1)D \rceil))$$

defines a birational map.

For any prime divisor P on X, since $q \mid m$ and $q \operatorname{mult}_P B \in \mathbb{Z}$, we have $m \operatorname{mult}_P B \in \mathbb{Z}$, so

$$\lceil (m - 1) \operatorname{mult}_P B \rceil = m \operatorname{mult}_P B - \lfloor \operatorname{mult}_P B \rfloor.$$

Since K_X is a Weil divisor, we obtain

$$K_X + \lceil (m - 1)D \rceil = mK_X + \lceil (m - 1)B \rceil = mD - \lfloor B \rfloor.$$

Since $B \geq 0$, $\lfloor B \rfloor \geq 0$. Multiplication by the section of $\mathcal{O}_X(\lfloor B \rfloor)$ with divisor $\lfloor B \rfloor$ embeds $H^0(X, \mathcal{O}_X(mD - \lfloor B \rfloor))$ into $H^0(X, \mathcal{O}_X(mD))$ and does not change the rational map defined by a linear subspace. Thus $|mD|$ contains a linear subsystem defining a birational map, so $|mD|$ defines a birational map. Since $mD = mK_X + \frac{m}{q} \cdot qB$ is a Weil divisor, $\lfloor mD \rfloor = mD$. The lemma follows. □

Theorem 6.6. *Let q be a positive integer, and let (X, B) be a projective lc pair of dimension 3 over an algebraically closed field k of characteristic $p > 5$ such that $B \in I_q$ and $D := K_X + B$ is big. Let b be a positive integer, and let*

$$n := c_q\big(2c_q(qb + 1) + 1\big),$$

where c_q is the integer of Definition-Lemma 6.3. Assume that

$$H^0(X, \mathcal{O}_X(\lfloor bD \rfloor)) \neq 0.$$

Then $||\lfloor mD \rfloor||$ defines a birational map for any positive integer m divisible by q with $m \geq 4qn + 2$.

Proof. We regard the section spaces on normal projective varieties birational to X as subspaces of their common function field $k(X)$ (Definition 2.3). For any $v \in k(X) \backslash \{0\}$ and any k-subspaces $0 \neq \Lambda \subset \Lambda'$ of $k(X)$ with $\dim \Lambda' < +\infty$,

$$k(v\Lambda) = k(\Lambda) \quad \text{and} \quad k(\Lambda) \subset k(\Lambda'). \tag{6.15}$$

Since D is an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor, D is $\mathbb{Q}$-Cartier by Lemma 2.6.

Step 1. Let (W, B_W) be a projective $\mathbb{Q}$-factorial lc pair of dimension 3 over k such that W is birational to X, $B_W \in I_q$, and $K_W + B_W$ is big. By [HNT20, Theorem 1.1], there exists a sequence of steps of a $(K_W + B_W)$-MMP $\psi : W \dashrightarrow Y$ that terminates. Since $K_W + B_W$ is big, ψ does not end with a Mori fiber space, so (Y, B_Y) is a $\mathbb{Q}$-factorial lc pair and $K_Y + B_Y$ is nef and big, where $B_Y := \psi_* B_W$. By [Xu24, Theorem 1.6], $K_Y + B_Y$ is semi-ample.

Let $g : Y \to Z$ be the contraction defined by $K_Y + B_Y$, and let

$$B_Z := g_* B_Y, \quad A_Z := K_Z + B_Z, \quad \text{and} \quad \varphi := g \circ \psi : W \dashrightarrow Z.$$

Since $K_Y + B_Y$ is big, g is birational, A_Z is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, and $K_Y + B_Y = g^*A_Z$. Thus (Z, B_Z) is a projective lc pair of dimension 3. Since ψ is a composite of divisorial contractions and flips and g is a birational morphism, neither ψ^{-1} nor φ^{-1} contracts a divisor. We show the following.

(1a) For any prime divisor Q on Z, there exists a unique prime divisor P on W with $\varphi_*P = Q$, and $\operatorname{mult}_Q B_Z = \operatorname{mult}_P B_W$. For any prime divisor Q on Y, there exists a unique prime divisor P on W with $\psi_*P = Q$, and $\operatorname{mult}_Q B_Y = \operatorname{mult}_P B_W$. For any $z \in k(X)\backslash\{0\}$, $\varphi_* \operatorname{div}_W(z) = \operatorname{div}_Z(z)$ and $\psi_* \operatorname{div}_W(z) = \operatorname{div}_Y(z)$. In particular, $B_Y \in I_q$ and $B_Z \in I_q$.

(1b) For any positive integer j,

$$H^0(Z, \mathcal{O}_Z(\lfloor jA_Z \rfloor)) \subset H^0(Y, \mathcal{O}_Y(\lfloor j(K_Y + B_Y)\rfloor)) = H^0(W, \mathcal{O}_W(\lfloor j(K_W + B_W)\rfloor)). \tag{6.16}$$

(1c) For any big $\mathbb{Q}$-divisor $E \geq 0$ on W, $\varphi_*E \neq 0$.

(1d) For any $z \in k(X)\backslash k$, write $\operatorname{div}_W(z) = F_0 - F_1$, where $F_0 \geq 0$ and $F_1 \geq 0$ are divisors on W without common components. Then $\varphi_*F_0 \neq 0$, $\varphi_*F_1 \neq 0$, and φ_*F_0 and φ_*F_1 have no common components.

(1e) Let $M \geq 0$ be a divisor on X with integer coefficients such that $M = \operatorname{div}_X(v) + cD$ for some $v \in k(X)\backslash\{0\}$ and $c \in \mathbb{N}^+$. Let $f : W \to X$ be a log resolution of $(X, B + M)$ [Cut09, Theorems 1.1–1.3], let $\mathcal{E}$ be the set of f-exceptional prime divisors, and let

$$R := \sum_{P \subset \operatorname{Supp} M} (1 - \operatorname{mult}_P B)P \quad \text{and} \quad B_W := f_*^{-1}(B + R) + \sum_{E \in \mathcal{E}} E.$$

Then (W, B_W) satisfies the conditions of **Step 1**, any component of f^*M has coefficient 1 in B_W, and for any $j \in \mathbb{N}^+$,

$$v^j H^0(W, \mathcal{O}_W(\lfloor j(K_W + B_W)\rfloor)) \subset H^0(X, \mathcal{O}_X(\lfloor j(c+1)D\rfloor)). \tag{6.17}$$

Since ψ^{-1} and φ^{-1} contract no divisor, (1a) and (1c) hold, and since ψ is a sequence of steps of a $(K_W + B_W)$-MMP, the equality in (1b) holds by the negativity lemma. The inclusion in (1b) follows from $K_Y + B_Y = g^*A_Z$.

For (1d), if $\varphi_*F_1 = 0$, then $\operatorname{div}_Z(z) = \varphi_*F_0 \geq 0$ by (1a), so $z \in k$, a contradiction. Replacing z with z^{-1}, we get $\varphi_*F_0 \neq 0$. Since F_0 and F_1 have no common components, by the uniqueness in (1a), φ_*F_0 and φ_*F_1 have no common components.

For (1e), since M has integer coefficients, $\operatorname{mult}_P M \geq 1 \geq \operatorname{mult}_P R$ for any prime divisor $P \subset \operatorname{Supp} M$, so

$$0 \leq R \leq M \quad \text{and} \quad f_*(K_W + B_W) = D + R. \tag{6.18}$$

Any coefficient of B_W is either 1 or a coefficient of B, so $B_W \in I_q$, and (W, B_W) is lc. Moreover, since $K_W + f_*^{-1}B = f^*D + \sum_{E \in \mathcal{E}}(a(E, X, B) - 1)E$ and $a(E, X, B) \geq 0$ for any $E \in \mathcal{E}$,

$$K_W + B_W = f^*D + f_*^{-1}R + \sum_{E \in \mathcal{E}} a(E, X, B)E \geq f^*D,$$

so $K_W + B_W$ is big. Thus (W, B_W) satisfies the conditions of **Step 1**. Since

$$\operatorname{Supp} f^*M \subset \operatorname{Supp} f_*^{-1}M \cup \bigcup_{E \in \mathcal{E}} E,$$

any component of f^*M has coefficient 1 in B_W. Let j be a positive integer and

$$s \in H^0(W, \mathcal{O}_W(\lfloor j(K_W + B_W)\rfloor)).$$

By (6.18),

$$\operatorname{div}_X(s) + j(D + R) = f_*\big(\operatorname{div}_W(s) + j(K_W + B_W)\big) \geq 0.$$

Adding $j(M - R) \geq 0$ and using $M = \operatorname{div}_X(v) + cD$, we get $\operatorname{div}_X(sv^j) + j(c+1)D \geq 0$, which implies (6.17).

Step 2. In this step we show that there exist $u_0, u_1 \in H^0(X, \mathcal{O}_X(\lfloor n_1 D \rfloor))$ with $u_0/u_1 \notin k$, where

$$n_1 := c_q(qb+1).$$

Let $u \in H^0(X, \mathcal{O}_X(\lfloor bD \rfloor))\backslash\{0\}$, $v := u^q$, and

$$M := \operatorname{div}_X(v) + qbD = q\big(\operatorname{div}_X(u) + bD\big).$$

Then $M \geq 0$, and since qB has integer coefficients, so do qbD and M. Let $f : W \to X$ and B_W be as in (1e) for M, v, and $c = qb$, and let Z, B_Z, A_Z, and φ be as in **Step 1** for (W, B_W).

Since $M \sim_{\mathbb{Q}} qbD$, the $\mathbb{Q}$-divisor $f^*M \geq 0$ is big. By (1c), $\varphi_* f^*M \neq 0$, so there exists a component P of f^*M with $\varphi_* P \neq 0$, and by (1a) and (1e), $\operatorname{mult}_{\varphi_* P} B_Z = \operatorname{mult}_P B_W = 1$. Thus $\lfloor B_Z \rfloor \neq 0$. Since $q \mid c_q$ and $c_q \geq 3N_q + 2$ (see Definition-Lemma 6.3), by Proposition 6.4(1) for (Z, B_Z) and $m = c_q$, the closure of the image of a prime component of $\lfloor B_Z \rfloor$ under the rational map defined by $|c_q A_Z|$ is a surface, so $k\big(H^0(Z, \mathcal{O}_Z(c_q A_Z))\big) \neq k$.

Let $\Lambda_1 := v^{c_q} H^0(Z, \mathcal{O}_Z(c_q A_Z))$. By (6.16) and (6.17) with $j = c_q$, $\Lambda_1 \subset H^0(X, \mathcal{O}_X(\lfloor n_1 D \rfloor))$, and by (6.15), $k(\Lambda_1) \neq k$, so there exist $u_0, u_1 \in \Lambda_1$ with $u_0/u_1 \notin k$.

Step 3. In this step we show that $|\lfloor nD \rfloor|$ defines a generically finite map. Let u_0, u_1 be as in **Step 2**, $z := u_0/u_1$, and

$$M_i := \operatorname{div}_X(u_i) + n_1 D \quad \text{for } i \in \{0, 1\}.$$

Since $q \mid n_1$, $n_1 D$ has integer coefficients, so $M_0 \geq 0$ and $M_1 \geq 0$ are divisors with integer coefficients, $\operatorname{div}_X(z) = M_0 - M_1$, and

$$M_0 + M_1 = \operatorname{div}_X(u_0 u_1) + 2n_1 D.$$

Let $f' : W' \to X$ and $B_{W'}$ be as in (1e) for $M_0 + M_1$, $v = u_0 u_1$, and $c = 2n_1$, and let Z', $B_{Z'}$, $A_{Z'}$, and φ' be as in **Step 1** for $(W', B_{W'})$.

Write $\operatorname{div}_{W'}(z) = F_0 - F_1$, where $F_0 \geq 0$ and $F_1 \geq 0$ are divisors on W' without common components. Since

$$F_0 - F_1 = \operatorname{div}_{W'}(z) = f'^* M_0 - f'^* M_1,$$

any component of $F_0 + F_1$ is a component of $f'^*(M_0 + M_1)$, hence has coefficient 1 in $B_{W'}$ by (1e). By (1d), $\varphi'_* F_0$ and $\varphi'_* F_1$ are non-zero and have no common components, and by (1a), their components have coefficient 1 in $B_{Z'}$. Thus $\lfloor B_{Z'} \rfloor$ has at least 2 prime components. By Proposition 6.4(2) for $(Z', B_{Z'})$ and $m = c_q$,

$$\operatorname{trdeg}_k k\big(H^0(Z', \mathcal{O}_{Z'}(c_q A_{Z'}))\big) = 3. \tag{6.19}$$

Since $n = c_q(2n_1 + 1)$, by (6.16) for $(W', B_{W'})$ and (6.17) for $M_0 + M_1$, both with $j = c_q$,

$$\Lambda := (u_0 u_1)^{c_q} H^0(Z', \mathcal{O}_{Z'}(c_q A_{Z'})) \subset H^0(X, \mathcal{O}_X(\lfloor nD \rfloor)).$$

By (6.15) and (6.19),

$$3 = \operatorname{trdeg}_k k(\Lambda) \leq \operatorname{trdeg}_k k\big(H^0(X, \mathcal{O}_X(\lfloor nD \rfloor))\big) \leq 3.$$

Thus $|\lfloor nD \rfloor|$ defines a generically finite map.

Step 4. Let $h : X' \to X$ be a $\mathbb{Q}$-factorial dlt modification of (X, B) such that $a(E, X, B) = 0$ for any h-exceptional prime divisor E [HNT20, Corollary 3.6], let $\mathcal{E}''$ be the set of h-exceptional prime divisors, and let $B' := h_*^{-1} B + \sum_{E \in \mathcal{E}''} E$. Then (X', B') is a projective $\mathbb{Q}$-factorial lc pair, $K_{X'} + B' = h^* D$ is big, and $B' \in I_q$. Let Y and B_Y be as in **Step 1** for (X', B'), and let $D_Y := K_Y + B_Y$. Then Y is a normal projective threefold, D_Y is $\mathbb{Q}$-Cartier, nef, and big, and by (1a), qB_Y is a Weil divisor.

Since $K_{X'} + B' = h^* D$ and $h_* \mathcal{O}_{X'} = \mathcal{O}_X$, by (6.16) for (X', B'),

$$H^0(X, \mathcal{O}_X(\lfloor jD \rfloor)) = H^0(X', \mathcal{O}_{X'}(\lfloor j(K_{X'} + B') \rfloor)) = H^0(Y, \mathcal{O}_Y(\lfloor jD_Y \rfloor)) \tag{6.20}$$

for any $j \in \mathbb{N}^+$. By **Step 3** and (6.20) with $j = n$, $|\lfloor nD_Y \rfloor|$ defines a generically finite map. By Lemma 6.5 for Y, B_Y, q, and n, $|\lfloor mD_Y \rfloor|$ defines a birational map for any positive integer m

divisible by q with $m \geq 4qn+2$. By (6.20) with $j=m$, $||\lfloor mD \rfloor||$ defines a birational map for any such m. The theorem follows. □

7. Birational linear systems from pencils of surfaces

In this section we prove Theorem 7.2.

Definition 7.1. Let $q \geq 2$ be an integer. We define the following integers.

(1) $q_1 := \min\{q' \in \mathbb{Z} \mid q' \geq 2$, and q' and some $\theta \in (0,1) \cap \mathbb{Q}$ have the property of Corollary 4.2 for $I = I_q\}$, and $q_2 := qq_1$.
(2) $q_3 := \min\{N' \in \mathbb{N}^+ \mid N'$ has the property of Proposition 5.7 for $\epsilon = \beta = \frac{1}{q_2}\}$.
(3) $q_4 := \min\{M' \in \mathbb{N}^+ \mid M'$ has the properties of Lemma 5.2 for $\epsilon = \frac{1}{q_2}$ and q_3 in place of $N\}$.
(4) $a_q := 6q_2q_4 + 2q_2$ and $b_q := 42q_2q_4 + 14q_2 - 5$.

Theorem 7.2. *Let $q \geq 2$ be an integer and a_q, b_q the integers of Definition 7.1. Let k be an algebraically closed field of characteristic $p \geq 11$, and let (X, B) be a projective lc pair of dimension 3 over k such that $B \in I_q$ and $D := K_X + B$ is $\mathbb{Q}$-Cartier, nef, and big. Let $g : X' \to X$ be a resolution, $h : X' \to C$ a contraction onto a smooth projective curve, H a Cartier divisor on C with $\deg H > 0$ and $|H|$ base point free, and $c > 0$ a real number such that*

$$(g^*D)^2 \cdot h^*H \leq cD^3.$$

Then for any positive integer m divisible by q the following hold.

(1) If $m > 9(a_q + b_q c) + 1$, then $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) \neq 0$.
(2) If $m > 12(a_q + b_q c) + 1$, then $||\lfloor mD \rfloor||$ defines a birational map.

Proof. Let $t := a_q - 1$. By Definition 7.1, q_2 divides $t+1$, $b_q = 7t + 2$, and, as $q_2 \geq 2$,

$$t \geq 6q_2q_4 + 3.$$

Let $\theta \in (0,1)$ be a rational number such that q_1 and θ have the property of Corollary 4.2 for $I = I_q$. Let η be the generic point of C. For a variety with a morphism to C and a $\mathbb{Q}$-divisor on it, the subscript η denotes the generic fiber over C, a variety over the F-finite field $k(C)$, and the restriction of the divisor to it.

Step 1: truncating the boundary on a log resolution. By [HNT20, Section 2.1], there exists a log resolution $\varphi_0 : U_0 \to X'$ of $(X', g_*^{-1} \operatorname{Supp} B \cup \operatorname{Exc}(g))$. Let $g_0 := g \circ \varphi_0$ and $B_0 := (g_0)_*^{-1} B + \operatorname{Exc}(g_0)$, where $\operatorname{Exc}(g_0)$ is the reduced exceptional divisor of g_0. Then $\operatorname{Supp} B_0$ is simple normal crossing and $B_0 \in I_q$. Since (X, B) is lc, $K_{U_0} + B_0 - g_0^* D \geq 0$, so $K_{U_0} + B_0$ is big. Write $B_0 = \sum_i b_i B_{0,i}$ with $b_i > 0$ and distinct prime divisors $B_{0,i}$, and let

$$B_0^- := \sum_i \min\{b_i, \theta\} B_{0,i}.$$

Since $p > 5$, by Corollary 4.2, $q_1 B_0^-$ is a Weil divisor, (U_0, B_0^-) is $\frac{1}{q_1}$-lc, and $K_{U_0} + B_0^-$ is big. As $q_2 = qq_1$, $q_2 B_0^-$ is a Weil divisor and (U_0, B_0^-) is $\frac{1}{q_2}$-lc, so $B_0^- \in \{\frac{i}{q_2} \mid 1 \leq i \leq q_2 - 1\}$. Moreover,

$$(g_0)_* B_0^- \leq (g_0)_* B_0 = B.$$

Step 2: a terminal generic surface. Let $\Delta_0 := B_0^-$. We construct blow-ups $\pi_j : U_j \to U_{j-1}$ and $\mathbb{Q}$-divisors Δ_j on U_j as follows. Suppose that U_j and Δ_j are constructed. Let

$$h_j := h \circ \varphi_0 \circ \pi_1 \circ \cdots \circ \pi_j : U_j \to C.$$

If there exist distinct prime components P, P' of Δ_j with $b + b' \geq 1$, where $b := \operatorname{mult}_P \Delta_j$ and $b' := \operatorname{mult}_{P'} \Delta_j$, and an irreducible component Σ_j of $P \cap P'$ with $h_j(\Sigma_j) = C$, let $\pi_{j+1} : U_{j+1} \to U_j$ be the blow-up of Σ_j, E_{j+1} the exceptional prime divisor, and

$$\Delta_{j+1} := (\pi_{j+1})_*^{-1} \Delta_j + (b + b' - 1) E_{j+1};$$

otherwise we stop. By induction on j, $\operatorname{Supp}\Delta_j$ is simple normal crossing, as Σ_j is a stratum of $\operatorname{Supp}\Delta_j$, and

$$\Delta_j \in \{\tfrac{i}{q_2} \mid 0 \le i \le q_2 - 1\},$$

as $0 \le b + b' - 1 \le 1 - \frac{2}{q_2}$. We have $K_{U_{j+1}} = \pi_{j+1}^* K_{U_j} + E_{j+1}$ and, as Σ_j is contained in exactly 2 components of Δ_j,

$$\pi_{j+1}^*\Delta_j = (\pi_{j+1})_*^{-1}\Delta_j + (b+b')E_{j+1},$$

so

$$K_{U_{j+1}} + \Delta_{j+1} = \pi_{j+1}^*(K_{U_j} + \Delta_j).$$

We show that the sequence $(U_j, \Delta_j)_j$ is finite. By induction on j, the center of $E_{j,\eta}$ on $U_{0,\eta}$ is a closed point x at which components $P_x \ne P_x'$ of $\Delta_{0,\eta}$ intersect, with coefficients b_x and b_x', and $E_{j,\eta}$ is the divisor of the monomial valuation with weights (u, v) on local equations of P_x and P_x', for some coprime positive integers u and v. The log discrepancy of $E_{j,\eta}$ with respect to $(U_{0,\eta}, \Delta_{0,\eta})$ is $u(1 - b_x) + v(1 - b_x')$, and it equals $1 - \operatorname{mult}_{E_j}\Delta_j \le 1$ by the crepant formula above. Since $1 - b_x \ge \frac{1}{q_2}$ and $1 - b_x' \ge \frac{1}{q_2}$, we have $u + v \le q_2$. As there exist only finitely many such x and (u, v), and distinct j correspond to distinct valuations, the sequence $(U_j, \Delta_j)_j$ stops at some U_n.

Let $U := U_n$, $\Delta_U := \Delta_n$, $\psi : U \to U_0$ the composite of the π_j, $g_U := g_0 \circ \psi$, and $h_U := h \circ \varphi_0 \circ \psi$. Then the following hold.

(1) (U, Δ_U) is $\frac{1}{q_2}$-lc, $q_2\Delta_U$ is a Weil divisor, and $K_U + \Delta_U = \psi^*(K_{U_0} + \Delta_0)$ is big.
(2) $(g_U)_*\Delta_U = (g_0)_*\Delta_0 \le B$.
(3) U_η is a regular projective surface over $k(C)$ with $H^0(U_\eta, \mathcal{O}_{U_\eta}) = k(C)$, as $(h_U)_*\mathcal{O}_U = \mathcal{O}_C$.
(4) $\operatorname{Supp}\Delta_{U,\eta}$ is simple normal crossing.
(5) $\operatorname{mult}_P \Delta_{U,\eta} < 1$ for any prime component P of $\Delta_{U,\eta}$.
(6) $\operatorname{mult}_P \Delta_{U,\eta} + \operatorname{mult}_{P'} \Delta_{U,\eta} < 1$ for any distinct prime components P, P' of $\Delta_{U,\eta}$ with $P \cap P' \ne \emptyset$.

Since U_η is regular and (4)–(6) hold, $(U_\eta, \Delta_{U,\eta})$ is terminal by the computation of [KM98, Corollary 2.31(3)], which only uses blow-ups of closed points of regular surfaces [Liu02, Theorem 9.2.2].

Step 3: pruning the generic surface. The field $k(C)$ has characteristic $p \ge 11$, $(U_\eta, \Delta_{U,\eta})$ is $\frac{1}{q_2}$-lc, the coefficients of $\Delta_{U,\eta}$ are $\ge \frac{1}{q_2}$, and $K_{U_\eta} + \Delta_{U,\eta}$ is big. By Proposition 5.7 for $\epsilon = \beta = \frac{1}{q_2}$, there exists a $\mathbb{Q}$-divisor Δ_U' on U with at most q_3 components such that $K_{U_\eta} + \Delta_{U,\eta}'$ is big and any component P of Δ_U' is a component of Δ_U dominating C with $\operatorname{mult}_P \Delta_U' = \operatorname{mult}_P \Delta_U$. Since $0 \le \Delta_U' \le \Delta_U$, the following hold.

(1) (U, Δ_U') is $\frac{1}{q_2}$-lc, $q_2\Delta_U'$ is a Weil divisor, and $\Delta_U' \in \{\frac{i}{q_2} \mid 1 \le i \le q_2 - 1\}$.
(2) $(g_U)_*\Delta_U' \le B$.
(3) $(U_\eta, \Delta_{U,\eta}')$ is a projective terminal $\frac{1}{q_2}$-lc pair, U_η is regular, $\Delta_{U,\eta}'$ has at most q_3 components, $q_2\Delta_{U,\eta}'$ is a Weil divisor, and $K_{U_\eta} + \Delta_{U,\eta}'$ is big.

Step 4: a relative minimal model over C. Since (U, Δ_U') is a $\mathbb{Q}$-factorial klt pair and $p > 5$, by [HNT20, Theorem 1.1], there exists a $(K_U + \Delta_U')$-MMP over C which terminates with a $\mathbb{Q}$-factorial klt pair (Y, Δ_Y) with a morphism $h_Y : Y \to C$, such that either $L_Y := K_Y + \Delta_Y$ is nef over C, or there exists an L_Y-Mori fiber space $Y \to Y_1$ over C. Since $U \dashrightarrow Y$ extracts no divisor, $L_{Y,\eta}$ is the pushforward of the big divisor $K_{U_\eta} + \Delta_{U,\eta}'$, hence big. In the second case, $Y_1 \to C$ is surjective and $\dim Y_1 < 3$, so $Y_\eta \to Y_{1,\eta}$ is a contraction of positive relative dimension with $-L_{Y,\eta}$ ample over $Y_{1,\eta}$, which contradicts the bigness of $L_{Y,\eta}$. Thus L_Y is nef over C. Moreover, $(h_Y)_*\mathcal{O}_Y = \mathcal{O}_C$, as $(h_U)_*\mathcal{O}_U = \mathcal{O}_C$.

A flip over C is an isomorphism over a neighborhood of η, since the flipping locus is a union of finitely many curves contracted to points of C. Let $Y' \to Y''$ be a divisorial contraction in the MMP with exceptional prime divisor E and let $\Delta_{Y'}$ be the boundary on Y'. If E does not

dominate C, then $Y' \to Y''$ is an isomorphism over a neighborhood of η. Otherwise, $Y'_\eta \to Y''_\eta$ is the contraction of the irreducible curve E_η and $(K_{Y'} + \Delta_{Y'}) \cdot E_\eta < 0$. Thus $U_\eta \to Y_\eta$ is the birational morphism of a $(K_{U_\eta} + \Delta'_{U,\eta})$-MMP over $k(C)$. Let

$$S := Y_\eta, \quad \Delta_S := \Delta_{Y,\eta}, \quad \text{and} \quad L_S := K_S + \Delta_S = L_{Y,\eta}.$$

Then S is a normal $\mathbb{Q}$-factorial projective surface over $k(C)$ with $H^0(S, \mathcal{O}_S) = k(C)$, $\Delta_S \geq 0$, and L_S is nef and big. By **Step 3**(3) and Lemma 5.2(2), for $\epsilon = \frac{1}{q_2}$, the integer q_3, and q_2 in place of q,

$$q_2 q_4 L_S \text{ is nef, big, and Cartier.}$$

We show that $L_Y + 6h_Y^* H$ is nef. Since (Y, Δ_Y) is a klt threefold pair over the algebraically closed field k and $p > 5$, by [HNT20, Theorem 1.3],

$$\overline{\mathrm{NE}}(Y) = \overline{\mathrm{NE}}(Y)_{L_Y \geq 0} + \sum_j \mathbb{R}_{\geq 0}[\Gamma_j]$$

for curves Γ_j with $-6 \leq L_Y \cdot \Gamma_j < 0$. Since L_Y is nef over C, $h_Y(\Gamma_j) = C$, so

$$h_Y^* H \cdot \Gamma_j = \deg H \cdot \deg(\Gamma_j \to C)$$

is a positive integer. Thus $(L_Y + 6h_Y^* H) \cdot \Gamma_j \geq 0$ for any j. As $h_Y^* H$ is nef, $L_Y + 6h_Y^* H$ is non-negative on $\overline{\mathrm{NE}}(Y)_{L_Y \geq 0}$, so $L_Y + 6h_Y^* H$ is nef. Since $(h_Y^* H)^2 \equiv 0$ and $L_Y^2 \cdot h_Y^* H = \deg H \cdot L_S^2 > 0$,

$$(L_Y + 7h_Y^* H)^3 = (L_Y + 6h_Y^* H)^3 + 3L_Y^2 \cdot h_Y^* H > 0,$$

so $L_Y + 7h_Y^* H$ is nef and big.

Step 5: birational Frobenius-stable sections on the generic surface. We show that $S^0_-(S, K_S + tL_S)$ defines a birational map. Let $w : S' \to S$ be the minimal resolution [Kol13, Theorem 2.25]. Since $K_{S'}$ is w-nef, the $\mathbb{Q}$-divisor $G_{S'} := w^* K_S - K_{S'}$ satisfies $w_* G_{S'} = 0$ and $-G_{S'}$ is w-nef, so $G_{S'} \geq 0$ by [Tan18, Lemma 2.11]. Thus

$$w^* L_S = K_{S'} + \Delta_{S'}, \quad \Delta_{S'} := G_{S'} + w^* \Delta_S \geq 0.$$

Let $\lambda : S'' \to S'$ be the composite of the normalization of $(S' \times_{k(C)} \overline{k(C)})_{\mathrm{red}}$ with the projection to S'. Since S' is regular, $K_{S'}$ is Cartier, and $\lambda^* K_{S'} - K_{S''}$ is the conductor, so $\lambda^* K_{S'} - K_{S''} \geq 0$ [Zha22, Section 2.4]. Let $z : \widetilde{S} \to S''$ be the minimal resolution. Then $\widetilde{S}$ is a smooth projective surface over the algebraically closed field $\overline{k(C)}$. The divisor $z^* \lambda^* K_{S'} - K_{\widetilde{S}}$ has pushforward $\lambda^* K_{S'} - K_{S''} \geq 0$ to S'', and $K_{\widetilde{S}} - z^* \lambda^* K_{S'}$ is z-nef, so $z^* \lambda^* K_{S'} - K_{\widetilde{S}} \geq 0$ by [Tan18, Lemma 2.11]. Let $L_{\widetilde{S}} := (w \circ \lambda \circ z)^* L_S$. Then

$$L_{\widetilde{S}} = K_{\widetilde{S}} + \Delta_{\widetilde{S}}, \quad \Delta_{\widetilde{S}} := (z^* \lambda^* K_{S'} - K_{\widetilde{S}}) + z^* \lambda^* \Delta_{S'} \geq 0,$$

and $A_1 := q_2 q_4 L_{\widetilde{S}}$ is a nef and big Cartier divisor with

$$(A_1 - K_{\widetilde{S}}) \cdot A_1 = q_2 q_4 (q_2 q_4 - 1) L_{\widetilde{S}}^2 + q_2 q_4 \Delta_{\widetilde{S}} \cdot L_{\widetilde{S}} \geq 0. \tag{7.1}$$

Let $a : \widetilde{S} \to \mathrm{Alb}(\widetilde{S})$ be the Albanese morphism. By [Zha22, Theorem 4.2] for the nef and big Cartier divisor A_1 and a big $\mathbb{Q}$-divisor A_2, the space $S^0_-(\widetilde{S}, K_{\widetilde{S}} + \Xi)$ defines a birational map in the following cases.

(1) If $\mathrm{Alb}(\widetilde{S})$ is a point: $\Xi = 6A_1 + A_2$, with $A_2 := (t - 6q_2 q_4) L_{\widetilde{S}}$. Here we use (7.1).
(2) If $\dim a(\widetilde{S}) = 2$: $\Xi = A_2 + 2(K_{\widetilde{S}} + A_1)$, with $A_2 := (t - 2q_2 q_4 - 2) L_{\widetilde{S}} + 2\Delta_{\widetilde{S}}$.
(3) If $\dim a(\widetilde{S}) = 1$ and $A_1 \cdot F \geq 2$ for a general fiber F of the Stein factorization of a: $\Xi = A_1 + A_2 + 2(K_{\widetilde{S}} + A_1)$, with $A_2 := (t - 3q_2 q_4 - 2) L_{\widetilde{S}} + 2\Delta_{\widetilde{S}}$.
(4) If $\dim a(\widetilde{S}) = 1$ and $A_1 \cdot F = 1$ for a general fiber F of the Stein factorization of a: $\Xi = 2A_1 + A_2 + 2(K_{\widetilde{S}} + 2A_1)$, with $A_2 := (t - 6q_2 q_4 - 2) L_{\widetilde{S}} + 2\Delta_{\widetilde{S}}$.

In (3) and (4), $A_1 \cdot F$ is a positive integer for a general fiber F of the Stein factorization of a, as A_1 is a nef and big Cartier divisor. Thus (1)–(4) cover all cases. In each case A_2 is big, as $t \geq 6q_2q_4+3$, and $\Xi = tL_{\widetilde{S}}$, as $K_{\widetilde{S}} = L_{\widetilde{S}} - \Delta_{\widetilde{S}}$. Thus $S^0_-(\widetilde{S}, K_{\widetilde{S}} + tL_{\widetilde{S}})$ defines a birational map. By [Zha22, Proposition 2.3(iv-1)] for z, so does $S^0_-(S'', K_{S''} + t\lambda^* w^* L_S)$. By [Zha22, Theorem 2.7], as $H^0(S', \mathcal{O}_{S'}) = k(C)$, so does $S^0_-(S', K_{S'} + tw^* L_S)$. By [Zha22, Proposition 2.3(iv-1)] for w, so does $S^0_-(S, K_S + tL_S)$.

Step 6: a birational divisor on Y. Let

$$V_Y := S^0_-(Y, K_Y + t(L_Y + 7h_Y^* H) + 2h_Y^* H) \quad \text{and} \quad R_Y := a_q L_Y + b_q h_Y^* H.$$

The $\mathbb{Q}$-divisor $t(L_Y + 7h_Y^* H)$ is nef, big, and $\mathbb{Q}$-Cartier, $t(L_Y + 7h_Y^* H)|_S = tL_S$, $|H|$ is base point free with $\deg H > 0$, and $S^0_-(S, K_S + tL_S)$ defines a birational map by **Step 5**. By [Zha22, Theorem 3.1(ii)] for h_Y, the divisor H, a very ample divisor $\widetilde{H}$ on C, and $s = 2$, V_Y defines a birational map. By Definition 6.1,

$$V_Y \subset H^0(Y, \mathcal{O}_Y(K_Y + \lceil tL_Y \rceil + (7t+2)h_Y^* H)).$$

For any coefficient b of Δ_Y, we have $b \in \{\frac{i}{q_2} \mid 1 \leq i \leq q_2 - 1\}$ by **Step 3**(1), so $(t+1)b \in \mathbb{Z}$ as q_2 divides $t+1$, and thus $\lceil tb \rceil = \lceil (t+1)b - b \rceil = (t+1)b$. Hence $K_Y + \lceil tL_Y \rceil = (t+1)L_Y = a_q L_Y$, and as $b_q = 7t+2$,

$$V_Y \subset H^0(Y, \mathcal{O}_Y(R_Y)).$$

Step 7: a birational Cartier divisor of bounded mixed degree. The divisor R_Y is a $\mathbb{Q}$-Cartier Weil divisor, as Y is $\mathbb{Q}$-factorial. By [Cut09, Theorems 1.1 and 1.3], there exist a smooth projective threefold Z and birational morphisms $\alpha : Z \to U$ and $\zeta : Z \to Y$ with $h_U \circ \alpha = h_Y \circ \zeta$, such that the base ideal of V_Y, regarded as a subspace of $H^0(Z, \mathcal{O}_Z(\lfloor \zeta^* R_Y \rfloor))$ by Lemma 2.4(2), is invertible. Write $\lfloor \zeta^* R_Y \rfloor = A_Z + F_Z$, where $F_Z \geq 0$ is the fixed divisor of V_Y. Then $V_Y \subset H^0(Z, \mathcal{O}_Z(A_Z))$ has no base points, so A_Z is a base point free Cartier divisor, $|A_Z|$ defines a birational map, and

$$\zeta^* R_Y - A_Z = (\zeta^* R_Y - \lfloor \zeta^* R_Y \rfloor) + F_Z \geq 0.$$

Let $D_Z := (g_U \circ \alpha)^* D$. Then D_Z is nef and $D_Z^3 = D^3$. Since $U \dashrightarrow Y$ is a composite of steps of a $(K_U + \Delta'_U)$-MMP, by the negativity lemma,

$$\alpha^*(K_U + \Delta'_U) = \zeta^* L_Y + G_Z \quad \text{for some } G_Z \geq 0.$$

Since $(g_U)_* K_U = K_X$ and prime divisors on U exceptional over X have zero intersection with $(g_U^* D)^2$, the projection formula and **Step 3**(2) imply

$$D_Z^2 \cdot \alpha^*(K_U + \Delta'_U) = D^2 \cdot (K_X + (g_U)_* \Delta'_U) = D^3 - D^2 \cdot (B - (g_U)_* \Delta'_U) \leq D^3,$$

where the inequality holds as D is nef and $B - (g_U)_* \Delta'_U \geq 0$. As D_Z is nef and $G_Z \geq 0$, $D_Z^2 \cdot \zeta^* L_Y \leq D^3$. Since h_U and g_U factor through the birational morphism $\varphi_0 \circ \psi : U \to X'$,

$$D_Z^2 \cdot \zeta^* h_Y^* H = D_Z^2 \cdot \alpha^* h_U^* H = (g^* D)^2 \cdot h^* H \leq cD^3.$$

Since D_Z is nef and $\zeta^* R_Y - A_Z \geq 0$, we obtain

$$D_Z^2 \cdot A_Z \leq D_Z^2 \cdot \zeta^* R_Y \leq (a_q + b_q c) D^3.$$

Step 8: returning to X. Let m be a positive integer divisible by q and $j \in \{3, 4\}$ an integer with $m - 1 > 3j(a_q + b_q c)$. Since D_Z and A_Z are nef, by [Cut15, Proposition 5.4] and **Step 7**,

$$\operatorname{vol}((m-1)D_Z - jA_Z) \geq (m-1)^3 D^3 - 3j(m-1)^2 D_Z^2 \cdot A_Z \geq (m-1)^2 D^3 \big(m - 1 - 3j(a_q + b_q c)\big) > 0.$$

Thus there exists a $\mathbb{Q}$-divisor $F'_Z \geq 0$ on Z such that $A_0 := (m-1)D_Z - jA_Z - F'_Z$ is ample. Since $|A_Z|$ defines a birational map, by Lemma 6.2 for the nef and big $\mathbb{Q}$-divisor A_0 and the

divisor A_Z, the space $S^0_-(Z, K_Z + A_0 + jA_Z)$ is non-zero, and it defines a birational map if $j = 4$. As jA_Z is a Weil divisor, by Definition 6.1,

$$S^0_-(Z, K_Z + A_0 + jA_Z) \subset H^0(Z, \mathcal{O}_Z(K_Z + \lceil (m-1)D_Z - F'_Z \rceil)) \\ \subset H^0(Z, \mathcal{O}_Z(K_Z + \lceil (m-1)D_Z \rceil)).$$

Let $\pi_Z := g_U \circ \alpha$, choose K_Z with $(\pi_Z)_* K_Z = K_X$, let P be a prime divisor on X, and let

$$b_P := \operatorname{mult}_P B \in I_q.$$

Since $mb_P \in \mathbb{Z}$ and $0 \le b_P \le 1$, the coefficient of P in $(\pi_Z)_*(K_Z + \lceil (m-1)D_Z \rceil)$ is

$$\operatorname{mult}_P K_X + \lceil (m-1) \operatorname{mult}_P D \rceil = m \operatorname{mult}_P K_X + \lceil mb_P - b_P \rceil = m \operatorname{mult}_P D - \lfloor b_P \rfloor.$$

Hence, as mD is a Weil divisor,

$$(\pi_Z)_*(K_Z + \lceil (m-1)D_Z \rceil) = mD - \lfloor B \rfloor \le \lfloor mD \rfloor.$$

By Lemma 2.4(1) for π_Z, the divisor $K_Z + \lceil (m-1)D_Z \rceil$ on Z, mD in place of D, and 1 in place of m,

$$H^0(Z, \mathcal{O}_Z(K_Z + \lceil (m-1)D_Z \rceil)) \subset H^0(X, \mathcal{O}_X(\lfloor mD \rfloor))$$

as subspaces of the function field of X. Taking $j = 3$, $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) \neq 0$ if $m - 1 > 9(a_q + b_q c)$. Taking $j = 4$, $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor))$ contains a subspace defining a birational map, hence $|\lfloor mD \rfloor|$ defines a birational map, if $m - 1 > 12(a_q + b_q c)$. The theorem follows. □

Lemma 7.3. *Let W be a smooth projective threefold over an algebraically closed field, L a Cartier divisor on W, and $s_0, s_1 \in H^0(W, \mathcal{O}_W(L))$ linearly independent sections. Then there exist a birational morphism $\mu : U \to W$ from a smooth projective threefold U, a contraction $h : U \to C$ onto a smooth projective curve C, a Cartier divisor H on C with $\deg H > 0$ and $|H|$ base point free, and a divisor $F \ge 0$ on U such that $\mu^* L \sim h^* H + F$.*

Proof. By [Cut09, Theorem 1.3], there exists a composite of blow-ups $\mu : U \to W$ with U smooth such that the base ideal of the pencil spanned by $\mu^* s_0$ and $\mu^* s_1$ is invertible. Let $F \ge 0$ be the fixed divisor of the pencil. Then $\mu^* s_0$ and $\mu^* s_1$ induce sections of $\mathcal{O}_U(\mu^* L - F)$ without common zeros, which define a morphism $f : U \to \mathbb{P}^1$ with $\mu^* L - F \sim f^*(\text{point})$. The morphism f is non-constant as s_0 and s_1 are linearly independent. Let $U \xrightarrow{h} C \xrightarrow{\nu} \mathbb{P}^1$ be the Stein factorization of f. Then C is a smooth projective curve, h is a contraction, $H := \nu^*(\text{point})$ is a Cartier divisor on C with $\deg H > 0$ and $|H|$ base point free, and $\mu^* L - F \sim h^* H$. □

Corollary 7.4. *Let $q \ge 2$ be an integer and k an algebraically closed field of characteristic $p \ge 11$. Let (X, B) be a projective lc pair of dimension 3 over k such that $B \in I_q$, $D := K_X + B$ is $\mathbb{Q}$-Cartier, nef, and big, and a resolution of X has trivial Albanese variety. Fix a projective embedding of X, let T be a positive-dimensional integral locally closed subscheme of the Hilbert scheme of X whose members V_z are prime divisors on X, and let $c > 0$ be a real number such that*

$$D^2 \cdot V_z \le cD^3$$

for a general point $z \in T$. Then $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) \neq 0$ for any positive integer m divisible by q with $m > 9(a_q + b_q c) + 1$, where a_q and b_q are the integers of Definition 7.1.

Proof. Let $g : W \to X$ be a resolution such that $\operatorname{Alb}(W)$ is trivial, and let $T^\circ \subset T$ be a dense open subset such that $D^2 \cdot V_z \le cD^3$ for any closed point $z \in T^\circ$. Since $\dim T \ge 1$, there exists an integral curve $T_1 \subset T$ with $T_1 \cap T^\circ \neq \emptyset$. Let C_1 be a non-empty open subset of the normalization of T_1 which maps isomorphically onto an open subset of $T_1 \cap T^\circ$. Then C_1 is a smooth curve. Let $\mathcal{V}_1 \subset X \times C_1$ be the pullback of the universal family, so that the fiber of $\mathcal{V}_1$ over a closed point $z \in C_1$ is V_z. Let $X^\circ \subset X$ be the largest open subset over which g is an isomorphism, and let $\mathcal{V}_W \subset W \times C_1$ be the closure of

$$(g \times \mathrm{id}_{C_1})^{-1}(\mathcal{V}_1 \cap (X^\circ \times C_1)).$$

Since $\mathcal{V}_1$ is flat over C_1 with fibers of dimension 2, any irreducible component of $\mathcal{V}_W$ is of dimension 3 and dominates C_1. Thus $\mathcal{V}_W$ is an effective Cartier divisor on the smooth variety $W \times C_1$, flat and proper over C_1.

Since no irreducible component of $\mathcal{V}_W$ is contained in $g^{-1}(X \backslash X^\circ) \times C_1$, possibly shrinking C_1, we may assume that any irreducible component of the fiber $\mathcal{V}_{W,z}$ of $\mathcal{V}_W$ over any closed point $z \in C_1$ intersects $g^{-1}(X^\circ)$. As $\mathcal{V}_{W,z}$ is an effective Cartier divisor on the smooth variety W, it has no embedded points, and $\mathcal{V}_{W,z} \cap g^{-1}(X^\circ) \cong V_z \cap X^\circ$ is integral. Thus $\mathcal{V}_{W,z} = g_*^{-1}V_z$ for any closed point $z \in C_1$.

By [Kle05, Theorem 9.4.8], the line bundle $\mathcal{O}_{W\times C_1}(\mathcal{V}_W)$ defines a morphism $\mu : C_1 \to \mathrm{Pic}_{W/k}$ which sends a closed point z to the class of $\mathcal{O}_W(g_*^{-1}V_z)$. By [Kle05, Remark 9.5.25], $(\mathrm{Pic}^0_{W/k})_{\mathrm{red}}$ is an abelian variety whose dual is $\mathrm{Alb}(W)$, so $(\mathrm{Pic}^0_{W/k})_{\mathrm{red}}$ is a point. Any connected component of $\mathrm{Pic}_{W/k}$ containing a k-point is a translate of $\mathrm{Pic}^0_{W/k}$, so the reduction of any such component is a point. Since C_1 is connected and reduced, μ is constant. Let $z_0 \neq z_1$ be closed points of C_1 and $V_i' := g_*^{-1}V_{z_i}$ for $i \in \{0,1\}$. Then V_0' and V_1' are distinct linearly equivalent prime divisors on W.

Let $s_0, s_1 \in H^0(W, \mathcal{O}_W(V_0'))$ be sections with $\mathrm{div}(s_i) = V_i'$ for $i \in \{0,1\}$. The sections s_0 and s_1 are linearly independent as $V_0' \neq V_1'$. By Lemma 7.3 for $L = V_0'$, there exist a birational morphism $\varphi : X' \to W$ from a smooth projective threefold X', a contraction $h : X' \to C$ onto a smooth projective curve C, a Cartier divisor H on C with $\deg H > 0$ and $|H|$ base point free, and a divisor $F_{X'} \geq 0$ on X' such that $\varphi^*V_0' \sim h^*H + F_{X'}$. Since $(g\circ\varphi)^*D$ is nef and $F_{X'} \geq 0$,

$$((g\circ\varphi)^*D)^2 \cdot h^*H \leq ((g\circ\varphi)^*D)^2 \cdot \varphi^*V_0' = D^2 \cdot g_*V_0' = D^2 \cdot V_{z_0} \leq cD^3.$$

By Theorem 7.2(1) for $g\circ\varphi : X' \to X$, $h : X' \to C$, and H, we have $H^0(X, \mathcal{O}_X(\lfloor mD \rfloor)) \neq 0$ for any positive integer m divisible by q with $m > 9(a_q + b_q c) + 1$. □

8. The first non-zero section on models with bounded denominators

Set-up 8.1. Let q and (X, B) be as follows.

(1) $q \geq 2$ is an integer.
(2) (X, B) is a projective $\mathbb{Q}$-factorial $\frac{1}{q}$-lc pair (Definition 2.2) of dimension 3 over an algebraically closed field k of characteristic p, and $B \in I_q \backslash \{1\}$.
(3) $D := K_X + B$ is nef and big.
(4) A resolution of X has trivial Albanese variety.

Let c_q be the integer of Definition-Lemma 6.3, and let a_q and b_q be the integers of Definition 7.1. Let

$$u_0 := 8c_q^2 q, \qquad u_1 := 4c_q^2 + 2c_q + 3,$$
$$\gamma_q := 1024 q u_0, \qquad \sigma_q := \max\{q+13, 9a_q + 3b_q + q + 1\},$$
$$\sigma_q^* := \max\{\sigma_q, 49q+1\}, \qquad U_q := \max\{2048 q u_1, 8\gamma_q\sigma_q^*, 8\},$$
$$P_q^* := \max\{64, 1024 q u_1, 2\gamma_q\sigma_q, (U_q+1)^2\},$$
$$J_q := \mathrm{lcm}(1, 2, \ldots, \lfloor 4q/3 \rfloor), \qquad M_q := 14(q u_0 + u_1),$$
$$E_q := \max\{2\sigma_q^*, q(3584 M_q + 900) + 4\}, \qquad Q_q := 8E_q,$$

and $P_q^\dagger := \max\{P_q^*, (4\gamma_q E_q + 1)^2, (M_q - 2)(M_q - 3) + 1\}$.

Let

$$b_0 := \min\{j \in \mathbb{N}^+ \mid H^0(X, \mathcal{O}_X(\lfloor jD \rfloor)) \neq 0\} \quad \text{and} \quad n_0 := c_q\big(2c_q(qb_0+1)+1\big),$$

and let r be an integer such that $r \geq \max\{2, J_q\}$ and $r^3D^3 > 216$.

In this section we show that $b_0 \leq Q_q r$ if $p > P_q^\dagger$ (see Theorem 8.18).

Lemma 8.2. *Let X be a $\mathbb{Q}$-factorial quasi-projective threefold over an algebraically closed field of characteristic $p > 5$, and let $B \geq 0$ be a $\mathbb{Q}$-divisor on X. Let $\pi : Y \to X$ be a projective birational morphism from a $\mathbb{Q}$-factorial normal variety Y such that π has exactly one exceptional prime divisor S, and $C := \pi(S)$ is a curve. Let $h := \pi|_S : S \to C$. Let $B_Y \geq 0$ be a $\mathbb{Q}$-divisor on Y with $S \not\subset \operatorname{Supp} B_Y$, and let L be a Weil divisor on X which is Cartier near the generic point of C. Assume that*

$$K_Y + S + B_Y = \pi^*(K_X + B)$$

and $(Y, S + B_Y)$ is plt. Then:

(1) $-S$ is π-ample, and the $\mathbb{Q}$-divisor π^*L has integer coefficients.
(2) $R^i\pi_*\mathcal{O}_Y(j\pi^*L - S) = 0$ for any integers $i > 0$ and j.
(3) S is normal, C is smooth, and $h_*\mathcal{O}_S = \mathcal{O}_C$.

Proof. We first prove (1). Since π is projective and X is quasi-projective, Y is quasi-projective. Let $H \geq 0$ be an ample Cartier divisor on Y with $S \not\subset \operatorname{Supp} H$. Since X is $\mathbb{Q}$-factorial, π_*H is $\mathbb{Q}$-Cartier, and

$$\pi^*\pi_*H = H + bS, \quad \text{where } b := \operatorname{mult}_S \pi^*\pi_*H \geq 0$$

since $\pi_*H \geq 0$. Since $\dim S = 2 > \dim C$, there exists a curve $\Sigma \subset S$ such that $\pi(\Sigma)$ is a point, and

$$0 < H \cdot \Sigma = -bS \cdot \Sigma.$$

Thus $b > 0$. Since $-bS \sim_{\mathbb{Q},X} H$, $-S$ is π-ample. Since L is principal near the generic point of C, $\operatorname{mult}_S \pi^*L \in \mathbb{Z}$. Any prime divisor $P \neq S$ on Y satisfies $P = \pi_*^{-1}\pi(P)$, so $\operatorname{mult}_P \pi^*L = \operatorname{mult}_{\pi(P)} L \in \mathbb{Z}$. Thus π^*L has integer coefficients.

We prove (2). Since $-S$ is π-ample, (Y, S) is plt, and $\pi(S) = C$ is a curve, by [Arv23, Propositions 8 and 13(1)] the following holds: for any Weil divisor A on Y and any $\mathbb{Q}$-divisor $\Gamma = \Gamma' + cS$ with $c \in [0, 1]$, $\Gamma' \geq 0$, and $S \not\subset \operatorname{Supp} \Gamma'$ such that $(Y, \Gamma' + S)$ is dlt and $A - (K_Y + \Gamma)$ is π-ample,

$$R^i\pi_*\mathcal{O}_Y(A) = 0 \quad \text{for any integer } i > 0.$$

Let $u \in (0, 1)$ be a rational number and j an integer. By (1), $j\pi^*L$ is a Weil divisor on Y. Since $(Y, S + B_Y)$ is plt, it is dlt. We have

$$(j\pi^*L - S) - (K_Y + B_Y + uS) = \pi^*(jL - K_X - B) - uS.$$

Since $-S$ is π-ample and $u > 0$, the right-hand side is π-ample. Taking

$$(A, \Gamma) = (j\pi^*L - S, B_Y + uS),$$

we obtain (2).

We prove (3). Since $(Y, S+B_Y)$ is plt, it is a 3-dimensional dlt pair over a field of characteristic $p > 5$ and S is a log canonical center of $(Y, S + B_Y)$, so S is normal by [BK23, Theorem 1.3, Theorem 3.1(4) and the paragraph following it]. Since Y is normal, $\mathcal{O}_Y(-S)$ is the ideal sheaf of S, and there exists an exact sequence

$$0 \to \mathcal{O}_Y(-S) \to \mathcal{O}_Y \to \mathcal{O}_S \to 0.$$

Since $\pi(S) = C$, a regular function ϕ on an open subset $V \subset X$ satisfies $\pi^*\phi \in \mathcal{O}_Y(-S)(\pi^{-1}V)$ if and only if $\phi|_{C\cap V} = 0$. Thus $\pi_*\mathcal{O}_Y(-S) = \mathcal{I}_C$, the ideal sheaf of C. Since X is normal, $\pi_*\mathcal{O}_Y = \mathcal{O}_X$. By (2) with $j = 0$, $R^1\pi_*\mathcal{O}_Y(-S) = 0$. Thus the push-forward of the exact sequence above implies

$$h_*\mathcal{O}_S = \pi_*\mathcal{O}_S = \mathcal{O}_X/\mathcal{I}_C = \mathcal{O}_C.$$

Since S is normal and $h_*\mathcal{O}_S = \mathcal{O}_C$, C is normal, hence smooth. □

Lemma 8.3. *Let X be the spectrum of a 2-dimensional normal local ring essentially of finite type over a field, let x be the closed point of X, and let $S \subset X$ be a prime divisor. Assume that $(X \ni x, S)$ is plt. Then S is regular, and there exists a positive integer N satisfying the following.*

(1) ND is Cartier for any Weil divisor D on X.

(2) The coefficient at x of the different $\mathrm{Diff}_S(0)$ *of* (X,S) *on* S [Kol13, Definition 4.2] *is* $1-\frac{1}{N}$.

Proof. The proof follows [HW19, Proof of Lemma 3.3]. By [Kol13, 3.35(1)], S is regular. If x is a regular point of X, then any Weil divisor on X is Cartier and $\mathrm{Diff}_S(0)=0$, so $N:=1$ satisfies (1) and (2). Thus we may assume that x is a singular point of X.

Let κ be the residue field of x. Since $(X\ni x,S)$ is plt, X is klt, so X has a rational singularity at x [Kol13, Proposition 2.28]. Let $f:Y\to X$ be the minimal resolution [Kol13, Theorem 2.25], let $E_1,\dots,E_n$ be the exceptional prime divisors of f, where $n\geq 1$, and let $S':=f_*^{-1}S$. Since S is regular, $f|_{S'}:S'\to S$ is an isomorphism, and the closed point s' of S' has residue field κ. Intersection numbers with E_i are degrees over κ. By [Kol13, Corollary 3.31(1) and 3.35(1)], possibly reordering indices, we may assume the following.

(i) $E_i\cong\mathbb{P}^1_\kappa$ for any i.
(ii) $E_i\cdot E_{i+1}=1$ for any $1\leq i\leq n-1$, and $E_i\cdot E_j=0$ for any i,j with $|i-j|\geq 2$.
(iii) $S'\cdot E_1=1$, and $S'\cap E_i=\emptyset$ for any $i\geq 2$.

In particular, the scheme-theoretic intersection $S'\cap E_1$ is the reduced point s'. Let

$$M:=(E_i\cdot E_j)_{1\leq i,j\leq n}.$$

Since M is negative definite [Lip69, Lemma (14.1)] and has integer entries,

$$N:=\det(-M)$$

is a positive integer, and the adjugate matrix of $-M$ has integer entries.

We prove (1). Let D be a Weil divisor on X, let $D':=f_*^{-1}D$, and let $(d_1,\dots,d_n)\in\mathbb{Q}^n$ be the solution of

$$\sum_{i=1}^n d_i(E_i\cdot E_j)=-D'\cdot E_j\quad\text{for any } 1\leq j\leq n.$$

Since the adjugate matrix of $-M$ has integer entries, $Nd_i\in\mathbb{Z}$ for any i. Thus

$$G:=N\Big(D'+\sum_{i=1}^n d_iE_i\Big)$$

is a Cartier divisor on the regular surface Y, and $G\cdot E_j=0$ for any j. Since X is affine and has a rational singularity, $H^1(Y,\mathcal{O}_Y)=0$. By [Lip69, Theorem (12.1)(ii)], $\mathcal{O}_Y(G)$ is generated by global sections. Let σ be a global section of $\mathcal{O}_Y(G)$ which does not vanish at a closed point of E_1. For any j, since $\deg_\kappa(\mathcal{O}_Y(G)|_{E_j})=0$ and E_j is integral, $\sigma|_{E_j}$ is either 0 or nowhere vanishing. Since $\sigma|_{E_1}\neq 0$ and $E_j\cap E_{j+1}\neq\emptyset$ for any $1\leq j\leq n-1$, σ does not vanish at any point of $f^{-1}(x)=\bigcup_{i=1}^n E_i$. Let $Z\subset Y$ be the zero locus of σ. Since f is proper, $f(Z)$ is a closed subset of X not containing x. Since X is local, $f(Z)=\emptyset$. Thus $\mathcal{O}_Y(G)\cong\mathcal{O}_Y$, that is, $G=\mathrm{div}(\varphi)$ for a rational function φ on Y. Thus

$$ND=f_*G=\mathrm{div}(\varphi)$$

is Cartier.

We prove (2). By (1), K_X+S is $\mathbb{Q}$-Cartier. Let K_Y be a canonical divisor on Y with $f_*K_Y=K_X$, and write

$$f^*(K_X+S)=K_Y+S'+\sum_{i=1}^n a_iE_i.$$

Since $E_j\cong\mathbb{P}^1_\kappa$, $(K_Y+E_j)\cdot E_j=-2$. Since $f^*(K_X+S)\cdot E_j=0$,

$$\sum_{i=1}^n(1-a_i)(E_i\cdot E_j)=S'\cdot E_j+\sum_{i\neq j}E_i\cdot E_j-2=\begin{cases}0 & \text{if } j<n,\\ -1 & \text{if } j=n,\end{cases}$$

where the second equality holds as $S'\cdot E_j+\sum_{i\neq j}E_i\cdot E_j$ is the number of neighbours of E_j in the chain $S',E_1,\dots,E_n$. Thus $(-M)(1-a_1,\dots,1-a_n)^T=(0,\dots,0,1)^T$. By Cramer's rule,

$1 - a_1 = \frac{\det M_1}{N}$, where M_1 is the matrix $-M$ with the first column replaced by $(0, \dots, 0, 1)^T$. The first column of M_1 has the single non-zero entry 1 in row n, and the submatrix of M_1 formed by the rows $1, \dots, n-1$ and the columns $2, \dots, n$ is lower triangular with all diagonal entries -1. Thus

$$\det M_1 = (-1)^{n+1}(-1)^{n-1} = 1, \quad \text{so} \quad a_1 = 1 - \frac{1}{N}.$$

Since Y and S' are regular along S', $(K_Y + S')|_{S'} = K_{S'}$. Since $S' \cap E_1$ is the reduced point s' and $S' \cap E_i = \emptyset$ for any $i \geq 2$,

$$\Big(K_Y + S' + \sum_{i=1}^{n} a_i E_i\Big)\Big|_{S'} = K_{S'} + a_1 s'.$$

Since f is a crepant morphism from $(Y, S' + \sum_{i=1}^n a_i E_i)$ to (X, S), S' is Cartier on Y, and $f|_{S'} : S' \to S$ is an isomorphism mapping s' to x, the coefficient of $\mathrm{Diff}_S(0)$ at x is $a_1 = 1 - \frac{1}{N}$ [Kol13, Propositions 4.5(4) and 4.6]. □

Set-up 8.4. Let (X, B), $f : W \to T$, and $g : W \to X$ be as follows.

(1) (X, B) is a projective $\mathbb{Q}$-factorial pair of dimension 3 over an algebraically closed field of characteristic $p > 5$, and B is a $\mathbb{Q}$-divisor.
(2) T is a smooth quasi-projective surface, W is a smooth threefold, f is a smooth projective morphism, and $F_t := f^{-1}(t)$ is an integral curve for any closed point $t \in T$.
(3) g is dominant and generically finite, and $g|_{F_t} : F_t \to C_t := g(F_t)$ is the normalization of C_t for any closed point $t \in T$.

Let $D := K_X + B$, and let

$$\lambda := \deg(g^* D|_{F_t})$$

for a general closed point $t \in T$. If g is birational, let ω be a non-zero rational differential 3-form on X with $\mathrm{div}_X(\omega) = K_X$, let $K_W := \mathrm{div}_W(g^*\omega)$, and let B_W be the $\mathbb{Q}$-divisor on W such that

$$K_W + B_W = g^* D.$$

Theorem 8.5. *Notation and conditions as in Set-up 8.4, with g birational. Let $\epsilon \in (0, 1]$ be a real number, let $t \in T$ be a closed point, let B_W^+ be the positive part of B_W, and let $G \geq 0$ be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X. Assume that the following hold.*

(i) g is an isomorphism near the generic points of F_t and C_t.
(ii) No component of B_W contains F_t.
(iii) $\mathrm{mult}_x(B_W^+|_{F_t}) \leq 1 - \epsilon$ for any closed point $x \in F_t$.
(iv) $G \cdot C_t < \epsilon$.
(v) Near the generic point of C_t, the pair $(X, B + G)$ is lc and has exactly one lc place, and the center on X of that lc place is C_t.

Then the following hold.

(1) $(X, B + G)$ is lc near C_t.
(2) C_t is smooth.
(3) For any rational number $\delta \in (0, \epsilon - G \cdot C_t]$ and any $\mathbb{Q}$-Cartier Weil divisor L on X which is Cartier near the generic point of C_t, the divisor $\mathrm{lcm}(1, 2, \dots, \lfloor 1/\delta \rfloor)L$ is Cartier near C_t.

Proof. We let $F := F_t$, $C := C_t$, $G_W := g^*G$, and $\alpha := G \cdot C$, and we let η_F and η_C be the generic points of F and C. Since $g|_F : F \to C$ is birational,

$$G_W \cdot F = \alpha < \epsilon \leq 1.$$

By (i), (ii), and (v) (by (i), $B_W = g^*B$ near η_F, so (ii) implies $C \not\subset \mathrm{Supp}\, B$), there exist open neighborhoods $W_F \subset W$ of η_F and $X_C \subset X \setminus \mathrm{Supp}\, B$ of η_C such that g restricts to an isomorphism $W_F \to X_C$, the pair $(X_C, G|_{X_C})$ is lc, and $(X_C, G|_{X_C})$ has exactly one lc place v,

with $\operatorname{center}_X v = C$. Thus $B_W|_{W_F} = 0$, the pair $(W_F, G_W|_{W_F})$ is lc, v is the only lc place of $(W_F, G_W|_{W_F})$, and $\operatorname{center}_W v = F$.

Step 1. In this step we show that v is the valuation of the preimage in $W \times_T T_N$ of the last exceptional curve of a sequence $T_N \to \cdots \to T_1 \to T$ of blow-ups at closed points over t.

Let $T_0 := T$, let $i \geq 0$, and assume that $T_i \to T$ is a composition $\mu_1 \circ \cdots \circ \mu_i$, where $\mu_j : T_j \to T_{j-1}$ is the blow-up of a closed point of T_{j-1} over t. For $1 \leq j \leq i$, let $\mu_{i,j} : T_i \to T_j$ be the composite morphism (with $\mu_{i,i} := \mathrm{id}_{T_i}$), let $E_j^\circ \subset T_j$ be the exceptional curve of μ_j, and let $E_j := (\mu_{i,j})_*^{-1} E_j^\circ$ (here i is fixed, and we omit i from the notation E_j and S_j). Let $W_i := W \times_T T_i$, let $f_i : W_i \to T_i$ and $\varphi_i : W_i \to W$ be the projections, and let $S_j := f_i^{-1}(E_j) \cong F \times E_j$ for $1 \leq j \leq i$. Since f_i is smooth, W_i is smooth, $S_j = f_i^* E_j$, and for any closed point $z \in T_i$ over t, φ_i maps the fiber $f_i^{-1}(z)$ isomorphically onto F. Since f_i is flat, the blow-up of W_i along $f_i^{-1}(z)$ is $W \times_T T_i'$, where $T_i' \to T_i$ is the blow-up of z. Let $G_i := (\varphi_i)_*^{-1} G_W$. Since $\varphi_i^* G_W - G_i$ is a linear combination of $S_1, \ldots, S_i$, and $S_j \cdot f_i^{-1}(z) = 0$ for any j,

$$G_i \cdot f_i^{-1}(z) = G_W \cdot F = \alpha \quad \text{for any closed point } z \in T_i \text{ over } t. \tag{8.1}$$

We claim that if $Z := \operatorname{center}_{W_i} v$ is a curve, then $Z = f_i^{-1}(z_i)$ for a closed point $z_i \in T_i$ over t. For $i = 0$, $Z = F = f^{-1}(t)$. Let $i \geq 1$. Since $\varphi_i(Z) = F$ and $\varphi_i^{-1}(F) = \bigcup_{j=1}^i S_j$, Z dominates F and there exists j with $Z \subset S_j$. Suppose that Z is not a fiber of f_i. Then

$$n_Z := \deg(f_i|_Z : Z \to E_j) \geq 1,$$

and $Z \not\subset S_k$ for any $k \neq j$, since $S_j \cap S_k = f_i^{-1}(E_j \cap E_k)$ is a union of fibers of f_i. Since $S_j \not\subset \operatorname{Supp} G_i$, $G_i|_{S_j} \geq 0$. Let $c_Z := \operatorname{mult}_Z(G_i|_{S_j})$. For a general closed point $z \in E_j$, the fiber $f_i^{-1}(z)$ is not contained in $\operatorname{Supp} G_i$, and $Z \cdot f_i^{-1}(z) = n_Z$ on $S_j \cong F \times E_j$, so by (8.1),

$$c_Z \leq c_Z n_Z \leq G_i \cdot f_i^{-1}(z) = \alpha < 1.$$

Any component G' of G_i containing Z has coefficient at most c_Z, since $G'|_{S_j} \geq Z$. Thus near the generic point η_Z of Z, $\lfloor S_j + G_i \rfloor = S_j$, and the different $G_i|_{S_j}$ of $(W_i, S_j + G_i)$ on the regular curve germ $(S_j \ni \eta_Z)$ is $c_Z \eta_Z$ with $c_Z < 1$. By [Das15, Lemma 3.3], $(W_i, S_j + G_i)$ is plt near η_Z. Write

$$K_{W_i} + \sum_{k=1}^{i} b_k S_k + G_i = \varphi_i^*(K_W + G_W).$$

Since $(W_F, G_W|_{W_F})$ is lc and $\operatorname{center}_W S_k = F$, $b_k \leq 1$ for any k. Since $Z \not\subset S_k$ for $k \neq j$, $b_j \leq 1$, and v is exceptional over W_i with center Z,

$$0 = a(v, W, G_W) = a\Big(v, W_i, \sum_{k=1}^{i} b_k S_k + G_i\Big) \geq a(v, W_i, S_j + G_i) > 0,$$

a contradiction. This proves the claim.

When $\operatorname{center}_{W_i} v$ is a curve, we let $T_{i+1} \to T_i$ be the blow-up of the point z_i of the claim, so that W_{i+1} is the blow-up of W_i along $\operatorname{center}_{W_i} v$. Since v is a divisorial valuation, there exists $N \geq 1$ such that $\operatorname{center}_{W_N} v$ is a divisor. Since $\operatorname{center}_{W_N} v$ maps into $f_{N-1}^{-1}(z_{N-1})$, $\operatorname{center}_{W_N} v = S_N = f_N^{-1}(E_N^\circ)$, that is, v is the prime divisor S_N on W_N.

Step 2. In this step we extract v alone and compute the different on the extracted divisor.

By [Lip69, Theorem (12.1)(ii), Propositions (1.2)(1) and (21.3)], possibly shrinking T to a neighborhood of t, there exists a projective birational morphism $\rho : T' \to T$ from a normal surface T' with rational singularities such that ρ is an isomorphism over $T \setminus \{t\}$, $\rho^{-1}(t)$ is a prime divisor E, and $T_N \to T$ factors through a morphism $T_N \to T'$ mapping E_N onto E. By [Lip69, Proposition (17.1)], T' is $\mathbb{Q}$-factorial. Let $V := W \times_T T'$, let $u : V \to W$ and $f_V : V \to T'$ be the projections, and let

$$S_V := f_V^{-1}(E) \cong F \times E.$$

Since f_V is smooth, V is normal and $S_V = f_V^* E$ is a $\mathbb{Q}$-Cartier prime divisor. The morphism u is an isomorphism over $W \backslash F$, S_V is the only exceptional prime divisor of u, and v is the prime divisor S_V on V, since $W_N \to V$ maps S_N onto S_V. Let $G_V := u_*^{-1} G_W$. Since $a(v, W, G_W) = 0$,

$$K_V + S_V + G_V = u^*(K_W + G_W). \tag{8.2}$$

Let $V_F := u^{-1}(W_F)$. Since v is the only lc place of $(W_F, G_W|_{W_F})$, the pair $(V_F, (S_V + G_V)|_{V_F})$ is plt.

For any projective birational morphism $\psi : T'' \to T'$ from a normal surface, let $\Delta_{T''}$ be the $\mathbb{Q}$-divisor with $K_{T''} + \Delta_{T''} = \psi^*(K_{T'} + E)$ and $\psi_* \Delta_{T''} = E$, and let $V'' := W \times_T T''$, with projections $f'' : V'' \to T''$ and $\psi_V : V'' \to V$. Since ψ_V is the base change of ψ by the smooth morphism f_V, the relative canonical divisor of ψ_V is the pullback by f'' of that of ψ, so

$$K_{V''} + f''^* \Delta_{T''} = \psi_V^*(K_V + S_V). \tag{8.3}$$

Let E' be an exceptional prime divisor over T' whose center on T' is contained in E, and choose ψ such that E' is a divisor on T''. By (8.3), the prime divisor $f''^{-1}(E') \cong F \times E'$ satisfies $a(f''^{-1}(E'), V, S_V) = a(E', T', E)$. The center of $f''^{-1}(E')$ on V dominates F, so it intersects V_F, and

$$a(E', T', E) = a(f''^{-1}(E'), V, S_V) \geq a(f''^{-1}(E'), V, S_V + G_V) > 0.$$

Since $T' \backslash E \cong T \backslash \{t\}$ is smooth, (T', E) is plt. By Lemma 8.3, E is regular and $\Delta_E := \mathrm{Diff}_E(0)$, the different of (T', E) on E, has coefficients in $[0, 1)$. In particular, $S_V \cong F \times E$ is a smooth projective surface. Let $\mathrm{pr}_F : S_V \to F$ and $\mathrm{pr}_E : S_V \to E$ be the projections, and for any closed point $z \in E$ let $F'_z := f_V^{-1}(z)$, so that pr_F maps F'_z isomorphically onto F.

We show that $\mathrm{Diff}_{S_V}(0) = \mathrm{pr}_E^* \Delta_E$. Let ψ be a resolution of T'. Since E is regular, $E'' := \psi_*^{-1} E \cong E$, so $\Delta_E = (\Delta_{T''} - E'')|_{E''}$ by [Kol13, Propositions 4.5(4) and 4.6]. Since V'' is smooth and $f''^* E'' \cong S_V$, (8.3) and [Kol13, Propositions 4.5(4) and 4.6] imply

$$\mathrm{Diff}_{S_V}(0) = \big(f''^*(\Delta_{T''} - E'')\big)|_{f''^* E''} = \mathrm{pr}_E^* \Delta_E.$$

Let $B_W^- := B_W^+ - B_W$, $B_V^\pm := u^* B_W^\pm$, and $\Lambda^\pm := B_W^\pm|_F$. By (ii), no component of $B_W^\pm$ contains F, so $\Lambda^\pm \geq 0$, $S_V \not\subset \mathrm{Supp}\, B_V^\pm$, and $B_V^\pm|_{S_V} = \mathrm{pr}_F^* \Lambda^\pm$, since $u|_{S_V} = \mathrm{pr}_F$. Since $u^* G_W - G_V$ is a multiple of $S_V = f_V^* E$, G_V is $\mathbb{Q}$-Cartier, and as u maps F'_z isomorphically onto F,

$$(G_V|_{S_V}) \cdot F'_z = G_W \cdot F = \alpha \quad \text{for any closed point } z \in E. \tag{8.4}$$

Since B_V^+ and G_V are $\mathbb{Q}$-Cartier,

$$\mathrm{Diff}_{S_V}(B_V^+ + G_V) = \mathrm{pr}_E^* \Delta_E + \mathrm{pr}_F^* \Lambda^+ + G_V|_{S_V}. \tag{8.5}$$

Step 3. In this step we show that $(V, S_V + B_V^+ + G_V)$ is plt near S_V and that

$$\big(S_V, \Gamma_V + c\, \mathrm{pr}_F^* y\big) \text{ is sub-klt} \tag{8.6}$$

for any closed point $y \in F$ and any rational number $c \in [0, \epsilon - \alpha)$, where

$$\Gamma_V := \mathrm{Diff}_{S_V}(u^* B_W + G_V).$$

Write

$$\mathrm{pr}_E^* \Delta_E + G_V|_{S_V} = R_0 + R_1,$$

where $R_0 \geq 0$ is supported on fibers of $f_V|_{S_V}$ and $R_1 \geq 0$ has no such component. For any closed point $z \in E$, F'_z dominates F, so the generic point of F'_z is contained in V_F, where $(V_F, (S_V + G_V)|_{V_F})$ is plt and $B_V^+ = 0$. By adjunction [Kol13, Lemma 4.8] and (8.5), the coefficients of R_0 are < 1. Since distinct fibers of f_V are disjoint, $F'^2_z = 0$ and $\mathrm{pr}_E^* \Delta_E \cdot F'_z = 0$, so (8.4) implies $R_1 \cdot F'_z = \alpha$. As $F'_z \not\subset \mathrm{Supp}\, R_1$, $R_1|_{F'_z} \geq 0$ is a $\mathbb{Q}$-divisor of degree α on F'_z, so

$$\mathrm{mult}_x(R_1|_{F'_z}) \leq \alpha \quad \text{for any closed points } z \in E \text{ and } x \in F'_z. \tag{8.7}$$

Fix a closed point $y \in F$ and a rational number $c \in [0, \epsilon - \alpha)$, and let

$$\Delta_{y,c} := R_0 + R_1 + \mathrm{pr}_F^*(\Lambda^+ + cy) = \mathrm{Diff}_{S_V}(B_V^+ + G_V) + c\, \mathrm{pr}_F^* y.$$

Fix a closed point $z \in E$, let $r_z < 1$ be the coefficient of F'_z in R_0, and let $\Delta'_{y,c} := \Delta_{y,c} - r_z F'_z$. Since S_V is smooth and the fibers of $f_V|_{S_V}$ other than F'_z are disjoint from F'_z, the different of $(S_V, F'_z + \Delta'_{y,c})$ on F'_z is

$$\Delta'_{y,c}|_{F'_z} = \mathrm{pr}_F^*(\Lambda^+ + cy)|_{F'_z} + R_1|_{F'_z}.$$

By (iii) and (8.7),

$$\mathrm{mult}_x(\Delta'_{y,c}|_{F'_z}) \le 1 - \epsilon + c + \alpha < 1 \quad \text{for any closed point } x \in F'_z. \tag{8.8}$$

Hence $(F'_z, \Delta'_{y,c}|_{F'_z})$ is strongly F-regular in the sense of [Das15, Definition 2.3(1)] (purely F-regular pairs are defined in [Das15, Definition 2.3(2)]). Let $x \in F'_z$ be a closed point, let w be a uniformizer of the regular local ring $\mathcal{O}_{F'_z,x}$, let $b < 1$ be the coefficient of x in $\Delta'_{y,c}|_{F'_z}$, and let $\theta = w^\kappa \theta_0$ be a non-zero element of $\mathcal{O}_{F'_z,x}$, with $\kappa \ge 0$ and θ_0 a unit. There exists e with $\kappa + \lceil (p^e - 1)b \rceil \le p^e - 1$, so multiplication by $F_*^e(\theta_0^{-1} w^{p^e - 1 - \kappa})$ maps $F_*^e \mathcal{O}_{F'_z,x}(\lceil (p^e - 1)b \rceil x)$ into $F_*^e \mathcal{O}_{F'_z,x}$. Let χ_e be the composite of that multiplication with the trace of the e-th Frobenius of $\mathcal{O}_{F'_z,x}$. Then $\chi_e(F_*^e \theta)$ is the trace of $F_*^e w^{p^e - 1}$, which is a unit, and $\chi_e(F_*^e\theta)^{-1}\chi_e$ is the splitting for θ in [Das15, Definition 2.3(1)]. Any prime divisor $P \ne F'_z$ on S_V intersecting F'_z at a point x satisfies $\mathrm{mult}_P \Delta'_{y,c} \le \mathrm{mult}_x(\Delta'_{y,c}|_{F'_z}) < 1$, so $\lfloor F'_z + \Delta'_{y,c} \rfloor = F'_z$ near F'_z. By [Das15, Theorem 4.1, Corollary 4.3], $(S_V, F'_z + \Delta'_{y,c})$ is purely F-regular and plt near F'_z. Since $r_z < 1$, the pair $(S_V, \Delta_{y,c}) = (S_V, r_z F'_z + \Delta'_{y,c})$ is klt near F'_z. Moreover $(S_V, \Delta_{y,0})$ is strongly F-regular near F'_z. Write a non-zero element of the local ring of S_V at a point of F'_z as $w_z^d \zeta$, where w_z is a local equation of F'_z, $d \ge 0$, and ζ does not vanish identically on F'_z. Composing a splitting for ζ for the purely F-regular pair $(S_V, F'_z + \Delta'_{y,0})$ with the sharp F-splittings of $(S_V, F'_z + \Delta'_{y,0})$ at all multiples of some e_0 [SS10, Proposition 3.8(b)] yields splittings for ζ for all e in an arithmetic progression, and for e with $\lfloor (p^e - 1)(1 - r_z) \rfloor \ge d$ the factor $w_z^{\lfloor (p^e-1)(1-r_z) \rfloor}$ absorbs w_z^d. As $S_V = \bigcup_z F'_z$, the pair $(S_V, \Delta_{y,c})$ is klt, and $(S_V, \Delta_{y,0})$ is strongly F-regular.

Let $Q \ne S_V$ be a prime divisor on V intersecting S_V, and let P be a curve in $Q \cap S_V$. If P is not a fiber of f_V, then V is smooth at the generic point of P, since the singular locus of V is contained in $f_V^{-1}(\mathrm{Sing}\, T')$ and $\mathrm{Sing}\, T'$ is finite. Hence $\mathrm{mult}_Q(B_V^+ + G_V) \le \mathrm{mult}_P \Delta_{y,0} < 1$. If P is a fiber of f_V, then the generic point of P is contained in V_F, where $(V_F, (S_V + G_V)|_{V_F})$ is plt and $B_V^+ = 0$, so $\mathrm{mult}_Q(B_V^+ + G_V) < 1$. Thus $\lfloor S_V + B_V^+ + G_V \rfloor = S_V$ near S_V. Since $K_V + S_V + B_V^+ + G_V = u^*(K_W + B_W^+ + G_W)$ is $\mathbb{Q}$-Cartier and $(S_V, \Delta_{y,0})$ is strongly F-regular, [Das15, Theorem 4.1, Corollary 4.3] imply that $(V, S_V + B_V^+ + G_V)$ is plt near S_V.

By (8.2) and $K_W + B_W = g^*(K_X + B)$,

$$K_V + S_V + u^* B_W + G_V = (g \circ u)^*(K_X + B + G). \tag{8.9}$$

For a sub-pair (X', Δ'), let $\mathrm{Nklt}(X', \Delta')$ be the closed subset of points of X' at which (X', Δ') is not sub-klt. Since $u^* B_W = B_V^+ - B_V^- \le B_V^+$, the sub-pair $(V, S_V + u^* B_W + G_V)$ is sub-plt near S_V, and $\mathrm{Nklt}(V, S_V + u^* B_W + G_V) = S_V$ near S_V. Moreover $\Gamma_V = \Delta_{y,0} - \mathrm{pr}_F^* \Lambda^-$, so

$$\Gamma_V + c\, \mathrm{pr}_F^* y = \Delta_{y,c} - \mathrm{pr}_F^* \Lambda^- \le \Delta_{y,c},$$

and (8.6) follows.

Step 4. In this step we prove (1) and (2) and construct the extraction of v over X.

Since $g \circ u : V \to X$ is birational and quasi-projective, there exist a normal variety $\overline{V}$ containing V as an open subset and a projective birational morphism $\overline{g} : \overline{V} \to X$ with $\overline{g}|_V = g \circ u$. Let $\overline{B}$ be the $\mathbb{Q}$-divisor on $\overline{V}$ with $K_{\overline{V}} + \overline{B} = \overline{g}^*(K_X + B + G)$ and $\overline{g}_* K_{\overline{V}} = K_X$. Then $\overline{g}_* \overline{B} = B + G \ge 0$, $-(K_{\overline{V}} + \overline{B})$ is $\overline{g}$-nef, and by (8.9), $\overline{B}|_V = S_V + u^* B_W + G_V$. Since S_V is projective, S_V is closed in $\overline{V}$, and by **Step 3** there exists an open subset $U \subset \overline{V}$ containing S_V such that $(U, \overline{B}|_U)$ is sub-plt and

$$\mathrm{Nklt}(\overline{V}, \overline{B}) \cap U = S_V.$$

By [NT20, Theorem 2.15], $\mathrm{Nklt}(\overline{V},\overline{B})\to X$ has connected fibers. For any closed point $x\in C$, $S_V\cap\overline{g}^{-1}(x)\neq\emptyset$ since $\overline{g}(S_V)=C$, and

$$S_V\cap\overline{g}^{-1}(x)=\mathrm{Nklt}(\overline{V},\overline{B})\cap U\cap\overline{g}^{-1}(x)$$

is open and closed in $\mathrm{Nklt}(\overline{V},\overline{B})\cap\overline{g}^{-1}(x)$. Hence

$$\mathrm{Nklt}(\overline{V},\overline{B})\cap\overline{g}^{-1}(C)\subset S_V.$$

Since $\overline{g}$ is proper, the set

$$X^\circ:=X\backslash\overline{g}\big(\mathrm{Nklt}(\overline{V},\overline{B})\backslash U\big)$$

is an open neighborhood of C. Over X°, $\mathrm{Nklt}(\overline{V},\overline{B})\subset U$, so any prime divisor E' over X° with $a(E',X,B+G)\le 0$ has center on $\overline{V}$ contained in U, where $(U,\overline{B}|_U)$ is sub-plt. Thus E' is S_V. Therefore $(X^\circ,B+G)$ is lc, v is the only lc place of $(X^\circ,B+G)$, and $\mathrm{Nklt}(X^\circ,B+G)=C$. This proves (1). Possibly replacing X with X°, we may assume that $(X,B+G)$ is lc, v is the only lc place of $(X,B+G)$, and $\mathrm{Nklt}(X,B+G)=C$.

By [Bir16, Theorem 1.6], there exists a crepant $\mathbb{Q}$-factorial dlt modification $\pi:Y\to X$ of $(X,B+G)$: Y is $\mathbb{Q}$-factorial, $K_Y+B'_Y=\pi^*(K_X+B+G)$, (Y,B'_Y) is dlt, and any exceptional prime divisor of π has coefficient 1 in B'_Y. Since π is crepant, $\pi^{-1}(C)=\mathrm{Nklt}(Y,B'_Y)=\lfloor B'_Y\rfloor$, and since $\mathrm{Nklt}(X,B+G)=C$ is a curve, any component of $\lfloor B'_Y\rfloor$ is an exceptional lc place of $(X,B+G)$. Hence π has exactly one exceptional prime divisor S, v is the prime divisor S on Y, $\pi(S)=C$, and

$$K_Y+S+B_Y+G_Y=\pi^*(K_X+B+G),$$

where $B_Y:=\pi_*^{-1}B$ and $G_Y:=\pi_*^{-1}G$. As $\lfloor S+B_Y+G_Y\rfloor=S$, the pair $(Y,S+B_Y+G_Y)$ is plt. By Lemma 8.2 for $B+G$ in place of B and B_Y+G_Y in place of B_Y, C is smooth, S is normal, and $h:=\pi|_S$ satisfies $h_*\mathcal{O}_S=\mathcal{O}_C$. This proves (2). Since $g|_F:F\to C$ is the normalization, $g|_F$ is an isomorphism.

Step 5. In this step we deduce a bound on the coefficients of $\mathrm{Diff}_S(B_Y+G_Y)$ from (8.6). Let $\Gamma_S:=\mathrm{Diff}_S(B_Y+G_Y)$. We show that

$$\mathrm{mult}_P(\Gamma_S+ch^*x)<1 \tag{8.10}$$

for any closed point $x\in C$, any rational number $c\in[0,\epsilon-\alpha)$, and any prime divisor P on S.

By (8.9), $(V,S_V+u^*B_W+G_V)$ and $(Y,S+B_Y+G_Y)$ are crepant birational over X, and $\mathrm{center}_V v=S_V$ and $\mathrm{center}_Y v=S$. By [Kol13, Proposition 4.6], (S_V,Γ_V) and (S,Γ_S) are crepant birational, via a birational map $S_V\dashrightarrow S$ over C, where S_V maps to C by $g\circ\mathrm{pr}_F$. Let $y:=(g|_F)^{-1}(x)$. Then $(g\circ\mathrm{pr}_F)^*x=\mathrm{pr}_F^*y$, so $(S_V,\Gamma_V+c\,\mathrm{pr}_F^*y)$ and (S,Γ_S+ch^*x) are crepant birational. By (8.6), (S,Γ_S+ch^*x) is sub-klt, and (8.10) follows.

Step 6. In this step we prove (3). For a Weil divisor L' on a normal variety, we regard local sections of $\mathcal{O}(L')$ as rational functions s with $\mathrm{div}(s)+L'\ge 0$. Fix δ and L as in (3), and let

$$J:=\mathrm{lcm}(1,2,\dots,\lfloor 1/\delta\rfloor),\qquad L_Y:=\pi^*L.$$

By Lemma 8.2, L_Y is a Weil divisor and

$$R^1\pi_*\mathcal{O}_Y(JL_Y-S)=0. \tag{8.11}$$

We show that for any prime divisor P on S not dominating C, with $x:=h(P)$ and $m:=\mathrm{mult}_P h^*x\ge 1$, there exists a positive integer N_P such that

$$N_PL_Y\text{ is Cartier at the generic point }\eta_P\text{ of }P,\quad\text{and}\quad N_Pm\mid J. \tag{8.12}$$

Since (Y,S) is plt, Lemma 8.3 for the local scheme of Y at η_P implies that there exists a positive integer N_P such that N_PL_Y is Cartier at η_P and $\mathrm{mult}_P\mathrm{Diff}_S(0)=1-\frac{1}{N_P}$. Since $B_Y+G_Y\ge 0$, $\Gamma_S\ge\mathrm{Diff}_S(0)$, so by (8.10),

$$1-\frac{1}{N_P}+cm<1\quad\text{for any rational number }c\in[0,\epsilon-\alpha).$$

Thus $N_P m \le \frac{1}{\epsilon-\alpha} \le \frac{1}{\delta}$, so $N_P m \le \lfloor 1/\delta \rfloor$ and $N_P m \mid J$. Since L is Cartier near η_C, L_Y is Cartier at the generic point of any prime divisor on S dominating C. Therefore JL_Y is Cartier at the generic point of any prime divisor on S.

Let

$$\mathcal{Q} := \mathcal{O}_Y(JL_Y)/\mathcal{O}_Y(JL_Y - S).$$

The sheaf $\mathcal{Q}$ is an $\mathcal{O}_S$-module, since

$$\mathcal{O}_Y(-S) \cdot \mathcal{O}_Y(JL_Y) \subset \mathcal{O}_Y(JL_Y - S).$$

By [BK23, Theorem 1.3, Theorem 3.1(2), and the paragraph following it], $\mathcal{O}_Y(JL_Y)$ and $\mathcal{O}_Y(JL_Y - S)$ are Cohen–Macaulay, so by the depth lemma [BH98, Proposition 1.2.9], $\mathcal{Q}$ is an S_2 sheaf on S. As JL_Y is Cartier at the generic point of any prime divisor on S, $\mathcal{Q}$ is invertible there, so $\mathcal{Q}$ is a reflexive sheaf of rank 1 on S [Har80, Proposition 1.3].

Let n be a positive integer such that nL is Cartier near C, and let φ be a generator of $\mathcal{O}_X(L)$ at η_C. Then φ^n is a rational section of the line bundle $\mathcal{O}_X(nL)|_C$, and we let

$$L_C := \frac{1}{n} \operatorname{div}(\varphi^n|_C),$$

a $\mathbb{Q}$-divisor on C. Let P and x be as in (8.12). Since $N_P L_Y$ is Cartier at η_P and $(\varphi^{N_P})^n = \varphi^{nN_P}$ restricts on S to the rational section of $h^*(\mathcal{O}_X(nN_P L)|_C)$ with divisor $nN_P h^* L_C$, the order along P of the restriction of φ^{N_P} is $N_P m \operatorname{mult}_x L_C \in \mathbb{Z}$. As $N_P m \mid J$ and $h^{-1}(x)$ is a curve, so that it contains a prime divisor P on S with $h(P) = x$, $J \operatorname{mult}_x L_C \in \mathbb{Z}$ for any closed point $x \in C$, so JL_C is a divisor on C. Since JL_Y is Cartier at η_P and $(\varphi^J)^n = \varphi^{nJ}$ restricts on S to the rational section of $h^*(\mathcal{O}_X(nJL)|_C)$ with divisor $nJh^* L_C$, the image σ of φ^J in $\mathcal{Q}$ has order $Jm \operatorname{mult}_x L_C = \operatorname{mult}_P h^*(JL_C)$ along P. Along a prime divisor P' on S dominating C, σ has order $0 = \operatorname{mult}_{P'} h^*(JL_C)$, as φ generates $\mathcal{O}_X(L)$ near η_C. Since $\mathcal{Q}$ and $h^*\mathcal{O}_C(JL_C)$ are reflexive of rank 1 on the normal surface S, and the divisor of the rational section σ of $\mathcal{Q}$ and the divisor of the rational section h^*1 of $h^*\mathcal{O}_C(JL_C)$ are equal to $h^*(JL_C)$,

$$\mathcal{Q} \cong h^*\mathcal{O}_C(JL_C).$$

Since $\pi_*\mathcal{O}_Y(JL_Y) = \mathcal{O}_X(JL)$ and $h_*\mathcal{O}_S = \mathcal{O}_C$, (8.11) and the projection formula imply that the restriction

$$\mathcal{O}_X(JL) \to \iota_*\mathcal{O}_C(JL_C)$$

is surjective, where $\iota : C \to X$ is the inclusion and the map sends $\beta\varphi^J$ to $\beta|_C$ for any $\beta \in \mathcal{O}_{X,\eta_C}$. Fix a closed point $x \in C$, let τ be a generator of $\mathcal{O}_C(JL_C)$ at x, and let s be a section of $\mathcal{O}_X(JL)$ near x restricting to τ. Then s^n is a section of $\mathcal{O}_X(nJL)$ near x, and the image of s^n in $\mathcal{O}_X(nJL)|_C = \mathcal{O}_C(nJL_C)$ is τ^n, since both are determined at η_C, where restriction is multiplicative. As τ^n generates $\mathcal{O}_C(nJL_C)$ at x and $\mathcal{O}_X(nJL)$ is invertible near x, s^n generates $\mathcal{O}_X(nJL)$ at x by Nakayama's lemma, that is, $\operatorname{div}(s^n) + nJL = 0$ near x. Thus $\operatorname{div}(s) + JL = 0$ near x, and JL is Cartier near x. Since $x \in C$ is arbitrary, (3) follows. □

For a scheme T, a *geometric point* of T is a morphism $t : \operatorname{Spec} \Omega \to T$ with Ω an algebraically closed field, whose image need not be a closed point of T, and for a T-scheme $\mathcal{V}$ we let $\mathcal{V}_t := \mathcal{V} \times_T \operatorname{Spec} \Omega$ be the *geometric fiber* of $\mathcal{V}$ over t.

Lemma 8.6. *Let X be a normal projective threefold over an algebraically closed field k of characteristic $p > 0$, let T be an integral variety over k, and let $\mathcal{C} \subset X \times T$ be an integral closed subscheme which is flat and projective over T, whose geometric fibers C_t are integral curves, and whose projection to X is dominant. Let M be a $\mathbb{Q}$-Cartier Weil divisor on X such that $|M|$ defines a birational map, and let e be a positive integer such that*

$$\deg(M|_{C_t^\nu}) < e + 1$$

for a general closed point $t \in T$, where C_t^ν is the normalization of C_t and $\deg(M|_{C_t^\nu})$ denotes $\frac{1}{a}$ times the degree of the pullback of $\mathcal{O}_X(aM)$ to C_t^ν for a positive integer a such that aM is Cartier. Assume that

$$p > (e-1)(e-2)+1.$$

Let $\mathcal{C}^\nu$ be the normalization of $\mathcal{C}$. Then there exists a non-empty open subset $T_0 \subset T$ satisfying the following, where $W := \mathcal{C}^\nu \times_T T_0$.

(1) T_0 and W are smooth.
(2) The morphism $W \to T_0$ is smooth and projective with geometrically integral fibers.
(3) For any geometric point t of T_0, the morphism $W_t \to C_t$ is the normalization of C_t.

Proof. Let $\rho : \mathcal{C}^\nu \to \mathcal{C}$ be the normalization morphism. The morphism ρ is finite. Possibly shrinking T, we may assume that T is smooth. Let C_η and C_η^ν be the generic fibers of $\mathcal{C}$ and $\mathcal{C}^\nu$ over T. Then C_η and C_η^ν are curves over $k(T)$. For a geometrically integral projective curve C over a field, let $p_a(C) := 1 - \chi(C, \mathcal{O}_C)$, computed over that field. We divide the proof into three steps.

Step 1. C_η^ν is a regular geometrically integral projective curve over $k(T)$ with

$$p_a(C_\eta^\nu) \le \frac{(e-1)(e-2)}{2}. \tag{8.13}$$

Since normalization commutes with localization, C_η^ν is the normalization of C_η. Thus C_η^ν is a normal, hence regular, integral projective curve over $k(T)$. Since C_η is geometrically integral and C_η^ν is birational to C_η, C_η^ν is geometrically integral [Gro65, Propositions 4.5.9 and 4.6.1].

Let a be a positive integer such that aM is Cartier, and let $\mathcal{L}$ be the pullback of $\mathcal{O}_X(aM)$ to $\mathcal{C}$. Since $\mathcal{C}$ is flat and projective over the integral scheme T and the normalization of an integral projective curve is finite and birational, for a general closed point $t \in T$,

$$\deg(M|_{C_\eta^\nu}) := \frac{1}{a}\deg(\mathcal{L}|_{C_\eta^\nu}) = \frac{1}{a}\deg(\mathcal{L}|_{C_\eta}) = \frac{1}{a}\deg(\mathcal{L}|_{C_t}) = \deg(M|_{C_t^\nu}) < e+1. \tag{8.14}$$

Let $s_0, \dots, s_m$ be a basis of $H^0(X, \mathcal{O}_X(M))$, regarded as rational functions on X with

$$D_i := \operatorname{div}_X(s_i) + M \ge 0.$$

The composite morphism $C_\eta^\nu \to \mathcal{C} \to X$ maps the generic point of C_η^ν to the generic point of X, so the pullbacks $D_i|_{C_\eta^\nu} \ge 0$ are $\mathbb{Q}$-divisors on C_η^ν. Let Φ be their coefficientwise minimum and let $A_i := D_i|_{C_\eta^\nu} - \Phi$. Since $A_i - A_0$ is the divisor of s_i/s_0 on C_η^ν, the $\mathbb{Q}$-divisors $A_0, \dots, A_m \ge 0$ have integer coefficients, are linearly equivalent, and have no common point. They define a morphism

$$\psi = [s_0 : \cdots : s_m] : C_\eta^\nu \to \mathbb{P}^m_{k(T)} \quad \text{with} \quad \psi^*\mathcal{O}(1) \cong \mathcal{O}_{C_\eta^\nu}(A_0).$$

Let $\Gamma := \psi(C_\eta^\nu)$ and $n := \deg_{k(T)} A_0$. Since $|M|$ is birational, $k(X)$ is generated over k by $s_1/s_0, \dots, s_m/s_0$, and since $\mathcal{C} \subset X \times T$, $k(\mathcal{C})$ is generated by $k(X)$ and $k(T)$. Hence $\psi : C_\eta^\nu \to \Gamma$ is birational. Thus Γ is a geometrically integral curve of degree $n \ge 1$ whose normalization is C_η^ν. Since $\Phi \ge 0$, (8.14) implies $n \le \deg(M|_{C_\eta^\nu}) < e+1$, so

$$1 \le n \le e.$$

Since $|M|$ is birational and $\dim X = 3$, $m \ge 3$.

Let $\overline{F}$ be an algebraic closure of $k(T)$ and fix a smooth closed point x_0 of $\Gamma_{\overline{F}}$. For a $3 \times (m+1)$ matrix Ξ of rank 3, let $\pi_\Xi : \mathbb{P}^m \dashrightarrow \mathbb{P}^2$ be the linear projection defined by Ξ. Consider the following conditions on Ξ over $\overline{F}$:

(i) The center of π_Ξ is disjoint from $\Gamma_{\overline{F}}$.
(ii) Only finitely many pairs (x_1, x_2) of distinct points of $\Gamma_{\overline{F}}$ satisfy $\pi_\Xi(x_1) = \pi_\Xi(x_2)$.
(iii) The differential of $\pi_\Xi|_{\Gamma_{\overline{F}}}$ at x_0 is non-zero.

Conditions (i) and (iii) define non-empty open subsets of the space of matrices. For fixed distinct points x_1, x_2 of $\Gamma_{\overline{F}}$, the matrices Ξ with $\pi_\Xi(x_1) = \pi_\Xi(x_2)$ form a closed subset of dimension $3(m+1)-2$. Thus the set Σ of triples (Ξ, x_1, x_2) with $x_1 \neq x_2$ and $\pi_\Xi(x_1) = \pi_\Xi(x_2)$ satisfies $\dim \Sigma \leq 3(m+1)$, so the fiber of Σ over a general Ξ is finite, that is, condition (ii) holds on a non-empty open subset. Since $k(T)$ is infinite, there exists Ξ with entries in $k(T)$ satisfying (i), (ii) and (iii). By (i), π_Ξ restricts to a finite morphism from Γ onto a plane curve $\Gamma' \subset \mathbb{P}^2_{k(T)}$. By (ii), $\Gamma_{\overline{F}} \to \Gamma'_{\overline{F}}$ has separable degree 1. By (iii), $\Gamma_{\overline{F}} \to \Gamma'_{\overline{F}}$ is separable, since an inseparable finite morphism of curves has zero differential at any smooth point. Thus $\Gamma \to \Gamma'$ is birational, Γ' is a geometrically integral plane curve of degree n, and C^ν_η is the normalization of Γ'. By the exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^2}(-n) \to \mathcal{O}_{\mathbb{P}^2} \to \mathcal{O}_{\Gamma'} \to 0,$$

we have $p_a(\Gamma') = \frac{(n-1)(n-2)}{2}$. Since $H^0(\mathcal{O}_{\Gamma'}) = H^0(\mathcal{O}_{C^\nu_\eta}) = k(T)$ and the normalization $C^\nu_\eta \to \Gamma'$ has a cokernel of finite length,

$$p_a(C^\nu_\eta) \leq p_a(\Gamma') = \frac{(n-1)(n-2)}{2} \leq \frac{(e-1)(e-2)}{2}.$$

Step 2. C^ν_η is smooth over $k(T)$. Suppose not. Then $(C^\nu_\eta)_{\overline{F}}$ is an integral curve over $\overline{F}$ which is not normal. Let Y be the normalization of $(C^\nu_\eta)_{\overline{F}}$. The cokernel of $\mathcal{O}_{(C^\nu_\eta)_{\overline{F}}} \to \mathcal{O}_Y$ has positive finite length, so $0 \leq p_a(Y) < p_a((C^\nu_\eta)_{\overline{F}}) = p_a(C^\nu_\eta)$. By Tate's genus change formula [Gu16, Corollary 2.4], $p-1$ divides $2(p_a(C^\nu_\eta) - p_a(Y))$. By (8.13),

$$p - 1 \leq 2\big(p_a(C^\nu_\eta) - p_a(Y)\big) \leq 2p_a(C^\nu_\eta) \leq (e-1)(e-2),$$

which contradicts $p > (e-1)(e-2)+1$.

Step 3. Conclusion. By generic flatness, possibly shrinking T, we may assume that $\mathcal{C}^\nu$ is flat over T. Since ρ is finite, $\mathcal{C}^\nu$ is projective over T. By [Gro66, Théorème 12.2.4(iv)(viii)], the points $t \in T$ such that the fiber of $\mathcal{C}^\nu$ over t is geometrically normal, resp. geometrically integral, form open subsets of T, which contain η by **Steps 1 and 2**. A geometrically normal curve is smooth. Possibly shrinking T, we may therefore assume that $\mathcal{C}^\nu \to T$ is smooth and projective with geometrically integral fibers. Since T is smooth, so is $\mathcal{C}^\nu$.

The finite birational morphism ρ is an isomorphism over a non-empty open subset $\mathcal{C}^\circ \subset \mathcal{C}$. Since $\mathcal{C}$ is flat over T, the image T_0 of $\mathcal{C}^\circ$ in T is open. Let t be a geometric point of T_0. Then $C_t \cap \mathcal{C}^\circ$ is a dense open subset of the integral curve C_t, and $W_t \to C_t$ is a finite morphism which is an isomorphism over $C_t \cap \mathcal{C}^\circ$. Since W_t is smooth and integral, $W_t \to C_t$ is the normalization of C_t. Thus T_0 satisfies (1), (2) and (3). □

Construction-Proposition 8.7. Notation and conditions as in Set-up 8.4. Assume that (X, B) is ϵ-lc for a real number $\epsilon \in (0, 1]$, and qB is a Weil divisor for a positive integer q. Let N be a positive integer such that $|jqD|$ defines a birational map for any integer $j \in [N, 2N]$ and

$$p > 2q(1 + 4qN)\lambda.$$

Then, after replacing T with a non-empty open subset, the following hold.

(1) (a) There exist a smooth threefold V, a generically finite separable morphism $\pi : V \to X$, and a finite purely inseparable morphism $\varphi : W \to V$ such that $g = \pi \circ \varphi$.
(b) If g is birational, then φ is an isomorphism.

(2) Let ω be a non-zero rational differential 3-form on X with $\mathrm{div}_X(\omega) = K_X$, let $K_V := \mathrm{div}_V(\pi^*\omega)$, let B_V be the $\mathbb{Q}$-divisor on V with

$$K_V + B_V = \pi^* D,$$

and let B^+_V be the positive part of B_V. Then for any closed point $t \in T$ and any closed point $x \in F_t$,

$$\mathrm{mult}_x\left((\varphi^* B^+_V)|_{F_t}\right) \leq 1 - \epsilon + \lambda.$$

Proof. We divide the proof into six steps. Since qK_X and qB are Weil divisors, so is jqD for any positive integer j.

Step 1. Construction of π and φ. Let $U \subset X \times T$ be the image of $(g, f) : W \to X \times T$, which is closed since f is proper. We show that $(g, f) : W \to U$ is birational. Since $U \subset X \times T$, this implies that $k(W)$ is generated by $g^*k(X)$ and $f^*k(T)$:

$$k(W) = k(X)k(T). \tag{8.15}$$

Since $g|_{F_t} : F_t \to C_t$ is finite for any closed point $t \in T$, $\dim U = 3$.

First, $k(W)$ is separable over $k(U)$. Let $\alpha : g^*\Omega_X \to \Omega_{W/T}$ be the composite of $g^*\Omega_X \to \Omega_W$ and $\Omega_W \to \Omega_{W/T}$. Since f is smooth of relative dimension 1, $\Omega_{W/T} = \Omega_W/f^*\Omega_T$ is a line bundle, and the cokernel of α is $\Omega_{W/X\times T}$, whose stalk at the generic point of W is $\Omega_{k(W)/k(U)}$. If $k(W)$ were inseparable over $k(U)$, then $\Omega_{k(W)/k(U)} \neq 0$, so α would vanish at the generic point of W, and hence $\alpha = 0$ since $\Omega_{W/T}$ is torsion free. For a closed point $t \in T$, $\alpha|_{F_t}$ is the morphism $(g|_{F_t})^*\Omega_X \to \Omega_{F_t}$ induced by $g|_{F_t}$, which is non-zero because $\Omega_X|_{C_t} \to \Omega_{C_t}$ is surjective and $F_t \to C_t$ is an isomorphism over a dense open subset of C_t. Thus $k(W)$ is separable over $k(U)$.

Second, $[k(W) : k(U)] = 1$. There exists a dense open subset $U^\circ \subset U$ over which (g, f) is finite and étale of degree $[k(W) : k(U)]$. Since U is the union of the curves $C_t \times \{t\}$ and $\dim(U \backslash U^\circ) \leq 2$, there exist a closed point $t \in T$ and a closed point $x \in C_t$ such that $(x, t) \in U^\circ$ and $F_t \to C_t$ is an isomorphism over a neighborhood of x. Then $(g, f)^{-1}(x, t) = (g|_{F_t})^{-1}(x)$ is one point, so $[k(W) : k(U)] = 1$. This proves (8.15).

Since $k(W)$ is finite over $k(X)$, there exists an integer $\ell \geq 0$ such that a^{p^ℓ} is separable over $k(X)$ for any $a \in k(W)$. Let $\overline{T}$ be a projective compactification of T, let $\overline{T}'$ be the Frobenius twist $\overline{T}^{(p^\ell)}$ with the open subset T' corresponding to T, and let $\mathrm{Fr} : \overline{T} \to \overline{T}'$ be the ℓ-fold relative Frobenius morphism. Since k is perfect, Fr is a finite homeomorphism, and for any open subset $T_1' \subset \overline{T}'$ and any regular function a on $\mathrm{Fr}^{-1}(T_1')$, a^{p^ℓ} is the pullback of a regular function on T_1'. Let $V_0 \subset X \times T'$ be the image of $(g, \mathrm{Fr} \circ f) : W \to X \times T'$. Then $k(V_0) = k(X)k(T)^{p^\ell}$ as subfields of $k(W)$. Thus $k(V_0)$ is separable over $k(X)$ by the choice of ℓ, and by (8.15),

$$k(W)^{p^\ell} = k(X)^{p^\ell}k(T)^{p^\ell} \subset k(V_0).$$

Let $\overline{V}_0$ be the closure of V_0 in $X \times \overline{T}'$, and let $\overline{V} \to \overline{V}_0$ be a resolution [Cut09, Theorem 1.1], which exists since $p > 5$. Then $\overline{V}$ is a smooth projective threefold with morphisms $\overline{\pi} : \overline{V} \to X$ and $\overline{V} \to \overline{T}'$, and $\overline{\pi}$ is generically finite and separable. Let $\overline{\varphi} : \overline{W} \to \overline{V}$ be the normalization of $\overline{V}$ in $k(W)$. The morphism $\overline{\varphi}$ is finite. For an affine open subset $\operatorname{Spec} R \subset \overline{V}$, the preimage $\overline{\varphi}^{-1}(\operatorname{Spec} R)$ is the spectrum of $\{b \in k(W) \mid b^{p^\ell} \in R\}$, since R is normal and $k(W)^{p^\ell} \subset k(\overline{V})$. Hence $\overline{\varphi}$ is a homeomorphism. For an affine open subset $T_1' \subset \overline{T}'$ and a regular function a on $\mathrm{Fr}^{-1}(T_1')$, a^{p^ℓ} is a regular function on the preimage of T_1' in $\overline{V}$, so a is a regular function on the preimage of T_1' in the normal variety $\overline{W}$. Thus there exists a morphism $\overline{W} \to \overline{T}$ inducing the inclusion $k(T) \subset k(W)$.

Let W_η and W_η' be the generic fibers of W and of $\overline{W} \times_{\overline{T}} T$ over T. Then W_η and W_η' are normal projective curves over $k(T)$ with function field $k(W)$, so $W_\eta \cong W_\eta'$ over $k(T)$. By [Gro66, Corollaire 8.8.2.5], possibly shrinking T, we may assume that $W \cong \overline{W} \times_{\overline{T}} T$ over T. The morphism $W \to X$ induced by $\overline{W} \to \overline{V} \to X$ coincides with g, since both coincide at the generic point of W and X is separated. Let $V := \overline{\varphi}(W)$, which is open in $\overline{V}$, and let $\varphi := \overline{\varphi}|_W : W \to V$ and $\pi := \overline{\pi}|_V$. Since $\overline{\varphi}^{-1}(V) = W$, φ is finite. Moreover, φ is purely inseparable and a homeomorphism, V is smooth, π is generically finite and separable, and $g = \pi \circ \varphi$. If g is birational, then $k(X) \subset k(V) \subset k(W) = k(X)$, so $\overline{\varphi}$ is a finite birational morphism to the normal variety $\overline{V}$, hence an isomorphism. This proves (1).

Step 2. $\mathrm{mult}_P B_V \leq 1 - \epsilon$ for any prime divisor P on V. Let P be a prime divisor on V, and regard $k(X)$ as a subfield of $k(V)$ via π^*. Since $k(V)$ is finite over $k(X)$, the function

$\zeta \to \mathrm{mult}_P \mathrm{div}_V(\zeta)$ on $k(X)^\times$ is r_0 times a discrete valuation of $k(X)$ whose residue field has transcendence degree 2 over k, for a positive integer r_0. By Zariski's lemma for the valuation ring of the discrete valuation $r_0^{-1}\,\mathrm{mult}_P \mathrm{div}_V$ of $k(X)$, followed by normalization, there exist a projective birational morphism $h : X' \to X$ from a normal variety X' and a prime divisor $P_{X'}$ on X' such that

$$\mathrm{mult}_P \mathrm{div}_V(\zeta) = r_0 \,\mathrm{mult}_{P_{X'}} \mathrm{div}_{X'}(\zeta) \quad \text{for any } \zeta \in k(X)^\times.$$

Let $K_{X'} := \mathrm{div}_{X'}(\omega)$ and $\beta := \mathrm{mult}_{P_{X'}}(h^*D - K_{X'})$. Since (X, B) is ϵ-lc, $\beta \le 1 - \epsilon$.

Since X' and V are smooth at the generic points of $P_{X'}$ and P, we may let τ be a uniformizer of $\mathcal{O}_{X',P_{X'}}$ and $u_1, u_2 \in \mathcal{O}_{X',P_{X'}}$ elements such that $d\tau, du_1, du_2$ form a basis of $\Omega_{\mathcal{O}_{X',P_{X'}}/k}$. Write $\omega = \psi d\tau \wedge du_1 \wedge du_2$ with $\psi \in k(X)$, and let z be a uniformizer of $\mathcal{O}_{V,P}$. Since $\mathrm{mult}_P \mathrm{div}_V(\tau) = r_0$, $\tau = wz^{r_0}$ for a unit w of $\mathcal{O}_{V,P}$, so

$$d\tau = z^{r_0} dw + r_0 w z^{r_0-1} dz \in z^{r_0-1}\Omega_{\mathcal{O}_{V,P}/k}.$$

Since $k(V)$ is separable over $k(X)$, the form $\pi^*(d\tau \wedge du_1 \wedge du_2)$ is non-zero, so it vanishes along P to a finite order ϱ, and $\varrho \ge r_0 - 1$ since $d\tau \in z^{r_0-1}\Omega_{\mathcal{O}_{V,P}/k}$. Since D is $\mathbb{Q}$-Cartier, $\mathrm{mult}_P \pi^*D = r_0 \,\mathrm{mult}_{P_{X'}} h^*D$. Hence

$$\begin{aligned}\mathrm{mult}_P B_V &= \mathrm{mult}_P \pi^*D - \mathrm{mult}_P \mathrm{div}_V(\psi) - \varrho \\ &= r_0\big(\mathrm{mult}_{P_{X'}} h^*D - \mathrm{mult}_{P_{X'}} \mathrm{div}_{X'}(\psi)\big) - \varrho \\ &= r_0\beta - \varrho \le r_0(1-\epsilon) - (r_0 - 1) \le 1 - \epsilon.\end{aligned}$$

Step 3. Fixed parts and the divisors Q_X and Q_V. For any integer $j \in [N, 2N]$, let

$$H_j := H^0(X, \mathcal{O}_X(jqD)),$$

regarded as a space of rational functions s on X with $D_s := \mathrm{div}_X(s) + jqD \ge 0$. Let Fix_j be the coefficientwise minimum of π^*D_s over $s \in H_j\backslash\{0\}$, and let

$$\mathrm{Mov}_j := \pi^*(jqD) - \mathrm{Fix}_j.$$

For any prime divisor P on V and $s \in H_j\backslash\{0\}$, since $\pi^*D_s = \mathrm{div}_V(\pi^*s) + jqK_V + jqB_V$ and K_V has integer coefficients,

$$\mathrm{mult}_P \pi^*D_s \equiv jq \,\mathrm{mult}_P B_V \pmod{\mathbb{Z}}.$$

Thus Mov_j has integer coefficients, the divisors $\mathrm{Mov}_j + \mathrm{div}_V(\pi^*s) = \pi^*D_s - \mathrm{Fix}_j$, $s \in H_j\backslash\{0\}$, satisfy $\pi^*D_s - \mathrm{Fix}_j \ge 0$ and have no common component, and

$$\mathrm{mult}_P \mathrm{Fix}_j \ge \{jq \,\mathrm{mult}_P B_V\}. \tag{8.16}$$

Since $|NqD|$ is birational, $k(X)$ is generated over k by the ratios of elements of H_N. Since k is perfect, there exist $s_0, s_1, s_2, s_3 \in H_N$ such that, for $y_i := s_i/s_0$, the rational 3-form $\omega_1 := dy_1 \wedge dy_2 \wedge dy_3$ on X is non-zero. Since π is separable, $\pi^*\omega_1 \ne 0$. Let

$$Q_X := \mathrm{div}_X(\omega_1) + 4D_{s_0} \quad \text{and} \quad Q_V := \mathrm{div}_V(\pi^*\omega_1) + 4\big(\mathrm{Mov}_N + \mathrm{div}_V(\pi^*s_0)\big).$$

We show that $Q_X \ge 0$ and $Q_V \ge 0$. Let P be a prime divisor on V and let τ be a uniformizer of $\mathcal{O}_{V,P}$. Then $\widetilde{s}_i := \tau^{\mathrm{mult}_P \mathrm{Mov}_N} \pi^* s_i \in \mathcal{O}_{V,P}$ for $i \in \{0,1,2,3\}$, $\pi^* y_i = \widetilde{s}_i/\widetilde{s}_0$, and

$$\mathrm{mult}_P \mathrm{div}_V(\widetilde{s}_0) = \mathrm{mult}_P(\mathrm{Mov}_N + \mathrm{div}_V(\pi^* s_0)).$$

A direct expansion implies

$$\widetilde{s}_0^4 \pi^*\omega_1 = \sum_{i=0}^{3} (-1)^i \widetilde{s}_i \bigwedge_{i' \ne i} d\widetilde{s}_{i'}, \tag{8.17}$$

where the wedge is taken in increasing order of i'. The right-hand side is regular at the generic point of P, so $\mathrm{mult}_P Q_V \ge 0$. Similarly, for a prime divisor S on X and a uniformizer τ_X of $\mathcal{O}_{X,S}$, the elements $s_i' := \tau_X^{\mathrm{mult}_S(NqD)} s_i \in \mathcal{O}_{X,S}$ satisfy $\mathrm{mult}_S \mathrm{div}_X(s_0') = \mathrm{mult}_S D_{s_0}$, and the identity

(8.17) with s_i' and ω_1 in place of $\widetilde{s}_i$ and $\pi^*\omega_1$ shows $\operatorname{mult}_S Q_X \geq 0$, since X is smooth at the generic point of S. Thus $Q_X \geq 0$.

Let $\psi_1 := \omega_1/\omega \in k(X)$. Then

$$\operatorname{div}_X(\omega_1) = K_X + \operatorname{div}_X(\psi_1) \quad \text{and} \quad \operatorname{div}_V(\pi^*\omega_1) = K_V + \operatorname{div}_V(\pi^*\psi_1). \tag{8.18}$$

Since $\pi^*D_{s_0} = \operatorname{Fix}_N + \operatorname{Mov}_N + \operatorname{div}_V(\pi^*s_0)$ and $\pi^*K_X - K_V = B_V - \pi^*B$,

$$\pi^*Q_X = Q_V + B_V - \pi^*B + 4\operatorname{Fix}_N.$$

Since $Q_X \geq 0$ and $B \geq 0$ are $\mathbb{Q}$-Cartier, $\pi^*Q_X \geq 0$ and $\pi^*B \geq 0$, so

$$B_V^- \leq Q_V + 4\operatorname{Fix}_N, \tag{8.19}$$

where $B_V^- := B_V^+ - B_V$ is the negative part of B_V.

Let $\mathcal{S}$ be the finite set of prime divisors on V contained in $\operatorname{Supp} B_V \cup \operatorname{Supp} Q_V \cup \bigcup_{j=N}^{2N} \operatorname{Supp}\operatorname{Fix}_j$. For any integer $j \in [N, 2N]$, the base locus Z_j of the divisors $\operatorname{Mov}_j + \operatorname{div}_V(\pi^*s)$, $s \in H_j\backslash\{0\}$, satisfies $\dim Z_j \leq 1$, and the rational map defined by $|jqD|$ maps a non-empty open subset $X_j^\circ \subset X$ isomorphically onto the image of X_j°. Since f is proper and flat, possibly shrinking T, we may assume the following.

(i) For any $P \in \mathcal{S}$, $\varphi^{-1}(P)$ is finite and flat over T.
(ii) For any integer $j \in [N, 2N]$ and any closed point $t \in T$, $F_t \cap \varphi^{-1}(Z_j) = \emptyset$ and $F_t \cap g^{-1}(X_j^\circ) \neq \emptyset$.

For $P \in \mathcal{S}$, $\varphi^{-1}(P)$ is a prime divisor P' on W since φ is a homeomorphism. Let

$$\varphi^*P = r_P P' \quad \text{and} \quad m_P := P' \cdot F_t.$$

By (i), m_P is the degree of the finite flat morphism $P' \to T$, so $m_P = [k(P') : k(T)]$ is a positive integer. Since $\varphi_*\varphi^*P = (\deg\varphi)P$, r_P divides $\deg\varphi = [k(W) : k(V)]$, which is a power of p. Thus r_P is a power of p. By (i), no component of a divisor supported on $\varphi^{-1}(\bigcup_{P\in\mathcal{S}} P)$ contains a fiber F_t. Since f is flat and projective and T is connected, intersection numbers of $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on W with F_t do not depend on t. In particular, $(g^*D) \cdot F_t = \lambda$ for any closed point $t \in T$.

Step 4. $(\varphi^*B_V^+) \cdot F_t \leq (1 + 4qN)\lambda$ for any closed point $t \in T$. Let $j \in [N, 2N]$ be an integer and let $t \in T$ be a closed point. By **Step 3**(ii), the divisors $(\varphi^*(\operatorname{Mov}_j + \operatorname{div}_V(\pi^*s)))|_{F_t}$, $s \in H_j\backslash\{0\}$, are the divisors of sections of the line bundle $\varphi^*\mathcal{O}_V(\operatorname{Mov}_j)|_{F_t}$ without common zero, and the morphism $F_t \to \mathbb{P}(H_j^\vee)$ which they define is the composite of $g|_{F_t}$ and the map defined by $|jqD|$ on the dense open subset $F_t \cap g^{-1}(X_j^\circ)$. In particular, the morphism $F_t \to \mathbb{P}(H_j^\vee)$ is non-constant, so

$$n_j := (\varphi^* \operatorname{Mov}_j) \cdot F_t$$

is a positive integer. Since $g^*(jqD) = \varphi^* \operatorname{Mov}_j + \varphi^* \operatorname{Fix}_j$,

$$(\varphi^* \operatorname{Fix}_j) \cdot F_t = jq\lambda - n_j. \tag{8.20}$$

Let $\kappa := (\varphi^*K_V) \cdot F_t$. The pullback

$$\varphi^* \operatorname{div}_V(\pi^*\psi_1) = g^* \operatorname{div}_X(\psi_1)$$

is principal, so (8.18) implies $(\varphi^*Q_V) \cdot F_t = \kappa + 4n_N$. By (8.19) and (8.20),

$$(\varphi^*B_V^-) \cdot F_t \leq \kappa + 4n_N + 4(Nq\lambda - n_N) = \kappa + 4qN\lambda.$$

Since

$$(\varphi^*B_V) \cdot F_t = (g^*D) \cdot F_t - (\varphi^*K_V) \cdot F_t = \lambda - \kappa,$$

we obtain

$$(\varphi^*B_V^+) \cdot F_t = (\varphi^*B_V) \cdot F_t + (\varphi^*B_V^-) \cdot F_t \leq (1 + 4qN)\lambda. \tag{8.21}$$

Step 5. Components of large degree on the fibers. For any prime divisor P on V, let $b_P := \operatorname{mult}_P B_V$. Let

$$B_1 := \sum_{P\in\mathcal{S},b_P>0,r_Pm_P<p} b_P P \quad \text{and} \quad B_2 := \sum_{P\in\mathcal{S},b_P>0,r_Pm_P\geq p} b_P P.$$

Then $B_V^+ = B_1 + B_2$. Fix $P \in \mathcal{S}$ with $b_P > 0$ and $r_Pm_P \geq p$, and let $c := qb_P$. Since $(\varphi^*P)\cdot F_t = r_Pm_P \geq p$, (8.21) and $p > 2q(1+4qN)\lambda$ imply

$$0 < c \leq \frac{q(1+4qN)\lambda}{p} < \frac{1}{2}.$$

For any integer $j \in [N, 2N]$, by (8.16), (8.20) and $4qN\lambda < 2q(1+4qN)\lambda < p$,

$$\{jc\} \leq \operatorname{mult}_P \operatorname{Fix}_j \leq \frac{(\varphi^*\operatorname{Fix}_j)\cdot F_t}{r_Pm_P} < \frac{2Nq\lambda}{p} < \frac{1}{2}.$$

For any integer $j \in [N, 2N-1]$, since $\{jc\} < \frac{1}{2}$ and $c < \frac{1}{2}$, we have $\{jc\} + c < 1$, so

$$\{(j+1)c\} = \{jc\} + c.$$

Thus

$$Nc \leq \{Nc\} + Nc = \{2Nc\} < \frac{1}{2},$$

so $\{Nc\} = Nc$, and (8.16) implies $\operatorname{mult}_P \operatorname{Fix}_N \geq Nqb_P$. By (8.20) with $j = N$ and $n_N \geq 1$,

$$Nq(\varphi^*B_2)\cdot F_t = \sum_{P\subset\operatorname{Supp} B_2} Nqb_Pr_Pm_P \leq \sum_{P\subset\operatorname{Supp} B_2} r_Pm_P \operatorname{mult}_P \operatorname{Fix}_N \leq (\varphi^*\operatorname{Fix}_N)\cdot F_t \leq Nq\lambda - 1.$$

Hence $(\varphi^*B_2)\cdot F_t \leq \lambda - \frac{1}{Nq} < \lambda$ for any closed point $t \in T$.

Step 6. Proof of (2). Let $P \subset \operatorname{Supp} B_1$ be a prime divisor. Since r_P is a power of p and $r_Pm_P < p$, $r_P = 1$ and $m_P < p$. The inseparable degree of $k(P')$ over $k(T)$ is a power of p dividing m_P, hence equal to 1, so the finite flat morphism $P' \to T$ is generically étale. Possibly shrinking T further, we may assume that $P' \to T$ is étale for any prime divisor $P \subset \operatorname{Supp} B_1$ and that the divisors P' are pairwise disjoint. Then $\varphi^*B_1 = \sum_{P\subset\operatorname{Supp} B_1} b_P P'$, and $\operatorname{mult}_E \varphi^*B_1 \leq 1-\epsilon$ for any prime divisor E on W by **Step 2**. Moreover, $\operatorname{Supp}\varphi^*B_1$ is a disjoint union of prime divisors which are finite and étale over T.

Let $t \in T$ and $x \in F_t$ be closed points. Since $P' \cap F_t$ is the fiber of the étale morphism $P' \to T$ over t, $P'|_{F_t}$ is reduced for any prime divisor $P \subset \operatorname{Supp} B_1$, and at most one of the divisors P' contains x. Thus $\operatorname{mult}_x((\varphi^*B_1)|_{F_t}) \leq 1-\epsilon$. Since $(\varphi^*B_2)|_{F_t} \geq 0$ has degree $(\varphi^*B_2)\cdot F_t < \lambda$, $\operatorname{mult}_x((\varphi^*B_2)|_{F_t}) < \lambda$. Hence

$$\operatorname{mult}_x\left((\varphi^*B_V^+)|_{F_t}\right) = \operatorname{mult}_x\left((\varphi^*B_1)|_{F_t}\right) + \operatorname{mult}_x\left((\varphi^*B_2)|_{F_t}\right) \leq 1 - \epsilon + \lambda.$$

This proves (2). □

Corollary 8.8. *Notation and conditions as in Set-up 8.4. Let q and N be positive integers such that $|jqD|$ defines a birational map for every integer $j \in [N, 2N]$. Assume that (X, B) is $\frac{1}{q}$-lc, $B \in I_q\backslash\{1\}$, g is birational, $0 < \lambda < \frac{1}{32q}$, $p > 2q(1+4qN)\lambda$, and for a general closed point $t \in T$ there exists a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $G_t \geq 0$ on X with $G_t \cdot C_t < \frac{1}{6q}$ such that $(X, B+G_t)$ is lc near the generic point of C_t and has exactly one lc place there, whose center is C_t. Then $D \cdot C_t \geq \frac{1}{qJ_q}$ for a general closed point $t \in T$, where $J_q := \operatorname{lcm}(1, 2, \ldots, \lfloor 4q/3 \rfloor)$.*

Proof. Since $B \in I_q$, qB is a Weil divisor. Thus the hypotheses of Construction-Proposition 8.7 hold with $\frac{1}{q}$ in place of ϵ. We replace T with the non-empty open subset given there. Since g is birational, φ in Construction-Proposition 8.7(1) is an isomorphism. Regarding φ as the identity

of W, we have $\pi = g$. Let ω be as in Set-up 8.4. Then $K_W = \mathrm{div}_W(g^*\omega)$, and thus $B_V = B_W$. By Construction-Proposition 8.7(2), for any closed point $t \in T$ and any closed point $x \in F_t$,

$$\mathrm{mult}_x(B_W^+|_{F_t}) \leq 1 - \frac{1}{q} + \lambda.$$

Let $t \in T$ be a general closed point. Since g is birational and the curves C_t cover a dense subset of X, g is an isomorphism near the generic points of F_t and C_t, no component of B_W contains F_t, and the generic point of C_t is a smooth point of X. Let $\epsilon := \frac{1}{q} - \lambda \in (0, 1]$. By $\lambda < \frac{1}{32q}$ and $G_t \cdot C_t < \frac{1}{6q}$,

$$\epsilon - G_t \cdot C_t > \frac{1}{q} - \frac{1}{32q} - \frac{1}{6q} = \frac{77}{96q} > \frac{3}{4q}.$$

Since X is smooth at the generic point of C_t, the $\mathbb{Q}$-Cartier Weil divisor $qD = qK_X + qB$ is Cartier near the generic point of C_t. Thus the hypotheses of Theorem 8.5 hold for t, ϵ, $G := G_t$, $\delta := \frac{3}{4q}$, and $L := qD$. Since $\lfloor 1/\delta \rfloor = \lfloor 4q/3 \rfloor$, by Theorem 8.5, the divisor $J_q qD$ is Cartier near C_t. Thus $J_q qD \cdot C_t$ is an integer. Since $g|_{F_t} : F_t \to C_t$ is birational, $D \cdot C_t = (g^*D) \cdot F_t = \lambda > 0$. Hence

$$D \cdot C_t \geq \frac{1}{qJ_q}.$$

□

Lemma 8.9 (cf. [MS14, Corollary 3.2(ii)] and [Sat24, Proposition 4.3 and Theorem 4.4])**.** *Let X be a normal projective variety of dimension $d \geq 2$ over an algebraically closed field of characteristic $p > 0$, and let L be a nef $\mathbb{Q}$-Cartier Weil divisor on X. For a smooth closed point $x \in X$, let*

$$\epsilon(L; x) := \inf \left\{ \frac{L \cdot C}{\mathrm{mult}_x C} \,\middle|\, C \subset X \text{ an integral curve with } x \in C \right\}.$$

Assume that there exist an ample $\mathbb{Q}$-divisor H, a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $E \geq 0$ with $L = H + E$, and a smooth closed point $x \in X \setminus \mathrm{Supp}\, E$ with $\epsilon(L; x) > 2d$. Then $|K_X + L|$ defines a birational map.

Proof of Lemma 8.9. Let $\delta \in (0, 1)$ be a rational number such that $(1 - \delta)\epsilon(L; x) > 2d$, and let

$$A := L - \delta E = (1 - \delta)L + \delta H.$$

For a positive rational number s and a smooth closed point $y \in X$, we define $\epsilon(sA; y)$ by the formula defining $\epsilon(L; x)$. Since L is nef and H is ample, A is ample. For any integral curve $C \subset X$ with $x \in C$, we have $H \cdot C > 0$, so

$$A \cdot C \geq (1 - \delta)L \cdot C \geq (1 - \delta)\epsilon(L; x)\, \mathrm{mult}_x C.$$

Thus $\epsilon(A; x) \geq (1 - \delta)\epsilon(L; x) > 2d$.

Step 1. Let m be a positive integer such that mA is Cartier. Since mA is ample, the augmented base locus of mA [Sat24, Definition 2.12] is empty [Sat24, Remark 2.13], and $\epsilon(mA; y) = m\epsilon(A; y)$ for any smooth closed point $y \in X$. By [Sat24, Lemma 2.18 and Proposition 2.20], the closed points y of the non-singular locus X_{reg} of X with $\epsilon(A; y) > 2d$ are the closed points of an open subset of X_{reg}. Since $x \notin \mathrm{Supp}\, E$, there exists an open subset $U \subset X_{\mathrm{reg}} \setminus \mathrm{Supp}\, E$ containing x such that

$$\epsilon(A; y) > 2d \quad \text{for any closed point } y \in U.$$

Step 2. Let $y \neq z$ be closed points of U. Let $\pi_y : Y_y \to X$ be the blow-up of y, and let F_y be the exceptional prime divisor. Let $\pi_{y,z} : Y_{y,z} \to X$ be the blow-up of $\{y, z\}$, and let F'_y and F'_z be the exceptional prime divisors of $\pi_{y,z}$ over y and z. Let $c_2 := \min\{\epsilon(A; y), \epsilon(A; z)\}$. We show that for any rational number c,

$$\begin{aligned} &\pi_y^* A - cF_y \text{ is ample if } 0 < c < \epsilon(A; y), \quad \text{and} \\ &\pi_{y,z}^* A - c(F'_y + F'_z) \text{ is ample if } 0 < c < \tfrac{c_2}{2}. \end{aligned} \tag{8.22}$$

For an integral curve $C' \subset Y_y$ with $C' \not\subset F_y$ and $C := \pi_y(C')$, we have

$$(\pi_y^* A - c' F_y) \cdot C' = A \cdot C - c' \operatorname{mult}_y C,$$

where $\operatorname{mult}_y C := 0$ if $y \notin C$, and $A \cdot C > 0$. The restriction of $\pi_y^* A - c' F_y$ to $F_y \cong \mathbb{P}^{d-1}$ is c' times the hyperplane class. Thus $\pi_y^* A - c' F_y$ is nef for any rational number $c' \in [0, \epsilon(A; y)]$. For $\pi_{y,z}$, any integral curve $C \subset X$ satisfies

$$A \cdot C \geq c_2 \max\{\operatorname{mult}_y C, \operatorname{mult}_z C\} \geq \frac{c_2}{2}(\operatorname{mult}_y C + \operatorname{mult}_z C),$$

so $\pi_{y,z}^* A - c'(F_y' + F_z')$ is nef for any rational number $c' \in [0, \frac{c_2}{2}]$.

Let (π, F, c_*) be $(\pi_y, F_y, \epsilon(A; y))$ in the first case and $(\pi_{y,z}, F_y' + F_z', \frac{c_2}{2})$ in the second. Since A is ample and $-F$ is π-ample, $\pi^* A - c_0 F$ is ample for any sufficiently small rational number $c_0 > 0$. For a rational number $c \in (0, c_*)$, we choose such a $c_0 < c$ and a rational number $c_1 \in (c, c_*)$. Then $\pi^* A - cF$ is a positive linear combination of the ample divisor $\pi^* A - c_0 F$ and the nef divisor $\pi^* A - c_1 F$, hence ample. Thus (8.22) holds.

Step 3. We fix one of the following two cases.

- Case 1: a closed point $y \in U$, $\pi := \pi_y$, $Y := Y_y$, $F := F_y$, and a rational number c with $d + 1 < c < \epsilon(A; y)$.
- Case 2: closed points $y \neq z$ of U, $\pi := \pi_{y,z}$, $Y := Y_{y,z}$, $F := F_y' + F_z'$, and a rational number c with $d < c < \frac{c_2}{2}$.

Such c exist since $\epsilon(A; y) > 2d$, $\epsilon(A; z) > 2d$, and $d \geq 2$. By (8.22), $\pi^* A - cF$ is ample. For an integer $j \geq 0$, let $\mathcal{I}_j := \pi_* \mathcal{O}_Y(-jF)$. Since X is smooth at y and z, $\mathcal{I}_j = \mathfrak{m}_y^j$ in Case 1 and $\mathcal{I}_j = \mathfrak{m}_y^j \cap \mathfrak{m}_z^j$ in Case 2. We show that for any integer $n \gg 0$, the natural map

$$H^0(X, \mathcal{O}_X(K_X + nL)) \to H^0(X, \mathcal{O}_X(K_X + nL) \otimes \mathcal{O}_X/\mathcal{I}_{\lfloor cn \rfloor}) \tag{8.23}$$

is surjective.

Let b be a positive integer such that bA is Cartier and $bc \in \mathbb{Z}$. Then $b\delta E = bL - bA$ is a divisor, and

$$M := \pi^*(bA) - bcF$$

is an ample Cartier divisor on Y. For $n_2 \in \{0, 1, \ldots, b-1\}$, let

$$\mathcal{G}_{n_2} := \pi^* \mathcal{O}_X(K_X + n_2 L) \otimes \mathcal{O}_Y(-\lfloor cn_2 \rfloor F).$$

By Serre vanishing, there exists a positive integer n_3 such that $H^1(Y, \mathcal{G}_{n_2} \otimes \mathcal{O}_Y(n_1 M)) = 0$ for any $n_2 \in \{0, 1, \ldots, b-1\}$ and any integer $n_1 \geq n_3$. Let $n = bn_1 + n_2$ with $n_1 \geq n_3$ and $0 \leq n_2 \leq b-1$. Then $\lfloor cn \rfloor = bcn_1 + \lfloor cn_2 \rfloor$. Since $\mathcal{O}_X(K_X + n_2 L)$ is invertible near y and z, π is an isomorphism over $X \setminus \{y, z\}$, and bA is Cartier, the projection formula implies

$$\pi_*(\mathcal{G}_{n_2} \otimes \mathcal{O}_Y(n_1 M)) = \mathcal{I}_{\lfloor cn \rfloor} \cdot \mathcal{O}_X(K_X + n_2 L + bn_1 A).$$

By the Leray spectral sequence,

$$H^1(X, \mathcal{I}_{\lfloor cn \rfloor} \cdot \mathcal{O}_X(K_X + n_2 L + bn_1 A))$$

injects into $H^1(Y, \mathcal{G}_{n_2} \otimes \mathcal{O}_Y(n_1 M)) = 0$. Thus

$$H^0(X, \mathcal{O}_X(K_X + n_2 L + bn_1 A)) \to H^0(X, \mathcal{O}_X(K_X + n_2 L + bn_1 A) \otimes \mathcal{O}_X/\mathcal{I}_{\lfloor cn \rfloor})$$

is surjective. Since

$$(K_X + nL) - (K_X + n_2 L + bn_1 A) = bn_1 \delta E \geq 0$$

and $y, z \notin \operatorname{Supp} E$, we have $\mathcal{O}_X(K_X + n_2 L + bn_1 A) \subset \mathcal{O}_X(K_X + nL)$, with equality near y and z. Thus (8.23) is surjective for any $n \geq bn_3$.

Step 4. Let e be a positive integer, and let

$$\operatorname{Tr}^e : F_*^e \omega_{X_{\mathrm{reg}}} \to \omega_{X_{\mathrm{reg}}}$$

be the e-th iterate of the Cartier trace Tr of [MS11, Section 3.3]. Then Tr^e is additive,

$$\mathrm{Tr}^e(g^{p^e}\alpha) = g\,\mathrm{Tr}^e(\alpha) \tag{8.24}$$

for local sections g of $\mathcal{O}_{X_{\mathrm{reg}}}$ and α of $\omega_{X_{\mathrm{reg}}}$, and Tr^e maps regular forms to regular forms. For coordinates $u_1,\dots,u_d$ on an open subset of X_{reg}, let $du := du_1\wedge\cdots\wedge du_d$, $u^\beta := u_1^{\beta_1}\cdots u_d^{\beta_d}$ for $\beta\in\mathbb{N}^d$, and $\mathbf{1} := (1,\dots,1)$. For $\beta\in\{0,1,\dots,p-1\}^d$, [MS11, Section 3.3, (6)] states that $\mathrm{Tr}(u^\beta du) = du$ if $\beta = (p-1)\mathbf{1}$ and $\mathrm{Tr}(u^\beta du) = 0$ otherwise. Since

$$u^{p^e\beta+(p^e-1)\mathbf{1}} = \left(u^{p^{e-1}\beta+(p^{e-1}-1)\mathbf{1}}\right)^p u^{(p-1)\mathbf{1}},$$

applying Tr e times implies

$$\mathrm{Tr}^e(u^{p^e\beta+(p^e-1)\mathbf{1}}du) = u^\beta du \quad \text{for any } \beta\in\mathbb{N}^d. \tag{8.25}$$

Since X_{reg} is smooth, $L|_{X_{\mathrm{reg}}}$ is Cartier, and Tr^e induces an additive map

$$\mathrm{Tr}^e_L : F^e_*\mathcal{O}_{X_{\mathrm{reg}}}(K_X+p^eL) \to \mathcal{O}_{X_{\mathrm{reg}}}(K_X+L), \quad \alpha\otimes\lambda^{p^e} \to \mathrm{Tr}^e(\alpha)\otimes\lambda$$

for local sections α of $\omega_{X_{\mathrm{reg}}}$ and λ of $\mathcal{O}_{X_{\mathrm{reg}}}(L)$. The map Tr^e_L is well defined by (8.24). Since X is normal and $\mathcal{O}_X(K_X+nL)$ is reflexive,

$$H^0(X,\mathcal{O}_X(K_X+nL)) = H^0(X_{\mathrm{reg}},\mathcal{O}_{X_{\mathrm{reg}}}(K_X+nL)) \quad \text{for any integer } n.$$

Thus Tr^e_L induces an additive map

$$\mathrm{Tr}^e_L : H^0(X,\mathcal{O}_X(K_X+p^eL)) \to H^0(X,\mathcal{O}_X(K_X+L))$$

compatible with the maps on stalks at the closed points of X_{reg}.

Let $y\in X_{\mathrm{reg}}$ be a closed point, let $u_1,\dots,u_d$ be a regular system of parameters at y, let λ be a generator of $\mathcal{O}_X(L)_y$, and let j be an integer. We show the following.

(a) If $j\ge dp^e-d+1$, then

$$\mathrm{Tr}^e_L(\mathfrak{m}_y^j\mathcal{O}_X(K_X+p^eL)_y) \subset \mathfrak{m}_y\mathcal{O}_X(K_X+L)_y,$$

and the induced map

$$\mathcal{O}_X(K_X+p^eL)_y/\mathfrak{m}_y^j \to \mathcal{O}_X(K_X+L)_y/\mathfrak{m}_y$$

is surjective.

(b) If $j\ge(d+1)p^e-d+1$, then

$$\mathrm{Tr}^e_L(\mathfrak{m}_y^j\mathcal{O}_X(K_X+p^eL)_y) \subset \mathfrak{m}_y^2\mathcal{O}_X(K_X+L)_y,$$

and the induced map

$$\mathcal{O}_X(K_X+p^eL)_y/\mathfrak{m}_y^j \to \mathcal{O}_X(K_X+L)_y/\mathfrak{m}_y^2$$

is surjective.

Any element of $\mathfrak{m}_y^j\mathcal{O}_X(K_X+p^eL)_y$ is a sum of elements $gu^\alpha du\otimes\lambda^{p^e}$ with $g\in\mathcal{O}_{X,y}$ and $\alpha\in\mathbb{N}^d$, $|\alpha|=j$. We write $\alpha = p^e\beta+\gamma$ with $\beta,\gamma\in\mathbb{N}^d$ and $\gamma\in\{0,1,\dots,p^e-1\}^d$. Then

$$|\beta| \ge \frac{j-d(p^e-1)}{p^e},$$

which is >0 in (a) and >1 in (b), and

$$\mathrm{Tr}^e(gu^\alpha du) = u^\beta\,\mathrm{Tr}^e(gu^\gamma du) \in \mathfrak{m}_y^{|\beta|}\omega_{X,y}.$$

Thus the inclusions hold. For $a\in k$ and $\beta\in\mathbb{N}^d$, the element $a^{p^e}u^{p^e\beta+(p^e-1)\mathbf{1}}du\otimes\lambda^{p^e}$ has image $au^\beta du\otimes\lambda$ by (8.25). Since the elements $au^\beta du\otimes\lambda$ with $|\beta|=0$ span $\mathcal{O}_X(K_X+L)_y/\mathfrak{m}_y$, and those with $|\beta|\le 1$ span $\mathcal{O}_X(K_X+L)_y/\mathfrak{m}_y^2$, the induced maps in (a) and (b) are surjective.

Step 5. In Case 1 of **Step 3**, since $c>d+1$, there exists a positive integer e such that (8.23) is surjective for $n=p^e$ and $\lfloor cp^e\rfloor\ge(d+1)p^e-d+1$. In Case 2 of **Step 3**, since $c>d$, there

exists a positive integer e such that (8.23) is surjective for $n = p^e$ and $\lfloor cp^e \rfloor \geq dp^e - d + 1$. Let $j := \lfloor cp^e \rfloor$. In Case 2,

$$\mathcal{O}_X/\mathcal{I}_j = \mathcal{O}_X/\mathfrak{m}_y^j \oplus \mathcal{O}_X/\mathfrak{m}_z^j.$$

Let $\mathcal{F} := \mathcal{O}_X(K_X + L)$. By **Step 4**(b) in Case 1 and **Step 4**(a) in Case 2, any element of $\mathcal{F}_y/\mathfrak{m}_y^2\mathcal{F}_y$ in Case 1, and any element of

$$\mathcal{F}_y/\mathfrak{m}_y\mathcal{F}_y \oplus \mathcal{F}_z/\mathfrak{m}_z\mathcal{F}_z$$

in Case 2, is the image under Tr_L^e of an element of

$$H^0(X, \mathcal{O}_X(K_X + p^e L) \otimes \mathcal{O}_X/\mathcal{I}_j),$$

which lifts to $H^0(X, \mathcal{O}_X(K_X + p^e L))$ by (8.23) with $n = p^e$. Since Tr_L^e is compatible with the maps on stalks, Tr_L^e of the lift is a section of $\mathcal{F}$ with the given image. Thus for any closed point $y \in U$, the map

$$H^0(X, \mathcal{F}) \to \mathcal{F}_y/\mathfrak{m}_y^2\mathcal{F}_y \text{ is surjective,} \tag{8.26}$$

and for any closed points $y \neq z$ of U, the map

$$H^0(X, \mathcal{F}) \to \mathcal{F}_y/\mathfrak{m}_y\mathcal{F}_y \oplus \mathcal{F}_z/\mathfrak{m}_z\mathcal{F}_z \text{ is surjective.} \tag{8.27}$$

Step 6. By (8.26), $|K_X + L|$ is base point free on U. Let $\phi : X \dashrightarrow \mathbb{P}^N$ be the map defined by $|K_X + L|$, let $\Gamma \subset X \times \mathbb{P}^N$ be the closure of the graph of $\phi|_U$, let $W \subset \mathbb{P}^N$ be the image of Γ, and let $g : \Gamma \to W$ be the projection. The projection $\Gamma \to X$ is birational and an isomorphism over U. Let $U_\Gamma \subset \Gamma$ be the preimage of U. By (8.27), g is injective on the closed points of U_Γ. For a closed point $y \in U$, let $s_0 \in H^0(X, \mathcal{F})$ with $s_0 \notin \mathfrak{m}_y\mathcal{F}_y$. The functions s/s_0 with $s \in H^0(X, \mathcal{F})$ are pullbacks of elements of $\mathcal{O}_{W,\phi(y)}$, and by (8.26) their classes span $\mathcal{O}_{X,y}/\mathfrak{m}_y^2$. Thus the natural map

$$\mathcal{O}_{W,\phi(y)} \to \mathcal{O}_{X,y}/\mathfrak{m}_y^2$$

is surjective. By [Sat24, Lemma 4.5], g is birational. Thus ϕ is birational. The lemma follows. □

Lemma 8.10. *Let (X, B) be a projective klt pair of dimension 3 over an algebraically closed field such that B is a $\mathbb{Q}$-divisor, let D be a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor on X, let $C \subset X$ be an integral curve, and let $G \geq 0$ be a $\mathbb{Q}$-divisor on X. Assume the following.*

(i) $D \sim_{\mathbb{Q}} H_D + E_D$ for an ample $\mathbb{Q}$-divisor H_D and a $\mathbb{Q}$-divisor $E_D \geq 0$.
(ii) $C \not\subset \mathrm{Supp}(B + E_D)$.
(iii) $G \sim_{\mathbb{Q}} \gamma D$ for a rational number γ.
(iv) $(X, B + G)$ is lc near the generic point of C, and C is an lc center of $(X, B + G)$.

Then for any real number $\eta > 0$, there exist a rational number γ' and a $\mathbb{Q}$-divisor $G' \geq 0$ satisfying the following.

(1) $|\gamma' - \gamma| < \eta$ and $G' \sim_{\mathbb{Q}} \gamma' D$.
(2) $G' \cdot C < G \cdot C + \eta$, and $\mathrm{mult}_S G' < \mathrm{mult}_S G + \eta$ for any prime divisor $S \subset X$ containing C.
(3) $(X, B + G')$ is lc near the generic point of C, and there exists exactly one lc place F of $(X, B + G')$ with $C \subset \mathrm{center}_X F$. Moreover, $\mathrm{center}_X F = C$.

Proof of Lemma 8.10. We follow [Kaw97, Proof of Theorem 1.10], written in characteristic 0. The argument works in any characteristic. Let $V := \mathrm{Supp}(B + G + E_D) \cup C$, and let $T \subset X$ be the closed subscheme defined by $\mathcal{I}_{\mathrm{Supp}(B+G+E_D)} \cdot \mathcal{I}_C$, where $\mathcal{I}_{\mathrm{Supp}(B+G+E_D)}$ and $\mathcal{I}_C$ are the ideal sheaves of the reduced subschemes $\mathrm{Supp}(B + G + E_D)$ and C. By [BMP+23, Proposition 2.14], there exists a projective birational morphism $\mu : Y \to X$ from a regular variety Y such that $\mu^{-1}(T)$ and the exceptional locus of μ are divisors, $\mathrm{Supp}\,\mu^{-1}(T) \cup \mathrm{Exc}(\mu)$ is SNC, and $-E$ is μ-ample for some μ-exceptional $\mathbb{Q}$-divisor E. Since the product of $\mathcal{I}_{\mathrm{Supp}(B+G+E_D)}\mathcal{O}_Y$ and $\mathcal{I}_C\mathcal{O}_Y$ is invertible and the local rings of Y are domains, $\mathcal{I}_C\mathcal{O}_Y$ is invertible, so $\mu^{-1}(C)$ is

a divisor. Moreover, $\operatorname{Supp}\mu^{-1}(T) = \mu^{-1}(V)$. By the negativity lemma, $E \geq 0$ and any μ-exceptional prime divisor has positive coefficient in E. Let $K_Y + B_Y := \mu^*(K_X + B)$, so that $a(F, X, B + \Delta) = 1 - \operatorname{mult}_F(B_Y + \mu^*\Delta)$ (Definition 2.1) for any prime divisor F on Y and any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $\Delta \geq 0$ on X. Let $X^\circ \subset X$ be an open subset intersecting C such that $(X, B + G)$ is lc over X° and $C \subset \mu(F')$ for any prime divisor $F' \subset \operatorname{Supp}\mu^{-1}(V) \cup \operatorname{Exc}(\mu)$ with $\mu(F') \cap X^\circ \neq \emptyset$. Such an open subset X° exists since $\operatorname{Supp}\mu^{-1}(V) \cup \operatorname{Exc}(\mu)$ has finitely many irreducible components. Over X°, $B_Y + \mu^*G$ has SNC support and coefficients at most 1.

Let $F_1, \dots, F_N$ be the μ-exceptional prime divisors F on Y with $\mu(F) = C$ and $a(F, X, B + G) = 0$. We show that $N \geq 1$. Since C is an lc center of $(X, B + G)$, there exists a prime divisor $\overline{F}$ over X with $\operatorname{center}_X(\overline{F}) = C$ and $a(\overline{F}, X, B + G) = 0$. Since $B_Y + \mu^*G$ has SNC support and coefficients at most 1 over X°, the center $\operatorname{center}_Y(\overline{F})$ is an irreducible component of $P_1 \cap \cdots \cap P_n$ for some prime divisors $P_1, \dots, P_n$ with coefficient 1 in $B_Y + \mu^*G$, $n \geq 1$. Since $\mu(\operatorname{center}_Y(\overline{F})) = C$, $\operatorname{center}_Y(\overline{F})$ is contained in an irreducible component F' of $\mu^{-1}(C)$, and F' is μ-exceptional. If no P_i were μ-exceptional, then $F' \notin \{P_1, \dots, P_n\}$, and $\operatorname{center}_Y(\overline{F}) \subset F' \cap P_1 \cap \cdots \cap P_n$, and $\dim(F' \cap P_1 \cap \cdots \cap P_n) = \dim Y - n - 1$ since $\operatorname{Supp}\mu^{-1}(V) \cup \operatorname{Exc}(\mu)$ is SNC. This contradicts $\dim \operatorname{center}_Y(\overline{F}) = \dim Y - n$. Thus some P_i is μ-exceptional, and $C \subset \mu(P_i)$ with $\dim \mu(P_i) \leq 1$, so $\mu(P_i) = C$ and $P_i \in \{F_1, \dots, F_N\}$.

For $1 \leq i \leq N$, let $a_i := a(F_i, X, B)$. Since (X, B) is klt, $a_i > 0$, and $\operatorname{mult}_{F_i}\mu^*G = a_i$. Since the μ-ample cone is open and the coefficients of E are positive, possibly changing the coefficients of E slightly, we may assume that

$$\frac{\operatorname{mult}_{F_1} E}{a_1}, \dots, \frac{\operatorname{mult}_{F_N} E}{a_N} \text{ are pairwise distinct.}$$

Since H_D is ample and $-E$ is μ-ample, possibly replacing E with a multiple, there exists a positive integer m such that mH_D is Cartier, E is a divisor, and $\mu^*(mH_D) - E$ is very ample. Let H'_Y be a general member of $|\mu^*(mH_D) - E|$. By Bertini, H'_Y is smooth, $H'_Y + \operatorname{Supp}\mu^{-1}(V) + \operatorname{Exc}(\mu)$ is SNC, and H'_Y contains none of the prime divisors $F_1, \dots, F_N$ and $\mu_*^{-1}S$ for the components S of $B + G$. Let

$$H := \mu_* H'_Y + mE_D.$$

Write $H'_Y = \mu^*(mH_D) - E + \operatorname{div}(f)$ with $f \in k(X)$. Since $\mu_* E = 0$, $\mu_* H'_Y = mH_D + \operatorname{div}(f)$, so $H \geq 0$, $H \sim_{\mathbb{Q}} mD$, and

$$\mu^* H = H'_Y + E + m\mu^* E_D. \tag{8.28}$$

For $1 \leq i \leq N$, $F_i \not\subset H'_Y$, and $F_i \not\subset \operatorname{Supp}\mu^* E_D$ since $\mu(F_i) = C \not\subset \operatorname{Supp} E_D$. By (8.28), $\operatorname{mult}_{F_i}\mu^* H = \operatorname{mult}_{F_i} E > 0$. Let

$$s := \min_{1 \leq i \leq N} \frac{a_i}{\operatorname{mult}_{F_i} E}.$$

Possibly reordering indices, $\frac{a_1}{\operatorname{mult}_{F_1} E} = s < \frac{a_i}{\operatorname{mult}_{F_i} E}$ for any $2 \leq i \leq N$.

For a rational number $\epsilon \in (0, 1)$, let

$$G_\epsilon := (1 - \epsilon)G + \epsilon s H \quad \text{and} \quad \gamma_\epsilon := (1 - \epsilon)\gamma + \epsilon s m.$$

Then $G_\epsilon \geq 0$, $\gamma_\epsilon \in \mathbb{Q}$, $G_\epsilon \sim_{\mathbb{Q}} \gamma_\epsilon D$, and for any prime divisor F on Y,

$$a(F, X, B + G_\epsilon) = (1 - \epsilon)a(F, X, B + G) + \epsilon\big(a(F, X, B) - s \operatorname{mult}_F \mu^* H\big). \tag{8.29}$$

We compute $a(F, X, B + G_\epsilon)$ for the finitely many prime divisors $F \subset \operatorname{Supp}\mu^{-1}(V) \cup \operatorname{Exc}(\mu) \cup H'_Y$ with $\mu(F) \cap X^\circ \neq \emptyset$.

- For $F = F_i$, (8.29) implies

$$a(F_i, X, B + G_\epsilon) = \epsilon \operatorname{mult}_{F_i} E\Big(\frac{a_i}{\operatorname{mult}_{F_i} E} - s\Big),$$

  which is 0 for $i = 1$ and positive for $i \geq 2$.
- If $a(F, X, B + G) > 0$, then $a(F, X, B + G_\epsilon) > 0$ for $0 < \epsilon \ll 1$ by (8.29).

- Assume that $a(F,X,B+G)=0$ and $F\notin\{F_1,\dots,F_N\}$. If F were μ-exceptional, then $C\subset\mu(F)$ and $\dim\mu(F)\le 1$, so $\mu(F)=C$ and $F\in\{F_1,\dots,F_N\}$. Thus $F=\mu_*^{-1}S$ for a prime divisor S on X with $\operatorname{mult}_S(B+G)=1$. Then $F\subset\mu^{-1}(V)$, so $C\subset S$. Since $C\not\subset\operatorname{Supp}(B+E_D)$, $S\not\subset\operatorname{Supp}(B+E_D)$. Thus $a(F,X,B)=1$ and $\operatorname{mult}_S G=1$. Since $H'_Y\not\supset\mu_*^{-1}S$, $\operatorname{mult}_S H=\operatorname{mult}_S\mu_*H'_Y+m\operatorname{mult}_S E_D=0$. By (8.29), $a(F,X,B+G_\epsilon)=\epsilon>0$, that is, S has coefficient $1-\epsilon<1$ in $B+G_\epsilon$.

Fix a rational number $\epsilon\in(0,1)$ satisfying the following.

- $\epsilon|sm-\gamma|<\eta$ and $\epsilon(sH\cdot C-G\cdot C)<\eta$.
- $\epsilon s\operatorname{mult}_S H<\eta$ for the finitely many prime divisors $S\subset X$ with $\operatorname{mult}_S H>0$.
- $a(F,X,B+G_\epsilon)>0$ for any prime divisor $F\subset\operatorname{Supp}\mu^{-1}(V)\cup\operatorname{Exc}(\mu)\cup H'_Y$ with $\mu(F)\cap X^\circ\ne\emptyset$ and $a(F,X,B+G)>0$.

Let $G':=G_\epsilon$ and $\gamma':=\gamma_\epsilon$. Over X°, the support of $B_Y+\mu^*G'$ is contained in

$$\operatorname{Supp}\mu^{-1}(V)\cup\operatorname{Exc}(\mu)\cup H'_Y,$$

which is SNC, and $\operatorname{mult}_F(B_Y+\mu^*G')\le 1$ for any prime divisor F on Y with $\mu(F)\cap X^\circ\ne\emptyset$, with equality if and only if $F=F_1$. Thus $(X,B+G')$ is lc over X°, and F_1 is the only lc place of $(X,B+G')$ whose center intersects X°. Since $X^\circ\cap C\ne\emptyset$ and $\mu(F_1)=C$, (3) holds.

Since $\gamma'-\gamma=\epsilon(sm-\gamma)$, $G'\cdot C=G\cdot C+\epsilon(sH\cdot C-G\cdot C)$, and $\operatorname{mult}_S G'\le\operatorname{mult}_S G+\epsilon s\operatorname{mult}_S H$ for any prime divisor $S\subset X$, (1) and (2) hold. The lemma follows. □

Lemma 8.11. *Let k be an algebraically closed field of characteristic $p>0$, let X be a normal threefold over k, let $B\ge 0$ be a $\mathbb{Q}$-divisor on X, and let R be a positive integer such that $L:=R(K_X+B)$ is a Weil divisor. For any integer $n\ge 0$, identify $H^0(X,\mathcal{O}_X(nL))$ with the set of elements fy^n of the rational function field $k(X)(y)$ such that $f=0$ or $\operatorname{div}(f)+nL\ge 0$. Let b be a positive integer with $p\nmid b$, and let $z_0,\dots,z_3\in H^0(X,\mathcal{O}_X(bL))$ with $z_0\ne 0$ such that $d(z_1/z_0)$, $d(z_2/z_0)$, and $d(z_3/z_0)$ are linearly independent over $k(X)$. Such $z_0,\dots,z_3$ exist if $|bL|$ defines a birational map. Then $z_0,\dots,z_3$ is a separating transcendence basis of $k(X)(y)$ over k. Let $\partial^{[I]}$ ($I\in\mathbb{N}^4$) be the Hasse–Schmidt derivatives of $k(X)(y)$ with respect to $z_0,\dots,z_3$, that is, $g\to\sum_{I\in\mathbb{N}^4}\partial^{[I]}(g)T^I$ is the unique k-algebra homomorphism $k(X)(y)\to k(X)(y)[[T_0,\dots,T_3]]$ that sends z_j to z_j+T_j for $0\le j\le 3$ and is the identity modulo $(T_0,\dots,T_3)$. Then there exists $\mathcal{J}\in H^0(X,\mathcal{O}_X((4bR+1)L))\setminus\{0\}$ such that*

$$\mathcal{J}^e z_0^{|I|}\partial^{[I]}(s)\in H^0(X,\mathcal{O}_X((n+(4bR+1)e)L))$$

for any integers $n,e\ge 0$, any $s\in H^0(X,\mathcal{O}_X(nL))$, and any $I\in\mathbb{N}^4$ with $Re\ge 2|I|-1$, where $|I|:=I_0+I_1+I_2+I_3$.

Proof. For a prime divisor Γ on X, let v_Γ be the Gauss valuation of $k(X)(y)$ with

$$v_\Gamma\Big(\sum_j f_jy^j\Big)=\min_j\big(\operatorname{mult}_\Gamma\operatorname{div}(f_j)+j\operatorname{mult}_\Gamma L\big)\quad\text{for } f_j\in k(X).$$

For $f\in k(X)\setminus\{0\}$ and $n\in\mathbb{Z}$, $v_\Gamma(fy^n)=\operatorname{mult}_\Gamma(\operatorname{div}(f)+nL)$. Thus for $n\ge 0$,

$$fy^n\in H^0(X,\mathcal{O}_X(nL))$$

if and only if

$$v_\Gamma(fy^n)\ge 0\quad\text{for any prime divisor }\Gamma\text{ on }X. \tag{8.30}$$

Step 1. If $|bL|$ defines a birational map, then for a non-zero $z_0\in H^0(X,\mathcal{O}_X(bL))$ the quotients z/z_0 with $z\in H^0(X,\mathcal{O}_X(bL))$ generate $k(X)$ over k, and since k is perfect and

$$\dim_{k(X)}\Omega_{k(X)/k}=3,$$

their differentials span $\Omega_{k(X)/k}$, so such $z_0,\dots,z_3$ exist. We show that $z_0,\dots,z_3$ is a separating transcendence basis. Since k is perfect,

$$\Omega_{k(X)(y)/k}=\big(\Omega_{k(X)/k}\otimes_{k(X)}k(X)(y)\big)\oplus k(X)(y)dy,$$

and $\dim_{k(X)(y)} \Omega_{k(X)(y)/k} = 4$. Since $z_j/z_0 \in k(X)$ and $dz_j = z_0 d(z_j/z_0) + (z_j/z_0)dz_0$, we have

$$\sum_{j=0}^{3} k(X)(y)dz_j = k(X)(y)dz_0 + \sum_{j=1}^{3} k(X)(y)d(z_j/z_0).$$

Write $z_0 = f_0 y^b$ with $f_0 \in k(X)\backslash\{0\}$. Then $dz_0 = y^b df_0 + b f_0 y^{b-1} dy$, and $b f_0 y^{b-1} \neq 0$ since $p \nmid b$. Since $d(z_1/z_0)$, $d(z_2/z_0)$, $d(z_3/z_0)$ belong to $\Omega_{k(X)/k} \otimes_{k(X)} k(X)(y)$ and are linearly independent, $dz_0, \dots, dz_3$ is a basis of $\Omega_{k(X)(y)/k}$. Thus $z_0, \dots, z_3$ is a separating transcendence basis of $k(X)(y)$ over k.

Step 2. We show that the homomorphism defining $\partial^{[I]}$ exists and is unique, and compute the weights. Since $k(X)(y)$ is separable algebraic over $k(z_0, \dots, z_3)$, it is 0-étale over $k(z_0, \dots, z_3)$ [Mat89, Theorem 25.3]. The k-algebra homomorphism

$$k(z_0, \dots, z_3) \to k(X)(y)[[T_0, \dots, T_3]], \quad g(z_0, \dots, z_3) \to g(z_0 + T_0, \dots, z_3 + T_3)$$

is well defined since $g(z + T)$ has constant term $g(z)$. Lifting along the surjections $k(X)(y)[[T]]/(T)^{i+1} \to k(X)(y)[[T]]/(T)^i$, whose kernels have square zero, by the definition of 0-étale [Mat89, Section 25] (for one variable, see [Mat89, Theorem 27.2]), there exists a unique k-algebra homomorphism

$$k(X)(y) \to k(X)(y)[[T_0, \dots, T_3]], \quad g \to \sum_{I \in \mathbb{N}^4} \partial^{[I]}(g) T^I,$$

that extends the homomorphism above and is the identity modulo $(T_0, \dots, T_3)$.

We say that $g \in k(X)(y)$ is *homogeneous of weight* $n' \in \mathbb{Z}$ if $g = f y^{n'}$ for some $f \in k(X)$. For $\lambda \in k^*$, let ρ_λ be the $k(X)$-automorphism of $k(X)(y)$ with $\rho_\lambda(y) = \lambda y$. Since k is infinite, $g \neq 0$ is homogeneous of weight n' if and only if $\rho_\lambda(g) = \lambda^{n'} g$ for any $\lambda \in k^*$: write $g = P_1/P_2$ with coprime $P_1, P_2 \in k(X)[y]$ and P_2 monic. Then $\rho_\lambda(g) = \lambda^{n'} g$ for infinitely many λ forces $P_2 = y^{\deg P_2}$ and $P_1 \in k(X) y^{n' + \deg P_2}$. Since $\rho_\lambda(z_j) = \lambda^b z_j$, the map

$$g \to \sum_I \lambda^{b|I|} \rho_\lambda\big(\partial^{[I]}(\rho_\lambda^{-1}(g))\big) T^I$$

is a k-algebra homomorphism that sends z_j to $z_j + T_j$ and is the identity modulo $(T_0, \dots, T_3)$. By uniqueness, it is $g \to \sum_I \partial^{[I]}(g) T^I$. Thus for $s \in H^0(X, \mathcal{O}_X(nL))$,

$$\rho_\lambda(\partial^{[I]}(s)) = \lambda^{n - b|I|} \partial^{[I]}(s) \quad \text{for any } \lambda \in k^*,$$

so

$$\partial^{[I]}(s) \text{ is homogeneous of weight } n - b|I|. \tag{8.31}$$

Step 3. We construct $\mathcal{J}$. Let ω be a non-zero element of $\Omega^3_{k(X)/k}$ with $\operatorname{div}(\omega) = K_X$. Let

$$\theta := \Big(\omega \wedge \frac{dy}{y}\Big)^{\otimes R} \Big/ y \quad \text{and} \quad \mathcal{J} := \frac{(dz_0 \wedge dz_1 \wedge dz_2 \wedge dz_3)^{\otimes R}}{\theta} \in k(X)(y)\backslash\{0\},$$

where θ and $(dz_0 \wedge \cdots \wedge dz_3)^{\otimes R}$ are non-zero elements of the 1-dimensional $k(X)(y)$-vector space $(\Omega^4_{k(X)(y)/k})^{\otimes R}$. Since $\rho_\lambda(dz_j) = \lambda^b dz_j$, $\rho_\lambda(\omega) = \omega$, $\rho_\lambda(dy/y) = dy/y$, and $\rho_\lambda(y) = \lambda y$, we have $\rho_\lambda(\mathcal{J}) = \lambda^{4bR+1} \mathcal{J}$. Thus $\mathcal{J}$ is homogeneous of weight $4bR + 1$.

Step 4. Fix a prime divisor Γ on X. We introduce coordinates at v_Γ. Since X is normal, X is smooth at the generic point of Γ. Let $t \in \mathcal{O}_{X,\Gamma}$ be a uniformizer, and let $x_1, x_2, x_3 \in \mathcal{O}_{X,\Gamma}$ such that dx_1, dx_2, dx_3 is a basis of $\Omega_{\mathcal{O}_{X,\Gamma}/k}$. There exists an affine open subset $X^\circ \subset X$ intersecting Γ such that $t, x_1, x_2, x_3 \in \mathcal{O}_X(X^\circ)$ and the morphism $X^\circ \to \mathbb{A}^3$ defined by x_1, x_2, x_3 is étale. Let

$$u := t^{-\operatorname{mult}_\Gamma L} y \quad \text{and} \quad x_0 := u.$$

Then

$$v_\Gamma\Big(\sum_j f_j u^j\Big) = \min_j \operatorname{mult}_\Gamma \operatorname{div}(f_j) \quad \text{for } f_j \in k(X),$$

and $v_\Gamma(u) = 0$, so the valuation ring $\mathcal{O}_{v_\Gamma}$ of v_Γ is the localization of $\mathcal{O}_X(X^\circ)[u]$ at the prime ideal generated by the ideal of $\Gamma \cap X^\circ$. Thus $\mathcal{O}_{v_\Gamma}$ is a localization of an étale $k[x_0,\dots,x_3]$-algebra, so it is 0-étale over $k[x_0,\dots,x_3]$ by [Gro67, Théorème 17.6.1] and [Mat89, Section 25], $\Omega_{\mathcal{O}_{v_\Gamma}/k}$ is free with basis $dx_0,\dots,dx_3$, and the lifting of **Step 2**, with $k[x_0,\dots,x_3]$ and $x_i \to x_i + S_i$ in place of $k(z_0,\dots,z_3)$ and $z_j \to z_j + T_j$, implies a k-algebra homomorphism

$$\mathcal{O}_{v_\Gamma} \to \mathcal{O}_{v_\Gamma}[[S_0,\dots,S_3]], \quad g \to \sum_{J\in\mathbb{N}^4} \partial_x^{[J]}(g) S^J,$$

that sends x_i to $x_i + S_i$ and is the identity modulo $(S_0,\dots,S_3)$. It extends to $k(X)(y)$ since non-zero elements of $\mathcal{O}_{v_\Gamma}$ are sent to invertible power series. In particular,

$$\partial_x^{[J]}(\mathcal{O}_{v_\Gamma}) \subset \mathcal{O}_{v_\Gamma} \quad \text{for any } J \in \mathbb{N}^4.$$

By (8.30), $z_j \in \mathcal{O}_{v_\Gamma}$. Let M be the 4×4 matrix with entries

$$M_{ji} := \partial_x^{[\mathbf{e}_i]}(z_j) \in \mathcal{O}_{v_\Gamma},$$

where $\mathbf{e}_0,\dots,\mathbf{e}_3$ is the standard basis of $\mathbb{N}^4$. Since $g \to \partial_x^{[\mathbf{e}_i]}(g)$ is the derivation dual to dx_i, we have $dz_j = \sum_i M_{ji} dx_i$, so

$$dz_0 \wedge \cdots \wedge dz_3 = \det(M) dx_0 \wedge \cdots \wedge dx_3, \quad \text{and} \quad \ell_\Gamma := v_\Gamma(\det M) \ge 0$$

is finite by **Step 1**. Write $\omega = h dx_1 \wedge dx_2 \wedge dx_3$ with $h \in k(X)$. Since $X^\circ \to \mathbb{A}^3$ is étale, $\operatorname{mult}_\Gamma \operatorname{div}(h) = \operatorname{mult}_\Gamma K_X$. Since $dy/y = du/u + \operatorname{mult}_\Gamma L \cdot dt/t$ and $dt \in \sum_i \mathcal{O}_{X,\Gamma} dx_i$, we have

$$\omega \wedge \frac{dy}{y} = h u^{-1} dx_1 \wedge dx_2 \wedge dx_3 \wedge du, \quad \text{so} \quad \mathcal{J} = \pm \frac{\det(M)^R u^R y}{h^R}.$$

Since $v_\Gamma(u) = 0$, $B \ge 0$, and

$$v_\Gamma(y) = \operatorname{mult}_\Gamma L = R \operatorname{mult}_\Gamma K_X + R \operatorname{mult}_\Gamma B,$$

we have

$$v_\Gamma(\mathcal{J}) = R\ell_\Gamma + R \operatorname{mult}_\Gamma B \ge R\ell_\Gamma \ge 0. \tag{8.32}$$

Since Γ is arbitrary, (8.30), (8.32), and **Step 3** imply $\mathcal{J} \in H^0(X, \mathcal{O}_X((4bR+1)L))\backslash\{0\}$.

Step 5. We show that for any $g \in \mathcal{O}_{v_\Gamma}$ and any $I \in \mathbb{N}^4$ with $|I| \ge 1$,

$$v_\Gamma(\partial^{[I]}(g)) \ge -(2|I|-1)\ell_\Gamma. \tag{8.33}$$

For $0 \le j \le 3$, write

$$\sum_J \partial_x^{[J]}(z_j) S^J = z_j + \sum_i M_{ji} S_i + Q_j(S_0,\dots,S_3),$$

where $Q_j \in \mathcal{O}_{v_\Gamma}[[S_0,\dots,S_3]]$ has no terms of degree less than 2. Since $\det M \ne 0$, there exists a unique $\sigma = (\sigma_0,\dots,\sigma_3)$, where each σ_i is a power series in $T_0,\dots,T_3$ over $k(X)(y)$ without constant term, such that

$$\sum_i M_{ji}\sigma_i + Q_j(\sigma) = T_j \quad \text{for } 0 \le j \le 3,$$

determined degree by degree by $\sigma = M^{-1}(T - Q(\sigma))$. The map $g \to \sum_J \partial_x^{[J]}(g)\sigma^J$ is a k-algebra homomorphism $k(X)(y) \to k(X)(y)[[T_0,\dots,T_3]]$ that sends z_j to $z_j + T_j$ and is the identity modulo $(T_0,\dots,T_3)$. By uniqueness in **Step 2**,

$$\partial^{[I]}(g) \text{ is the coefficient of } T^I \text{ in } \sum_J \partial_x^{[J]}(g)\sigma^J. \tag{8.34}$$

Let $\sigma_{(a)}$ be the homogeneous part of degree a of σ in $T_0,\dots,T_3$. Since $M^{-1}=\det(M)^{-1}\operatorname{adj}(M)$ and M has entries in $\mathcal{O}_{v_\Gamma}$, the entries of M^{-1} have $v_\Gamma\ge-\ell_\Gamma$. We show by induction on $a\ge1$ that the coefficients of $\sigma_{(a)}$ have $v_\Gamma\ge-(2a-1)\ell_\Gamma$. For $a=1$, $\sigma_{(1)}=M^{-1}T$. For $a\ge2$, $\sigma_{(a)}=-M^{-1}Q(\sigma)_{(a)}$, and each coefficient of $Q(\sigma)_{(a)}$ is a sum of products of an element of $\mathcal{O}_{v_\Gamma}$ with coefficients of $\sigma_{(a_1)},\dots,\sigma_{(a_m)}$, where $m\ge2$ and $a_1+\cdots+a_m=a$. By induction, such a product has

$$v_\Gamma\ge-\big((2a_1-1)+\cdots+(2a_m-1)\big)\ell_\Gamma=-(2a-m)\ell_\Gamma\ge-(2a-2)\ell_\Gamma.$$

Multiplying by M^{-1} completes the induction. For $g\in\mathcal{O}_{v_\Gamma}$ and $|I|\ge1$, by (8.34), $\partial^{[I]}(g)$ is a sum of products of $\partial_x^{[J]}(g)\in\mathcal{O}_{v_\Gamma}$, $J\ne0$, with $|J|$ coefficients of $\sigma_{(a_1)},\dots,\sigma_{(a_{|J|})}$, where $a_1+\cdots+a_{|J|}=|I|$. Such a product has $v_\Gamma\ge-(2|I|-|J|)\ell_\Gamma\ge-(2|I|-1)\ell_\Gamma$. Thus (8.33) holds.

Step 6. Let $n,e\ge0$ be integers, $s\in H^0(X,\mathcal{O}_X(nL))$, and $I\in\mathbb{N}^4$ with $Re\ge2|I|-1$. By (8.31) and **Step 3**, $\mathcal{J}^e z_0^{|I|}\partial^{[I]}(s)$ is homogeneous of weight $n+(4bR+1)e$. For any prime divisor Γ on X, $v_\Gamma(s)\ge0$ and $v_\Gamma(z_0)\ge0$ by (8.30). If $I=0$, then $v_\Gamma(\mathcal{J}^e s)\ge0$ by (8.32). If $|I|\ge1$, then by (8.32) and (8.33),

$$v_\Gamma\big(\mathcal{J}^e z_0^{|I|}\partial^{[I]}(s)\big)\ge eR\ell_\Gamma-(2|I|-1)\ell_\Gamma\ge0.$$

By (8.30),

$$\mathcal{J}^e z_0^{|I|}\partial^{[I]}(s)\in H^0(X,\mathcal{O}_X((n+(4bR+1)e)L)).$$

The lemma follows. □

Set-up 8.12. Notation and conditions as in Set-up 8.1. Let

$$w:=\lfloor\sqrt{p}\rfloor,\quad r':=2r\left(1+\left\lceil\frac{\gamma_q b_0}{rw}\right\rceil\right),\quad R:=r'q,\quad\text{and}\quad L:=RD.$$

Since $1\le\lceil\gamma_q b_0/(rw)\rceil<1+\gamma_q b_0/(rw)$, we have

$$4r\le r'\le4r+\frac{2\gamma_q b_0}{w}.$$

For a field extension k' of k, let $X_{k'}:=X\times_k\operatorname{Spec}k'$, and let $L_{k'}$ be the pullback of L to $X_{k'}$. For a point $x\in X_{k'}(k')$ with ideal sheaf $\mathfrak{m}_x$, a positive integer n, and a real number $\tau>0$, let

$$\mathcal{W}_n(\tau;x):=H^0(X_{k'},\mathfrak{m}_x^{\lceil n\tau\rceil}\cdot\mathcal{O}_{X_{k'}}(nL_{k'})),$$

where

$$\mathfrak{m}_x^{\lceil n\tau\rceil}\cdot\mathcal{O}_{X_{k'}}(nL_{k'})\subset\mathcal{O}_{X_{k'}}(nL_{k'})$$

is the image of the multiplication map. Let $\mathbb{K}:=k(X)$. We regard $\mathbb{K}$ as the function field of the second factor of $X\times X$, so that $X_{\mathbb{K}}$ is the generic fiber of the second projection $X\times X\to X$. Let $\xi\in X_{\mathbb{K}}(\mathbb{K})$ be the point defined by the generic point of X, with ideal sheaf $\mathfrak{m}_\xi$. For any positive integer n and any real number $\tau>0$, let

$$\mathcal{W}_n(\tau):=\mathcal{W}_n(\tau;\xi)\quad\text{and}\quad\mathbf{B}_\tau:=\bigcap_{n\ge1}\operatorname{Bs}(\mathcal{W}_n(\tau))_{\mathrm{red}},$$

where $\operatorname{Bs}(0):=X_{\mathbb{K}}$.

Construction-Proposition 8.13. Notation and conditions as in Set-up 8.12. Assume that $p>P_q^*$ and $b_0>16\sigma_q^* r$. Then the following hold.

(1) There exists a positive integer b' with $p\nmid b'$ such that $|b'L|$ defines a birational map and

$$b'\le\frac{u_0b_0}{r'}+u_1<\frac{w}{1024q}.$$

(2) $\dim_\xi\mathbf{B}_{2q}\le1$.

Proof. We divide the proof into three steps. Let pr_1 and pr_2 be the first and second projections $X \times X \to X$. Since the image of ξ in X is the generic point of X and k is perfect, $\xi \notin \operatorname{Supp} L_{\mathbb{K}}$, $X_{\mathbb{K}}$ is smooth over $\mathbb{K}$ at ξ, and $\mathcal{O}_{X_{\mathbb{K}},\xi}$ is a regular local ring of dimension 3 with residue field $\mathbb{K}$.

Step 1. In this step we prove (1) and the inequality

$$\sigma_q^* r' < \frac{b_0}{2}. \tag{8.35}$$

Since $B \in I_q$ and $q \mid R$, $L = RD$ is a Weil divisor and $\lfloor jRD \rfloor = jL$ for any $j \in \mathbb{N}^+$. Let

$$b' := \min\left\{ i \in \mathbb{Z} \;\middle|\; p \nmid i,\ i \geq \frac{4n_0}{r'} + \frac{2}{R} \right\}.$$

Then $q \mid b'R$ and $b'R \geq 4qn_0 + 2$. Since (X, B) is lc, D is $\mathbb{Q}$-Cartier and big, $p > P_q^* \geq 64$, and $H^0(X, \mathcal{O}_X(\lfloor b_0 D \rfloor)) \neq 0$, Theorem 6.6 implies that $|b'L| = |\lfloor b'RD \rfloor|$ defines a birational map. Since $r' \geq 4r \geq 8$, we have $R \geq 8q$. Since one of any two consecutive integers is not divisible by p and $4n_0 = u_0 b_0 + 8c_q^2 + 4c_q$,

$$b' < \frac{4n_0}{r'} + \frac{2}{R} + 2 \leq \frac{u_0 b_0}{r'} + c_q^2 + \frac{c_q}{2} + \frac{1}{8} + 2 < \frac{u_0 b_0}{r'} + u_1.$$

Since $p > P_q^* \geq (U_q + 1)^2$, we have $w > U_q$. Since $r' > 2r\lceil \gamma_q b_0/(rw) \rceil \geq 2\gamma_q b_0/w$, $\gamma_q = 1024 q u_0$, and $U_q \geq 2048 q u_1$,

$$\frac{u_0 b_0}{r'} + u_1 < \frac{u_0 w}{2\gamma_q} + \frac{U_q}{2048q} < \frac{w}{2048q} + \frac{w}{2048q} = \frac{w}{1024q}.$$

Thus (1) holds. Moreover, $r' \leq 4r + 2\gamma_q b_0/w$ (Set-up 8.12) and $w > U_q \geq 8\gamma_q \sigma_q^*$, so

$$\sigma_q^* r' < 4\sigma_q^* r + \frac{b_0}{4} < \frac{b_0}{2},$$

where the last inequality follows from $b_0 > 16\sigma_q^* r$. This is (8.35).

Step 2. In this step we show that a section of high order at ξ has no prime divisor of large multiplicity through ξ.

We use the identification of Lemma 8.11. Let $\mathfrak{A} := k(X) \otimes_k \mathbb{K}$. By flat base change, for any $m \in \mathbb{N}$,

$$H^0(X_{\mathbb{K}}, \mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}})) = H^0(X, \mathcal{O}_X(mL)) \otimes_k \mathbb{K},$$

so any global section s of $\mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}})$ is written as $s = \varphi_s y^m$ with $\varphi_s \in \mathfrak{A}$. For $s \neq 0$, let

$$E_s := \operatorname{div}(\varphi_s) + mL_{\mathbb{K}} \geq 0.$$

Since $\xi \notin \operatorname{Supp} L_{\mathbb{K}}$, $\operatorname{mult}_\xi E_s$ is the order of φ_s in $\mathcal{O}_{X_{\mathbb{K}},\xi}$.

Claim 8.14. *Let m be a positive integer, let*

$$s \in H^0(X_{\mathbb{K}}, \mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}}))\backslash\{0\},$$

and let $\mu := \operatorname{mult}_\xi E_s$. Assume that $\mu > 6qm$. Then $\operatorname{mult}_V E_s \leq \mu/2$ for any prime divisor V on $X_{\mathbb{K}}$ with $\xi \in V$.

Proof. Suppose that $a_V := \operatorname{mult}_V E_s > \mu/2$ for a prime divisor V on $X_{\mathbb{K}}$ with $\xi \in V$. Since $\mathcal{O}_{X_{\mathbb{K}},\xi}$ is factorial, $\mu \geq a_V \operatorname{mult}_\xi V$, so $\operatorname{mult}_\xi V = 1$ and V is regular at ξ. Since ξ is a $\mathbb{K}$-rational point, V is smooth over $\mathbb{K}$ at ξ [Gro67, Proposition 17.15.1], so the integral $\mathbb{K}$-scheme V is geometrically integral over $\mathbb{K}$ [Sta, Tag 0CDW].

Let $\mathcal{V}$ be the closure of V in $X \times X$, so that V is the generic fiber of $\mathrm{pr}_2|_{\mathcal{V}}$. Since $\xi \in V$ and the image of ξ in $X \times X$ is the generic point of the diagonal, the diagonal is contained in $\mathcal{V}$. Write $s = \sum_i s_i \otimes \kappa_i$ with $s_i \in H^0(X, \mathcal{O}_X(mL))$ and $\kappa_i \in \mathbb{K}$ by flat base change. By [Sta, Tags 0578 and 0559] and generic flatness, there exists a non-empty open subset U of the non-singular locus of X satisfying the following.

(1) $\mathcal{V}_U := \mathcal{V} \cap (X \times U) \to U$ is flat with geometrically integral fibers. For any closed point $x \in U$, we let $S_x \subset X$ be the fiber of $\mathcal{V}_U$ over x, a prime divisor on X with $x \in S_x$.
(2) s is the restriction to $X_{\mathbb{K}}$ of a section

$$s_U \in H^0(X \times U, \mathcal{O}_{X\times U}(m \operatorname{pr}_1^* L))$$

whose restriction $s_x \in H^0(X, \mathcal{O}_X(mL))$ to $X \times \{x\} = X$ is non-zero for any closed point $x \in U$.

Let $x \in U$ be a closed point, and let $E_x := \operatorname{div}(s_x) + mL$. Since $X \times U$ is smooth at the generic point of $S_x \times \{x\}$ and

$$\operatorname{mult}_{\mathcal{V}_U}(\operatorname{div}(s_U) + m \operatorname{pr}_1^* L) = a_V,$$

the Weil divisor

$$\operatorname{div}(s_U) + m \operatorname{pr}_1^* L - a_V \mathcal{V}_U \geq 0$$

is Cartier near that point. Restricting it to $X \times \{x\}$, we obtain $E_x \geq a_V S_x$. Since D is nef, $E_x \sim mL = mRD$, and $a_V > \mu/2 > 3qm$,

$$D^2 \cdot S_x \leq \frac{D^2 \cdot E_x}{a_V} = \frac{mRD^3}{a_V} < \frac{R}{3q} D^3 = \frac{r'}{3} D^3. \tag{8.36}$$

Fix a projective embedding of X. By [Kol96, Theorem I.1.4], the flat family $\mathcal{V}_U \to U$ defines a morphism $U \to \operatorname{Hilb}(X)$ whose member over a closed point x is S_x. The morphism $U \to \operatorname{Hilb}(X)$ is not constant: otherwise $U \subset S_x$ for a closed point $x \in U$, which is impossible as U is dense in X. By Chevalley's theorem, the closure of the image of U in $\operatorname{Hilb}(X)$ has a non-empty open subset Z contained in the image of U. Then Z is a positive-dimensional integral locally closed subscheme of $\operatorname{Hilb}(X)$ whose members over closed points are prime divisors S_x, and by (8.36) they satisfy $D^2 \cdot S_x < \frac{r'}{3} D^3$. Since (X, B) is lc, $p \geq 11$, and a resolution of X has trivial Albanese variety, Corollary 7.4 with $c = r'/3$ implies $H^0(X, \mathcal{O}_X(\lfloor jD \rfloor)) \neq 0$ for $j := \min\{i \in q\mathbb{Z} \mid i > 9a_q + 3b_q r' + 1\}$. Thus

$$b_0 \leq j \leq 9a_q + 3b_q r' + q + 1 \leq \sigma_q r' \leq \sigma_q^* r',$$

which contradicts (8.35). □

Step 3. In this step we prove (2).

Let $z_0, \dots, z_3 \in H^0(X, \mathcal{O}_X(b'L))$ with $z_0 \neq 0$ such that $d(z_1/z_0)$, $d(z_2/z_0)$, $d(z_3/z_0)$ are linearly independent over $k(X)$. They exist by Lemma 8.11, since $|b'L|$ defines a birational map by (1). Since $p \nmid b'$, Lemma 8.11 with $b = b'$ implies that $z_0, \dots, z_3$ is a separating transcendence basis of $k(X)(y)$ over k, and that, for the Hasse–Schmidt derivatives $\partial^{[I]}$ ($I \in \mathbb{N}^4$) of $k(X)(y)$ with respect to $z_0, \dots, z_3$, there exists $\mathcal{J} \in H^0(X, \mathcal{O}_X((4b'R + 1)L)) \setminus \{0\}$ with

$$\mathcal{J}^e z_0^{|I|} \partial^{[I]}(s) \in H^0(X, \mathcal{O}_X((n + (4b'R + 1)e)L)) \otimes_k \mathbb{K} \tag{8.37}$$

for any integers $n, e \geq 0$, any $s \in H^0(X, \mathcal{O}_X(nL)) \otimes_k \mathbb{K}$, and any I with $Re \geq 2|I| - 1$. Here $\partial^{[I]}$ acts on the first factor of $k(X)(y) \otimes_k \mathbb{K}$, and (8.37) follows from the lemma by $\mathbb{K}$-linearity.

For $j \in \{1, 2, 3\}$, let $x_j := z_j/z_0 \in k(X)$, and let $t_j \in \mathbb{K}$ be x_j regarded as an element of $\mathbb{K} = k(X)$. Then $k(X)$ is separable algebraic over $k(x_1, x_2, x_3)$, so $(x_1, x_2, x_3) : X \dashrightarrow \mathbb{A}^3$ is étale on a non-empty open subset of X. Hence the base change $X_{\mathbb{K}} \dashrightarrow \mathbb{A}^3_{\mathbb{K}}$ of (x_1, x_2, x_3) is étale at ξ, which maps to the $\mathbb{K}$-point (t_1, t_2, t_3), so the maximal ideal $\mathfrak{m}_\xi$ of $\mathcal{O}_{X_{\mathbb{K}},\xi}$ is generated by $u_j := x_j \otimes 1 - 1 \otimes t_j$ ($j \in \{1, 2, 3\}$). For ξ and for any prime divisor V on $X_{\mathbb{K}}$ with $\xi \in V$, the images of ξ and of the generic point of V under pr_1 and pr_2 are the generic point of X, so $\mathcal{O}_{X_{\mathbb{K}},\xi}$ and $\mathcal{O}_{X_{\mathbb{K}},V}$ are localizations of $\mathfrak{A}$ at prime ideals. Let

$$\partial_{x_j} := \frac{\partial}{\partial x_j} \otimes \mathrm{id}_{\mathbb{K}} \quad \text{and} \quad \partial_{t_j} := \mathrm{id}_{k(X)} \otimes \frac{\partial}{\partial t_j}$$

be the induced derivations of $\mathfrak{A}$ over k.

We use the following Taylor homomorphisms. Let T be a variable, and let $\mathbf{e}_0, \dots, \mathbf{e}_3$ be the standard basis of $\mathbb{N}^4$. For $j \in \{1,2,3\}$:

(1) Let

$$\chi_j : k(X)(y) \otimes_k \mathbb{K} \to (k(X)(y) \otimes_k \mathbb{K})[[T]], \quad g \to \sum_{a \geq 0} z_0^a \partial^{[a\mathbf{e}_j]}(g) T^a.$$

It is the composite of the ring homomorphism

$$g \to \sum_I \partial^{[I]}(g) T_0^{I_0} \cdots T_3^{I_3},$$

which sends z_i to $z_i + T_i$, with $T_j \to z_0 T$ and $T_i \to 0$ $(i \neq j)$. Hence χ_j is a ring homomorphism with $\chi_j(x_j) = x_j + T$ and $\chi_j(x_{j'}) = x_{j'}$ $(j' \neq j)$. Since $k(X)$ is separable algebraic over $k(x_1, x_2, x_3)$, χ_j maps $k(X)$ into $k(X)[[T]]$, and the coefficient of T in $\chi_j|_{k(X)}$ is $\frac{\partial}{\partial x_j}$ [Mat89, Theorems 25.3 and 27.2]. Let $f_0 := z_0 y^{-b'} \in k(X)$. Since $\chi_j(z_0) = z_0$ and $p \nmid b'$, $\chi_j(y) = y\beta_j$, where $\beta_j \in k(X)[[T]]$ is the unique b'-th root of $f_0/\chi_j(f_0)$ with constant term 1.

(2) Let $\psi_j : \mathbb{K} \to \mathbb{K}[[T]]$ be the ring homomorphism over k with $\psi_j(t_j) = t_j + T$, $\psi_j(t_{j'}) = t_{j'}$ $(j' \neq j)$, and constant term $\mathrm{id}_{\mathbb{K}}$, which exists and is unique as $\mathbb{K}$ is separable algebraic over $k(t_1, t_2, t_3)$ [Mat89, Theorems 25.3 and 27.2]. We let ψ_j also denote $\mathrm{id}_{k(X)(y)} \otimes \psi_j$ on $k(X)(y) \otimes_k \mathbb{K}$, and let $\partial_{t_j}^{[a]}$ be the coefficient of T^a in ψ_j. Then $\psi_j(y) = y$.

Let $\iota \in \{\chi_j, \psi_j \mid j = 1,2,3\}$, and let $\iota^{(a)}$ be the coefficient of T^a in ι. Then $\iota(\mathfrak{A}) \subset \mathfrak{A}[[T]]$, $\iota(g) \equiv g$ modulo T, the coefficient of T in $\iota|_{\mathfrak{A}}$ is the derivation $\delta_\iota := \partial_{x_j}$ for $\iota = \chi_j$ and $\delta_\iota := \partial_{t_j}$ for $\iota = \psi_j$, and $\iota(y) = y\beta$ with $\beta \in k(X)[[T]]$ of constant term 1. Since an element of $\mathfrak{A}$ outside a prime ideal $\mathfrak{p}$ is mapped to a unit of $\mathfrak{A}_{\mathfrak{p}}[[T]]$, ι extends to $\mathcal{O}_{X_{\mathbb{K}},\xi}$ and to $\mathcal{O}_{X_{\mathbb{K}},V}$. Let $m \in \mathbb{N}^+$, let s be a global section of $\mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}})$, and let $a \in \mathbb{N}$. Then $\iota^{(a)}(s) = \varphi' y^m$ with $\varphi' \in \mathfrak{A}$, and:

(a) If $\varphi_s \in \mathfrak{m}_\xi^d$, then $\varphi' \in \mathfrak{m}_\xi^{d-a}$, since $\beta^m \in k(X)[[T]]$ and

$$\iota(\mathfrak{m}_\xi) \subset \mathfrak{m}_\xi + T\mathcal{O}_{X_{\mathbb{K}},\xi}[[T]], \quad \text{so} \quad \iota(\mathfrak{m}_\xi^d) \subset \sum_{c \geq 0} \mathfrak{m}_\xi^{d-c} T^c.$$

(b) If V is a prime divisor on $X_{\mathbb{K}}$ with $\xi \in V$, $f \in \mathcal{O}_{X_{\mathbb{K}},V}$ is a uniformizer, $\varphi_s = f^a g$ for a unit g of $\mathcal{O}_{X_{\mathbb{K}},V}$, and $\delta_\iota(f) \notin f\mathcal{O}_{X_{\mathbb{K}},V}$, then $\varphi' \in \mathcal{O}_{X_{\mathbb{K}},V}^\times$: we have

$$\iota(\varphi_s y^m) = \iota(f)^a \iota(g) \beta^m y^m,$$

$$\text{where} \quad \iota(f)^a \equiv (\delta_\iota(f)T + T^2(\cdots))^a \pmod{f} \quad \text{and} \quad \iota(g)\beta^m \equiv g \pmod{T},$$

so $\varphi' \equiv \delta_\iota(f)^a g$ modulo $f\mathcal{O}_{X_{\mathbb{K}},V}$.

For $\iota = \chi_j$, we have $\iota^{(a)}(s) = z_0^a \partial^{[a\mathbf{e}_j]}(s)$. For $\iota = \psi_j$, write $s = \sum_i s_i \otimes \kappa_i$ with $\kappa_i \in \mathbb{K}$ and global sections s_i of $\mathcal{O}_X(mL)$. Then

$$\iota^{(a)}(s) = \sum_i s_i \otimes \partial_{t_j}^{[a]}(\kappa_i) \in H^0(X_{\mathbb{K}}, \mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}})).$$

Let n_L be a positive integer such that $n_L L$ is Cartier. Since $n_L L$ is a nef Cartier divisor, $h^0(X, \mathcal{O}_X(mL)) = \frac{L^3}{6} m^3 + O(m^2)$ for positive integers m divisible by n_L [Kol96, Theorem VI.2.15]. Since $r' \geq 4r$ and $r^3 D^3 > 216$,

$$L^3 = r'^3 q^3 D^3 > 64 \cdot 216 q^3 > (6q)^3.$$

The global sections s of $\mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}})$ with $s = 0$ or $\mathrm{mult}_\xi E_s \geq 6qm + 1$ form the kernel of the $\mathbb{K}$-linear map $s \to \varphi_s$ to $\mathcal{O}_{X_{\mathbb{K}},\xi}/\mathfrak{m}_\xi^{6qm+1}$, and

$$\dim_{\mathbb{K}} \mathcal{O}_{X_{\mathbb{K}},\xi}/\mathfrak{m}_\xi^{6qm+1} = \binom{6qm+3}{3} = \frac{(6q)^3 m^3}{6} + O(m^2).$$

Since $L^3 > (6q)^3$, for any sufficiently large m divisible by n_L we have

$$\dim_{\mathbb{K}} H^0(X_{\mathbb{K}}, \mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}})) = h^0(X, \mathcal{O}_X(mL)) > \dim_{\mathbb{K}} \mathcal{O}_{X_{\mathbb{K}},\xi}/\mathfrak{m}_\xi^{6qm+1}.$$

Thus we may fix a positive integer m divisible by n_L with $m \geq 3(4b'R+1)$ and an element

$$s \in H^0(X_{\mathbb{K}}, \mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}}))\backslash\{0\} \quad \text{with} \quad \mu := \operatorname{mult}_\xi E_s > 6qm.$$

Let Λ be the finite set of prime divisors V on $X_{\mathbb{K}}$ with $\xi \in V \subset \operatorname{Supp} E_s$. For $V \in \Lambda$, let $a_V := \operatorname{mult}_V E_s$, and let $f_V \in \mathcal{O}_{X_{\mathbb{K}},\xi}$ be a prime element defining V near ξ. Since $\mathcal{O}_{X_{\mathbb{K}},\xi}$ is factorial, φ_s is a unit times $\prod_{V\in\Lambda} f_V^{a_V}$ in $\mathcal{O}_{X_{\mathbb{K}},\xi}$, and

$$\mu = \sum_{V\in\Lambda} a_V \operatorname{mult}_\xi V.$$

Let $a := \max_{V\in\Lambda} a_V$. By Claim 8.14, $a \leq \mu/2$. Let

$$\Lambda' := \{V \in \Lambda \mid \partial_{t_j} f_V \in f_V \mathcal{O}_{X_{\mathbb{K}},\xi} \text{ for any } j \in \{1,2,3\}\}.$$

We prove $\operatorname{mult}_\xi V \geq p$ for any $V \in \Lambda'$. The subfield $1 \otimes \mathbb{K} \subset \mathcal{O}_{X_{\mathbb{K}},\xi}$ maps isomorphically onto the residue field of ξ, so

$$\widehat{\mathcal{O}}_{X_{\mathbb{K}},\xi} = \mathbb{K}[[u_1, u_2, u_3]].$$

Since

$$\partial_{t_j}(\mathfrak{m}_\xi^d) \subset \mathfrak{m}_\xi^{d-1},$$

the derivation ∂_{t_j} extends continuously to $\mathbb{K}[[u_1,u_2,u_3]]$, and since $\partial_{t_j}(u_{j'}) = -1$ for $j' = j$ and 0 for $j' \neq j$, it equals $\partial_j^{\mathrm{c}} - \frac{\partial}{\partial u_j}$, where ∂_j^{c} is $\frac{\partial}{\partial t_j}$ on the coefficients. Let $F_V \neq 0$ be the homogeneous part of degree $\operatorname{mult}_\xi V$ of f_V. Since $\partial_{t_j} f_V \in f_V \mathcal{O}_{X_{\mathbb{K}},\xi}$ has order at least $\operatorname{mult}_\xi V$ and $\partial_j^{\mathrm{c}} f_V$ has order at least $\operatorname{mult}_\xi V$, the homogeneous part of degree $\operatorname{mult}_\xi V - 1$ of $\partial_{t_j} f_V$, which is $-\partial F_V/\partial u_j$, vanishes. Thus any monomial of F_V has all exponents divisible by p, so $\operatorname{mult}_\xi V \geq p$. Let $a' := \max\{a_V \mid V \in \Lambda'\}$ if $\Lambda' \neq \emptyset$, and $a' := 0$ otherwise. Then $a' \leq \mu/p$ as $\mu \geq a_V \operatorname{mult}_\xi V \geq p a_V$ for any $V \in \Lambda'$. Let $e := 0$ if $a' = 0$, and $e := \lceil (2a'-1)/R \rceil$ otherwise. Then $Re \geq 2a_V - 1$ for any $V \in \Lambda'$, and $e < 2\mu/(pR) + 1$.

For $V \in \Lambda$ we choose ι_V as follows. If $V \notin \Lambda'$, there exists j with $\partial_{t_j} f_V \notin f_V \mathcal{O}_{X_{\mathbb{K}},\xi}$, hence $\partial_{t_j} f_V \notin f_V \mathcal{O}_{X_{\mathbb{K}},V}$ as $\mathcal{O}_{X_{\mathbb{K}},\xi}$ is factorial. Let $\iota_V := \psi_j$ for such j. If $V \in \Lambda'$, let $\mathcal{V}$ be the closure of V in $X \times X$. Since k is perfect, $\mathcal{V}$ is smooth over k at the generic point, so $df_V \neq 0$ in $\Omega_{X\times X/k} \otimes \kappa(\mathcal{V})$, where $\kappa(\mathcal{V})$ is the residue field of the generic point of $\mathcal{V}$. Since $\mathcal{V}$ dominates both factors and (x_1, x_2, x_3) is étale on a non-empty open subset of X, $dx_1, dx_2, dx_3, dt_1, dt_2, dt_3$ form a basis of $\Omega_{X\times X/k}$ at the generic point of $\mathcal{V}$, and

$$df_V = \sum_{j=1}^3 \partial_{x_j} f_V dx_j + \sum_{j=1}^3 \partial_{t_j} f_V dt_j.$$

Since $\partial_{t_j} f_V \in f_V \mathcal{O}_{X_{\mathbb{K}},V}$ for any j, there exists j with $\partial_{x_j} f_V \notin f_V \mathcal{O}_{X_{\mathbb{K}},V}$. Let $\iota_V := \chi_j$ for such j. Let $N := m + (4b'R+1)e$, and for any $V \in \Lambda$ let $s_V := \mathcal{J}^e \iota_V^{(a_V)}(s)$. By (8.37) for $\iota_V = \chi_j$, and since $\iota_V^{(a_V)}(s)$ is a global section of $\mathcal{O}_{X_{\mathbb{K}}}(mL_{\mathbb{K}})$ for $\iota_V = \psi_j$, we have

$$s_V \in H^0(X_{\mathbb{K}}, \mathcal{O}_{X_{\mathbb{K}}}(NL_{\mathbb{K}})).$$

Since $\mathcal{J} = f_{\mathcal{J}} y^{4b'R+1}$ with $f_{\mathcal{J}} \in k(X)\backslash\{0\}$, and non-zero elements of $k(X) \otimes 1$ are units in $\mathcal{O}_{X_{\mathbb{K}},\xi}$ and $\mathcal{O}_{X_{\mathbb{K}},V}$, (a) and (b) imply

$$\operatorname{mult}_\xi E_{s_V} \geq \mu - a_V \geq \mu - a \quad \text{and} \quad V \notin \operatorname{Supp} E_{s_V}.$$

Since $\mathbb{K}$ is infinite and, for any $V \in \Lambda$, $V \notin \operatorname{Supp} E_{s_V}$, there exist $\lambda_V \in \mathbb{K}$ $(V \in \Lambda)$ such that $s' := \sum_{V\in\Lambda} \lambda_V s_V$ satisfies $V \notin \operatorname{Supp} E_{s'}$ for any $V \in \Lambda$. Then $\operatorname{mult}_\xi E_{s'} \geq \mu - a$. Moreover, $\mathcal{J}^e s$ is a global section of $\mathcal{O}_{X_{\mathbb{K}}}(NL_{\mathbb{K}})$, $\operatorname{mult}_\xi E_{\mathcal{J}^e s} = \mu$, and the prime divisors V with $\xi \in V \subset \operatorname{Supp} E_{\mathcal{J}^e s}$ are exactly the elements of Λ. Thus $E_{\mathcal{J}^e s}$ and $E_{s'}$ have no common prime divisor through ξ.

Since $b' < w/(1024q) \le \sqrt{p}/(1024q)$ and $\sqrt{p} > 8$, we have $8b'/p < 1/(1024q)$. Since $R \ge 8q$ and $p > 64$, we have $2/(pR) < 1/(256q)$. Since $m \ge 3(4b'R+1)$ and $\mu > 6qm$, we have $(4b'R+1)/\mu < 1/(16q)$. Hence, by $e < 2\mu/(pR)+1$ and $m < \mu/(6q)$,

$$\frac{N}{\mu} < \frac{1}{6q} + \frac{8b'}{p} + \frac{2}{pR} + \frac{4b'R+1}{\mu} < \left(\frac{1}{6} + \frac{1}{1024} + \frac{1}{256} + \frac{1}{16}\right)\frac{1}{q} < \frac{1}{4q}.$$

Thus $2qN < \mu/2 \le \mu - a$, so $\mathcal{J}^e s, s' \in \mathcal{W}_N(2q)$. Since $\mathcal{O}_{X_{\mathbb{K}},\xi}$ is a regular local ring of dimension 3 and the local equations of $E_{\mathcal{J}^e s}$ and $E_{s'}$ at ξ have no common prime factor,

$$\dim_\xi \mathbf{B}_{2q} \le \dim_\xi \operatorname{Bs}(\mathcal{W}_N(2q)) \le \dim_\xi (\operatorname{Supp} E_{\mathcal{J}^e s} \cap \operatorname{Supp} E_{s'}) \le 1.$$

□

Construction-Proposition 8.15. Notation and conditions as in Set-up 8.12. Let

$$\delta_q := 1/(512M_q + 128),$$

and assume that $p > P_q^\dagger$ and $b_0 > Q_q r$. Then the following objects exist.

(1) A smooth quasi-projective surface T and morphisms $f : W \to T$ and $g : W \to X$ as in Set-up 8.4 for the pair (X, B) satisfying the following.
   (a) The image $\mathcal{C}$ of $(g, f) : W \to X \times T$ is flat over T with fibers C_t, and the morphism $T \to \operatorname{Hilb}(X)$ defined by $\mathcal{C}$ is an immersion.
   (b) $k(W)/k(X)$ is purely inseparable of degree p^e with $e \ge 1$.
   (c) $\ell := \deg(g^*L|_{F_t})$ satisfies $0 < \ell < \delta_q$ for any closed point $t \in T$.

(2) For any real number $\eta > 0$, after shrinking T to a non-empty open subset, a $\mathbb{Q}$-divisor $G_t \ge 0$ on X for any closed point $t \in T$ satisfying the following.
   (a) $G_t \cdot C_t < 1/(8q)$.
   (b) $(X, B + G_t)$ is lc near the generic point of C_t and has exactly one lc place there, whose center is C_t.
   (c) $\operatorname{mult}_S G_t < \eta$ for any prime divisor $S \subset X$ containing C_t.

Proof. We divide the proof into six steps. Let Ω be an algebraic closure of $\mathbb{K}$. We write X_Ω, L_Ω, and $\xi_\Omega \in X_\Omega(\Omega)$ for the base changes of $X_{\mathbb{K}}$, $L_{\mathbb{K}}$, and ξ. For a field $\Omega_1 \supset k$ and a point $x \in X(\Omega_1)$, we regard x as a point of $X_{\Omega_1}(\Omega_1)$. Let $\tau_q := q/(32M_q + 8)$, so that $\delta_q = \tau_q/(16q)$.

Step 1. In this step we find a curve $\Gamma \subset X_\Omega$ through ξ_Ω with small Seshadri ratio.

Since $b_0 > Q_q r = 8E_q r \ge 16\sigma_q^* r$ and $p > P_q^\dagger \ge P_q^*$, Construction-Proposition 8.13(1) provides a positive integer b with $p \nmid b$ such that $|bL|$ defines a birational map and

$$b \le u_0 b_0/r' + u_1 < w/(1024q).$$

Since $p > (4\gamma_q E_q + 1)^2$, we have $w \ge 4\gamma_q E_q + 1 > 4\gamma_q E_q$. If $b_0 \le E_q r'$, then by Set-up 8.12,

$$b_0 \le 4E_q r + \frac{2\gamma_q E_q}{w} b_0 < 4E_q r + \frac{b_0}{2},$$

so $b_0 < 8E_q r = Q_q r$, a contradiction. Thus, since $E_q \ge q(3584M_q + 900) + 4$ and $R = r'q$,

$$b_0 > E_q r' > (3584M_q + 900)R. \tag{8.38}$$

Let $m_0 := \lfloor \frac{b_0 - 2}{R} \rfloor$. Since $R \ge 2$ and $m_0 > \frac{b_0-2}{R} - 1$, (8.38) implies

$$m_0 \ge 7(512M_q + 128), \qquad m_0 \ge \frac{b_0}{2R}, \qquad \text{and} \qquad m_0 R + 1 \le b_0 - 1. \tag{8.39}$$

Since D is nef and big and X is $\mathbb{Q}$-factorial, by Kodaira's lemma we may write $D = H_D + E_D$ with an ample $\mathbb{Q}$-divisor H_D and a $\mathbb{Q}$-divisor $E_D \ge 0$. Then $\xi_\Omega \notin (\operatorname{Supp} E_D)_\Omega$. Since k is perfect, X_Ω is a normal projective threefold and ξ_Ω is a smooth closed point of X_Ω. Suppose that $\epsilon(L_\Omega; \xi_\Omega) \ge 7/m_0$. Then

$$m_0 L_\Omega = m_0 R(H_D)_\Omega + m_0 R(E_D)_\Omega, \quad \text{and} \quad \epsilon(m_0 L_\Omega; \xi_\Omega) \ge 7 > 6.$$

By Lemma 8.9,

$$H^0(X_\Omega, \mathcal{O}_{X_\Omega}(K_{X_\Omega} + m_0 L_\Omega)) \neq 0,$$

so $H^0(X, \mathcal{O}_X(K_X + m_0 L)) \neq 0$ by flat base change. Since $B \in I_q \backslash \{1\}$ and $q \mid R$, RB is a divisor and $\lfloor B \rfloor = 0$, so

$$\lfloor (m_0 R + 1) D \rfloor = K_X + m_0 L.$$

Thus $H^0(X, \mathcal{O}_X(\lfloor (m_0 R + 1) D \rfloor)) \neq 0$ with $m_0 R + 1 < b_0$ by (8.39), contradicting the definition of b_0. Hence $\epsilon(L_\Omega; \xi_\Omega) < \frac{7}{m_0}$, and there exists an integral curve $\Gamma \subset X_\Omega$ with $\xi_\Omega \in \Gamma$ such that

$$\ell := L_\Omega \cdot \Gamma \quad \text{and} \quad \mu := \operatorname{mult}_{\xi_\Omega} \Gamma \quad \text{satisfy} \quad \frac{\ell}{\mu} < \frac{7}{m_0}.$$

Since $\Gamma \not\subset (\operatorname{Supp} E_D)_\Omega$, we have $\ell \geq R(H_D)_\Omega \cdot \Gamma > 0$. By (8.39) and the bound on b, and since $R < b_0$,

$$\frac{\ell}{\mu} < \frac{7}{m_0} \leq \delta_q \qquad \text{and} \qquad \frac{b\ell}{\mu} < \frac{14bR}{b_0} \leq 14\left(q u_0 + \frac{u_1 R}{b_0}\right) < M_q. \tag{8.40}$$

Let $n' \geq 1$ and $j \geq 0$ be integers, and let s be an element of

$$H^0(X_\Omega, \mathfrak{m}_{\xi_\Omega}^j \cdot \mathcal{O}_{X_\Omega}(n' L_\Omega))$$

which does not vanish identically on Γ. Then $E_s := \operatorname{div}(s) + n' L_\Omega \geq 0$ and $\operatorname{mult}_{\xi_\Omega} E_s \geq j$. Since ξ_Ω is a smooth point of X_Ω and $\Gamma \not\subset \operatorname{Supp} E_s$, we have

$$n'\ell = E_s \cdot \Gamma \geq \operatorname{mult}_{\xi_\Omega} E_s \cdot \operatorname{mult}_{\xi_\Omega} \Gamma \geq j\mu. \tag{8.41}$$

Let $\tau \in \{\tau_q, 2q\}$ and $n \geq 1$. Since $\frac{\ell}{\mu} < \delta_q = \frac{\tau_q}{16q} < \tau$, (8.41) with $j = \lceil n\tau \rceil$ implies that any element of

$$\mathcal{W}_n(\tau; \xi_\Omega) = \mathcal{W}_n(\tau) \otimes_{\mathbb{K}} \Omega$$

vanishes on Γ. Let $C_\xi \subset X_{\mathbb{K}}$ be the image of Γ, an integral curve with $\xi \in C_\xi$. Thus

$$C_\xi \subset \operatorname{Bs}(\mathcal{W}_n(\tau)) \quad \text{for any } n \geq 1 \text{ and } \tau \in \{\tau_q, 2q\}. \tag{8.42}$$

In particular $C_\xi \subset \mathbf{B}_{2q}$. Since C_ξ is a curve through ξ and $\dim_\xi \mathbf{B}_{2q} \leq 1$ by Construction-Proposition 8.13(2), we have $\dim_\xi \mathbf{B}_{2q} = 1$, and C_ξ is an irreducible component of $\mathbf{B}_{2q}$.

By Noetherianity there exist positive integers $n_1, \ldots, n_{k_0}$ with

$$\mathbf{B}_{2q} = \bigcap_{i=1}^{k_0} \operatorname{Bs}(\mathcal{W}_{n_i}(2q))_{\mathrm{red}}.$$

Since

$$\mathcal{W}_n(2q) \cdot \mathcal{W}_{n'}(2q) \subset \mathcal{W}_{n+n'}(2q),$$

we have $\operatorname{Bs}(\mathcal{W}_{jn}(2q)) \subset \operatorname{Bs}(\mathcal{W}_n(2q))$ for any positive integers j and n. Let $N_0 := n_1 \cdots n_{k_0}$. Then

$$\operatorname{Bs}(\mathcal{W}_n(2q))_{\mathrm{red}} = \mathbf{B}_{2q} \quad \text{for any positive multiple } n \text{ of } N_0. \tag{8.43}$$

Let $V \subset X \times X$ be the closure of C_ξ under $X_{\mathbb{K}} = X \times_k \operatorname{Spec} \mathbb{K} \to X \times X$. Then V is integral, $\dim V = 4$, the generic fiber of the second projection $V \to X$ is C_ξ, and the first and second projections $V \to X$ are dominant, since $\xi \in C_\xi$ maps to the generic point of X under the first projection. For a field $\Omega_1 \supset k$ and a point $z \in X(\Omega_1)$, let $V_z \subset X_{\Omega_1}$ be the fiber of the second projection $V \to X$ over z. Thus $V_{\xi_\Omega} = (C_\xi)_\Omega$.

Step 2. In this step we prove that, for any integer $n \geq 4(4bR + 1)$ and $a_n := \lfloor \tau_q n \rfloor + 1$,

$$\operatorname{mult}_{C_\xi} s \geq a_n \quad \text{for any } s \in \mathcal{W}_n(2q) \backslash \{0\}, \tag{8.44}$$

where $\operatorname{mult}_{C_\xi} s$ is the multiplicity of s along C_ξ, that is, the order of vanishing of s at the generic point of C_ξ.

Since $|bL|$ defines a birational map, by Lemma 8.11 there exist $z_0, \ldots, z_3 \in H^0(X, \mathcal{O}_X(bL))$ with $z_0 \neq 0$ such that $d(z_1/z_0)$, $d(z_2/z_0)$, and $d(z_3/z_0)$ are linearly independent over $k(X)$. Let

$\partial^{[I]}$ ($I \in \mathbb{N}^4$) be the Hasse–Schmidt derivatives of $k(X)(y)$ with respect to $z_0, \dots, z_3$. Since $p \nmid b$, by Lemma 8.11 there exists $\mathcal{J} \in H^0(X, \mathcal{O}_X((4bR+1)L))\backslash\{0\}$ with

$$\mathcal{J}^j z_0^{|I|} \partial^{[I]}(s) \in H^0(X, \mathcal{O}_X((n+(4bR+1)j)L)) \tag{8.45}$$

for any integers $n, j \geq 0$, any $s \in H^0(X, \mathcal{O}_X(nL))$, and any I with $Rj \geq 2|I| - 1$.

Let x_1, x_2, x_3 be a separating transcendence basis of $\mathbb{K}/k$ and let $\partial_{\mathbb{K}}^{[I']}$ ($I' \in \mathbb{N}^3$) be the Hasse–Schmidt derivatives of $\mathbb{K}$ with respect to x_1, x_2, x_3. We have

$$H^0(X_{\mathbb{K}}, \mathcal{O}_{X_{\mathbb{K}}}(nL_{\mathbb{K}})) = H^0(X, \mathcal{O}_X(nL)) \otimes_k \mathbb{K} \subset k(X)(y) \otimes_k \mathbb{K}.$$

We let $\partial^{[I]}$ act on the first factor and $\partial_{\mathbb{K}}^{[I']}$ on the second factor. For $\alpha = (I, I') \in \mathbb{N}^7$, the operators

$$\partial^{[\alpha]} := \partial^{[I]} \partial_{\mathbb{K}}^{[I']}$$

are the Hasse–Schmidt derivatives of the fraction field of $k(X)(y) \otimes_k \mathbb{K}$ with respect to $z_0, \dots, z_3, x_1, x_2, x_3$. Recall that $f \to \sum_\alpha \partial^{[\alpha]}(f) v^\alpha$ is a ring homomorphism into the formal power series ring in $v_1, \dots, v_7$ which sends the i-th coordinate c_i to $c_i + v_i$, and that

$$\partial^{[\beta]} \partial^{[\alpha]} = \binom{\alpha+\beta}{\alpha} \partial^{[\alpha+\beta]}.$$

Let $s \in \mathcal{W}_n(2q)\backslash\{0\}$, write $s = \sum_i s_i \otimes \theta_i$ with $s_i \in H^0(X, \mathcal{O}_X(nL))$ and $\theta_i \in \mathbb{K}$, and let $m := \operatorname{mult}_{C_\xi} s$. Then $m \geq 1$ by (8.42). Let $X_1 \subset X$ be a non-empty smooth affine open subset with $X_1 \cap \operatorname{Supp} L = \emptyset$ on which z_0 and $\mathcal{J}$ have no zeros and such that $z_0, \dots, z_3$ are étale coordinates on $X_1 \times \mathbb{G}_m$, where y is the coordinate of $\mathbb{G}_m$. Such X_1 exists since, $z_0, \dots, z_3$ being a separating transcendence basis of $k(X)(y)$ by Lemma 8.11, the zero locus of $dz_0 \wedge \cdots \wedge dz_3$ is a proper closed subset of $X \times \mathbb{G}_m$ stable under $\mathbb{G}_m$, hence the preimage of a proper closed subset of X. Let $X_2 \subset X$ be a non-empty affine open subset on which x_1, x_2, x_3 are étale coordinates and θ_i are regular. Since $X_1 \times \mathbb{G}_m$ and X_2 are formally étale over the affine spaces with coordinates $z_0, \dots, z_3$ and x_1, x_2, x_3, the Taylor homomorphisms extend uniquely to their coordinate rings [Mat89, Section 25], so the operators $\partial^{[\alpha]}$ preserve the coordinate ring of $\mathcal{Y} := X_1 \times \mathbb{G}_m \times X_2$. For a closed point $Q \in \mathcal{Y}$, composing the Taylor homomorphism with $f \to f(Q)$ on coefficients yields a homomorphism $\mathcal{O}_{\mathcal{Y},Q} \to k[[v_1, \dots, v_7]]$ sending $c_i - c_i(Q)$ to v_i, which induces the isomorphism $\widehat{\mathcal{O}}_{\mathcal{Y},Q} \cong k[[v_1, \dots, v_7]]$. Thus, for any $f \in \mathcal{O}_{\mathcal{Y},Q}$ and any integer $N \geq 0$,

$$f \in \mathfrak{m}_Q^N \iff \partial^{[\alpha]}(f)(Q) = 0 \text{ for any } |\alpha| < N, \quad \text{hence} \quad \partial^{[\alpha]}(\mathfrak{m}_Q^N) \subset \mathfrak{m}_Q^{N-|\alpha|}. \tag{8.46}$$

Let $\widetilde{V}$ and $\widetilde{\Delta}$ be the preimages of V and of the diagonal under the morphism

$$\mathcal{Y} \to X_1 \times X_2 \subset X \times X$$

forgetting the factor $\mathbb{G}_m$. Since V and the diagonal dominate each factor of $X \times X$, $\widetilde{V}$ and $\widetilde{\Delta}$ are non-empty. On $\mathcal{Y}$, s is a regular function with $\operatorname{mult}_{\widetilde{V}} s = m$ and $\operatorname{mult}_{\widetilde{\Delta}} s \geq 2qn$. Suppose that $m \leq \tau_q n$. Let $Q \in \widetilde{V}$ be a general closed point. Then $s \in \mathfrak{m}_Q^m \backslash \mathfrak{m}_Q^{m+1}$, and by (8.46) there exists $\alpha = (I, I')$ with $|I| + |I'| = m$ and $\partial^{[\alpha]}(s)(Q) \neq 0$. Let

$$j := \left\lceil \frac{2m-1}{R} \right\rceil, \quad N' := n + (4bR+1)j, \quad \text{and} \quad s' := \mathcal{J}^j z_0^{|I|} \partial^{[\alpha]}(s) = \sum_i \mathcal{J}^j z_0^{|I|} \partial^{[I]}(s_i) \otimes \partial_{\mathbb{K}}^{[I']}(\theta_i).$$

Since $|I| \leq m$, we have $Rj \geq 2|I| - 1$, so

$$s' \in H^0(X_{\mathbb{K}}, \mathcal{O}_{X_{\mathbb{K}}}(N' L_{\mathbb{K}}))$$

by (8.45). Since $\mathcal{J}$ and z_0 are units on $\mathcal{Y}$, $s'(Q) \neq 0$, so s' does not vanish identically on C_ξ. By (8.46) at general closed points of $\widetilde{\Delta}$, $\operatorname{mult}_{\widetilde{\Delta}} s' \geq 2qn - m$, that is,

$$s' \in H^0(X_{\mathbb{K}}, \mathfrak{m}_\xi^{2qn-m} \cdot \mathcal{O}_{X_{\mathbb{K}}}(N' L_{\mathbb{K}})).$$

Since Γ dominates C_ξ, $s' \otimes_{\mathbb{K}} 1$ does not vanish identically on Γ, so (8.41) implies

$$2qn - m \le \frac{\ell}{\mu} N'. \tag{8.47}$$

On the other hand,

$$(4bR+1)j \le (4bR+1)\Big(\frac{2m-1}{R}+1\Big) \le 8bm + \frac{2m}{R} + 4bR + 1.$$

By (8.40), we have $\frac{\ell}{\mu} < 1$ and $\frac{b\ell}{\mu} < M_q$. Since $m \le \tau_q n$, $(8M_q+1)\tau_q < (8M_q+2)\tau_q = \frac{q}{4}$, $n \ge 4(4bR+1)$, and $q \ge 2$,

$$\frac{\ell}{\mu} N' < n + 8M_q\tau_q n + \tau_q n + 4bR + 1 < \Big(1+\frac{q}{4}\Big)n + 4bR + 1 \le \Big(\frac{5}{4}+\frac{q}{4}\Big)n < qn.$$

But $2qn - m \ge (2q - \tau_q)n > qn$, contradicting (8.47). Thus $m > \tau_q n$, and (8.44) holds.

Step 3. In this step we prove the following for any positive multiple n of N_0 with $n \ge 4(4bR+1)$. Let $\Omega_1 \subset \Omega_2$ be algebraically closed fields containing k, $x \in X(\Omega_1)$ a point mapping to the generic point of X, C_1 an irreducible component of $(V_x)_{\mathrm{red}}$, and $x' \in X(\Omega_2)$ a point mapping to the generic point of $C_1 \subset X_{\Omega_1}$. Then

$$\mathcal{W}_n(2q;x) \subset \mathcal{W}_n(\tau_q;x'), \qquad (V_{x'})_{\mathrm{red}} = C_1 \times_{\Omega_1} \Omega_2, \qquad \text{and} \qquad V_x \text{ is irreducible,} \tag{8.48}$$

where x is regarded as a point of $X(\Omega_2)$ in the first inclusion.

First, for any $z \in X(\Omega_1)$ mapping to the generic point of X, the morphism $\operatorname{Spec}\Omega_1 \to \operatorname{Spec}\mathbb{K}$ defined by z satisfies $z = \xi \otimes_{\mathbb{K}} \Omega_1$, and by flat base change

$$V_z = C_\xi \otimes_{\mathbb{K}} \Omega_1, \quad \mathcal{W}_n(\tau;z) = \mathcal{W}_n(\tau) \otimes_{\mathbb{K}} \Omega_1, \quad \text{and} \quad \operatorname{Bs}(\mathcal{W}_n(\tau;z)) = \operatorname{Bs}(\mathcal{W}_n(\tau)) \otimes_{\mathbb{K}} \Omega_1. \tag{8.49}$$

Next we show that z belongs to every irreducible component of V_z. Let $\mathbb{K}^{\mathrm{s}} \subset \Omega_1$ be the separable closure of $\mathbb{K}$. Since C_ξ is integral, $\mathrm{Gal}(\mathbb{K}^{\mathrm{s}}/\mathbb{K})$ acts transitively on the irreducible components of $C_\xi \otimes_{\mathbb{K}} \mathbb{K}^{\mathrm{s}}$ [Sta, Tag 04KY], and it fixes the point $\xi \otimes_{\mathbb{K}} \mathbb{K}^{\mathrm{s}}$, so any irreducible component of $C_\xi \otimes_{\mathbb{K}} \mathbb{K}^{\mathrm{s}}$ contains $\xi \otimes_{\mathbb{K}} \mathbb{K}^{\mathrm{s}}$. Since $\mathbb{K}^{\mathrm{s}}$ is separably closed, the irreducible components of $C_\xi \otimes_{\mathbb{K}} \mathbb{K}^{\mathrm{s}}$ are geometrically irreducible [Sta, Tag 020J], so the irreducible components of V_z are their base changes to Ω_1, and each of them contains $z = \xi \otimes_{\mathbb{K}} \Omega_1$.

Since $C_1 \subset X_{\Omega_1}$ is an irreducible component of $(V_x)_{\mathrm{red}}$, which contains x, and x maps to the generic point of X, the image of C_1 in X is dense. Hence x' maps to the generic point of X. By (8.44), any $s \in \mathcal{W}_n(2q)$ belongs to the a_n-th power of the maximal ideal of the local ring of $X \times X$ at the generic point of V. The morphism $X_{\Omega_2} \to X \times X$ which is the projection on the first factor and x on the second factor maps x' to the generic point of V. Thus $s \otimes_{\mathbb{K}} \Omega_2$ vanishes to order at least $a_n \ge \lceil \tau_q n \rceil$ at x', and the first relation in (8.48) holds.

By (8.42) and (8.49) with Ω_2 and x' in place of Ω_1 and z,

$$V_{x'} = C_\xi \otimes_{\mathbb{K}} \Omega_2 \subset \operatorname{Bs}(\mathcal{W}_n(\tau_q)) \otimes_{\mathbb{K}} \Omega_2 = \operatorname{Bs}(\mathcal{W}_n(\tau_q;x')).$$

Thus

$$V_{x'} \subset \operatorname{Bs}(\mathcal{W}_n(2q;x)) = \operatorname{Bs}(\mathcal{W}_n(2q)) \otimes_{\mathbb{K}} \Omega_2,$$

whose underlying set is that of $\mathbf{B}_{2q} \otimes_{\mathbb{K}} \Omega_2$ by (8.43). The irreducible components of $\mathbf{B}_{2q} \otimes_{\mathbb{K}} \Omega_2$ are the base changes to Ω_2 of the irreducible components of $\mathbf{B}_{2q} \otimes_{\mathbb{K}} \Omega_1$, where $\mathbb{K} \to \Omega_1$ is defined by x. Let Z be an irreducible component of $\mathbf{B}_{2q} \otimes_{\mathbb{K}} \Omega_1$ with $x' \in Z \times_{\Omega_1} \Omega_2$. Then Z contains the generic point of C_1, so $C_1 \subset Z$ and $x \in Z$. Since

$$\dim_x(\mathbf{B}_{2q} \otimes_{\mathbb{K}} \Omega_1) = \dim_\xi \mathbf{B}_{2q} = 1,$$

we have $\dim Z \le 1$, so $Z = C_1$. Any irreducible component of $V_{x'}$ contains x' and is contained in $\mathbf{B}_{2q} \otimes_{\mathbb{K}} \Omega_2$, so it is contained in, hence equal to, the curve $C_1 \times_{\Omega_1} \Omega_2$. Thus $(V_{x'})_{\mathrm{red}} = C_1 \times_{\Omega_1} \Omega_2$. In particular $V_{x'}$ is irreducible. The number of irreducible components of $C_\xi \otimes_{\mathbb{K}} \Omega_1$ for an algebraically closed field Ω_1 does not depend on the homomorphism $\mathbb{K} \to \Omega_1$, so V_x is irreducible. This proves (8.48).

Applying (8.48) to $x = \xi_\Omega$, we see that $(C_\xi)_\Omega$ is irreducible. Since Γ is an irreducible component of $(C_\xi)_\Omega$,

$$((C_\xi)_\Omega)_{\mathrm{red}} = \Gamma. \tag{8.50}$$

Step 4. In this step we construct T, $\mathcal{C}$, W, f, and g, and prove (1)(a) and (1)(c).

Since Ω is algebraic over $\mathbb{K}$, by (8.50) there exist a finite extension $\mathbb{K}' \subset \Omega$ of $\mathbb{K}$ and a geometrically integral curve $C' \subset X_{\mathbb{K}'}$ with $C' \otimes_{\mathbb{K}'} \Omega = \Gamma$. Let $\phi : Y \to X$ be the normalization of X in $\mathbb{K}'$, and let $\mathcal{C}_Y \subset X \times Y$ be the closure of C' under $X_{\mathbb{K}'} \to X \times Y$. Since $\xi_\Omega \in \Gamma$, $\mathcal{C}_Y$ contains the graph of ϕ. By generic flatness and [Sta, Tags 0578 and 0559], there exists a non-empty open subset $Y^\circ \subset Y$ such that $\mathcal{C}_Y \cap (X \times Y^\circ)$ is flat over Y° with geometrically integral fibers. Let $\kappa : Y^\circ \to \mathrm{Hilb}(X)$ be the classifying morphism of the family $\mathcal{C}_Y \cap (X \times Y^\circ) \to Y^\circ$ [Kol96, Theorem I.1.4], let $\overline{T} \subset \mathrm{Hilb}(X)$ be the closure of $\kappa(Y^\circ)$ with the reduced structure, let $T \subset \overline{T}$ be a non-empty smooth open subset, and let $\mathcal{C} \subset X \times T$ be the pullback of the universal family. Then T is quasi-projective, $T \to \mathrm{Hilb}(X)$ is an immersion, and $\mathcal{C}$ is flat and projective over T. Since κ maps the generic point of Y° to the generic point of T, the generic fiber of $\mathcal{C} \to T$ becomes C' after a field extension, so it is geometrically integral. By [Sta, Tags 0578 and 0559], possibly replacing T with a non-empty open subset, any fiber of $\mathcal{C} \to T$ is geometrically integral. Since $\mathcal{C}$ is flat over the integral scheme T with integral generic fiber, $\mathcal{C}$ is integral, and we let $\zeta_\mathcal{C}$ be the generic point of $\mathcal{C}$. For a point $t \in T(\Omega_1)$ with Ω_1 a field, let $C_t \subset X_{\Omega_1}$ be the fiber of $\mathcal{C}$ over t.

Let $h : \mathcal{C} \to X$ be the first projection, and let $\psi : \kappa^{-1}(T) \to \mathcal{C}$ be the morphism sending y' to $(\phi(y'), \kappa(y'))$, which is defined since $\mathcal{C}_Y$ contains the graph of ϕ. Then $h \circ \psi = \phi|_{\kappa^{-1}(T)}$ is dominant, so h is dominant.

Let $y'_1 \in \kappa^{-1}(T)(\Omega)$ be the point defined by $\mathbb{K}' \subset \Omega$, and let $t_1 := \kappa(y'_1) \in T(\Omega)$, which maps to the generic point of T. Then

$$C_{t_1} = C' \otimes_{\mathbb{K}'} \Omega = \Gamma = (V_{\xi_\Omega})_{\mathrm{red}}$$

by (8.50). Let Ω_2 be an algebraic closure of the function field of Γ, let $x' \in X(\Omega_2)$ be the point defined by the generic point of Γ, and let $t \in T(\Omega_2)$ be the point induced by t_1. Since C_{t_1} is the base change of the integral generic fiber of $\mathcal{C} \to T$, $(x', t) \in \mathcal{C}(\Omega_2)$ maps to $\zeta_\mathcal{C}$, and $(V_{x'})_{\mathrm{red}} = \Gamma \times_\Omega \Omega_2 = C_t$ by (8.48) with $x := \xi_\Omega$ and $C_1 := \Gamma$.

Let $\mathcal{A} := \mathcal{C} \times_T \mathcal{C}$ and $\mathcal{B} := \{(z, u) \in X \times \mathcal{C} \mid (z, h(u)) \in V\}$, closed subsets of $X \times \mathcal{C}$ whose fibers over $u = (x', t)$ are C_t and $V_{x'}$. By the last paragraph, the fibers $\mathcal{A}_{\zeta_\mathcal{C}}$ and $\mathcal{B}_{\zeta_\mathcal{C}}$ of $\mathcal{A}$ and $\mathcal{B}$ over $\zeta_\mathcal{C}$ satisfy

$$(\mathcal{A}_{\zeta_\mathcal{C}})_{\mathrm{red}} = (\mathcal{B}_{\zeta_\mathcal{C}})_{\mathrm{red}}.$$

By Chevalley's theorem the images in $\mathcal{C}$ of $\mathcal{A} \backslash \mathcal{B}$ and $\mathcal{B} \backslash \mathcal{A}$ are constructible and do not contain $\zeta_\mathcal{C}$, so there exists a dense open subset $\mathcal{C}^\circ \subset \mathcal{C}$ such that

$$C_t = (V_{x'})_{\mathrm{red}} \quad \text{for any point } (x', t) \in \mathcal{C}^\circ(\Omega_2) \text{ with } \Omega_2 \text{ an algebraically closed field.} \tag{8.51}$$

If $(x', t), (x', t') \in \mathcal{C}^\circ(\Omega_2)$, then $C_t = C_{t'}$ by (8.51), so $t = t'$ since $T \to \mathrm{Hilb}(X)$ is an immersion. Thus the generic fiber of $h|_{\mathcal{C}^\circ}$ is finite. Since h is dominant, $\dim \mathcal{C} = 3$ and $\dim T = 2$. Then $\dim(\mathcal{C} \backslash \mathcal{C}^\circ) \le 2$, so the geometric generic fiber of h is contained in $\mathcal{C}^\circ$ and consists of one point. Since the number of points of the geometric generic fiber of h is the separable degree of $k(\mathcal{C})/k(X)$,

$$k(\mathcal{C})/k(X) \text{ is finite and purely inseparable.} \tag{8.52}$$

Since $h \circ \psi$ is dominant and $\dim \mathcal{C} = 3$, ψ is dominant, so ψ maps the generic point of Y° to $\zeta_\mathcal{C}$. The generic fiber of $\mathcal{C} \to T$ is geometrically integral, so $\mathcal{C} \to T$ is smooth at $\zeta_\mathcal{C}$. Since

$$\mathcal{C}_Y \cap (X \times \kappa^{-1}(T)) = \mathcal{C} \times_T \kappa^{-1}(T)$$

and the point of C' defined by ξ maps to $\zeta_\mathcal{C}$, it is a smooth point of the curve C' over $\mathbb{K}'$, so ξ_Ω is a smooth point of $\Gamma = C' \otimes_{\mathbb{K}'} \Omega$, that is, $\mu = 1$. By (8.40),

$$0 < \ell < \delta_q \qquad \text{and} \qquad b\ell < M_q. \tag{8.53}$$

Let N_L be a positive integer such that $N_L L$ is Cartier. Since $\mathcal{C}$ is flat over T, $N_L L \cdot C_t$ is constant in t, so $L \cdot C_t = L_\Omega \cdot \Gamma = \ell$ for any closed point $t \in T$.

The integral closed subscheme $\mathcal{C} \subset X \times T$ is flat and projective over the integral surface T, the geometric fibers C_t are integral curves, h is dominant, $|bL|$ defines a birational map, and $\deg(bL|_{C_t^\nu}) = b\ell < M_q$ for any closed point $t \in T$, where C_t^ν is the normalization of C_t. Since $M_q \geq 2$ is an integer and $p > (M_q - 2)(M_q - 3) + 1$, by Lemma 8.6 with $M := bL$ and $e := M_q - 1$, possibly replacing T with a non-empty open subset, T and $W := \mathcal{C}^\nu$ are smooth, where $\mathcal{C}^\nu$ is the normalization of $\mathcal{C}$, the morphism $f : W \to T$ is smooth and projective with geometrically integral fibers, and $F_t \to C_t$ is the normalization for any closed point $t \in T$. Let $g : W \to X$ be the composite of $W \to \mathcal{C}$ and h. Then g is dominant and generically finite, $g(F_t) = C_t$, the image of (g, f) is $\mathcal{C}$, and $k(W) = k(\mathcal{C})$. Since (X, B) is a projective $\mathbb{Q}$-factorial pair of dimension 3 over k with $p > 5$ and B is a $\mathbb{Q}$-divisor, T, f, and g are as in Set-up 8.4. Moreover $\deg(g^*L|_{F_t}) = L \cdot C_t = \ell$ for any closed point $t \in T$. Thus (1)(a) holds, and (1)(c) holds by (8.53). Replacing T with a non-empty open subset does not affect them.

Step 5. In this step we prove (2). Let $\eta > 0$. Since (2) for $\min\{\eta, 1\}$ implies (2) for η, we may assume $\eta \leq 1$. Since $R = r'q \geq 4rq$ and $r^3D^3 > 216$, we have $L^3 = R^3D^3 > 8q^3$. By [Kol96, Theorem VI.2.15], $h^0(X, \mathcal{O}_X(nL)) = \frac{L^3}{6}n^3 + O(n^2)$. Let n be a positive multiple of N_0 with

$$n \geq 4(4bR + 1), \qquad \frac{2}{a_n} < \frac{\eta}{2}, \qquad \text{and} \qquad h^0(X, \mathcal{O}_X((n-b)L)) > \binom{2qn+2}{3}.$$

Then for any smooth closed point $x \in X$ there exists

$$u_x \in H^0(X, \mathfrak{m}_x^{2qn} \cdot \mathcal{O}_X((n-b)L)) \backslash \{0\}. \tag{8.54}$$

Let $s_1, \ldots, s_{N_n}$ be a basis of $\mathcal{W}_n(2q)$ over $\mathbb{K}$. By generic flatness and upper semicontinuity, there exists a non-empty open subset $X^\circ \subset X$ such that the coefficients of $s_1, \ldots, s_{N_n}$ in $\mathbb{K}$ are regular on X°, and for any closed point $x \in X^\circ$ the specializations $s_1(x), \ldots, s_{N_n}(x)$ form a basis of $\mathcal{W}_n(2q; x)$. By (8.44) there exists an open subset $V^\circ \subset V$ with $V^\circ \neq \emptyset$ such that $s_1, \ldots, s_{N_n}$ belong to the a_n-th power of the ideal of V at any point of V°. Since $\dim_\xi \mathrm{Bs}(\mathcal{W}_n(2q)) = 1$ by (8.43) and **Step 1**, by the upper semicontinuity of fiber dimension, possibly shrinking X°, we have $\dim_x \mathrm{Bs}(\mathcal{W}_n(2q; x)) \leq 1$ for any closed point $x \in X^\circ$. Possibly replacing T with a non-empty open subset, since $\mathcal{C} \to T$ is open, we may assume that for any closed point $t \in T$ we have $C_t \not\subset \operatorname{Sing} X \cup \operatorname{Supp}(B + E_D)$ and there exists a closed point $x \in C_t \backslash \operatorname{Sing} X$ such that

(a) $(x, t) \in \mathcal{C}^\circ$.
(b) $x \in X^\circ$.
(c) $V^\circ \cap (V_x \times \{x\}) \neq \emptyset$.

Fix a closed point $t \in T$ and such a point x. By (8.51), $(V_x)_{\mathrm{red}} = C_t$. The points $z \in C_t$ with $(z, x) \in V^\circ$ form a non-empty open subset of C_t, and $s_1(x), \ldots, s_{N_n}(x)$ vanish to order at least a_n at any such z. Thus any member of $|\mathcal{W}_n(2q; x)|$ has multiplicity at least a_n along C_t. By (8.54), $u_x \cdot H^0(X, \mathcal{O}_X(bL)) \subset \mathcal{W}_n(2q; x)$, so $|\mathcal{W}_n(2q; x)|$ defines a birational map. Let Fix_n be the fixed part of $|\mathcal{W}_n(2q; x)|$. Since $\operatorname{Supp} \mathrm{Fix}_n \subset \mathrm{Bs}(\mathcal{W}_n(2q; x))$ and $\dim_x \mathrm{Bs}(\mathcal{W}_n(2q; x)) \leq 1$, no component of Fix_n contains x, so $C_t \not\subset \operatorname{Supp} \mathrm{Fix}_n$. The linear system $|\mathcal{W}_n(2q; x)| - \mathrm{Fix}_n$ has no fixed prime divisor and defines a birational map, so a general member S' of it is irreducible by [Sta, Tag 0G4F]. Moreover S' is reduced, since by Bertini it is reduced on the open subset of the non-singular locus of X on which the map is an isomorphism onto the image, and since no prime divisor is a fixed component. Let $\Delta_n := \mathrm{Fix}_n + S' \in |\mathcal{W}_n(2q; x)|$. Then $S' \not\subset \operatorname{Supp} \mathrm{Fix}_n$, S' is the only prime divisor in $\operatorname{Supp} \Delta_n$ containing C_t, and $\operatorname{mult}_{C_t} S' = \operatorname{mult}_{C_t} \Delta_n \geq a_n$.

Near the generic point of C_t, X is smooth and $B = 0$. Let $c > 0$ be the lc threshold of Δ_n with respect to (X, B) near the generic point of C_t. Blowing up C_t near the generic point of C_t yields a prime divisor with log discrepancy $2 - c \operatorname{mult}_{C_t} \Delta_n$ with respect to $(X, B + c\Delta_n)$, so $c \leq \frac{2}{a_n} < \frac{\eta}{2}$. Since the coefficient of S' in $c\Delta_n$ is $c < 1$, the lc places of $(X, B + c\Delta_n)$ near the generic point of C_t have center C_t, and C_t is an lc center. Let $G := c\Delta_n \sim_{\mathbb{Q}} cnRD$. Since

$a_n > \tau_q n$ and $\ell < \delta_q = \frac{\tau_q}{16q}$,

$$cn \le \frac{2n}{a_n} < \frac{2}{\tau_q} \qquad \text{and} \qquad G \cdot C_t = cn\ell < \frac{2}{\tau_q}\delta_q = \frac{1}{8q}.$$

Let

$$\eta_t := \min\left\{\frac{\eta}{2}, \frac{1}{8q} - G \cdot C_t\right\} > 0.$$

Since (X, B) is klt, by Lemma 8.10 with $D = H_D + E_D$, $C := C_t$, G, $\gamma := cnR$, and η_t, let G_t be the $\mathbb{Q}$-divisor G' there. Then $(X, B + G_t)$ is lc near the generic point of C_t and has exactly one lc place there, whose center is C_t, $G_t \cdot C_t < G \cdot C_t + \eta_t \le 1/(8q)$, and, since $\mathrm{mult}_S G \in \{0, c\}$ for any prime divisor S containing C_t, $\mathrm{mult}_S G_t < c + \eta/2 < \eta$. Thus (2) holds.

Step 6. In this step we prove $e \ge 1$. Suppose that $k(W) = k(X)$. Then g is birational. Applying (2) with $\eta = 1$, we may assume that for any closed point $t \in T$ there exists a $\mathbb{Q}$-divisor $G_t \ge 0$ with $G_t \cdot C_t < 1/(8q) < 1/(6q)$ such that $(X, B + G_t)$ is lc near the generic point of C_t and has exactly one lc place there, whose center is C_t. Here G_t is $\mathbb{Q}$-Cartier since X is $\mathbb{Q}$-factorial. Since $L = RD$, $\lambda = \ell/R$. Since $R = r'q \ge 8q$ and $0 < \ell < \delta_q < 1/4$, we have $0 < \lambda < 1/(32q)$.

Let $N := 4n_0 + 1$, where n_0 is the integer of Set-up 8.1. By Theorem 6.6 with b_0, $|jqD|$ defines a birational map for any integer $j \ge N$, since $jq \ge 4qn_0 + 2$ and qB is a divisor. Since $N = u_0 b_0 + 2u_1 - 5$ and $r' \ge 2$, we have $N/r' < u_0 b_0/r' + u_1 < w/(1024q)$ by Construction-Proposition 8.13(1). Thus

$$2q(1 + 4qN)\lambda = \frac{2(1 + 4qN)\ell}{r'} < 1 + 8q\ell\frac{N}{r'} < 1 + w < p.$$

Since (X, B) is $\frac{1}{q}$-lc and $B \in I_q\backslash\{1\}$, by Corollary 8.8, $D \cdot C_t \ge \frac{1}{qJ_q}$ for a general closed point $t \in T$. But, since $r \ge J_q$,

$$D \cdot C_t = \lambda < \frac{1}{r'q} \le \frac{1}{4rq} \le \frac{1}{4qJ_q}.$$

Thus $[k(W) : k(X)] > 1$. By (8.52), $[k(W) : k(X)] = p^e$ with $e \ge 1$, which proves (1)(b). □

Proposition 8.16. *Notation and conditions as in Construction-Proposition 8.15. After replacing T with a non-empty open subset, the following hold.*

(1) The morphism g factors as

$$W = Y_0 \xrightarrow{\rho_0} Y_1 \xrightarrow{\rho_1} \cdots \xrightarrow{\rho_{e-1}} Y_e \xrightarrow{\pi} X,$$

where every Y_i is a smooth threefold, every ρ_i is finite and purely inseparable of degree p, and π is birational. For every $i < e$, the saturation $\mathcal{F}_i$ of the kernel of the differential of $Y_i \to X$ is a regular foliation of rank 1 on Y_i (see [Pos26, Sections 2.4 and 2.5]*), and ρ_i is the quotient of Y_i by $\mathcal{F}_i$. Let $\varphi : W \to Y_e$ be the composite morphism. Then there exists a projective morphism $a : Y_e \to T^{(p^e)}$ with $a \circ \varphi = \mathrm{Fr}^e_{T/k} \circ f$, where $T^{(p^e)}$ is the base change of T by the p^e-th power map of k and $\mathrm{Fr}^e_{T/k} : T \to T^{(p^e)}$ is the relative Frobenius morphism.*

(2) Let B_{Y_e} be the $\mathbb{Q}$-divisor with $K_{Y_e} + B_{Y_e} = \pi^* D$ and $\pi_* K_{Y_e} = K_X$, and write

$$B_{Y_e} = B^{\ge 0}_{Y_e} - B^{<0}_{Y_e},$$

where $B^{\ge 0}_{Y_e}, B^{<0}_{Y_e} \ge 0$ have no common components. Then every component of $B^{<0}_{Y_e}$ is π-exceptional and

$$(\varphi^* B^{<0}_{Y_e}) \cdot F_t \ge p - 3 - \lambda$$

for every closed point $t \in T$.

Proof. We divide the proof into five steps. For a scheme Z over k and an integer $i \geq 0$, we let $Z^{(p^i)}$ be the base change of Z by the p^i-th power map of k and $\mathrm{Fr}^i_{Z/k} : Z \to Z^{(p^i)}$ the relative Frobenius morphism, which is finite and a homeomorphism. Let $f^{(p^i)} : W^{(p^i)} \to T^{(p^i)}$ be the base change of f, so that

$$f^{(p^i)} \circ \mathrm{Fr}^i_{W/k} = \mathrm{Fr}^i_{T/k} \circ f.$$

Since f is flat and projective and T is connected, the degree $M \cdot F_t$ of a $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor M on W does not depend on the closed point $t \in T$.

Step 1. In this step we construct the morphisms in (1) and make every Y_i smooth.

For any integer $i \in [0, e]$, the morphism $(g, \mathrm{Fr}^i_{W/k}) : W \to X \times W^{(p^i)}$ is finite, since $\mathrm{Fr}^i_{W/k}$ is finite. We let Y_i be the normalization of the image of $(g, \mathrm{Fr}^i_{W/k})$ and $\varphi_i : W \to Y_i$ the induced finite morphism. Thus Y_i is finite over $W^{(p^i)}$, $Y_0 = W$, $\varphi_0 = \mathrm{id}_W$, and, as subfields of $k(W)$,

$$k(Y_i) = k(X) \cdot k(W)^{p^i}.$$

Let $g_i : Y_i \to X$ be the morphism induced by the first projection, and $a_i : Y_i \to T^{(p^i)}$ the composite of the morphism $Y_i \to W^{(p^i)}$ induced by the second projection with $f^{(p^i)}$. Then

$$g = g_i \circ \varphi_i \quad \text{and} \quad a_i \circ \varphi_i = \mathrm{Fr}^i_{T/k} \circ f.$$

For $i < e$, the morphism $\mathrm{id}_X \times \mathrm{Fr}_{W^{(p^i)}/k}$ maps the image of W in $X \times W^{(p^i)}$ onto the image of W in $X \times W^{(p^{i+1})}$. Since Y_i is normal, it induces a morphism $\rho_i : Y_i \to Y_{i+1}$ with $\rho_i \circ \varphi_i = \varphi_{i+1}$ and $g_{i+1} \circ \rho_i = g_i$. In particular, $\varphi = \varphi_e$. The morphism ρ_i is finite since Y_i is finite over $W^{(p^{i+1})}$. Moreover, $a := a_i$ for $i = e$ is projective, since $Y_e \to W^{(p^e)}$ is finite and $f^{(p^e)}$ is projective.

Since $k(W)/k(X)$ is purely inseparable of degree p^e, $k(W)^{p^e} \subset k(X)$. Thus $k(Y_e) = k(X)$ and $\pi := g_e$ is birational. Since $k(Y_i)^p \subset k(Y_{i+1})$ and Y_{i+1} is normal, the p-th power of any local section of $(\rho_i)_* \mathcal{O}_{Y_i}$ is a local section of $\mathcal{O}_{Y_{i+1}}$. Hence the finite surjective morphism ρ_i is a homeomorphism. Since Y_{i+1} is normal and ρ_i is finite,

$$\mathcal{O}_{Y_{i+1}} = (\rho_i)_* \mathcal{O}_{Y_i} \cap k(Y_{i+1}). \tag{8.55}$$

Since Y_i is normal, $\dim \mathrm{Sing}(Y_i) \leq 1$, so the closure $\overline{a_i(\mathrm{Sing}(Y_i))}$ is a proper closed subset of the surface $T^{(p^i)}$. For a non-empty open subset $T^\circ \subset T$, the normalization of the image of $(g, \mathrm{Fr}^i_{W/k})|_{f^{-1}(T^\circ)}$ is the open subset $a_i^{-1}((T^\circ)^{(p^i)}) \subset Y_i$. Thus, possibly replacing T with

$$T \setminus \bigcup_{i=0}^{e} (\mathrm{Fr}^i_{T/k})^{-1}\big(\overline{a_i(\mathrm{Sing}(Y_i))}\big),$$

we may assume that every Y_i is a smooth threefold.

Step 2. In this step we use the Hilbert scheme to show that $\mathrm{rank}(dg) \geq 2$.

For $i \in [0, e]$, let

$$dg_i : T_{Y_i} \to \mathcal{H}om(g_i^* \Omega_X, \mathcal{O}_{Y_i})$$

be the differential of g_i, that is, the dual of $g_i^* \Omega_X \to \Omega_{Y_i}$, and let $dg := dg_0$. By $\mathrm{rank}(dg_i)$ we mean the rank of dg_i at the generic point of Y_i. For a closed point $y \in W$, let $dg_y : T_{W,y} \to T_{X,g(y)}$ and $df_y : T_{W,y} \to T_{T,f(y)}$ be the maps of Zariski tangent spaces. At the generic point of Y_i, dg_i is the restriction map

$$\mathrm{Der}_k(k(Y_i)) \to \mathrm{Der}_k(k(X), k(Y_i)),$$

whose kernel is $\mathrm{Der}_{k(X)}(k(Y_i))$. Since $\dim_{k(Y_i)} \mathrm{Der}_k(k(Y_i)) = 3$,

$$\mathrm{rank}(dg_i) = 3 - \dim_{k(Y_i)} \mathrm{Der}_{k(X)}(k(Y_i)). \tag{8.56}$$

Claim 8.17. *Let $t \in T$ be a closed point and $v \in T_{T,t}$. Assume that for a general closed point $y \in F_t$ there exists $v_y \in T_{W,y}$ with $df_y(v_y) = v$ and $dg_y(v_y) = 0$. Then $v = 0$.*

Proof. Let $\mathcal{I} \subset \mathcal{O}_X$ be the ideal sheaf of C_t. We view v as a morphism $\operatorname{Spec} k[\epsilon] \to T$ with image t, and let

$$\mathcal{C}_v := \mathcal{C} \times_T \operatorname{Spec} k[\epsilon] \subset X \times \operatorname{Spec} k[\epsilon].$$

By [Kol96, Theorem I.2.8], the tangent map of the morphism $T \to \operatorname{Hilb}(X)$ defined by $\mathcal{C}$ sends v to the homomorphism $h : \mathcal{I}/\mathcal{I}^2 \to \mathcal{O}_{C_t}$ defined by $h(\alpha) := \alpha'|_{C_t}$, where α is a local section of $\mathcal{I}$ and $\alpha + \epsilon\alpha'$ is a local section of the ideal sheaf of $\mathcal{C}_v$. Since $T \to \operatorname{Hilb}(X)$ is an immersion, $h \neq 0$ if $v \neq 0$.

Let $y \in F_t$ be a general closed point, $x := g(y)$, and $y_\epsilon : \operatorname{Spec} k[\epsilon] \to W$ the morphism defined by v_y. Since $dg_y(v_y) = 0$, $g \circ y_\epsilon$ is the constant morphism to x. Since $(g, f)(W) \subset \mathcal{C}$ and $f \circ y_\epsilon = v$, the morphism

$$(x, \mathrm{id}_{\operatorname{Spec} k[\epsilon]}) : \operatorname{Spec} k[\epsilon] \to X \times \operatorname{Spec} k[\epsilon]$$

factors through $\mathcal{C}_v$. Thus $\alpha(x) + \epsilon\alpha'(x) = 0$ for any local section α of $\mathcal{I}$ near x and any local section $\alpha + \epsilon\alpha'$ of the ideal sheaf of $\mathcal{C}_v$ near x, so $h(\alpha)$ vanishes at x. Since C_t is integral and x is a general point of C_t, $h = 0$. Hence $v = 0$. □

Let $t \in T$ be a general closed point and $y \in F_t$ a general closed point. Since $g|_{F_t} : F_t \to C_t$ is birational, $dg_y(T_{F_t,y}) \neq 0$. Suppose that $\operatorname{rank}(dg) \leq 1$. Since $g(y)$ is a non-singular point of X, $\operatorname{rank}(dg_y) \leq 1$, so $\dim \operatorname{Ker}(dg_y) \geq 2$ and $\operatorname{Ker}(dg_y) \cap T_{F_t,y} = 0$. Since $T_{F_t,y} = \operatorname{Ker}(df_y)$ and $\dim T_{T,t} = 2$, we have $df_y(\operatorname{Ker}(dg_y)) = T_{T,t}$. By Claim 8.17, $T_{T,t} = 0$, a contradiction. Thus $\operatorname{rank}(dg) \geq 2$.

Step 3. In this step we prove (1).

Let $m := \min\{j \in [0, e] \mid k(Y_j) = k(X)\}$, and let $i < m$. Since $g = g_i \circ \varphi_i$,

$$\operatorname{rank}(dg_i) \geq \operatorname{rank}(dg) \geq 2.$$

Since $k(Y_i)/k(X)$ is a non-trivial purely inseparable extension, it is not separable, so $\operatorname{Der}_{k(X)}(k(Y_i)) \neq 0$ [Jac89, Section 8.16]. By (8.56),

$$\operatorname{rank}(dg_i) = 2 \quad \text{and} \quad \dim_{k(Y_i)} \operatorname{Der}_{k(X)}(k(Y_i)) = 1.$$

The stalk of $\mathcal{F}_i$ at the generic point of Y_i is $\operatorname{Der}_{k(X)}(k(Y_i))$, which is closed under Lie brackets and p-th powers. Thus $\mathcal{F}_i$ is a 1-foliation of rank 1 [Pos26, Definition 2.14 and Remark 2.15]. A local section of $\mathcal{O}_{Y_i}$ is annihilated by $\mathcal{F}_i$ if and only if it is a constant of $\operatorname{Der}_{k(X)}(k(Y_i))$, and by [Jac89, Section 8.17] the field of constants of $\operatorname{Der}_{k(X)}(k(Y_i))$ is

$$k(X) \cdot k(Y_i)^p = k(Y_{i+1}).$$

By (8.55), $\mathcal{O}_{Y_{i+1}}$ is the annihilator of $\mathcal{F}_i$ in $\mathcal{O}_{Y_i}$, that is, ρ_i is the quotient of Y_i by $\mathcal{F}_i$. By [Pos26, Theorem 2.39], $\deg \rho_i = p$. Thus $p^e = [k(W) : k(X)] = p^m$, so $m = e$. Since Y_i and Y_{i+1} are smooth, $\mathcal{F}_i$ is regular [Pos26, Lemma 2.37]. Since $k(Y_i)/k(Y_{i+1})$ is purely inseparable, (1) holds.

Step 4. In this step we bound $(\varphi^* K_{Y_e}) \cdot F_t$ from below.

Since $\mathcal{F}_i$ is regular of rank 1, it is a line bundle. For $i < e$ and a closed point $t \in T$, let

$$\kappa_i := -\deg\big((\varphi_i^* \mathcal{F}_i)|_{F_t}\big),$$

which does not depend on t.

We show $\kappa_0 \geq 1$. Let $t \in T$ be a general closed point. For a general closed point $y \in F_t$, $\operatorname{rank}(dg_y) = 2$ by **Step 3**, and dg_y vanishes on the fiber $\mathcal{F}_0(y)$ of the line subbundle $\mathcal{F}_0 \subset T_W$, so $\mathcal{F}_0(y) = \operatorname{Ker}(dg_y)$. Let

$$\sigma : \mathcal{F}_0|_{F_t} \to T_{T,t} \otimes_k \mathcal{O}_{F_t}$$

be the composite of the inclusion $\mathcal{F}_0|_{F_t} \subset T_W|_{F_t}$ with df. If $\sigma = 0$, then for general y,

$$\operatorname{Ker}(dg_y) \subset \operatorname{Ker}(df_y) = T_{F_t,y},$$

so $dg_y(T_{F_t,y}) = 0$, contrary to **Step 2**. Thus σ is a non-zero map from $\mathcal{F}_0|_{F_t}$ to a trivial bundle of rank 2, and $\kappa_0 \geq 0$. If $\kappa_0 = 0$, then $\mathcal{F}_0|_{F_t} \cong \mathcal{O}_{F_t}$, and since $H^0(F_t, \mathcal{O}_{F_t}) = k$, σ maps a nowhere

vanishing section of $\mathcal{F}_0|_{F_t}$ to $v\otimes 1$ for a non-zero $v\in T_{T,t}$. Thus for a general closed point $y\in F_t$ there exists $v_y\in\mathcal{F}_0(y)=\mathrm{Ker}(dg_y)$ with $df_y(v_y)=v$, contrary to Claim 8.17. Hence $\kappa_0\geq 1$.

Let $1\leq j<e$. By [Pos26, Proposition 2.41], there exists an exact sequence

$$0\to\mathcal{F}_{j-1}\to T_{Y_{j-1}}\xrightarrow{d\rho_{j-1}}\rho_{j-1}^*T_{Y_j}\to\mathcal{F}_{j-1}^{\otimes p}\to 0,$$

whose last term is the Frobenius pullback of the line bundle $\mathcal{F}_{j-1}$. Let

$$\sigma_j:\rho_{j-1}^*\mathcal{F}_j\to\mathcal{F}_{j-1}^{\otimes p}$$

be the composite of $\rho_{j-1}^*\mathcal{F}_j\subset\rho_{j-1}^*T_{Y_j}$ with the last map. If $\sigma_j=0$, then at the generic point of Y_{j-1} the line $\rho_{j-1}^*\mathcal{F}_j$ is contained in the 2-dimensional image of $d\rho_{j-1}$, and $\rho_{j-1}^*dg_j$ vanishes on $\rho_{j-1}^*\mathcal{F}_j$. Since $dg_{j-1}=(\rho_{j-1}^*dg_j)\circ d\rho_{j-1}$, we get $\mathrm{rank}(dg_{j-1})\leq 1$, contrary to **Step 3**. Thus $\sigma_j\neq 0$, so $\varphi_{j-1}^*\sigma_j\neq 0$, and $(\varphi_{j-1}^*\sigma_j)|_{F_t}\neq 0$ for a general closed point $t\in T$. Since $\varphi_{j-1}^*\rho_{j-1}^*=\varphi_j^*$,

$$\kappa_j=p\kappa_{j-1}+z_j,$$

where $z_j\geq 0$ is the degree of the zero divisor of $(\varphi_{j-1}^*\sigma_j)|_{F_t}$. Hence $\kappa_i\geq p^i\kappa_0\geq p^i$ for any $i<e$.

By [Pos26, Proposition 2.41], $\rho_i^*K_{Y_{i+1}}\sim K_{Y_i}+(p-1)K_{\mathcal{F}_i}$ with $\mathcal{O}_{Y_i}(-K_{\mathcal{F}_i})\cong\mathcal{F}_i$. Pulling back by φ_i and restricting to F_t, we obtain

$$(\varphi_{i+1}^*K_{Y_{i+1}})\cdot F_t=(\varphi_i^*K_{Y_i})\cdot F_t+(p-1)\kappa_i.$$

Since F_t is a smooth projective curve and a fiber of the smooth morphism f,

$$K_W\cdot F_t=\deg K_{F_t}\geq -2.$$

Thus

$$(\varphi^*K_{Y_e})\cdot F_t=K_W\cdot F_t+(p-1)\sum_{i=0}^{e-1}\kappa_i\geq -2+(p-1)\sum_{i=0}^{e-1}p^i=p^e-3\geq p-3. \tag{8.57}$$

Step 5. In this step we prove (2).

Let P be a prime divisor on Y_e which is not π-exceptional. Since X is normal and π is birational, $\mathcal{O}_{Y_e,P}=\mathcal{O}_{X,\pi(P)}$. Since $\pi_*K_{Y_e}=K_X$, $\mathrm{mult}_P B_{Y_e}=\mathrm{mult}_{\pi(P)}B\geq 0$. Thus every component of $B_{Y_e}^{<0}$ is π-exceptional. Since $B_{Y_e}^{\geq 0}\geq 0$, $(\varphi^*B_{Y_e}^{\geq 0})\cdot F_t\geq 0$, computing on a general fiber. Since $\varphi^*\pi^*D=g^*D$ and $(g^*D)\cdot F_t=\lambda$, (8.57) implies

$$(\varphi^*B_{Y_e}^{<0})\cdot F_t=(\varphi^*B_{Y_e}^{\geq 0})\cdot F_t-\lambda+(\varphi^*K_{Y_e})\cdot F_t\geq p-3-\lambda.$$

The proposition follows. □

Theorem 8.18. *Notation and conditions as in Set-up 8.1. Assume that $p>P_q^\dagger$. Then $b_0\leq Q_qr$.*

Proof. Suppose that $b_0>Q_qr$. We use the notation of Set-up 8.12, derive a contradiction, and divide the proof into seven steps.

Step 1. In this step we fix the family and the tower.

Let $f:W\to T$, $g:W\to X$, e, and ℓ be as in Construction-Proposition 8.15(1), and let F_t, C_t, and λ be as in Set-up 8.4. Since f is flat and projective and T is connected, $M\cdot F_u$ does not depend on the closed point $u\in T$ for any $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor M on W. Since $L=RD$ and $R=r'q$, we have $\ell=r'q\lambda$. Since $\ell<\delta_q<\frac{1}{8}$,

$$0<\lambda<\frac{1}{8r'q}. \tag{8.58}$$

Let $N:=4n_0+1=u_0b_0+2u_1-5$, where n_0 is the integer of Set-up 8.1. For any integer $j\in[N,2N]$, $jq\geq q(4n_0+1)\geq 4qn_0+2$, and jqD is a Weil divisor since qB is. Since $H^0(X,\mathcal{O}_X(\lfloor b_0D\rfloor))\neq 0$, Theorem 6.6 implies that $|jqD|$ defines a birational map. Since

$$p>P_q^*\geq(U_q+1)^2\geq(2048qu_1+1)^2,$$

we have $w > 2048qu_1$. By the definition of r' in Set-up 8.12,

$$r' \geq 2\gamma_q b_0/w \quad \text{and} \quad r' \geq 4r \geq 8.$$

Since $\gamma_q = 1024qu_0$, (8.58) implies

$$2q(1+4qN)\lambda < \frac{1}{4r'} + \frac{qN}{r'} < \frac{1}{32} + \frac{qu_0b_0}{r'} + \frac{2qu_1}{r'} \leq \frac{1}{32} + \frac{w}{2048} + \frac{qu_1}{4} < \frac{1}{32} + \frac{w}{2048} + \frac{w}{8192} < p.$$

Since (X, B) is $\frac{1}{q}$-lc and qB is a Weil divisor, the hypotheses of Construction-Proposition 8.7 hold with $\epsilon := \frac{1}{q}$ and N. Possibly replacing T with a non-empty open subset, we may assume that the conclusions of Construction-Proposition 8.7 and Proposition 8.16 hold. Let Y_i, ρ_i, $\mathcal{F}_i$, π, and φ be as in Proposition 8.16(1), let ω be a non-zero rational differential 3-form on X with $\mathrm{div}_X(\omega) = K_X$, let $K_{Y_e} := \mathrm{div}_{Y_e}(\pi^*\omega)$, so that $\pi_* K_{Y_e} = K_X$, let B_{Y_e} be the $\mathbb{Q}$-divisor with $K_{Y_e} + B_{Y_e} = \pi^* D$, let $B_{Y_e}^{\geq 0}, B_{Y_e}^{<0} \geq 0$ be the $\mathbb{Q}$-divisors without common components with $B_{Y_e} = B_{Y_e}^{\geq 0} - B_{Y_e}^{<0}$, and let $\varphi_i : W \to Y_i$ be the composite morphisms.

Let $V \to X$ and $W \to V$ be the morphisms of Construction-Proposition 8.7(1). Since $k(W)/k(X)$ is purely inseparable and $k(V)/k(X)$ is separable, $k(V) = k(X)$. Since $W \to V$ and φ are finite, surjective, and purely inseparable, they are homeomorphisms. Since V and Y_e are normal with function field $k(X)$, identifying the underlying topological spaces of V and Y_e with that of W via $W \to V$ and φ, we have

$$\mathcal{O}_V = \mathcal{O}_{Y_e} = \varphi_* \mathcal{O}_W \cap k(X). \tag{8.59}$$

Thus $V = Y_e$, $V \to X$ is π, and $W \to V$ is φ, so $K_V = K_{Y_e}$ in Construction-Proposition 8.7(2) for this ω, $B_V = B_{Y_e}$, and $B_{Y_e}^{\geq 0}$ is the positive part of B_V. Hence, by Construction-Proposition 8.7(2), for any closed points $t \in T$ and $x \in F_t$,

$$\mathrm{mult}_x\big((\varphi^* B_{Y_e}^{\geq 0})|_{F_t}\big) \leq 1 - \frac{1}{q} + \lambda. \tag{8.60}$$

Step 2. In this step we take the projective morphism a from Y_e to a Frobenius twist of T and construct a compactification of π.

For a scheme Z over k, let $Z^{(p^e)}$ be the base change of Z by the p^e-th power map of k and $\mathrm{Fr}^e_{Z/k} : Z \to Z^{(p^e)}$ the relative Frobenius morphism, which is finite and a homeomorphism. By Proposition 8.16(1), there exists a projective morphism $a : Y_e \to T^{(p^e)}$ with $a \circ \varphi = \mathrm{Fr}^e_{T/k} \circ f$. Since $T^{(p^e)}$ is quasi-projective, Y_e is quasi-projective.

Let $\overline{T} \supset T$ be a projective compactification and $Y_e \subset \mathbb{P}^n$ a locally closed embedding. Let $\overline{Y}_e$ be the normalization of the closure of the image of Y_e under the morphism $Y_e \to \mathbb{P}^n \times X \times \overline{T}^{(p^e)}$ defined by the embedding, π, and a, and let $\overline{\pi} : \overline{Y}_e \to X$ and $\overline{a} : \overline{Y}_e \to \overline{T}^{(p^e)}$ be the induced morphisms. Since a is proper, the immersion $Y_e \to \mathbb{P}^n \times X \times T^{(p^e)}$ is proper, hence a closed immersion. Since Y_e is normal,

$$Y_e = \overline{a}^{-1}(T^{(p^e)}),$$

and $\overline{\pi}$ is a projective birational morphism extending π. Let $B_{\overline{Y}_e}$ be the $\mathbb{Q}$-divisor with

$$K_{\overline{Y}_e} + B_{\overline{Y}_e} = \overline{\pi}^* D \quad \text{and} \quad \overline{\pi}_* K_{\overline{Y}_e} = K_X.$$

Then $B_{\overline{Y}_e}|_{Y_e} = B_{Y_e}$.

Below, replacing T with a non-empty open subset T° means replacing W with $f^{-1}(T^\circ)$ and every Y_i with the open subset $\varphi_i(f^{-1}(T^\circ))$, so that Y_e is replaced with

$$a^{-1}((T^\circ)^{(p^e)}) = \overline{a}^{-1}((T^\circ)^{(p^e)}),$$

while $\overline{T}$ and $\overline{Y}_e$ are kept.

There exist finitely many π-exceptional prime divisors and finitely many components of B_{Y_e}. Let $\Lambda \subset X$ be the image of the exceptional locus of $\overline{\pi}$. Since X is normal, $\dim \Lambda \leq 1$ by Zariski's main theorem. Since $T \to \mathrm{Hilb}(X)$ is an immersion, $C_{t_1} \neq C_{t_2}$ for any closed points $t_1 \neq t_2$ in T,

so $C_t \subset \Lambda$ for only finitely many t. Possibly replacing T with a non-empty open subset, we may assume the following.

(i) Any prime divisor on Y_e which is π-exceptional or a component of B_{Y_e} dominates $T^{(p^e)}$ via a.
(ii) For any closed point $t \in T$, $C_t \not\subset \Lambda$.

By (ii), $\overline{\pi}$ is an isomorphism over a neighborhood of the generic point of C_t. Since $\pi(\varphi(F_t)) = C_t$, the generic point of the curve $\varphi(F_t)$ is the only point of $\overline{Y}_e$ over the generic point of C_t.

Step 3. In this step we fix the $\mathbb{Q}$-divisor G on X.

By Construction-Proposition 8.15(2) with $\eta := \frac{1}{4p^e}$, possibly replacing T with a non-empty open subset, for any closed point $t \in T$ there exists a $\mathbb{Q}$-divisor $G_t \geq 0$ as in Construction-Proposition 8.15(2). Let G'_t be the sum of the components of G_t containing C_t, with the coefficients of G_t. Since $G_t - G'_t \geq 0$ does not contain C_t, the following hold.

(a) $(X, B + G'_t)$ is not klt at the generic point of C_t, since $G'_t = G_t$ near the generic point of C_t and $(X, B + G_t)$ has an lc place with center C_t.
(b) Every coefficient of G'_t is less than $\frac{1}{4p^e}$.
(c) The degree $\beta_t := (g^*G'_t) \cdot F_t$ satisfies $0 \leq \beta_t < \frac{1}{8q}$. The upper bound holds since $(g^*(G_t - G'_t))|_{F_t} \geq 0$ and $(g^*G_t) \cdot F_t = G_t \cdot C_t < \frac{1}{8q}$, as $F_t \to C_t$ is birational. The lower bound holds since $\beta_t = (g^*G'_t) \cdot F_u$ for any closed point $u \in T$ (**Step 1**) and $F_u \not\subset g^{-1}(\operatorname{Supp} G'_t)$ for a general closed point $u \in T$.

Fix two closed points $t_1 \neq t_2$ in T, and let $G := G'_{t_1} + G'_{t_2}$. Then $(X, B + G)$ is not klt at the generic points of C_{t_1} and C_{t_2}, every coefficient of G is less than $\frac{1}{2p^e}$, and

$$(g^*G) \cdot F_u = \beta := \beta_{t_1} + \beta_{t_2} < \frac{1}{4q} \quad \text{for any closed point } u \in T. \tag{8.61}$$

Step 4. In this step we show that $f(\operatorname{Nklt}(W, \Delta_W))$ is finite for the $\mathbb{Q}$-divisor Δ_W below.

Let

$$\Delta_{Y_e} := B_{Y_e}^{\geq 0} + \pi^*G \quad \text{and} \quad \Delta_W := \varphi^*\Delta_{Y_e} = \varphi^*B_{Y_e}^{\geq 0} + g^*G.$$

Then $\Delta_{Y_e} \geq 0$ and $\Delta_W \geq 0$. A prime divisor on W is horizontal if it dominates T and vertical otherwise. A vertical prime divisor is disjoint from F_u for general u, so the degree on any F_u of a vertical prime divisor is 0 (**Step 1**). For a vertical prime divisor P on W, $a(\varphi(P)) = \operatorname{Fr}^e_{T/k}(f(P))$ is not dense, so by (i), $\varphi(P)$ is neither π-exceptional nor a component of B_{Y_e}. Write $g^*G = G^{\mathrm{h}}_W + G^{\mathrm{v}}_W$ and $\Delta_W = \Delta^{\mathrm{h}}_W + \Delta^{\mathrm{v}}_W$ for the horizontal and vertical parts. Then

$$\Delta^{\mathrm{h}}_W = \varphi^*B_{Y_e}^{\geq 0} + G^{\mathrm{h}}_W, \quad \Delta^{\mathrm{v}}_W = G^{\mathrm{v}}_W, \quad \text{and} \quad G^{\mathrm{h}}_W \cdot F_u = \beta.$$

Let $u \in T$ be a closed point with $F_u \not\subset \operatorname{Supp} \Delta^{\mathrm{h}}_W$. All but finitely many closed points $u \in T$ satisfy $F_u \not\subset \operatorname{Supp} \Delta^{\mathrm{h}}_W$. Since $G^{\mathrm{h}}_W|_{F_u} \geq 0$ and $\deg(G^{\mathrm{h}}_W|_{F_u}) = \beta$, (8.60), (8.58), and (8.61) imply that for any $x \in F_u$,

$$\operatorname{mult}_x(\Delta^{\mathrm{h}}_W|_{F_u}) \leq 1 - \frac{1}{q} + \lambda + \beta < 1 - \frac{1}{q} + \frac{1}{8r'q} + \frac{1}{4q} < 1. \tag{8.62}$$

Let P be a vertical prime divisor on W and $P_Y := \varphi(P)$. Since P_Y is not π-exceptional, $\mathcal{O}_{Y_e, P_Y} = \mathcal{O}_{X, \pi(P_Y)}$, so $\operatorname{mult}_{P_Y} \pi^*G = \operatorname{mult}_{\pi(P_Y)} G < \frac{1}{2p^e}$. Since φ is a homeomorphism,

$$\varphi^*P_Y = m_P P \quad \text{with} \quad m_P \leq \deg \varphi = p^e.$$

Hence

$$\operatorname{mult}_P \Delta_W = m_P \operatorname{mult}_{P_Y} \pi^*G < \frac{1}{2}. \tag{8.63}$$

We show that (W, Δ_W) is klt at the generic point of any closed subvariety whose image in T is positive-dimensional. For a horizontal prime divisor P, a general closed point $u \in T$, and

$x \in P \cap F_u$, (8.62) implies $\operatorname{mult}_P \Delta_W \le \operatorname{mult}_x(\Delta_W^{\mathrm{h}}|_{F_u}) < 1$. For a vertical prime divisor P, $\operatorname{mult}_P \Delta_W < \frac{1}{2}$ by (8.63).

Let $\Gamma \subset W$ be an integral curve such that $f(\Gamma)$ is a curve, and let $\Sigma := f^{-1}(f(\Gamma))$. Since the fibers of $\Sigma \to f(\Gamma)$ are integral, Σ is a prime divisor, it is the only vertical prime divisor containing Γ, and Σ is smooth near the generic point of Γ since f is smooth. Let $c_\Sigma := \operatorname{mult}_\Sigma \Delta_W$, so $c_\Sigma < \frac{1}{2}$ by (8.63) ($c_\Sigma = 0$ if $\Sigma \not\subset \operatorname{Supp} \Delta_W$), and let c_Γ be the coefficient of Γ in $\Delta_W^{\mathrm{h}}|_\Sigma$. For a general closed point $u \in f(\Gamma)$ and $x \in \Gamma \cap F_u$, Σ is smooth at x and $F_u \ne \Gamma$, so by (8.62),

$$c_\Gamma \le c_\Gamma(\Gamma \cdot F_u)_x \le (\Delta_W^{\mathrm{h}}|_\Sigma \cdot F_u)_x = \operatorname{mult}_x(\Delta_W^{\mathrm{h}}|_{F_u}) < 1,$$

where $(\cdot)_x$ is the local intersection number at x on Σ. Let $\mu > 1$ be a rational number with $\mu c_\Gamma < 1$ and $\mu c_\Sigma < 1$. There exists an open neighborhood $W_\Gamma^\circ \subset W$ of the generic point of Γ such that $\Sigma \cap W_\Gamma^\circ$ and $\Gamma \cap W_\Gamma^\circ$ are smooth,

$$\Delta_W|_{W_\Gamma^\circ} = c_\Sigma(\Sigma \cap W_\Gamma^\circ) + \Delta_W^{\mathrm{h}}|_{W_\Gamma^\circ}, \quad \text{and} \quad \Delta_W^{\mathrm{h}}|_{\Sigma \cap W_\Gamma^\circ} = c_\Gamma(\Gamma \cap W_\Gamma^\circ).$$

Then the pair

$$(\Sigma \cap W_\Gamma^\circ, \mu \Delta_W^{\mathrm{h}}|_{\Sigma \cap W_\Gamma^\circ})$$

is lc. By the lc inversion of adjunction [BMP+23, Corollary 10.1], the pair

$$(W_\Gamma^\circ, \Sigma \cap W_\Gamma^\circ + \mu \Delta_W^{\mathrm{h}}|_{W_\Gamma^\circ})$$

is lc near $\Sigma \cap W_\Gamma^\circ$. Since $(W_\Gamma^\circ, 0)$ is klt and log discrepancies are affine functions of the boundary, the pair

$$(W_\Gamma^\circ, \mu^{-1}(\Sigma \cap W_\Gamma^\circ) + \Delta_W^{\mathrm{h}}|_{W_\Gamma^\circ})$$

is klt near $\Sigma \cap W_\Gamma^\circ$. Since $c_\Sigma < \mu^{-1}$, (W, Δ_W) is klt at the generic point of Γ.

Thus no prime divisor on W and no integral curve Γ with $\dim f(\Gamma) = 1$ is contained in $\operatorname{Nklt}(W, \Delta_W)$. Hence every irreducible component of $\operatorname{Nklt}(W, \Delta_W)$ is contained in a fiber of f, and $f(\operatorname{Nklt}(W, \Delta_W))$ is finite.

Step 5. In this step we descend **Step 4** along the tower.

For $i \in [0, e]$, let $\Delta_i := (\rho_{e-1} \circ \cdots \circ \rho_i)^* \Delta_{Y_e}$ on Y_i, so that $\Delta_0 = \Delta_W$, $\Delta_e = \Delta_{Y_e}$, and $\Delta_i = \rho_i^* \Delta_{i+1}$ for $i < e$. We show that for any integer $i \in [0, e-1]$,

$$\operatorname{Nklt}(Y_{i+1}, \Delta_{i+1}) \subset \rho_i(\operatorname{Nklt}(Y_i, \Delta_i)).$$

Let $Y_{i+1}^\circ := Y_{i+1} \setminus \rho_i(\operatorname{Nklt}(Y_i, \Delta_i))$, which is open as ρ_i is finite. Then $(\rho_i^{-1}(Y_{i+1}^\circ), \Delta_i)$ is klt, and by Proposition 8.16(1), $\rho_i^{-1}(Y_{i+1}^\circ) \to Y_{i+1}^\circ$ is the quotient by $\mathcal{F}_i$. The foliation $\mathcal{F}_i$ is regular, so it is a line bundle and is canonical [Pos26, Lemma 3.3]. Let P be a prime divisor on Y_{i+1} and $P' := \rho_i^{-1}(P)$. By [Pos26, Lemma 4.7], $\rho_i^* P = P'$ if P' is $\mathcal{F}_i$-invariant and $\rho_i^* P = pP'$ otherwise, so $\operatorname{mult}_{P'} \Delta_i$ equals $\operatorname{mult}_P \Delta_{i+1}$ or $p \operatorname{mult}_P \Delta_{i+1}$, respectively. The boundary of [Pos26, Definition 4.6] on the quotient Y_{i+1}° of $(\rho_i^{-1}(Y_{i+1}^\circ), \Delta_i)$ has coefficient $\operatorname{mult}_{P'} \Delta_i$ along P if P' is $\mathcal{F}_i$-invariant and $\frac{1}{p} \operatorname{mult}_{P'} \Delta_i$ otherwise, so it is $\Delta_{i+1}|_{Y_{i+1}^\circ}$. By [Pos26, Theorem 4.10(1)], $(Y_{i+1}^\circ, \Delta_{i+1}|_{Y_{i+1}^\circ})$ is klt, which proves the inclusion.

Thus $\operatorname{Nklt}(Y_e, \Delta_{Y_e}) \subset \varphi(\operatorname{Nklt}(W, \Delta_W))$. Since $B_{Y_e} \le B_{Y_e}^{\ge 0}$ and $a \circ \varphi = \operatorname{Fr}_{T/k}^e \circ f$, **Step 4** implies that

$$a\big(\operatorname{Nklt}(Y_e, B_{Y_e} + \pi^* G)\big) \subset \operatorname{Fr}_{T/k}^e\big(f(\operatorname{Nklt}(W, \Delta_W))\big) \quad \text{is finite.} \tag{8.64}$$

Step 6. In this step we show $C_{t_1} \cap C_{t_2} = \emptyset$ by relative connectedness.

Suppose that $x_0 \in C_{t_1} \cap C_{t_2}$ for a closed point $x_0 \in X$. Let $\overline{B} := B_{\overline{Y}_e} + \overline{\pi}^* G$. We have

$$K_{\overline{Y}_e} + \overline{B} = \overline{\pi}^*(K_X + B + G), \quad \overline{\pi}_* \overline{B} = B + G \ge 0.$$

By the last paragraph of **Step 2**, $\overline{\pi}$ is an isomorphism over a neighborhood of the generic point of C_{t_1}, and the generic point of $\varphi(F_{t_1})$ is the only point of $\overline{Y}_e$ over it. Since $(X, B+G)$ is not klt at the generic point of C_{t_1}, the sub-pair $(\overline{Y}_e, \overline{B})$ is not sub-klt at the generic point of $\varphi(F_{t_1})$.

Hence $\varphi(F_{t_1}) \subset \mathrm{Nklt}(\overline{Y}_e, \overline{B})$, and likewise $\varphi(F_{t_2}) \subset \mathrm{Nklt}(\overline{Y}_e, \overline{B})$. Since $\pi(\varphi(F_{t_1})) = C_{t_1}$ and $\pi(\varphi(F_{t_2})) = C_{t_2}$, there exist $y_{t_1} \in \varphi(F_{t_1})$ and $y_{t_2} \in \varphi(F_{t_2})$ over x_0, and

$$\overline{a}(y_{t_1}) = \mathrm{Fr}^e_{T/k}(t_1) \neq \mathrm{Fr}^e_{T/k}(t_2) = \overline{a}(y_{t_2}).$$

By [NT20, Theorem 2.15],

$$\mathrm{Nklt}(\overline{Y}_e, \overline{B}) \cap \overline{\pi}^{-1}(x_0)$$

is connected. Since $\overline{a}$ is projective, the subset

$$\Xi := \overline{a}\big(\mathrm{Nklt}(\overline{Y}_e, \overline{B}) \cap \overline{\pi}^{-1}(x_0)\big) \subset \overline{T}^{(p^e)}$$

is closed and connected, and contains $\mathrm{Fr}^e_{T/k}(t_1)$ and $\mathrm{Fr}^e_{T/k}(t_2)$. Since

$$Y_e = \overline{a}^{-1}(T^{(p^e)}) \quad \text{and} \quad B_{\overline{Y}_e}|_{Y_e} = B_{Y_e},$$

(8.64) implies that $\Xi \cap T^{(p^e)}$ is a finite set of closed points. Since $\Xi \cap T^{(p^e)}$ is open in Ξ, the point $\mathrm{Fr}^e_{T/k}(t_1)$ is open and closed in Ξ, so $\Xi = \{\mathrm{Fr}^e_{T/k}(t_1)\}$, a contradiction. Hence $C_{t_1} \cap C_{t_2} = \emptyset$.

Step 7. In this step we derive the contradiction.

Since t_1 and t_2 in **Step 3** are arbitrary, $C_{t_1} \cap C_{t_2} = \emptyset$ for any closed points $t_1 \neq t_2$ in T. For a closed point $x_0 \in X$, $g^{-1}(x_0)$ is thus contained in a single fiber F_t, on which g is the finite morphism $F_t \to C_t$. Hence every closed fiber of g is finite, and g is quasi-finite. Let E be a π-exceptional prime divisor on Y_e. Since φ is a finite homeomorphism, $\varphi^{-1}(E)$ is a prime divisor on W with $g(\varphi^{-1}(E)) = \pi(E)$ and $\dim \pi(E) \leq 1$, contrary to the quasi-finiteness of g. Thus Y_e has no π-exceptional prime divisor, and $B^{<0}_{Y_e} = 0$ by Proposition 8.16(2). By Proposition 8.16(2) and (8.58),

$$0 = (\varphi^* B^{<0}_{Y_e}) \cdot F_t \geq p - 3 - \lambda > p - 4 > 0,$$

a contradiction. Hence $b_0 \leq Q_q r$. □

9. DCC of volumes and log birational boundedness

Volumes of projective lc threefold pairs over algebraically closed fields of characteristic > 5 with coefficients in a DCC set that are log birationally bounded over $\mathbb{Z}[1/30]$ (Definition 9.1) form a DCC set (see Theorem 9.9).

Definition 9.1. Let $\mathcal{C}$ be a set of couples (X, Σ), each consisting of a normal projective threefold X over an algebraically closed field k depending on (X, Σ) and a reduced divisor Σ on X. We say that $\mathcal{C}$ is *log birationally bounded over* $\mathbb{Z}[1/30]$ if there exist

(1) a scheme T of finite type over $\mathbb{Z}[1/30]$,
(2) a projective flat morphism $\pi : \mathcal{Y} \to T$ whose geometric fibers are normal integral threefolds, and
(3) a closed subset $\mathcal{Q} \subset \mathcal{Y}$ containing no geometric fiber of π,

such that for any $(X, \Sigma) \in \mathcal{C}$ there exist a geometric point $t : \operatorname{Spec} k \to T$ and a birational map $f : \mathcal{Y}_t \dashrightarrow X$, where $\mathcal{Y}_t$ and $\mathcal{Q}_t$ are the fibers of $\mathcal{Y}$ and $\mathcal{Q}$ over t, such that $\mathcal{Q}_t$ contains $\operatorname{Supp} f_*^{-1}\Sigma$ and any prime divisor on $\mathcal{Y}_t$ contracted by f. A set of pairs (X, B) is *log birationally bounded over* $\mathbb{Z}[1/30]$ if the set of couples $(X, \operatorname{Supp} B)$ is log birationally bounded over $\mathbb{Z}[1/30]$.

In [HMX13, Definition 2.4.1] the base T is of finite type over an algebraically closed field of characteristic 0. In Definition 9.1, since any residue field of $\mathbb{Z}[1/30]$ has characteristic 0 or $p \geq 7$, the field k has characteristic 0 or $p > 5$, and one family π may have geometric fibers in several characteristics.

Set-up 9.2. Let T, $\pi : \mathcal{Y} \to T$, and $\Sigma = \sum_{j=1}^n \Sigma_j$ be as follows.

(1) T is a scheme of finite type over $\mathbb{Z}[1/30]$.
(2) π is a smooth projective morphism whose geometric fibers are integral threefolds.
(3) Σ is a divisor on $\mathcal{Y}$ with prime components $\Sigma_1, \ldots, \Sigma_n$.

(4) For any non-empty subset $\Lambda \subset \{1, \dots, n\}$, the scheme $\bigcap_{j\in\Lambda} \Sigma_j$ is either empty or smooth over T and a disjoint union of finitely many closed subschemes whose non-empty geometric fibers are integral with $\dim = 3 - |\Lambda|$.

For a geometric point t of T, we let $\mathcal{Y}_t$ be the fiber of π over t, and for an $\mathbb{R}$-divisor D on $\mathcal{Y}$ supported on Σ, we let D_t be the restriction of D to $\mathcal{Y}_t$.

Lemma 9.3. *Notation and conditions as in Set-up 9.2. Assume that T is integral. Let $b_1, \dots, b_n \in [0,1]$ be real numbers and $B := \sum_{j=1}^n b_j \Sigma_j$. Then there exist a non-empty open subset $T^\circ \subset T$ and $v \in \mathbb{R}$ such that $\mathrm{vol}(K_{\mathcal{Y}_t} + B_t) = v$ for any geometric point t of T°.*

Proof. We use the following properties of volumes. Let Y be a normal projective threefold over an algebraically closed field and D an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor on Y.

(V1) If D' is an $\mathbb{R}$-divisor on Y with $D' \geq D$, then $\mathrm{vol}(D') \geq \mathrm{vol}(D)$. If $\psi : Y' \to Y$ is a birational morphism from a normal projective variety Y' and $\Gamma \geq 0$ is a ψ-exceptional $\mathbb{R}$-divisor on Y', then $\mathrm{vol}(\psi^* D + \Gamma) = \mathrm{vol}(D)$. Both follow from Definition 2.3 and Lemma 2.4.

(V2) If D is nef, then $\mathrm{vol}(D) = D^3$. For an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D this follows from the asymptotic Riemann–Roch formula. Any nef class in $N^1(Y)_{\mathbb{R}}$ is a limit of ample $\mathbb{Q}$-classes, so the continuity of vol on $N^1(Y)_{\mathbb{R}}$ [Cut15, Theorem 2.5] implies $\mathrm{vol}(D) = D^3$ for nef D.

Let $\overline{k(T)}$ be an algebraic closure of the function field $k(T)$, $\overline{\eta} : \operatorname{Spec} \overline{k(T)} \to T$ the geometric generic point, and $X := \mathcal{Y}_{\overline{\eta}}$. Since T is of finite type over $\mathbb{Z}[1/30]$,

$$\operatorname{char} \overline{k(T)} = 0 \quad \text{or} \quad \operatorname{char} \overline{k(T)} = p > 5.$$

For $c = (c_1, \dots, c_n) \in \mathbb{R}^n$, let $B(c) := \sum_{j=1}^n c_j \Sigma_j$ and $B_X(c) := B(c)_{\overline{\eta}}$. Let $b := (b_1, \dots, b_n)$, so $B = B(b)$. By Set-up 9.2(4), $\Sigma_{\overline{\eta}}$ has simple normal crossing support on the smooth projective threefold X, so $(X, B_X(c))$ is a $\mathbb{Q}$-factorial dlt pair for any $c \in [0,1]^n$.

Step 1. In this step we assume that $K_X + B_X(b)$ is pseudo-effective, and we construct rational points $c^{(1)}, \dots, c^{(N)} \in [0,1]^n$ and real numbers $\lambda_1, \dots, \lambda_N > 0$ such that

$$\sum_{i=1}^N \lambda_i = 1 \quad \text{and} \quad \sum_{i=1}^N \lambda_i c^{(i)} = b, \tag{9.1}$$

together with birational morphisms $f : X' \to X$ and $g : X' \to X''$ of normal projective threefolds over $\overline{k(T)}$, semi-ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors $L_1, \dots, L_N$ on X'', and g-exceptional $\mathbb{Q}$-divisors $E_1, \dots, E_N \geq 0$ on X' such that for any $1 \leq i \leq N$,

$$f^*(K_X + B_X(c^{(i)})) = g^* L_i + E_i. \tag{9.2}$$

By [HNT20, Theorem 1.1] if $\operatorname{char} \overline{k(T)} > 0$, and by [Sho96, Theorem 5.2 and 5.1.3] if $\operatorname{char} \overline{k(T)} = 0$, there exists a $(K_X + B_X(b))$-MMP $\varphi : X \dashrightarrow X''$ which terminates. Let $B''(c) := \varphi_* B_X(c)$ for $c \in \mathbb{R}^n$. Since $K_X + B_X(b)$ is pseudo-effective, the MMP does not end with a Mori fiber space, so $(X'', B''(b))$ is $\mathbb{Q}$-factorial dlt and $K_{X''} + B''(b)$ is nef. Since $(X'', B''(b))$ is dlt and X'' is $\mathbb{Q}$-factorial, X'' is klt. Let X' be the normalization of the closure of the graph of φ, with projections $f : X' \to X$ and $g : X' \to X''$. For $c \in \mathbb{R}^n$, let

$$E(c) := f^*(K_X + B_X(c)) - g^*(K_{X''} + B''(c)).$$

Since φ is a composition of steps of a $(K_X + B_X(b))$-MMP, $E(b) \geq 0$ by the negativity lemma. Since φ^{-1} contracts no divisor, any prime divisor P on X' which is not g-exceptional is $f_*^{-1} P_X$ for a prime divisor P_X on X with $g(P) = \varphi_* P_X$, so $\operatorname{mult}_P E(c) = 0$. Thus $\operatorname{Supp} E(c) \subset \operatorname{Exc}(g)$ for any c, and since X'' is $\mathbb{Q}$-factorial,

$$E(c) = E(0) + \sum_{j=1}^n c_j \left(f^* \Sigma_{j,\overline{\eta}} - g^* \varphi_* \Sigma_{j,\overline{\eta}} \right),$$

where $E(0)$ and $f^*\Sigma_{j,\overline{\eta}} - g^*\varphi_*\Sigma_{j,\overline{\eta}}$ are $\mathbb{Q}$-divisors supported on $\mathrm{Exc}(g)$.

Let $\mathcal{V}$ be the $\mathbb{R}$-vector space of $\mathbb{R}$-divisors on X'' supported on $\operatorname{Supp}\varphi_*\Sigma_{\overline{\eta}}$, and

$$\mathcal{N} := \{\Delta \in \mathcal{V} \mid (X'', \Delta) \text{ is lc and } K_{X''} + \Delta \text{ is nef}\}.$$

Since X'' is a $\mathbb{Q}$-factorial klt threefold, $\mathcal{N}$ is a rational polytope by [HNT20, Proposition 4.9(3)] and [BW17, Proposition 3.8(3)] if $\operatorname{char}\overline{k(T)} > 0$, and by [Bir11, Remark 3.1, Proposition 3.2(3)] if $\operatorname{char}\overline{k(T)} = 0$. Since $B''(c)$ and $E(c)$ are affine functions of c with rational coefficients,

$$\mathcal{P} := \{c \in [0,1]^n \mid B''(c) \in \mathcal{N} \text{ and } E(c) \geq 0\}$$

is a rational polytope, and $b \in \mathcal{P}$. Let $c^{(1)}, \dots, c^{(N)}$ be the vertices of $\mathcal{P}$ with positive weight in an expression of b as a convex combination of vertices of $\mathcal{P}$, and let $\lambda_1, \dots, \lambda_N > 0$ be the weights, so (9.1) holds and $c^{(i)} \in \mathbb{Q}^n$. Let $L_i := K_{X''} + B''(c^{(i)})$ and $E_i := E(c^{(i)})$. Then (9.2) holds by the definition of $E(c)$, $E_i \geq 0$ is g-exceptional, $(X'', B''(c^{(i)}))$ is lc, and L_i is nef. Since $B''(c^{(i)})$ is a $\mathbb{Q}$-divisor, L_i is semi-ample by Lemma 2.7 if $\operatorname{char}\overline{k(T)} > 0$, and by [KMM94, Theorem 1.1] and [KMM04, Section 6] if $\operatorname{char}\overline{k(T)} = 0$.

Step 2. In this step we assume that $K_X + B_X(b)$ is not pseudo-effective. Since the set of $c \in [0,1]^n$ with $K_X + B_X(c)$ pseudo-effective is closed and rational points are dense in $\{c \in [0,1]^n \mid c_j \geq b_j \text{ for any } j\}$, there exists $c^+ \in [0,1]^n \cap \mathbb{Q}^n$ with $c^+_j \geq b_j$ for any j such that $K_X + B_X(c^+)$ is not pseudo-effective. By [HNT20, Theorem 1.1] if $\operatorname{char}\overline{k(T)} > 0$, and by [Sho96, Theorem 5.2 and 5.1.3] if $\operatorname{char}\overline{k(T)} = 0$, there exists a $(K_X + B_X(c^+))$-MMP $\varphi : X \dashrightarrow X''$ which terminates with a Mori fiber space $h : X'' \to Z$, since $K_X + B_X(c^+)$ is not pseudo-effective. Thus $\dim Z \leq 2$, and $-L_1$ is h-ample, where $L_1 := K_{X''} + \varphi_* B_X(c^+)$. Let X', f, and g be as in **Step 1**, now for the MMP φ of this step. Let $N := 1$ and $c^{(1)} := c^+$. By the negativity lemma, (9.2) holds for $i = 1$ with a g-exceptional $\mathbb{Q}$-divisor $E_1 \geq 0$ on X'.

Step 3. In this step we spread out the data of **Step 1** or **Step 2**. Fix $r \in \mathbb{N}^+$ such that $rc^{(i)} \in \mathbb{Z}^n$, rL_i is Cartier, and, in the case of **Step 1**, $|rL_i|$ is base point free, for any $1 \leq i \leq N$. Then $rE_i = f^*(r(K_X + B_X(c^{(i)}))) - g^*(rL_i)$ is Cartier for any i. By [Gro66, Théorème 8.5.2, Proposition 8.5.5, Théorème 8.8.2], there exists a finite extension $k' \subset \overline{k(T)}$ of $k(T)$ such that X', X'', f, g, the divisors rE_i and rL_i, finitely many sections generating $\mathcal{O}_{X''}(rL_i)$ in the case of **Step 1**, and Z and h in the case of **Step 2**, are base changes of data over k'.

Let T' be an integral scheme of finite type over T with function field k' through which $\overline{\eta}$ factors, $\tau : T' \to T$ the structure morphism, and $\mathcal{Y}_{T'} := \mathcal{Y} \times_T T'$. By [Gro66, Théorème 8.5.2, Proposition 8.5.5, Théorème 8.8.2, Théorème 8.10.5(xiii)], possibly replacing T' with a non-empty open subset, there exist

- projective morphisms $\mathcal{X}' \to T'$ and $\mathcal{X}'' \to T'$, and T'-morphisms $f_{T'} : \mathcal{X}' \to \mathcal{Y}_{T'}$ and $g_{T'} : \mathcal{X}' \to \mathcal{X}''$,
- for any $1 \leq i \leq N$, a line bundle $\mathcal{M}_i$ on $\mathcal{X}''$, a Cartier divisor $\mathcal{E}_i \geq 0$ on $\mathcal{X}'$, and an isomorphism

$$f^*_{T'}\left(\omega^{\otimes r}_{\mathcal{Y}_{T'}/T'} \otimes \mathcal{O}_{\mathcal{Y}_{T'}}(rB(c^{(i)})_{T'})\right) \cong g^*_{T'}\mathcal{M}_i \otimes \mathcal{O}_{\mathcal{X}'}(\mathcal{E}_i),$$

- in the case of **Step 2**, a projective T'-scheme $\mathcal{Z}$ and a T'-morphism $h_{T'} : \mathcal{X}'' \to \mathcal{Z}$,

whose base changes along $\operatorname{Spec}\overline{k(T)} \to T'$ are X', X'', f, g, $\mathcal{O}_{X''}(rL_i)$, rE_i, the isomorphism given by (9.2), Z, and h. The following properties hold over the generic point of T', so by generic flatness, [Gro66, Théorème 8.10.5, Théorème 12.2.4, Corollaire 13.1.5], and [Gro61, Théorème 4.7.1], possibly replacing T' with a non-empty open subset, they hold over T'.

(S1) $\mathcal{X}'' \to T'$ is flat, and the geometric fibers of $\mathcal{X}' \to T'$ and $\mathcal{X}'' \to T'$ are normal integral threefolds.

(S2) The restrictions of $f_{T'}$ and $g_{T'}$ to any geometric fiber are birational.

(S3) For any $1 \le i \le N$, the image $g_{T'}(\operatorname{Supp} \mathcal{E}_i)$ has fibers over T' with $\dim \le 1$, and in the case of **Step 1**, the restriction of $\mathcal{M}_i$ to any geometric fiber of $\mathcal{X}'' \to T'$ is globally generated.

(S4) In the case of **Step 2**, the fibers of $\mathcal{Z} \to T'$ have $\dim \le 2$, and the restriction of $\mathcal{M}_1^{-1}$ to any fiber of $h_{T'}$ is ample.

For global generation in (S3), the image in T' of the support of the cokernel of the evaluation map of the spread sections is closed and does not contain the generic point of T'. Since τ is dominant and generically finite, there exists a non-empty open subset $T^\circ \subset T$ such that $\tau^{-1}(T^\circ) \to T^\circ$ is finite. Since a finite dominant morphism is closed, $\tau^{-1}(T^\circ) \to T^\circ$ is surjective.

Step 4. In this step we show that T° satisfies the lemma. Let $t : \operatorname{Spec} k \to T^\circ$ be a geometric point. Since $\tau^{-1}(T^\circ) \to T^\circ$ is finite and surjective and k is algebraically closed, t lifts to a geometric point $t' : \operatorname{Spec} k \to \tau^{-1}(T^\circ)$. Let $X'_{t'}$, $X''_{t'}$, $f_{t'}$, and $g_{t'}$ be the fibers of $\mathcal{X}'$, $\mathcal{X}''$, $f_{T'}$, and $g_{T'}$ over t'. Then $\mathcal{Y}_{T'} \times_{T'} t' = \mathcal{Y}_t$, and since π is smooth, $\omega_{\mathcal{Y}_{T'}/T'}$ restricts to $\omega_{\mathcal{Y}_t}$. For any $1 \le i \le N$, let $M_{i,t'}$ be a Cartier divisor on $X''_{t'}$ with

$$\mathcal{O}_{X''_{t'}}(M_{i,t'}) \cong \mathcal{M}_i|_{X''_{t'}},$$

and let $E_{i,t'} := r^{-1}\mathcal{E}_i|_{X'_{t'}}$. By (S1)–(S3), $f_{t'}$ and $g_{t'}$ are birational morphisms of normal projective threefolds, $E_{i,t'} \ge 0$ is $g_{t'}$-exceptional, and

$$f_{t'}^*(K_{\mathcal{Y}_t} + B(c^{(i)})_t) \sim_{\mathbb{Q}} g_{t'}^*(r^{-1}M_{i,t'}) + E_{i,t'}. \tag{9.3}$$

Assume that we are in the case of **Step 1**. By (S3), $|M_{i,t'}|$ is base point free for any i, so

$$L_{t'} := \sum_{i=1}^N \lambda_i r^{-1} M_{i,t'}$$

is nef. Let $E_{t'} := \sum_{i=1}^N \lambda_i E_{i,t'} \ge 0$, which is $g_{t'}$-exceptional. By (9.1), $\sum_{i=1}^N \lambda_i B(c^{(i)})_t = B_t$, so (9.3) implies

$$f_{t'}^*(K_{\mathcal{Y}_t} + B_t) \sim_{\mathbb{R}} g_{t'}^* L_{t'} + E_{t'}.$$

By (V1), (V2), and [FKL16, Theorem 3.5(iv)],

$$\begin{aligned}\operatorname{vol}(K_{\mathcal{Y}_t} + B_t) &= \operatorname{vol}(f_{t'}^*(K_{\mathcal{Y}_t} + B_t)) = \operatorname{vol}(g_{t'}^* L_{t'} + E_{t'}) = \operatorname{vol}(L_{t'}) \\ &= L_{t'}^3 = r^{-3} \sum_{i_1,i_2,i_3=1}^N \lambda_{i_1}\lambda_{i_2}\lambda_{i_3}(M_{i_1,t'} \cdot M_{i_2,t'} \cdot M_{i_3,t'}).\end{aligned}$$

The intersection number $(M_{i_1,t'} \cdot M_{i_2,t'} \cdot M_{i_3,t'})$ is the coefficient of $a_1a_2a_3$ in the polynomial

$$\chi_{t'}(a_1,a_2,a_3) := \chi(X''_{t'}, \mathcal{M}_{i_1}^{\otimes a_1} \otimes \mathcal{M}_{i_2}^{\otimes a_2} \otimes \mathcal{M}_{i_3}^{\otimes a_3}|_{X''_{t'}})$$

in (a_1,a_2,a_3). By (S1), $\mathcal{X}'' \to T'$ is flat and projective over the connected scheme $\tau^{-1}(T^\circ)$, so $\chi_{t'}(a_1,a_2,a_3)$ does not depend on t'. Thus $\operatorname{vol}(K_{\mathcal{Y}_t} + B_t)$ does not depend on t.

Assume that we are in the case of **Step 2**. Let $h_{t'} : X''_{t'} \to \mathcal{Z}_{t'}$ be the fiber of $h_{T'}$ over t'. By (S4), $\dim h_{t'}(X''_{t'}) \le 2$ and $-M_{1,t'}$ is ample on the general fibers of $h_{t'}$, so $H^0(X''_{t'}, \mathcal{O}_{X''_{t'}}(mM_{1,t'})) = 0$ for any $m \in \mathbb{N}^+$ and $\operatorname{vol}(M_{1,t'}) = 0$. Since $c_j^+ \ge b_j$ for any j, $B_t \le B(c^+)_t$. By (V1), (9.3), [FKL16, Theorem 3.5(iv)], and the homogeneity of volumes [Cut15, Theorem 2.5 and (3)],

$$0 \le \operatorname{vol}(K_{\mathcal{Y}_t} + B_t) \le \operatorname{vol}(K_{\mathcal{Y}_t} + B(c^+)_t) = \operatorname{vol}(g_{t'}^*(r^{-1}M_{1,t'}) + E_{1,t'}) = r^{-3}\operatorname{vol}(M_{1,t'}) = 0.$$

Thus $\operatorname{vol}(K_{\mathcal{Y}_t} + B_t) = 0$ for any geometric point t of T°. The lemma follows. □

Definition 9.4. Let X be a normal variety and B an $\mathbb{R}$-divisor on X such that $K_X + B$ is $\mathbb{R}$-Cartier. For a prime divisor E over X, let $h : Y \to X$ be a proper birational morphism from a normal variety Y such that E is a prime divisor on Y, write $K_Y + B_Y := h^*(K_X + B)$, and let

$$a(E, X, B) := 1 - \operatorname{mult}_E B_Y.$$

Then $a(E, X, B)$ does not depend on Y, and it coincides with the log discrepancy of Definition 2.1 when (X, B) has a log resolution. For an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor D on X we write

$$\operatorname{mult}_E D := \operatorname{mult}_E(h^* D) \quad \text{and} \quad \operatorname{center}_X E := h(E).$$

For a smooth variety X, a reduced divisor $\Sigma = \sum_{j=1}^n \Sigma_j$ on X with simple normal crossing support and prime components $\Sigma_1, \dots, \Sigma_n$, and a non-empty subset $\Lambda \subset \{1, \dots, n\}$, an irreducible component of $\bigcap_{j\in\Lambda} \Sigma_j$ is called a *stratum* of Σ.

Lemma 9.5 (cf. [HMX13, Lemma 5.4])**.** *Let X be a smooth variety over an algebraically closed field, $\Sigma = \sum_{j=1}^n \Sigma_j$ a reduced divisor on X with simple normal crossing support and prime components $\Sigma_1, \dots, \Sigma_n$, $b_1, \dots, b_n \in [0, 1]$ real numbers, and $B := \sum_{j=1}^n b_j \Sigma_j$. Let E be a prime divisor over X such that $a(E, X, B) < 1$. Then:*

(1) $\operatorname{center}_X E$ *is a stratum of* Σ.
(2) There exist an integer $N \geq 0$, smooth varieties $X_0 := X, X_1, \dots, X_N$, and reduced divisors $\Sigma_{X_0} := \Sigma, \Sigma_{X_1}, \dots, \Sigma_{X_N}$ with simple normal crossing support on $X_0, X_1, \dots, X_N$ respectively, such that E is a component of Σ_{X_N} and for any $1 \leq i \leq N$ the following hold.
  (a) $h_{i-1} : X_i \to X_{i-1}$ is the blow-up of a stratum W_{i-1} of $\Sigma_{X_{i-1}}$ of codimension at least 2.
  (b) $\Sigma_{X_i} = h_{i-1*}^{-1} \Sigma_{X_{i-1}} + F_i$, where F_i is the exceptional prime divisor of h_{i-1}.
(3) $a(E, X, B) = \sum_{j=1}^n (1 - b_j) \operatorname{mult}_E \Sigma_j$.

Proof. By the definition of log discrepancies,

$$a(E, X, B) = a(E, X, \Sigma) + \operatorname{mult}_E(\Sigma - B) = a(E, X, \Sigma) + \sum_{j=1}^n (1 - b_j) \operatorname{mult}_E \Sigma_j. \tag{9.4}$$

Since $b_j \leq 1$ for any j, $\Sigma - B \geq 0$, so $a(E, X, \Sigma) \leq a(E, X, B) < 1$. Since $K_X + \Sigma$ is Cartier, $a(E, X, \Sigma) \in \mathbb{Z}$. Since (X, Σ) is lc, $a(E, X, \Sigma) \geq 0$. Thus

$$a(E, X, \Sigma) = 0, \tag{9.5}$$

and (9.4) implies (3).

We prove (1) for any X and Σ as in the lemma and any prime divisor E over X satisfying (9.5). Let $W := \operatorname{center}_X E$, $c := \operatorname{codim}_X W$, and $\Lambda' := \{j \mid W \subset \Sigma_j\}$. If $\Lambda' \neq \emptyset$, then $\bigcap_{j\in\Lambda'} \Sigma_j$ is smooth of pure codimension $|\Lambda'|$ and contains W, so

$$|\Lambda'| \leq c.$$

If $c = 1$, then $E = W$ and $0 = a(E, X, \Sigma) = 1 - \operatorname{mult}_W \Sigma$, so $W = \Sigma_j$ for some j, which is a stratum of Σ. Assume that $c \geq 2$. Possibly shrinking X to a neighborhood of the generic point of W, we may assume that W is smooth and $\Sigma_j \cap W = \emptyset$ for any $j \notin \Lambda'$. Let $h : X' \to X$ be the blow-up of W, and let F be the exceptional prime divisor. Since $K_{X'} = h^* K_X + (c - 1)F$, $h^* \Sigma_j = h_*^{-1} \Sigma_j + F$ for $j \in \Lambda'$, and $h^* \Sigma_j = h_*^{-1} \Sigma_j$ for $j \notin \Lambda'$,

$$h^*(K_X + \Sigma) = K_{X'} + h_*^{-1} \Sigma + F - (c - |\Lambda'|)F. \tag{9.6}$$

Since $W \subset \bigcap_{j\in\Lambda'} \Sigma_j$ and both are smooth, at any closed point $w \in W$ there exists a regular system of parameters $y_1, \dots, y_{\dim X}$ of $\mathcal{O}_{X,w}$ such that $W = (y_1 = \cdots = y_c = 0)$ and the components Σ_j with $j \in \Lambda'$ are $(y_1 = 0), \dots, (y_{|\Lambda'|} = 0)$ near w. Thus $h_*^{-1} \Sigma + F$ has simple normal crossing support, so $(X', h_*^{-1} \Sigma + F)$ is lc. By (9.6),

$$a(E, X, \Sigma) = a(E, X', h_*^{-1} \Sigma + F) + (c - |\Lambda'|) \operatorname{mult}_E F.$$

Since $h(\operatorname{center}_{X'} E) = W$, $\operatorname{center}_{X'} E \subset F$, so $\operatorname{mult}_E F \geq 1$, also if $E = F$. Since $(X', h_*^{-1} \Sigma + F)$ is lc, $a(E, X', h_*^{-1} \Sigma + F) \geq 0$. Thus if $|\Lambda'| < c$, then $a(E, X, \Sigma) \geq 1$, which contradicts (9.5). Thus $|\Lambda'| = c$. Since W is irreducible, $\operatorname{codim}_X W = c$, and W is contained in $\bigcap_{j\in\Lambda'} \Sigma_j$ of pure

codimension c, W is an irreducible component of $\bigcap_{j\in\Lambda'}\Sigma_j$, that is, a stratum of Σ. This proves (1).

We prove (2). Let $X_0 := X$ and $\Sigma_{X_0} := \Sigma$. Assume that for some $i \geq 0$ we have constructed $X_0, \dots, X_i$ and $\Sigma_{X_0}, \dots, \Sigma_{X_i}$ satisfying (a) and (b) up to i, and that $a(E, X_i, \Sigma_{X_i}) = 0$, which holds for $i = 0$ by (9.5). If E is a divisor on X_i, then by (1) for (X_i, Σ_{X_i}), E is a stratum of Σ_{X_i} with $\operatorname{codim}_{X_i} E = 1$, that is, a component of Σ_{X_i}, and we let $N := i$. Otherwise, by (1) for (X_i, Σ_{X_i}), $W_i := \operatorname{center}_{X_i} E$ is a stratum of Σ_{X_i} with $c_i := \operatorname{codim}_{X_i} W_i \geq 2$. Thus W_i is smooth and exactly c_i components of Σ_{X_i} contain W_i. Let $h_i : X_{i+1} \to X_i$ be the blow-up of W_i with exceptional prime divisor F_{i+1}, and let $\Sigma_{X_{i+1}} := h_{i*}^{-1}\Sigma_{X_i} + F_{i+1}$. At any closed point $w \in W_i$ there exists a regular system of parameters $y_1, \dots, y_{\dim X}$ of $\mathcal{O}_{X_i,w}$ such that $\Sigma_{X_i} = (y_1 \cdots y_m = 0)$ near w for some $m \geq c_i$ and $W_i = (y_1 = \cdots = y_{c_i} = 0)$, so $\Sigma_{X_{i+1}}$ has simple normal crossing support. Since exactly c_i components of Σ_{X_i} contain W_i, the computation giving (9.6), with $c = |\Lambda'| = c_i$, shows that

$$h_i^*(K_{X_i} + \Sigma_{X_i}) = K_{X_{i+1}} + \Sigma_{X_{i+1}},$$

so $a(E, X_{i+1}, \Sigma_{X_{i+1}}) = a(E, X_i, \Sigma_{X_i}) = 0$. Since $X_{i+1} \to X_i$ is the blow-up of $\operatorname{center}_{X_i} E$ for any i, by Zariski's lemma, E is a divisor on X_i for some i, so N is defined. This proves (2). □

Definition 9.6. Notation and conditions as in Set-up 9.2. For a non-empty subset $\Lambda \subset \{1, \dots, n\}$, the connected components of $\bigcap_{j\in\Lambda}\Sigma_j$ are called the *strata* of Σ, and for a stratum W we write $\Lambda_W := \{j \mid W \subset \Sigma_j\}$. A *blow-up of strata* of $\mathcal{Y}$ is a composite

$$\mathcal{Y}_m \to \cdots \to \mathcal{Y}_0 := \mathcal{Y}$$

such that, with $\Sigma_{\mathcal{Y}_0} := \Sigma$ and $\Sigma_{\mathcal{Y}_{i+1}}$ the reduced total transform of $\Sigma_{\mathcal{Y}_i}$, the morphism $\mathcal{Y}_i \to T$ and $\Sigma_{\mathcal{Y}_i}$ satisfy the conditions of Set-up 9.2 and $\mathcal{Y}_{i+1} \to \mathcal{Y}_i$ is the blow-up along a stratum of $\Sigma_{\mathcal{Y}_i}$ for any $i < m$. For a smooth variety Y over an algebraically closed field and a reduced divisor Σ_Y on Y with simple normal crossing support, a *blow-up of strata* of Y is a composite of finitely many blow-ups, each along a stratum of the reduced total transform of Σ_Y, and a prime divisor over Y is *toroidal* if it is a component of the reduced total transform of Σ_Y on a blow-up of strata of Y.

Lemma 9.7. *Notation and conditions as in Set-up 9.2.*

(1) *Let W be a stratum of Σ, $\mathcal{Y}' \to \mathcal{Y}$ the blow-up along W, and Σ' the reduced total transform of Σ. Then $\mathcal{Y}' \to T$ and Σ' satisfy the conditions of Set-up 9.2, and for any geometric point t of T, $\mathcal{Y}'_t \to \mathcal{Y}_t$ is the blow-up along W_t and Σ'_t is the reduced total transform of Σ_t.*

(2) *Let t be a geometric point of T, and let $g_k : Y_k \to \mathcal{Y}_t$, $1 \leq k \leq m$, be blow-ups of strata of $\mathcal{Y}_t$ with respect to Σ_t or normalized blow-ups of $\mathcal{Y}_t$ along ideal sheaves locally generated by monomials in local equations of the components of Σ_t. Then there exists a blow-up of strata $h : \mathcal{Y}' \to \mathcal{Y}$ such that the fiber $h_t : \mathcal{Y}'_t \to \mathcal{Y}_t$ of h over t factors through g_k for any k.*

(3) *Let t be a geometric point of T and W a stratum of Σ with $W_t \neq \emptyset$. Then the map*

$$E \to (\operatorname{mult}_E \Sigma_{j,t})_{j\in\Lambda_W}$$

is a bijection from the set of toroidal prime divisors over $\mathcal{Y}_t$ with center W_t onto the set of primitive vectors in $(\mathbb{N}^+)^{\Lambda_W}$.

(4) *Let t be a geometric point of T. Then $W \to W_t$ is a bijection from the set of strata W of Σ with $W_t \neq \emptyset$ onto the set of strata of Σ_t, and W_t is an irreducible component of $\bigcap_{j\in\Lambda_W}\Sigma_{j,t}$.*

(5) *Let $h : \mathcal{Y}' \to \mathcal{Y}$ be a blow-up of strata and $\Sigma' = \sum_{j=1}^{n'}\Sigma'_j$ the reduced total transform of Σ. Then for any geometric point t of T, a prime divisor over $\mathcal{Y}_t$ is toroidal with respect to Σ_t if and only if it is toroidal with respect to Σ'_t. If moreover $W_t \neq \emptyset$ for any stratum W of Σ and any geometric point t of T, then $W'_t \neq \emptyset$ for any stratum W' of Σ' and any geometric*

point t of T, and for any stratum W of Σ and any primitive vector $u \in (\mathbb{N}^+)^{\Lambda_W}$, the stratum W' of Σ' with $W'_t = \operatorname{center}_{\mathcal{Y}'_t} E$ and the vector $(\operatorname{mult}_E \Sigma'_{j,t})_{1\le j\le n'}$ of the toroidal prime divisor E over $\mathcal{Y}_t$ with center W_t and $(\operatorname{mult}_E \Sigma_{j,t})_{j\in\Lambda_W} = u$ do not depend on t.

Proof. (4) Let W be a stratum of Σ, a connected component of $\bigcap_{j\in\Lambda}\Sigma_j$. The closed subschemes of $\bigcap_{j\in\Lambda}\Sigma_j$ in Set-up 9.2(4) are open and closed in $\bigcap_{j\in\Lambda}\Sigma_j$, so W is contained in one of them, Z say, and for any geometric point t of T, W_t is open and closed in the integral or empty scheme Z_t, hence $W_t = Z_t$ or $W_t = \emptyset$. For $k \notin \Lambda$, the non-empty geometric fibers of $\bigcap_{j\in\Lambda\cup\{k\}}\Sigma_j$ have dimension $2 - \#\Lambda < \dim W_t$ whenever $W_t \neq \emptyset$, so $W_t \not\subset \Sigma_{k,t}$. Thus $\Lambda_W = \Lambda$, and $W_t \subset \Sigma_{j,t}$ if and only if $j \in \Lambda_W$, provided $W_t \neq \emptyset$. Since $\bigcap_{j\in\Lambda}\Sigma_{j,t}$ is the disjoint union of the W_t over the connected components W of $\bigcap_{j\in\Lambda}\Sigma_j$, each integral or empty, the connected components of $\bigcap_{j\in\Lambda}\Sigma_{j,t}$, which are the strata of Σ_t for Λ, are exactly the non-empty W_t, and they are the irreducible components of $\bigcap_{j\in\Lambda}\Sigma_{j,t}$. For strata $W \neq W'$ of Σ with $W_t \neq \emptyset$ and $W'_t \neq \emptyset$, $W_t \neq W'_t$: if $\Lambda_W = \Lambda_{W'}$, then $W_t \cap W'_t = \emptyset$, and otherwise W_t and W'_t are contained in different sets of components of Σ_t.

(1) Let W be a stratum of Σ. Then W is smooth over T, being open and closed in $\bigcap_{j\in\Lambda_W}\Sigma_j$, with integral or empty geometric fibers by (4), and in étale local coordinates over T the components of Σ are coordinate hyperplanes and W is a coordinate subspace. Thus the blow-up $\mathcal{Y}' \to \mathcal{Y}$ along W is smooth and projective over T, for any geometric point t of T the fiber $\mathcal{Y}'_t$ is the blow-up of $\mathcal{Y}_t$ along W_t and Σ'_t is the reduced total transform of Σ_t, the exceptional divisor of $\mathcal{Y}' \to \mathcal{Y}$ is a projective bundle over W, and for any non-empty subset $\Lambda' \subset \{1,\dots,n\}$ the components of the intersections of the components of Σ' are projective bundles over the components of $W \cap \bigcap_{j\in\Lambda'}\Sigma_j$ or strict transforms of the components of $\bigcap_{j\in\Lambda'}\Sigma_j$. Hence $\mathcal{Y}' \to T$ and Σ' satisfy the conditions of Set-up 9.2.

(2) We first treat a single ideal. Let X be a smooth variety over an algebraically closed field, Σ_X a reduced divisor on X with simple normal crossing support, and $\mathcal{I} \neq 0$ an ideal sheaf on X locally generated by monomials in local equations of the components of Σ_X. We show that there exists a blow-up of strata $X'' \to X$ such that $\mathcal{I}\cdot\mathcal{O}_{X''}$ is invertible. By [Kol19, Theorem 10], there exist blow-ups $f_i : X_{i+1} \to X_i$ for $0 \le i < N$ with $X_0 := X$, each along a stratum or a disjoint union of strata of Σ_{X_i}, the reduced total transform of Σ_X on X_i, and integers $c_i \ge 0$, such that, with E_{i+1} the exceptional divisor of f_i and

$$\mathcal{I}_0 := \mathcal{I}, \quad \mathcal{I}_{i+1} := \mathcal{O}_{X_{i+1}}(c_i E_{i+1}) \cdot f_i^* \mathcal{I}_i,$$

the co-support of $\mathcal{I}_N$ contains no stratum of Σ_{X_N}. The blow-up along a disjoint union of strata is the composite of the blow-ups along them one at a time, so $X_N \to X$ is a blow-up of strata. The pullback of a monomial in local equations of the components of Σ_{X_i} is a unit times a monomial in local equations of the components of $\Sigma_{X_{i+1}}$, so each $\mathcal{I}_i$ is locally generated by monomials. The co-support of a non-zero ideal sheaf locally generated by monomials, if non-empty, contains a stratum, so $\mathcal{I}_N = \mathcal{O}_{X_N}$ (cf. [Kol19, Corollary 18]). Hence $\mathcal{I}\cdot\mathcal{O}_{X_N} = \mathcal{O}_{X_N}(-F)$, where F is the sum of the pullbacks of the divisors $c_i E_{i+1}$ to X_N, and $X'' := X_N$ works. By the universal property of the blow-up, $X'' \to X$ factors through the blow-up of X along $\mathcal{I}$, and through its normalization since X'' is smooth.

Now let t and $g_1,\dots,g_m$ be as in (2). For a blow-up of strata $Y' \to \mathcal{Y}_t$ we denote by $\Sigma_{Y'}$ the reduced total transform of Σ_t on Y'. If $Y' \to \mathcal{Y}_t$ factors through a blow-up of strata $Z \to \mathcal{Y}_t$, then the pullback to Y' of a local equation of a component of Σ_Z is supported on $\Sigma_{Y'}$, hence locally a unit times a monomial in local equations of the components of $\Sigma_{Y'}$. We construct by induction on k a blow-up of strata $Y'_k \to \mathcal{Y}_t$ factoring through $g_1,\dots,g_k$, with $Y'_0 := \mathcal{Y}_t$. Let $k \ge 1$. If g_k is the normalized blow-up along an ideal sheaf $\mathcal{I}$, then $\mathcal{I}\cdot\mathcal{O}_{Y'_{k-1}}$ is locally generated by monomials in local equations of the components of $\Sigma_{Y'_{k-1}}$, and the first part, applied to $(Y'_{k-1}, \Sigma_{Y'_{k-1}})$ and $\mathcal{I}\cdot\mathcal{O}_{Y'_{k-1}}$, gives Y'_k. If g_k is the composite of the blow-ups $Y_{k,i+1} \to Y_{k,i}$ for $0 \le i < N$ with $Y_{k,0} := \mathcal{Y}_t$ and $Y_{k,N} = Y_k$, each along a stratum $W_{k,i}$ of $\Sigma_{Y_{k,i}}$, we show by induction on i that

there exists a blow-up of strata $Y' \to \mathcal{Y}_t$ factoring through Y'_{k-1} and $Y_{k,i}$. For $i = 0$ we take $Y' := Y'_{k-1}$. Given such Y' for $i < N$, the ideal sheaf $\mathcal{I}_{W_{k,i}}$ of $W_{k,i}$ is locally generated by local equations of components of $\Sigma_{Y_{k,i}}$, so $\mathcal{I}_{W_{k,i}} \cdot \mathcal{O}_{Y'}$ is locally generated by monomials in local equations of the components of $\Sigma_{Y'}$. The first part gives a blow-up of strata $Y'' \to Y'$ with $\mathcal{I}_{W_{k,i}} \cdot \mathcal{O}_{Y''}$ invertible, and $Y'' \to Y_{k,i}$ factors through $Y_{k,i+1}$ by the universal property of the blow-up, so Y'' works for $i+1$. For $i = N$ we obtain Y'_k.

Let

$$Y'_m = Y'_{m,N} \to \cdots \to Y'_{m,0} = \mathcal{Y}_t$$

be the blow-ups along strata W'_i of $\Sigma_{Y'_{m,i}}$, $0 \le i < N$. We construct by induction on i a blow-up of strata $\mathcal{Y}_i \to \mathcal{Y}$, with reduced total transform $\Sigma_{\mathcal{Y}_i}$ of Σ, such that $\mathcal{Y}_{i,t} \to \mathcal{Y}_t$ is $Y'_{m,i} \to \mathcal{Y}_t$ and $\Sigma_{\mathcal{Y}_{i,t}} = \Sigma_{Y'_{m,i}}$, with $\mathcal{Y}_0 := \mathcal{Y}$. Given $\mathcal{Y}_i$ for $i < N$, by (4) for $\mathcal{Y}_i \to T$ and $\Sigma_{\mathcal{Y}_i}$, $W'_i = W_t$ for a stratum W of $\Sigma_{\mathcal{Y}_i}$, and by (1), the blow-up $\mathcal{Y}_{i+1}$ of $\mathcal{Y}_i$ along W is a blow-up of strata of $\mathcal{Y}$ with $\mathcal{Y}_{i+1,t} = Y'_{m,i+1}$ over $\mathcal{Y}_t$ and $\Sigma_{\mathcal{Y}_{i+1,t}} = \Sigma_{Y'_{m,i+1}}$. Then $h : \mathcal{Y}_N \to \mathcal{Y}$ is a blow-up of strata, and h_t, being $Y'_m \to \mathcal{Y}_t$, factors through g_k for any k.

(3) Let $\mathfrak{C}_t$ be the conical polyhedral complex of $(\mathcal{Y}_t, \Sigma_t)$, and $e_1, \dots, e_n$ the standard basis of $\mathbb{R}^n$. By (4), $\mathfrak{C}_t$ has one cone $\mathbb{R}^{\Lambda_V}_{\ge 0}$ for each stratum V of Σ with $V_t \ne \emptyset$. A blow-up of strata of $\mathcal{Y}_t$ is a toroidal modification of $(\mathcal{Y}_t, \Sigma_t)$, since the blow-up along a stratum V_t is the toroidal modification given by the star subdivision of $\mathbb{R}^{\Lambda_V}_{\ge 0}$ at $\sum_{j \in \Lambda_V} e_j$ [KKMS73, Chapter II, §2, Theorem 10*], the toroidal modifications correspond to the subdivisions of $\mathfrak{C}_t$, and a toroidal prime divisor E on such a modification with center W_t corresponds to a ray in the interior of $\mathbb{R}^{\Lambda_W}_{\ge 0}$ whose primitive vector is $(\operatorname{mult}_E \Sigma_{j,t})_{j \in \Lambda_W}$ [KKMS73, Chapter II, §2, Theorem 6*]. Hence the map in (3) is well defined.

It is injective: let E and E' be toroidal prime divisors over $\mathcal{Y}_t$ with center W_t such that $\operatorname{mult}_E \Sigma_{j,t} = \operatorname{mult}_{E'} \Sigma_{j,t}$ for any $j \in \Lambda_W$. By (2), (1), and (4) there exists a blow-up of strata Y'' of $\mathcal{Y}_t$ dominating blow-ups of strata on which E and E' are divisors, so E and E' are divisors on Y'', both corresponding to the ray spanned by $(\operatorname{mult}_E \Sigma_{j,t})_{j \in \Lambda_W}$ in the subdivision of Y'', and $E = E'$.

It is surjective: for $j \in \Lambda_W$ let s_j be a local equation of $\Sigma_{j,t}$, and let $u \in (\mathbb{N}^+)^{\Lambda_W}$ be a primitive vector and l a positive integer divisible by u_j for any $j \in \Lambda_W$. The star subdivision of $\mathfrak{C}_t$ at u corresponds to the normalized blow-up Y_u of $\mathcal{Y}_t$ along the ideal sheaf $\mathcal{I}_u$ which is generated near W_t by s_j^{l/u_j}, $j \in \Lambda_W$, and equals $\mathcal{O}_{\mathcal{Y}_t}$ away from W_t. The order function of $\mathcal{I}_u$ is $x \to l \min_{j \in \Lambda_W} x_j/u_j$ on the cones of $\mathfrak{C}_t$ containing $\mathbb{R}^{\Lambda_W}_{\ge 0}$ and 0 on the other cones, and the ray of u corresponds to a prime divisor E on Y_u [KKMS73, Chapter II, §2, Theorem 10*]. By (2), Y_u is dominated by h_t for a blow-up of strata h of $\mathcal{Y}$, and h_t is a blow-up of strata of $\mathcal{Y}_t$ by (1) and (4), so E is a toroidal prime divisor with center W_t and vector u.

(5) Let t be a geometric point of T. If a prime divisor E over $\mathcal{Y}_t$ is toroidal with respect to Σ'_t, then E is a component of the reduced total transform of Σ'_t on a blow-up of strata Y of $\mathcal{Y}'_t$. By (1) and (4), h_t is a blow-up of strata of $\mathcal{Y}_t$, hence so is $Y \to \mathcal{Y}_t$, and the reduced total transforms of Σ'_t and of Σ_t on Y are equal, so E is toroidal with respect to Σ_t. Conversely, let E be toroidal with respect to Σ_t. For any blow-up of strata $Y \to \mathcal{Y}_t$ with reduced total transform Σ_Y of Σ_t, $K_Y + \Sigma_Y$ is the pullback of $K_{\mathcal{Y}_t} + \Sigma_t$, since a stratum of codimension c is contained in exactly c components of the reduced total transform. As h_t is a blow-up of strata of $\mathcal{Y}_t$ by (1) and (4), $a(E, \mathcal{Y}'_t, \Sigma'_t) = a(E, \mathcal{Y}_t, \Sigma_t) = 0$, so E is toroidal with respect to Σ'_t by Lemma 9.5(2) applied to $\mathcal{Y}'_t$, Σ'_t, and $b_j = 1$.

Assume moreover that $W_t \ne \emptyset$ for any stratum W of Σ and any geometric point t of T. We show by induction on the number of blow-ups in h that the strata of Σ' map onto T and that, for E as in (5), the stratum W' of Σ' with $W'_t = \operatorname{center}_{\mathcal{Y}'_t} E$ and the vector $(\operatorname{mult}_E \Sigma'_{j,t})_{1 \le j \le n'}$ do not depend on t. Thus we may assume that h is the blow-up along a stratum V of Σ. Any stratum of Σ' is a component of a projective bundle over a stratum of Σ contained in V or the strict transform of a stratum of Σ, so its image in T is the image of a stratum of Σ, which is T. By (4),

for any geometric point t of T the cones of $\mathfrak{C}_t$ are the $\mathbb{R}^{\Lambda_W}_{\geq 0}$ for the strata W of Σ, and h_t is the toroidal modification given by the star subdivision of $\mathfrak{C}_t$ at $\sum_{j\in\Lambda_V} e_j$ [KKMS73, Chapter II, §2, Theorem 10*], which does not depend on t. Let C_u be the smallest cone of the star subdivision containing u. The prime divisor E corresponds to the ray of u, so $(\operatorname{mult}_E \Sigma'_{j,t})_{1\leq j\leq n'}$ is the vector of coordinates of u with respect to the primitive vectors of the rays of C_u, and by (4) for $\mathcal{Y}' \to T$ and Σ', $\operatorname{center}_{\mathcal{Y}'_t} E = W'_t$ for the stratum W' of Σ' corresponding to C_u. Since C_u does not depend on t, W' and $(\operatorname{mult}_E \Sigma'_{j,t})_{1\leq j\leq n'}$ do not depend on t. □

Proposition 9.8. *Notation and conditions as in Set-up 9.2. Let $I \subset [0,1]$ be a DCC set and $\mathcal{S}$ a set of projective lc pairs (X,B) of dimension 3 over algebraically closed fields of characteristic > 5 such that $B \in I$. Assume that for any $(X,B) \in \mathcal{S}$ there exist a geometric point t of T and a birational map $f : \mathcal{Y}_t \dashrightarrow X$ such that $\operatorname{Supp}\Sigma_t$ contains $\operatorname{Supp} f_*^{-1}B$ and any prime divisor on $\mathcal{Y}_t$ contracted by f. Then $\{\operatorname{vol}(K_X+B) \mid (X,B)\in\mathcal{S}\}$ is a DCC set.*

Proof. **Step 1.** In this step we fix notation for log pullbacks and toroidal divisors.

By Set-up 9.2(3) and (4) with $\#\Lambda = 1$, $\Sigma_{j,t}$ is integral or empty for any j and any geometric point t of T.

Let Y be a normal projective variety over an algebraically closed field and D a boundary on Y such that $K_Y + D$ is $\mathbb{R}$-Cartier. For any prime divisor E over Y we let

$$\mathbf{L}_D(E) := \max\{0, 1 - a(E,Y,D)\} \quad \text{and} \quad \mathbf{M}_D(E) := \begin{cases} \operatorname{mult}_E D & \text{if } E \text{ is a divisor on } Y,\\ 1 & \text{otherwise.}\end{cases}$$

For a $\boldsymbol{b}$-divisor $\mathbf{N}$ over Y, that is, a function from the set of prime divisors over Y to $\mathbb{R}$, and a birational morphism $g : Y' \to Y$ from a normal projective variety, we let $\mathbf{N}_{Y'} := \sum_P \mathbf{N}(P)P$, the sum running over the prime divisors P on Y', whenever the sum is finite. We write $\mathbf{L}_{D,Y'} := (\mathbf{L}_D)_{Y'}$ and $\mathbf{M}_{D,Y'} := (\mathbf{M}_D)_{Y'}$. For two such $\boldsymbol{b}$-divisors $\mathbf{N}$ and $\mathbf{N}'$, $\mathbf{N}\wedge\mathbf{N}'$ denotes the pointwise minimum, and for two $\mathbb{R}$-divisors D and D', $D\wedge D'$ denotes the coefficientwise minimum. Then $g_*\mathbf{L}_{D,Y'} = D$ and

$$K_{Y'} + \mathbf{L}_{D,Y'} = g^*(K_Y + D) + F \quad \text{with } F \geq 0 \text{ } g\text{-exceptional},$$

since the coefficient of F at a g-exceptional prime divisor E is $\max\{0, a(E,Y,D)-1\}$. We use the following two properties (cf. [HMX13, Lemma 5.3]). Let $g : Y' \to Y$ be a birational morphism between normal projective varieties, D' a boundary on Y', and D a boundary on Y such that $K_Y + D$ is $\mathbb{R}$-Cartier.

(C1) If $g_*D' = D$ and $D' \geq \mathbf{L}_{D,Y'}$, then $\operatorname{vol}(K_{Y'} + D') = \operatorname{vol}(K_Y + D)$.

(C2) If $g_*D' \leq D$, then for any integer $m \geq 0$,

$$H^0(Y', \mathcal{O}_{Y'}(\lfloor m(K_{Y'} + D')\rfloor)) = H^0(Y', \mathcal{O}_{Y'}(\lfloor m(K_{Y'} + D' \wedge \mathbf{L}_{D,Y'})\rfloor)).$$

In particular, $\operatorname{vol}(K_{Y'} + D') = \operatorname{vol}(K_{Y'} + D' \wedge \mathbf{L}_{D,Y'})$.

For (C1), $K_{Y'} + D' \geq K_{Y'} + \mathbf{L}_{D,Y'} \geq g^*(K_Y + D)$ and $g_*(K_{Y'} + D') = K_Y + D$, so (C1) follows from the last sentence of Lemma 2.4. For (C2), Lemma 2.4(1) and (2) imply that the left side is contained in $H^0(Y, \mathcal{O}_Y(\lfloor m(K_Y + D)\rfloor))$, which is contained in $H^0(Y', \mathcal{O}_{Y'}(\lfloor m(K_{Y'} + \mathbf{L}_{D,Y'})\rfloor))$. Since the coefficients of $K_{Y'}$ are integers and

$$\lfloor m \min\{x,y\}\rfloor = \min\{\lfloor mx\rfloor, \lfloor my\rfloor\} \quad \text{for any } x, y \in \mathbb{R}, \tag{9.7}$$

a rational function contained in both

$$H^0(Y', \mathcal{O}_{Y'}(\lfloor m(K_{Y'} + D')\rfloor)) \quad \text{and} \quad H^0(Y', \mathcal{O}_{Y'}(\lfloor m(K_{Y'} + \mathbf{L}_{D,Y'})\rfloor))$$

belongs to the right side. The reverse inclusion holds since $D' \wedge \mathbf{L}_{D,Y'} \leq D'$.

Let t be a geometric point of T. For a toroidal prime divisor E over $\mathcal{Y}_t$ (Definition 9.6) we let

$$u(E) := (\operatorname{mult}_E \Sigma_{1,t}, \dots, \operatorname{mult}_E \Sigma_{n,t}) \in \mathbb{N}^n.$$

For a smooth variety Y and a reduced divisor Σ_Y on Y with simple normal crossing support, the blow-up $Y_1 \to Y$ along a stratum of Σ_Y pulls back $K_Y + \Sigma_Y$ to $K_{Y_1} + \Sigma_{Y_1}$, where Σ_{Y_1} is the reduced total transform of Σ_Y, so $a(E, \mathcal{Y}_t, \Sigma_t) = 0$. Thus, for any toroidal E and any $D = \sum_{j=1}^n d_j \Sigma_{j,t}$ with $d_1, \dots, d_n \in [0,1]$, we have

$$a(E, \mathcal{Y}_t, D) = a(E, \mathcal{Y}_t, \Sigma_t) + \mathrm{mult}_E(\Sigma_t - D) = \sum_{j=1}^n u_j(E)(1 - d_j),$$

hence

$$\mathbf{L}_D(E) = \max\Big\{0, 1 - \sum_{j=1}^n u_j(E)(1-d_j)\Big\}, \tag{9.8}$$

and $\mathbf{L}_D(E) = 0$ for any prime divisor E over $\mathcal{Y}_t$ which is exceptional over $\mathcal{Y}_t$ and not toroidal, by Lemma 9.5(2).

The toroidal modifications of $(\mathcal{Y}_t, \Sigma_t)$ correspond to the subdivisions of the conical polyhedral complex $\mathfrak{C}_t$ of $(\mathcal{Y}_t, \Sigma_t)$, whose cones are $\mathbb{R}^{\Lambda_W}_{\geq 0} \subset \mathbb{R}^n$, one for each stratum W of Σ with $W_t \neq \emptyset$ (Lemma 9.7(4)) [KKMS73, Chapter II, §2, Theorem 6*]. Let $\mathcal{R}$ be the set of pairs $\rho = (W, u)$ of a stratum W of Σ and a primitive vector $u \in (\mathbb{N}^+)^{\Lambda_W}$, where we regard $\mathbb{R}^{\Lambda_W}$ as a coordinate subspace of $\mathbb{R}^n$. For $\rho = (W, u) \in \mathcal{R}$ and a geometric point t of T with $W_t \neq \emptyset$, let $E_{\rho,t}$ be the toroidal prime divisor over $\mathcal{Y}_t$ with center W_t and $u(E_{\rho,t}) = u$ (Lemma 9.7(3)). A toroidal modification Y'' of $(\mathcal{Y}_t, \Sigma_t)$ dominates another one Y' if and only if each cone of Y'' is contained in a cone of Y' [KKMS73, Chapter II, §2, Theorem 7*].

Step 2. In this step we choose a decreasing sequence of volumes and replace the pairs by blow-ups of strata of the fibers of $\mathcal{Y}$.

Suppose that the proposition does not hold. Then there exist pairs $(X_i, B_i) \in \mathcal{S}$ such that

$$v_1 > v_2 > \cdots, \quad \text{where } v_i := \mathrm{vol}(K_{X_i} + B_i) \text{ for } i \in \mathbb{N}^+.$$

Let t_i and $f_i : \mathcal{Y}_{t_i} \dashrightarrow X_i$ be the geometric points and birational maps given by the assumption. Since T is noetherian, there exists a closed subset $T' \subset T$ minimal among the closed subsets containing the images of t_i for infinitely many i. We equip T' with the reduced structure. By minimality, T' is irreducible, hence integral, and any proper closed subset of T' contains the images of only finitely many t_i. Smoothness over the base and the conditions on geometric fibers in Set-up 9.2 are preserved by base change, so the base change of $(\mathcal{Y} \to T, \Sigma)$ to T' satisfies the conditions of Set-up 9.2, after removing the indices j with $\Sigma_j \times_T T' = \emptyset$. The other $\Sigma_j \times_T T'$ are prime since they are smooth over the integral scheme T' with integral geometric fibers. Possibly passing to a subsequence and replacing T with T', we may assume that T is integral and that for any non-empty open subset $U \subset T$,

$$\text{the image of } t_i \text{ belongs to } U \text{ for any } i \gg 0. \tag{9.9}$$

Since any stratum W of Σ is smooth over T, the image of W in T is open. Possibly replacing T with the intersection of the finitely many non-empty such images and truncating the sequence, we may assume that $W_t \neq \emptyset$ for any non-empty stratum W of Σ and any geometric point t of T, and that (9.9) holds.

Let $\mathbf{M}_i := \mathbf{M}_{B_i}$, a $\boldsymbol{b}$-divisor over X_i, equivalently over $\mathcal{Y}_{t_i}$. For any smooth projective variety X_i' with a birational morphism $X_i' \to X_i$, since (X_i, B_i) is lc, we have $\mathbf{L}_{B_i, X_i'} \leq (\mathbf{M}_i)_{X_i'}$, so (C1) implies

$$\mathrm{vol}(K_{X_i'} + (\mathbf{M}_i)_{X_i'}) = v_i. \tag{9.10}$$

Let $B_{i,\mathcal{Y}} := (\mathbf{M}_i)_{\mathcal{Y}_{t_i}}$. A prime divisor on $\mathcal{Y}_{t_i}$ with positive coefficient in $B_{i,\mathcal{Y}}$ is either contracted by f_i or a component of $f_{i*}^{-1} B_i$, hence it is contained in $\mathrm{Supp}\, \Sigma_{t_i}$. Thus $B_{i,\mathcal{Y}}$ is a boundary supported on Σ_{t_i}. Let $\mathcal{P}_i$ be the set of prime divisors E on X_i which are exceptional over $\mathcal{Y}_{t_i}$ and satisfy $\mathbf{L}_{B_{i,\mathcal{Y}}}(E) > 0$. By Lemma 9.5(2), any $E \in \mathcal{P}_i$ is toroidal, so by Lemma 9.7(2), applied

at t_i, there exists a blow-up of strata $\widetilde{X}_i \to \mathcal{Y}_{t_i}$ on which any of the finitely many $E \in \mathcal{P}_i$ is a divisor. Let $\widetilde{B}_i := (\mathbf{M}_i)_{\widetilde{X}_i}$.

We show that $\mathrm{vol}(K_{\widetilde{X}_i} + \widetilde{B}_i) = v_i$. Let X_i' be a resolution of the closure of the graph of $X_i \dashrightarrow \widetilde{X}_i$ [Cut09, Theorem 1.1]. By (9.10) and Lemma 2.4(1), $v_i \le \mathrm{vol}(K_{\widetilde{X}_i} + \widetilde{B}_i)$. For the reverse inequality we show

$$(\mathbf{M}_i)_{X_i'} \ge (\mathbf{M}_{\widetilde{B}_i})_{X_i'} \wedge \mathbf{L}_{B_{i,\mathcal{Y}}, X_i'}. \tag{9.11}$$

Let P be a prime divisor on X_i'. If P is a divisor on $\widetilde{X}_i$, then $\mathbf{M}_{\widetilde{B}_i}(P) = \mathbf{M}_i(P)$. If P is exceptional over $\widetilde{X}_i$ and $\mathbf{M}_i(P) < 1$, then P is a divisor on X_i, and P is exceptional over $\mathcal{Y}_{t_i}$ since any prime divisor on $\mathcal{Y}_{t_i}$ is a divisor on $\widetilde{X}_i$. As $P \notin \mathcal{P}_i$, $\mathbf{L}_{B_{i,\mathcal{Y}}}(P) = 0$, so (9.11) holds at P. If $\mathbf{M}_i(P) = 1$, (9.11) holds at P since $\mathbf{M}_{\widetilde{B}_i}(P) \le 1 = \mathbf{M}_i(P)$. Since $(\mathbf{M}_{\widetilde{B}_i})_{X_i'}$ pushes forward to $(\mathbf{M}_i)_{\mathcal{Y}_{t_i}} = B_{i,\mathcal{Y}}$, (9.10), (9.11), (C2), and (C1) imply

$$v_i \ge \mathrm{vol}(K_{X_i'} + (\mathbf{M}_{\widetilde{B}_i})_{X_i'} \wedge \mathbf{L}_{B_{i,\mathcal{Y}}, X_i'}) = \mathrm{vol}(K_{X_i'} + (\mathbf{M}_{\widetilde{B}_i})_{X_i'}) = \mathrm{vol}(K_{\widetilde{X}_i} + \widetilde{B}_i).$$

Possibly replacing (X_i, B_i) with $(\widetilde{X}_i, \widetilde{B}_i)$ and $\mathbf{M}_i$ with $\mathbf{M}_{\widetilde{B}_i}$, we may assume the following for any i:

(i) X_i is a blow-up of strata of $\mathcal{Y}_{t_i}$, $\mathbf{M}_i = \mathbf{M}_{B_i}$, and (9.10) holds for any smooth projective variety X_i' with a birational morphism $X_i' \to X_i$.
(ii) $\mathbf{M}_i$ takes values in $I \cup \{0, 1\}$, $\mathbf{M}_i(E) = 0$ for any prime divisor E on $\mathcal{Y}_{t_i}$ not contained in $\mathrm{Supp}\, \Sigma_{t_i}$, and $\mathbf{M}_i(E) = 1$ for any prime divisor E exceptional over $\mathcal{Y}_{t_i}$ which is not toroidal.

Condition (i) holds by (C1), since (X_i, B_i) is lc, B_i being a boundary supported on the reduced total transform of Σ_{t_i}. Condition (ii) holds since the coefficients of the original B_i belong to I, since $B_{i,\mathcal{Y}}$ is supported on Σ_{t_i}, and since any prime divisor on X_i exceptional over $\mathcal{Y}_{t_i}$ is toroidal.

Step 3. In this step we take limits of the coefficients.

Let I' be the closure of $I \cup \{0, 1\}$ in $[0, 1]$. Then I' is a DCC set, and any sequence in I' has an increasing subsequence.

For $\rho = (W, u) \in \mathcal{R}$, $E_{\rho,t}$ is defined for any geometric point t of T since $W_t \ne \emptyset$ (**Step 2**), and we let $\mathbf{M}_i(\rho) := \mathbf{M}_i(E_{\rho,t_i})$. Since $\mathbf{M}_i(\rho) \in I'$, applying the existence of an increasing subsequence of any sequence in I' to $(\mathbf{M}_i(\rho))_{i \ge 1}$ for ρ in an enumeration of the countable set $\mathcal{R}$, possibly passing to subsequences one ρ at a time and then to the diagonal subsequence, we may assume that for any $\rho \in \mathcal{R}$ the sequence $(\mathbf{M}_i(\rho))_{i \ge 1}$ is increasing for $i \gg 0$. We let $\mathbf{M}(\rho) \in I'$ be the limit of $(\mathbf{M}_i(\rho))_{i \ge 1}$. Thus $\mathbf{M}_i(\rho) \le \mathbf{M}(\rho)$ for $i \gg 0$. For any geometric point t of T we extend $\mathbf{M}$ to the prime divisors E over $\mathcal{Y}_t$ by $\mathbf{M}(E_{\rho,t}) := \mathbf{M}(\rho)$ for $\rho \in \mathcal{R}$, by $\mathbf{M}(E) := 0$ if E is a divisor on $\mathcal{Y}_t$ not contained in $\mathrm{Supp}\, \Sigma_t$, and by $\mathbf{M}(E) := 1$ if E is exceptional over $\mathcal{Y}_t$ and not toroidal. By **Step 2**(ii), $\mathbf{M}_i$ satisfies the last two rules.

Step 4. In this step we construct a blow-up of strata $\mathcal{Y}' \to \mathcal{Y}$ such that $\mathbf{L}_{B'}(\rho) \le \mathbf{M}(\rho)$ for any $\rho \in \mathcal{R}$, where B' is the $\mathbb{R}$-divisor on $\mathcal{Y}'$ defined below.

We replace $\mathcal{Y}'$, the pairs (X_i, B_i), $\mathbf{M}_i$, $\mathbf{M}$, and I' finitely many times. At each stage $\mathcal{Y}' \to \mathcal{Y}$ is a blow-up of strata defined over T, with Σ' the reduced total transform of Σ, and the following hold. At the start, $\mathcal{Y}' = \mathcal{Y}$.

(a) For any i, X_i is a blow-up of strata of $\mathcal{Y}_{t_i}$ dominating $\mathcal{Y}'_{t_i}$, and **Step 2**(i) and the second and third clauses of **Step 2**(ii) hold.
(b) $I' \subset [0, 1]$ is a DCC set containing $\mathbf{M}(\mathcal{R})$, and for any $\rho \in \mathcal{R}$, $(\mathbf{M}_i(\rho))_{i \ge 1}$ is increasing for $i \gg 0$ with limit $\mathbf{M}(\rho)$.

By Lemma 9.7(1) and (5), $(\mathcal{Y}' \to T, \Sigma')$ satisfies the conditions of Set-up 9.2, the strata of Σ' have non-empty fibers over any geometric point of T, and the toroidal prime divisors over $\mathcal{Y}'_t$

with respect to Σ'_t are those over $\mathcal{Y}_t$ with respect to Σ_t. Let $\Sigma'_1, \dots, \Sigma'_{n'}$ be the components of Σ'. For $\rho \in \mathcal{R}$ we let

$$u'(\rho) := (\mathrm{mult}_{E_{\rho,t}} \Sigma'_{j,t})_{1 \le j \le n'},$$

which does not depend on t by Lemma 9.7(5). We regard $u'(\rho)$ as a point of the cone $\mathbb{R}^{\Lambda_{W'}}_{\ge 0}$ attached to the stratum W' of Σ' with $W'_t = \mathrm{center}_{\mathcal{Y}'_t} E_{\rho,t}$ in the conical polyhedral complex of $(\mathcal{Y}'_t, \Sigma'_t)$. Strata $W'_1 \ne W'_2$ of Σ' with $\Lambda_{W'_1} = \Lambda_{W'_2}$ have different cones. Let $b_j := \mathbf{M}(\Sigma'_{j,t})$ for any geometric point t of T, which does not depend on t since $\Sigma'_{j,t} = E_{\rho,t}$ for the $\rho \in \mathcal{R}$ such that $u'(\rho)$ is the j-th standard basis vector of $\mathbb{R}^{n'}$, and let $B' := \sum_{j=1}^{n'} b_j \Sigma'_j$ and $B_{i,\mathcal{Y}'} := (\mathbf{M}_i)_{\mathcal{Y}'_{t_i}}$. Thus the conclusions of **Step 1** hold for $(\mathcal{Y}', \Sigma')$, and $\mathbf{L}_{B'_t}(E_{\rho,t})$ is computed by (9.8) with $u'(\rho)$ in place of $u(E)$. For an $\mathbb{R}$-divisor D on a blow-up of strata $\mathcal{Y}'$ of $\mathcal{Y}$ supported on Σ' with coefficients in $[0,1]$ and $\rho \in \mathcal{R}$, we write $\mathbf{L}_D(\rho) := \mathbf{L}_{D_t}(E_{\rho,t})$ and $\mathbf{M}_D(\rho) := \mathbf{M}_{D_t}(E_{\rho,t})$, which do not depend on t by (9.8) and Lemma 9.7(5), and for a blow-up of strata $\mathcal{Y}''$ of $\mathcal{Y}'$ we let $\mathbf{L}_{D,\mathcal{Y}''}$ be the $\mathbb{R}$-divisor on $\mathcal{Y}''$ whose restriction to $\mathcal{Y}''_t$ is $\mathbf{L}_{D_t,\mathcal{Y}''_t}$ for any t.

We call $\rho \in \mathcal{R}$ *violating* if $\mathbf{L}_{B'}(\rho) > \mathbf{M}(\rho)$. If $E_{\rho,t}$ is a divisor on $\mathcal{Y}'_t$, then

$$\mathbf{L}_{B'}(\rho) = \mathrm{mult}_{E_{\rho,t}} B' = \mathbf{M}(\rho),$$

so $E_{\rho,t}$ is exceptional over $\mathcal{Y}'_t$ for any violating ρ. For a stratum W' of Σ' let

$$\Lambda_1(W') := \{j \in \Lambda_{W'} \mid b_j = 1\}, \quad \Lambda_{<1}(W') := \Lambda_{W'} \backslash \Lambda_1(W'). \tag{9.12}$$

For $\rho \in \mathcal{R}$ with $E_{\rho,t}$ exceptional over $\mathcal{Y}'_t$, let W' be the stratum of Σ' with $W'_t = \mathrm{center}_{\mathcal{Y}'_t} E_{\rho,t}$ (Lemma 9.5 and Lemma 9.7(4)), so that $u'_j(\rho) > 0$ for $j \in \Lambda_{W'}$, and call $\#\Lambda_1(W')$ the *weight* of ρ. Let $w \in \{-1, 0, 1, 2, 3\}$ be the maximal weight of violating elements of $\mathcal{R}$, with $w = -1$ if no element of $\mathcal{R}$ is violating. We show that the following replacement decreases w. Hence after at most 4 replacements, no element of $\mathcal{R}$ is violating.

Let W' be any stratum of Σ' with $\#\Lambda_{W'} \ge 2$. The set $F_{W'}$ of $x \in (\mathbb{N}^+)^{\Lambda_{<1}(W')}$ with $\sum_{j \in \Lambda_{<1}(W')} x_j(1 - b_j) < 1$ is finite, since $b_j < 1$ for $j \in \Lambda_{<1}(W')$. For any $x \in F_{W'}$ such that some $\rho \in \mathcal{R}$ satisfies $\mathrm{center}_{\mathcal{Y}'_t} E_{\rho,t} = W'_t$ and $u'_j(\rho) = x_j$ for $j \in \Lambda_{<1}(W')$, we choose such a ρ minimizing $\mathbf{M}(\rho)$, which exists since I' is a DCC set, and denote it by $\sigma = \sigma(W', x)$.

For a geometric point t of T, let $Y_{\sigma,t}$ be the toroidal modification of $\mathcal{Y}'_t$ given by the star subdivision at $u'(\sigma)$ [KKMS73, Chapter II, §2, Theorem 6*]. The variety $Y_{\sigma,t}$ is projective, since it is the normalized blow-up of $\mathcal{Y}'_t$ along a toroidal ideal [KKMS73, Chapter II, §2, Theorem 10*], and any toroidal Weil divisor on $Y_{\sigma,t}$ is $\mathbb{Q}$-Cartier, since the cones of $Y_{\sigma,t}$ are simplicial [KKMS73, Chapter II, §2, Theorem 9*, III(c)]. Inside $\mathbb{R}^{\Lambda_{W'}}_{\ge 0}$ the cones of $Y_{\sigma,t}$ are

$$C_k := \langle u'(\sigma), e_j : j \in \Lambda_{W'} \backslash \{k\} \rangle, \quad k \in \Lambda_{W'},$$

where e_j is the j-th standard basis vector of $\mathbb{R}^{\Lambda_{W'}}$, and $v \in \mathbb{R}^{\Lambda_{W'}}_{\ge 0}$ belongs to C_k if and only if

$$v = \lambda u'(\sigma) + \sum_{j \in \Lambda_{W'} \backslash \{k\}} \lambda_j e_j \quad \text{with} \quad \lambda = \frac{v_k}{u'_k(\sigma)} = \min_{j \in \Lambda_{W'}} \frac{v_j}{u'_j(\sigma)}, \ \lambda_j = v_j - \lambda u'_j(\sigma) \ge 0. \tag{9.13}$$

Let $E_{\sigma,t}$ be the exceptional divisor of $h_{\sigma,t} : Y_{\sigma,t} \to \mathcal{Y}'_t$. Since $a(E_{\sigma,t}, \mathcal{Y}'_t, \Sigma'_t) = 0$, we have

$$K_{Y_{\sigma,t}} + E_{\sigma,t} + h^{-1}_{\sigma,t*} \Sigma'_t = h^*_{\sigma,t}(K_{\mathcal{Y}'_t} + \Sigma'_t),$$

so by **Step 1**, for any $\rho \in \mathcal{R}$,

$$a(E_{\rho,t}, Y_{\sigma,t}, E_{\sigma,t} + h^{-1}_{\sigma,t*} \Sigma'_t) = a(E_{\rho,t}, \mathcal{Y}'_t, \Sigma'_t) = 0.$$

Since moreover the multiplicities of a toroidal $\mathbb{Q}$-Cartier divisor on $Y_{\sigma,t}$ along toroidal prime divisors are linear on each cone [KKMS73, Chapter II, §2, Theorem 9*, III(c)], for any boundary

$$B_{\sigma,t} = \gamma_\sigma E_{\sigma,t} + \sum_j \gamma_j h^{-1}_{\sigma,t*} \Sigma'_{j,t}$$

on $Y_{\sigma,t}$ and for any $\rho \in \mathcal{R}$ with $u'(\rho) \in C_k$, with λ and λ_j given by (9.13) for $v = u'(\rho)$,

$$a(E_{\rho,t}, Y_{\sigma,t}, B_{\sigma,t}) = \lambda(1-\gamma_\sigma) + \sum_{j \in \Lambda_{W'} \setminus \{k\}} \lambda_j(1-\gamma_j). \tag{9.14}$$

We let

$$B_{\sigma,t} := (\mathbf{L}_{B'_t} \wedge \mathbf{M})_{Y_{\sigma,t}},$$

which has non-zero coefficients only along $E_{\sigma,t}$ and the $h^{-1}_{\sigma,t*}\Sigma'_{j,t}$ since B' is supported on Σ'. Then $\gamma_j = b_j$ and

$$\gamma_\sigma = \min\{\mathbf{L}_{B'}(\sigma), \mathbf{M}(\sigma)\}.$$

By Lemma 9.7(2), applied at a geometric point t of T, there exists a blow-up of strata $\mathcal{Y}''$ of $\mathcal{Y}'$ such that $\mathcal{Y}''_t$ dominates $Y_{\sigma,t}$ for that t and any of the finitely many $\sigma = \sigma(W', x)$. By Lemma 9.7(1), for any geometric point t of T, $\mathcal{Y}''_t \to \mathcal{Y}'_t$ is the composite of the blow-ups along the fibers over t of the strata blown up by $\mathcal{Y}'' \to \mathcal{Y}'$, so by Lemma 9.7(4) and (5) the cones of $\mathcal{Y}''_t$ and of $Y_{\sigma,t}$ do not depend on t, and $\mathcal{Y}''_t$ dominates $Y_{\sigma,t}$ for any geometric point t of T [KKMS73, Chapter II, §2, Theorem 7*]. Let $\Sigma''_1, \dots, \Sigma''_{n''}$ be the components of the reduced total transform Σ'' of Σ' and $B'' := \sum_j b''_j \Sigma''_j$, where $b''_j := \min_\sigma \mathbf{L}_{B_{\sigma,t}}(\Sigma''_{j,t})$ for a geometric point t of T, the minimum running over the finitely many $\sigma = \sigma(W', x)$. By (9.14), b''_j does not depend on t. We replace $\mathcal{Y}'$ by $\mathcal{Y}''$ and $\mathbf{M}$ by $\mathbf{M} \wedge \mathbf{M}_{B''}$, so the new B' is $\widehat{B}' := \mathbf{M}_{\mathcal{Y}''} \wedge B''$, and $\mathbf{M}$ changes only at the prime divisors on $\mathcal{Y}''$.

For the sequence, let

$$B_{i,\sigma} := (\mathbf{L}_{B_{i,\mathcal{Y}'}} \wedge \mathbf{M}_i)_{Y_{\sigma,t_i}}$$

on Y_{σ,t_i}, let B''_i be the $\mathbb{R}$-divisor on $\mathcal{Y}''_{t_i}$ with $\operatorname{mult}_P B''_i := \min_\sigma \mathbf{L}_{B_{i,\sigma}}(P)$ for any prime divisor P on $\mathcal{Y}''_{t_i}$, the minimum running over the finitely many $\sigma = \sigma(W', x)$, and let $\widehat{X}_i$ be a blow-up of strata of $\mathcal{Y}_{t_i}$ dominating X_i and $\mathcal{Y}''_{t_i}$ (Lemma 9.7(2), applied at t_i). We replace X_i by $\widehat{X}_i$ and B_i by

$$\widehat{B}_i := (\mathbf{M}_i \wedge \mathbf{M}_{B''_i})_{\widehat{X}_i}.$$

Then $\mathbf{M}_{\widehat{B}_i}$ equals $\mathbf{M}_i \wedge \mathbf{M}_{B''_i}$, since both equal 1 at the prime divisors exceptional over $\widehat{X}_i$. We show that $\operatorname{vol}(K_{\widehat{X}_i} + \widehat{B}_i) = v_i$. Let $D := (\mathbf{M}_i)_{\widehat{X}_i}$ and $m \geq 0$ an integer, and let s be an element of

$$H^0(\widehat{X}_i, \mathcal{O}_{\widehat{X}_i}(\lfloor m(K_{\widehat{X}_i} + D)\rfloor)).$$

By Lemma 2.4(1),

$$s \in H^0(Y_{\sigma,t_i}, \mathcal{O}_{Y_{\sigma,t_i}}(\lfloor m(K_{Y_{\sigma,t_i}} + (\mathbf{M}_i)_{Y_{\sigma,t_i}})\rfloor)).$$

Since $(\mathbf{M}_i)_{Y_{\sigma,t_i}}$ pushes forward to $B_{i,\mathcal{Y}'}$ on $\mathcal{Y}'_{t_i}$, (C2) implies

$$s \in H^0(Y_{\sigma,t_i}, \mathcal{O}_{Y_{\sigma,t_i}}(\lfloor m(K_{Y_{\sigma,t_i}} + B_{i,\sigma})\rfloor)),$$

and since $K_{Y_{\sigma,t_i}} + B_{i,\sigma}$ is $\mathbb{R}$-Cartier, Lemma 2.4(2) implies

$$s \in H^0(\widehat{X}_i, \mathcal{O}_{\widehat{X}_i}(\lfloor m(K_{\widehat{X}_i} + \mathbf{L}_{B_{i,\sigma},\widehat{X}_i})\rfloor)).$$

Since

$$\widehat{B}_i \geq D \wedge \min_\sigma \mathbf{L}_{B_{i,\sigma},\widehat{X}_i},$$

the coefficients of $K_{\widehat{X}_i}$ are integers, and (9.7) holds, we obtain

$$s \in H^0(\widehat{X}_i, \mathcal{O}_{\widehat{X}_i}(\lfloor m(K_{\widehat{X}_i} + \widehat{B}_i)\rfloor)).$$

Since $\widehat{B}_i \leq D$,

$$H^0(\widehat{X}_i, \mathcal{O}_{\widehat{X}_i}(\lfloor m(K_{\widehat{X}_i} + \widehat{B}_i)\rfloor)) = H^0(\widehat{X}_i, \mathcal{O}_{\widehat{X}_i}(\lfloor m(K_{\widehat{X}_i} + D)\rfloor)),$$

and (9.10) implies $\operatorname{vol}(K_{\widehat{X}_i} + \widehat{B}_i) = v_i$. Since $(\widehat{X}_i, \widehat{B}_i)$ is lc, (C1) then implies (9.10) for any smooth projective variety with a birational morphism to $\widehat{X}_i$, and (a) holds after the replacement.

The second and third clauses of **Step 2**(ii) hold for $\mathbf{M}_i \wedge \mathbf{M}_{B_i''}$, since they hold for $\mathbf{M}_i$ and $\mathbf{M}_{B_i''} = 1$ at the prime divisors exceptional over $\mathcal{Y}_{t_i}''$.

For (b), let $\rho \in \mathcal{R}$ with $E_{\rho,t}$ a divisor on $\mathcal{Y}_t''$. By (9.8) and (9.14), there exist a finite subset $\mathcal{R}_\rho \subset \mathcal{R}$ and a composite $\phi_\rho : [0,1]^{\mathcal{R}_\rho} \to [0,1]$ of minima, maxima, and increasing affine functions such that

$$\min\{\mathbf{M}_i(\rho), \operatorname{mult}_{E_{\rho,t_i}} B_i''\} = \phi_\rho\big((\mathbf{M}_i(\rho'))_{\rho' \in \mathcal{R}_\rho}\big) \quad \text{and}$$
$$\min\{\mathbf{M}(\rho), \operatorname{mult}_{E_{\rho,t}} B''\} = \phi_\rho\big((\mathbf{M}(\rho'))_{\rho' \in \mathcal{R}_\rho}\big).$$

Since ϕ_ρ is continuous and increasing in each coordinate, the new $\mathbf{M}_i(\rho)$ is increasing for $i \gg 0$ with limit the new $\mathbf{M}(\rho)$, and I' together with the finitely many new values of $\mathbf{M}$ is a DCC set.

We show that the replacement decreases w. Let ρ be violating after the replacement. Then $E_{\rho,t}$ is exceptional over $\mathcal{Y}_t''$. Since $\widehat{B}' \le B'' \le \mathbf{L}_{B_{\sigma,t},\mathcal{Y}_t''}$ and $B_{\sigma,t} \le \mathbf{L}_{B_t',Y_{\sigma,t}}$, and since

$$K_{Y'} + \mathbf{L}_{D,Y'} = g^*(K_Y + D) + F$$

with $F \ge 0$ g-exceptional for any birational morphism $g : Y' \to Y$ and any boundary D on Y with $K_Y + D$ $\mathbb{R}$-Cartier (**Step 1**), where $F = 0$ for $g = h_{\sigma,t}$ and $D = B_t'$ because, for $\sigma = \sigma(W', x)$,

$$a(E_{\sigma,t}, \mathcal{Y}_t', B_t') = \sum_{j \in \Lambda_{<1}(W')} x_j(1 - b_j) < 1,$$

and where for $g : \mathcal{Y}_t'' \to Y_{\sigma,t}$ and $D = B_{\sigma,t}$ the $\mathbb{R}$-divisor $B'' \ge \widehat{B}'$ has coefficient 0 along the components of F, at which $\mathbf{L}_{B_{\sigma,t}} = 0$, so that on the smooth $\mathcal{Y}_t''$ any prime divisor with center in $\operatorname{Supp} F$ has log discrepancy at least 1 with respect to $\widehat{B}_t'$ by **Step 1**,

$$\mathbf{L}_{\widehat{B}'}(\rho) \le \mathbf{L}_{B_{\sigma,t}}(E_{\rho,t}) \le \mathbf{L}_{B'}(\rho) \quad \text{and} \quad \operatorname{mult}_{E_{\kappa,t}} \widehat{B}' \le \mathbf{L}_{B_{\sigma,t}}(E_{\kappa,t}) \le \mathbf{L}_{B'}(\kappa) \tag{9.15}$$

for any $\sigma = \sigma(W', x)$ and any $\kappa \in \mathcal{R}$ with $E_{\kappa,t}$ a divisor on $\mathcal{Y}_t''$. As the new $\mathbf{M}(\rho)$ equals the old one, ρ was violating before the replacement. Let W' be as in the definition of the weight of ρ for the old $\mathcal{Y}'$. Let C be the smallest cone of $\mathcal{Y}_t''$ containing $u'(\rho)$. Then $C \subset \mathbb{R}^{\Lambda_{W'}}_{\ge 0}$, and the new weight of ρ is the number of $\kappa \in \mathcal{R}$ such that $u'(\kappa)$ spans a ray of C and $\operatorname{mult}_{E_{\kappa,t}} \widehat{B}' = 1$, since $\operatorname{center}_{\mathcal{Y}_t''} E_{\rho,t}$ is the stratum corresponding to C, whose components of Σ_t'' correspond to the rays of C. For such κ, (9.15) and (9.8) imply $\mathbf{L}_{B'}(\kappa) = 1$, so $u_j'(\kappa) = 0$ for $j \in \Lambda_{<1}(W')$. Since $\mathcal{Y}_t''$ is smooth, the rays of C are linearly independent, and the new weight of ρ is at most $\#\Lambda_1(W') \le w$.

Suppose that the new weight of ρ is $\#\Lambda_1(W')$. Let $x := (u_j'(\rho))_{j \in \Lambda_{<1}(W')}$. Since $\mathbf{L}_{B'}(\rho) > 0$, $x \in F_{W'}$ by (9.8). Let $\sigma := \sigma(W', x)$. Then

$$\gamma_\sigma \le \mathbf{M}(\sigma) \le \mathbf{M}(\rho) < \mathbf{L}_{B'}(\rho) \le 1.$$

Since $\mathcal{Y}_t''$ dominates $Y_{\sigma,t}$, $C \subset C_k$ for some $k \in \Lambda_{W'}$. For a ray of C spanned by

$$u'(\kappa) = \lambda' u'(\sigma) + \sum_{j \ne k} \lambda_j' e_j$$

with $\operatorname{mult}_{E_{\kappa,t}} \widehat{B}' = 1$, (9.15) implies $\mathbf{L}_{B_{\sigma,t}}(E_{\kappa,t}) = 1$, so $\lambda' = 0$ and $\lambda_j' = 0$ for $j \in \Lambda_{<1}(W')$ by (9.14) and $\gamma_\sigma < 1$. Thus the $\#\Lambda_1(W')$ such rays are spanned by linearly independent elements of $\mathbb{R}^{\Lambda_1(W') \setminus \{k\}}_{\ge 0}$, so $k \in \Lambda_{<1}(W')$. Hence $u_k'(\sigma) = x_k = u_k'(\rho)$, so

$$\lambda = u_k'(\rho)/u_k'(\sigma) = 1,$$

and (9.14) implies

$$\mathbf{L}_{B_{\sigma,t}}(E_{\rho,t}) = \max\Big\{0, \gamma_\sigma - \sum_{j \ne k} \lambda_j(1 - b_j)\Big\} \le \gamma_\sigma \le \mathbf{M}(\rho).$$

By (9.15), $\mathbf{L}_{\widehat{B}'}(\rho) \le \mathbf{M}(\rho)$, which contradicts that ρ is violating. Therefore the new weight of ρ is less than $\#\Lambda_1(W') \le w$, and w decreases.

After the replacements, $\mathbf{L}_{B'}(\rho) \le \mathbf{M}(\rho)$ for any $\rho \in \mathcal{R}$. Moreover, let t be a geometric point of T and E a prime divisor over $\mathcal{Y}'_t$ which is not of the form $E_{\rho,t}$. Then either E is exceptional over $\mathcal{Y}'_t$ and not toroidal, so $\mathbf{L}_{B'_t}(E) = 0$ by **Step 1**, or E is a divisor on $\mathcal{Y}'_t$ not contained in $\operatorname{Supp}\Sigma'_t$, so $\mathbf{L}_{B'_t}(E) = \mathbf{M}(E) = 0$. Hence

$$\mathbf{L}_{B'_t}(E) \le \mathbf{M}(E) \quad \text{for any prime divisor } E \text{ over } \mathcal{Y}'_t. \tag{9.16}$$

Step 5. In this step we bound v_i from above.

Each non-zero coefficient of $B_{i,\mathcal{Y}'}$ is $\mathbf{M}_i(\rho)$ for one of the finitely many $\rho \in \mathcal{R}$ with E_{ρ,t_i} a component of Σ'_{t_i}, so by (b), $B_{i,\mathcal{Y}'} \le B'_{t_i}$ for any $i \gg 0$. By Lemma 9.3 for $(\mathcal{Y}' \to T, \Sigma')$ and B', there exist a non-empty open subset $T^\circ \subset T$ and a real number V_0 such that $\operatorname{vol}(K_{\mathcal{Y}'_t} + B'_t) = V_0$ for any geometric point t of T°. By (9.9), t_i is a geometric point of T° for $i \gg 0$. Since X_i dominates $\mathcal{Y}'_{t_i}$ and $B_{i,\mathcal{Y}'}$ is the pushforward of B_i, Lemma 2.4(1) implies

$$v_i \le \operatorname{vol}(K_{\mathcal{Y}'_{t_i}} + B_{i,\mathcal{Y}'}) \le \operatorname{vol}(K_{\mathcal{Y}'_{t_i}} + B'_{t_i}) = V_0 \quad \text{for any } i \gg 0. \tag{9.17}$$

Step 6. In this step we bound v_i from below and conclude.

Let $\epsilon \in (0,1)$ be a real number. Any coefficient b of B' satisfies $1-(1-\epsilon)b \ge \epsilon$, so by (9.8) the set of $\rho \in \mathcal{R}$ with $E_{\rho,t}$ exceptional over $\mathcal{Y}'_t$ and $\mathbf{L}_{(1-\epsilon)B'}(\rho) > 0$ is finite: such ρ satisfy $\sum_j u'_j(\rho) < 1/\epsilon$. By Lemma 9.7(2), applied at one geometric point of T, there exists a blow-up of strata $\mathcal{Y}^\epsilon \to \mathcal{Y}'$ such that $E_{\rho,t}$ is a divisor on $\mathcal{Y}^\epsilon_t$ at that point for any such ρ. By Lemma 9.7(1), (4), and (5), the cones of $\mathcal{Y}^\epsilon_t$ do not depend on t, so this holds for any geometric point t of T [KKMS73, Chapter II, §2, Theorems 6* and 7*]. Let

$$B^\epsilon := \mathbf{L}_{(1-\epsilon)B',\mathcal{Y}^\epsilon},$$

an $\mathbb{R}$-divisor on $\mathcal{Y}^\epsilon$ whose coefficients do not depend on t by (9.8). By the choice of $\mathcal{Y}^\epsilon$ and **Step 1**, $\mathbf{L}_{(1-\epsilon)B'}(E) = 0$ for any prime divisor E exceptional over $\mathcal{Y}^\epsilon_t$. Since $K_{\mathcal{Y}^\epsilon} + B^\epsilon$ is the pullback of $K_{\mathcal{Y}'} + (1-\epsilon)B'$ plus an exceptional $\mathbb{R}$-divisor $F \ge 0$, along whose components B^ϵ has coefficient 0, any prime divisor over the smooth $\mathcal{Y}^\epsilon_t$ with center in $\operatorname{Supp} F_t$ has log discrepancy at least 1 with respect to B^ϵ_t by **Step 1**, so $\mathbf{L}_{B^\epsilon} \le \mathbf{L}_{(1-\epsilon)B'}$. Hence $\mathbf{L}_{B^\epsilon}(E) = 0$ for any such E.

Let P be a prime divisor on $\mathcal{Y}^\epsilon_t$ with $\operatorname{mult}_P B^\epsilon > 0$. Then $P = E_{\rho,t}$ for some $\rho \in \mathcal{R}$,

$$\operatorname{mult}_P B^\epsilon = 1 - \sum_j u'_j(\rho)(1-(1-\epsilon)b_j) > 0,$$

and $\sum_j u'_j(\rho) b_j > 0$, since otherwise $\operatorname{mult}_P B^\epsilon = 1 - \sum_j u'_j(\rho) \le 0$ for exceptional P and $\operatorname{mult}_P B^\epsilon = 0$ for $P = \Sigma'_{j,t}$. Hence $\operatorname{mult}_P B^\epsilon < \mathbf{L}_{B'}(P) \le \mathbf{M}(P)$ by (9.8) and (9.16). By (b), at each of the finitely many components P of B^ϵ, $\mathbf{M}_i(P)$ converges to $\mathbf{M}(P) > \operatorname{mult}_P B^\epsilon$, so

$$B^\epsilon_{t_i} \le (\mathbf{M}_i)_{\mathcal{Y}^\epsilon_{t_i}} \quad \text{for any } i \gg 0.$$

Let $i \gg 0$ and let X'_i be a blow-up of strata of $\mathcal{Y}_{t_i}$ dominating X_i and $\mathcal{Y}^\epsilon_{t_i}$ (Lemma 9.7(2), applied at t_i). At a prime divisor P on X'_i which is a divisor on $\mathcal{Y}^\epsilon_{t_i}$,

$$\mathbf{L}_{B^\epsilon_{t_i}}(P) = \operatorname{mult}_P B^\epsilon_{t_i} \le \mathbf{M}_i(P).$$

At the other prime divisors on X'_i, $\mathbf{L}_{B^\epsilon_{t_i}}(P) = 0$. Thus $\mathbf{L}_{B^\epsilon_{t_i},X'_i} \le (\mathbf{M}_i)_{X'_i}$. Moreover

$$\mathbf{L}_{B^\epsilon_{t_i},X'_i} = \mathbf{L}_{(1-\epsilon)B'_{t_i},X'_i},$$

since both sides have the coefficients of $B^\epsilon_{t_i}$ along the divisors on $\mathcal{Y}^\epsilon_{t_i}$ and vanish elsewhere. By (9.10), the monotonicity of the volume, and (C1),

$$v_i = \operatorname{vol}(K_{X'_i} + (\mathbf{M}_i)_{X'_i}) \ge \operatorname{vol}(K_{X'_i} + \mathbf{L}_{(1-\epsilon)B'_{t_i},X'_i}) = \operatorname{vol}(K_{\mathcal{Y}'_{t_i}} + (1-\epsilon)B'_{t_i}) \quad \text{for any } i \gg 0. \tag{9.18}$$

Let $\overline{\eta}$ be a geometric point of T over the generic point. By Lemma 9.3 for $(1-\epsilon)B'$ and by (9.9), the right side of (9.18) equals

$$V_\epsilon := \operatorname{vol}(K_{\mathcal{Y}'_{\overline{\eta}}} + (1-\epsilon)B'_{\overline{\eta}})$$

for any $i \gg 0$. Thus $\lim_{i\to+\infty} v_i \geq V_\epsilon$ for any $\epsilon \in (0,1)$. By Lemma 9.3 for B', $V_0 = \mathrm{vol}(K_{\mathcal{Y}'_{\overline{\eta}}} + B'_{\overline{\eta}})$. Since $\mathcal{Y}'_{\overline{\eta}}$ is a smooth projective variety over an algebraically closed field, the volume is continuous on $N^1(\mathcal{Y}'_{\overline{\eta}})_{\mathbb{R}}$ [Cut15, Theorem 2.5], and it coincides with the volume defined by $\lfloor\cdot\rfloor$ [FKL16, Theorem 3.5(ii)]. Hence $V_\epsilon \to V_0$ as $\epsilon \to 0$, and $\lim_{i\to+\infty} v_i \geq V_0$. Since $(v_i)_{i\geq 1}$ is strictly decreasing, (9.17) implies

$$V_0 \geq v_i > v_{i+1} \geq \lim_{j\to+\infty} v_j \geq V_0 \quad \text{for any } i \gg 0,$$

a contradiction. The proposition follows. □

Theorem 9.9. *Let $I \subset [0,1]$ be a DCC set and $\mathcal{S}$ a set of projective lc pairs (X,B) of dimension 3 over algebraically closed fields of characteristic > 5 such that $B \in I$. Assume that $\mathcal{S}$ is log birationally bounded over $\mathbb{Z}[1/30]$. Then $\{\mathrm{vol}(K_X + B) \mid (X,B) \in \mathcal{S}\}$ is a DCC set.*

Proof. Let T, $\pi : \mathcal{Y} \to T$, and $\mathcal{Q}$ be as in Definition 9.1 for the set of couples $(X, \operatorname{Supp} B)$ with $(X,B) \in \mathcal{S}$.

Step 1. Let $T' \subset T$ be an integral closed subscheme. In this step we construct a dense open subset $U' \subset T'$, a scheme T''_0 of finite type over $\mathbb{Z}[1/30]$ with a morphism $\rho_0 : T''_0 \to T'$, a smooth projective morphism $\mathcal{Z} \to T''_0$, and a divisor $\Sigma = \sum_{j=1}^n \Sigma_j$ on $\mathcal{Z}$ satisfying the following.

(1) T''_0, $\mathcal{Z} \to T''_0$, and Σ satisfy the conditions of Set-up 9.2.
(2) For any geometric point $t : \operatorname{Spec} k \to U'$, there exist a geometric point $t'' : \operatorname{Spec} k \to T''_0$ with $\rho_0 \circ t'' = t$ and a birational morphism $q_{t''} : \mathcal{Z}_{t''} \to \mathcal{Y}_t$ such that $\operatorname{Supp} \Sigma_{t''}$ contains the exceptional locus of $q_{t''}$ and $q_{t''}^{-1}(\mathcal{Q}_t)$.

Let $\overline{\eta} : \operatorname{Spec} \overline{k(T')} \to T'$ be a geometric generic point of T', where $\overline{k(T')}$ is an algebraic closure of the function field $k(T')$. Since T is of finite type over $\mathbb{Z}[1/30]$, $\operatorname{char} k(T') = 0$ or $\operatorname{char} k(T') > 5$. The fiber $\mathcal{Y}_{\overline{\eta}}$ is a normal integral projective threefold and $\mathcal{Q}_{\overline{\eta}} \neq \mathcal{Y}_{\overline{\eta}}$. Let

$$q_{\overline{\eta}} : \mathcal{Z}_{\overline{\eta}} \to \mathcal{Y}_{\overline{\eta}}$$

be a log resolution of $(\mathcal{Y}_{\overline{\eta}}, \mathcal{Q}_{\overline{\eta}})$ [FW24, Proposition 2.4], whose exceptional locus is the support of a divisor $\operatorname{Exc}(q_{\overline{\eta}})$, let $\Sigma_{\overline{\eta}}$ be the reduced divisor with support

$$\operatorname{Exc}(q_{\overline{\eta}}) \cup q_{\overline{\eta}}^{-1}(\mathcal{Q}_{\overline{\eta}})$$

and prime components $\Sigma_{\overline{\eta},1}, \dots, \Sigma_{\overline{\eta},n}$, and let

$$\mathcal{Q}'_{\overline{\eta}} := \mathcal{Q}_{\overline{\eta}} \cup q_{\overline{\eta}}(\operatorname{Exc}(q_{\overline{\eta}})).$$

Since $\mathcal{Y}_{\overline{\eta}}$ is normal, $q_{\overline{\eta}}$ is an isomorphism over the open subset

$$\mathcal{Y}_{\overline{\eta}} \setminus \mathcal{Q}'_{\overline{\eta}} \neq \emptyset,$$

and

$$q_{\overline{\eta}}^{-1}(\mathcal{Q}'_{\overline{\eta}}) = \operatorname{Supp} \Sigma_{\overline{\eta}}.$$

For any non-empty subset $\Lambda \subset \{1, \dots, n\}$, since $\Sigma_{\overline{\eta}}$ is a simple normal crossing divisor, $\bigcap_{j\in\Lambda} \Sigma_{\overline{\eta},j}$ is either empty or the disjoint union of finitely many smooth integral closed subschemes of dimension $3 - |\Lambda|$.

The finitely many schemes, morphisms, and closed subschemes above are defined over a finite extension k' of $k(T')$ in $\overline{k(T')}$ [Gro66, Théorèmes 8.5.2, 8.8.2, and 8.10.5]. Let T'' be the normalization of T' in k'. Then the induced morphism $\rho : T'' \to T'$ is finite [Gro65, Scholie 7.8.3] and surjective. By [Gro66, Théorèmes 8.5.2, 8.8.2, and 8.10.5] again, there exist a dense open subset $T''_0 \subset T''$, a projective morphism $q : \mathcal{Z} \to \mathcal{Y} \times_T T''_0$, prime divisors $\Sigma_1, \dots, \Sigma_n$ on $\mathcal{Z}$, and a closed subset $\mathcal{Q}' \subset \mathcal{Y} \times_T T''_0$, whose base changes to $\overline{k(T')}$ are $q_{\overline{\eta}}$, $\Sigma_{\overline{\eta},1}, \dots, \Sigma_{\overline{\eta},n}$, and $\mathcal{Q}'_{\overline{\eta}}$, such that q is an isomorphism over the complement of $\mathcal{Q}'$. For any non-empty subset $\Lambda \subset \{1, \dots, n\}$, the connected components of $\bigcap_{j\in\Lambda} \Sigma_{\overline{\eta},j}$ also spread to closed subschemes of $\bigcap_{j\in\Lambda} \Sigma_j$. Let $\rho_0 := \rho|_{T''_0}$ and $\Sigma := \sum_{j=1}^n \Sigma_j$. Each of the following properties holds over the generic point of T''_0. By generic freeness [Mat89, Theorem 24.1], the openness of geometrically regular and geometrically

integral fibers [Gro66, Théorème 12.2.4(iii)(viii)], and the upper semicontinuity of fiber dimension [Gro66, Corollaire 13.1.5], possibly replacing T_0'' with a dense open subset, we may assume the following.

(a) $\mathcal{Z} \to T_0''$ is smooth and projective, and the geometric fibers of $\mathcal{Z} \to T_0''$ are integral threefolds.
(b) For any non-empty subset $\Lambda \subset \{1, \dots, n\}$, the scheme $\bigcap_{j\in\Lambda} \Sigma_j$ is either empty or smooth over T_0'' and the disjoint union of finitely many closed subschemes whose geometric fibers over T_0'' are integral of dimension $3 - |\Lambda|$.
(c) For any geometric point t'' of T_0'', $\dim \mathcal{Q}'_{t''} \le 2$.

The subsets $(\mathcal{Q} \times_T T_0'')\backslash\mathcal{Q}'$ and $q^{-1}(\mathcal{Q}')\backslash \operatorname{Supp}\Sigma$ are constructible, and their images in T_0'' are constructible subsets not containing the generic point of T_0'', hence not dense. Thus, possibly replacing T_0'' with a dense open subset, we may further assume the following.

(d) $\mathcal{Q} \times_T T_0'' \subset \mathcal{Q}'$ and $q^{-1}(\mathcal{Q}') \subset \operatorname{Supp}\Sigma$.

By (a) and (b), T_0'', $\mathcal{Z} \to T_0''$, and Σ satisfy the conditions of Set-up 9.2. This proves (1). Let $t'' : \operatorname{Spec} k \to T_0''$ be a geometric point and $t := \rho_0 \circ t''$. The morphism $q_{t''} : \mathcal{Z}_{t''} \to \mathcal{Y}_t$ is an isomorphism over $\mathcal{Y}_t\backslash\mathcal{Q}'_{t''}$, which is dense in $\mathcal{Y}_t$ by (c), so $q_{t''}$ is birational and the exceptional locus of $q_{t''}$ is contained in $q_{t''}^{-1}(\mathcal{Q}'_{t''})$. By (d),

$$q_{t''}^{-1}(\mathcal{Q}_t) \subset q_{t''}^{-1}(\mathcal{Q}'_{t''}) \subset \operatorname{Supp}\Sigma_{t''}.$$

Let

$$U' := T'\backslash\rho(T''\backslash T_0'').$$

Since ρ is finite and ρ^{-1} of the generic point of T' is the generic point of T'', U' is a dense open subset of T' and $\rho^{-1}(U') \subset T_0''$. For any geometric point $t : \operatorname{Spec} k \to U'$, the fiber $T'' \times_{T'} \operatorname{Spec} k$ is a non-empty finite scheme over k, so it has a k-point t'', and t'' factors through T_0''. This proves (2).

Step 2. We apply **Step 1** to the irreducible components T' of T with reduced structure, then to the irreducible components of $T'\backslash U'$ with reduced structure, and so on. Since $\dim T < +\infty$, the dimensions strictly decrease, and any closed subset of T has finitely many irreducible components, after finitely many applications we obtain finitely many schemes $T_1, \dots, T_N$ of finite type over $\mathbb{Z}[1/30]$, morphisms $\rho_i : T_i \to T$, smooth projective morphisms $\mathcal{Z}_i \to T_i$, and divisors $\Sigma^{(i)}$ on $\mathcal{Z}_i$ satisfying the following.

(1) For any i, T_i, $\mathcal{Z}_i \to T_i$, and $\Sigma^{(i)}$ satisfy the conditions of Set-up 9.2.
(2) For any geometric point $t : \operatorname{Spec} k \to T$, there exist $i \in \{1, \dots, N\}$, a geometric point $t'' : \operatorname{Spec} k \to T_i$ with $\rho_i \circ t'' = t$, and a birational morphism $q_{t''} : \mathcal{Z}_{i,t''} \to \mathcal{Y}_t$ such that $\operatorname{Supp}\Sigma^{(i)}_{t''}$ contains the exceptional locus of $q_{t''}$ and $q_{t''}^{-1}(\mathcal{Q}_t)$.

Step 3. Let $(X, B) \in \mathcal{S}$ over a field k, and let $t : \operatorname{Spec} k \to T$ be a geometric point and $f : \mathcal{Y}_t \dashrightarrow X$ a birational map such that $\mathcal{Q}_t$ contains $\operatorname{Supp} f_*^{-1}B$ and any prime divisor on $\mathcal{Y}_t$ contracted by f. Let i, t'', and $q_{t''}$ be as in **Step 2**(2), and let $g := f \circ q_{t''} : \mathcal{Z}_{i,t''} \dashrightarrow X$. Let P be a prime divisor on $\mathcal{Z}_{i,t''}$ which is either contracted by g or a component of $g_*^{-1}B$. If P is contained in the exceptional locus of $q_{t''}$, then $P \subset \operatorname{Supp}\Sigma^{(i)}_{t''}$. Otherwise $q_{t''}(P)$ is a prime divisor on $\mathcal{Y}_t$ which is either contracted by f or a component of $f_*^{-1}B$, so $q_{t''}(P) \subset \mathcal{Q}_t$ and

$$P \subset q_{t''}^{-1}(\mathcal{Q}_t) \subset \operatorname{Supp}\Sigma^{(i)}_{t''}.$$

Thus $\operatorname{Supp}\Sigma^{(i)}_{t''}$ contains $\operatorname{Supp} g_*^{-1}B$ and any prime divisor on $\mathcal{Z}_{i,t''}$ contracted by g.

For any $i \in \{1, \dots, N\}$, let $\mathcal{S}_i$ be the set of $(X, B) \in \mathcal{S}$ for which there exist a geometric point t'' of T_i and a birational map $g : \mathcal{Z}_{i,t''} \dashrightarrow X$ such that $\operatorname{Supp}\Sigma^{(i)}_{t''}$ contains $\operatorname{Supp} g_*^{-1}B$ and any prime divisor on $\mathcal{Z}_{i,t''}$ contracted by g. Then $\mathcal{S} = \bigcup_{i=1}^N \mathcal{S}_i$, and by Proposition 9.8 for T_i, $\mathcal{Z}_i \to T_i$, $\Sigma^{(i)}$, and $\mathcal{S}_i$, $\{\operatorname{vol}(K_X + B) \mid (X, B) \in \mathcal{S}\}$ is a finite union of DCC sets, hence a DCC set. □

Lemma 9.10 (cf. [Tan24, Proposition 5.3]). *Let n and N be positive integers. Then there exist a scheme T of finite type over $\mathbb{Z}[1/30]$, a projective flat morphism $\pi : \mathcal{Y} \to T$ whose geometric fibers are normal integral threefolds, and a closed subset $\mathcal{Q} \subset \mathcal{Y}$ containing no geometric fiber of π, with the following property. For any algebraically closed field k of characteristic > 5, any integral threefold $Y \subset \mathbb{P}^N_k$ with $\deg Y \leq n$, and any reduced closed subscheme $\Sigma \subset Y$ of pure dimension 2 whose irreducible components have degrees adding up to at most n, there exist a geometric point $t : \operatorname{Spec} k \to T$ and an isomorphism $\mathcal{Y}_t \cong Y^\nu$ onto the normalization $\nu : Y^\nu \to Y$ under which $\nu^{-1}(\Sigma) \subset \mathcal{Q}_t$.*

Proof. Let Ω be an algebraically closed field and $V \subset \mathbb{P}^M_\Omega$ an integral closed subvariety of dimension d and degree e. Then the linear span $\langle V \rangle$ of V satisfies

$$\dim\langle V\rangle \leq e + d - 1. \tag{9.19}$$

Since linear embeddings of projective spaces do not change Hilbert polynomials, the Hilbert polynomial of V equals the Hilbert polynomial of the image of V under a linear embedding $\langle V\rangle \to \mathbb{P}^{e+d-1}_\Omega$.

Step 1. For $d \in \{2, 3\}$ and an integer $e \geq 1$, let $\Phi_{d,e}$ be the set of Hilbert polynomials of integral closed subvarieties of $\mathbb{P}^{e+d-1}_\Omega$ of dimension d and degree e, where Ω runs over all algebraically closed fields. By the previous paragraph, the Hilbert polynomial of any integral closed subvariety of dimension d and degree e of $\mathbb{P}^M_\Omega$, for any M and any algebraically closed field Ω, belongs to $\Phi_{d,e}$. In this step we show that $\Phi_{d,e}$ is finite.

For $\varphi \in \Phi_{d,e}$, let

$$\mathcal{H}^\varphi := \operatorname{Hilb}^\varphi(\mathbb{P}^{e+d-1}_\mathbb{Z}/\mathbb{Z}),$$

which is projective over $\mathbb{Z}$ [Kol96, Theorem I.1.4]. Let $\mathcal{H}^\varphi_\circ \subset \mathcal{H}^\varphi$ be the open subset of points over which the fiber of the universal family is geometrically integral [Gro66, Théorème 12.2.4(viii)], and let $\mathcal{H}^\varphi_\nu$ be the normalization of $(\mathcal{H}^\varphi_\circ)_{\mathrm{red}}$, which is finite over $(\mathcal{H}^\varphi_\circ)_{\mathrm{red}}$ [Gro65, Scholie 7.8.3]. Since $\mathcal{H}^\varphi_\nu$ is normal, the pullback of the universal family to $\mathcal{H}^\varphi_\nu$ is a well-defined family of algebraic cycles [Kol96, Theorem I.3.17]. Let $\operatorname{Chow}_{d,e}$ be the Chow variety of d-dimensional cycles of degree e on $\mathbb{P}^{e+d-1}_\mathbb{Z}$, which is projective over $\mathbb{Z}$ [Kol96, Theorem I.4.13]. By [Kol96, Theorem I.4.13], there exists a morphism

$$c_\varphi : \mathcal{H}^\varphi_\nu \to \operatorname{Chow}_{d,e}$$

which maps any Ω-point of $\mathcal{H}^\varphi_\nu$ over the point of $\mathcal{H}^\varphi$ corresponding to $V \subset \mathbb{P}^{e+d-1}_\Omega$ to the cycle $[V]$, and the image of c_φ is contained in the subset $\operatorname{Chow}^{\mathrm{int}}_{d,e}$ of $\operatorname{Chow}_{d,e}$ of geometrically integral cycles. The subset $\operatorname{Chow}^{\mathrm{int}}_{d,e}$ is open: by [Kol96, Lemma I.3.24.2], the irreducible components of a cycle γ correspond to the irreducible factors of the Chow form $\operatorname{ch}(\gamma)$ of γ, and

$$\operatorname{ch}\Big(\sum_i a_i[V_i]\Big) = \prod_i \operatorname{ch}([V_i])^{a_i},$$

so a cycle over an algebraically closed field is integral of multiplicity 1 if and only if its Chow form is irreducible and reduced, and the reducible or non-reduced forms of a fixed multidegree form a closed subset (the union of the images of the multiplication maps from products of spaces of forms of smaller multidegrees). Let

$$\mathrm{Ch}_\varphi := c_\varphi(\mathcal{H}^\varphi_\nu),$$

a constructible subset of $\operatorname{Chow}^{\mathrm{int}}_{d,e}$. Then:

(i) For any $\varphi \in \Phi_{d,e}$, $\mathrm{Ch}_\varphi \neq \emptyset$, since any Ω-point of $(\mathcal{H}^\varphi_\circ)_{\mathrm{red}}$ lifts to $\mathcal{H}^\varphi_\nu$.

(ii) We have

$$\bigcup_{\varphi\in\Phi_{d,e}} \mathrm{Ch}_\varphi = \operatorname{Chow}^{\mathrm{int}}_{d,e},$$

since for any point $x \in \operatorname{Chow}^{\mathrm{int}}_{d,e}$ and an algebraically closed field Ω containing the residue field of x, the cycle over Ω defined by x is $[V]$ for an integral closed subvariety $V \subset \mathbb{P}^{e+d-1}_\Omega$

of dimension d and degree e, and the Hilbert polynomial φ of V belongs to $\Phi_{d,e}$, so $x \in \mathrm{Ch}_\varphi$.

(iii) For any $\varphi \neq \varphi'$ in $\Phi_{d,e}$,
$$\mathrm{Ch}_\varphi \cap \mathrm{Ch}_{\varphi'} = \emptyset :$$
a point of $\mathrm{Ch}_\varphi \cap \mathrm{Ch}_{\varphi'}$ would define, over an algebraically closed field Ω containing the residue fields of a preimage in $\mathcal{H}_\nu^\varphi$ and of a preimage in $\mathcal{H}_\nu^{\varphi'}$, two integral closed subvarieties $V, V' \subset \mathbb{P}^{e+d-1}_\Omega$ with Hilbert polynomials φ, φ' and $[V] = [V']$. Since $V = \mathrm{Supp}[V]$ and $V' = \mathrm{Supp}[V']$, $V = V'$, so $\varphi = \varphi'$.

Suppose that $\Phi_{d,e}$ is infinite. Since $\mathrm{Chow}^{\mathrm{int}}_{d,e}$ is noetherian, by (i) there exists a closed subset $Z \subset \mathrm{Chow}^{\mathrm{int}}_{d,e}$ which is minimal among the closed subsets intersecting Ch_φ for infinitely many $\varphi \in \Phi_{d,e}$. By minimality, Z is irreducible. By (ii), the generic point of Z belongs to Ch_{φ_0} for some $\varphi_0 \in \Phi_{d,e}$. Since $\mathrm{Ch}_{\varphi_0} \cap Z$ is constructible and contains the generic point of Z, it contains a dense open subset U of Z. By (iii), $\mathrm{Ch}_\varphi \cap Z \subset Z \backslash U$ for any $\varphi \neq \varphi_0$, so the proper closed subset $Z \backslash U$ of Z intersects Ch_φ for infinitely many φ, contradicting the minimality of Z. Thus $\Phi_{d,e}$ is finite.

Step 2. In this step we parametrize the pairs (Y, Σ) in the lemma by a flat family of embedded threefolds. For $d \in \{2, 3\}$, let
$$\mathcal{H}_d := \coprod_{e=1}^{n} \coprod_{\varphi \in \Phi_{d,e}} \mathrm{Hilb}^\varphi(\mathbb{P}^N_{\mathbb{Z}}/\mathbb{Z})$$
and let $\mathcal{V}_d \subset \mathbb{P}^N_{\mathcal{H}_d}$ be the universal family. By **Step 1** and [Kol96, Theorem I.1.4], $\mathcal{H}_d$ is projective over $\mathbb{Z}$. For any integer $0 \leq m \leq n$, let
$$\mathcal{T}_m := \mathcal{H}_3 \times_{\mathbb{Z}} \mathcal{H}_2^{\times m} \times_{\mathbb{Z}} \mathrm{Spec}\, \mathbb{Z}[1/30],$$
let $\mathcal{X}_m \to \mathcal{T}_m$ be the pullback of $\mathcal{V}_3$ under the first projection, and let $\mathcal{C}_m \subset \mathcal{X}_m$ be the intersection of $\mathcal{X}_m$ with the union of the pullbacks of $\mathcal{V}_2$ under the m projections $\mathcal{T}_m \to \mathcal{H}_2$, as closed subsets of $\mathbb{P}^N_{\mathcal{T}_m}$. Let $\mathcal{T}_m^\circ \subset \mathcal{T}_m$ be the open subset of points over which the fiber of $\mathcal{X}_m$ is geometrically integral [Gro66, Théorème 12.2.4(viii)]. Let
$$\mathcal{T} := \coprod_{m=0}^{n} \mathcal{T}_m^\circ,$$
and let $\mathcal{X} \to \mathcal{T}$ and $\mathcal{C} \subset \mathcal{X}$ be the restrictions of the $\mathcal{X}_m$ and the $\mathcal{C}_m$. Then $\mathcal{T}$ is of finite type over $\mathbb{Z}[1/30]$, $\mathcal{X} \to \mathcal{T}$ is flat and projective, the geometric fibers of $\mathcal{X} \to \mathcal{T}$ are integral threefolds, and for any geometric point t' of $\mathcal{T}$, $\dim \mathcal{C}_{t'} \leq 2$, since the geometric fibers of $\mathcal{V}_2 \to \mathcal{H}_2$ are of dimension 2.

Let k, Y, and Σ be as in the lemma, and let $\Sigma_1, \ldots, \Sigma_m$ be the irreducible components of Σ. Since $\deg \Sigma_j \geq 1$ for any j, $m \leq n$. By **Step 1**, the Hilbert polynomials of Y and of $\Sigma_1, \ldots, \Sigma_m$ belong to $\bigcup_{e=1}^n \Phi_{3,e}$ and $\bigcup_{e=1}^n \Phi_{2,e}$ respectively. Since $\mathrm{char}\, k > 5$, the Hilbert points of $Y, \Sigma_1, \ldots, \Sigma_m$ define a geometric point $t' : \mathrm{Spec}\, k \to \mathcal{T}_m^\circ \subset \mathcal{T}$ with
$$\mathcal{X}_{t'} = Y \quad \text{and} \quad \mathcal{C}_{t'} = \Sigma \text{ as closed subsets.}$$

Step 3. In this step we normalize the fibers of $\mathcal{X} \to \mathcal{T}$. Let $T' \subset \mathcal{T}$ be an integral closed subscheme and $\overline{\eta} : \mathrm{Spec}\, \overline{k(T')} \to T'$ a geometric generic point, where $\overline{k(T')}$ is an algebraic closure of the function field $k(T')$. The fiber $\mathcal{X}_{\overline{\eta}}$ is an integral projective threefold, and the normalization $\mathcal{N}_{\overline{\eta}} \to \mathcal{X}_{\overline{\eta}}$ is finite [Gro65, Scholie 7.8.3] and an isomorphism over a dense open subset of $\mathcal{X}_{\overline{\eta}}$. The scheme $\mathcal{N}_{\overline{\eta}}$ and the morphism $\mathcal{N}_{\overline{\eta}} \to \mathcal{X}_{\overline{\eta}}$ are defined over a finite extension k' of $k(T')$ in $\overline{k(T')}$ [Gro66, Théorèmes 8.5.2, 8.8.2, and 8.10.5]. Let T'' be the normalization of T' in k', and $\rho : T'' \to T'$ the induced finite [Gro65, Scholie 7.8.3] surjective morphism. By

[Gro66, Théorèmes 8.5.2, 8.8.2, and 8.10.5], there exist a dense open subset $T_0'' \subset T''$ and a finite morphism

$$\mu : \mathcal{N} \to \mathcal{X} \times_{\mathcal{T}} T_0''$$

whose base change to $\overline{k(T')}$ is $\mathcal{N}_{\overline{\eta}} \to \mathcal{X}_{\overline{\eta}}$. Let $\mathcal{W} \subset \mathcal{X} \times_{\mathcal{T}} T_0''$ be the complement of the largest open subset over which μ is an isomorphism. Then $\dim \mathcal{W}_{\overline{\eta}} \le 2$. By generic freeness [Mat89, Theorem 24.1], the openness of geometrically normal and geometrically integral fibers [Gro66, Théorème 12.2.4(iv)(viii)], and the upper semicontinuity of fiber dimension [Gro66, Corollaire 13.1.5], possibly replacing T_0'' with a dense open subset, we may assume the following.

(a) $\mathcal{N} \to T_0''$ is flat and the geometric fibers of $\mathcal{N} \to T_0''$ are normal and integral.
(b) For any geometric point t'' of T_0'', $\dim \mathcal{W}_{t''} \le 2$.

Let t'' be a geometric point of T_0'' and $t' := \rho \circ t''$. By (b), $\mu_{t''} : \mathcal{N}_{t''} \to \mathcal{X}_{t'}$ is a finite morphism which is an isomorphism over the dense open subset $\mathcal{X}_{t'} \backslash \mathcal{W}_{t''}$ of the integral threefold $\mathcal{X}_{t'}$. Since $\mathcal{N}_{t''}$ is normal and integral by (a), $\mu_{t''}$ is the normalization of $\mathcal{X}_{t'}$. Let $\mathcal{Q}_0 := \mu^{-1}(\mathcal{C} \times_{\mathcal{T}} T_0'')$. Since $\mu_{t''}$ is finite and $\dim \mathcal{C}_{t'} \le 2$, $\mathcal{Q}_0$ contains no geometric fiber of $\mathcal{N} \to T_0''$. Let

$$U' := T' \backslash \rho(T'' \backslash T_0'').$$

Since ρ is finite and ρ^{-1} of the generic point of T' is the generic point of T'', U' is a dense open subset of T' and $\rho^{-1}(U') \subset T_0''$. Since ρ is surjective and finite, any geometric point $\operatorname{Spec} \Omega \to U'$ lifts to a geometric point $\operatorname{Spec} \Omega \to T_0''$.

We take for T' the irreducible components of $\mathcal{T}$ with reduced structure, then the irreducible components of $T' \backslash U'$ with reduced structure, and so on. Since $\dim \mathcal{T} < +\infty$, after finitely many steps we obtain finitely many schemes T_0'' with morphisms to $\mathcal{T}$, whose images cover $\mathcal{T}$, together with the corresponding $\mathcal{N} \to T_0''$, μ, and $\mathcal{Q}_0$. Let T be the disjoint union of these T_0'', $\pi : \mathcal{Y} \to T$ the disjoint union of the corresponding morphisms $\mathcal{N} \to T_0''$, and $\mathcal{Q} \subset \mathcal{Y}$ the union of the corresponding $\mathcal{Q}_0$. Then T is of finite type over $\mathbb{Z}[1/30]$, π is projective and flat, the geometric fibers of π are normal integral threefolds, and $\mathcal{Q}$ contains no geometric fiber of π.

Let k, Y, and Σ be as in the lemma, and let $t' : \operatorname{Spec} k \to \mathcal{T}$ be the geometric point of **Step 2**. There exists a geometric point $t : \operatorname{Spec} k \to T$ over t'. Then $\mu_t : \mathcal{Y}_t \to \mathcal{X}_{t'} = Y$ is the normalization of Y, so it induces an isomorphism $\mathcal{Y}_t \cong Y^\nu$, under which

$$\nu^{-1}(\Sigma) = \mu_t^{-1}(\mathcal{C}_{t'}) = \mathcal{Q}_t.$$

The lemma follows. □

Proposition 9.11. *Let $I \subset [0,1]$ be a DCC set and $V > 0$ a real number. Let $\mathcal{S}_V$ be the set of projective pairs (X, B) of dimension 3 over algebraically closed fields of characteristic > 5 such that $B \in I$ and there exists a positive integer m such that $|\lfloor m(K_X + B) \rfloor|$ defines a birational map and $m^3 \operatorname{vol}(K_X + B) \le V$. Then $\mathcal{S}_V$ is log birationally bounded over $\mathbb{Z}[1/30]$.*

Proof. Since $I \cup \{1\}$ is a DCC set, the set of positive elements of $I \cup \{1\}$ has a minimum $\alpha \in (0, 1]$. Let

$$n := \max\left\{1, \left\lfloor \frac{8}{196} \alpha^{-3} (5 + 10\alpha)^3 V \right\rfloor\right\} \quad \text{and} \quad N := \lfloor V \rfloor + 2.$$

Since $\alpha \le 1$,

$$\frac{8}{196} \alpha^{-3} (5 + 10\alpha)^3 \ge \frac{8}{196} \cdot 15^3 > 1,$$

so $\lfloor V \rfloor \le n$. By Lemma 9.10 for n and N, there exist a scheme T of finite type over $\mathbb{Z}[1/30]$, a projective flat morphism $\pi : \mathcal{Y} \to T$ whose geometric fibers are normal integral threefolds, and a closed subset $\mathcal{Q} \subset \mathcal{Y}$ containing no geometric fiber of π, satisfying the conclusion of Lemma 9.10. Here T, π, and $\mathcal{Q}$ depend only on I and V.

Fix $(X, B) \in \mathcal{S}_V$ over a field k and a positive integer m as in the definition of $\mathcal{S}_V$, and let

$$v := \operatorname{vol}(K_X + B),$$

so $m^3 v \le V$. We divide the proof into four steps.

Step 1. In this step we bound the degree of the image of $||\lfloor m(K_X+B)\rfloor||$. Let $a_0,\dots,a_{N'}$ be rational functions on X which form a basis of $H^0(X,\mathcal{O}_X(\lfloor m(K_X+B)\rfloor))$, so that

$$\mathrm{div}_X(a_i)+\lfloor m(K_X+B)\rfloor\geq 0 \quad \text{for any } i.$$

Let $\phi: X \dashrightarrow \mathbb{P}^{N'}_k$ be the birational map onto the image defined by $[a_0:\cdots:a_{N'}]$, and let $Y\subset\mathbb{P}^{N'}_k$ be the closure of the image of ϕ, a non-degenerate integral threefold. Let

$$F_X:=\min_{0\leq i\leq N'}(\mathrm{div}_X(a_i)+\lfloor m(K_X+B)\rfloor)\geq 0$$

be the fixed part of $||\lfloor m(K_X+B)\rfloor||$, where the minimum is taken coefficientwise. By [Cut09, Theorems 1.1–1.3], there exists a resolution $f: W\to X$ such that $h:=\phi\circ f: W\to Y$ is a morphism, the exceptional locus of f is the support of a divisor $\mathrm{Exc}(f)$, and $f_*^{-1}B+\mathrm{Exc}(f)$ has simple normal crossing support. Let K_W be a canonical divisor of W with $f_*K_W=K_X$. Let $x_0,\dots,x_{N'}$ be the homogeneous coordinates of $\mathbb{P}^{N'}_k$ and

$$H:=h^*\{x_0=0\}-\mathrm{div}_W(a_0).$$

Then H is a Cartier divisor with $\mathcal{O}_W(H)\cong h^*\mathcal{O}_Y(1)$, $|H|$ is base point free and defines a birational map, and $\mathrm{div}_W(a_i)+H=h^*\{x_i=0\}\geq 0$ for any i. Since f is an isomorphism over the generic point of any prime divisor on X,

$$f_*H=\lfloor m(K_X+B)\rfloor-F_X. \tag{9.20}$$

Since h is birational onto Y, $\mathcal{O}_W(H)\cong h^*\mathcal{O}_Y(1)$, and $f_*H\leq\lfloor m(K_X+B)\rfloor$ by (9.20), we have

$$\deg Y=\mathrm{vol}(H)\leq\mathrm{vol}(m(K_X+B))=m^3v\leq V.$$

Since $Y\subset\mathbb{P}^{N'}_k$ is a non-degenerate threefold, $N'\leq\deg Y+2\leq N$. Via a linear inclusion $\mathbb{P}^{N'}_k\subset\mathbb{P}^N_k$, Y is an integral threefold in $\mathbb{P}^N_k$ with $\deg Y\leq\lfloor V\rfloor\leq n$.

Step 2. In this step we show that there exists a divisor $D\geq 0$ on W with $D\sim K_W+4H$. Since k is algebraically closed, $\Omega^1_{k(W)/k}$ is a $k(W)$-vector space of dimension 3 spanned by the differentials of any set of generators of the field $k(W)$ over k. Since ϕ is birational, $k(W)$ is generated over k by $a_1/a_0,\dots,a_{N'}/a_0$. Possibly reordering indices, we may assume that

$$\omega:=d(a_1/a_0)\wedge d(a_2/a_0)\wedge d(a_3/a_0)\neq 0.$$

Let $w\in W$ be a point, and let u_w be a rational function on W with $\mathrm{div}_W(u_w)=H$ on a neighborhood W_w of w. Then $\sigma_i:=a_iu_w$ is regular on W_w for any i, $a_i/a_0=\sigma_i/\sigma_0$, and

$$\sigma_0^4\omega=\sigma_0d\sigma_1\wedge d\sigma_2\wedge d\sigma_3-\sigma_1d\sigma_0\wedge d\sigma_2\wedge d\sigma_3+\sigma_2d\sigma_0\wedge d\sigma_1\wedge d\sigma_3-\sigma_3d\sigma_0\wedge d\sigma_1\wedge d\sigma_2$$

is a regular 3-form on W_w. Thus

$$\mathrm{div}_W(\omega)+4(\mathrm{div}_W(a_0)+H)\geq 0 \quad \text{on } W_w.$$

Since w is arbitrary,

$$D:=\mathrm{div}_W(a_0^4\omega)+4H\geq 0,$$

and $D\sim K_W+4H$ since $\mathrm{div}_W(a_0^4\omega)$ is a canonical divisor of W.

Step 3. In this step we bound the degree of the marking. Let $B_W:=f_*^{-1}B+\mathrm{Exc}(f)$. Then $\lceil B_W\rceil$ is a reduced simple normal crossing divisor, and $B_W\geq\alpha\lceil B_W\rceil$ since the non-zero coefficients of B_W are positive elements of $I\cup\{1\}$. By **Step 2**, we have

$$K_W+\lceil B_W\rceil+14H\sim_{\mathbb{R}}\alpha^{-1}\left(K_W+B_W+(4+10\alpha)H\right)-\alpha^{-1}\left((1-\alpha)D+B_W-\alpha\lceil B_W\rceil\right),$$

and $(1-\alpha)D+B_W-\alpha\lceil B_W\rceil\geq 0$. Since the volume is invariant under $\mathbb{R}$-linear equivalence [FKL16, Theorem 3.5(iv)], homogeneous of degree 3 (see Subsection 2.2), and $\mathrm{vol}(D_1-D_2)\leq\mathrm{vol}(D_1)$ for any $\mathbb{R}$-divisors D_1 and $D_2\geq 0$,

$$\mathrm{vol}(K_W+\lceil B_W\rceil+14H)\leq\alpha^{-3}\,\mathrm{vol}\left(K_W+B_W+(4+10\alpha)H\right).$$

By (9.20),

$$\begin{aligned} f_*\left(K_W + B_W + (4+10\alpha)H\right) &= K_X + B + (4+10\alpha)\left(\lfloor m(K_X+B)\rfloor - F_X\right) \\ &\leq \left(1 + (4+10\alpha)m\right)(K_X+B). \end{aligned}$$

Since f is birational,

$$\operatorname{vol}\left(K_W + B_W + (4+10\alpha)H\right) \leq \operatorname{vol}\left((1+(4+10\alpha)m)(K_X+B)\right) = \left(1+(4+10\alpha)m\right)^3 v.$$

By [Sat24, Proposition 5.2],

$$(14H)^2 \cdot \lceil B_W \rceil \leq 8 \operatorname{vol}(K_W + \lceil B_W \rceil + 14H).$$

Since $m \geq 1$, $1 + (4+10\alpha)m \leq (5+10\alpha)m$. Combining the three inequalities above with $m^3 v \leq V$, we obtain

$$H^2 \cdot \lceil B_W \rceil \leq \frac{8}{196}\alpha^{-3}(5+10\alpha)^3 m^3 v \leq \frac{8}{196}\alpha^{-3}(5+10\alpha)^3 V. \tag{9.21}$$

Let $\nu : Y^\nu \to Y$ be the normalization. Since W is normal, $h = \nu \circ g$ for a birational morphism $g : W \to Y^\nu$. Let $\psi := f \circ g^{-1} : Y^\nu \dashrightarrow X$ and

$$\Sigma := \nu(\operatorname{Supp} g_* \lceil B_W \rceil) \subset Y$$

with reduced structure, a reduced closed subscheme of pure dimension 2. Any irreducible component of Σ is $h(P_W)$ for a component P_W of $\lceil B_W \rceil$ with $\dim h(P_W) = 2$. By the projection formula and (9.21), the degrees of the irreducible components of Σ add up to at most

$$\mathcal{O}_Y(1)^2 \cdot h_* \lceil B_W \rceil = H^2 \cdot \lceil B_W \rceil \leq n,$$

where the last inequality holds since $H^2 \cdot \lceil B_W \rceil$ is an integer. Let P be a prime divisor on Y^ν which is either contracted by ψ or a component of $\psi_*^{-1}B$, and let $P_W := g_*^{-1}P$. In the first case P_W is a component of $\operatorname{Exc}(f)$, and in the second case P_W is a component of $f_*^{-1}B$. In both cases

$$P = g(P_W) \subset \operatorname{Supp} g_* \lceil B_W \rceil \subset \nu^{-1}(\Sigma).$$

Thus

$$\nu^{-1}(\Sigma) \supset \operatorname{Supp} \psi_*^{-1} B \cup \bigcup \{P \mid P \text{ is a prime divisor on } Y^\nu \text{ contracted by } \psi\}.$$

Step 4. In this step we conclude the proof. By **Step 1**, $Y \subset \mathbb{P}_k^N$ is an integral threefold with $\deg Y \leq n$, and by **Step 3**, $\Sigma \subset Y$ is a reduced closed subscheme of pure dimension 2 whose irreducible components have degrees adding up to at most n. Since $\operatorname{char} k > 5$, by Lemma 9.10, there exist a geometric point $t : \operatorname{Spec} k \to T$ and an isomorphism $\iota : \mathcal{Y}_t \to Y^\nu$ with $\iota^{-1}(\nu^{-1}(\Sigma)) \subset \mathcal{Q}_t$. By **Step 3**, the birational map $\psi \circ \iota : \mathcal{Y}_t \dashrightarrow X$ satisfies the following: $\mathcal{Q}_t$ contains $\operatorname{Supp}(\psi \circ \iota)_*^{-1} B$ and any prime divisor on $\mathcal{Y}_t$ contracted by $\psi \circ \iota$. Thus $\pi : \mathcal{Y} \to T$ and $\mathcal{Q}$ satisfy Definition 9.1 for $\mathcal{S}_V$. □

10. Effective birationality for reduced boundaries

In this section, we prove Definition-Theorem 10.4, which defines the integers n_{red} and P_{red} for reduced boundaries. We first treat pairs (X, B) such that a resolution of X admits a non-constant morphism to an abelian variety.

Theorem 10.1. *There exists a positive integer m satisfying the following. Let k be an algebraically closed field of characteristic $p \geq 11$, and let (X, B) be a projective lc pair of dimension 3 over k such that B is a reduced divisor. Assume that $K_X + B$ is big and a resolution of X admits a non-constant morphism to an abelian variety. Then $|m(K_X + B)|$ defines a birational map.*

Proof. Let $\theta \in (0,1)$ and $q \geq 2$ be the numbers of Corollary 4.2 for $I = \{0,1\}$. Let $N := N(\frac{1}{q}, \frac{1}{q})$ be the integer of Proposition 5.7, and let $M := M(\frac{1}{q}, N)$ be the integer of Lemma 5.2. Let

$$u := 5qM + q - 1, \quad v := 9qM + q - 1, \quad \text{and} \quad m := 19qM + 3q.$$

Then $m = (v+1) + 2(u+1)$, and q divides $u+1$, $v+1$, and m.

Step 1. In this step, we construct a log resolution $g : W \to X$ and the Stein factorization $W \to T \to A$.

Let $X_1 \to X$ be a resolution and $a_1 : X_1 \to A$ a non-constant morphism to an abelian variety A. By [Cut09, Theorems 1.2 and 1.3] on X_1 for the preimage of $\operatorname{Supp} B$ and the exceptional locus of $X_1 \to X$, there exists a log resolution $g : W \to X$ of (X, B) which factors through X_1. Let $a_W : W \to A$ be the induced morphism, and let

$$\Gamma := g_*^{-1} B + \operatorname{Exc}(g).$$

Then Γ is a reduced divisor with simple normal crossing support. Since (X, B) is a pair and B is reduced, $K_X + B$ is an $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor, so $K_X + B$ is $\mathbb{Q}$-Cartier by Lemma 2.6. Since (X, B) is lc,

$$K_W + \Gamma = g^*(K_X + B) + E_g$$

for some g-exceptional $\mathbb{Q}$-divisor $E_g \geq 0$, so $K_W + \Gamma$ is big. By Corollary 4.2, $q\theta\Gamma$ is a Weil divisor, $(W, \theta\Gamma)$ is $\frac{1}{q}$-lc, and $K_W + \theta\Gamma$ is big. Since $\theta \in (0,1)$, any coefficient of $\theta\Gamma$ belongs to $\{\frac{1}{q}, \dots, \frac{q-1}{q}\}$.

Let $W \to T \to A$ be the Stein factorization of a_W: $W \to T$ is a contraction onto a normal projective variety T and $T \to A$ is finite. Since a_W is non-constant, $\dim T \in \{1,2,3\}$. Let η be the generic point of T, $k(T)$ the residue field of η, $\overline{k(T)}$ an algebraic closure of $k(T)$, $k(T)^{\mathrm{sep}} \subset \overline{k(T)}$ the separable closure of $k(T)$, and W_η the generic fiber of $W \to T$. Then W_η is regular, $H^0(W_\eta, \mathcal{O}_{W_\eta}) = k(T)$, and the field $k(T)$ is F-finite since it is finitely generated over the perfect field k. Definition 6.1 and the results of [Zha22, Section 2] that we use below hold over any F-finite field, in particular over $k(T)$. We use [Zha22, Theorems 3.4 and 4.1 and Corollary 4.3] below only over algebraically closed fields.

Step 2. In this step, we construct a $\mathbb{Q}$-divisor Δ on W. If $\dim T \in \{2,3\}$, we let $\Delta := \theta\Gamma$.

Assume that $\dim T = 1$, let $W_0 := W$, and let $\Delta := \theta\Gamma$ at the start. Suppose that there exist prime divisors $R_1 \neq R_2$ in $\operatorname{Supp}\Delta$ and an irreducible component C_1 of $R_1 \cap R_2$ dominating T such that $b_1 + b_2 \geq 1$, where $b_j := \operatorname{mult}_{R_j} \Delta$ for $j \in \{1,2\}$. Since $\operatorname{Supp}\Delta$ is simple normal crossing, C_1 is a smooth curve. Let $\pi : W' \to W$ be the blow-up of C_1, let E_1 be the exceptional prime divisor, and let

$$\Delta' := \pi_*^{-1}\Delta + (b_1 + b_2 - 1)E_1.$$

Then W' is smooth, $\operatorname{Supp}\Delta'$ is simple normal crossing, and $K_{W'} + \Delta' = \pi^*(K_W + \Delta)$. Since $0 \leq b_1 + b_2 - 1 < \min\{b_1, b_2\}$, any coefficient of Δ' belongs to $\{0, \frac{1}{q}, \dots, \frac{q-1}{q}\}$. We replace (W, Δ) by (W', Δ') and repeat.

The number of blow-ups is finite. By induction on the number of blow-ups, C_1 maps onto an irreducible component C_0 of $\Gamma_1 \cap \Gamma_2$ dominating T for some components $\Gamma_1 \neq \Gamma_2$ of Γ. The local equations of Γ_1 and Γ_2 form a regular system of parameters of $\mathcal{O}_{W_0, C_0}$, no other component of Γ contains C_0, and the blow-ups above the generic point of C_0 are toric. Thus any prime divisor E over W_0 extracted over C_0 corresponds to a primitive vector $(r_1, r_2) \in (\mathbb{N}^+)^2$, distinct prime divisors correspond to distinct vectors, and the log discrepancy of E with respect to $(W_0, \theta\Gamma)$ is $(r_1 + r_2)(1 - \theta)$. Since the coefficient of E in Δ is $1 - (r_1 + r_2)(1 - \theta) \geq 0$ and $1 - \theta \geq \frac{1}{q}$ (as $q\theta\Gamma$ is a Weil divisor and $\theta < 1$), we have $r_1 + r_2 \leq q$. Hence the number of blow-ups over C_0 is at most $\#\{(r_1, r_2) \in (\mathbb{N}^+)^2 \mid r_1 + r_2 \leq q\}$.

We replace g and a_W by their composites with the blow-ups. The Stein factorization of a_W is still $W \to T \to A$. Now for any prime divisors $R_1 \neq R_2$ in $\operatorname{Supp}\Delta$ such that $R_1 \cap R_2$ has an

irreducible component dominating T, we have

$$\mathrm{mult}_{R_1}\Delta + \mathrm{mult}_{R_2}\Delta \leq 1 - \frac{1}{q}.$$

Let $T^\circ \subset T$ be a non-empty open subset satisfying the following:

(1) No component of $\Delta|_{W^\circ}$ is vertical over T°, where W° is the preimage of T° in W.
(2) Any irreducible component of the intersection of two components of $\Delta|_{W^\circ}$ dominates T.

Since W° is smooth, $\mathrm{Supp}\,\Delta|_{W^\circ}$ is simple normal crossing, any coefficient of Δ is at most $1-\frac{1}{q}$, and by (2) any two components $R_1 \neq R_2$ of $\Delta|_{W^\circ}$ meeting in W° satisfy $\mathrm{mult}_{R_1}\Delta + \mathrm{mult}_{R_2}\Delta \leq 1-\frac{1}{q}$, the discrepancy computation for simple normal crossing pairs on smooth varieties shows that $(W^\circ, \Delta|_{W^\circ})$ is terminal and $\frac{1}{q}$-lc. Any exceptional prime divisor E over W_η is induced by an exceptional prime divisor E° over W° whose center dominates T, and

$$a(E, W_\eta, \Delta|_{W_\eta}) = a(E^\circ, W^\circ, \Delta|_{W^\circ}).$$

Thus $(W_\eta, \Delta|_{W_\eta})$ is terminal and $\frac{1}{q}$-lc.

In all cases, let $D_W := K_W + \Delta$ and $\Delta_\eta := \Delta|_{W_\eta}$. Then D_W is big and (W, Δ) is klt. Any coefficient of Δ belongs to $\{0, \frac{1}{q}, \dots, \frac{q-1}{q}\}$, so $q\Delta$ is a Weil divisor, and

$$g_* D_W = K_X + \theta B \leq K_X + B.$$

Step 3. In this step, we run a D_W-MMP over T, construct a higher model V of W, and reduce the theorem to a statement on W_η.

By [HNT20, Theorem 1.1], there exists a D_W-MMP over T which terminates. Let $\phi : W \dashrightarrow Y$ be the induced birational map, $\Delta_Y := \phi_*\Delta$, and $D_Y := K_Y + \Delta_Y$. Since (W, Δ) is $\mathbb{Q}$-factorial klt, so is (Y, Δ_Y). Since D_W is big, $D_Y = \phi_* D_W$ is big, so the MMP does not end with a Mori fiber space and D_Y is nef over T. Any coefficient of Δ_Y belongs to $\{0, \frac{1}{q}, \dots, \frac{q-1}{q}\}$.

We show that D_Y is nef. Otherwise, by [HNT20, Theorem 2.8], there exists a rational curve $C \subset Y$ such that $D_Y \cdot C < 0$. Since the normalization of C is $\mathbb{P}^1$, the composite $C \to Y \to T \to A$ is constant [Mil86, Corollary 3.9]. Since $T \to A$ is finite, C maps to a point of T, which contradicts the nefness of D_Y over T.

Let Y_η be the generic fiber of $Y \to T$. Let $\Sigma \subset W$ be the indeterminacy locus of ϕ. If $\dim T \in \{2, 3\}$, then $\dim \Sigma \leq 1$, so Σ does not dominate T. If $\dim T = 1$, then the flipping curves and the flipped curves of any flip over T map to closed points of T, and any divisorial contraction over T restricts to a birational morphism of generic fibers. In all cases, $\Sigma \cap W_\eta = \emptyset$, so ϕ restricts to a birational morphism $\tau : W_\eta \to Y_\eta$, and there exists a non-empty open subset $T_1 \subset T$ such that ϕ is a morphism on the preimage W_1 of T_1 in W. The closure of the graph of ϕ is isomorphic to W_1 over T_1, so the preimage of T_1 in it is non-singular. By [Cut09, Theorem 1.1], there exists a resolution V of the closure of the graph of ϕ which is an isomorphism over the non-singular locus of the closure. Let $g_V : V \to W$ and $h : V \to Y$ be the induced morphisms. Then g_V is an isomorphism over T_1, $V_\eta = W_\eta$, and the Stein factorization of $a_V := a_W \circ g_V$ is $V \to T \to A$. Let

$$D_V := h^* D_Y,$$

which is nef and big. Since ϕ is a composite of D_W-negative steps of the MMP, the negativity lemma implies that

$$g_V^* D_W = D_V + E_h$$

for some h-exceptional $\mathbb{Q}$-divisor $E_h \geq 0$. Let $D_{Y_\eta} := D_Y|_{Y_\eta}$ and $E_\tau := E_h|_{W_\eta}$. Restricting to $W_\eta = V_\eta$, we obtain

$$K_{W_\eta} + \Delta_\eta = \tau^* D_{Y_\eta} + E_\tau \quad \text{and} \quad D_V|_{W_\eta} = \tau^* D_{Y_\eta}, \tag{10.1}$$

where $E_\tau \geq 0$ is τ-exceptional, since any component of E_h dominating T maps onto a curve on Y.

We claim that

$$S^0_-(W_\eta, K_{W_\eta} + uD_V|_{W_\eta}) \neq 0 \quad \text{and} \quad S^0_-(W_\eta, K_{W_\eta} + vD_V|_{W_\eta}) \text{ defines a birational map.} \tag{10.2}$$

Step 4. In this step, we prove (10.2) when $\dim T \in \{2, 3\}$.

Assume that $\dim T = 3$. Then $W_\eta = \operatorname{Spec} k(T)$ and any divisor on W_η is zero. For any integer $e \geq 1$, the inclusion $k(T) \to k(T)^{1/p^e}$ splits as a $k(T)$-linear map, so the Frobenius trace maps on W_η are surjective and $S^0_-(W_\eta, K_{W_\eta}) = k(T)$. Thus (10.2) holds.

Assume that $\dim T = 2$. Then τ is a birational morphism of regular projective curves over $k(T)$, so τ is an isomorphism, $E_\tau = 0$, and $D_V|_{W_\eta} = K_{W_\eta} + \Delta_\eta$ by (10.1). Since $q(K_{W_\eta} + \Delta_\eta)$ is a Weil divisor on the regular curve W_η, it is Cartier. Since D_V is big, $D_V|_{W_\eta}$ has positive degree, so $D_V|_{W_\eta}$ is ample. Let W^ν_η be the normalization of $((W_\eta)_{\overline{k(T)}})_{\mathrm{red}}$, where $(W_\eta)_{\overline{k(T)}} := W_\eta \times_{k(T)} \overline{k(T)}$, and let $\psi : W^\nu_\eta \to W_\eta$ be the induced morphism. Since W_η is normal and $H^0(W_\eta, \mathcal{O}_{W_\eta}) = k(T)$, W^ν_η is a smooth projective curve over $\overline{k(T)}$. Since $\psi^*(qD_V|_{W_\eta})$ is a Cartier divisor of positive degree on W^ν_η, $\deg \psi^*(D_V|_{W_\eta}) \geq \frac{1}{q}$. Since $M \geq 1$,

$$\deg \psi^*(uD_V|_{W_\eta}) \geq \frac{u}{q} = 5M + 1 - \frac{1}{q} > 2 \quad \text{and} \quad \deg \psi^*(vD_V|_{W_\eta}) \geq \frac{v}{q} = 9M + 1 - \frac{1}{q} > 2.$$

By [Zha22, Theorem 4.1], $S^0_-(W^\nu_\eta, K_{W^\nu_\eta} + \psi^*(uD_V|_{W_\eta}))$ and $S^0_-(W^\nu_\eta, K_{W^\nu_\eta} + \psi^*(vD_V|_{W_\eta}))$ are non-zero and define embeddings. Since $D_V|_{W_\eta}$ is ample, [Zha22, Theorem 2.7] implies (10.2).

Step 5. Assume that $\dim T = 1$. In this step, we construct a nef big Cartier divisor H on W_η and show that

$$\begin{gathered} S^0_-(W_\eta, K_{W_\eta} + 5H + (q-1)D_V|_{W_\eta}) \neq 0 \quad \text{and} \\ S^0_-(W_\eta, K_{W_\eta} + 9H + (q-1)D_V|_{W_\eta}) \text{ defines a birational map.} \end{gathered} \tag{10.3}$$

By Proposition 5.7 for $\epsilon = \beta = \frac{1}{q}$, there exists a $\mathbb{Q}$-divisor $0 \leq \widetilde{\Delta}_\eta \leq \Delta_\eta$ with at most N components such that $\operatorname{mult}_R \widetilde{\Delta}_\eta = \operatorname{mult}_R \Delta_\eta$ for any component R of $\widetilde{\Delta}_\eta$, and $K_{W_\eta} + \widetilde{\Delta}_\eta$ is big. Then $(W_\eta, \widetilde{\Delta}_\eta)$ is terminal and $\frac{1}{q}$-lc, and $q\widetilde{\Delta}_\eta$ is a Weil divisor. Let $f : W_\eta \to W'_\eta$ be the composition of the steps of a $(K_{W_\eta} + \widetilde{\Delta}_\eta)$-MMP ending with a minimal model [Tan18, Theorem 1.1], and let $\widetilde{\Delta}'_\eta := f_*\widetilde{\Delta}_\eta$. By Lemma 5.2(2),

$$H' := qM(K_{W'_\eta} + \widetilde{\Delta}'_\eta)$$

is a nef big Cartier divisor. Let $H := f^*H'$. By the negativity lemma [Tan18, Lemma 2.11],

$$K_{W_\eta} + \widetilde{\Delta}_\eta = \frac{1}{qM}H + E_f \tag{10.4}$$

for some f-exceptional $\mathbb{Q}$-divisor $E_f \geq 0$. Since $(W'_\eta, \widetilde{\Delta}'_\eta)$ is $\frac{1}{q}$-lc, W'_η is klt.

Let W^ν_η and $(W'_\eta)^\nu$ be the normalizations of $((W_\eta)_{\overline{k(T)}})_{\mathrm{red}}$ and $((W'_\eta)_{\overline{k(T)}})_{\mathrm{red}}$, and let

$$\psi : W^\nu_\eta \to W_\eta \quad \text{and} \quad \psi' : (W'_\eta)^\nu \to W'_\eta$$

be the induced morphisms. Since W_η is normal and $H^0(W_\eta, \mathcal{O}_{W_\eta}) = k(T)$, W^ν_η and $(W'_\eta)^\nu$ are normal projective surfaces over $\overline{k(T)}$. Let $\rho : Z \to W^\nu_\eta$ be the minimal resolution and $\lambda := \psi \circ \rho$. The morphism f induces a birational morphism $f^\nu : W^\nu_\eta \to (W'_\eta)^\nu$ such that $\psi' \circ f^\nu = f \circ \psi$. Let $\sigma := f^\nu \circ \rho : Z \to (W'_\eta)^\nu$, so that $\psi' \circ \sigma = f \circ \lambda$. Let

$$H_Z := \lambda^*H = \sigma^*\psi'^*H' \quad \text{and} \quad D_Z := \lambda^*(D_V|_{W_\eta}).$$

[Zha22, Corollary 4.3] requires that the Albanese morphism of Z factor through the birational morphism σ. We first show that any σ-exceptional curve $C \subset Z$ is rational. If ρ contracts C, then C is rational by [Zha22, Proposition 2.6]. Before treating the other curves, let $C_0 \subset W_\eta$ be an f-exceptional curve. We show that any component of the normalization of $((C_0)_{\overline{k(T)}})_{\mathrm{red}}$ is

isomorphic to $\mathbb{P}^1_{\overline{k(T)}}$. Let $\mu : W''_\eta \to W'_\eta$ be the minimal resolution [Kol13, Theorem 2.25] (see also [Sat25, Definition 2.11]). Since $K_{W''_\eta}$ is μ-nef, no μ-exceptional curve is an exceptional divisor in the sense of [Liu02, Definition 9.3.1] by [Liu02, Proposition 9.3.10(a)], so f factors through μ [Liu02, Proposition 9.3.32]. Since W'_η is klt, any μ-exceptional curve E satisfies $H^1(E, \mathcal{O}_E) = 0$ by [Sat25, Theorem A.3(1)] (see also [Kol13, Section 3.3]). By [Liu02, Theorem 9.2.2], $W_\eta \to W''_\eta$ is a composite of blow-ups of closed points, and the exceptional curve of the blow-up of a closed point x is isomorphic to $\mathbb{P}^1$ over the residue field of x [Liu02, Theorem 8.1.19(b)]. Thus C_0 is birational to a projective curve C''_0 over $k(T)$ such that $H^1(C''_0, \mathcal{O}_{C''_0}) = 0$. Since $H^1(\mathcal{O})$ of a projective curve stays zero under base field extension, reduction, and normalization, the claim on C_0 follows.

Now assume that ρ does not contract C, and let $C' := \rho(C)$. Let $W_\eta^{\mathrm{sep}} := W_\eta \times_{k(T)} k(T)^{\mathrm{sep}}$, which is regular by [Gro65, Proposition 6.7.4]. The induced morphism $W_\eta^\nu \to W_\eta^{\mathrm{sep}}$ is integral, and the induced extension of function fields is purely inseparable since $\overline{k(T)}$ is purely inseparable over $k(T)^{\mathrm{sep}}$. Let C^{sep} and C_0 be the images of C' in W_η^{sep} and in W_η. Since W_η^{sep} is normal, the local ring of W_η^{sep} at the generic point of C^{sep} is a discrete valuation ring dominated by the local ring of W_η^ν at the generic point of C'. Hence the function field of C' is purely inseparable over the function field of C^{sep}. Since σ contracts C and $f \circ \lambda = \psi' \circ \sigma$, C_0 is an f-exceptional curve. The subfield of the function field of C' generated by $\overline{k(T)}$ and the function field of C^{sep} is the function field of a component of the normalization of $((C_0)_{\overline{k(T)}})_{\mathrm{red}}$, so it is $\overline{k(T)}(t)$ for some t by the last paragraph. Thus, for some integer $e \geq 0$,

$$\overline{k(T)}(t) \subset \text{the function field of } C' \subset \overline{k(T)}(t^{1/p^e}),$$

and C' is rational by Lüroth's theorem. Since $C \to C'$ is birational, C is rational.

By [Mil86, Corollary 3.9], the Albanese morphism of Z is constant on any σ-exceptional curve, hence on any fiber of σ. Since σ is a birational projective morphism onto the normal surface $(W'_\eta)^\nu$, the rigidity lemma [Deb01, Lemma 1.15] implies that the Albanese morphism of Z factors through the birational morphism $\sigma : Z \to (W'_\eta)^\nu$.

By [Zha22, Section 2.4], $\psi^* K_{W_\eta} - K_{W''_\eta} \geq 0$. Since K_Z is ρ-nef, $\lambda^* K_{W_\eta}$ is ρ-trivial, and

$$\rho_*(\lambda^* K_{W_\eta} - K_Z) = \psi^* K_{W_\eta} - K_{W_\eta^\nu} \geq 0,$$

the negativity lemma [Tan18, Lemma 2.11] implies that $K_Z \leq \lambda^* K_{W_\eta}$. Let $E_Z := \lceil \lambda^* E_f \rceil$. Since $f \circ \lambda = \psi' \circ \sigma$ and the fibers of the integral morphism ψ' are 0-dimensional, E_Z is a σ-exceptional divisor. By (10.4),

$$2H_Z + E_Z - K_Z \geq 2H_Z + \lambda^* E_f - \lambda^* K_{W_\eta} = \Big(2 - \frac{1}{qM}\Big) H_Z + \lambda^* \widetilde{\Delta}_\eta,$$

which is big. By [Zha22, Corollary 4.3], with $r = 2$, the nef big Cartier divisor $\psi'^* H'$ on $(W'_\eta)^\nu$, the big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor $H_Z + (q-1)D_Z$, and the divisor E_Z,

$$S^0_-(Z, K_Z + 5H_Z + (q-1)D_Z) \neq 0 \quad \text{and}$$
$$S^0_-(Z, K_Z + 9H_Z + (q-1)D_Z) \text{ defines a birational map.}$$

The $\mathbb{Q}$-divisors $5H + (q-1)D_V|_{W_\eta}$ and $9H + (q-1)D_V|_{W_\eta}$ are nef big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on W_η whose pullbacks under λ are $5H_Z + (q-1)D_Z$ and $9H_Z + (q-1)D_Z$. By [Zha22, Proposition 2.3(iv)] for ρ, and then [Zha22, Theorem 2.7] for W_η, we obtain (10.3).

Step 6. In this step, we prove (10.2) when $\dim T = 1$.

Since Y is $\mathbb{Q}$-factorial, so is Y_η. By (10.4), $E_f \geq 0$, $\widetilde{\Delta}_\eta \leq \Delta_\eta$, and (10.1),

$$H \leq qM(K_{W_\eta} + \widetilde{\Delta}_\eta) \leq qM(K_{W_\eta} + \Delta_\eta) = qMD_V|_{W_\eta} + qME_\tau.$$

Let

$$G_5 := 5\big(qM(K_{W_\eta} + \Delta_\eta) - H\big) \quad \text{and} \quad G_9 := 9\big(qM(K_{W_\eta} + \Delta_\eta) - H\big).$$

Since $H \le qM(K_{W_\eta}+\Delta_\eta)$, $G_5 \ge 0$ and $G_9 \ge 0$. Since K_{W_η} and $q\Delta_\eta$ are Weil divisors and H is Cartier, G_5 and G_9 are Weil divisors. Since $u = 5qM+q-1$, $v = 9qM+q-1$, and $K_{W_\eta}+\Delta_\eta = D_V|_{W_\eta}+E_\tau$ by (10.1),

$$5H+(q-1)D_V|_{W_\eta}+G_5 = uD_V|_{W_\eta}+5qME_\tau,$$
$$9H+(q-1)D_V|_{W_\eta}+G_9 = vD_V|_{W_\eta}+9qME_\tau.$$

Thus [Zha22, Proposition 2.3(i)] with $E = G_5$ (resp. G_9) implies that

$$S^0_-(W_\eta, K_{W_\eta}+5H+(q-1)D_V|_{W_\eta}) \subset S^0_-(W_\eta, K_{W_\eta}+uD_V|_{W_\eta}+5qME_\tau) \tag{10.5}$$

and

$$S^0_-(W_\eta, K_{W_\eta}+9H+(q-1)D_V|_{W_\eta}) \subset S^0_-(W_\eta, K_{W_\eta}+vD_V|_{W_\eta}+9qME_\tau). \tag{10.6}$$

By (10.1), $D_V|_{W_\eta} = \tau^*D_{Y_\eta}$, and uD_{Y_η} and vD_{Y_η} are nef big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisors on the $\mathbb{Q}$-factorial surface Y_η. By [Zha22, Proposition 2.3(iv)] for τ and the τ-exceptional $\mathbb{Q}$-divisors $5qME_\tau \ge 0$ and $9qME_\tau \ge 0$, as subspaces of the function field of W_η,

$$S^0_-(W_\eta, K_{W_\eta}+uD_V|_{W_\eta}+5qME_\tau) = S^0_-(W_\eta, K_{W_\eta}+uD_V|_{W_\eta}),$$
$$S^0_-(W_\eta, K_{W_\eta}+vD_V|_{W_\eta}+9qME_\tau) = S^0_-(W_\eta, K_{W_\eta}+vD_V|_{W_\eta}).$$

Together with (10.3), (10.5), and (10.6), (10.2) follows.

Step 7. In this step, we deduce the theorem from (10.2).

Since $W_\eta = V_\eta$ is the generic fiber of the Stein factorization $V \to T$ of a_V and uD_V is a nef big $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, (10.2) and [Zha22, Theorem 3.4(i)] imply that

$$H^0(V, \mathcal{O}_V(K_V+\lceil uD_V\rceil)\otimes a_V^*\mathcal{L}) \ne 0 \tag{10.7}$$

for any $\mathcal{L} \in \mathrm{Pic}^0(A)$. By [Zha22, Theorem 3.4(iii)], with the nef big $\mathbb{Q}$-divisor vD_V and with both divisors there equal to $K_V+\lceil uD_V\rceil$, (10.2) and (10.7) imply that $S^0_-(V, K_V+vD_V+2K_V+2\lceil uD_V\rceil)$ defines a birational map. The space $S^0_-(V, K_V+vD_V+2K_V+2\lceil uD_V\rceil)$ is a subspace of $H^0(V, \mathcal{O}_V(L_V))$, where

$$L_V := 3K_V+\lceil vD_V\rceil+2\lceil uD_V\rceil.$$

Let α be a coefficient of Δ_Y. Since $\alpha \in \{0, \frac{1}{q}, \dots, \frac{q-1}{q}\}$ and q divides $u+1$ and $v+1$, we have $(u+1)\alpha, (v+1)\alpha \in \mathbb{Z}$ and $\alpha \in [0,1)$, so

$$\lceil u\alpha\rceil = (u+1)\alpha \quad \text{and} \quad \lceil v\alpha\rceil = (v+1)\alpha.$$

Since K_Y is a Weil divisor,

$$K_Y+\lceil uD_Y\rceil = (u+1)D_Y \quad \text{and} \quad K_Y+\lceil vD_Y\rceil = (v+1)D_Y.$$

For any prime divisor R on Y, $\mathrm{mult}_{h_*^{-1}R} D_V = \mathrm{mult}_R D_Y$. Thus

$$\begin{aligned} h_*L_V &= 3K_Y+\lceil vD_Y\rceil+2\lceil uD_Y\rceil \\ &= (v+1)D_Y+2(u+1)D_Y = mD_Y. \end{aligned} \tag{10.8}$$

We regard all section spaces below as subspaces of the function field of X. We have

$$\begin{aligned} H^0(V, \mathcal{O}_V(L_V)) &\subset H^0(Y, \mathcal{O}_Y(mD_Y)) \subset H^0(V, \mathcal{O}_V(\lfloor mg_V^*D_W\rfloor)) \\ &\subset H^0(W, \mathcal{O}_W(\lfloor mD_W\rfloor)) \subset H^0(X, \mathcal{O}_X(m(K_X+B))). \end{aligned}$$

The first inclusion follows from (10.8) and Lemma 2.4(1) for h, D_Y, and $\frac{1}{m}L_V$. The second follows from Lemma 2.4(2) for h and $g_V^*D_W = D_V+E_h \ge D_V$, where D_Y is $\mathbb{Q}$-Cartier as Y is $\mathbb{Q}$-factorial. The third follows from Lemma 2.4(1) for g_V, and the fourth from Lemma 2.4(1) for g and $g_*D_W \le K_X+B$. Since the first space contains a subspace which defines a birational map, $|m(K_X+B)|$ defines a birational map. □

Lemma 10.2. *Let X be a normal projective threefold, D an $\mathbb{R}$-divisor on X, and m a positive integer.*

(1) *If* $||\lfloor mD\rfloor||$ *defines a birational map, then* $||\lfloor amD\rfloor||$ *defines a birational map for every positive integer* a.
(2) *If* $||\lfloor mD\rfloor||$ *defines a generically finite map, then* $\mathrm{vol}(D) \geq m^{-3}$.

Proof. Let $s_0, \dots, s_N$ be a basis of $H^0(X, \mathcal{O}_X(\lfloor mD\rfloor))$, and let a be a positive integer. Since $a\lfloor mD\rfloor \leq \lfloor amD\rfloor$, the products of a elements of $H^0(X, \mathcal{O}_X(\lfloor mD\rfloor))$ belong to $H^0(X, \mathcal{O}_X(\lfloor amD\rfloor))$. Thus (1) holds. If $||\lfloor mD\rfloor||$ defines a generically finite map, then, possibly reordering indices, s_1/s_0, s_2/s_0, and s_3/s_0 are algebraically independent over k, so the $\binom{a+3}{3}$ monomials of degree a in $s_0, \dots, s_3$ are linearly independent, and

$$\mathrm{vol}(D) \geq \lim_{a\to+\infty} \frac{6\binom{a+3}{3}}{(am)^3} = m^{-3}.$$

□

Lemma 10.3. *Let* $\alpha \in (0,1]$ *and* $\beta \in [0,1)$ *be real numbers,* q *an integer with* $q > \frac{4}{\alpha(1-\beta)}$, *and* $\theta := \frac{2\beta}{1+\beta}$. *Let* (X,B) *be a projective lc pair of dimension* 3 *over an algebraically closed field of characteristic* $p > 5$ *such that* $K_X + B$ *is big and* $\mathrm{mult}_S B \geq \alpha$ *for any prime divisor* $S \subset \mathrm{Supp}\, B$. *Let* $f : X' \to X$ *be a log resolution of* (X,B) *and* $B' := f_*^{-1}B + \mathrm{Exc}(f)$. *Assume that* $K_{X'} + \beta B'$ *is pseudo-effective. Then the pair* $(X', \frac{1}{q}\lfloor \frac{q(3+\beta)}{4} B'\rfloor)$ *has a log minimal model* (Y, B_Y) *satisfying the following:*

(1) Y *is projective and* $\mathbb{Q}$*-factorial, and* (Y, B_Y) *is* $\frac{1}{q}$*-lc.*
(2) $B_Y \in I_q \backslash \{1\}$.
(3) $K_Y + B_Y$ *is nef and big.*
(4) $K_Y + \theta B_Y$ *is pseudo-effective.*
(5) *For any integer* $m \geq 0$,

$$H^0(Y, \mathcal{O}_Y(\lfloor m(K_Y + B_Y)\rfloor)) \subset H^0(X, \mathcal{O}_X(\lfloor m(K_X + B)\rfloor))$$

as subspaces of the function field of X.

Proof. The proof is similar to [HMX14, Proof of Lemma 7.4]. Let

$$c := \frac{1+\beta}{2} \quad \text{and} \quad \Delta' := \frac{1}{q}\lfloor \frac{q(3+\beta)}{4} B'\rfloor.$$

Let $\gamma > 0$ be a coefficient of B'. Since (X,B) is lc, $\mathrm{mult}_S B \geq \alpha$ for any prime divisor $S \subset \mathrm{Supp}\, B$, and $\alpha \leq 1$, we have $\alpha \leq \gamma \leq 1$. Since $\frac{3+\beta}{4} - \frac{1-\beta}{4} = c$ and

$$\frac{1}{q} < \frac{\alpha(1-\beta)}{4} \leq \frac{(1-\beta)\gamma}{4},$$

we have

$$\frac{1}{q}\lfloor \frac{q(3+\beta)}{4}\gamma\rfloor > \frac{3+\beta}{4}\gamma - \frac{1}{q} > \frac{3+\beta}{4}\gamma - \frac{1-\beta}{4}\gamma = c\gamma.$$

Thus

$$cB' \leq \Delta' \leq \frac{3+\beta}{4}B' \leq B'. \tag{10.9}$$

Since $\frac{3+\beta}{4} < 1$, $\lfloor \frac{q(3+\beta)}{4}\gamma\rfloor \leq q-1$, so $\Delta' \in I_q\backslash\{1\}$. Since X' is smooth, $\mathrm{Supp}\, B'$ is a simple normal crossing divisor, and the coefficients of Δ' are at most $\frac{q-1}{q}$, the pair (X', Δ') is $\frac{1}{q}$-lc.

Since (X,B) is lc, $K_{X'} + B' = f^*(K_X + B) + G$ for an f-exceptional $\mathbb{R}$-divisor $G \geq 0$, so $K_{X'} + B'$ is big. Since

$$K_{X'} + cB' = \frac{1}{2}(K_{X'} + B') + \frac{1}{2}(K_{X'} + \beta B')$$

and $K_{X'}+\beta B'$ is pseudo-effective, $K_{X'}+cB'$ is big. By (10.9), $\Delta'-cB'\geq 0$, so $K_{X'}+\Delta'$ is big. Since $\theta\geq 0$ and $\theta c=\beta$, (10.9) implies $\theta\Delta'-\beta B'\geq \theta cB'-\beta B'=0$, so

$$K_{X'}+\theta\Delta' \text{ is pseudo-effective.} \tag{10.10}$$

The pair (X',Δ') is $\mathbb{Q}$-factorial klt. By [HNT20, Theorem 1.1], there exists a $(K_{X'}+\Delta')$-MMP which terminates. Since $K_{X'}+\Delta'$ is big, the MMP does not end with a Mori fiber space, so it ends with a log minimal model (Y,B_Y) of (X',Δ'), where Y is projective and $\mathbb{Q}$-factorial, K_Y+B_Y is nef, and $B_Y=\phi_*\Delta'$ for the induced birational map $\phi: X'\dashrightarrow Y$. Since each step of the MMP is a divisorial contraction or a flip, any prime divisor on Y is ϕ_*S for a prime divisor S on X'. Thus the coefficients of B_Y are coefficients of Δ', and (2) holds. By the negativity lemma,

$$a(E,X',\Delta')\leq a(E,Y,B_Y) \text{ for any prime divisor } E \text{ over } X'. \tag{10.11}$$

Since (X',Δ') is $\frac{1}{q}$-lc, (Y,B_Y) is $\frac{1}{q}$-lc, and (1) holds.

Let $g: W\to X'$ and $h: W\to Y$ be a common resolution [Cut09, Theorems 1.1–1.3]. Since ϕ is a $(K_{X'}+\Delta')$-MMP, by the negativity lemma $g^*(K_{X'}+\Delta')=h^*(K_Y+B_Y)+F$ for an h-exceptional $\mathbb{Q}$-divisor $F\geq 0$, so by Lemma 2.4, for any integer $m\geq 0$,

$$\begin{aligned} H^0(Y,\mathcal{O}_Y(\lfloor m(K_Y+B_Y)\rfloor)) &= H^0(W,\mathcal{O}_W(\lfloor mg^*(K_{X'}+\Delta')\rfloor)) \\ &= H^0(X',\mathcal{O}_{X'}(\lfloor m(K_{X'}+\Delta')\rfloor)). \end{aligned} \tag{10.12}$$

Thus $\operatorname{vol}(K_Y+B_Y)=\operatorname{vol}(K_{X'}+\Delta')>0$, so K_Y+B_Y is big, and (3) holds.

By (10.9),

$$f_*(K_{X'}+\Delta')=K_X+f_*\Delta'\leq K_X+f_*B'=K_X+B.$$

By Lemma 2.4(1), for any integer $m\geq 0$,

$$H^0(X',\mathcal{O}_{X'}(\lfloor m(K_{X'}+\Delta')\rfloor))\subset H^0(X,\mathcal{O}_X(\lfloor m(K_X+B)\rfloor)).$$

Together with (10.12), this proves (5).

Since any prime divisor on Y is ϕ_*S for a prime divisor S on X' and $\phi_*(K_{X'}+\theta\Delta')=K_Y+\theta B_Y$, (10.10) implies that $K_Y+\theta B_Y$ is pseudo-effective, and (4) holds. The lemma follows. □

Definition-Theorem 10.4. Let $\mathcal{P}$ be the set of $(n,P_0)\in(\mathbb{N}^+)^2$ with $P_0\geq 11$ satisfying the following: for any algebraically closed field k of characteristic $p>P_0$ and any projective lc pair (X,B) of dimension 3 over k such that B is a reduced divisor and K_X+B is big, the linear system $|n(K_X+B)|$ defines a birational map. Then $\mathcal{P}\neq\emptyset$. We let

$$P_{\mathrm{red}}:=\min\{P_0\mid (n,P_0)\in\mathcal{P}\} \quad\text{and}\quad n_{\mathrm{red}}:=\min\{n\mid (n,P_{\mathrm{red}})\in\mathcal{P}\}.$$

Proof. We first fix constants. Let $\eta\in(0,\frac{1}{2})$ be the rational number of Proposition 4.1 for $J=\{0\}$, and let $q:=\lfloor\frac{4}{\eta}\rfloor+1$. Let c_q be the integer of Definition-Lemma 6.3, let Q_q, $P_q^\dagger$, and J_q be the integers of Set-up 8.1, and let m_A be the integer of Theorem 10.1. Let

$$r_{\mathrm{base}}:=\max\{2,J_q\} \quad\text{and}\quad s_q:=2c_q^2qQ_q+2c_q^2+c_q.$$

Let $\mathcal{S}_q$ be the set of lc pairs in the set $\mathcal{S}_V$ of Proposition 9.11 for $I=I_q$ and $V=125q^3s_q^3(r_{\mathrm{base}}+7)^3$. By Proposition 9.11, $\mathcal{S}_q$ is log birationally bounded over $\mathbb{Z}[1/30]$. Thus by Theorem 9.9,

$$\mathcal{V}_q:=\{\operatorname{vol}(K_X+B)\mid (X,B)\in\mathcal{S}_q\}$$

is a DCC set. Since $\mathcal{V}_q\cup\{1\}$ is a DCC set, it has a minimum. Let

$$\begin{aligned} v_q&:=\min(\mathcal{V}_q\cup\{1\}), \quad r_q:=\max\{r_{\mathrm{base}},\lfloor 6v_q^{-1/3}\rfloor+1\},\\ n_q&:=c_q(2c_q(qQ_qr_q+1)+1), \quad\text{and}\quad m_q:=q(4n_q+1). \end{aligned}$$

Since any element of $\mathcal{V}_q$ is the volume of a big divisor, $v_q>0$. Since η depends only on $\{0\}$ and the other constants depend only on q, the numbers q, v_q, m_q, $P_q^\dagger$, and m_A are absolute constants.

Claim 10.5. *Let (Y, B_Y) be a pair satisfying the conditions on (X, B) in Set-up 8.1 with (Y, B_Y) in place of (X, B), and let $D_Y := K_Y + B_Y$. If $p > P_q^\dagger$, then $D_Y^3 \geq v_q$ and $||\lfloor m_q D_Y \rfloor||$ defines a birational map.*

Proof. The proof of $D_Y^3 \geq v_q$ is similar to that of [HMX14, Lemma 7.2]. Let b_0 and n_0 be as in Set-up 8.1, and let

$$v := D_Y^3 = \mathrm{vol}(D_Y),$$

which is positive by [Kol96, Theorem VI.2.15].

We show $v \geq v_q$. If $v \geq 1$, then $v \geq v_q$ since $v_q \leq 1$. Assume $v < 1$, and let

$$r := \max\{r_{\mathrm{base}}, \lfloor 6v^{-1/3} \rfloor + 1\}.$$

Then $r \geq r_{\mathrm{base}}$ and $r^3 v > 216$. Since $v^{-1/3} > 1$,

$$r \leq (r_{\mathrm{base}} + 7)v^{-1/3}.$$

Since $r \geq r_{\mathrm{base}}$ and $r^3 v > 216$, Theorem 8.18 implies $b_0 \leq Q_q r$. Let $m_0 := q(4n_0 + 1)$. Since $q \mid m_0$ and $q \geq 2$, $m_0 \geq 4qn_0 + 2$, so $||\lfloor m_0 D_Y \rfloor||$ defines a birational map by Theorem 6.6 for $b = b_0$. Since $r \geq 1$ and $b_0 \leq Q_q r$,

$$n_0 = 2c_q^2 q b_0 + 2c_q^2 + c_q \leq s_q r \leq s_q(r_{\mathrm{base}} + 7)v^{-1/3}.$$

Since $m_0 \leq 5qn_0$,

$$m_0^3 \,\mathrm{vol}(D_Y) = m_0^3 v \leq 125 q^3 n_0^3 v \leq 125 q^3 s_q^3 (r_{\mathrm{base}} + 7)^3.$$

Thus $(Y, B_Y) \in \mathcal{S}_q$, so $v \in \mathcal{V}_q$ and $v \geq v_q$.

Since $r_q \geq r_{\mathrm{base}}$ and $r_q^3 v \geq r_q^3 v_q > 216$, Theorem 8.18 for $r = r_q$ implies $b_0 \leq Q_q r_q$. Then $n_0 \leq n_q$, so

$$m_q = 4qn_q + q \geq 4qn_0 + 2 \quad \text{and} \quad q \mid m_q.$$

By Theorem 6.6 for $b = b_0$, $||\lfloor m_q D_Y \rfloor||$ defines a birational map. □

Let $P_0 := \max\{11, P_q^\dagger\}$. We show $(m_A m_q, P_0) \in \mathcal{P}$. Let k be an algebraically closed field of characteristic $p > P_0$, and let (X, B) be a projective lc pair of dimension 3 over k such that B is a reduced divisor and $L := K_X + B$ is big. Since B is reduced, $|jL| = ||\lfloor jL \rfloor||$ for any $j \in \mathbb{N}^+$.

If a resolution of X admits a non-constant morphism to an abelian variety, then since $p > P_0 \geq 11$, $|m_A L|$ defines a birational map by Theorem 10.1, so $|m_A m_q L|$ defines a birational map by Lemma 10.2(1).

Assume that no resolution of X admits a non-constant morphism to an abelian variety. Let $f : X' \to X$ be a log resolution of (X, B) [Cut09, Theorems 1.1–1.3] and $B' := f_*^{-1}B + \mathrm{Exc}(f)$. Since (X, B) is lc, $K_{X'} + B' = f^*L + G$ for an f-exceptional $\mathbb{R}$-divisor $G \geq 0$, so $K_{X'} + B'$ is big. Since $p > 5$, Proposition 4.1 for $J = \{0\}$, with 0 and B' in place of B and S, implies that $K_{X'} + (1 - \eta)B'$ is pseudo-effective. Since B is reduced and $q > \frac{4}{\eta}$, Lemma 10.3 with $\alpha = 1$ and $\beta = 1 - \eta$ implies that $(X', \frac{1}{q}\lfloor \frac{q(4-\eta)}{4} B' \rfloor)$ has a log minimal model (Y, B_Y) satisfying Lemma 10.3(1)–(5).

Let Z be a resolution of the graph of the birational map $Y \dashrightarrow X$ [Cut09, Theorems 1.1–1.3]. Then Z is a resolution of X and a resolution of Y. Any morphism from Z to an abelian variety is constant, so the Albanese variety of Z is trivial [Kle05, Remark 9.5.25]. By Lemma 10.3(1)–(3), (Y, B_Y) satisfies the conditions on (X, B) in Set-up 8.1 with (Y, B_Y) in place of (X, B). Since $p > P_q^\dagger$, Claim 10.5 implies that $||\lfloor m_q(K_Y + B_Y) \rfloor||$ defines a birational map. By Lemma 10.3(5),

$$H^0(Y, \mathcal{O}_Y(\lfloor m_q(K_Y + B_Y) \rfloor)) \subset H^0(X, \mathcal{O}_X(m_q L))$$

as subspaces of the function field of X. Thus the rational map defined by $||\lfloor m_q(K_Y + B_Y) \rfloor||$ is the composite of $Y \dashrightarrow X$, the rational map defined by $|m_q L|$, and a linear projection, so $|m_q L|$ defines a birational map. By Lemma 10.2(1), $|m_A m_q L|$ defines a birational map.

Thus $(m_A m_q, P_0) \in \mathcal{P}$, so $\mathcal{P} \neq \emptyset$ and $P_{\mathrm{red}} \leq P_0$. □

11. Effective birationality for DCC coefficients

In this section we prove Theorem 1.3 from Theorem 11.3 and Proposition 11.4.

Lemma 11.1. *Let $I \subset [0,+\infty)$ be a DCC set. Then the following hold.*

(1) For any positive integer i, let $J_i \subset I$ be a finite set and $r_i : J_i \to \mathbb{R}$ a map such that

$$\max\left\{0, a-\frac{1}{i}\right\} \leq r_i(a) \leq a \quad \text{for any } a \in J_i.$$

Then $\bigcup_{i\geq 1} r_i(J_i)$ is a DCC set.

(2) Let $I' \subset [0,+\infty)$ be a DCC set. Then $\{aa' \mid a \in I,\ a' \in I'\}$ is a DCC set.

(3) Assume that $I \subset [c,+\infty)$ for some real number $c > 0$. Then

$$\left\{\sum_{j=1}^{n} a_j \;\middle|\; n \in \mathbb{N}^+,\ a_1,\dots,a_n \in I\right\}$$

is a DCC set, and for any real number v,

$$\#\left\{(a_1,\dots,a_n) \;\middle|\; n \in \mathbb{N}^+,\ a_1,\dots,a_n \in I,\ \sum_{j=1}^{n} a_j = v\right\} < +\infty.$$

Proof. (1) Let $J := \bigcup_{i\geq 1} r_i(J_i)$, and suppose that $(d_k)_{k\geq 1}$ is a strictly decreasing sequence in J. For any k, pick a positive integer i_k and $a_k \in J_{i_k}$ such that $d_k = r_{i_k}(a_k)$. Since the numbers d_k are pairwise distinct and $\#\bigcup_{i=1}^{N} r_i(J_i) < +\infty$ for any positive integer N, we have $\lim_{k\to+\infty} i_k = +\infty$. Since $a_k \in I$, possibly passing to a subsequence, we may assume that $(a_k)_{k\geq 1}$ is increasing. Let $t := \lim_{k\to+\infty} d_k$. Then $t < d_k$ for any k. Since

$$0 \leq a_k - d_k \leq \frac{1}{i_k},$$

we have $\lim_{k\to+\infty} a_k = t$, so $a_k \leq t < d_k$ for any k. This contradicts $d_k = r_{i_k}(a_k) \leq a_k$.

(2) follows since any sequence in a DCC set has an increasing subsequence and $a, a' \geq 0$.

(3) Suppose that $(v_k)_{k\geq 1}$ is strictly decreasing, where

$$v_k = \sum_{j=1}^{n_k} a_{k,j}, \quad n_k \in \mathbb{N}^+, \quad \text{and} \quad a_{k,1},\dots,a_{k,n_k} \in I.$$

Since $a_{k,j} \geq c$, we have $n_k c \leq v_k \leq v_1$, so $n_k \leq v_1/c$ for any k. Possibly passing to a subsequence, we may assume that $n_k = n$ for any k. Possibly passing to a subsequence n times, we may assume that $(a_{k,j})_{k\geq 1}$ is increasing for any $j \in \{1,\dots,n\}$. Then $(v_k)_{k\geq 1}$ is increasing, a contradiction. Since $I \subset [c,+\infty)$ is a DCC set of positive real numbers, the finiteness of the set of tuples in (3) follows from [HMX18, Lemma 2.4.1]. The lemma follows. □

Lemma 11.2. *Let $J \subset [0,1] \cap \mathbb{Q}$ be a DCC set such that $1 \in J$. For any positive integer i, let X_i be a smooth projective threefold over an algebraically closed field k_i of characteristic $p_i > 5$, let $B_i \geq 0$ be a $\mathbb{Q}$-divisor on X_i with simple normal crossing support whose coefficients belong to J, and let $t_i < 1$ be a real number. Assume that $K_{X_i} + B_i$ is big and $K_{X_i} + t_i B_i$ is not pseudo-effective for any i, and that*

$$\lim_{i\to+\infty} t_i = 1.$$

Then, possibly passing to a subsequence, there exist

- *projective $\mathbb{Q}$-factorial klt threefolds X_i' over k_i such that $\rho(X_i') = 1$ and $-K_{X_i'}$ is ample,*
- *$\mathbb{Q}$-divisors $B_i' \geq 0$ on X_i' whose coefficients belong to J, and*
- *strictly increasing rational numbers $\lambda_i \in [\frac{1}{2}, 1)$ such that $\lim_{i\to+\infty} \lambda_i = 1$,*

such that for any i, (X_i', B_i') is lc, $(X_i', \lambda_i B_i')$ is klt, and

$$K_{X_i'} + \lambda_i B_i' \equiv 0.$$

Proof. **Step 1.** In this step we construct Mori fiber spaces $\pi_i : X'_i \to Z_i$ and the numbers λ_i.

Possibly passing to a subsequence, we may assume that $t_i \geq \frac{1}{2}$ for any i. Since X_i is smooth, $\operatorname{Supp} B_i$ is simple normal crossing, and $t_i B_i \in [0,1)$, $(X_i, t_i B_i)$ is klt. By [HNT20, Theorem 1.1], we may run a $(K_{X_i} + t_i B_i)$-MMP which terminates. Since $K_{X_i} + t_i B_i$ is not pseudo-effective, the MMP $\phi_i : X_i \dashrightarrow X'_i$ terminates with a Mori fiber space $\pi_i : X'_i \to Z_i$. Let $B'_i := (\phi_i)_* B_i$. Moreover, X'_i is a projective $\mathbb{Q}$-factorial threefold over k_i, $(X'_i, t_i B'_i)$ is klt, and $B'_i \in J$.

Since $X_i \dashrightarrow X'_i$ does not extract divisors, for any positive integer m we have

$$H^0(X_i, \mathcal{O}_{X_i}(\lfloor m(K_{X_i} + B_i) \rfloor)) \subset H^0(X'_i, \mathcal{O}_{X'_i}(\lfloor m(K_{X'_i} + B'_i) \rfloor)),$$

so $K_{X'_i} + B'_i$ is big. We write $K_{X'_i} + B'_i \sim_{\mathbb{Q}} A_i + E_i$, where A_i is an ample $\mathbb{Q}$-divisor and $E_i \geq 0$. Since $\dim X'_i > \dim Z_i$, there exists a curve $C_i \subset X'_i$ contracted by π_i and not contained in $\operatorname{Supp} E_i$. Then

$$(K_{X'_i} + B'_i) \cdot C_i \geq A_i \cdot C_i > 0.$$

Since $\rho(X'_i/Z_i) = 1$, $K_{X'_i} + B'_i$ is π_i-ample. Thus

$$(1 - t_i)B'_i = (K_{X'_i} + B'_i) - (K_{X'_i} + t_i B'_i)$$

is π_i-ample, so B'_i is π_i-ample as $t_i < 1$. Since the general fiber of π_i has positive dimension, some prime component of B'_i intersects the general fiber of π_i, hence dominates Z_i.

Since $(K_{X'_i} + t_i B'_i) \cdot C_i < 0 < (K_{X'_i} + B'_i) \cdot C_i$, the number

$$\lambda_i := -\frac{K_{X'_i} \cdot C_i}{B'_i \cdot C_i}$$

satisfies $t_i < \lambda_i < 1$, and $\lambda_i \in \mathbb{Q}$ because X'_i is $\mathbb{Q}$-factorial. Since $\rho(X'_i/Z_i) = 1$,

$$K_{X'_i} + \lambda_i B'_i \equiv_{Z_i} 0. \tag{11.1}$$

Step 2. In this step we prove that (X'_i, B'_i) is lc and $(X'_i, \lambda_i B'_i)$ is klt, and we fix a DCC set containing the coefficients of $\lambda_i B'_i$.

Since $B'_i \in J$, $(X'_i, t_i B'_i)$ is klt, and $t_i \to 1$, (X'_i, B'_i) is lc for $i \gg 0$ by Lemma 3.7. Possibly passing to a subsequence, we may assume that (X'_i, B'_i) is lc for any i. Since

$$\lambda_i B'_i = \frac{1 - \lambda_i}{1 - t_i} t_i B'_i + \frac{\lambda_i - t_i}{1 - t_i} B'_i$$

and $\frac{1-\lambda_i}{1-t_i} > 0$, $(X'_i, \lambda_i B'_i)$ is klt.

Since $t_i < \lambda_i < 1$ and $t_i \to 1$, possibly passing to a subsequence, we may assume that λ_i is strictly increasing, $\lim_{i \to +\infty} \lambda_i = 1$, and $\dim Z_i$ does not depend on i. Since J is a DCC set and $1 \in J$, $\alpha_J := \min(J \setminus \{0\})$ exists. We let

$$J_\lambda := \{\lambda_i b \mid i \geq 1,\ b \in J,\ b > 0\}.$$

Since λ_i is strictly increasing, $\{\lambda_i \mid i \geq 1\}$ is a DCC set, so J_λ is a DCC set by Lemma 11.1(2). Since $\lambda_i > t_i \geq \frac{1}{2}$,

$$J_\lambda \subset \left[\frac{\alpha_J}{2}, 1\right),$$

and the coefficients of $\lambda_i B'_i$ belong to J_λ.

Step 3. In this step we prove that $\dim Z_i \neq 2$.

Suppose that $\dim Z_i = 2$ for any i. Let η_i be the generic point of Z_i, and let X'_{η_i} be the generic fiber of π_i, a projective curve over the function field $k_i(Z_i)$. By [Das21, Lemma 3.4], X'_{η_i} is normal, hence regular. Since $(\pi_i)_* \mathcal{O}_{X'_i} = \mathcal{O}_{Z_i}$, we have $H^0(X'_{\eta_i}, \mathcal{O}_{X'_{\eta_i}}) = k_i(Z_i)$. Let $B'_{i1}, \ldots, B'_{in_i}$ be the prime components of B'_i dominating Z_i, and let $b_{ij} \in J$ be the coefficient of B'_{ij} in B'_i. By **Step 1**, $n_i \geq 1$. For any $1 \leq j \leq n_i$, $B'_{ij}|_{X'_{\eta_i}}$ is a closed point of X'_{η_i}, and we let r_{ij} be the

degree of $B'_{ij}|_{X'_{\eta_i}}$ over $k_i(Z_i)$, a positive integer. The prime components of B'_i not dominating Z_i do not meet X'_{η_i}. Since $K_{X'_{\eta_i}} = K_{X'_i}|_{X'_{\eta_i}}$, (11.1) implies that

$$\deg_{k_i(Z_i)} K_{X'_{\eta_i}} = -\lambda_i \sum_{j=1}^{n_i} r_{ij} b_{ij} < 0.$$

By [Das21, Lemma 5.1], $\deg_{k_i(Z_i)} K_{X'_{\eta_i}} = -2$. Thus

$$\sum_{j=1}^{n_i} r_{ij} \lambda_i b_{ij} = 2. \tag{11.2}$$

Since $\lambda_i b_{ij} \in J_\lambda \subset [\frac{\alpha_J}{2}, 1)$, we have $n_i \leq \frac{4}{\alpha_J}$ and $r_{ij} \leq \frac{4}{\alpha_J}$. Possibly passing to a subsequence and reordering indices, we may assume that $n_i = n$ and $r_{ij} = r_j$ for any i and any $1 \leq j \leq n$. Since J is a DCC set, possibly passing to a subsequence, we may assume that b_{ij} is increasing in i for any $1 \leq j \leq n$. Since λ_i is strictly increasing and $b_{ij} > 0$, $\sum_{j=1}^{n} r_j \lambda_i b_{ij}$ is strictly increasing in i, which contradicts (11.2).

Step 4. In this step we prove that $\dim Z_i \neq 1$.

Suppose that $\dim Z_i = 1$ for any i. By [Wal23, Proposition 5.2] and (11.1), there exists a finite set J_0 depending only on J_λ such that for any i, the coefficient in $\lambda_i B'_i$ of any prime component of B'_i dominating Z_i belongs to J_0. By **Step 1**, for any i there exists a prime component B'_{i1} of B'_i dominating Z_i. Let $b_i \in J$ be the coefficient of B'_{i1} in B'_i. Since J is a DCC set, possibly passing to a subsequence, we may assume that b_i is increasing. Since λ_i is strictly increasing and $b_i > 0$, $\lambda_i b_i$ is strictly increasing, which contradicts $\lambda_i b_i \in J_0$ and $\#J_0 < +\infty$.

Step 5. In this step we conclude the proof.

By **Steps 3 and 4**, Z_i is a point for any i. Thus $\rho(X'_i) = \rho(X'_i/Z_i) = 1$, and (11.1) reads $K_{X'_i} + \lambda_i B'_i \equiv 0$. Since B'_i is π_i-ample, B'_i is ample, so $-K_{X'_i} \equiv \lambda_i B'_i$ is ample. Since $(X'_i, t_i B'_i)$ is klt and $B'_i \geq 0$, X'_i is klt. The lemma follows. □

Theorem 11.3. *Let $I \subset [0,1]$ be a DCC set. There exists a rational number $\beta \in [0,1)$ depending only on I with the following property. Let k be an algebraically closed field of characteristic $p > P_{\mathrm{red}}$ (see Definition-Theorem 10.4), X a smooth projective threefold over k, and $B \geq 0$ an $\mathbb{R}$-divisor with simple normal crossing support on X whose coefficients belong to $I \cup \{1\}$. Assume that $K_X + B$ is big. Then $K_X + \beta B$ is pseudo-effective.*

Proof. The proof follows [HMX14, Proof of Lemma 7.3]. Suppose that the theorem fails for I. For any $i \in \mathbb{N}^+$ let

$$\beta_i := 1 - \frac{1}{i} \in [0,1) \cap \mathbb{Q}.$$

Since $\beta = \beta_i$ does not have the property of the theorem, there exist an algebraically closed field k_i of characteristic $p_i > P_{\mathrm{red}}$, a smooth projective threefold X_i over k_i, and an $\mathbb{R}$-divisor $B_i \geq 0$ with simple normal crossing support on X_i whose coefficients belong to $I \cup \{1\}$, such that

$$K_{X_i} + B_i \text{ is big} \quad \text{and} \quad K_{X_i} + \beta_i B_i \text{ is not pseudo-effective.}$$

We divide the proof into four steps.

Step 1. In this step we reduce to the case when the coefficients of all B_i belong to one DCC set $J \subset [0,1] \cap \mathbb{Q}$ with $1 \in J$.

Fix $i \in \mathbb{N}^+$ and let

$$J_i := \{\mathrm{mult}_S B_i \mid S \subset \operatorname{Supp} B_i \text{ a prime divisor}\} \subset I \cup \{1\}.$$

Since the big cone of X_i is open [Laz04, Theorem 2.2.26] and $\operatorname{Supp} B_i$ has finitely many components, there exists a real number $\epsilon_i \in (0, \frac{1}{i}]$ such that $K_{X_i} + B_i - G$ is big for any $\mathbb{R}$-divisor G with $\operatorname{Supp} G \subset \operatorname{Supp} B_i$ and coefficients in $[0, \epsilon_i]$. Let $r_i : J_i \to \mathbb{Q}$ be a map with

$$\max\{0, a - \epsilon_i\} < r_i(a) \leq a \quad \text{for any } a \in J_i,$$

and let

$$B_i^{\mathbb{Q}} := \sum_{S \subset \operatorname{Supp} B_i} r_i(\operatorname{mult}_S B_i)S,$$

where S runs over the prime divisors contained in $\operatorname{Supp} B_i$. Since $0 \notin J_i$, $\operatorname{Supp} B_i^{\mathbb{Q}} = \operatorname{Supp} B_i$, and $B_i - B_i^{\mathbb{Q}} \in [0, \epsilon_i]$. Thus $K_{X_i} + B_i^{\mathbb{Q}}$ is big. If $K_{X_i} + \beta_i B_i^{\mathbb{Q}}$ were pseudo-effective, then so would be

$$K_{X_i} + \beta_i B_i = (K_{X_i} + \beta_i B_i^{\mathbb{Q}}) + \beta_i(B_i - B_i^{\mathbb{Q}}),$$

a contradiction. Let

$$J := \{1\} \cup \bigcup_{i \in \mathbb{N}^+} r_i(J_i) \subset [0,1] \cap \mathbb{Q}.$$

Since $\epsilon_i \le \frac{1}{i}$, Lemma 11.1(1) for the DCC set $I \cup \{1\}$ implies that J is a DCC set. Possibly replacing B_i with $B_i^{\mathbb{Q}}$ for any i, we may assume that B_i is a $\mathbb{Q}$-divisor whose coefficients belong to J.

Step 2. In this step we pass to Fano threefolds of Picard number 1 and show that the volumes of the adjoint divisors tend to 0.

Since $p_i > 5$ and $\lim_{i\to+\infty} \beta_i = 1$, Lemma 11.2 for J and $t_i = \beta_i$ implies that, possibly passing to a subsequence, there exist projective $\mathbb{Q}$-factorial klt threefolds X_i' over k_i with $\rho(X_i') = 1$ and $-K_{X_i'}$ ample, $\mathbb{Q}$-divisors $B_i' \ge 0$ on X_i' whose coefficients belong to J, and strictly increasing rational numbers $\lambda_i \in [\frac{1}{2}, 1)$ with $\lim_{i\to+\infty} \lambda_i = 1$, such that for any i, (X_i', B_i') is lc, $(X_i', \lambda_i B_i')$ is klt, and

$$K_{X_i'} + \lambda_i B_i' \equiv 0.$$

Since $B_i' \equiv -\lambda_i^{-1} K_{X_i'}$,

$$K_{X_i'} + uB_i' \equiv \Big(\frac{u}{\lambda_i} - 1\Big)(-K_{X_i'}) \quad \text{for any } u \in \mathbb{Q}. \tag{11.3}$$

Let

$$J_\lambda := \{\lambda_i b \mid i \in \mathbb{N}^+,\ b \in J,\ b > 0\} \subset (0,1) \cap \mathbb{Q}.$$

Since $\{\lambda_i \mid i \in \mathbb{N}^+\}$ is a strictly increasing sequence, it is a DCC set, so J_λ is a DCC set by Lemma 11.1(2). Since $\lambda_i B_i' \in J_\lambda$, by Theorem 3.8 there exists a real number $c > 0$ depending only on J_λ such that

$$(-K_{X_i'})^3 \le c \quad \text{for any } i. \tag{11.4}$$

By (11.3) with $u = 1$, $K_{X_i'} + B_i'$ is numerically equivalent to the ample $\mathbb{Q}$-divisor $\frac{1-\lambda_i}{\lambda_i}(-K_{X_i'})$. Since the volume of an $\mathbb{R}$-Cartier $\mathbb{R}$-divisor depends only on the numerical class [FKL16, Theorem 3.5(iv)] and equals the self-intersection number for a nef divisor [Laz04, Corollary 1.4.41], (11.4) implies that

$$0 < v_i := \operatorname{vol}(K_{X_i'} + B_i') = \Big(\frac{1-\lambda_i}{\lambda_i}\Big)^3 (-K_{X_i'})^3 \le \Big(\frac{1-\lambda_i}{\lambda_i}\Big)^3 c, \tag{11.5}$$

and thus $\lim_{i\to+\infty} v_i = 0$.

Step 3. In this step we bound the volumes of reduced log resolutions of (X_i', B_i').

For any i, let $f_i : Z_i \to X_i'$ be a log resolution of $(X_i', \operatorname{Supp} B_i')$ [Cut09, Theorems 1.1–1.3], and let

$$\Sigma_i := (f_i)_*^{-1} \operatorname{Supp} B_i' + \operatorname{Exc}(f_i).$$

Then (Z_i, Σ_i) is a projective lc pair with reduced boundary. Since (X_i', B_i') is lc and $B_i' \le \operatorname{Supp} B_i'$, $K_{Z_i} + \Sigma_i \ge f_i^*(K_{X_i'} + B_i')$, so $K_{Z_i} + \Sigma_i$ is big.

Since $(f_i)_*(K_{Z_i} + \Sigma_i) = K_{X_i'} + \operatorname{Supp} B_i'$, Lemma 2.4(1) implies that

$$H^0(Z_i, \mathcal{O}_{Z_i}(m(K_{Z_i} + \Sigma_i))) \subset H^0(X_i', \mathcal{O}_{X_i'}(\lfloor m(K_{X_i'} + \operatorname{Supp} B_i')\rfloor))$$

for any integer $m \geq 0$, so $\mathrm{vol}(K_{Z_i}+\Sigma_i) \leq \mathrm{vol}(K_{X_i'}+\mathrm{Supp}\, B_i')$. Let $\alpha_J := \min(J\backslash\{0\})$, which exists since J is a DCC set containing 1. Since $B_i' \in J$, $\mathrm{Supp}\, B_i' \leq \alpha_J^{-1} B_i'$. Since $\frac{1}{\alpha_J \lambda_i} - 1 > 0$, $-K_{X_i'}$ is ample, and $\lambda_i \geq \frac{1}{2}$, by (11.3) with $u = \alpha_J^{-1}$, [FKL16, Theorem 3.5(iv)], [Laz04, Corollary 1.4.41], and (11.4),

$$\begin{aligned} \mathrm{vol}(K_{Z_i} + \Sigma_i) &\leq \mathrm{vol}(K_{X_i'} + \mathrm{Supp}\, B_i') \leq \mathrm{vol}(K_{X_i'} + \alpha_J^{-1} B_i') \\ &= \Big(\frac{1}{\alpha_J \lambda_i} - 1\Big)^3 (-K_{X_i'})^3 \leq \Big(\frac{2}{\alpha_J} - 1\Big)^3 c. \end{aligned} \tag{11.6}$$

Step 4. In this step we show that $\{(X_i', B_i')\}_{i\in\mathbb{N}^+}$ is log birationally bounded over $\mathbb{Z}[1/30]$ and conclude.

By Definition-Theorem 10.4, $|n_{\mathrm{red}}(K_{Z_i} + \Sigma_i)|$ defines a birational map. Thus by (11.6), Proposition 9.11 for the DCC set $\{1\}$, with $m = n_{\mathrm{red}}$ and $V = n_{\mathrm{red}}^3(\frac{2}{\alpha_J} - 1)^3 c$, implies that $\{(Z_i, \Sigma_i)\}_{i\in\mathbb{N}^+}$ is log birationally bounded over $\mathbb{Z}[1/30]$.

Let $\mathcal{Y} \to T$ and $\mathcal{Q} \subset \mathcal{Y}$ be as in Definition 9.1 for the set $\{(Z_i, \Sigma_i)\}_{i\in\mathbb{N}^+}$. Then for any i there exist a geometric point t_i of T and a birational map $\psi_i : \mathcal{Y}_{t_i} \dashrightarrow Z_i$ such that $\mathcal{Q}_{t_i}$ contains $\mathrm{Supp}\, \psi_{i*}^{-1}\Sigma_i$ and any prime divisor on $\mathcal{Y}_{t_i}$ contracted by ψ_i. Let R be a prime divisor on $\mathcal{Y}_{t_i}$ contracted by $f_i \circ \psi_i$. If ψ_i contracts R, then $R \subset \mathcal{Q}_{t_i}$. Otherwise $\psi_{i*}R$ is a prime divisor on Z_i contracted by f_i, so $\psi_{i*}R \subset \mathrm{Exc}(f_i) \subset \Sigma_i$ and $R \subset \mathrm{Supp}\, \psi_{i*}^{-1}\Sigma_i \subset \mathcal{Q}_{t_i}$. Moreover,

$$\mathrm{Supp}(f_i \circ \psi_i)_*^{-1} B_i' = \psi_{i*}^{-1}(f_i)_*^{-1}\, \mathrm{Supp}\, B_i' \subset \mathrm{Supp}\, \psi_{i*}^{-1}\Sigma_i \subset \mathcal{Q}_{t_i}.$$

Thus $\mathcal{Y} \to T$, $\mathcal{Q}$, and the birational maps $f_i \circ \psi_i : \mathcal{Y}_{t_i} \dashrightarrow X_i'$ show that $\{(X_i', B_i')\}_{i\in\mathbb{N}^+}$ is log birationally bounded over $\mathbb{Z}[1/30]$ (see Definition 9.1).

By Theorem 9.9, $\{v_i \mid i \in \mathbb{N}^+\}$ is a DCC set. By (11.5), $v_i > 0$ for any i and $\lim_{i\to+\infty} v_i = 0$, so $\{v_i\}_{i\in\mathbb{N}^+}$ contains a strictly decreasing subsequence, a contradiction. The theorem follows. □

Proposition 11.4. *Let $\alpha \in (0, 1]$ and $\beta \in [0, 1)$ be real numbers. There exists a positive integer m depending only on α and β with the following property. Let k be an algebraically closed field of characteristic $p > P_{\mathrm{red}}$ (see Definition-Theorem 10.4), (X, B) a projective lc pair of dimension 3 over k, $f : X' \to X$ a log resolution of (X, B), and $B' := f_*^{-1}B + \mathrm{Exc}(f)$. Assume that $K_X + B$ is big, $\mathrm{mult}_S B \geq \alpha$ for any prime divisor $S \subset \mathrm{Supp}\, B$, and $K_{X'} + \beta B'$ is pseudo-effective. Then $|\lfloor m(K_X + B)\rfloor|$ defines a birational map and $\mathrm{vol}(K_X + B) \geq m^{-3}$.*

Proof. Let $q := \min\{n \in \mathbb{Z} \mid n > \frac{4}{\alpha(1-\beta)}\}$, let a_q and b_q be the integers of Definition 7.1, and let

$$\theta := \frac{2\beta}{1+\beta}, \quad c := n_{\mathrm{red}}\Big(1 + \frac{q-1}{1-\theta}\Big), \quad \text{and} \quad m := q\Big\lceil \frac{12(a_q + b_q c) + 2}{q} \Big\rceil.$$

Then $q \geq 2$, $\theta \in [0, 1)$, m depends only on α and β, q divides m, and

$$m > 12(a_q + b_q c) + 1. \tag{11.7}$$

Step 1. In this step we pass to a $\frac{1}{q}$-lc model and bound a mixed intersection number on it.

By Lemma 10.3 for α, β, and q, there exists a pair (Y, B_Y) satisfying Lemma 10.3(1)–(5). Let $D := K_Y + B_Y$. Since Y is $\mathbb{Q}$-factorial and B_Y is a $\mathbb{Q}$-divisor, D is $\mathbb{Q}$-Cartier. Since D is nef (Lemma 10.3(3)), $D^2 \cdot S \geq 0$ for any prime divisor S on Y [Laz04, Theorem 1.4.9 and Remark 1.4.34]. Since $K_Y + \theta B_Y$ is pseudo-effective (Lemma 10.3(4)), its class in $N^1(Y)_{\mathbb{R}}$ is a limit of classes of $\mathbb{R}$-divisors $G \geq 0$. Thus

$$0 \leq D^2 \cdot (K_Y + \theta B_Y) = D^3 - (1-\theta) D^2 \cdot B_Y.$$

Since $\theta < 1$,

$$D^2 \cdot B_Y \leq \frac{D^3}{1-\theta}. \tag{11.8}$$

Step 2. In this step we construct a pencil from the reduced boundary and bound the mixed degree in Theorem 7.2.

Let $g : Z \to Y$ be a log resolution of (Y, B_Y) [Cut09, Theorems 1.1–1.3], and let

$$\Sigma_Z := \operatorname{Supp}(g_*^{-1}B_Y + \operatorname{Exc}(g)).$$

Since (Y, B_Y) is lc,

$$K_Z + g_*^{-1}B_Y + \operatorname{Exc}(g) = g^*D + E_Z, \quad E_Z := \sum_E a(E, Y, B_Y)E \geq 0,$$

where E runs over the g-exceptional prime divisors. Since any positive coefficient of B_Y belongs to $[\frac{1}{q}, 1)$,

$$0 \leq \operatorname{Supp} B_Y - B_Y \leq (q-1)B_Y, \tag{11.9}$$

and

$$K_Z + \Sigma_Z = g^*D + E_Z + g_*^{-1}(\operatorname{Supp} B_Y - B_Y).$$

In particular, $K_Z + \Sigma_Z$ is big. Since E_Z is g-exceptional, the projection formula implies that

$$(g^*D)^2 \cdot E_Z = 0 \quad \text{and} \quad (g^*D)^2 \cdot g_*^{-1}(\operatorname{Supp} B_Y - B_Y) = D^2 \cdot (\operatorname{Supp} B_Y - B_Y).$$

Since D is nef, (11.9) and (11.8) imply that

$$\begin{aligned}(g^*D)^2 \cdot (K_Z + \Sigma_Z) &= D^3 + D^2 \cdot (\operatorname{Supp} B_Y - B_Y) \\ &\leq D^3 + (q-1)D^2 \cdot B_Y \leq \Big(1 + \frac{q-1}{1-\theta}\Big)D^3.\end{aligned} \tag{11.10}$$

Since $p > P_{\mathrm{red}}$ and (Z, Σ_Z) is a projective lc pair with reduced boundary and big $K_Z + \Sigma_Z$, Definition-Theorem 10.4 implies that $|n_{\mathrm{red}}(K_Z + \Sigma_Z)|$ defines a birational map. In particular, there exist linearly independent sections $s_0, s_1 \in H^0(Z, \mathcal{O}_Z(n_{\mathrm{red}}(K_Z + \Sigma_Z)))$. By Lemma 7.3 for $L = n_{\mathrm{red}}(K_Z + \Sigma_Z)$, there exist a birational morphism $\mu : U \to Z$ from a smooth projective threefold U, a contraction $h : U \to C$ onto a smooth projective curve C, a Cartier divisor H on C with $\deg H > 0$ and $|H|$ base point free, and a Cartier divisor $F \geq 0$ on U such that

$$\mu^*(n_{\mathrm{red}}(K_Z + \Sigma_Z)) \sim h^*H + F.$$

Since μ^*g^*D is nef and $F \geq 0$, (11.10) implies that

$$(\mu^*g^*D)^2 \cdot h^*H \leq (\mu^*g^*D)^2 \cdot (h^*H + F) = n_{\mathrm{red}}(g^*D)^2 \cdot (K_Z + \Sigma_Z) \leq cD^3. \tag{11.11}$$

Step 3. In this step we conclude.

By (11.11), (11.7), and Theorem 7.2, $||\lfloor mD \rfloor||$ defines a birational map. Since

$$H^0(Y, \mathcal{O}_Y(\lfloor mD \rfloor)) \subset H^0(X, \mathcal{O}_X(\lfloor m(K_X + B) \rfloor))$$

as subspaces of the function field of X (Lemma 10.3(5)), $||\lfloor m(K_X + B) \rfloor||$ defines a birational map. By Lemma 10.2(2), $\operatorname{vol}(K_X + B) \geq m^{-3}$. □

Proof of Theorem 1.3. Let $P := P_{\mathrm{red}}$. Let $I \subset [0, 1]$ be a DCC set, let

$$\alpha := \min\big((I \cup \{1\}) \backslash \{0\}\big) \in (0, 1],$$

let $\beta \in [0, 1)$ be the rational number of Theorem 11.3 for I, and let m be the positive integer of Proposition 11.4 for α and β. Then m depends only on I.

Let k be an algebraically closed field of characteristic $p > P$ and (X, B) a projective lc pair of dimension 3 over k such that $B \in I$ and $K_X + B$ is big. Let $f : X' \to X$ be a log resolution of (X, B) [Cut09, Theorems 1.1–1.3] and $B' := f_*^{-1}B + \operatorname{Exc}(f)$. Since (X, B) is lc, $K_{X'} + B' \geq f^*(K_X + B)$, so $K_{X'} + B'$ is big. By Theorem 11.3, $K_{X'} + \beta B'$ is pseudo-effective. Since $\operatorname{mult}_S B \geq \alpha$ for any prime divisor $S \subset \operatorname{Supp} B$, Proposition 11.4 implies that $||\lfloor m(K_X + B) \rfloor||$ defines a birational map and $\operatorname{vol}(K_X + B) \geq m^{-3}$. □

12. The descending chain condition for volumes

In this section we prove Theorem 1.4.

Definition 12.1. Let X be a reduced projective scheme of pure dimension 3 over an algebraically closed field which is S_2 and nodal in codimension 1, that is, X is demi-normal [Kol13, Definition 5.1]. Let $\nu : Y \to X$ be the normalization of X, and D_Y the conductor divisor on Y, which is reduced [Kol13, 5.2]. Let $B \geq 0$ be an $\mathbb{R}$-divisor on X none of whose components is contained in the non-normal locus of X, and let $B_Y := \nu_*^{-1}B$. Let $Y_1, \dots, Y_n$ be the connected components of Y, and let $D_i := D_Y|_{Y_i}$ and $B_i := B_Y|_{Y_i}$. We say that (X, B) is an *slc pair* if $K_X + B$ is $\mathbb{R}$-Cartier and $(Y_i, D_i + B_i)$ is lc for any i, where $\nu^*(K_X + B) = K_Y + D_Y + B_Y$ [Kol13, 5.7]. For an slc pair (X, B), we define

$$\operatorname{vol}(K_X + B) := \sum_{i=1}^{n} \operatorname{vol}(K_{Y_i} + D_i + B_i).$$

The equality $\nu^*(K_X + B) = K_Y + D_Y + B_Y$ in Definition 12.1 follows by comparing coefficients at the prime divisors of Y. Indeed, ν is an isomorphism over the normal locus of X, which contains the generic point of the image of any prime divisor of Y not contained in D_Y. At the generic point of a component of D_Y, X has a node, $\nu^*\omega_X \cong \omega_Y(D_Y)$ [Kol13, 5.7], and B_Y has coefficient 0, since no component of B is contained in the non-normal locus of X. In particular, $K_{Y_i} + D_i + B_i = \nu^*(K_X + B)|_{Y_i}$ is $\mathbb{R}$-Cartier for any i.

Lemma 12.2. *Let (X, B) be a projective slc pair of dimension 3 over an algebraically closed field, and let $\nu : Y \to X$, $Y_1, \dots, Y_n$, D_i, and B_i be as in Definition 12.1. Then*

$$(K_X + B)^3 = \sum_{i=1}^{n} (K_{Y_i} + D_i + B_i)^3.$$

Proof. Since $\nu_*[Y] = \sum_{i=1}^{n} [X_i] = [X]$, where $X_i := \nu(Y_i)$ are the irreducible components of X, the lemma follows from the projection formula for intersections of Cartier divisors with cycles under proper morphisms [Ful98, Proposition 2.5(c)] and the multilinearity of intersection numbers. □

Proof of Theorem 1.4. Let $P := P_{\mathrm{red}}$ (Definition-Theorem 10.4). Then $P \geq 11$, and Theorem 1.3 holds for P by its proof. Let m be the integer of Theorem 1.3 for the DCC set $I \cup \{1\} \subset [0, 1]$, and let

$$c := m^{-3}.$$

Then c depends only on I. Let $\mathcal{V}_{I\cup\{1\}}$ be the set of volumes $\operatorname{vol}(K_X + B)$ of all projective lc pairs (X, B) of dimension 3 over all algebraically closed fields of characteristic $> P$ such that $B \in I \cup \{1\}$ and $K_X + B$ is big. Since $\mathcal{V}_{\mathrm{lc}} \subset \mathcal{V}_{I\cup\{1\}}$, it suffices to prove (1) for $\mathcal{V}_{I\cup\{1\}}$, that is,

$$\mathcal{V}_{I\cup\{1\}} \subset [c, +\infty) \quad \text{and} \quad \mathcal{V}_{I\cup\{1\}} \text{ is a DCC set.}$$

(1) Let (X, B) be a projective lc pair of dimension 3 over an algebraically closed field of characteristic $> P$ such that $B \in I \cup \{1\}$ and $K_X + B$ is big. By Theorem 1.3, $|\lfloor m(K_X + B) \rfloor|$ defines a birational map and $\operatorname{vol}(K_X + B) \geq c$. Thus $\mathcal{V}_{I\cup\{1\}} \subset [c, +\infty)$. Moreover, since $P \geq 11$, for any real number $w \geq \operatorname{vol}(K_X + B)$, (X, B) belongs to the set $\mathcal{S}_{m^3 w}$ of Proposition 9.11 for the DCC set $I \cup \{1\}$.

Suppose that $\mathcal{V}_{I\cup\{1\}}$ contains a strictly decreasing sequence $w_1 > w_2 > \cdots$. For any j, let (X_j, B_j) be a projective lc pair of dimension 3 over an algebraically closed field of characteristic $> P$ such that $B_j \in I \cup \{1\}$, $K_{X_j} + B_j$ is big, and $\operatorname{vol}(K_{X_j} + B_j) = w_j$. Since $w_j \leq w_1$, $(X_j, B_j) \in \mathcal{S}_{m^3 w_1}$ for any j. By Proposition 9.11, $\mathcal{S}_{m^3 w_1}$ is log birationally bounded over $\mathbb{Z}[1/30]$, so its subset $\{(X_j, B_j) \mid j \geq 1\}$, which consists of lc pairs, is log birationally bounded over $\mathbb{Z}[1/30]$, and by Theorem 9.9, $\{w_j \mid j \geq 1\}$ is a DCC set. This contradicts $w_1 > w_2 > \cdots$. Thus $\mathcal{V}_{I\cup\{1\}}$ is a DCC set, and (1) follows.

(2) Let (X, B) be a projective slc pair of dimension 3 over an algebraically closed field k of characteristic $> P$ such that $B \in I$ and $\mathrm{vol}(K_X + B) > 0$. Let $\nu : Y \to X$, D_Y, B_Y, $Y_1, \dots, Y_n$, D_i, and B_i be as in Definition 12.1, and let

$$v_i := \mathrm{vol}(K_{Y_i} + D_i + B_i)$$

for any i. Any component of D_Y maps into the non-normal locus of X, and no component of B_Y does. Since D_Y is reduced, $D_i + B_i \in I \cup \{1\}$. By Definition 12.1, $(Y_i, D_i + B_i)$ is lc, and it is a projective pair of dimension 3 over k since ν is finite. Thus $v_i \in \mathcal{V}_{I\cup\{1\}}$ for any i with $v_i > 0$. Since $\sum_{i=1}^n v_i = \mathrm{vol}(K_X + B) > 0$, the set $\{i \mid v_i > 0\}$ is non-empty, and

$$\mathrm{vol}(K_X + B) = \sum_{i:v_i>0} v_i \in \left\{ \sum_{j=1}^N a_j \,\middle|\, N \in \mathbb{N}^+,\ a_1, \dots, a_N \in \mathcal{V}_{I\cup\{1\}} \right\}.$$

Since $\mathcal{V}_{I\cup\{1\}}$ is a DCC set contained in $[c, +\infty)$, by Lemma 11.1(3) for $\mathcal{V}_{I\cup\{1\}}$, the right-hand side is a DCC set, and it is contained in $[c, +\infty)$. (2) follows.

(3) Let (X, B) be as in (3), and keep the notation of (2). Since ν is finite and $K_X + B$ is ample, $K_{Y_i} + D_i + B_i = \nu^*(K_X + B)|_{Y_i}$ is ample, so

$$v_i = (K_{Y_i} + D_i + B_i)^3 > 0$$

by [Kol96, Theorem VI.2.15] and the continuity and homogeneity of vol on $N^1(Y_i)_{\mathbb{R}}$ (Subsection 2.2), and $v_i \in \mathcal{V}_{I\cup\{1\}} \subset [c, +\infty)$ for any i. By Lemma 12.2,

$$\sum_{i=1}^n v_i = (K_X + B)^3 = V.$$

Thus $nc \le V$, that is, $n \le V/c$. Moreover, $(v_1, \dots, v_n)$ belongs to the set of tuples $(a_1, \dots, a_N)$ with $N \in \mathbb{N}^+$, $a_1, \dots, a_N \in \mathcal{V}_{I\cup\{1\}}$, and $\sum_{j=1}^N a_j = V$, which is finite by Lemma 11.1(3) for $\mathcal{V}_{I\cup\{1\}}$ and depends only on I and V. (3) follows. □

13. Boundedness at a fixed volume

In this section we prove Theorem 1.5.

Definition 13.1. Let $\mathcal{S}$ be a set of couples (X, B), each consisting of a projective scheme X of pure dimension 3 over an algebraically closed field F depending on (X, B) and an $\mathbb{R}$-divisor $B \ge 0$ on X, and let $I_0 \subset \mathbb{R}$ be a finite set. We say that $\mathcal{S}$ is *bounded over* $\mathbb{Z}[1/30]$ *with coefficients in* I_0 if there exist finitely many projective flat morphisms $\mathcal{X}_j \to T_j$ onto schemes T_j of finite type over $\mathbb{Z}[1/30]$, closed subschemes $\mathcal{B}_{j,i} \subset \mathcal{X}_j$ flat over T_j for $1 \le i \le n_j$, and numbers $b_{j,i} \in I_0$, such that for any $(X, B) \in \mathcal{S}$ there exist j, a geometric point $t : \mathrm{Spec}\, F \to T_j$, and an isomorphism of F-schemes $X \cong \mathcal{X}_{j,t}$ under which $\mathcal{B}_{j,1,t}, \dots, \mathcal{B}_{j,n_j,t}$ are the pairwise distinct prime components of B and

$$B = \sum_{i=1}^{n_j} b_{j,i} \mathcal{B}_{j,i,t}.$$

Lemma 13.2. *Notation and conditions as in Set-up 9.2, with T integral. Let $B := \sum_{j=1}^n b_j \Sigma_j$ with $b_1, \dots, b_n \in [0, 1]$, and assume that $\mathrm{vol}(K_{\mathcal{Y}_{\overline{\eta}}} + B_{\overline{\eta}}) > 0$ for a geometric generic point $\overline{\eta}$ of T. Then there exist*

- *a non-empty open subset $T^\circ \subset T$ and a finite surjective morphism $T' \to T^\circ$ from an integral scheme T',*
- *a projective flat morphism $\mathcal{X} \to T'$ whose geometric fibers are normal integral threefolds and a birational map $\phi : \mathcal{Y}_{T'} \dashrightarrow \mathcal{X}$ over T', where $\mathcal{Y}_{T'} := \mathcal{Y} \times_T T'$, and*
- *invertible sheaves $\mathcal{H}_1, \dots, \mathcal{H}_m$ on $\mathcal{X}$ generated by global sections over T' such that $\bigotimes_{k=1}^m \mathcal{H}_k$ is ample over T', and real numbers $\mu_1, \dots, \mu_m > 0$,*

such that for any geometric point t of T', with $\mathcal{Y}_t$ and B_t the fibers of $\mathcal{Y}$ and B over the image of t in T, the following hold.

(1) ϕ restricts to a birational map $\phi_t : \mathcal{Y}_t \dashrightarrow \mathcal{X}_t$ whose inverse contracts no divisor.
(2) $(\mathcal{X}_t, \phi_{t*}B_t)$ is lc and

$$K_{\mathcal{X}_t} + \phi_{t*}B_t \sim_{\mathbb{R}} \sum_{k=1}^{m} \mu_k H_{k,t},$$

where $H_{k,t}$ is a Cartier divisor on $\mathcal{X}_t$ with

$$\mathcal{O}_{\mathcal{X}_t}(H_{k,t}) \cong \mathcal{H}_k|_{\mathcal{X}_t}.$$

In particular, $K_{\mathcal{X}_t} + \phi_{t*}B_t$ is ample.
(3) For any common resolution $f : W \to \mathcal{Y}_t$ and $g : W \to \mathcal{X}_t$ of ϕ_t, we have

$$f^*(K_{\mathcal{Y}_t} + B_t) = g^*(K_{\mathcal{X}_t} + \phi_{t*}B_t) + E$$

with $E \geq 0$ and g-exceptional.

Proof. Let $\overline{k(\eta)}$ be an algebraic closure of the residue field $k(\eta)$ of the generic point η of T, and let $\overline{\eta} : \operatorname{Spec} \overline{k(\eta)} \to T$ be the corresponding geometric generic point. Since T is of finite type over $\mathbb{Z}[1/30]$, either $\operatorname{char} k(\eta) = 0$ or $\operatorname{char} k(\eta) > 5$. Let $Y := \mathcal{Y}_{\overline{\eta}}$ and $B_Y := B_{\overline{\eta}}$. By Set-up 9.2, Y is a smooth projective threefold and $\Sigma_{\overline{\eta}}$ is a simple normal crossing divisor, so (Y, B_Y) is dlt.

We run a $(K_Y + B_Y)$-MMP, which terminates by [HNT20, Theorem 1.1] if $\operatorname{char} k(\eta) > 5$ and by [Sho96, Theorem 5.2 and 5.1.3] if $\operatorname{char} k(\eta) = 0$. Since $K_Y + B_Y$ is big, the MMP ends with a $\mathbb{Q}$-factorial dlt pair (Y_1, B_1) such that $K_{Y_1} + B_1$ is nef and big, where B_1 is the pushforward of B_Y. In particular, Y_1 is klt. Let $f : W \to Y$ and $g_1 : W \to Y_1$ be a common resolution of $Y \dashrightarrow Y_1$, which exists by [Bir16, Section 2.4] if $\operatorname{char} k(\eta) > 5$ and by Hironaka's theorem if $\operatorname{char} k(\eta) = 0$. For $\mathbf{c} = (c_1, \dots, c_n) \in [0,1]^n$, let $C(\mathbf{c}) := \sum_{j=1}^{n} c_j \Sigma_{j,\overline{\eta}}$, let $C_1(\mathbf{c})$ be the pushforward of $C(\mathbf{c})$ to Y_1, and let

$$E(\mathbf{c}) := f^*(K_Y + C(\mathbf{c})) - g_1^*(K_{Y_1} + C_1(\mathbf{c})).$$

Since Y_1 is $\mathbb{Q}$-factorial, $E(\mathbf{c})$ is an affine function of $\mathbf{c}$ with rational coefficients. Since the inverse of $Y \dashrightarrow Y_1$ contracts no divisor, $g_{1*}E(\mathbf{c}) = 0$, that is, $E(\mathbf{c})$ is g_1-exceptional. Let $\mathbf{b} := (b_1, \dots, b_n)$. Since $Y \dashrightarrow Y_1$ is a composition of steps of the $(K_Y + B_Y)$-MMP, $E(\mathbf{b}) \geq 0$ by the negativity lemma.

Let $\mathcal{P}$ be the set of $\mathbf{c} \in [0,1]^n$ satisfying the following.

(i) $(Y_1, C_1(\mathbf{c}))$ is lc and $K_{Y_1} + C_1(\mathbf{c})$ is nef.
(ii) $E(\mathbf{c}) \geq 0$.

The set of $\mathbf{c} \in [0,1]^n$ satisfying (i) is a rational polytope by [HNT20, Proposition 4.9(3)] if $\operatorname{char} k(\eta) > 5$ and by [Bir11, Remark 3.1, Proposition 3.2(3)] if $\operatorname{char} k(\eta) = 0$, and (ii) consists of finitely many linear inequalities with rational coefficients. Thus $\mathcal{P}$ is a rational polytope, and $\mathbf{b} \in \mathcal{P}$. Taking the vertices of the face $\mathcal{F}$ of $\mathcal{P}$ such that $\mathbf{b}$ belongs to the relative interior of $\mathcal{F}$, we obtain $\mathbf{c}_1, \dots, \mathbf{c}_m \in \mathcal{P} \cap \mathbb{Q}^n$ and real numbers $\mu'_1, \dots, \mu'_m > 0$ such that

$$\mathbf{b} = \sum_{k=1}^{m} \mu'_k \mathbf{c}_k \quad \text{and} \quad \sum_{k=1}^{m} \mu'_k = 1. \tag{13.1}$$

For any k, $C_1(\mathbf{c}_k)$ is a $\mathbb{Q}$-divisor, $(Y_1, C_1(\mathbf{c}_k))$ is lc, and $K_{Y_1} + C_1(\mathbf{c}_k)$ is nef, so $K_{Y_1} + C_1(\mathbf{c}_k)$ is semi-ample by Lemma 2.7 if $\operatorname{char} k(\eta) > 5$ and by [KMM94, Theorem 1.1] and [KMM04, Section 6] if $\operatorname{char} k(\eta) = 0$. Let r be a positive integer such that $r(K_{Y_1} + C_1(\mathbf{c}_k))$ is a base point free Cartier divisor for any k, let $\psi : Y_1 \to \prod_{k=1}^{m} \mathbb{P}^{N_k}$ be the product of the morphisms defined by the linear systems $|r(K_{Y_1} + C_1(\mathbf{c}_k))|$, and let $\psi = \beta \circ \alpha$ be the Stein factorization, where $\alpha : Y_1 \to X$ is a contraction onto a normal projective variety X and β is finite. Let H_k be the

pullback under β of a hyperplane of the k-th factor. Then H_k is a base point free Cartier divisor, $r(K_{Y_1}+C_1(\mathbf{c}_k))\sim\alpha^*H_k$, and $\sum_{k=1}^m H_k$ is ample since β is finite. Let $\mu_k:=\mu'_k/r$. By (13.1),

$$K_{Y_1}+B_1=\sum_{k=1}^m\mu'_k(K_{Y_1}+C_1(\mathbf{c}_k))\sim_{\mathbb{R}}\alpha^*\Big(\sum_{k=1}^m\mu_kH_k\Big).$$

Since $K_{Y_1}+B_1$ is big, $\dim X=3$, and α is birational. Let $C_X(\mathbf{c}):=\alpha_*C_1(\mathbf{c})$. For any k, $r(K_X+C_X(\mathbf{c}_k))\sim H_k$, and $K_{Y_1}+C_1(\mathbf{c}_k)-\alpha^*(K_X+C_X(\mathbf{c}_k))$ is α-exceptional and numerically trivial over X, hence zero by the negativity lemma. Let $g:=\alpha\circ g_1$ and $\phi_{\overline{\eta}}:=\alpha\circ(Y\dashrightarrow Y_1)$. Then for any k,

$$f^*(K_Y+C(\mathbf{c}_k))=g^*(K_X+C_X(\mathbf{c}_k))+E(\mathbf{c}_k),\tag{13.2}$$

where $E(\mathbf{c}_k)\geq 0$ is g-exceptional.

The schemes X and W, the morphisms f, g, β, the Cartier divisors H_k, the $\mathbb{Q}$-divisors $E(\mathbf{c}_k)$ and $C_X(\mathbf{c}_k)$, sections generating $\mathcal{O}_X(H_k)$, and rational functions realizing the linear equivalences $r(K_X+C_X(\mathbf{c}_k))\sim H_k$ are finitely many data, hence are defined over a finite extension $k_1\subset\overline{k(\eta)}$ of $k(\eta)$ [Gro66, Théorème 8.5.2, Proposition 8.5.5, Théorème 8.8.2, Théorème 8.10.5]. Let $T^\circ\subset T$ be a non-empty open affine subset and T' the normalization of T° in k_1. Since T is of finite type over $\mathbb{Z}[1/30]$, $T'\to T^\circ$ is finite, and it is surjective. By [Gro66, §8, §9, Théorème 12.2.4], possibly shrinking T° and replacing T' with the preimage of T°, the data spread out to the following:

(a) A smooth projective morphism $\mathcal{W}\to T'$ whose geometric fibers are integral and birational morphisms $f:\mathcal{W}\to\mathcal{Y}_{T'}$ and $g:\mathcal{W}\to\mathcal{X}$ over T', where $\mathcal{X}\to T'$ is projective and flat and the geometric fibers of $\mathcal{X}$ are normal integral threefolds, and we let $\phi:=g\circ f^{-1}$, a birational map $\mathcal{Y}_{T'}\dashrightarrow\mathcal{X}$.

(b) Invertible sheaves $\mathcal{H}_k=\mathcal{O}_{\mathcal{X}}(H_k)$ generated by global sections over T' and a finite morphism $\beta:\mathcal{X}\to\prod_k\mathbb{P}^{N_k}_{T'}$ with $\bigotimes_k\mathcal{H}_k\cong\beta^*\mathcal{O}(1,\dots,1)$, so that $\bigotimes_k\mathcal{H}_k$ is ample over T'.

(c) $\mathbb{Q}$-divisors $\mathcal{E}_k\geq 0$ on $\mathcal{W}$ and $\mathcal{C}_k$ on $\mathcal{X}$ such that for any geometric point t of T', we have $\mathcal{C}_{k,t}=\phi_{t*}C(\mathbf{c}_k)_t$, $r(K_{\mathcal{X}_t}+\mathcal{C}_{k,t})\sim H_{k,t}$, and

$$f_t^*(K_{\mathcal{Y}_t}+C(\mathbf{c}_k)_t)=g_t^*(K_{\mathcal{X}_t}+\mathcal{C}_{k,t})+\mathcal{E}_{k,t}.$$

Let $\mathcal{U}\subset\mathcal{Y}_{T'}$ be the largest open subset over which f is an isomorphism. Let $\mathcal{Z}_1:=\mathcal{Y}_{T'}\backslash\mathcal{U}$, $\mathcal{Z}_2:=g(f^{-1}(\mathcal{Z}_1))$, and $\mathcal{Z}_3:=\bigcup_k g(\operatorname{Supp}\mathcal{E}_k)$, and let ξ be the generic point of T'. Then $\dim\mathcal{Z}_{j,\xi}\leq 1$ for $j\in\{1,2,3\}$, since the inverse of $\phi_{\overline{\eta}}$, a composite of steps of an MMP and the birational morphism α, contracts no divisor, so that any f-exceptional prime divisor on W is g-exceptional, and since $E(\mathbf{c}_k)$ is g-exceptional by (13.2). By [Gro66, Corollaire (13.1.5)], possibly shrinking T° again, we have $\dim\mathcal{Z}_{j,s}\leq 1$ for $j\in\{1,2,3\}$ and any point s of T'. Since $\mathcal{X}_\xi$ is normal, the closed subset of $\mathcal{X}$ over which g is not an isomorphism has fiber of dimension ≤ 1 over ξ, so, possibly shrinking T° again [Gro66, Corollaire (13.1.5)], its fibers over all points of T' have dimension ≤ 1. Since the geometric fibers of $\mathcal{W}$, $\mathcal{Y}_{T'}$, and $\mathcal{X}$ over T' are integral of dimension 3 and f, g are isomorphisms over dense open subsets meeting every fiber, f_t and g_t are birational for any geometric point t of T'.

Let t be a geometric point of T'. Any f_t-exceptional prime divisor on $\mathcal{W}_t$ is contained in $f_t^{-1}(\mathcal{Y}_t\backslash\mathcal{U}_t)$, hence is g_t-exceptional. Thus any prime divisor on $\mathcal{X}_t$ is the image of a prime divisor on $\mathcal{W}_t$ which is not f_t-exceptional, which is (1). By (13.1) and (c),

$$f_t^*(K_{\mathcal{Y}_t}+B_t)=g_t^*(K_{\mathcal{X}_t}+\phi_{t*}B_t)+E_t,\quad E_t:=\sum_{k=1}^m\mu'_k\mathcal{E}_{k,t}\geq 0,$$

and E_t is g_t-exceptional since $\dim g_t(\operatorname{Supp}E_t)\leq 1$. For another common resolution $f':W'\to\mathcal{Y}_t$ and $g':W'\to\mathcal{X}_t$, let $h':W''\to W'$ and $h:W''\to\mathcal{W}_t$ be a common resolution. Then

$$f'^*(K_{\mathcal{Y}_t}+B_t)-g'^*(K_{\mathcal{X}_t}+\phi_{t*}B_t)=h'_*h^*E_t,$$

and $h'_*h^*E_t \geq 0$ is g'-exceptional. This proves (3). By (c),

$$K_{\mathcal{X}_t} + \phi_{t*}B_t = \sum_k \mu'_k(K_{\mathcal{X}_t} + \mathcal{C}_{k,t}) \sim_{\mathbb{R}} \sum_k \mu_k H_{k,t}.$$

Since $\bigotimes_k \mathcal{H}_k$ is ample over T' and each $\mathcal{H}_k$ is generated by global sections over T', $\sum_k H_{k,t}$ is ample and each $H_{k,t}$ is nef, so $\sum_k \mu_k H_{k,t}$, the sum of $(\min_k \mu_k)\sum_k H_{k,t}$ and a nef divisor, is ample. For a prime divisor D over $\mathcal{X}_t$, choose a common resolution $f' : W' \to \mathcal{Y}_t$ and $g' : W' \to \mathcal{X}_t$ of ϕ_t such that D is a divisor on W', and write

$$K_{W'} + \Gamma_Y = f'^*(K_{\mathcal{Y}_t} + B_t) \quad \text{and} \quad K_{W'} + \Gamma_X = g'^*(K_{\mathcal{X}_t} + \phi_{t*}B_t).$$

By (3), $\Gamma_X \leq \Gamma_Y$. Since Σ_t is a simple normal crossing divisor and $b_j \in [0,1]$ for any j, $(\mathcal{Y}_t, B_t)$ is lc, so $\operatorname{mult}_D \Gamma_X \leq \operatorname{mult}_D \Gamma_Y \leq 1$. Thus $(\mathcal{X}_t, \phi_{t*}B_t)$ is lc, which proves (2). □

Set-up 13.3. Let I, v, and $\mathcal{S}$ be as follows.

(1) $I \subset [0,1]$ is a DCC set and $v > 0$ is a real number.
(2) $\mathcal{S}$ is a set of projective lc pairs (X, B) of dimension 3 over algebraically closed fields of characteristic > 5 such that $B \in I$, $K_X + B$ is ample, and $(K_X + B)^3 = v$.
(3) $\mathcal{S}$ is log birationally bounded over $\mathbb{Z}[1/30]$.

Theorem 13.4. *Notation and conditions as in Set-up 13.3. Then there exists a finite set $I_0 \subset I$ such that $\mathcal{S}$ is bounded over $\mathbb{Z}[1/30]$ with coefficients in I_0.*

Proof. We divide the proof into several steps. For a normal projective threefold X over an algebraically closed field F and a reduced divisor D on X, let $\operatorname{cpx}(X, D)$ be the least positive integer n such that X has a closed embedding into $\mathbb{P}^n_F$ with $\deg X \leq n$ and $\deg D \leq n$.

Step 1. In this step we show that there exist finitely many triples $(T, \pi : \mathcal{Y} \to T, \Sigma)$ as in Set-up 9.2, with T integral, satisfying the following.

(i) For any $(X, B) \in \mathcal{S}$ there exist one of the triples, a geometric point $t : \operatorname{Spec} F \to T$, and a birational map $\psi : \mathcal{Y}_t \dashrightarrow X$ such that Σ_t contains $\operatorname{Supp} \psi_*^{-1}B$ and every prime divisor on $\mathcal{Y}_t$ contracted by ψ.

Let T_0, $\pi_0 : \mathcal{Y}_0 \to T_0$, and $\mathcal{Q}_0$ be as in Definition 9.1 for $\mathcal{S}$. By noetherian induction on T_0, it suffices to show that for any integral closed subscheme $T_1 \subset T_0$ there exist a non-empty open subset $T_1^\circ \subset T_1$, a triple (T, π, Σ) as in Set-up 9.2 with T integral, a finite surjective morphism $T \to T_1^\circ$, and a birational morphism $h : \mathcal{Y} \to \mathcal{Y}_0 \times_{T_0} T$ over T such that for any geometric point t of T the following hold.

(a) $h_t^{-1}(\mathcal{Q}_{0,t}) \subset \operatorname{Supp} \Sigma_t$.
(b) There exists a closed subset $Z_t \subset \mathcal{Y}_{0,t}$ such that h_t is an isomorphism over $\mathcal{Y}_{0,t} \backslash Z_t$ and $h_t^{-1}(Z_t) \subset \operatorname{Supp} \Sigma_t$.

Given such data, let $(X, B) \in \mathcal{S}$, and let t_0 and $\psi_0 : \mathcal{Y}_{0,t_0} \dashrightarrow X$ be as in Definition 9.1 for (X, B), with t_0 a geometric point of T_1°. Since $T \to T_1^\circ$ is finite and surjective and the field of definition of t_0 is algebraically closed, t_0 lifts to a geometric point t of T. Let $\psi := \psi_0 \circ h_t$ and let S be a prime divisor on $\mathcal{Y}_t$ which is contracted by ψ or is a component of $\psi_*^{-1}B$. If S is h_t-exceptional, then $S \subset h_t^{-1}(Z_t) \subset \operatorname{Supp} \Sigma_t$ by (b). Otherwise $h_{t*}S$ is contracted by ψ_0 or is a component of $\psi_{0*}^{-1}B$, so $h_{t*}S \subset \mathcal{Q}_{0,t}$ and $S \subset \operatorname{Supp} \Sigma_t$ by (a).

Let $\overline{\eta}$ be a geometric generic point of T_1. Then $\operatorname{char} \overline{\eta} = 0$ or $\operatorname{char} \overline{\eta} > 5$. Let $h : \widetilde{Y} \to \mathcal{Y}_{0,\overline{\eta}}$ be a log resolution of $(\mathcal{Y}_{0,\overline{\eta}}, \mathcal{Q}_{0,\overline{\eta}})$ [Bir16, Section 2.4] (by Hironaka's theorem in characteristic zero), and let $\Sigma_{\widetilde{Y}}$ be the reduced divisor with

$$\operatorname{Supp} \Sigma_{\widetilde{Y}} = \operatorname{Exc}(h) \cup h^{-1}(\mathcal{Q}_{0,\overline{\eta}}).$$

Since $\mathcal{Y}_{0,\overline{\eta}}$ is normal, h is an isomorphism over $\mathcal{Y}_{0,\overline{\eta}} \backslash h(\operatorname{Exc}(h))$. The variety $\widetilde{Y}$, the morphism h, the components of $\Sigma_{\widetilde{Y}}$, and the connected components of all intersections of components of $\Sigma_{\widetilde{Y}}$ are defined over a finite extension of the residue field of the generic point of T_1 [Gro66,

Théorème 8.8.2, Théorème 8.10.5]. Normalizing a non-empty open subset of T_1 in the extension, spreading out, and shrinking [Gro66, §8, §9, Théorème 12.2.4], we obtain T_1°, T, $(\mathcal{Y}, \Sigma)$, and h satisfying (a) and (b), where Z_t is the fiber over t of the closure of $h(\mathrm{Exc}(h))$.

Since the conclusion of Theorem 13.4 for finitely many subsets of $\mathcal{S}$ implies the conclusion for their union, we may assume that one triple (T, π, Σ) satisfies (i) for all pairs in $\mathcal{S}$. We write $\Sigma = \sum_{j=1}^n \Sigma_j$.

Step 2. In this step we introduce the $\boldsymbol{b}$-divisors and the blow-ups of strata of $\mathcal{Y}$, and we fix a sequence. We show the following statement in **Steps 2–5**.

($\star$) For any sequence $(X_i, B_i)_{i\in\mathbb{N}}$ in $\mathcal{S}$ there exist a subsequence, a finite set $R_0 \subset \mathbb{R}$, and a positive integer n_0 such that $B_i \in R_0$ and $\mathrm{cpx}(X_i, \operatorname{Supp} B_i) \le n_0$ for any i in the subsequence.

Let Z be a normal projective variety over an algebraically closed field. A $\boldsymbol{b}$-divisor $\mathbf{N}$ over Z assigns a real number $\mathbf{N}(E)$ to any prime divisor E over Z. For a higher model U of Z, the trace of $\mathbf{N}$ on U is $\mathbf{N}_U := \sum_E \mathbf{N}(E)E$, where E runs over the prime divisors on U, and $\wedge$ denotes the minimum at any prime divisor. Let D be an $\mathbb{R}$-divisor on Z with coefficients in $[0,1]$ such that $K_Z + D$ is $\mathbb{R}$-Cartier. As in [HMX13, Definitions 5.2 and 5.5], let $\mathbf{M}_D(E) := \mathrm{mult}_E D$ if E is a divisor on Z and $\mathbf{M}_D(E) := 1$ otherwise, and let $\mathbf{L}_D(E) := \max\{0, 1 - a(E, Z, D)\}$. If $D \le D'$, then $\mathbf{L}_D \le \mathbf{L}_{D'}$. Let $h : U \to Z$ be a smooth higher model and $m \ge 0$ an integer, and regard the section spaces on U and on Z below as subspaces of the function field of Z. Since $h_*(K_U + \mathbf{L}_{D,U}) = K_Z + D$ and $K_U + \mathbf{L}_{D,U} \ge h^*(K_Z + D)$, Lemma 2.4(1)(2) imply the following for any $\boldsymbol{b}$-divisor $\mathbf{N}$ over Z with $\mathbf{N}_Z = D$:

$$H^0(U, \mathcal{O}_U(\lfloor m(K_U + \mathbf{N}_U)\rfloor)) = H^0(U, \mathcal{O}_U(\lfloor m(K_U + (\mathbf{N} \wedge \mathbf{L}_D)_U)\rfloor)), \tag{13.3}$$

because the left side is contained in

$$H^0(Z, \mathcal{O}_Z(\lfloor m(K_Z + D)\rfloor)) \quad \text{hence in} \quad H^0(U, \mathcal{O}_U(\lfloor m(K_U + \mathbf{L}_{D,U})\rfloor)).$$

If moreover Z is smooth and $\operatorname{Supp} D$ is a simple normal crossing divisor, then (Z, D) is lc, so $K_U + \mathbf{M}_{D,U} \ge h^*(K_Z + D)$, and Lemma 2.4 implies

$$H^0(U, \mathcal{O}_U(\lfloor m(K_U + \mathbf{M}_{D,U})\rfloor)) = H^0(Z, \mathcal{O}_Z(\lfloor m(K_Z + D)\rfloor)). \tag{13.4}$$

Let V be a stratum of Σ (Definition 9.6), with Λ_V as there. To V we attach the cone $\mathbb{R}_{\ge 0}^{\Lambda_V}$ with rays e_j, $j \in \Lambda_V$. For a geometric point t of T, the cones attached to the strata V with $V_t \ne \emptyset$, glued along the faces given by the inclusions of strata (the face $\mathbb{R}_{\ge 0}^{\Lambda_{V'}}$ of the cone of V being the cone of the stratum $V' \supset V$), form the conical polyhedral complex of $(\mathcal{Y}_t, \Sigma_t)$ [KKMS73, Chapter II, §1] (Lemma 9.7(3)). Two different strata V_1, V_2 with $\Lambda_{V_1} = \Lambda_{V_2}$ give two different cones. Let $\mathcal{R}$ be the set of pairs $\rho = (V, u)$ of a stratum V and a primitive vector $u \in (\mathbb{N}^+)^{\Lambda_V}$. We identify $\rho = (V, u)$ with the vector u in the cone attached to V. For $\rho = (V, u) \in \mathcal{R}$ and a geometric point t of T with $V_t \ne \emptyset$, let $E_{\rho,t}$ be the toroidal prime divisor (Definition 9.6) over $\mathcal{Y}_t$ with center V_t and $\mathrm{mult}_{E_{\rho,t}} \Sigma_{j,t} = u_j$ for any $j \in \Lambda_V$ (Lemma 9.7(3)). For a $\boldsymbol{b}$-divisor $\mathbf{N}$ over $\mathcal{Y}_t$ we write $\mathbf{N}(\rho) := \mathbf{N}(E_{\rho,t})$.

In this proof, the modifications of $\mathcal{Y}$ over T are the blow-ups of strata $Y \to \mathcal{Y}$ (Definition 9.6). For such Y, by Lemma 9.7(1), $Y \to T$ and the reduced total transform Σ_Y of Σ satisfy Set-up 9.2, and for any geometric point t of T, $Y_t \to \mathcal{Y}_t$ is a blow-up of strata of $\mathcal{Y}_t$ with respect to Σ_t, that is, the toroidal modification of $(\mathcal{Y}_t, \Sigma_t)$ defined by a regular subdivision of the conical polyhedral complex of $(\mathcal{Y}_t, \Sigma_t)$ [KKMS73, Chapter II, §2, Theorems 6* and 10*]. By induction on the number of blow-ups, there exists a finite set $\mathcal{R}_Y \subset \mathcal{R}$ such that the components of Σ_Y are divisors E_κ, $\kappa \in \mathcal{R}_Y$, whose fiber over any geometric point t of T is $E_{\kappa,t}$: the blow-up along a stratum of Σ_Y adds the sum of the vectors κ of the components E_κ containing that stratum. For any geometric point t of T with $V_t \ne \emptyset$ for any stratum V of Σ, the rays $\mathcal{R}_Y$ span the conical polyhedral complex of $(Y_t, \Sigma_{Y,t})$, which is regular and does not depend on such t. We call it the *complex of* Y.

If $D = \sum_{\kappa\in\mathcal{R}_Y} d_\kappa E_\kappa$ and $\rho = \sum_k \lambda_k\kappa_k$ with $\lambda_k \geq 0$ in a cone of the complex of Y with rays κ_k, then

$$a(E_{\rho,t}, Y_t, D_t) = \sum_k \lambda_k(1 - d_{\kappa_k}), \tag{13.5}$$

because the log discrepancy of $E_{\rho,t}$ is linear on the cone and equals $1 - d_\kappa$ at κ. If $d_\kappa \in [0, 1]$ for any κ, then $\mathbf{L}_{D_t}(E) = 0$ for any prime divisor E over Y_t which is not of the form $E_{\rho,t}$, by Lemma 9.5 applied on Y_t, since Y_t is smooth and $\Sigma_{Y,t}$ is a simple normal crossing divisor. Thus $\mathbf{L}_{D_t}$ is determined by the numbers $\mathbf{L}_D(\rho) := \max\{0, 1 - a(E_{\rho,t}, Y_t, D_t)\}$, which do not depend on such t.

For a primitive vector ω in a cone of the complex of Y and a geometric point t of T with $V_t \neq \emptyset$ for any stratum V of Σ, let $Y_{\omega,t} \to Y_t$ be the toroidal modification of $(Y_t, \Sigma_{Y,t})$ defined by the star subdivision at ω [KKMS73, Chapter II, §2, Theorem 6*], which is the normalized blow-up of Y_t along a monomial ideal sheaf [KKMS73, Chapter II, §2, Theorem 10*]. The cones of the star subdivision are simplicial, hence the toroidal Weil divisors on $Y_{\omega,t}$ are $\mathbb{Q}$-Cartier [KKMS73, Chapter II, §2, Theorem 9*, III(c)], so (13.5) also holds on $Y_{\omega,t}$. The rays of the complex of $Y_{\omega,t}$ are $\mathcal{R}_Y \cup \{\omega\}$. For a function $\mathbf{N} : \mathcal{R} \to [0, 1]$ let

$$\mathbf{N}_Y := \sum_{\kappa\in\mathcal{R}_Y} \mathbf{N}(\kappa)E_\kappa \quad \text{and} \quad \mathbf{N}_{Y_{\omega,t}} := \sum_{\kappa\in\mathcal{R}_Y\cup\{\omega\}} \mathbf{N}(\kappa)E_{\kappa,t}.$$

Let t be a geometric point of T, $Y'' \to Y_t$ a toroidal modification of $(Y_t, \Sigma_{Y,t})$ whose cones are simplicial, D an $\mathbb{R}$-divisor on Y'' supported on the toroidal divisors with coefficients in $[0, 1]$, $Y''' \to Y''$ a toroidal modification whose cones are simplicial, and D^{c} the crepant pullback of D to Y''', so that $(\mathbf{L}_D)_{Y'''}$ is the positive part of D^{c}. We claim that

$$\mathbf{L}_{(\mathbf{L}_D)_{Y'''}} = \mathbf{L}_D \quad \text{if } D^{\mathrm{c}} \geq 0 \text{ or } Y''' \text{ is smooth.} \tag{13.6}$$

Since the cones of Y'' and Y''' are simplicial, (13.5) holds on Y'' and on Y'''. If $D^{\mathrm{c}} \geq 0$, then $(\mathbf{L}_D)_{Y'''} = D^{\mathrm{c}}$, and $\mathbf{L}_{D^{\mathrm{c}}} = \mathbf{L}_D$ since D^{c} is the crepant pullback of D. If Y''' is smooth, then, with $\mathbf{L}_{D^{\mathrm{c}}}(E) := \max\{0, 1 - a(E, Y''', D^{\mathrm{c}})\}$, which is defined although D^{c} may have negative coefficients, $D^{\mathrm{c}} \leq (\mathbf{L}_D)_{Y'''}$ and $K_{Y'''} + D^{\mathrm{c}} = (Y''' \to Y'')^*(K_{Y''} + D)$ give

$$\mathbf{L}_{(\mathbf{L}_D)_{Y'''}} \geq \mathbf{L}_{D^{\mathrm{c}}} = \mathbf{L}_D.$$

Conversely, let $\rho \in \mathcal{R}$ with $\mathbf{L}_{(\mathbf{L}_D)_{Y'''}}(\rho) > 0$, and write $\rho = \sum_k \lambda_k\kappa_k$ as in (13.5) on Y'''. Since Y''' is smooth, its complex is regular, so $\lambda_k \in \mathbb{N}$. If $\lambda_k \geq 1$ for some k with $\mathrm{mult}_{E_{\kappa_k}} D^{\mathrm{c}} < 0$, then E_{κ_k} has coefficient 0 in $(\mathbf{L}_D)_{Y'''}$, so $a(E_{\rho,t}, Y''', (\mathbf{L}_D)_{Y'''}) \geq \lambda_k \geq 1$, contradicting $\mathbf{L}_{(\mathbf{L}_D)_{Y'''}}(\rho) > 0$. Thus $\lambda_k = 0$ for any such k, and hence

$$a(E_{\rho,t}, Y''', (\mathbf{L}_D)_{Y'''}) = a(E_{\rho,t}, Y''', D^{\mathrm{c}}).$$

Let $(X_i, B_i)_{i\in\mathbb{N}}$ be a sequence in $\mathcal{S}$, and for any i let $t_i : \operatorname{Spec} F_i \to T$ and $\psi_i : \mathcal{Y}_{t_i} \dashrightarrow X_i$ be as in (i) for (X_i, B_i). Choose a closed subset of T which is minimal among those containing t_i for infinitely many i. It is irreducible, and Set-up 9.2 is preserved by base change to integral closed subschemes of T. Possibly passing to a subsequence and replacing T with the minimal closed subset, we may assume that

$$\text{any nonempty open subset of } T \text{ contains } t_i \text{ for any } i \gg 0. \tag{13.7}$$

Since any stratum V of Σ is smooth over T, its image in T is open. Possibly replacing T with the intersection of the non-empty such images and truncating the sequence, we may assume that $V_t \neq \emptyset$ for any non-empty stratum V of Σ and any geometric point t of T, and that (13.7) holds. Then $V'_t \neq \emptyset$ for any stratum V' of Σ_Y, any blow-up of strata Y of $\mathcal{Y}$, and any geometric point t of T, since such V' are projective bundles over strata or strict transforms of strata. For any i, let $f_i : W_i \to \mathcal{Y}_{t_i}$ and $q_i : W_i \to X_i$ be a common resolution of ψ_i such that $\mathrm{Exc}(q_i) \cup q_i^{-1}(\mathrm{Supp}\, B_i)$ is a simple normal crossing divisor [Bir16, Section 2.4], and let $\Delta_i := q_{i*}^{-1}B_i + \mathrm{Exc}(q_i)$, where

$\mathrm{Exc}(q_i)$ denotes the reduced divisor. Since (X_i, B_i) is lc, $K_{W_i} + \Delta_i - q_i^*(K_{X_i} + B_i)$ is effective and q_i-exceptional. By Lemma 2.4, for any m,

$$H^0(W_i, \mathcal{O}_{W_i}(\lfloor m(K_{W_i} + \Delta_i)\rfloor)) = H^0(X_i, \mathcal{O}_{X_i}(\lfloor m(K_{X_i} + B_i)\rfloor)),$$

so $\mathrm{vol}(K_{W_i} + \Delta_i) = (K_{X_i} + B_i)^3 = v$.

For any $\rho \in \mathcal{R}$, $\mathbf{M}_{\Delta_i}(\rho) \in J := I \cup \{0, 1\}$: it is 1 if E_{ρ,t_i} is not a divisor on W_i or is q_i-exceptional, a coefficient of B_i if $q_{i*}E_{\rho,t_i}$ is a component of B_i, and 0 otherwise. A prime divisor on $\mathcal{Y}_{t_i}$ not contained in $\operatorname{Supp}\Sigma_{t_i}$ is not contracted by ψ_i and is not a component of $\psi_{i*}^{-1}B_i$, so the coefficient of $\mathbf{M}_{\Delta_i}$ at such a prime divisor is 0. Let $Y_i^{(0)}$ be a blow-up of strata of $\mathcal{Y}$ such that $\mathcal{R}_{Y_i^{(0)}}$ contains the finitely many ρ for which E_{ρ,t_i} is a divisor on W_i, which exists by Lemma 9.7(2) since each such E_{ρ,t_i} is a divisor on a blow-up of strata of $\mathcal{Y}_{t_i}$, and let $\mathbf{M}_i := \mathbf{M}_{\Delta_i'}$ with $\Delta_i' := (\mathbf{M}_{\Delta_i})_{Y_{i,t_i}^{(0)}}$. Then $\mathbf{M}_i(\rho) = \mathbf{M}_{\Delta_i}(\rho)$ for any $\rho \in \mathcal{R}$, and

$$(\mathbf{M}_i)_{\mathcal{Y}_{t_i}} = (\mathbf{M}_{\Delta_i})_{\mathcal{Y}_{t_i}}.$$

Let $\Phi_i^{(0)} := f_{i*}\Delta_i$, which equals $(\mathbf{M}_i)_{\mathcal{Y}_{t_i}}$. Since $\mathbf{L}_{\Phi_i^{(0)}}$ vanishes at prime divisors over $\mathcal{Y}_{t_i}$ not of the form E_{ρ,t_i},

$$\mathbf{M}_{\Delta_i} \wedge \mathbf{L}_{\Phi_i^{(0)}} = \mathbf{M}_i \wedge \mathbf{L}_{\Phi_i^{(0)}}. \tag{13.8}$$

Let U be a smooth higher model of W_i and $Y_{i,t_i}^{(0)}$. By (13.3) with $Z = \mathcal{Y}_{t_i}$ and (13.8), for any m,

$$\begin{aligned} H^0(U, \mathcal{O}_U(\lfloor m(K_U + (\mathbf{M}_{\Delta_i})_U)\rfloor)) &= H^0(U, \mathcal{O}_U(\lfloor m(K_U + (\mathbf{M}_i \wedge \mathbf{L}_{\Phi_i^{(0)}})_U)\rfloor)) \\ &= H^0(U, \mathcal{O}_U(\lfloor m(K_U + (\mathbf{M}_i)_U)\rfloor)). \end{aligned}$$

Thus by (13.4) with $Z = W_i$ and with $Z = Y_{i,t_i}^{(0)}$, for any m,

$$H^0(Y_{i,t_i}^{(0)}, \mathcal{O}(\lfloor m(K_{Y_{i,t_i}^{(0)}} + \Delta_i')\rfloor)) = H^0(W_i, \mathcal{O}_{W_i}(\lfloor m(K_{W_i} + \Delta_i)\rfloor)). \tag{13.9}$$

Any sequence in the DCC set J has an increasing subsequence. Since $\mathcal{R}$ is countable, by passing to subsequences for the elements of $\mathcal{R}$ one by one and taking a diagonal subsequence, we may assume that for any $\rho \in \mathcal{R}$ the sequence $(\mathbf{M}_i(\rho))_i$ is increasing from some index i_ρ on, and let $\mathbf{M}_\infty(\rho) := \lim_i \mathbf{M}_i(\rho)$. The closure $\overline{J}$ of J is a DCC set, since a strictly decreasing sequence in $\overline{J}$ yields, by choosing an element of J close to each term, a strictly decreasing sequence in J. Thus the values of $\mathbf{M}_\infty : \mathcal{R} \to [0, 1]$ form a DCC set.

Step 3. In this step we find a blow-up of strata Y_∞ of $\mathcal{Y}$ such that $\mathbf{L}_\Phi \le \mathbf{M}_\infty$ on $\mathcal{R}$ for $\Phi := (\mathbf{M}_\infty)_{Y_\infty}$, after modifying $\mathbf{M}_i$ and $\mathbf{M}_\infty$. We keep the following conditions.

(I1) For any i there exists a blow-up of strata Y_i of $\mathcal{Y}$ such that $\mathbf{M}_i = \mathbf{M}_{(\mathbf{M}_i)_{Y_{i,t_i}}}$ and, for any m,

$$H^0(Y_{i,t_i}, \mathcal{O}(\lfloor m(K_{Y_{i,t_i}} + (\mathbf{M}_i)_{Y_{i,t_i}})\rfloor)) = H^0(W_i, \mathcal{O}_{W_i}(\lfloor m(K_{W_i} + \Delta_i)\rfloor)).$$

(II2) For any $\rho \in \mathcal{R}$, $(\mathbf{M}_i(\rho))_i$ is increasing from some index on, with limit $\mathbf{M}_\infty(\rho)$, and the values of $\mathbf{M}_\infty$ form a DCC set.

By (13.9) and **Step 2**, (I1) with $Y_i := Y_i^{(0)}$ and (I2) hold for $\mathbf{M}_i$, $\mathbf{M}_\infty$.

Let Y be a blow-up of strata of $\mathcal{Y}$, $\Phi := (\mathbf{M}_\infty)_Y$, and $\Phi_i := (\mathbf{M}_i)_Y$. We say that $\rho \in \mathcal{R}$ is *violating* if $\mathbf{L}_\Phi(\rho) > \mathbf{M}_\infty(\rho)$. Then $\rho \notin \mathcal{R}_Y$, since

$$\mathbf{L}_\Phi(\kappa) = \Phi(\kappa) = \mathbf{M}_\infty(\kappa) \quad \text{for any } \kappa \in \mathcal{R}_Y.$$

For $\rho \in \mathcal{R}$ let Q_ρ be the cone of the complex of Y whose relative interior contains ρ, and let the *weight* of ρ and $w(Y)$ be

$$w_Y(\rho) := \#\{\kappa \text{ a ray of } Q_\rho \mid \Phi(\kappa) = 1\}, \qquad w(Y) := \max\{w_Y(\rho) \mid \rho \in \mathcal{R} \text{ violating}\},$$

and $w(Y) := -1$ if no ρ is violating. Then $w(Y) \in \{-1, 0, 1, 2, 3\}$.

Assume that $w := w(Y) \geq 0$. Let $\mathcal{C}$ be the set of pairs (Q, x), where Q is a cone of the complex of Y with rays $\kappa_1, \dots, \kappa_{\dim Q}$ such that $\Phi(\kappa_j) < 1$ if and only if $j \leq \dim Q - w$, and $x = (u_1, \dots, u_{\dim Q - w})$ for a violating $\rho = \sum_{j=1}^{\dim Q} u_j \kappa_j$ in the relative interior of Q. By (13.5), $\mathbf{L}_\Phi(\rho) > 0$ implies $\sum_{j \leq \dim Q - w} u_j(1 - \Phi(\kappa_j)) < 1$, so $\mathcal{C}$ is finite. For $(Q, x) \in \mathcal{C}$, by (I2) we may choose a primitive vector $\omega = \omega(Q, x)$ in the relative interior of Q whose first $\dim Q - w$ coordinates are x and which minimizes $\mathbf{M}_\infty$ among such vectors. Then for any violating ρ in the relative interior of Q whose first $\dim Q - w$ coordinates are x,

$$a(E_{\omega,t}, Y_t, \Phi_t) = \sum_{j \leq \dim Q - w} u_j(1 - \Phi(\kappa_j)) < 1 \quad \text{and} \quad \mathbf{M}_\infty(\omega) \leq \mathbf{M}_\infty(\rho) < \mathbf{L}_\Phi(\rho) \leq 1. \tag{13.10}$$

For any $\omega = \omega(Q, x)$, $(Q, x) \in \mathcal{C}$, and any geometric point t of T, let $Y_{\omega,t} \to Y_t$ be the toroidal modification defined by the star subdivision at ω, and let

$$\Gamma_{\omega,t} := (\mathbf{L}_\Phi \wedge \mathbf{M}_\infty)_{Y_{\omega,t}} \quad \text{and} \quad \Gamma_{i,\omega,t} := (\mathbf{L}_{\Phi_i} \wedge \mathbf{M}_i)_{Y_{\omega,t}},$$

and let $\Gamma_\omega(\kappa)$ be the coefficient of $E_{\kappa,t}$ in $\Gamma_{\omega,t}$, which does not depend on t. By (13.5) on $Y_{\omega,t}$ and Lemma 9.5 applied to the positive part of the crepant pullback of $\Gamma_{\omega,t}$ to a blow-up of strata of Y_t through which $Y_{\omega,t} \to Y_t$ factors (Lemma 9.7(2)), $\mathbf{L}_{\Gamma_{\omega,t}}$ vanishes at the prime divisors over $Y_{\omega,t}$ not of the form $E_{\rho,t}$, and the value $\mathbf{L}_{\Gamma_\omega}(\rho)$ of $\mathbf{L}_{\Gamma_{\omega,t}}$ at $E_{\rho,t}$ does not depend on t, and similarly for $\mathbf{L}_{\Gamma_{i,\omega}}$. Let Y' be a blow-up of strata of Y such that $Y'_t \to Y_t$ factors through $Y_{\omega,t}$ for all ω and any geometric point t of T. It exists by Lemma 9.7(2) applied at one geometric point, and the factorization holds at any t since the complexes of Y'_t and $Y_{\omega,t}$ do not depend on t [KKMS73, Chapter II, §2, Theorem 7*]. Let

$$\Phi' := \Big(\mathbf{M}_\infty \wedge \bigwedge_\omega \mathbf{L}_{\Gamma_\omega}\Big)_{Y'}, \quad \mathbf{M}'_\infty := \mathbf{M}_\infty \wedge \mathbf{M}_{\Phi'},$$

$$\Phi'_i := \Big(\mathbf{M}_i \wedge \bigwedge_\omega \mathbf{L}_{\Gamma_{i,\omega}}\Big)_{Y'}, \quad \text{and} \quad \mathbf{M}'_i := \mathbf{M}_i \wedge \mathbf{M}_{\Phi'_i}.$$

Thus $\mathbf{M}'_\infty = \Phi'$ on $\mathcal{R}_{Y'}$ and $\mathbf{M}'_\infty = \mathbf{M}_\infty$ on $\mathcal{R} \backslash \mathcal{R}_{Y'}$, and similarly for $\mathbf{M}'_i$.

We show that (I1) and (I2) hold for $\mathbf{M}'_i$, $\mathbf{M}'_\infty$. Let Y'_i be a blow-up of strata of $\mathcal{Y}$ such that $Y'_{i,t_i} \to \mathcal{Y}_{t_i}$ factors through Y_{i,t_i} and Y'_{t_i} (Lemma 9.7(1)(2)). Then $\mathbf{M}'_i = \mathbf{M}_{(\mathbf{M}'_i)_{Y'_{i,t_i}}}$. Let U be a smooth higher model of Y'_{i,t_i} and of Y_{ω,t_i} for all ω. Let s be a section of $\lfloor m(K_U + (\mathbf{M}_i)_U) \rfloor$. By (13.3) with $Z = Y_{t_i}$, s is a section of

$$\lfloor m(K_U + (\mathbf{M}_i \wedge \mathbf{L}_{\Phi_i})_U) \rfloor.$$

Since the trace of $\mathbf{M}_i \wedge \mathbf{L}_{\Phi_i}$ on Y_{ω,t_i} is Γ_{i,ω,t_i}, Lemma 2.4(1)(2) imply that s is a section of $\lfloor m(K_U + (\mathbf{L}_{\Gamma_{i,\omega}})_U) \rfloor$ for any ω. Thus s is a section of

$$\Big\lfloor m\Big(K_U + \Big(\mathbf{M}_i \wedge \bigwedge_\omega \mathbf{L}_{\Gamma_{i,\omega}}\Big)_U\Big) \Big\rfloor,$$

whose pushforward to Y'_{t_i} is $\lfloor m(K_{Y'_{t_i}} + \Phi'_{i,t_i}) \rfloor$. By Lemma 2.4(1) and (13.4), s is a section of $\lfloor m(K_U + (\mathbf{M}_{\Phi'_i})_U) \rfloor$, hence of $\lfloor m(K_U + (\mathbf{M}'_i)_U) \rfloor$. Thus

$$H^0(U, \mathcal{O}_U(\lfloor m(K_U + (\mathbf{M}'_i)_U) \rfloor)) = H^0(U, \mathcal{O}_U(\lfloor m(K_U + (\mathbf{M}_i)_U) \rfloor)),$$

and (I1) for $\mathbf{M}'_i$ follows from (13.4) and (I1) for $\mathbf{M}_i$. For $\kappa \in \mathcal{R}_{Y'}$, the definitions of Φ' and Φ'_i and (13.5) on Y and on the $Y_{\omega,t}$ give a finite set $\mathcal{R}_\kappa \subset \mathcal{R}$ and a minimum Ψ_κ of finitely many increasing continuous functions, built from minima and the functions $\max\{0, 1 - \sum_k \lambda_k(1 - y_k)\}$, such that

$$\Phi'_i(\kappa) = \Psi_\kappa((\mathbf{M}_i(\rho))_{\rho \in \mathcal{R}_\kappa}) \quad \text{and} \quad \Phi'(\kappa) = \Psi_\kappa((\mathbf{M}_\infty(\rho))_{\rho \in \mathcal{R}_\kappa}).$$

Thus (I2) for $\mathbf{M}_i$, $\mathbf{M}_\infty$ implies that $(\mathbf{M}'_i(\rho))_i$ is increasing from some index on, with limit $\mathbf{M}'_\infty(\rho)$ for any ρ, and the values of $\mathbf{M}'_\infty$ are those of $\mathbf{M}_\infty$ together with finitely many numbers, hence form a DCC set.

We show that $w(Y') < w$, where $w(Y')$ is defined with Φ' and $\mathbf{M}'_\infty$. Let $\rho' \in \mathcal{R}$ be violating for Y', that is, $\mathbf{L}_{\Phi'}(\rho') > \mathbf{M}'_\infty(\rho')$. Then $\rho' \notin \mathcal{R}_{Y'}$, so $\mathbf{M}'_\infty(\rho') = \mathbf{M}_\infty(\rho')$. Fix ω and a geometric point t of T.

(a) By (13.10), the crepant pullback Φ^{c}_ω of Φ_t to $Y_{\omega,t}$ satisfies $\Phi^{\mathrm{c}}_\omega \geq 0$, so $(\mathbf{L}_\Phi)_{Y_{\omega,t}} = \Phi^{\mathrm{c}}_\omega$, and $\Gamma_{\omega,t} \leq (\mathbf{L}_\Phi)_{Y_{\omega,t}}$ and (13.6) with $Y'' = Y_t$, $D = \Phi_t$, and $Y''' = Y_{\omega,t}$ imply $\mathbf{L}_{\Gamma_\omega} \leq \mathbf{L}_\Phi$. Since $\Phi' \leq (\mathbf{L}_{\Gamma_\omega})_{Y'}$ and Y' is smooth, (13.6) with $Y'' = Y_{\omega,t}$, $D = \Gamma_{\omega,t}$, and $Y''' = Y'_t$ implies

$$\mathbf{L}_{\Phi'} \leq \mathbf{L}_{\Gamma_\omega} \leq \mathbf{L}_\Phi, \quad \text{so} \quad \mathbf{L}_\Phi(\rho') \geq \mathbf{L}_{\Phi'}(\rho') > \mathbf{M}_\infty(\rho'),$$

that is, ρ' is violating for Y, and $w_Y(\rho') \leq w$.

(b) Let Q' be the cone of the complex of Y' whose relative interior contains ρ' and $\tau_1, \dots, \tau_{w'}$ the rays of Q' with $\Phi'(\tau_k) = 1$, where $w' = w_{Y'}(\rho')$. Then $\mathbf{L}_{\Gamma_\omega}(\tau_k) = 1$. Let Q_ω be the smallest cone of the complex of $Y_{\omega,t}$ containing Q'. By (13.5) on $Y_{\omega,t}$, $a(E_{\tau_k,t}, Y_{\omega,t}, \Gamma_{\omega,t}) = 0$ implies that τ_k belongs to the face G_ω of Q_ω spanned by the rays κ of Q_ω with $\Gamma_\omega(\kappa) = 1$. Since the complex of $Y_{\omega,t}$ refines that of Y, $Q_\omega \subset Q_{\rho'}$, and the rays of Q_ω are rays of $Q_{\rho'}$ or ω. For a ray κ of $Q_{\rho'}$, $\Gamma_\omega(\kappa) = \Phi(\kappa)$, and $\Gamma_\omega(\omega) \leq \mathbf{M}_\infty(\omega) < 1$ by (13.10). Since $\tau_1, \dots, \tau_{w'}$ are linearly independent, $w' \leq \dim G_\omega \leq w_Y(\rho') \leq w$.

(c) Assume that $w' = w$. Then $w_Y(\rho') = w$, so $(Q_{\rho'}, x) \in \mathcal{C}$, where x is the vector of the first $\dim Q_{\rho'} - w$ coordinates of $\rho' = \sum_j u_j \kappa_j$ in $Q_{\rho'}$. Let $\omega := \omega(Q_{\rho'}, x)$ in (b). Since ω belongs to the relative interior of $Q_{\rho'}$, there exists j_0 with $Q_\omega \subset \mathrm{Cone}(\omega, \kappa_j \mid j \neq j_0)$, and $\rho' = \lambda\omega + \sum_{j \neq j_0} \lambda_j \kappa_j$ with $\lambda, \lambda_j \geq 0$. By (b), $\dim G_\omega = w$, so the w rays κ_j with $j > \dim Q_{\rho'} - w$ are rays of Q_ω, and $j_0 \leq \dim Q_{\rho'} - w$. Comparing the j_0-th coordinates, $u_{j_0} = \lambda u_{j_0}$, so $\lambda = 1$. By (13.5) on $Y_{\omega,t}$,

$$a(E_{\rho',t}, Y_{\omega,t}, \Gamma_{\omega,t}) \geq 1 - \Gamma_\omega(\omega),$$

so by (a) and (13.10),

$$\mathbf{L}_{\Phi'}(\rho') \leq \mathbf{L}_{\Gamma_\omega}(\rho') \leq \Gamma_\omega(\omega) \leq \mathbf{M}_\infty(\omega) \leq \mathbf{M}_\infty(\rho') = \mathbf{M}'_\infty(\rho'),$$

which contradicts the assumption that ρ' is violating for Y'.

Thus $w(Y') < w$. Starting with $Y = \mathcal{Y}$ and replacing $(Y, \mathbf{M}_i, \mathbf{M}_\infty)$ with $(Y', \mathbf{M}'_i, \mathbf{M}'_\infty)$ at most four times, we obtain a blow-up of strata Y_∞ of $\mathcal{Y}$ with $w(Y_\infty) = -1$. Let $\Phi := (\mathbf{M}_\infty)_{Y_\infty}$ and $\Phi_i := (\mathbf{M}_i)_{Y_\infty}$. Then (I1) and (I2) hold. Since $\mathcal{R}_{Y_\infty}$ is finite, after truncating the sequence, $\mathbf{M}_i \leq \mathbf{M}_\infty$ on $\mathcal{R}_{Y_\infty}$ for any i by (I2), so since $w(Y_\infty) = -1$,

$$\mathbf{L}_\Phi(\rho) \leq \mathbf{M}_\infty(\rho) \quad \text{and} \quad \Phi_i \leq \Phi \quad \text{for any } \rho \in \mathcal{R} \text{ and any } i. \tag{13.11}$$

Moreover $Y_\infty \to T$ with the reduced total transform Σ_{Y_∞} of Σ satisfies Set-up 9.2 by Lemma 9.7(1), and Φ is supported on Σ_{Y_∞} with coefficients in $[0, 1]$.

Step 4. In this step we show that $\mathrm{vol}(K_{Y_{\infty,t}} + \Phi_t) = v$ for any geometric point t of a non-empty open subset of T, and we construct the ample models of $(Y_{\infty,t}, \Phi_t)$.

For any i, let U be a smooth higher model of W_i, of the variety Y_{i,t_i} of (I1), and of Y_{∞,t_i}. By (I1) and (13.4), $\mathrm{vol}(K_U + (\mathbf{M}_i)_U) = \mathrm{vol}(K_{W_i} + \Delta_i) = v$. By Lemma 2.4(1) and (13.11),

$$v \leq \mathrm{vol}(K_{Y_{\infty,t_i}} + \Phi_{i,t_i}) \leq \mathrm{vol}(K_{Y_{\infty,t_i}} + \Phi_{t_i}). \tag{13.12}$$

Since $Y_\infty \to T$ with Σ_{Y_∞} satisfies Set-up 9.2 and T is integral, Lemma 9.3 implies that there exist a non-empty open subset $T^\circ_0 \subset T$ and a real number V_0 such that $\mathrm{vol}(K_{Y_{\infty,t}} + \Phi_t) = V_0$ for any geometric point t of T°_0. By (13.7) and (13.12), $v \leq V_0$.

Fix $\epsilon \in (0, 1)$. Let $\rho \in \mathcal{R}$ with $\mathbf{L}_{(1-\epsilon)\Phi}(\rho) > 0$, and write $\rho = \sum_k \lambda_k \kappa_k$ in a cone of the complex of Y_∞ with rays κ_k, where $\lambda_k \in \mathbb{N}$ since the complex of Y_∞ is regular. By (13.5),

$$\sum_k \lambda_k (1 - (1 - \epsilon)\Phi(\kappa_k)) < 1.$$

Since $1-(1-\epsilon)\Phi(\kappa_k)\geq\epsilon$ for any k, we have $\sum_k\lambda_k<1/\epsilon$, so there exist only finitely many such ρ. Since $\sum_k\lambda_k\geq 1$, the last inequality implies $\sum_k\lambda_k\Phi(\kappa_k)>0$, so by (13.11)

$$\mathbf{L}_{(1-\epsilon)\Phi}(\rho)=1-\sum_k\lambda_k(1-\Phi(\kappa_k))-\epsilon\sum_k\lambda_k\Phi(\kappa_k)<\mathbf{L}_\Phi(\rho)\leq\mathbf{M}_\infty(\rho).$$

By (I2), for any $i\gg 0$, $\mathbf{L}_{(1-\epsilon)\Phi}(\rho)\leq\mathbf{M}_i(\rho)$ for the finitely many ρ above. Since $\mathbf{L}_{(1-\epsilon)\Phi_{t_i}}$ vanishes at the other prime divisors over Y_{∞,t_i}, $\mathbf{L}_{(1-\epsilon)\Phi_{t_i}}\leq\mathbf{M}_i$. For a smooth higher model U of W_i, Y_{i,t_i}, and Y_{∞,t_i}, Lemma 2.4(2) implies

$$\begin{aligned}H^0(Y_{\infty,t_i},\mathcal{O}(\lfloor m(K_{Y_{\infty,t_i}}+(1-\epsilon)\Phi_{t_i})\rfloor))&\subset H^0(U,\mathcal{O}_U(\lfloor m(K_U+(\mathbf{L}_{(1-\epsilon)\Phi_{t_i}})_U)\rfloor))\\&\subset H^0(U,\mathcal{O}_U(\lfloor m(K_U+(\mathbf{M}_i)_U)\rfloor))\end{aligned}$$

for any m, so $\mathrm{vol}(K_{Y_{\infty,t_i}}+(1-\epsilon)\Phi_{t_i})\leq v$ for any $i\gg 0$. By Lemma 9.3, there exist a non-empty open subset $T^\circ_\epsilon\subset T$ and a real number V_ϵ such that $\mathrm{vol}(K_{Y_{\infty,t}}+(1-\epsilon)\Phi_t)=V_\epsilon$ for any geometric point t of T°_ϵ. By (13.7), $V_\epsilon\leq v$. Let $\overline{\eta}$ be a geometric generic point of T. Then

$$V_\epsilon=\mathrm{vol}(K_{Y_{\infty,\overline{\eta}}}+(1-\epsilon)\Phi_{\overline{\eta}})\quad\text{for any }\epsilon\in[0,1),$$

and since $Y_{\infty,\overline{\eta}}$ is smooth, by [FKL16, Theorem 3.5(ii)] and the continuity of the volume on $N^1(Y_{\infty,\overline{\eta}})_{\mathbb{R}}$ [Cut15, Theorem 2.5], $V_0=\lim_{\epsilon\to 0}V_\epsilon\leq v$. Thus

$$\mathrm{vol}(K_{Y_{\infty,t}}+\Phi_t)=v\quad\text{for any geometric point }t\text{ of }T^\circ_0.\tag{13.13}$$

By (13.13), $\mathrm{vol}(K_{Y_{\infty,\overline{\eta}}}+\Phi_{\overline{\eta}})=v>0$. Thus Lemma 13.2 applied to $Y_\infty\to T$, Σ_{Y_∞}, and Φ implies that there exist T°, $T'\to T^\circ$, $\mathcal{X}'\to T'$, $\phi:Y_{\infty,T'}\dashrightarrow\mathcal{X}'$, the sheaves $\mathcal{H}_k$, and the numbers μ_k as there. By Lemma 13.2(2), $K_{\mathcal{X}'_{t'}}+\phi_{t'*}\Phi_{t'}$ is ample for any geometric point t' of T'. Let $\mathcal{G}\subset\mathcal{X}'$ be the closure of the union of the images under ϕ of the components of $\mathrm{Supp}\,\Phi_{T'}$ which are not contracted by ϕ, that is, $\mathcal{G}:=\mathrm{Supp}\,\phi_*\Phi_{T'}$. Possibly shrinking T°, we may assume that $\mathcal{G}_{t'}=\mathrm{Supp}\,\phi_{t'*}\Phi_{t'}$ for any geometric point t' of T' [Gro66, Proposition 9.5.3]. By (13.7), t_i is a point of $T^\circ\cap T^\circ_0$ for any $i\gg 0$, and since $T'\to T^\circ$ is finite and surjective, t_i lifts to a geometric point t'_i of T' with $Y_{\infty,T',t'_i}=Y_{\infty,t_i}$. Let $B'_i:=\phi_{t'_i*}\Phi_{t_i}$.

Step 5. In this step we show that $(X_i,B_i)\cong(\mathcal{X}'_{t'_i},B'_i)$ for any $i\gg 0$, which implies $(\star)$.

For a big $\mathbb{R}$-divisor D on a smooth projective variety U and a prime divisor S on U, let $\sigma_S(D)$ be the asymptotic order of D along S, let $N_\sigma(D):=\sum_S\sigma_S(D)S$, and let $P_\sigma(D):=D-N_\sigma(D)$ (see [FKL16, Section 4, (4.1)]). If $D'\leq D$ are big $\mathbb{R}$-divisors on U with $\mathrm{vol}(D')=\mathrm{vol}(D)$, then by [FKL16, Theorem A],

$$H^0(U,\mathcal{O}_U(\lfloor mD'\rfloor))=H^0(U,\mathcal{O}_U(\lfloor mD\rfloor))\quad\text{for any }m,$$

and by [FKL16, Lemma 4.1(i)], $N_\sigma(D')=N_\sigma(D)-(D-D')$, that is,

$$P_\sigma(D')=P_\sigma(D).\tag{13.14}$$

We claim that for a birational morphism $q:U\to X$ onto a normal projective variety X, an ample $\mathbb{R}$-Cartier $\mathbb{R}$-divisor L on X, and a q-exceptional $\mathbb{R}$-divisor $E\geq 0$ on U,

$$N_\sigma(q^*L+E)=E.\tag{13.15}$$

Let S be a prime divisor on U. By Lemma 2.4,

$$H^0(U,\mathcal{O}_U(\lfloor m(q^*L+E)\rfloor))=H^0(X,\mathcal{O}_X(\lfloor mL\rfloor))\quad\text{for any }m,$$

and any non-zero $s\in H^0(X,\mathcal{O}_X(\lfloor mL\rfloor))$ satisfies $\mathrm{div}(s)+mL\geq 0$, hence $\mathrm{div}(s)+mq^*L\geq 0$. Thus $\sigma_S(q^*L+E)\geq\mathrm{mult}_S E$. Conversely, sections of $\lfloor mq^*L\rfloor$ are sections of $\lfloor m(q^*L+E)\rfloor$, so $\sigma_S(q^*L+E)\leq\sigma_S(q^*L)+\mathrm{mult}_S E$. Since q^*L is nef and big, $\sigma_S(q^*L)=0$ [FKL16, Remark 4.6]. This proves (13.15).

Fix $i\gg 0$ as in **Step 4**, and write $t:=t_i$ and $t':=t'_i$. Let $\mathbf{M}^{(0)}_i$ be the $\boldsymbol{b}$-divisor $\mathbf{M}_i$ of **Step 2** and $\mathbf{M}_i$ the one after **Step 3**, so that $\mathbf{M}_i\leq\mathbf{M}^{(0)}_i$. Let U be a smooth projective variety

with birational morphisms to W_i, to $Y_{i,t}^{(0)}$, to the variety $Y_{i,t}$ of (I1), to $Y_{\infty,t}$, and to $\mathcal{X}'_{t'}$ [Bir16, Section 2.4], and let $q : U \to X_i$, $h : U \to Y_{\infty,t}$, and $g : U \to \mathcal{X}'_{t'}$ be the induced morphisms. Let

$$D_0 := K_U + (\mathbf{M}_{\Delta_i})_U, \qquad D_1 := K_U + (\mathbf{M}_{\Delta_i} \wedge \mathbf{L}_{\Phi_i^{(0)}})_U, \qquad D_2 := K_U + (\mathbf{M}_i^{(0)})_U,$$
$$D_3 := K_U + (\mathbf{M}_i)_U, \qquad D_4 := K_U + (\mathbf{M}_{\Phi_t})_U.$$

By (13.8), $D_1 \le D_0$ and $D_1 \le D_2$, and $D_3 \le D_2$. Since $\mathbf{M}_i$ and $\mathbf{M}_{\Phi_t}$ vanish at the prime divisors on $Y_{\infty,t}$ not contained in $\operatorname{Supp} \Sigma_{Y_{\infty,t}}$, (13.11) implies $D_3 \le D_4$. By (13.4), (13.3), (13.9), and (I1), for any m,

$$H^0(U, \mathcal{O}_U(\lfloor mD_k \rfloor)) = H^0(W_i, \mathcal{O}_{W_i}(\lfloor m(K_{W_i} + \Delta_i) \rfloor)) \quad \text{for } k \in \{0,1,2,3\},$$

so $\operatorname{vol}(D_k) = v$ for $k \in \{0,1,2,3\}$. By (13.4) and (13.13), $\operatorname{vol}(D_4) = \operatorname{vol}(K_{Y_{\infty,t}} + \Phi_t) = v$. By (13.14) applied to the pairs (D_1, D_0), (D_1, D_2), (D_3, D_2), and (D_3, D_4),

$$P_\sigma(D_0) = P_\sigma(D_4). \tag{13.16}$$

Let $K_{X_i} := q_* K_U$, $K_{Y_{\infty,t}} := h_* K_U$, and $K_{\mathcal{X}'_{t'}} := g_* K_U$. Since $(\mathbf{M}_{\Delta_i})_U = q_*^{-1} B_i + \operatorname{Exc}(q)$, where $\operatorname{Exc}(q)$ denotes the reduced divisor, and (X_i, B_i) is lc, $D_0 = q^*(K_{X_i} + B_i) + E_X$ with $E_X \ge 0$ q-exceptional. Since $(Y_{\infty,t}, \Phi_t)$ is lc, $D_4 = h^*(K_{Y_{\infty,t}} + \Phi_t) + E_h$ with $E_h \ge 0$ h-exceptional. By Lemma 13.2(3),

$$h^*(K_{Y_{\infty,t}} + \Phi_t) = g^*(K_{\mathcal{X}'_{t'}} + B'_i) + E$$

with $E \ge 0$ g-exceptional, and by Lemma 13.2(1), any h-exceptional prime divisor is g-exceptional. Thus $D_4 = g^*(K_{\mathcal{X}'_{t'}} + B'_i) + E_{X'}$ with $E_{X'} := E + E_h \ge 0$ g-exceptional. Since $K_{X_i} + B_i$ and $K_{\mathcal{X}'_{t'}} + B'_i$ are ample, (13.15) and (13.16) imply

$$q^*(K_{X_i} + B_i) = g^*(K_{\mathcal{X}'_{t'}} + B'_i). \tag{13.17}$$

By (13.17), a curve $C \subset U$ is contracted by q if and only if $q^*(K_{X_i} + B_i) \cdot C = 0$, if and only if C is contracted by g. Since $q_* \mathcal{O}_U = \mathcal{O}_{X_i}$, by the rigidity lemma there exists a morphism $\chi : X_i \to \mathcal{X}'_{t'}$ with $g = \chi \circ q$. If χ contracts a curve C_X, then a curve $C \subset U$ with $q(C) = C_X$ is contracted by g but not by q, which is impossible. Thus χ is finite and birational, hence an isomorphism since $\mathcal{X}'_{t'}$ is normal. Since $\chi^* K_{\mathcal{X}'_{t'}} = K_{X_i}$, applying q_* to (13.17) implies $K_{X_i} + B_i = \chi^*(K_{\mathcal{X}'_{t'}} + B'_i)$, so

$$B_i = \chi^* B'_i = \chi^* \phi_{t'*} \Phi_t.$$

Thus the coefficients of B_i belong to the finite set $R_0 := \{\Phi(\kappa) \mid \kappa \in \mathcal{R}_{Y_\infty}\}$, and χ maps $\operatorname{Supp} B_i$ onto $\mathcal{G}_{t'}$. Since $\mathcal{X}' \to T'$ is projective and T' is of finite type, there exists a positive integer n_0 such that

$$\operatorname{cpx}(\mathcal{X}'_{t'}, (\mathcal{G}_{t'})_{\mathrm{red}}) \le n_0 \quad \text{for any geometric point } t' \text{ of } T',$$

as one sees by taking a finite affine open cover of T', a relatively very ample invertible sheaf over each member, and finitely many locally closed subsets of T' over which $\mathcal{X}'$ and $\mathcal{G}$ are flat. Thus $\operatorname{cpx}(X_i, \operatorname{Supp} B_i) \le n_0$ for any $i \gg 0$, which proves $(\star)$.

Step 6. In this step we prove the theorem. Assume that the set of coefficients of B, for $(X, B) \in \mathcal{S}$, is infinite. Then there exist $(X_i, B_i) \in \mathcal{S}$ and pairwise distinct real numbers c_i such that c_i is a coefficient of B_i for any $i \in \mathbb{N}$, which contradicts $(\star)$. Thus there exists a finite set $I_0 \subset I$ containing the coefficients of B for any $(X, B) \in \mathcal{S}$. Similarly, if $\{\operatorname{cpx}(X, \operatorname{Supp} B) \mid (X, B) \in \mathcal{S}\}$ is unbounded, then there exist $(X_i, B_i) \in \mathcal{S}$ with $\operatorname{cpx}(X_i, \operatorname{Supp} B_i) > i$ for any i, which contradicts $(\star)$. Thus there exists a positive integer n_0 such that $\operatorname{cpx}(X, \operatorname{Supp} B) \le n_0$ for any $(X, B) \in \mathcal{S}$.

Let $(X, B) \in \mathcal{S}$ over F and write $B = \sum_{k=1}^N b_k B_k$, where $B_1, \dots, B_N$ are pairwise distinct prime divisors. By $\operatorname{cpx}(X, \operatorname{Supp} B) \le n_0$, there exists a closed embedding $X \subset \mathbb{P}_F^{n_0}$ with $\deg X \le n_0$ and $\deg \operatorname{Supp} B \le n_0$, so $N \le \sum_{k=1}^N \deg B_k = \deg \operatorname{Supp} B \le n_0$ and $\deg B_k \le n_0$ for any k. By [HK19, Theorem 1.8], there exists a positive integer β depending only on n_0 such that X and $B_1, \dots, B_N$ are cut out by homogeneous forms of degree β. Let S_β be the free $\mathbb{Z}[1/30]$-module of homogeneous forms of degree β in $n_0 + 1$ variables, and let G_N be the product of $N + 1$

copies of the disjoint union of the Grassmannians parametrizing the subbundles (locally direct summands) of S_β of each rank. Over G_N there exist the closed subschemes $\mathcal{X}_{G_N} \subset \mathbb{P}^{n_0}_{G_N}$ and $\mathcal{B}_{G_N,k} := \mathcal{X}_{G_N} \cap \mathcal{B}'_{G_N,k}$, $1 \leq k \leq N$, where $\mathcal{X}_{G_N}$ and $\mathcal{B}'_{G_N,k}$ are cut out by the universal subspaces, and there exists a geometric point t of G_N such that

$$(X, B_1, \dots, B_N) \cong (\mathcal{X}_{G_N,t}, \mathcal{B}_{G_N,1,t}, \dots, \mathcal{B}_{G_N,N,t}).$$

By generic flatness, G_N is a finite disjoint union of locally closed subschemes over each of which $\mathcal{X}_{G_N}$ and all $\mathcal{B}_{G_N,k}$ are flat. Replacing each of them by the image of $\mathcal{X}_{G_N}$, which is open since $\mathcal{X}_{G_N}$ is flat, we obtain finitely many projective flat morphisms. Attaching to each of them, each $N \leq n_0$, and each $(b_1, \dots, b_N) \in I_0^N$ the numbers $b_1, \dots, b_N$, we obtain finitely many families as in Definition 13.1, and any $(X, B) \in \mathcal{S}$ satisfies the condition of Definition 13.1 with one of them. Thus $\mathcal{S}$ is bounded over $\mathbb{Z}[1/30]$ with coefficients in I_0. □

Theorem 13.8 needs the log canonical divisor of any fiber to be a positive real combination of globally generated line bundles given on the family. Proposition 13.6 replaces the bounded families of Theorem 13.4 by such families.

Set-up 13.5. For $j = 1, \dots, n$, let $\mathcal{Y}_j \to T_j$, $\mathcal{D}_j$, $\mathcal{B}_{j,1}, \dots, \mathcal{B}_{j,n_j}$, $b_{j,1}, \dots, b_{j,n_j}$, $\mathcal{H}_{j,1}, \dots, \mathcal{H}_{j,m_j}$, and $c_{j,1}, \dots, c_{j,m_j}$ be as follows.

(a) $\mathcal{Y}_j \to T_j$ is a projective flat morphism onto a scheme T_j of finite type over $\mathbb{Z}[1/30]$ whose geometric fibers are disjoint unions of normal integral threefolds.
(b) $\mathcal{D}_j$ and $\mathcal{B}_{j,1}, \dots, \mathcal{B}_{j,n_j}$ are reduced closed subschemes of $\mathcal{Y}_j$ flat over T_j.
(c) $b_{j,1}, \dots, b_{j,n_j} \in (0, 1]$ are real numbers.
(d) $\mathcal{H}_{j,1}, \dots, \mathcal{H}_{j,m_j}$ are invertible sheaves on $\mathcal{Y}_j$ generated by global sections over T_j such that $\bigotimes_{k=1}^{m_j} \mathcal{H}_{j,k}$ is ample over T_j.
(e) $c_{j,1}, \dots, c_{j,m_j} > 0$ are real numbers.

For a geometric point t of T_j, let

$$\mathcal{B}_{j,t} := \sum_{i=1}^{n_j} b_{j,i} \mathcal{B}_{j,i,t}.$$

Assume that for any j and any geometric point t of T_j, the following hold.

(1) $\mathcal{D}_{j,t}$ is a reduced divisor on $\mathcal{Y}_{j,t}$, and $\mathcal{B}_{j,1,t}, \dots, \mathcal{B}_{j,n_j,t}$ are pairwise distinct prime divisors on $\mathcal{Y}_{j,t}$ not contained in $\mathcal{D}_{j,t}$.
(2) Any connected component of $(\mathcal{Y}_{j,t}, \mathcal{D}_{j,t} + \mathcal{B}_{j,t})$ is lc.
(3) We have

$$K_{\mathcal{Y}_{j,t}} + \mathcal{D}_{j,t} + \mathcal{B}_{j,t} \sim_{\mathbb{R}} \sum_{k=1}^{m_j} c_{j,k} H_{j,k,t},$$

where $H_{j,k,t}$ is a Cartier divisor on $\mathcal{Y}_{j,t}$ with

$$\mathcal{O}_{\mathcal{Y}_{j,t}}(H_{j,k,t}) \cong \mathcal{H}_{j,k}|_{\mathcal{Y}_{j,t}}.$$

Proposition 13.6. *Notation and conditions as in Set-up 13.3. Then there exist families as in Set-up 13.5 with $\mathcal{D}_1 = \cdots = \mathcal{D}_n = \emptyset$ such that for any $(X, B) \in \mathcal{S}$ there exist j, a geometric point $t : \operatorname{Spec} F \to T_j$, where F is the ground field of X, and an isomorphism of F-schemes $X \cong \mathcal{Y}_{j,t}$ identifying B with $\mathcal{B}_{j,t}$.*

Proof. By Theorem 13.4, there exist finitely many projective flat morphisms $\mathcal{X}_j \to Q_j$ onto schemes of finite type over $\mathbb{Z}[1/30]$, closed subschemes $\mathcal{B}_{j,1}, \dots, \mathcal{B}_{j,n_j}$ of $\mathcal{X}_j$ flat over Q_j, and numbers $b_{j,i} \in I_0$ as in Definition 13.1. We say that $(X, B) \in \mathcal{S}$ over F is *represented* at a (not necessarily closed) point $s \in Q_j$ via a geometric point $t : \operatorname{Spec} F \to Q_j$ if s is the image of t and there exists an isomorphism $X \cong \mathcal{X}_{j,t}$ as in Definition 13.1. Let $\Pi_j \subset Q_j$ be the set of points at which some element of $\mathcal{S}$ is represented. By Definition 13.1, any element of $\mathcal{S}$ is represented at a point of Π_j for some j. Discarding the indices j with $\Pi_j = \emptyset$, we may assume that $\Pi_j \neq \emptyset$ for

any j. Since $b_{j,i}$ is then the coefficient of a prime component of a boundary with coefficients in $I \subset [0,1]$, we have $b_{j,i} \in (0,1]$.

Claim 13.7. *Let j be an index and let $T \subset Q_j$ be an integral closed subscheme such that $\Pi_j \cap T$ is dense in T. Then there exist a non-empty open subset $T^\circ \subset T$, a finite surjective morphism $T' \to T^\circ$ from an integral scheme T', a projective flat morphism $\mathcal{X}' \to T'$, reduced closed subschemes $\mathcal{C}_1, \dots, \mathcal{C}_{n_j} \subset \mathcal{X}'$, invertible sheaves $\mathcal{H}_1, \dots, \mathcal{H}_m$ on $\mathcal{X}'$, and real numbers $\mu_1, \dots, \mu_m > 0$ satisfying the following.*

(1) The morphism $\mathcal{X}' \to T'$ with $\mathcal{D} = \emptyset$, the subschemes $\mathcal{C}_1, \dots, \mathcal{C}_{n_j}$, the numbers $b_{j,1}, \dots, b_{j,n_j}$, the sheaves $\mathcal{H}_1, \dots, \mathcal{H}_m$, and the numbers $\mu_1, \dots, \mu_m$ is a family as in Set-up 13.5.

(2) For any $(X, B) \in \mathcal{S}$ represented at a point of $\Pi_j \cap T^\circ$ via a geometric point $t : \operatorname{Spec} F \to Q_j$, there exist a geometric point $t' : \operatorname{Spec} F \to T'$ and an isomorphism of F-schemes $X \cong \mathcal{X}'_{t'}$ under which

$$B = \sum_{i=1}^{n_j} b_{j,i} \mathcal{C}_{i,t'}.$$

Proof. **Step 1.** In this step we show that, possibly shrinking T, for any geometric point t of T,

(1.a) $\mathcal{X}_{j,t}$ is a normal integral threefold, and

(1.b) $\mathcal{B}_{j,1,t}, \dots, \mathcal{B}_{j,n_j,t}$ are pairwise distinct prime divisors on $\mathcal{X}_{j,t}$.

Since $\mathcal{X}_{j,T} \to T$ and $\mathcal{B}_{j,i,T} \to T$ are proper and flat, by [Gro66, Théorème 12.2.4(iv)(viii)] the set of points $s \in T$ such that $\mathcal{X}_{j,s}$ is geometrically normal and geometrically integral and $\mathcal{B}_{j,1,s}, \dots, \mathcal{B}_{j,n_j,s}$ are geometrically integral is open. It contains $\Pi_j \cap T$, so we may assume that it is T. Since T is integral, the Hilbert polynomials of the fibers of $\mathcal{X}_{j,T}$ and of $\mathcal{B}_{j,i,T}$ over T are constant, so $\dim \mathcal{X}_{j,t} = 3$ and $\dim \mathcal{B}_{j,i,t} = 2$ for any t, as at the points of $\Pi_j \cap T$. For $1 \le i < i' \le n_j$, the set of $s \in T$ with $\dim(\mathcal{B}_{j,i} \cap \mathcal{B}_{j,i'})_s \ge 2$ is closed by [Gro66, Corollaire 13.1.5] and does not meet $\Pi_j \cap T$. Removing it, we obtain (1.b).

Step 2. In this step we construct a family as in Set-up 9.2 which resolves $\mathcal{X}_j$.

Let $\overline{k(T)}$ be an algebraic closure of the function field $k(T)$ of T, and let $\overline{\eta} : \operatorname{Spec} \overline{k(T)} \to T$ be the induced geometric generic point. Since Q_j is of finite type over $\mathbb{Z}[1/30]$, $\operatorname{char} k(T) = 0$ or $\operatorname{char} k(T) > 5$. Let $\overline{h} : \overline{Y} \to \mathcal{X}_{j,\overline{\eta}}$ be a projective log resolution of

$$\Big(\mathcal{X}_{j,\overline{\eta}}, \bigcup_{i=1}^{n_j} \mathcal{B}_{j,i,\overline{\eta}}\Big).$$

It exists by Hironaka's theorem if $\operatorname{char} k(T) = 0$ and by [Bir16, Section 2.4] if $\operatorname{char} k(T) > 5$. Let $\overline{\Sigma}_i := \overline{h}_*^{-1} \mathcal{B}_{j,i,\overline{\eta}}$ for $1 \le i \le n_j$, and let $\overline{\Sigma}_{n_j+1}, \dots, \overline{\Sigma}_N$ be the $\overline{h}$-exceptional prime divisors. Then $\sum_{i=1}^N \overline{\Sigma}_i$ is a simple normal crossing divisor, so for any non-empty subset $\Lambda \subset \{1, \dots, N\}$, the scheme $\bigcap_{i \in \Lambda} \overline{\Sigma}_i$ is empty or smooth of pure dimension $3 - |\Lambda|$, hence the disjoint union of finitely many integral connected components. Since $\mathcal{X}_{j,\overline{\eta}}$ is normal, by Zariski's main theorem $\overline{h}$ restricts to an isomorphism

$$\overline{Y} \setminus \bigcup_{i=n_j+1}^{N} \overline{\Sigma}_i \to \mathcal{X}_{j,\overline{\eta}} \setminus \overline{h}\Big(\bigcup_{i=n_j+1}^{N} \overline{\Sigma}_i\Big).$$

There exist a finite extension $k_1 \subset \overline{k(T)}$ of $k(T)$, a projective k_1-morphism $h_{k_1} : \mathcal{Y}'_{k_1} \to \mathcal{X}_{j,k_1}$, and prime divisors $\Sigma'_{1,k_1}, \dots, \Sigma'_{N,k_1}$ on $\mathcal{Y}'_{k_1}$ whose base changes to $\overline{k(T)}$ are $\overline{h}, \overline{\Sigma}_1, \dots, \overline{\Sigma}_N$, such that the connected components of the schemes $\bigcap_{i \in \Lambda} \Sigma'_{i,k_1}$ are geometrically integral. Possibly shrinking T to an affine open subset $\operatorname{Spec} R$ over which generators $\alpha_1, \dots, \alpha_r$ of k_1 over $k(T)$ are integral, $T_1 := \operatorname{Spec} R[\alpha_1, \dots, \alpha_r]$ is an integral scheme with function field k_1 and a finite surjective morphism $T_1 \to T$. By [Gro66, §8, Théorèmes 8.8.2, 8.10.5], generic flatness, [Gro66, Théorème 12.2.4(iii)(viii)], and [Gro66, Corollaire 13.1.5], possibly shrinking T, $\mathcal{Y}'_{k_1}$, h_{k_1}, and Σ'_{i,k_1} extend over T_1: there exist a projective morphism $\mathcal{Y}' \to T_1$, prime divisors $\Sigma'_1, \dots, \Sigma'_N$

on $\mathcal{Y}'$, and a projective T_1-morphism $h : \mathcal{Y}' \to \mathcal{X}_{j,T_1}$, whose generic fibers are $\mathcal{Y}'_{k_1}$, Σ'_{i,k_1}, h_{k_1}, satisfying the following.

(2.a) T_1, $\mathcal{Y}' \to T_1$, and $\Sigma' := \sum_{i=1}^N \Sigma'_i$ satisfy the conditions of Set-up 9.2.

(2.b) For any geometric point t of T_1, h_t restricts to an isomorphism

$$\mathcal{Y}'_t \setminus \bigcup_{i>n_j} \Sigma'_{i,t} \to \mathcal{X}_{j,t} \setminus h_t\Big(\bigcup_{i>n_j} \Sigma'_{i,t}\Big),$$

$h_t(\Sigma'_{i,t}) = \mathcal{B}_{j,i,t}$ for $i \le n_j$, and $\dim h_t(\Sigma'_{i,t}) \le 1$ for $i > n_j$.

By (2.b), for any geometric point t of T_1, the preimage of $h_t(\bigcup_{i>n_j} \Sigma'_{i,t})$ is $\bigcup_{i>n_j} \Sigma'_{i,t}$, so the h_t-exceptional prime divisors are exactly $\Sigma'_{n_j+1,t}, \dots, \Sigma'_{N,t}$, and

$$h_{t*}\Sigma'_{i,t} = \begin{cases} \mathcal{B}_{j,i,t} & \text{if } 1 \le i \le n_j, \\ 0 & \text{if } n_j < i \le N. \end{cases} \tag{13.18}$$

Step 3. In this step we show that the geometric generic fiber of $\mathcal{Y}'$ with the boundary

$$\mathcal{B}' := \sum_{i=1}^{n_j} b_{j,i}\Sigma'_i + \sum_{i=n_j+1}^{N} \Sigma'_i$$

has volume v. Since $b_{j,i} \in (0,1]$, we have $\mathcal{B}' \in [0,1]$.

Let $(X,B) \in \mathcal{S}$ be represented at a point of $\Pi_j \cap T$ via $t : \operatorname{Spec} F \to T$, let $t_1 : \operatorname{Spec} F \to T_1$ be a geometric point over t, let $Y := \mathcal{Y}'_{t_1}$, and let $L := K_X + B$. We regard h_{t_1} as a morphism $Y \to X$. By (13.18), $h_{t_1*}(K_Y + \mathcal{B}'_{t_1}) = L$, and for $i > n_j$ the coefficient of Σ'_{i,t_1} in $K_Y + \mathcal{B}'_{t_1} - h_{t_1}^* L$ is the log discrepancy $a(\Sigma'_{i,t_1}, X, B) \ge 0$, as (X,B) is lc. Thus

$$K_Y + \mathcal{B}'_{t_1} = h_{t_1}^* L + G, \quad G \ge 0 \text{ is } h_{t_1}\text{-exceptional}. \tag{13.19}$$

Since $G \ge 0$ is h_{t_1}-exceptional and L is ample, by (13.19),

$$\operatorname{vol}(K_Y + \mathcal{B}'_{t_1}) = \operatorname{vol}(h_{t_1}^* L + G) = \operatorname{vol}(L) = L^3 = v.$$

By (2.a), since T_1 is integral, Lemma 9.3 for $\mathcal{Y}' \to T_1$ and $\mathcal{B}'$ shows that $\operatorname{vol}(K_{\mathcal{Y}'_t} + \mathcal{B}'_t)$ is constant for t over a non-empty open subset of T_1. Possibly shrinking T, we may assume that it is constant for t over T_1. Since $\Pi_j \cap T \ne \emptyset$, there exists $(X,B) \in \mathcal{S}$ represented at a point of $\Pi_j \cap T$ via some $t : \operatorname{Spec} F \to T$, and there exists a geometric point $t_1 : \operatorname{Spec} F \to T_1$ over t, as $T_1 \to T$ is finite and surjective. Thus for a geometric generic point $\overline{\eta}_1$ of T_1,

$$\operatorname{vol}(K_{\mathcal{Y}'_{\overline{\eta}_1}} + \mathcal{B}'_{\overline{\eta}_1}) = v > 0.$$

Step 4. In this step we construct $\mathcal{X}' \to T'$, $\mathcal{H}_k$, μ_k, and show that $X \cong \mathcal{X}'_{t'}$ for any $(X,B) \in \mathcal{S}$ represented at a point of $\Pi_j \cap T$ via t and any geometric point t' of T' over t.

By **Step 3** and Lemma 13.2 for $\mathcal{Y}' \to T_1$ and $\mathcal{B}'$, there exist a non-empty open subset $T_1^\circ \subset T_1$, a finite surjective morphism $T' \to T_1^\circ$ from an integral scheme T', a projective flat morphism $\mathcal{X}' \to T'$ whose geometric fibers are normal integral threefolds, a birational map $\phi : \mathcal{Y}'_{T'} \dashrightarrow \mathcal{X}'$ over T', invertible sheaves $\mathcal{H}_1, \dots, \mathcal{H}_m$ on $\mathcal{X}'$ generated by global sections over T' such that $\bigotimes_{k=1}^m \mathcal{H}_k$ is ample over T', and real numbers $\mu_1, \dots, \mu_m > 0$, satisfying (1)–(3) of Lemma 13.2. Possibly shrinking T, we may assume that $T_1^\circ = T_1$. For a geometric point t of T', let $L'_t := K_{\mathcal{X}'_t} + \phi_{t*}\mathcal{B}'_t$, which is ample by (2) of Lemma 13.2.

Let $(X,B) \in \mathcal{S}$ be represented at a point of $\Pi_j \cap T$ via $t : \operatorname{Spec} F \to T$, let $t' : \operatorname{Spec} F \to T'$ be a geometric point over t, and let t_1 be the image of t' in T_1. Let $Y := \mathcal{Y}'_{t'} = \mathcal{Y}'_{t_1}$, let L and G be as in **Step 3**, and let $X' := \mathcal{X}'_{t'}$ and $L' := L'_{t'}$. Since $\operatorname{char} F > 5$, there exists a resolution W of the closure of the graph of $\phi_{t'}$ in $Y \times X'$ [Bir16, Section 2.4]. Let $f : W \to Y$ and $g : W \to X'$ be the induced birational morphisms and let $f_X := h_{t_1} \circ f : W \to X$. By (3) of Lemma 13.2,

$f^*(K_Y + \mathcal{B}'_{t'}) = g^*L' + E$ with $E \geq 0$ and g-exceptional. Let $G_W := f^*G$. Then $G_W \geq 0$ and G_W is f_X-exceptional, as G is h_{t_1}-exceptional. By (13.19),

$$f_X^*L + G_W = g^*L' + E. \tag{13.20}$$

Since $-(E - G_W) = g^*L' - f_X^*L \equiv_{f_X} g^*L'$ is f_X-nef and $f_{X*}(E - G_W) = f_{X*}E \geq 0$, the negativity lemma implies $E \geq G_W$. Since $-(G_W - E) \equiv_g f_X^*L$ is g-nef and $g_*(G_W - E) = g_*G_W \geq 0$, the negativity lemma implies $G_W \geq E$. Thus $E = G_W$ and

$$f_X^*L = g^*L'.$$

As L and L' are ample, for a curve $C \subset W$, $f_X(C)$ is a point if and only if $f_X^*L \cdot C = 0$, and $g(C)$ is a point if and only if $g^*L' \cdot C = 0$. Thus $f_X(C)$ is a point if and only if $g(C)$ is a point. Since X and X' are normal, $f_{X*}\mathcal{O}_W = \mathcal{O}_X$ and $g_*\mathcal{O}_W = \mathcal{O}_{X'}$, so by the rigidity lemma there exist morphisms $\iota : X \to X'$ and $\iota' : X' \to X$ with $g = \iota \circ f_X$ and $f_X = \iota' \circ g$. Since f_X and g are surjective, ι is an isomorphism with inverse ι'. Since $\phi_{t'} \circ f = g = \iota \circ h_{t_1} \circ f$, we have $\phi_{t'} = \iota \circ h_{t_1}$ as birational maps, hence $\phi_{t'*} = \iota_* \circ h_{t_1*}$ on divisors. By (13.18),

$$\phi_{t'*}\Sigma'_{i,t'} = \begin{cases} \iota_*\mathcal{B}_{j,i,t} & \text{if } 1 \leq i \leq n_j, \\ 0 & \text{if } n_j < i \leq N, \end{cases} \qquad \iota_*B = \phi_{t'*}\mathcal{B}'_{t'}. \tag{13.21}$$

Step 5. In this step we construct $\mathcal{C}_1, \dots, \mathcal{C}_{n_j}$ and finish the proof of the claim.

For $1 \leq i \leq N$, by (2.a) and Set-up 9.2(4) with $\Lambda = \{i\}$, Σ'_i is smooth over T_1, and its geometric fibers are integral as $\overline{\Sigma}_i$ is integral. Thus $\Sigma'_{i,T'}$ is flat over T' with geometrically integral fibers, hence irreducible. Let ξ_i be the generic point of $\Sigma'_{i,T'}$. Since ξ_i maps to the generic point of T', the local ring of $\mathcal{Y}'_{T'}$ at ξ_i is the local ring of the smooth $k(T')$-scheme $\mathcal{Y}'_{k(T')}$ at the generic point of the prime divisor $\Sigma'_{i,k(T')}$, which is a discrete valuation ring. As $\mathcal{X}'$ is proper over T', ϕ is defined at ξ_i. Let $\mathcal{C}_i \subset \mathcal{X}'$ be the closure of $\phi(\xi_i)$ with the reduced structure, and let $d_i := \dim \mathcal{C}_{i,k(T')}$. Since ϕ is birational and the local ring at ξ_i is not a field, $d_i \leq 2$. Since $\Sigma'_{i,k(T')}$ is geometrically integral and dominates $\mathcal{C}_{i,k(T')}$, $\mathcal{C}_{i,k(T')}$ is geometrically integral.

Let $U \subset \mathcal{Y}'_{T'}$ be an open subset containing $\xi_1, \dots, \xi_N$ on which ϕ is a morphism. By Chevalley's theorem, $\phi(U \cap \Sigma'_{i,T'})$ contains a dense open subset $\mathcal{W}_i$ of $\mathcal{C}_i$. By generic flatness, [Gro66, Théorème 12.2.4(viii)], [Gro66, Corollaire 13.1.5], and (1) of Lemma 13.2, possibly shrinking T, we may assume that for any i and any geometric point t of T',

(5.a) $\mathcal{C}_i$ is flat over T', $\dim \mathcal{C}_{i,t} = d_i$, and $\mathcal{C}_{i,t}$ is integral if $d_i = 2$, and
(5.b) $\dim(\mathcal{C}_i \backslash \mathcal{W}_i)_t < d_i$, $\dim(\Sigma'_{i,T'} \backslash U)_t \leq 1$, and ϕ_t is the birational map defined by $\phi|_{U_t}$.

By (5.b), the generic point of $\Sigma'_{i,t}$ belongs to U_t, and $\phi_t(U_t \cap \Sigma'_{i,t}) \subset \mathcal{C}_{i,t}$ contains $\mathcal{W}_{i,t}$, which is a non-empty open subset of $\mathcal{C}_{i,t}$ if $d_i = 2$. By (5.a), for any geometric point t of T',

$$\phi_{t*}\Sigma'_{i,t} = \begin{cases} \mathcal{C}_{i,t} & \text{if } d_i = 2, \\ 0 & \text{if } d_i \leq 1. \end{cases} \tag{13.22}$$

As $T' \to T$ is finite and surjective, there exists t' as in **Step 4**. By (13.22) at t' and (13.21), for $i \leq n_j$ we have $d_i = 2$ and $\mathcal{C}_{i,t'} = \iota(\mathcal{B}_{j,i,t})$, and for $i > n_j$ we have $d_i \leq 1$. In particular, $\mathcal{C}_{1,t'}, \dots, \mathcal{C}_{n_j,t'}$ are pairwise distinct. For $1 \leq i < i' \leq n_j$, the set of $s \in T'$ with $\dim(\mathcal{C}_i \cap \mathcal{C}_{i'})_s \geq 2$ is closed by [Gro66, Corollaire 13.1.5], and it contains no point over $\Pi_j \cap T$, since $T' \to T$ is finite, so any point of T' over the image of t is the image of some t' as in **Step 4**. Since the image of the set in T is closed and does not meet $\Pi_j \cap T$, possibly shrinking T, we may assume that the set is empty. Thus for any geometric point t of T',

(5.c) $\mathcal{C}_{1,t}, \dots, \mathcal{C}_{n_j,t}$ are pairwise distinct prime divisors on $\mathcal{X}'_t$, and

$$\phi_{t*}\mathcal{B}'_t = \sum_{i=1}^{n_j} b_{j,i}\mathcal{C}_{i,t}.$$

Let $T^\circ \subset T$ be the non-empty open subset obtained by the shrinkings above. We verify (1). The scheme T' is of finite type over $\mathbb{Z}[1/30]$, as T' is finite over $T^\circ \subset Q_j$. The morphism $\mathcal{X}' \to T'$ and the sheaves $\mathcal{H}_k$ are as in Set-up 13.5 by Lemma 13.2, $\mathcal{C}_i$ is reduced and flat over T' by (5.a), and $b_{j,i} \in (0,1]$. Condition (1) of Set-up 13.5 holds by (5.c), and conditions (2) and (3) hold by (5.c) and (2) of Lemma 13.2. We verify (2). For (X,B) represented at a point of $\Pi_j \cap T^\circ$ via t, there exists t' over t, as $T' \to T^\circ$ is finite and surjective, and by (13.21) and (5.c), $\iota : X \to \mathcal{X}'_{t'}$ is an isomorphism under which $B = \sum_{i=1}^{n_j} b_{j,i}\mathcal{C}_{i,t'}$. The claim follows. □

Fix j. We show by Noetherian induction on the closed subset $V \subset Q_j$ that there exist finitely many families as in Set-up 13.5 with $\mathcal{D} = \emptyset$ such that for any $(X,B) \in \mathcal{S}$ over F represented at a point of $\Pi_j \cap V$, there exist a family $\mathcal{Y} \to T''$ among them, a geometric point $t'' : \operatorname{Spec} F \to T''$, and an isomorphism of F-schemes $X \cong \mathcal{Y}_{t''}$ identifying B with $\mathcal{B}_{t''}$. Let V_0 be the closure of $\Pi_j \cap V$. We may assume that $V_0 \neq \emptyset$. Let T be an irreducible component of V_0 with the reduced structure. Since $\Pi_j \cap V_0$ is dense in V_0 and T minus the other irreducible components of V_0 is a non-empty open subset of V_0, $\Pi_j \cap T$ is dense in T. By Claim 13.7, there exist a non-empty open subset $T^\circ \subset T$ and a family $\mathcal{Y} \to T''$ as in Set-up 13.5 with $\mathcal{D} = \emptyset$ such that for any $(X,B) \in \mathcal{S}$ over F represented at a point of $\Pi_j \cap T^\circ$, there exist a geometric point $t'' : \operatorname{Spec} F \to T''$ and an isomorphism of F-schemes $X \cong \mathcal{Y}_{t''}$ identifying B with $\mathcal{B}_{t''}$. Possibly shrinking T°, we may assume that T° does not meet the other irreducible components of V_0, so that T° is open in V_0. Then the closed subset $V_1 := V_0 \backslash T^\circ$ is properly contained in $V_0 \subset V$, and $\Pi_j \cap V \subset T^\circ \cup V_1$. The induction hypothesis for V_1 finishes the induction step.

Taking $V = Q_j$ for each of the finitely many indices j, the proposition follows. □

Theorem 13.8. *Notation and conditions as in Set-up 13.5. Let $\mathcal{S}$ be the set of projective slc pairs (X,B) (see Definition 12.1) over algebraically closed fields F of characteristic > 5 such that, for the normalization $\nu : Y \to X$ and the reduced conductor divisor D_Y on Y, the following hold.*

(i) $K_X + B$ is ample.
(ii) There exist $j \in \{1,\dots,n\}$, a geometric point $t : \operatorname{Spec} F \to T_j$, and an isomorphism of F-schemes $Y \cong \mathcal{Y}_{j,t}$ under which $D_Y = \mathcal{D}_{j,t}$ and $\nu_*^{-1}B = \mathcal{B}_{j,t}$.

Then the following hold.

(1) $\mathcal{S}$ is bounded over $\mathbb{Z}[1/30]$ with coefficients in $\{b_{j,i} \mid 1 \le j \le n,\ 1 \le i \le n_j\}$ (see Definition 13.1).
(2) If $b_{j,i} \in \mathbb{Q}$ for any j and i, then there exists a positive integer N such that $N(K_X + B)$ is a very ample Cartier divisor for any $(X,B) \in \mathcal{S}$.

Proof. Since T_j is a scheme over $\mathbb{Z}[1/30]$, the characteristic of F is 0 or a prime > 5 for any geometric point $t : \operatorname{Spec} F \to T_j$.

For $j \in \{1,\dots,n\}$ we let $\boldsymbol{b}_j := (b_{j,1},\dots,b_{j,n_j}) \in (0,1]^{n_j}$. For a geometric point $t : \operatorname{Spec} F \to T_j$ we use the following notation. We let $Y := \mathcal{Y}_{j,t}$, $D_Y := \mathcal{D}_{j,t}$, $B_{Y,i} := \mathcal{B}_{j,i,t}$, and

$$B_Y(x) := \sum_{i=1}^{n_j} x_i B_{Y,i}, \qquad L_Y(x) := K_Y + D_Y + B_Y(x) \qquad \text{for } x = (x_1,\dots,x_{n_j}) \in \mathbb{R}^{n_j},$$

so that $B_Y(\boldsymbol{b}_j) = \mathcal{B}_{j,t}$. We let $\iota : D_Y^n \to Y$ be the composite of the normalization $D_Y^n \to D_Y$ and the inclusion $D_Y \subset Y$, and $D_{Y,\alpha}^n$ the connected components of D_Y^n. Let

$$V_Y := \{x \in \mathbb{R}^{n_j} \mid L_Y(x) \text{ is } \mathbb{R}\text{-Cartier}\}.$$

For $x \in V_Y$ we let $\operatorname{Diff}(x) := \operatorname{Diff}_{D_Y^n}(B_Y(x))$ be the different [Kol13, Definition 4.2], extended linearly to $\mathbb{R}$-divisors, so that $\iota^* L_Y(x) \sim_{\mathbb{R}} K_{D_Y^n} + \operatorname{Diff}(x)$. When $(X,B) \in \mathcal{S}$ and t is as in (ii), this Y is the normalization of X (see **Step 3**).

Step 1. In this step we prove the following. Let X be a reduced scheme of finite type over an algebraically closed field of characteristic $\neq 2$ which is S_2 and nodal in codimension 1, $\nu : Y \to X$

the normalization, $D_X \subset X$ and $D_Y \subset Y$ the conductors, and $\iota : D_Y^n \to Y$ the composite of the normalization of D_Y and the inclusion. By [Kol13, 5.2], D_X and D_Y are reduced of pure codimension 1, and, for the normalization D_X^n of D_X, the induced morphism $D_Y^n \to D_X^n$ is the quotient by an involution τ of D_Y^n [Pos24, Lemma 2.3.5]. We call τ the *involution of D_Y^n associated with X*. Then

$$\mathcal{O}_X = \operatorname{Ker}\left(\delta : \nu_*\mathcal{O}_Y \to (\nu\circ\iota)_*\mathcal{O}_{D_Y^n}\right), \qquad \delta(u) := \iota^*u - \tau^*\iota^*u. \tag{13.23}$$

Since $\nu\circ\iota\circ\tau = \nu\circ\iota$, we have $\mathcal{O}_X \subset \operatorname{Ker}\delta$. Conversely, let $U \subset X$ be an open subset and $u \in (\operatorname{Ker}\delta)(U)$. Let $x \in U$ with $\dim \mathcal{O}_{X,x} \le 1$. If $x \notin D_X$, then ν is an isomorphism over a neighborhood of x, and $u_x \in \mathcal{O}_{X,x}$. If $x \in D_X$, then x is a node. Let $\mathfrak{c} \subset \mathcal{O}_{X,x}$ be the conductor ideal and $R := (\nu_*\mathcal{O}_Y)_x$. Since D_X and D_Y are reduced, $\mathfrak{c}$ is a radical ideal of both $\mathcal{O}_{X,x}$ and R. Since $\dim \mathcal{O}_{X,x} = 1$ and $\mathfrak{c}$ contains a non-zero-divisor, $\mathfrak{c} = \mathfrak{m}_x$, so

$$\mathcal{O}_{X,x}/\mathfrak{c} = k(x), \qquad R/\mathfrak{c} = \left((\nu\circ\iota)_*\mathcal{O}_{D_Y^n}\right)_x,$$

and $R/\mathfrak{c}$ is a $k(x)$-algebra of dimension 2 on which τ acts as the non-trivial $k(x)$-automorphism. As the characteristic is $\neq 2$, the τ-invariant part of $R/\mathfrak{c}$ is $k(x)$. Since $\delta(u) = 0$, the image of u_x in $R/\mathfrak{c}$ is τ-invariant, hence equal to the image of some $w \in \mathcal{O}_{X,x}$, and $u_x - w \in \mathfrak{c} \subset \mathcal{O}_{X,x}$. Thus $u_x \in \mathcal{O}_{X,x}$ for any $x \in U$ with $\dim \mathcal{O}_{X,x} \le 1$, so the support of the image of u in the coherent sheaf $\nu_*\mathcal{O}_Y/\mathcal{O}_X$ is a closed subset of U of codimension ≥ 2. Let $U_0 \subset U$ be its complement. Since X is S_2, $u|_{U_0}$ extends to some $u' \in \mathcal{O}_X(U)$ [Har94, Proposition 1.11]. Since Y is reduced and $\nu^{-1}(U_0)$ is dense in $\nu^{-1}(U)$, $u = u' \in \mathcal{O}_X(U)$.

Step 2. In this step we fix j and a geometric point $t : \operatorname{Spec} F \to T_j$. We show that V_Y is an affine subspace of $\mathbb{R}^{n_j}$ defined over $\mathbb{Q}$ containing $\boldsymbol{b}_j$, that the coefficient of $\operatorname{Diff}(x)$ along any prime divisor of D_Y^n is an affine function of $x \in V_Y$ defined over $\mathbb{Q}$, and the following. Let $V \subset V_Y$ be an affine subspace defined over $\mathbb{Q}$ with $\boldsymbol{b}_j \in V$, and let V° be an open neighborhood of $\boldsymbol{b}_j$ in V. Then there exist $\boldsymbol{b}^{(0)}, \dots, \boldsymbol{b}^{(d)} \in V^\circ \cap \mathbb{Q}^{n_j}$ and real numbers $\mu_0, \dots, \mu_d > 0$ satisfying the following:

(a) $\sum_{v=0}^d \mu_v = 1$ and $\boldsymbol{b}_j = \sum_{v=0}^d \mu_v \boldsymbol{b}^{(v)}$.
(b) For any v, $\boldsymbol{b}^{(v)} \in (0,1]^{n_j}$, $L_Y(\boldsymbol{b}^{(v)})$ is $\mathbb{Q}$-Cartier, and any connected component of $(Y, D_Y + B_Y(\boldsymbol{b}^{(v)}))$ is lc.

Since $L_Y(\boldsymbol{b}_j)$ is $\mathbb{R}$-Cartier by Set-up 13.5(3), Lemma 2.6 shows that V_Y is an affine subspace defined over $\mathbb{Q}$ and that $L_Y(x)$ is $\mathbb{Q}$-Cartier for any $x \in V_Y \cap \mathbb{Q}^{n_j}$. Let $f : Y' \to Y$ be a log resolution of $(Y, D_Y + \sum_i B_{Y,i})$, which exists by [Bir16, Section 2.4] if $\operatorname{char} F > 5$, and by Hironaka's theorem if $\operatorname{char} F = 0$. For $x \in V_Y$ we define $B_{Y'}(x)$ by

$$K_{Y'} + B_{Y'}(x) = f^*L_Y(x), \qquad f_*B_{Y'}(x) = D_Y + B_Y(x).$$

Since V_Y is the affine span of $V_Y \cap \mathbb{Q}^{n_j}$ and f^* is linear, the coefficient of $B_{Y'}(x)$ along any prime divisor of Y' is an affine function of $x \in V_Y$ defined over $\mathbb{Q}$. Since Y' is smooth and $\operatorname{Supp} B_{Y'}(x)$ is contained in the simple normal crossing divisor $\operatorname{Exc}(f) \cup f_*^{-1}(D_Y + \sum_i B_{Y,i})$, for $x \in V_Y \cap [0,1]^{n_j}$ any connected component of $(Y, D_Y + B_Y(x))$ is lc if and only if $\operatorname{mult}_P B_{Y'}(x) \le 1$ for any prime divisor P on Y'. For any α let $E_\alpha := f_*^{-1}\iota(D_{Y,\alpha}^n)$. Since E_α is smooth, $f|_{E_\alpha}$ factors through a birational morphism $g_\alpha : E_\alpha \to D_{Y,\alpha}^n$. Adjunction on Y' and $g_{\alpha*}K_{E_\alpha} = K_{D_{Y,\alpha}^n}$ imply

$$\operatorname{Diff}(x)|_{D_{Y,\alpha}^n} = g_{\alpha*}\left((B_{Y'}(x) - E_\alpha)|_{E_\alpha}\right) \quad \text{for any } \alpha \text{ and any } x \in V_Y, \tag{13.24}$$

so the coefficients of $\operatorname{Diff}(x)$ are affine functions of $x \in V_Y$ defined over $\mathbb{Q}$.

Let $\Pi \subset V$ be the set of $x \in V$ with $0 \le x_i \le 1$ for any i and $\operatorname{mult}_P B_{Y'}(x) \le 1$ for any component P of $\operatorname{Exc}(f) \cup f_*^{-1}(D_Y + \sum_i B_{Y,i})$. Then Π is a polyhedron defined by finitely many affine inequalities over $\mathbb{Q}$, and $\boldsymbol{b}_j \in \Pi$ by Set-up 13.5(2). Let Φ be the face of Π whose relative interior contains $\boldsymbol{b}_j$, and $d := \dim \Phi$. Since Φ is defined over $\mathbb{Q}$, the rational points are dense in the affine span $\langle\Phi\rangle$ of Φ. Since $b_{j,i} > 0$ for any i, there exists an open neighborhood $\Phi^\circ \subset V^\circ \cap \Phi \cap (0,1]^{n_j}$ of $\boldsymbol{b}_j$ in $\langle\Phi\rangle$. We choose $\boldsymbol{b}^{(0)}, \dots, \boldsymbol{b}^{(d)} \in \Phi^\circ \cap \mathbb{Q}^{n_j}$ spanning a d-simplex in

$\langle\Phi\rangle$ whose relative interior contains $\boldsymbol{b}_j$, and we let $\mu_0, \dots, \mu_d$ be the barycentric coordinates of $\boldsymbol{b}_j$. Then $\mu_v > 0$ for any v, (a) holds, and (b) holds since $\boldsymbol{b}^{(v)} \in \Pi \cap \mathbb{Q}^{n_j}$.

Step 3. In this step we let $(X, B) \in \mathcal{S}$ be over F, let $\nu : Y \to X$ be the normalization and D_Y the reduced conductor divisor on Y, and fix j, a geometric point $t : \operatorname{Spec} F \to T_j$, and an isomorphism of F-schemes $h : Y \to \mathcal{Y}_{j,t}$ with $h(D_Y) = \mathcal{D}_{j,t}$ and $h_*\nu_*^{-1}B = \mathcal{B}_{j,t}$, which exist by condition (ii) on $\mathcal{S}$. We identify Y with $\mathcal{Y}_{j,t}$ via h. Let τ be the involution of D_Y^n associated with X (**Step 1**). We show the following:

(a) $\tau^* \operatorname{Diff}(\boldsymbol{b}_j) = \operatorname{Diff}(\boldsymbol{b}_j)$, and the fixed locus $\operatorname{Fix}(\tau) \subset D_Y^n$ of τ satisfies $\dim \operatorname{Fix}(\tau) \leq 1$.
(b) For any morphism $q : Y \to Z'$ to a separated F-scheme Z' with $q \circ \iota = q \circ \iota \circ \tau$, there exists a morphism $\psi : X \to Z'$ with $q = \psi \circ \nu$.

Since ν is an isomorphism over $X \backslash D_X$ and $B_{Y,i} \not\subset D_Y$ for any i, the images $B_i := \nu(B_{Y,i})$ are pairwise distinct prime divisors on X not contained in D_X, and $B = \sum_i b_{j,i} B_i$. For $x \in \mathbb{R}^{n_j}$ let $B(x) := \sum_i x_i B_i$ and

$$V_X := \{x \in \mathbb{R}^{n_j} \mid K_X + B(x) \text{ is } \mathbb{R}\text{-Cartier}\} \ni \boldsymbol{b}_j.$$

Since X is S_2 and nodal in codimension 1 (see Definition 12.1), Lemma 2.6 shows that V_X is an affine subspace defined over $\mathbb{Q}$ and that $K_X + B(x)$ is $\mathbb{Q}$-Cartier for any $x \in V_X \cap \mathbb{Q}^{n_j}$. For $x \in V_X$ we have $\nu^*(K_X + B(x)) = L_Y(x)$ (see Definition 12.1), so $V_X \subset V_Y$. The set V_X° of $x \in V_X$ with $K_X + B(x)$ ample is an open neighborhood of $\boldsymbol{b}_j$ in V_X [Laz04, Example 1.3.14]. Let $\boldsymbol{b}^{(v)}$ and μ_v be as in **Step 2** for $V = V_X$ and $V^\circ = V_X^\circ$. For any v, $(X, B(\boldsymbol{b}^{(v)}))$ is an slc pair with $\mathbb{Q}$-boundary and ample $\mathbb{Q}$-Cartier $K_X + B(\boldsymbol{b}^{(v)})$, and $\operatorname{char} F > 5$. By [Pos24, Lemma 2.3.5], $\tau^* \operatorname{Diff}(\boldsymbol{b}^{(v)}) = \operatorname{Diff}(\boldsymbol{b}^{(v)})$ for any v. Since $\operatorname{Diff}(x)$ is affine in x (**Step 2**) and $\boldsymbol{b}_j = \sum_v \mu_v \boldsymbol{b}^{(v)}$, the first part of (a) holds. By [Pos24, Theorem 5.3.6], τ is generically fixed point free, so $\dim \operatorname{Fix}(\tau) \leq 1$. By [Pos24, Theorem 5.3.6, Proposition 3.4.1], ν is the categorical quotient of Y by the finite equivalence relation generated by $(\iota, \iota \circ \tau) : D_Y^n \to Y \times Y$. Since $q \circ \iota = q \circ \iota \circ \tau$, the morphism q coequalizes the projections to Y of the finite equivalence relation generated by $(\iota, \iota \circ \tau)$, so (b) follows from the universal property of the categorical quotient.

Step 4. In this step we fix j, a geometric point $t : \operatorname{Spec} F \to T_j$, and an involution τ of D_Y^n with $\dim \operatorname{Fix}(\tau) \leq 1$ and $\tau^* \operatorname{Diff}(\boldsymbol{b}_j) = \operatorname{Diff}(\boldsymbol{b}_j)$. We show that there exists a finite surjective morphism $q : Y \to Z$ onto a projective F-scheme Z satisfying the following:

(a) Z is reduced, S_2 and nodal in codimension 1, q is the normalization of Z, D_Y is the conductor in Y, and τ is the involution of D_Y^n associated with Z.
(b) Let $B_Z := q_* B_Y(\boldsymbol{b}_j)$. Then (Z, B_Z) is an slc pair and $K_Z + B_Z$ is ample.

By **Step 2**, the set

$$V := \{x \in V_Y \mid \tau^* \operatorname{Diff}(x) = \operatorname{Diff}(x)\}$$

is cut out in V_Y by finitely many affine equations over $\mathbb{Q}$, one for each prime divisor P' in the union of the supports of the $\operatorname{Diff}(x)$, $x \in V_Y$, and their images under τ. Thus V is an affine subspace defined over $\mathbb{Q}$, and $\boldsymbol{b}_j \in V$. By Set-up 13.5(3), $L_Y(\boldsymbol{b}_j) \sim_{\mathbb{R}} \sum_k c_{j,k} H_{j,k,t}$, which is ample since $\bigotimes_k \mathcal{H}_{j,k}$ is ample over T_j and any $\mathcal{H}_{j,k}$ is generated by global sections. Thus the set V° of $x \in V$ with $L_Y(x)$ ample is an open neighborhood of $\boldsymbol{b}_j$ in V [Laz04, Example 1.3.14]. Let $\boldsymbol{b}^{(v)}$ and μ_v be as in **Step 2**. For any v, any connected component of $(Y, D_Y + B_Y(\boldsymbol{b}^{(v)}))$ is lc, $L_Y(\boldsymbol{b}^{(v)})$ is $\mathbb{Q}$-Cartier and ample, and τ is a generically fixed point free involution of D_Y^n with $\tau^* \operatorname{Diff}(\boldsymbol{b}^{(v)}) = \operatorname{Diff}(\boldsymbol{b}^{(v)})$. By [Pos24, Theorem 5.3.6, Propositions 5.3.2, 5.3.5 and 3.4.1] if $\operatorname{char} F > 5$, and by [Kol16, Theorem 23 and paragraph 22] if $\operatorname{char} F = 0$, the quotient $q : Y \to Z$ of Y by the finite equivalence relation generated by $(\iota, \iota \circ \tau)$ exists, Z is projective and satisfies (a), and $K_Z + q_* B_Y(\boldsymbol{b}^{(v)})$ is $\mathbb{Q}$-Cartier and ample. The equivalence relation generated by $(\iota, \iota \circ \tau)$ depends only on (Y, D_Y, τ), so q does not depend on v. Since

$$K_Z + B_Z = \sum_{v=0}^{d} \mu_v \big(K_Z + q_* B_Y(\boldsymbol{b}^{(v)})\big),$$

$K_Z + B_Z$ is $\mathbb{R}$-Cartier and ample. Since $q^*(K_Z + B_Z) = L_Y(\boldsymbol{b}_j)$ and any connected component of $(Y, D_Y + B_Y(\boldsymbol{b}_j))$ is lc by Set-up 13.5(2), (b) holds.

Step 5. In this step we fix j and show that there exist finitely many morphisms $\phi : \widetilde{T} \to T_j$ from integral schemes $\widetilde{T}$ of finite type over $\mathbb{Z}[1/30]$, and for each of them a scheme $\mathcal{D}_j^n$ projective and flat over $\widetilde{T}$ with a finite morphism

$$\iota : \mathcal{D}_j^n \to \mathcal{Y}_j \times_{T_j} \widetilde{T},$$

satisfying the following:

(a) For any geometric point $\widetilde{t}$ of $\widetilde{T}$, $\iota_{\widetilde{t}}$ is the composite of the normalization of $\mathcal{D}_{j,\phi\circ\widetilde{t}}$ and the inclusion into $\mathcal{Y}_{j,\phi\circ\widetilde{t}}$.

(b) Any geometric point of T_j is $\phi \circ \widetilde{t}$ for some ϕ and some geometric point $\widetilde{t}$ of $\widetilde{T}$.

By Noetherian induction on the closed subsets of T_j, it suffices to find, for any integral closed subscheme $\Lambda \subset T_j$, a dense open subset $\Lambda^\circ \subset \Lambda$ and a finite surjective morphism $\widetilde{T} \to \Lambda^\circ$ from an integral scheme $\widetilde{T}$ with (a).

Let η be the generic point of Λ and $\overline{\eta}$ a geometric point over η. There exist a finite extension k_1 of $k(\eta)$ and a finite morphism

$$\iota_{k_1} : D'_{k_1} \to \mathcal{D}_j \times_{T_j} \operatorname{Spec} k_1$$

whose base change to $\overline{\eta}$ is the normalization of $\mathcal{D}_{j,\overline{\eta}}$ [Gro66, Théorèmes 8.8.2, 8.10.5]. Let $\widetilde{T}_0$ be the normalization of Λ in k_1, which is finite over Λ since Λ is excellent. By [Gro66, Théorèmes 8.8.2, 8.10.5], there exists a dense open subset $\widetilde{T} \subset \widetilde{T}_0$ such that ι_{k_1} extends to a finite morphism

$$\iota : \mathcal{D}_j^n \to \mathcal{Y}_j \times_{T_j} \widetilde{T}$$

that is an isomorphism over an open subset $\mathcal{D}^\circ \subset \mathcal{D}_j \times_{T_j} \widetilde{T}$ with $\mathcal{D}^\circ_{\overline{\eta}}$ dense in $\mathcal{D}_{j,\overline{\eta}}$. Possibly shrinking $\widetilde{T}$ further, $\mathcal{D}_j^n$ is flat over $\widetilde{T}$ [Mat89, Theorem 24.1] with normal geometric fibers of pure dimension 2 [Gro66, Théorème 12.2.4], and the fibers of the complements of $\mathcal{D}^\circ$ and of $\iota^{-1}(\mathcal{D}^\circ)$ are of dimension ≤ 1 [Gro66, Corollaire 13.1.5]. Then for any geometric point $\widetilde{t}$ of $\widetilde{T}$, $\iota_{\widetilde{t}}$ is a finite morphism from the normal scheme $\mathcal{D}^n_{j,\widetilde{t}}$ onto the reduced surface $\mathcal{D}_{j,\widetilde{t}}$ (Set-up 13.5(1)) which is an isomorphism over a dense open subset and whose exceptional set is nowhere dense. Thus (a) holds. We let $\Lambda^\circ \subset \Lambda$ be a dense open subset disjoint from the image of the closed subset $\widetilde{T}_0 \backslash \widetilde{T}$, which is closed and does not contain η since $\widetilde{T}_0 \to \Lambda$ is finite and the generic point of $\widetilde{T}_0$ is its only point over η, and replace $\widetilde{T}$ with the inverse image of Λ° in $\widetilde{T}_0$.

Step 6. In this step we fix j and $\phi : \widetilde{T} \to T_j$, ι as in **Step 5**, and we bound the graphs of the involutions. For a geometric point $\widetilde{t}$ of $\widetilde{T}$ we use the notation above for $t = \phi \circ \widetilde{t}$, so $D_Y^n = \mathcal{D}^n_{j,\widetilde{t}}$. Since ι is finite and $\bigotimes_k \mathcal{H}_{j,k}$ is ample over T_j, $\bigotimes_k \iota^*\mathcal{H}_{j,k}$ is ample over $\widetilde{T}$. Possibly replacing $\widetilde{T}$ with the members of a finite affine open cover, there exist a positive integer a and a closed immersion $\mathcal{D}_j^n \subset \mathbb{P}^M_{\widetilde{T}}$ over $\widetilde{T}$ with

$$\mathcal{O}_{\mathbb{P}^M_{\widetilde{T}}}(1)|_{\mathcal{D}_j^n} \cong \bigotimes_{k=1}^{m_j} \iota^*\mathcal{H}_{j,k}^{\otimes a}.$$

Let $\lambda := a/\min_k c_{j,k}$. For a geometric point $\widetilde{t}$ and any k, let J_k be a Cartier divisor on D_Y^n with

$$\mathcal{O}_{D_Y^n}(J_k) \cong \iota^*\mathcal{H}_{j,k}|_{D_Y^n},$$

and let

$$A := a\sum_{k=1}^{m_j} J_k, \qquad L_D := \sum_{k=1}^{m_j} c_{j,k} J_k.$$

Intersection numbers on D_Y^n are sums over the components $D_{Y,\alpha}^n$. The intersection numbers $J_k \cdot J_{k'}$ on the fibers of the flat projective morphism $\mathcal{D}_j^n \to \widetilde{T}$ are locally constant on $\widetilde{T}$ by [Kle66, Chapter I, §2, Definition 1] and [Gro63, Théorème 7.9.4], so there exists a real number β_0 with $L_D^2 \le \beta_0$ for any geometric point $\widetilde{t}$ of $\widetilde{T}$.

For $(X, B) \in \mathcal{S}$ over F, let $\widetilde{t} : \operatorname{Spec} F \to \widetilde{T}$ be a geometric point satisfying the following.

(†) There exists an isomorphism of F-schemes $Y \cong \mathcal{Y}_{j,\phi\circ\widetilde{t}}$ under which $D_Y = \mathcal{D}_{j,\phi\circ\widetilde{t}}$ and $\nu_*^{-1}B = \mathcal{B}_{j,\phi\circ\widetilde{t}}$, where $\nu : Y \to X$ is the normalization and D_Y is the reduced conductor divisor on Y.

We then regard the involution τ of D_Y^n associated with X as an involution of $\mathcal{D}_{j,\widetilde{t}}^n$. Let $\widetilde{t}$ be as in (†). By Set-up 13.5(3) and **Step 3**(a),

$$L_D \sim_{\mathbb{R}} \iota^* L_Y(\boldsymbol{b}_j) \sim_{\mathbb{R}} K_{D_Y^n} + \operatorname{Diff}(\boldsymbol{b}_j) \quad \text{and} \quad \tau^*\big(K_{D_Y^n} + \operatorname{Diff}(\boldsymbol{b}_j)\big) \sim_{\mathbb{R}} K_{D_Y^n} + \operatorname{Diff}(\boldsymbol{b}_j),$$

so $\tau^* L_D \equiv L_D$. Any J_k is nef since $\mathcal{H}_{j,k}$ is generated by global sections, and $\lambda c_{j,k} \ge a$ for any k. Thus $\lambda L_D - A = \sum_k (\lambda c_{j,k} - a) J_k$ and $\lambda L_D - \tau^* A \equiv \tau^*(\lambda L_D - A)$ are nef. Since $2\lambda L_D = (A + \tau^* A) + (2\lambda L_D - A - \tau^* A)$ with $A + \tau^* A$ and $2\lambda L_D - A - \tau^* A$ nef, and $N_1 \cdot N_2 \ge 0$ for nef $\mathbb{R}$-Cartier $\mathbb{R}$-divisors N_1 and N_2 on a projective surface, we obtain

$$(A + \tau^* A)^2 \le (2\lambda L_D)^2 \le 4\lambda^2 \beta_0 =: \beta_1. \tag{13.25}$$

Let $M' := (M+1)^2 - 1$ and embed

$$\mathcal{D}_j^n \times_{\widetilde{T}} \mathcal{D}_j^n \subset \mathbb{P}^{M'}_{\widetilde{T}}$$

by the Segre embedding, so that on any geometric fiber $\mathcal{O}(1)$ restricts to $p_1^*\mathcal{O}(A) \otimes p_2^*\mathcal{O}(A)$. For the graph $\Gamma_\tau \subset D_Y^n \times D_Y^n$ of τ, the projection p_1 restricts to an isomorphism $\Gamma_\tau \cong D_Y^n$, and $\mathcal{O}(1)|_{\Gamma_\tau}$ corresponds to $\mathcal{O}_{D_Y^n}(A + \tau^* A)$, so $\deg \Gamma_\tau \le \beta_1$ by (13.25). The connected components of Γ_τ are the graphs of the restrictions of τ to the $D_{Y,\alpha}^n$, which are integral surfaces of degree ≥ 1. Thus Γ_τ has at most β_1 connected components, each of degree $\le \beta_1$.

By [HK19, Theorem 1.8], there exists a positive integer β such that the homogeneous ideal of any integral surface of degree $\le \beta_1$ in $\mathbb{P}^{M'}$ over any field is generated in degrees $\le \beta$. Such a surface is then the zero locus of the degree-β part of the homogeneous ideal of the surface, which is a point of a Grassmannian of subspaces of $H^0(\mathbb{P}^{M'}, \mathcal{O}(\beta))$. For $0 \le m' \le \beta_1$ and positive integers $e_1, \dots, e_{m'}$, let

$$\mathcal{R} \subset \widetilde{T} \times \prod_{\gamma=1}^{m'} \operatorname{Gr}\left(e_\gamma, H^0(\mathbb{P}^{M'}_{\mathbb{Z}[1/30]}, \mathcal{O}(\beta))\right)$$

be the open subset over which the pullbacks of the m' universal closed subschemes of $\mathbb{P}^{M'}$ are pairwise disjoint, and let $\mathcal{G}_{\mathcal{R}} \subset \mathbb{P}^{M'}_{\mathcal{R}}$ be their union. By generic flatness [Mat89, Theorem 24.1] and Noetherian induction, $\mathcal{R}$ is a finite union of reduced locally closed subschemes $\widehat{T}$ such that

$$\mathcal{G} := \mathcal{G}_{\mathcal{R}} \times_{\mathcal{R}} \widehat{T}$$

is flat over $\widehat{T}$. For any $(X, B) \in \mathcal{S}$ and any $\widetilde{t} : \operatorname{Spec} F \to \widetilde{T}$ as in (†), with τ the involution of D_Y^n associated with X, the graph Γ_τ is the disjoint union of at most β_1 integral surfaces of degree $\le \beta_1$, so there exist $\widehat{T}$ and a geometric point $\widehat{t} : \operatorname{Spec} F \to \widehat{T}$ over $\widetilde{t}$ with $\mathcal{G}_{\widehat{t}} = \Gamma_\tau$. We call such $\widehat{t}$ *representative* for (X, B). By **Step 5**(b), for any $(X, B) \in \mathcal{S}$ there exist j, $\widehat{T}$ and a geometric point of $\widehat{T}$ representative for (X, B).

Step 7. In this step we fix j, $\widehat{T}$, and $\mathcal{G}$ as in **Step 6**, a set $\Sigma \subset \widehat{T}$ of images of representative geometric points, and an irreducible component W of the closure $\overline{\Sigma}$ with reduced structure. We show that there exist a dense open subset $W^\circ \subset W$, an integral scheme T' with a finite surjective morphism $T' \to W^\circ$, a projective flat morphism $\mathcal{Z} \to T'$, and closed subschemes $\mathcal{B}'_1, \dots, \mathcal{B}'_{n_j} \subset \mathcal{Z}$ flat over T', satisfying the following. Let $\widehat{t} : \operatorname{Spec} F \to \widehat{T}$ be representative for (X, B) with image

in $\Sigma \cap W^\circ$, and $t' : \operatorname{Spec} F \to T'$ a geometric point over $\widehat{t}$. Then there exists an isomorphism $X \cong \mathcal{Z}_{t'}$ under which $\mathcal{B}'_{1,t'}, \dots, \mathcal{B}'_{n_j,t'}$ are the pairwise distinct prime components of B and

$$B = \sum_{i=1}^{n_j} b_{j,i} \mathcal{B}'_{i,t'}. \tag{13.26}$$

Moreover, if $\boldsymbol{b}_j \in \mathbb{Q}^{n_j}$, there exists a positive integer $N_{\mathcal{Z}}$ such that $N_{\mathcal{Z}}(K_X + B)$ is a very ample Cartier divisor for any such (X, B).

Let η be the generic point of W and $\overline{\eta}$ a geometric point over η. We use the notation above for the geometric point of T_j induced by $\overline{\eta}$. We start with $W^\circ = T' = W$. We may assume that W° is disjoint from the other irreducible components of $\overline{\Sigma}$. To *shrink* T' means to replace W° with a dense open subset $W^{\circ\circ} \subset W^\circ$ and T' with the inverse image of $W^{\circ\circ}$ in the normalization of T' in a finite extension of $k(T')$.

The involution at $\overline{\eta}$. The set of points $s \in \widehat{T}$ such that $\mathcal{G}_{\overline{s}}$ is the graph of an involution of $\mathcal{D}^n_{j,\overline{s}}$ with fixed locus of dimension ≤ 1, for a geometric point $\overline{s}$ over s, is constructible [Gro66, Proposition 9.6.1, Corollaire 13.1.5]. By **Step 3**(a) the constructible set contains $\Sigma \cap W^\circ$, which is dense in W. Thus it contains η. Let $\tau_{\overline{\eta}}$ be the involution of $\mathcal{D}^n_{j,\overline{\eta}}$ with graph $\mathcal{G}_{\overline{\eta}}$.

The different at $\overline{\eta}$. Let $f : Y' \to \mathcal{Y}_{j,\overline{\eta}}$, E_α, and g_α be as in **Step 2** at $\overline{\eta}$. Let $\boldsymbol{b}^{(v)}$ and μ_v be as in **Step 2** at $\overline{\eta}$ for $V = V^\circ = V_Y$, and let I_1 be a positive integer such that $I_1 L_Y(\boldsymbol{b}^{(v)})$ is Cartier for any v. After shrinking T', the morphisms f and g_α, the involution $\tau_{\overline{\eta}}$, the Cartier divisors $I_1 L_Y(\boldsymbol{b}^{(v)})$, and the equalities

$$f^*\big(I_1 L_Y(\boldsymbol{b}^{(v)})\big) = I_1\big(K_{Y'} + B_{Y'}(\boldsymbol{b}^{(v)})\big), \tag{13.27}$$

with $K_{Y'}$ the divisor of a rational section of the relative canonical sheaf, extend over T' [Gro66, Proposition 8.5.5, Théorèmes 8.5.2, 8.8.2, 8.10.5] so that the following hold for any geometric point t' of T':

(7.a) $f_{t'}$ is a log resolution of

$$\Big(\mathcal{Y}_{j,t'}, \mathcal{D}_{j,t'} + \sum_i \mathcal{B}_{j,i,t'}\Big).$$

(7.b) The components of $\operatorname{Exc}(f) \cup f_*^{-1}(\mathcal{D}_j + \sum_i \mathcal{B}_{j,i})$, the irreducible components C of $E_\alpha \cap P$ for such components $P \ne E_\alpha$, the prime divisors P' of $\mathcal{D}^n_j$ of the form $g_\alpha(C)$, and the prime divisors $\tau(P')$ have geometrically integral and pairwise distinct fibers, and the closed subsets $g_\alpha(C)$ that are not divisors at $\overline{\eta}$ have fibers of dimension 0.

(7.c) The graph of $\tau_{t'}$ is $\mathcal{G}_{t'}$.

(7.d) The equalities (13.27) restrict to the fibers over t'.

Then by (13.24), for any geometric point t' of T', $\operatorname{Diff}(\boldsymbol{b}_j)$ on $\mathcal{D}^n_{j,t'}$ is supported on the fibers $P'_{t'}$ of the prime divisors P', and

$$\operatorname{mult}_{P'_{t'}} \operatorname{Diff}(\boldsymbol{b}_j) = \operatorname{mult}_{P'_{\overline{\eta}}} \operatorname{Diff}(\boldsymbol{b}_j) \quad \text{for any } P'.$$

If $\tau^*_{\overline{\eta}} \operatorname{Diff}(\boldsymbol{b}_j) \ne \operatorname{Diff}(\boldsymbol{b}_j)$, then $\tau^*_{t'} \operatorname{Diff}(\boldsymbol{b}_j) \ne \operatorname{Diff}(\boldsymbol{b}_j)$ for any geometric point t' of T'. Since $\Sigma \cap W^\circ \ne \emptyset$, there exists a geometric point t' of T' over a representative geometric point $\widehat{t}$ for some (X, B), and $\tau_{t'}$ is the involution of D^n_Y associated with X since the graph of $\tau_{t'}$ is $\mathcal{G}_{\widehat{t}}$, contradicting **Step 3**(a). Thus $\tau^*_{\overline{\eta}} \operatorname{Diff}(\boldsymbol{b}_j) = \operatorname{Diff}(\boldsymbol{b}_j)$.

The quotient family. By **Step 4** at $\overline{\eta}$, there exists a finite surjective morphism

$$q_{\overline{\eta}} : \mathcal{Y}_{j,\overline{\eta}} \to Z_{\overline{\eta}}$$

satisfying **Step 4**(a) and (b) for $\tau_{\overline{\eta}}$. If $\boldsymbol{b}_j \in \mathbb{Q}^{n_j}$, let $B_{Z_{\overline{\eta}}}$ be as in **Step 4**(b) at $\overline{\eta}$. By **Step 4**(b), $K_{Z_{\overline{\eta}}} + B_{Z_{\overline{\eta}}}$ is an ample $\mathbb{R}$-Cartier $\mathbb{Q}$-divisor. Since the prime divisors $q_{\overline{\eta}}(\mathcal{B}_{j,i,\overline{\eta}})$ are not contained in the non-normal locus $q_{\overline{\eta}}(\mathcal{D}_{j,\overline{\eta}})$ and $Z_{\overline{\eta}}$ is S_2 and nodal in codimension 1 (**Step**

4(a)), Lemma 2.6 shows that $K_{Z_{\overline{\eta}}} + B_{Z_{\overline{\eta}}}$ is $\mathbb{Q}$-Cartier. We let $N_{\mathcal{Z}}$ be a positive integer such that $N_{\mathcal{Z}} b_{j,i} \in \mathbb{Z}$ for any i and $N_{\mathcal{Z}}(K_{Z_{\overline{\eta}}} + B_{Z_{\overline{\eta}}})$ is a very ample Cartier divisor, and

$$\mathcal{L}_{\overline{\eta}} := \mathcal{O}_{Z_{\overline{\eta}}}\big(N_{\mathcal{Z}}(K_{Z_{\overline{\eta}}} + B_{Z_{\overline{\eta}}})\big).$$

Otherwise let $\mathcal{L}_{\overline{\eta}}$ be any very ample line bundle on $Z_{\overline{\eta}}$. After shrinking T', there exist a closed subscheme $\mathcal{Z} \subset \mathbb{P}^{M''}_{T'}$, where $M'' := h^0(Z_{\overline{\eta}}, \mathcal{L}_{\overline{\eta}}) - 1$, flat over T' and a finite surjective morphism $q : \mathcal{Y}_j \times_{T_j} T' \to \mathcal{Z}$ with $q \circ \iota = q \circ \iota \circ \tau$, whose base change to $\overline{\eta}$ is $q_{\overline{\eta}}$ followed by the embedding of $Z_{\overline{\eta}}$ by $\mathcal{L}_{\overline{\eta}}$ [Gro66, Théorèmes 8.8.2, 8.10.5], [Mat89, Theorem 24.1]. Let

$$\mathcal{L} := \mathcal{O}_{\mathbb{P}^{M''}_{T'}}(1)|_{\mathcal{Z}}.$$

Consider

$$0 \to \mathcal{O}_{\mathcal{Z}} \to q_*\mathcal{O}_{\mathcal{Y}_j \times_{T_j} T'} \xrightarrow{\delta} (q \circ \iota)_*\mathcal{O}_{\mathcal{D}^n_j \times_{\widetilde{T}} T'}, \qquad \delta(u) := \iota^* u - \tau^* \iota^* u. \tag{13.28}$$

By **Step 1** and **Step 4**(a), and since $\operatorname{char} k(\overline{\eta}) \neq 2$, (13.28) is exact on the geometric generic fiber, hence on the generic fiber. The sheaves in (13.28) are coherent since q and $q \circ \iota$ are finite, and the homology sheaves of (13.28) at the first two terms have supports proper over T' and disjoint from the generic fiber. After shrinking T', the homology sheaves of (13.28) at the first two terms vanish. After shrinking T' further, $\operatorname{Im}\delta$ and $\operatorname{Coker}\delta$ are flat over T' [Mat89, Theorem 24.1]. Then for any geometric point t' of T', the exact sequences

$$0 \to \mathcal{O}_{\mathcal{Z}} \to q_*\mathcal{O}_{\mathcal{Y}_j \times_{T_j} T'} \to \operatorname{Im}\delta \to 0, \qquad 0 \to \operatorname{Im}\delta \to (q \circ \iota)_*\mathcal{O}_{\mathcal{D}^n_j \times_{\widetilde{T}} T'} \to \operatorname{Coker}\delta \to 0$$

stay exact after base change to t', and q_* and $(q \circ \iota)_*$ commute with base change since q and $q \circ \iota$ are affine. Thus

$$\mathcal{O}_{\mathcal{Z}_{t'}} = \operatorname{Ker}\left(\delta_{t'} : q_{t'*}\mathcal{O}_{\mathcal{Y}_{j,t'}} \to (q_{t'} \circ \iota_{t'})_*\mathcal{O}_{\mathcal{D}^n_{j,t'}}\right) \quad \text{for any geometric point } t' \text{ of } T'. \tag{13.29}$$

The representative fibers. Let $\widehat{t}$ be representative for (X, B) with image in $\Sigma \cap W^\circ$, and t' a geometric point of T' over $\widehat{t}$. Then $Y = \mathcal{Y}_{j,t'}$, $D^n_Y = \mathcal{D}^n_{j,t'}$, and $\tau_{t'}$ is the involution of D^n_Y associated with X since the graph of $\tau_{t'}$ is $\mathcal{G}_{\widehat{t}}$. By **Step 3**(b), $q_{t'} = \psi \circ \nu$ for a morphism $\psi : X \to \mathcal{Z}_{t'}$. Since X is proper and $q_{t'}$ is finite, ψ is proper with finite fibers, hence finite [Sta, Tag 02LS]. Since $\operatorname{char} F > 5$, **Step 1** for X and the left exactness of ψ_* imply that the subsheaf $\psi_*\mathcal{O}_X$ of

$$\psi_*\nu_*\mathcal{O}_Y = q_{t'*}\mathcal{O}_Y$$

is the kernel of ψ_* of the homomorphism δ in (13.23), which is $\delta_{t'}$. By (13.29), the homomorphism $\mathcal{O}_{\mathcal{Z}_{t'}} \to \psi_*\mathcal{O}_X$ is an isomorphism. Since ψ is finite, ψ is an isomorphism.

The boundary. For any i, let $\mathcal{B}'_i \subset \mathcal{Z}$ be the image of $\mathcal{B}_{j,i} \times_{T_j} T'$ under q with reduced structure. After shrinking T', $\mathcal{B}'_i$ is flat over T' with geometrically integral fibers of dimension 2 [Mat89, Theorem 24.1], [Gro66, Théorème 9.7.7], and

$$\mathcal{B}'_{i,\overline{\eta}} = q_{\overline{\eta}}(\mathcal{B}_{j,i,\overline{\eta}}).$$

Let $\widehat{t}$ be representative for (X, B) with image in $\Sigma \cap W^\circ$, t' a geometric point of T' over $\widehat{t}$, and $\psi : X \to \mathcal{Z}_{t'}$ the isomorphism with $q_{t'} = \psi \circ \nu$ (*The representative fibers*). Then the integral surface $\mathcal{B}'_{i,t'}$ contains the prime divisor $q_{t'}(B_{Y,i}) = \psi(B_i)$, so $\mathcal{B}'_{i,t'} = \psi(B_i)$, where $B_i = \nu(B_{Y,i})$ as in **Step 3**. Since $B = \sum_i b_{j,i} B_i$ and $B_1, \dots, B_{n_j}$ are pairwise distinct prime divisors (**Step 3**), the isomorphism ψ satisfies (13.26).

Step 8. In this step we assume $\boldsymbol{b}_j \in \mathbb{Q}^{n_j}$ and prove the last statement of **Step 7**. Let $\mathcal{U} \subset \mathcal{Z}$ be the open subset [Sta, Tag 0C09] of points at which $\mathcal{Z} \to T'$ is Gorenstein and $\mathcal{B}'_1, \dots, \mathcal{B}'_{n_j}$ are Cartier. A point of $Z_{\overline{\eta}}$ of codimension ≤ 1 is a regular point or a node, hence Gorenstein, and it is contained in some $\mathcal{B}'_{i,\overline{\eta}}$ only if it is a regular point, since $\mathcal{B}'_{i,\overline{\eta}}$ is not contained in $q_{\overline{\eta}}(\mathcal{D}_{j,\overline{\eta}})$. Thus $\dim(Z_{\overline{\eta}} \backslash \mathcal{U}_{\overline{\eta}}) \leq 1$. After shrinking T', $\dim(\mathcal{Z}_{t'} \backslash \mathcal{U}_{t'}) \leq 1$ for any geometric point t' of T' [Gro66,

Corollaire 13.1.5]. Since $\mathcal{B}'_i$ is flat over T', $\mathcal{B}'_i \cap \mathcal{U}$ is a relative Cartier divisor on $\mathcal{U}$ over T' with fiber $\mathcal{B}'_{i,t'} \cap \mathcal{U}_{t'}$ at t'. The sheaf

$$\mathcal{M} := \omega_{\mathcal{U}/T'}^{\otimes N_{\mathcal{Z}}} \otimes \mathcal{O}_{\mathcal{U}}\Big(\sum_{i=1}^{n_j} N_{\mathcal{Z}} b_{j,i} \mathcal{B}'_i\Big)$$

is invertible, and for any geometric point t' of T' [Sta, Tag 0E2Y],

$$\mathcal{M}|_{\mathcal{U}_{t'}} \cong \omega_{\mathcal{U}_{t'}}^{\otimes N_{\mathcal{Z}}} \otimes \mathcal{O}_{\mathcal{U}_{t'}}\Big(\sum_{i=1}^{n_j} N_{\mathcal{Z}} b_{j,i} \mathcal{B}'_{i,t'}\Big).$$

By the choice of $\mathcal{L}_{\overline{\eta}}$ and since $B_{Z_{\overline{\eta}}} = \sum_i b_{j,i} \mathcal{B}'_{i,\overline{\eta}}$ (*The boundary* in **Step 7**),

$$\mathcal{L}|_{\mathcal{U}_{\overline{\eta}}} \cong \mathcal{M}|_{\mathcal{U}_{\overline{\eta}}}.$$

After shrinking T', the isomorphism over $\mathcal{U}_{\overline{\eta}}$ extends to an isomorphism $\mathcal{L}|_{\mathcal{U}} \cong \mathcal{M}$ [Gro66, Théorème 8.5.2(i)].

Let $\widehat{t}$, t', and $\psi : X \to \mathcal{Z}_{t'}$ be as in **Step 7**, and identify $X = \mathcal{Z}_{t'}$ by ψ. By (13.26), for the divisorial sheaf $\mathcal{O}_X(N_{\mathcal{Z}}(K_X + B))$ [Kol13, 5.6] we have

$$\mathcal{L}|_{\mathcal{U}_{t'}} \cong \mathcal{O}_X(N_{\mathcal{Z}}(K_X + B))|_{\mathcal{U}_{t'}}.$$

Since X is S_2 of pure dimension 3 and $\dim(X \backslash \mathcal{U}_{t'}) \leq 1$, [Har94, Proposition 1.11] implies $\mathcal{L}|_X \cong \mathcal{O}_X(N_{\mathcal{Z}}(K_X + B))$. Thus $N_{\mathcal{Z}}(K_X + B)$ is a very ample Cartier divisor.

Step 9. In this step we conclude. Fix j and $\widehat{T}$, $\mathcal{G}$ as in **Step 6**, and let $\Sigma_0 \subset \widehat{T}$ be the set of images of all representative geometric points. Let W be an irreducible component of $\overline{\Sigma_0}$, and W°, T', $\mathcal{Z}$, $\mathcal{B}'_i$ as in **Step 7** for $\Sigma = \Sigma_0$. Since W° is open in $\overline{\Sigma_0}$, the closure of $\Sigma_1 := \Sigma_0 \backslash W^\circ$ is a proper closed subset of $\overline{\Sigma_0}$. Repeating with Σ_1 in place of Σ_0, by Noetherian induction we obtain finitely many families

$$(\mathcal{Z} \to T', \mathcal{B}'_1, \dots, \mathcal{B}'_{n_j})$$

as in **Step 7**, each for one of the sets $\Sigma_0 \supset \Sigma_1 \supset \cdots$ and the open subset $W^\circ \subset \overline{\Sigma}$ of **Step 7**, such that Σ_0 is the union of the sets $\Sigma \cap W^\circ$ for Σ and W° attached to the families. Since j and $\widehat{T}$ run over finite sets, and since any $(X, B) \in \mathcal{S}$ has a representative geometric point of some $\widehat{T}$ (**Step 6**), the finitely many projective flat morphisms $\mathcal{Z} \to T'$ onto schemes T', which are of finite type over $\mathbb{Z}[1/30]$ since T' is finite over $W^\circ \subset \widehat{T}$ and $\widehat{T}$ is of finite type over $\widetilde{T}$ (**Steps 5** and **6**), with the closed subschemes $\mathcal{B}'_i$ flat over T' and the numbers $b_{j,i}$, satisfy Definition 13.1 for $\mathcal{S}$ by (13.26). This proves (1).

Assume that $b_{j,i} \in \mathbb{Q}$ for any j and i, and let N be the least common multiple of the integers $N_{\mathcal{Z}}$ of **Step 7** for the finitely many families above. Any $(X, B) \in \mathcal{S}$ satisfies $X \cong \mathcal{Z}_{t'}$ as in **Step 7** for one of the families $\mathcal{Z} \to T'$ above, so $N_{\mathcal{Z}}(K_X + B)$ is a very ample Cartier divisor and $N(K_X + B) = (N/N_{\mathcal{Z}}) \cdot N_{\mathcal{Z}}(K_X + B)$ is a very ample Cartier divisor. This proves (2). □

Proof of Theorem 1.5. Let $P := P_{\mathrm{red}}$ (Definition-Theorem 10.4). Then $P \geq 11$, and Theorems 1.3 and 1.4 hold for P by their proofs. Let $I' := I \cup \{1\}$, a DCC subset of $[0, 1]$. For any $(X, B) \in \mathcal{S}$ over a field F, we let $\nu : Y \to X$, D_Y, B_Y, $Y_1, \dots, Y_n$, D_i, and B_i be as in Definition 12.1, and let

$$w_i := (K_{Y_i} + D_i + B_i)^3 \quad \text{for any } 1 \leq i \leq n.$$

We divide the proof into four steps.

Step 1. Let $(X, B) \in \mathcal{S}$ over F. Since ν is finite and $K_X + B$ is ample, the $\mathbb{R}$-divisor

$$K_{Y_i} + D_i + B_i = \nu^*(K_X + B)|_{Y_i}$$

is ample for any i. Thus $w_i > 0$ for any i, and $\mathrm{vol}(K_X + B) = \sum_{i=1}^n w_i > 0$. By Definition 12.1, $(Y_i, D_i + B_i)$ is a projective lc pair of dimension 3 over F. The divisor D_i is reduced, $B_i \in I$,

and since no component of B is contained in the non-normal locus of X, no component of B_i is contained in D_i. Thus $D_i + B_i \in I'$.

By Theorem 1.4(3),

$$\sum_{i=1}^{n} w_i = V,$$

and the tuple $(w_1, \dots, w_n)$ belongs to a finite set depending only on I and V. Let

$$\mathcal{W}_{\mathcal{S}} := \{(w_1, \dots, w_n) \mid (X, B) \in \mathcal{S}\},$$

and

$$\Lambda := \{w_i \mid (w_1, \dots, w_n) \in \mathcal{W}_{\mathcal{S}},\ 1 \le i \le n\}.$$

Then $\mathcal{W}_{\mathcal{S}}$ and Λ are finite, and $\Lambda \subset (0, +\infty)$.

For $w \in \Lambda$, let

$$\mathcal{S}_w := \{(Y_i, D_i + B_i) \mid (X, B) \in \mathcal{S},\ 1 \le i \le n,\ w_i = w\}.$$

Let $m_{I'}$ be the integer of Theorem 1.3 for I'. For any $(X, B) \in \mathcal{S}$ over F and any $1 \le i \le n$, since the characteristic of F is $> P$ and $K_{Y_i} + D_i + B_i$ is big, $|\lfloor m_{I'}(K_{Y_i} + D_i + B_i)\rfloor|$ defines a birational map, and

$$m_{I'}^3 \operatorname{vol}(K_{Y_i} + D_i + B_i) = m_{I'}^3 w_i \le m_{I'}^3 V.$$

By Proposition 9.11 for I' and $m_{I'}^3 V$, the set $\bigcup_{w \in \Lambda} \mathcal{S}_w$ is log birationally bounded over $\mathbb{Z}[1/30]$, hence so is each $\mathcal{S}_w$. Thus for any $w \in \Lambda$, I', w, and $\mathcal{S}_w$ satisfy the conditions of Set-up 13.3.

By Proposition 13.6, for any $w \in \Lambda$ there exist a finite set J'_w and, for any $j \in J'_w$, data

$$\mathcal{Y}'_j \to T'_j, \quad \mathcal{D}'_j = \emptyset, \quad \mathcal{B}'_{j,1}, \dots, \mathcal{B}'_{j,n'_j}, \quad b'_{j,1}, \dots, b'_{j,n'_j}, \quad \mathcal{H}'_{j,1}, \dots, \mathcal{H}'_{j,m'_j}, \quad \text{and} \quad c'_{j,1}, \dots, c'_{j,m'_j}$$

as in Set-up 13.5, satisfying the following: for any $(Y_i, D_i + B_i) \in \mathcal{S}_w$ over F, there exist $j \in J'_w$, a geometric point $t : \operatorname{Spec} F \to T'_j$, and an isomorphism of F-schemes $Y_i \cong \mathcal{Y}'_{j,t}$ under which

$$D_i + B_i = \mathcal{B}'_{j,t}.$$

Step 2. In this step we mark the conductor in the families of **Step 1**.

Fix $w \in \Lambda$, $j \in J'_w$, and a subset $\Omega \subset \{1, \dots, n'_j\}$ with $b'_{j,i'} = 1$ for any $i' \in \Omega$. For any locally closed subset $W \subset T'_j$ with the reduced structure, let $\mathcal{D}^{\Omega}_W \subset \mathcal{Y}'_j \times_{T'_j} W$ be the reduced closed subscheme with support $\bigcup_{i' \in \Omega} \mathcal{B}'_{j,i'} \times_{T'_j} W$. If W is irreducible, then the fiber of $\mathcal{D}^{\Omega}_W$ over the generic point of W is geometrically reduced, being the union of the geometrically integral generic fibers of the $\mathcal{B}'_{j,i'} \times_{T'_j} W$ with $i' \in \Omega$. Thus by [Gro66, Théorème 9.7.7], generic flatness [Mat89, Theorem 24.1], and Noetherian induction, there exists a finite set $\mathcal{T}_{j,\Omega}$ of pairwise disjoint irreducible locally closed subsets of T'_j covering T'_j, each endowed with the reduced structure, such that for any $T \in \mathcal{T}_{j,\Omega}$:

(1) $\mathcal{D}^{\Omega}_T$ is flat over T.
(2) The fiber $\mathcal{D}^{\Omega}_t$ of $\mathcal{D}^{\Omega}_T$ is reduced for any geometric point t of T.

Since $\mathcal{B}'_{j,i',t}$ with $i' \in \Omega$ are pairwise distinct prime divisors (Set-up 13.5(1)) and

$$\operatorname{Supp} \mathcal{D}^{\Omega}_t = \bigcup_{i' \in \Omega} \mathcal{B}'_{j,i',t},$$

we have

$$\mathcal{D}^{\Omega}_t = \sum_{i' \in \Omega} \mathcal{B}'_{j,i',t} \quad \text{for any geometric point } t \text{ of } T.$$

Over each $T \in \mathcal{T}_{j,\Omega}$, consider $\mathcal{Y}'_j \times_{T'_j} T \to T$, the reduced closed subscheme $\mathcal{D}^{\Omega}_T$, the base changes of $\mathcal{B}'_{j,i'}$ with $i' \notin \Omega$, which are reduced [Mat89, Corollary to Theorem 23.9], with weights $b'_{j,i'}$,

and the pullbacks of $\mathcal{H}'_{j,k}$ with weights $c'_{j,k}$ for $1 \le k \le m'_j$. Since $b'_{j,i'} = 1$ for any $i' \in \Omega$,

$$\mathcal{D}^{\Omega}_t + \sum_{i' \notin \Omega} b'_{j,i'} \mathcal{B}'_{j,i',t} = \mathcal{B}'_{j,t} \quad \text{for any geometric point } t \text{ of } T,$$

and the prime divisors $\mathcal{B}'_{j,i',t}$ with $i' \notin \Omega$ are pairwise distinct and distinct from the components of $\mathcal{D}^{\Omega}_t$. Thus conditions (1)–(3) of Set-up 13.5 for the new data follow from those for the family indexed by j, and the new data form a family as in Set-up 13.5. We keep such a family only if $b'_{j,i'} \in I$ for any $i' \notin \Omega$.

For any $w \in \Lambda$, let J_w be the finite set indexing the kept families over the triples (j', Ω, T), where $j' \in J'_w$, $\Omega \subset \{1, \dots, n'_{j'}\}$ satisfies $b'_{j',i'} = 1$ for any $i' \in \Omega$, and $T \in \mathcal{T}_{j',\Omega}$. We choose $J_w \cap J_{w'} = \emptyset$ for any $w \ne w'$ in Λ. For $j \in J_w$ indexing the kept family over (j', Ω, T), let $T_j := T$, $\mathcal{Y}_j := \mathcal{Y}'_{j'} \times_{T'_{j'}} T$, and $\mathcal{D}_j := \mathcal{D}^{\Omega}_T$, let $\mathcal{B}_{j,1}, \dots, \mathcal{B}_{j,n_j}$ be the base changes of the $\mathcal{B}'_{j',i'}$ with $i' \notin \Omega$ in increasing order of i', with weights $b_{j,1}, \dots, b_{j,n_j}$ the corresponding $b'_{j',i'}$, and let $m_j := m'_{j'}$ and, for $1 \le k \le m_j$, $\mathcal{H}_{j,k}$ be the pullback of $\mathcal{H}'_{j',k}$ with weight $c_{j,k} := c'_{j',k}$. Let

$$I_0 := \{ b_{j,i'} \mid w \in \Lambda,\ j \in J_w,\ 1 \le i' \le n_j \}.$$

Since Λ and each J_w are finite, I_0 is a finite subset of I.

We show that for any $(X, B) \in \mathcal{S}$ over F and any $1 \le i \le n$, there exist $j \in J_{w_i}$, a geometric point $t : \operatorname{Spec} F \to T_j$, and an isomorphism of F-schemes $Y_i \cong \mathcal{Y}_{j,t}$ under which

$$D_i = \mathcal{D}_{j,t} \quad \text{and} \quad B_i = \mathcal{B}_{j,t}.$$

By **Step 1**, there exist $j' \in J'_{w_i}$, a geometric point $t' : \operatorname{Spec} F \to T'_{j'}$, and an isomorphism

$$Y_i \cong \mathcal{Y}'_{j',t'}$$

under which $D_i + B_i = \mathcal{B}'_{j',t'}$. Since $\mathcal{B}'_{j',1,t'}, \dots, \mathcal{B}'_{j',n'_{j'},t'}$ are pairwise distinct prime divisors and $b'_{j',1}, \dots, b'_{j',n'_{j'}} > 0$, they are the prime components of $D_i + B_i$, and $b'_{j',i'} = \operatorname{mult}_{\mathcal{B}'_{j',i',t'}}(D_i + B_i)$ for any i'. Let Ω be the set of i' with $\mathcal{B}'_{j',i',t'} \subset \operatorname{Supp} D_i$. Since D_i is reduced and no component of B_i is contained in D_i, $b'_{j',i'} = 1$ for any $i' \in \Omega$, and $b'_{j',i'}$ is a coefficient of B_i, hence belongs to I, for any $i' \notin \Omega$. The point t' factors through some $T \in \mathcal{T}_{j',\Omega}$, and the family over T is kept. Let $j \in J_{w_i}$ index the kept family over (j', Ω, T). Then, on the fiber over t', $\mathcal{D}_{j,t'} = \mathcal{D}^{\Omega}_{t'} = D_i$ and

$$\mathcal{B}_{j,t'} = \sum_{i' \notin \Omega} b'_{j',i'} \mathcal{B}'_{j',i',t'} = B_i.$$

Step 3. In this step we assemble the families of **Step 2** into families of normalizations.

Let $(w'_1, \dots, w'_{n'}) \in \mathcal{W}_{\mathcal{S}}$ and

$$\mathbf{j} = (j_1, \dots, j_{n'}) \in J_{w'_1} \times \cdots \times J_{w'_{n'}}.$$

Since the sets J_w are pairwise disjoint, $\mathbf{j}$ determines $(w'_1, \dots, w'_{n'})$. Let

$$T_{\mathbf{j}} := \left(T_{j_1} \times_{\mathbb{Z}[1/30]} \cdots \times_{\mathbb{Z}[1/30]} T_{j_{n'}} \right)_{\mathrm{red}}, \qquad \mathcal{Y}_{\mathbf{j}} := \bigsqcup_{i=1}^{n'} \mathcal{Y}_{j_i} \times_{T_{j_i}} T_{\mathbf{j}} \to T_{\mathbf{j}},$$

where $\mathcal{Y}_{j_i} \times_{T_{j_i}} T_{\mathbf{j}}$ is formed with the projection $T_{\mathbf{j}} \to T_{j_i}$. On $\mathcal{Y}_{\mathbf{j}}$ we take the following data:

(a) Let $\mathcal{D}_{\mathbf{j}}$ be the disjoint union of the pullbacks of $\mathcal{D}_{j_1}, \dots, \mathcal{D}_{j_{n'}}$.

(b) Let $n_{\mathbf{j}} := \sum_{i=1}^{n'} n_{j_i}$, and let $\mathcal{B}_{\mathbf{j},1}, \dots, \mathcal{B}_{\mathbf{j},n_{\mathbf{j}}}$ be the pullbacks of $\mathcal{B}_{j_1,1}, \dots, \mathcal{B}_{j_1,n_{j_1}}$ to the first piece, …, and of $\mathcal{B}_{j_{n'},1}, \dots, \mathcal{B}_{j_{n'},n_{j_{n'}}}$ to the n'-th piece, with the corresponding weights $b_{\mathbf{j},1}, \dots, b_{\mathbf{j},n_{\mathbf{j}}}$.

(c) Let $m_{\mathbf{j}} := \sum_{i=1}^{n'} m_{j_i}$, and let $\mathcal{H}_{\mathbf{j},1}, \dots, \mathcal{H}_{\mathbf{j},m_{\mathbf{j}}}$ be the invertible sheaves which are the pullback of $\mathcal{H}_{j_i,k}$ on the i-th piece and $\mathcal{O}$ on the other pieces, for $1 \le i \le n'$ and $1 \le k \le m_{j_i}$, with the corresponding weights $c_{\mathbf{j},1}, \dots, c_{\mathbf{j},m_{\mathbf{j}}}$.

A geometric point $t : \operatorname{Spec} F \to T_{\mathbf{j}}$ is a tuple $(t_1, \dots, t_{n'})$ of geometric points $t_i : \operatorname{Spec} F \to T_{j_i}$, and

$$\mathcal{Y}_{\mathbf{j},t} = \bigsqcup_{i=1}^{n'} \mathcal{Y}_{j_i,t_i}.$$

The subschemes in (a) and (b) are reduced [Mat89, Corollary to Theorem 23.9]. Since Set-up 13.5(1)–(3) hold on each $\mathcal{Y}_{j_i,t_i}$ by **Step 2**, and the sheaves in (c) attached to $j_{i'}$ with $i' \neq i$ are trivial on $\mathcal{Y}_{j_i,t_i}$, the data (a)–(c) form a family as in Set-up 13.5. The weights in (b) belong to I_0.

Step 4. In this step we conclude. Let $\mathcal{S}'$ be the set of Theorem 13.8 for the finitely many families of **Step 3**, indexed by $\mathbf{j} \in J_{w'_1} \times \cdots \times J_{w'_{n'}}$ with $(w'_1, \dots, w'_{n'}) \in \mathcal{W}_{\mathcal{S}}$.

Let $(X, B) \in \mathcal{S}$ over F. By **Step 2**, for any $1 \leq i \leq n$ there exist $j_i \in J_{w_i}$, a geometric point $t_i : \operatorname{Spec} F \to T_{j_i}$, and an isomorphism $Y_i \cong \mathcal{Y}_{j_i,t_i}$ under which $D_i = \mathcal{D}_{j_i,t_i}$ and $B_i = \mathcal{B}_{j_i,t_i}$. Let $\mathbf{j} := (j_1, \dots, j_n)$ and $t := (t_1, \dots, t_n) : \operatorname{Spec} F \to T_{\mathbf{j}}$. Then

$$Y = \bigsqcup_{i=1}^{n} Y_i \cong \mathcal{Y}_{\mathbf{j},t},$$

and under the isomorphism $D_Y = \mathcal{D}_{\mathbf{j},t}$ and $B_Y = \mathcal{B}_{\mathbf{j},t}$. Since (X, B) is a projective slc pair of dimension 3 over F of characteristic $> P > 5$ and $K_X + B$ is ample, $(X, B) \in \mathcal{S}'$. Thus

$$\mathcal{S} \subset \mathcal{S}'.$$

By Theorem 13.8(1), $\mathcal{S}'$ is bounded over $\mathbb{Z}[1/30]$ with coefficients in the set of weights of **Step 3**, which is contained in I_0. Thus $\mathcal{S}$ is bounded over $\mathbb{Z}[1/30]$ with coefficients in I_0, and (1) holds.

If $I \subset \mathbb{Q}$, then $I_0 \subset \mathbb{Q}$, so the weights of **Step 3** are rational. By Theorem 13.8(2), there exists a positive integer N such that $N(K_X + B)$ is a very ample Cartier divisor for any $(X, B) \in \mathcal{S}'$, in particular for any $(X, B) \in \mathcal{S}$. Thus (2) holds. □

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